Big Three Comparison: Federer, Nadal, and Djokovic

Performance, pressure, playing style, and tactical evolution

A Data Science Project by Vishal Srikanth

Executive overview

A great serve can protect a lead; a great return can make every service game uncomfortable. Longer rallies, a change of surface, or one break point can ask a different question of the same player. The Big Three debate becomes more revealing when those questions are kept separate before they are combined.

How do Federer, Nadal, and Djokovic compare through performance, pressure, playing style, and adaptability—and how does the answer change with what we value?

This is an observational comparison of the supplied **charted-match sample**, not a reconstruction of complete professional careers. Individual coverage windows, unknown endings, and metric-specific eligibility accompany the results. The implementation pools event counts, reconstructs pressure situations, fits two explanatory logistic models with match-clustered uncertainty, and uses 2,000 paired match-bootstrap draws (seed 42). Style and tactical change remain descriptive; they receive no automatic greatness credit.

The principal findings, checked again at release: Federer has the highest observed overall match-win rate and DR; Djokovic has the highest primary Surface Adaptability Score; Nadal has the highest observed clay win rate. Federer has a narrow balanced-score point lead over Djokovic, while the adaptability-priority scenario reverses their order. None of the six planned pressure-model player comparisons survives Holm correction at 5%. These statements describe this sample and these definitions, not universally superior ability. The generated synthesis gives current magnitudes, paired difference intervals, rank uncertainty, and the tactical/rivalry context.

Unequal opponents, career stages, surfaces, chart selection, and missing outcomes can shift every comparison. Career achievements outside the files—including ranking weeks—remain unmeasured. The result is a defensible comparison of observed tennis and explicit preferences, not an all-history GOAT verdict.

Read the report: I. Scope · II. Sources · III. Audit · IV. Events · V. Performance · VI. Pressure · VII. Shots · VIII. Adaptability and evolution · IX. Profiles, rivalries and preferences · computed synthesis · X. Reproduction and Tableau.

Headline evidence is visible below each analysis; large audit tables remain embedded in expandable panels. Saved outputs support reading without execution; reproduction requires the identified local inputs.

I. Project scope and KPI specifications

1.1 Purpose and comparison scope

This report compares **Roger Federer, Rafael Nadal, and Novak Djokovic** using only the CSV files supplied in `Raw Data`, interpreted with the accompanying local data dictionary and charting workbook. The objective is to compare their observed performance, pressure outcomes, surface adaptability, playing styles, and tactical evolution, then explore how explicit preferences affect their relative ranking.

Each player is followed from their **earliest to latest eligible charted match available in these files**. These are individual coverage windows, not a shared calendar period and not necessarily their actual professional debut or final career match. The report applies no common calendar cutoff. Time-based sections may use shorter supported periods, which must be identified and justified.

The analysis concerns the Big Three and this charted sample. It cannot establish an all-history GOAT across players outside the comparison. Total wins and match counts are sample totals, not complete career records. Different opponents, surfaces, ages, and charting coverage can influence comparisons.

Performance, style, and evolution are separate analytical questions. Shot diversity and tactical change are descriptive; greater variety or greater change does not automatically mean better tennis. Any preference-based overall score must disclose its included performance metrics, normalization, weights, uncertainty, and limitations.

All analysis code, definitions, tables, charts, and interpretation belong in this notebook. Saved outputs make the report readable without execution; reproducing it requires the supplied local source files and Python dependencies. Documented optional Tableau exports are generated by notebook code in Section X. Weeks at No. 1 remain unavailable until separately sourced ranking history is explicitly authorized.

1.2 KPI dictionary: performance and player context

Notation: all sums use eligible observations for the stated player, period, and surface. Rates are displayed as percentages unless identified as ratios or indices. Numerators and denominators remain available beside displayed rates. Unknown outcomes are excluded from outcome-rate denominators, never counted as losses.

KPI	Definition and denominator	Surface applicability / interpretation
Charted coverage	Earliest and latest eligible match dates; eligible matches and matches with known outcomes	Overall and per surface; does not establish actual career boundaries
Charted wins	Count of distinct matches with a verified player win	Overall and per surface; no extrapolation to career wins
Match win percentage	Known wins / matches with known outcomes	Overall and per surface; show unknown outcomes separately
Wins per surface	Charted win counts on Clay, Grass, and Hard	Counts plus within-surface win percentages
Handedness	Consistent player metadata; audit conflicting values before selecting a label	Player attribute, not a surface metric
Weeks at No. 1	Not available in the supplied files	Overall; display unavailable, not zero or an estimate
Total point-win percentage	Points won / points with a valid winner in eligible charted sequences	Overall and per surface; report invalid or missing point outcomes
First-serve in percentage	First serves in / service points played	Overall and per surface
First-serve win percentage	First-serve points won / first serves in	Overall and per surface; conditional on the first serve landing in
Second-serve win percentage	Second-serve points won / second-serve opportunities, including double faults	Overall and per surface; validate the source field semantics
Service-point win percentage	Service points won / service points played	Overall and per surface
Return-point win percentage	Return points won / return points played	Overall and per surface
Ace percentage	Aces / service points played	Overall and per surface
Double-fault percentage	Double faults / service points played	Overall and per surface; optional second-serve-opportunity rate labeled separately
Dominance Ratio (DR)	Return-point win rate / (1 – service-point win rate), calculated from pooled counts	Overall and per surface; undefined at zero service-point loss rate, flag near-zero denominators
Break-point save percentage	Break points saved / break points faced	Overall and per surface; a pressure component, not proof of psychological toughness
Break-point conversion percentage	Break points won while returning / break-point opportunities while returning	Overall and per surface; validate event/source reconciliation

KPI	Definition and denominator	Surface applicability / interpretation
Service hold / return break percentage	Service games held / completed service games; return games won / completed return games	Overall and per surface if game reconstruction is validated; exclude tiebreaks and incomplete games
Tiebreak win percentage	Completed tiebreaks won / completed tiebreaks played	Overall and per surface; distinguish tiebreaks from individual tiebreak points
Deciding-set win percentage	Completed deciding sets won / completed deciding sets played	Overall and per surface; distinguish match format and identify special match-tiebreak formats
Clutch Index (CI)	0.50 × break-point save rate + 0.30 × tiebreak win rate + 0.20 × deciding-set win rate	Overall and per surface when all three components are supported; show each component and its sample size

CI weights are a declared project preference, not empirically established measures of mental strength. Any comparison with ordinary service/return performance is a separate analysis. Serve speed is not measured in these CSVs: remove supplied speed estimates from evidence-based comparisons and do not score them.

1.2 KPI dictionary: adaptability, shot patterns, and style

KPI	Definition	Surface applicability / interpretation
Surface win-rate evidence	Win percentage and known-outcome match count on each of Clay, Grass, and Hard	Required evidence for the overall adaptability score
Surface parity	Let p be overall known-outcome win rate and p_s each surface rate. Parity = 100 × [1 – sqrt(mean_s((p_s – p)²))]	One overall measure requiring all three surfaces; consistency alone does not measure strength
Surface Adaptability Score (SAS)	0.60 × parity + 0.40 × [100 × min_s(p_s)]	Primary overall adaptability score, 0–100; show parity, weakest-surface rate, and sensitivity to weights
Surface performance profile	Clay, Grass, and Hard win rates, DR, and serve/return measures	Per-surface evidence; do not call a single-surface score cross-surface adaptability
Shot-type mix	Shares of parsed rally strokes by stroke family: groundstroke, slice, volley, overhead, drop shot, lob, half-volley, swinging volley, and other/unknown; retain forehand/backhand detail where coded	Overall and per surface; exclude serves, preserve unknown categories, distinguish attempted error strokes from successful strokes
Directional mix	Crosscourt, middle, down-the-line, inside-out, and inside-in counts / classified directional strokes from the appropriate statistical slice	Overall and per surface; direction-code target zones are a separate classification
Shot-type diversity	Normalized Shannon entropy H = –sum(p_k log(p_k)) / log(K), using the same fixed, mutually exclusive known shot categories for every player	Overall and per surface; 0–1 descriptive index, unknown share reported separately; fixed K > 1
Directional diversity	The same entropy formula over the five fixed directional categories, with shares calculated from pooled counts	Overall and per surface; display the underlying shares and classification coverage

KPI	Definition	Surface applicability / interpretation
Conditional directional diversity	Directional distribution and entropy within forehands, backhands, and returns where the available source grain supports it	Overall and per surface; do not invent shot-type/direction combinations unavailable in the data
Shot placement percentages	Counts in known chart-code target zones / strokes with classified target zones; serve zones reported separately	Overall and per surface; preserve unknown-direction share and actual depth coverage
Placement illustrations	Schematic categorical court strips and recorded return-depth/target heatmaps; no fabricated coordinates	Overall and per surface; not observed tracking, movement, or landing precision; numerical zone shares use observed codes
Net approach / net-point frequency	Approach points or net points / all eligible player points on matched source coverage	Overall and per surface; keep Approach and NetPoints slices distinct
Net-point success	Net points won / net points played, using the same source slice	Overall and per surface
Serve-and-volley frequency	Serve-and-volley points / service points on supported coverage	Overall and per surface; missing source rows are not assumed zero without validation
Winner and unforced-error rates	Winners and unforced errors per eligible player point, reported separately with explicit ace inclusion/exclusion from source semantics	Overall and per surface; avoid treating a winner/error ratio as a complete aggression measure
Rally profile	Shares of valid rallies in fixed bins: 1–4, 5–8, and 9+ shots, using the validated common counting convention	Overall and per surface; if direct reconstruction is unsupported, label any aggregate proxy separately
Tactical evolution	Changes over career periods in approach, serve-and-volley, drop-shot and slice shares, serve placement, and rally profile	Overall trends plus within-surface comparisons where supported; counts and denominator definitions accompany every change
Playing-style profile	Evidence from serving, returning, net activity, aggression, rally tendency, and shot diversity	Overall and per surface; transparent descriptions rather than inferred athleticism or hard-coded model labels

Definition choices: SAS is the primary adaptability metric; parity is its supporting component, not a second competing headline score. Tactical evolution replaces the unsupported full tactical-innovation composite: these files cannot establish invention, equipment change, measured serve-speed variation, or influence on the sport. Evolution is expressed first as component changes, not a score that rewards change for its own sake. Career periods use fixed four-year bins anchored to each player’s first charted year, with the final period truncated to available coverage; boundaries, samples, and surface composition are reported in Section VIII. Any later scoring or period-definition change will be documented here.

1.3 Eligibility, missing data, sample sizes, and uncertainty

1. **Shared eligibility and grain.** Deduplicate exact repeats. Quarantine genuinely conflicting match, point, and player-match keys at the documented affected scope; never choose a conflicting row arbitrarily. Preserve all statistical slices needed by downstream analyses. Match metadata must be unique by match ID; the master table must be unique by player and match; point keys and statistical keys must be validated at their actual grain. Publish exclusions and checks against source totals.
2. **Outcomes and missingness.** Verify inferred match, set, and game outcomes against complete sequence state where possible. Exclude unknown outcomes only from metrics needing those outcomes; do not discard otherwise usable statistics automatically. Distinguish structural zero, unknown value, unavailable source, and ineligible observation. A zero denominator returns unavailable, not zero. Full CI requires all three valid components; any exploratory available-component score is separately named and never silently substituted.
3. **Pooled counts and matched coverage.** Compute headline rates from summed numerators and denominators, not unweighted averages of match percentages. Use matched eligible observations for each metric's numerator and denominator. Display coverage as eligible contributing matches / relevant eligible player-matches, along with point/event counts and unknown-category shares. Statistical slices with different meanings must not be combined or relabeled interchangeably.
4. **Surface and period scope.** Primary surface comparisons cover Clay, Grass, and Hard. Audit any other or unknown surface and disclose its treatment; do not silently lose it from overall totals. SAS requires all three named surfaces and uses an overall win rate restricted to that same three-surface universe. Player career windows remain individual. Show exact coverage for each analysis and flag partial recent periods. Evolution uses the fixed bins and reporting thresholds stated in Section VIII.
5. **Reporting thresholds.** Always show valid counts and underlying rates, even for small samples. Flag match-based rates below 30 known-outcome matches, point-based rates below 200 valid points, and event-based rates below 30 events as **small sample**. These are transparent reporting heuristics, not guarantees of precision or hypothesis-test cutoffs. Mark full CI small-sample if any of its three components has fewer than 30 events. Mark SAS small-sample if any surface has fewer than 30 known-outcome matches. Mark shot diversity small-sample below 200 classified strokes. Unsupported definitions or zero denominators remain unavailable regardless of sample size.
6. **Uncertainty.** Report 95% intervals for headline comparative rates and indices where estimable. Use match-cluster resampling for point/event aggregates and preserve both player observations when resampling a shared rivalry match. Recompute derived metrics in each replicate; disclose replicate count, seed, invalid replicates, and whether normalization bounds are held fixed. Use the same replicates across preference scenarios. Counts and categorical metadata need no inferential interval. Sparse metrics may have unstable or unavailable intervals, which must be shown explicitly.

7. **Model interpretation.** Validate event reconstruction before fitting pressure models. Account for serving status in tiebreak-point models and show supported opponent/context adjustments. Report direct player contrasts and their uncertainty rather than deciding differences from overlap of separate confidence intervals. Describe odds as odds, probabilities as probabilities, and logistic point-outcome models as such. A small collection of player summaries cannot establish intrinsic archetypes or athletic ability.
8. **Comparison and scoring.** Unequal charting coverage, opponents, career stages, and surface schedules constrain conclusions. Raw win totals are descriptive. Publish raw metrics before any normalized preference score; justify performance inputs, avoid rewarding descriptive variety/evolution automatically, disclose overlap between inputs, and test weight and surface-composition sensitivity. Conditional bootstrap intervals do not eliminate chart-selection bias. Report scenario-dependent rankings and uncertainty instead of declaring an objectively proven GOAT.

II. Setup and source ingestion

2.1 Libraries and configuration

Load the libraries used by the current analyses and declare the player, surface, and reproducibility settings. Configuration, package versions, and the reporting formatter are shared throughout the analysis. Warnings remain visible.

In [1]:
```
from pathlib import Path
import re
import platform
from importlib.metadata import version

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
from sklearn.linear_model import LogisticRegression
from IPython.display import display, HTML, Markdown
from html import escape

BIG_THREE = ["Roger Federer", "Rafael Nadal", "Novak Djokovic"]
PROJECT_SURFACES = ["Clay", "Grass", "Hard"]
RANDOM_SEED = 42
BOOTSTRAP_REPLICATES = 2000  # Target for the final validated comparison.
SMALL_SAMPLE_THRESHOLDS = {"matches": 30, "points": 200, "events": 30, "shots": 200}
raw_data_dir = Path("Raw Data")



def report_table(frame, caption=None, expanded=None):
```

```
    """Render a complete readable table; large evidence tables remain embedded in details panels."""
    if not isinstance(frame, pd.DataFrame):
        display(frame)
        return
    if frame.empty:
        display(HTML("<p><em>No records in this review table.</em></p>"))
        return
    p_formatters = {column: (lambda value: f"{value:.3g}" if pd.notna(value) else "Unavailable")
                    for column in frame.columns if "p_value" in str(column)}
    html = frame.to_html(index=not isinstance(frame.index, pd.RangeIndex), border=0, na_rep="Unavailable",
        float_format=lambda value: f"{value:.3g}" if 0 < abs(value) < .001 else f"{value:,.3f}", formatters=p_formatters,
        max_rows=None, max_cols=None, escape=True, classes="report-table")
    label = escape(caption or f"Complete evidence table: {len(frame):,} rows × {len(frame.columns)} columns")
    wrapped = '<div class="screen-table" style="overflow-x:auto;max-width:100%">'+html+'</div>'
    if len(frame.columns)>10:
        key_columns=[column for column in ["player","surface","scope","period","metric","distribution"] if column in frame.columns][:3]
        remaining=[column for column in frame.columns if column not in key_columns]
        chunks=[]
        for offset in range(0,len(remaining),7):
            columns=key_columns+remaining[offset:offset+7]
            chunks.append(frame[columns].to_html(index=not isinstance(frame.index,pd.RangeIndex),border=0,na_rep="Unavailable",
                float_format=lambda value:f"{value:.3g}" if 0<abs(value)<.001 else f"{value:,.3f}",
                formatters={column:formatter for column,formatter in p_formatters.items() if column in columns},escape=True,max_rows=None,max_cols=None))
        wrapped+='<div class="print-table">'+''.join(chunks)+'</div>'
    else:
        wrapped=wrapped.replace('class="screen-table"','class="single-table"')
    collapse = len(frame)>40 or len(frame.columns)>25 if expanded is None else not expanded
    display(HTML('<details><summary>'+label+'</summary>'+wrapped+'</details>' if collapse else (('<p class="table-caption"><strong>'+escape(caption)+'</strong>
</p>') if caption else '')+wrapped))

REPORT_CSS = """
<style>
.print-table {display:none}
.report-table th {white-space:normal; overflow-wrap:anywhere}
details summary {cursor:pointer; margin:0.5em 0; font-weight:600}
@media print {
  @page {size:A4 landscape; margin:12mm}
  .screen-table {display:none!important} .print-table {display:block!important}
  .print-table table {margin:0 0 10mm 0; table-layout:fixed; width:100%; font-size:7pt}
  table {max-width:100%; font-size:8pt; border-collapse:collapse}
  th,td {overflow-wrap:anywhere; white-space:normal!important; padding:3px!important}
  thead {display:table-header-group} tr {break-inside:avoid}
  img {max-width:100%!important; max-height:170mm; object-fit:contain}
  h1,h2,h3,h4,summary {break-after:avoid}
  pre {white-space:pre-wrap!important; overflow-wrap:anywhere; font-size:7pt!important}
  .jp-OutputArea-output,.jp-Cell-outputWrapper {overflow:visible!important}
}
</style>
"""
display(HTML(REPORT_CSS))
runtime_versions = pd.DataFrame(
    [{"package": "Python", "version": platform.python_version()}]
    + [{"package": package, "version": version(package)}
       for package in ["pandas", "numpy", "matplotlib", "seaborn", "scipy", "scikit-learn"]]
)
print("Runtime versions for reproducing this saved execution:")
report_table(runtime_versions)
```

Runtime versions for reproducing this saved execution:

package	version
Python	3.10.7
pandas	2.2.2
numpy	1.26.4
matplotlib	3.8.4
seaborn	0.13.2
scipy	1.13.0
scikit-learn	1.4.2

The runtime table records the environment used for these saved setup outputs. Shared constants specify the executed analysis configuration. Warnings remain visible.

2.2 Load and preserve source tables

Require all 19 expected CSV files and check the columns needed for ingestion and coverage reporting. Each source is loaded without filtering or deduplication. `raw_dataframes_unfiltered` preserves the original inputs; `raw_dataframes` starts as independent working copies for compatibility with the existing downstream audit. No source CSV is modified.

In [2]:
```
csv_filenames = ['charting-m-matches.csv', 'charting-m-points-2010s.csv', 'charting-m-points-2020s.csv', 'charting-m-points-to-2009.csv', 'charting-m-stats-
KeyPointsReturn.csv', 'charting-m-stats-KeyPointsServe.csv', 'charting-m-stats-NetPoints.csv', 'charting-m-stats-Overview.csv', 'charting-m-stats-Rally.csv',
'charting-m-stats-ReturnDepth.csv', 'charting-m-stats-ReturnOutcomes.csv', 'charting-m-stats-ServeBasics.csv', 'charting-m-stats-ServeDirection.csv', 'charting-
m-stats-ServeInfluence.csv', 'charting-m-stats-ShotDirection.csv', 'charting-m-stats-ShotDirOutcomes.csv', 'charting-m-stats-SnV.csv', 'charting-m-stats-
SvBreakSplit.csv', 'charting-m-stats-SvBreakTotal.csv']

missing_files = [str(raw_data_dir / name) for name in csv_filenames
                 if not (raw_data_dir / name).is_file()]
if missing_files:
    raise FileNotFoundError("Missing expected source files: " + ", ".join(missing_files))

required_source_columns = {
    "charting_m_matches": {"match_id", "Player 1", "Player 2", "Date", "Surface",
                           "Pl 1 hand", "Pl 2 hand", "Best of"},
    "points": {"match_id", "Pt", "Set1", "Set2", "Gm1", "Gm2", "Pts", "Gm#",
               "TbSet", "Svr", "1st", "2nd", "PtWinner"},
    "statistics": {"match_id", "player"},
    "charting_m_stats_Rally": {"match_id", "server", "returner", "row", "pts"},
}
loaded_sources = {}
for filename in csv_filenames:
    dataset_key = Path(filename).stem.replace("-", "_")
    frame = pd.read_csv(raw_data_dir / filename, low_memory=False)
```

```
    source_kind = (dataset_key if dataset_key in {"charting_m_matches", "charting_m_stats_Rally"} else
                   "points" if dataset_key.startswith("charting_m_points") else "statistics")
    missing_columns = required_source_columns[source_kind] - set(frame.columns)
    if missing_columns:
        raise ValueError(f"{filename}: missing required columns {sorted(missing_columns)}")
    if frame.empty:
        raise ValueError(f"{filename}: source contains no rows")
    loaded_sources[dataset_key] = frame

if len(loaded_sources) != 19:
    raise AssertionError("Expected 19 distinct source tables")
raw_dataframes_unfiltered = loaded_sources
raw_dataframes = {key: frame.copy(deep=True)
                  for key, frame in raw_dataframes_unfiltered.items()}
del loaded_sources, frame

source_inventory = pd.DataFrame([
    {"dataset": key, "rows": len(frame), "columns": frame.shape[1],
     "distinct_nonmissing_match_ids": frame["match_id"].nunique(),
     "missing_match_id_rows": int(frame["match_id"].isna().sum()),
     "missing_player_rows": int(frame["player"].isna().sum()) if "player" in frame else pd.NA}
    for key, frame in raw_dataframes_unfiltered.items()
]).sort_values("dataset").reset_index(drop=True)
print("Loaded 19 original source tables; independent working copies created.")
print("Raw inventory: row counts include duplicates and conflicting records pending Step 3.")
report_table(source_inventory)
```

```
Loaded 19 original source tables; independent working copies created.
Raw inventory: row counts include duplicates and conflicting records pending Step 3.
```

dataset	rows	columns	distinct_nonmissing_match_ids	missing_match_id_rows	missing_player_rows
charting_m_matches	7566	15	7565	0	<NA>
charting_m_points_2010s	357229	14	2232	0	<NA>
charting_m_points_2020s	547478	14	3337	0	<NA>
charting_m_points_to_2009	379569	14	1963	0	<NA>
charting_m_stats_KeyPointsReturn	60464	8	7545	0	0
charting_m_stats_KeyPointsServe	60464	12	7545	0	0
charting_m_stats_NetPoints	59396	11	7541	0	0
charting_m_stats_Overview	56850	20	7545	0	0
charting_m_stats_Rally	96706	13	7545	0	<NA>
charting_m_stats_ReturnDepth	270002	12	7545	0	0
charting_m_stats_ReturnOutcomes	310496	11	7545	0	0
charting_m_stats_ServeBasics	45341	12	7544	0	0
charting_m_stats_ServeDirection	45341	15	7544	0	0
charting_m_stats_ServeInfluence	30235	14	7546	0	0
charting_m_stats_ShotDirOutcomes	174687	10	7491	0	0
charting_m_stats_ShotDirection	59997	8	7547	0	0
charting_m_stats_SnV	51300	14	6327	0	0
charting_m_stats_SvBreakSplit	164871	15	7544	0	0
charting_m_stats_SvBreakTotal	164871	11	7544	0	0

The inventory reports source rows, columns, unique nonmissing match IDs, and missing identifier fields. Different statistical slices can produce multiple rows per match, so source row totals are not interchangeable with match counts. These are pre-audit inputs; eligibility and exclusions are determined in Section III.

2.3 Source and player coverage

Coverage previews use unambiguous match metadata and distinct player-match membership. Statistical-source presence means at least one row exists for that player-match; it does not guarantee the required metric slice is available. Point-source percentages describe each era file separately and must not be added without checking overlap. Final analytical coverage follows the shared audit.

In [3]:
```
# Coverage is descriptive only: ambiguous metadata is isolated, not resolved here.
source_matches = raw_dataframes_unfiltered["charting_m_matches"].copy()
```

```
metadata_conflict_mask = source_matches.drop_duplicates().duplicated("match_id", keep=False)
metadata_conflict_ids = set(source_matches.drop_duplicates().loc[metadata_conflict_mask, "match_id"])
coverage_matches = source_matches.loc[
    source_matches["match_id"].notna()
    & ~source_matches["match_id"].isin(metadata_conflict_ids)
].drop_duplicates("match_id").copy()
coverage_matches["match_date"] = pd.to_datetime(
    coverage_matches["Date"].astype("string").str.strip(), format="%Y%m%d", errors="coerce")
coverage_matches["surface"] = coverage_matches["Surface"].astype("string").str.strip().str.title()
coverage_matches["surface"] = coverage_matches["surface"].replace(
    {"Hard Court": "Hard", "Grass Court": "Grass", "Clay Court": "Clay"}).fillna("Unknown")
coverage_participation = pd.concat([
    coverage_matches[["match_id", "match_date", "surface", column]].rename(columns={column: "player"})
    for column in ["Player 1", "Player 2"]
], ignore_index=True)
coverage_participation = coverage_participation[
    coverage_participation["player"].isin(BIG_THREE)
].drop_duplicates(["match_id", "player"])
player_source_coverage = coverage_participation.groupby("player", as_index=False).agg(
    charted_matches=("match_id", "nunique"), earliest_charted_date=("match_date", "min"),
    latest_charted_date=("match_date", "max"), missing_date_matches=("match_date", lambda x: int(x.isna().sum())))
player_surface_source_coverage = coverage_participation.groupby(["player", "surface"], as_index=False).agg(
    charted_matches=("match_id", "nunique"), earliest_charted_date=("match_date", "min"),
    latest_charted_date=("match_date", "max"))

# Membership sets avoid concatenating million-row point tables just to describe coverage.
source_match_id_sets = {key: set(frame["match_id"].dropna())
                        for key, frame in raw_dataframes_unfiltered.items()}
source_player_match_sets = {
    key: set(frame[["match_id", "player"]].dropna().itertuples(index=False, name=None))
    for key, frame in raw_dataframes_unfiltered.items() if "player" in frame
}
rally_source = raw_dataframes_unfiltered["charting_m_stats_Rally"]
source_player_match_sets["charting_m_stats_Rally"] = (
    set(rally_source[["match_id", "server"]].dropna().itertuples(index=False, name=None))
    | set(rally_source[["match_id", "returner"]].dropna().itertuples(index=False, name=None))
)
coverage_rows = []
for player in BIG_THREE:
    player_ids = set(coverage_participation.loc[coverage_participation["player"].eq(player), "match_id"])
    for key in raw_dataframes_unfiltered:
        supported_ids = ({mid for mid in player_ids if (mid, player) in source_player_match_sets[key]}
                         if key in source_player_match_sets else player_ids & source_match_id_sets[key])
        coverage_rows.append({"player": player, "dataset": key,
                              "metadata_matches": len(player_ids),
                              "matches_present_in_source": len(supported_ids),
                              "source_presence_pct": 100 * len(supported_ids) / len(player_ids) if player_ids else np.nan})
player_source_presence = pd.DataFrame(coverage_rows)
source_date_quality = pd.DataFrame([{
    "raw_metadata_rows": len(source_matches),
    "ambiguous_metadata_match_ids": len(metadata_conflict_ids),
    "unambiguous_metadata_matches": len(coverage_matches),
    "unparseable_or_missing_date_matches": int(coverage_matches["match_date"].isna().sum()),
    "earliest_source_match_date": coverage_matches["match_date"].min(),
    "latest_source_match_date": coverage_matches["match_date"].max(),
}])
print("Metadata date coverage (ambiguous match IDs excluded from this preview):")
```

```
report_table(source_date_quality)
print("Individual Big Three charted coverage; dates are not verified professional debut/retirement dates:")
report_table(player_source_coverage)
print("Big Three coverage by surface, including any unknown/other surfaces:")
report_table(player_surface_source_coverage)
print("Source presence by player (any statistical slice; not validated KPI coverage):")
report_table(player_source_presence)
```

```
Metadata date coverage (ambiguous match IDs excluded from this preview):
```

raw_metadata_rows	ambiguous_metadata_match_ids	unambiguous_metadata_matches	unparseable_or_missing_date_matches	earliest_source_match_date	latest_source_match_date
7566	1	7564	0	1960-05-29	2026-05-21

```
Individual Big Three charted coverage; dates are not verified professional debut/retirement dates:
```

player	charted_matches	earliest_charted_date	latest_charted_date	missing_date_matches
Novak Djokovic	553	2005-01-17	2026-03-11	0
Rafael Nadal	425	2003-05-15	2024-11-19	0
Roger Federer	723	1998-07-04	2021-07-05	0

```
Big Three coverage by surface, including any unknown/other surfaces:
```

player	surface	charted_matches	earliest_charted_date	latest_charted_date
Novak Djokovic	Clay	129	2006-04-19	2025-06-06
Novak Djokovic	Grass	58	2007-07-05	2025-07-11
Novak Djokovic	Hard	366	2005-01-17	2026-03-11
Rafael Nadal	Clay	166	2003-05-15	2024-07-29
Rafael Nadal	Grass	40	2003-06-27	2022-07-04
Rafael Nadal	Hard	219	2004-03-28	2024-11-19
Roger Federer	Clay	130	1999-07-16	2021-06-05
Roger Federer	Grass	119	1998-07-04	2021-07-05
Roger Federer	Hard	474	1998-10-05	2021-03-10

```
Source presence by player (any statistical slice; not validated KPI coverage):
```

▼ Complete evidence table: 57 rows × 5 columns

player	dataset	metadata_matches	matches_present_in_source	source_presence_pct
Roger Federer	charting_m_matches	723	723	100.000
Roger Federer	charting_m_points_2010s	723	344	47.580
Roger Federer	charting_m_points_2020s	723	14	1.936
Roger Federer	charting_m_points_to_2009	723	364	50.346
Roger Federer	charting_m_stats_KeyPointsReturn	723	722	99.862
Roger Federer	charting_m_stats_KeyPointsServe	723	722	99.862
Roger Federer	charting_m_stats_NetPoints	723	722	99.862
Roger Federer	charting_m_stats_Overview	723	722	99.862
Roger Federer	charting_m_stats_Rally	723	722	99.862
Roger Federer	charting_m_stats_ReturnDepth	723	722	99.862
Roger Federer	charting_m_stats_ReturnOutcomes	723	722	99.862
Roger Federer	charting_m_stats_ServeBasics	723	722	99.862
Roger Federer	charting_m_stats_ServeDirection	723	722	99.862
Roger Federer	charting_m_stats_ServeInfluence	723	722	99.862
Roger Federer	charting_m_stats_ShotDirection	723	722	99.862
Roger Federer	charting_m_stats_ShotDirOutcomes	723	719	99.447
Roger Federer	charting_m_stats_SnV	723	669	92.531
Roger Federer	charting_m_stats_SvBreakSplit	723	722	99.862
Roger Federer	charting_m_stats_SvBreakTotal	723	722	99.862
Rafael Nadal	charting_m_matches	425	425	100.000
Rafael Nadal	charting_m_points_2010s	425	262	61.647
Rafael Nadal	charting_m_points_2020s	425	51	12.000
Rafael Nadal	charting_m_points_to_2009	425	110	25.882
Rafael Nadal	charting_m_stats_KeyPointsReturn	425	424	99.765
Rafael Nadal	charting_m_stats_KeyPointsServe	425	424	99.765
Rafael Nadal	charting_m_stats_NetPoints	425	425	100.000
Rafael Nadal	charting_m_stats_Overview	425	424	99.765
Rafael Nadal	charting_m_stats_Rally	425	424	99.765
Rafael Nadal	charting_m_stats_ReturnDepth	425	424	99.765
Rafael Nadal	charting_m_stats_ReturnOutcomes	425	423	99.529

player	dataset	metadata_matches	matches_present_in_source	source_presence_pct
Rafael Nadal	charting_m_stats_ServeBasics	425	423	99.529
Rafael Nadal	charting_m_stats_ServeDirection	425	423	99.529
Rafael Nadal	charting_m_stats_ServeInfluence	425	425	100.000
Rafael Nadal	charting_m_stats_ShotDirection	425	425	100.000
Rafael Nadal	charting_m_stats_ShotDirOutcomes	425	422	99.294
Rafael Nadal	charting_m_stats_SnV	425	174	40.941
Rafael Nadal	charting_m_stats_SvBreakSplit	425	423	99.529
Rafael Nadal	charting_m_stats_SvBreakTotal	425	423	99.529
Novak Djokovic	charting_m_matches	553	553	100.000
Novak Djokovic	charting_m_points_2010s	553	283	51.175
Novak Djokovic	charting_m_points_2020s	553	197	35.624
Novak Djokovic	charting_m_points_to_2009	553	72	13.020
Novak Djokovic	charting_m_stats_KeyPointsReturn	553	550	99.458
Novak Djokovic	charting_m_stats_KeyPointsServe	553	550	99.458
Novak Djokovic	charting_m_stats_NetPoints	553	551	99.638
Novak Djokovic	charting_m_stats_Overview	553	550	99.458
Novak Djokovic	charting_m_stats_Rally	553	550	99.458
Novak Djokovic	charting_m_stats_ReturnDepth	553	550	99.458
Novak Djokovic	charting_m_stats_ReturnOutcomes	553	550	99.458
Novak Djokovic	charting_m_stats_ServeBasics	553	550	99.458
Novak Djokovic	charting_m_stats_ServeDirection	553	550	99.458
Novak Djokovic	charting_m_stats_ServeInfluence	553	551	99.638
Novak Djokovic	charting_m_stats_ShotDirection	553	552	99.819
Novak Djokovic	charting_m_stats_ShotDirOutcomes	553	548	99.096
Novak Djokovic	charting_m_stats_SnV	553	328	59.313
Novak Djokovic	charting_m_stats_SvBreakSplit	553	550	99.458
Novak Djokovic	charting_m_stats_SvBreakTotal	553	550	99.458

2.4 Handedness: source evidence and explicit adjudication

The original all-values-must-agree rule made Nadal unavailable because three records say `R`. Stripping whitespace cannot resolve that genuine conflict. We use a declared player-profile rule: at least 30 recognized records and at least 99% agreement, otherwise unavailable. This is **source-consensus adjudication**, not independent historical verification. Missing values do not vote; only unambiguous labels are normalized. All contradictory and missing records remain visible, and the original CSV and match-level codes remain unchanged.

The result is a player attribute shared across surface scorecards. Changing this label does not reverse parsed target codes or recompute source directional classifications. The three contradictory hand records remain a potential limitation of source-derived directional summaries; the code-based target-side analysis is independent of handedness. No external profile dataset is introduced.

In [4]:
```
# Profile metadata is adjudicated separately from match records; raw contradictions survive.
def normalize_hand(value):
    if pd.isna(value) or not str(value).strip():
        return pd.NA
    return {"R":"R", "RIGHT":"R", "RIGHT-HANDED":"R", "L":"L", "LEFT":"L", "LEFT-HANDED":"L"}.get(str(value).strip().upper(), pd.NA)
hand_parts=[]
for slot in [1,2]:
    part=source_matches.loc[source_matches[f"Player {slot}"].isin(BIG_THREE),
        ["match_id",f"Player {slot}",f"Pl {slot} hand"]].copy()
    part.columns=["match_id","player","raw_hand"]
    part["source_column"]=f"Pl {slot} hand"
    hand_parts.append(part)
handedness_source_records=pd.concat(hand_parts,ignore_index=True)
handedness_source_records["raw_representation"]=handedness_source_records["raw_hand"].map(repr)
handedness_source_records["normalized_hand"]=handedness_source_records["raw_hand"].map(normalize_hand)
handedness_raw_values=handedness_source_records.groupby(
    ["player","source_column","raw_representation"],as_index=False,dropna=False).size().rename(columns={"size":"records"})
profile_rows=[]
for player in BIG_THREE:
    group=handedness_source_records.loc[handedness_source_records["player"].eq(player)]
    counts=group["normalized_hand"].value_counts()
    dominant=counts.index[0] if len(counts) else None
    supporting=int(counts.iloc[0]) if len(counts) else 0
    known=int(counts.sum())
    # Declared source-consensus adjudication; no majority rule for weak/disputed metadata.
    supported=known>=30 and supporting/known>=.99 and (len(counts)==1 or counts.iloc[0]>counts.iloc[1])
    profile_rows.append({"player":player,"hand_code":dominant if supported else pd.NA,
        "handedness": {"R":"Right-handed","L":"Left-handed"}.get(dominant,"Unavailable") if supported else "Unavailable/conflicting",
        "supporting_records":supporting,"recognized_records":known,"conflicting_records":known-supporting,
        "missing_or_unrecognized_records":int(group["normalized_hand"].isna().sum()),
        "consensus_pct":100*supporting/known if known else np.nan,
        "resolution_status":"documented source consensus; contradictions retained" if supported and len(counts)>1 else "consistent recognized source records" if
supported else "unresolved",
        "evidence_source":"Raw Data/charting-m-matches.csv; both participant slots"})
player_metadata=pd.DataFrame(profile_rows)
```

```
handedness_source_records=handedness_source_records.merge(player_metadata[["player","hand_code"]],on="player",validate="many_to_one")
handedness_source_records["exception_type"]=np.select(
    [handedness_source_records["normalized_hand"].isna(),
     handedness_source_records["normalized_hand"].fillna("").ne(handedness_source_records["hand_code"].fillna(""))],
    ["missing/unrecognized","contradicts adjudicated player profile"],default="consistent")
handedness_exceptions=handedness_source_records.loc[handedness_source_records["exception_type"].ne("consistent"),
    ["match_id","player","source_column","raw_representation","normalized_hand","hand_code","exception_type"]].copy()
handedness_conflicts=handedness_exceptions.loc[handedness_exceptions["exception_type"].eq("contradicts adjudicated player profile")]
assert normalize_hand(" l ")=="L" and normalize_hand("RIGHT-HANDED")=="R" and pd.isna(normalize_hand("?"))
assert not player_metadata.duplicated("player").any()
assert not handedness_source_records.duplicated(["match_id","player","source_column"]).any()
print("Raw handedness values by original column (repr exposes whitespace):")
report_table(handedness_raw_values,expanded=True)
print("Adjudicated player profiles; no raw hand record is overwritten:")
report_table(player_metadata,expanded=True)
print("All handedness exceptions, including genuine contradictory codes and missing records:")
report_table(handedness_exceptions,expanded=True)
```

```
Raw handedness values by original column (repr exposes whitespace):
```

player	source_column	raw_representation	records
Novak Djokovic	Pl 1 hand	'R'	419
Novak Djokovic	Pl 1 hand	nan	1
Novak Djokovic	Pl 2 hand	'R'	133
Rafael Nadal	Pl 1 hand	'L '	1
Rafael Nadal	Pl 1 hand	'L'	198
Rafael Nadal	Pl 1 hand	'R'	1
Rafael Nadal	Pl 1 hand	nan	2
Rafael Nadal	Pl 2 hand	'L '	1
Rafael Nadal	Pl 2 hand	'L'	220
Rafael Nadal	Pl 2 hand	'R'	2
Roger Federer	Pl 1 hand	'R '	1
Roger Federer	Pl 1 hand	'R'	487
Roger Federer	Pl 1 hand	nan	1
Roger Federer	Pl 2 hand	'R'	234

```
Adjudicated player profiles; no raw hand record is overwritten:
```

player	hand_code	handedness	supporting_records	recognized_records	conflicting_records	missing_or_unrecognized_records	consensus_pct	resolution_status	evidence_source
Roger Federer	R	Right-handed	722	722	0	1	100.000	consistent recognized source records	Raw Data/charting-m-matches.csv; both participant slots
Rafael Nadal	L	Left-handed	420	423	3	2	99.291	documented source consensus; contradictions retained	Raw Data/charting-m-matches.csv; both participant slots
Novak Djokovic	R	Right-handed	552	552	0	1	100.000	consistent recognized source records	Raw Data/charting-m-matches.csv; both participant slots

All handedness exceptions, including genuine contradictory codes and missing records:

	match_id	player	source_column	raw_representation	normalized_hand	hand_code	exception_type
186	20210210-M-Australian_Open-R64-Novak_Djokovic-Francis_Tiafoe	Novak Djokovic	Pl 1 hand	nan	<NA>	R	missing/unrecognized
270	20190122-M-Australian_Open-QF-Rafael_Nadal-Francis_Tiafoe	Rafael Nadal	Pl 1 hand	nan	<NA>	L	missing/unrecognized
349	20170611-M-Roland_Garros-F-Rafael_Nadal-Stan_Wawrinka	Rafael Nadal	Pl 1 hand	'R'	R	L	contradicts adjudicated player profile
372	20170327-M-Miami_Masters-R32-Roger_Federer-Juan_Martin_Del_Potro	Roger Federer	Pl 1 hand	nan	<NA>	R	missing/unrecognized
526	20140624-M-Wimbledon-R128-Rafael_Nadal-Martin_Klizan	Rafael Nadal	Pl 1 hand	nan	<NA>	L	missing/unrecognized
1497	20100831-M-US_Open-R128-Teymuraz_Gabashvili-Rafael_Nadal	Rafael Nadal	Pl 2 hand	'R'	R	L	contradicts adjudicated player profile
1507	20100512-M-Madrid_Masters-R16-John_Isner-Rafael_Nadal	Rafael Nadal	Pl 2 hand	'R'	R	L	contradicts adjudicated player profile

The source preview identifies the observed windows for Federer, Nadal, and Djokovic. The table reports their exact earliest and latest charted dates. These are source coverage boundaries, not verified career boundaries. Later analytical samples may be smaller because of missing statistics, missing points, or audit exclusions.

One ambiguous match ID is excluded only from this coverage preview; preserved sources are unchanged. The Rally source identifies participants through `server` and `returner`, unlike the player-keyed statistical tables. Source presence respects that distinction but does not establish metric-specific coverage.

III. Shared data audit

Audit question: Can each observation be traced to one coherent match, participant, and statistical slice? Duplicate chart rows would otherwise over-weight selected matches and produce falsely precise comparisons.

3.1 Full-grain duplicate and eligibility policy

Audit every statistical slice, not only Total rows: retain `row` and `set` in the key wherever present. Rally uses `(match_id, server, returner, row)` rather than a player key. Exact repeats are removed before identifying conflicts. Conflicting match metadata or point sequences quarantine the entire affected match across sources; conflicting statistics quarantine the affected player-match across player-keyed statistics and Rally. Point records for an otherwise coherent match remain available, with player-statistic exclusions recorded separately.

Missing key rows and orphan source records are excluded and counted. Invalid point numbers or participant/winner codes make their match ineligible because sequence-based outcomes require coherent points. Metadata requires valid dates and distinct named participants; invalid identities/dates quarantine that match. Handedness whitespace/case is standardized only in working metadata; unknown handedness and match formats are flagged for metric-specific handling, not grounds to discard unrelated statistics. Unknown surfaces are reported rather than silently filtered. Numerical statistic anomalies are retained as explicit review flags, not silently repaired. All rules run from preserved originals and are repeatable. Full excluded source rows are retained in `audit_quarantined_rows`; point duplicates preserve their source provenance.

In [5]:
```
def filter_stat_slice(frame, row_value=None, set_value=None):
    # A compatibility helper for downstream sections; the shared audit never slices.
    if isinstance(frame, dict):
        frame = frame["charting_m_stats_Overview"]
    selected = frame.copy()
    for column, value in [("row", row_value), ("set", set_value)]:
        if value is not None and column in selected:
            selected = selected[selected[column].astype("string").str.strip().str.casefold().eq(value.casefold())]
    return selected.copy()


def audit_duplicate_key_frame(frame, key_columns, exact_columns=None):
    exact = frame.duplicated(subset=exact_columns, keep="first")
    dedup = frame.loc[~exact].copy()
    missing = dedup[key_columns].isna().any(axis=1)
    for column in key_columns:
        missing |= dedup[column].astype("string").str.strip().eq("").fillna(False)
```

```
    conflicts = ~missing & dedup.duplicated(key_columns, keep=False)
    return dedup.loc[~missing & ~conflicts].copy(), exact, dedup.loc[missing].copy(), dedup.loc[conflicts].copy()


# Explicit provenance and full statistical grain, built from originals on every run.
point_source_keys = sorted(key for key in raw_dataframes_unfiltered if key.startswith("charting_m_points"))
STAT_SOURCE_CONFIG = {key: {"row": None, "set": None}
                      for key in raw_dataframes_unfiltered if "_stats_" in key}
stat_key_columns = {}
analysis_dataframes = {}
audit_quarantined_rows = {}
audit_exact_duplicate_rows = {}
audit_records = {}
clean_sources = {}
for key, original in raw_dataframes_unfiltered.items():
    if key in point_source_keys:
        continue
    if key == "charting_m_matches":
        grain = ["match_id"]
    elif key == "charting_m_stats_Rally":
        grain = ["match_id", "server", "returner", "row"]
    else:
        grain = ["match_id", "player"] + [c for c in ["row", "set"] if c in original]
        stat_key_columns[key] = grain
    clean, exact, missing, conflicts = audit_duplicate_key_frame(original, grain)
    clean_sources[key] = clean
    audit_exact_duplicate_rows[key] = original.loc[exact].copy()
    audit_quarantined_rows[key] = pd.concat([
        missing.assign(_audit_reason="missing key"),
        conflicts.assign(_audit_reason="conflicting key")], ignore_index=True)
    audit_records[key] = {"dataset": key, "key": ", ".join(grain), "raw_rows": len(original),
        "exact_duplicates_removed": int(exact.sum()), "missing_key_rows": len(missing),
        "conflicting_key_rows": len(conflicts), "conflicting_keys": len(conflicts[grain].drop_duplicates())}

matches_source = raw_dataframes_unfiltered["charting_m_matches"]
malformed_match_records = audit_quarantined_rows["charting_m_matches"].drop(columns="_audit_reason")
conflicted_match_ids = set(audit_quarantined_rows["charting_m_matches"].loc[
    lambda x: x["_audit_reason"].eq("conflicting key"), "match_id"].dropna().astype(str))
metadata = clean_sources["charting_m_matches"].copy()
for hand_column in ["Pl 1 hand", "Pl 2 hand"]:
    metadata[hand_column] = metadata[hand_column].astype("string").str.strip().str.upper()
clean_sources["charting_m_matches"] = metadata
metadata_dates = pd.to_datetime(metadata["Date"].astype("string"), format="%Y%m%d", errors="coerce")
metadata_invalid = (metadata_dates.isna()
    | metadata["Player 1"].isna() | metadata["Player 2"].isna()
    | metadata["Player 1"].astype("string").str.strip().eq("")
    | metadata["Player 2"].astype("string").str.strip().eq("")
    | metadata["Player 1"].eq(metadata["Player 2"]))
invalid_metadata_ids = set(metadata.loc[metadata_invalid, "match_id"].astype(str))

point_columns = list(raw_dataframes_unfiltered[point_source_keys[0]].columns)
if any(list(raw_dataframes_unfiltered[k].columns) != point_columns for k in point_source_keys):
    raise ValueError("Point sources have different schemas; reconcile before combining")
points_with_origin = pd.concat([
    raw_dataframes_unfiltered[key].assign(_audit_source=key, _audit_row=np.arange(len(raw_dataframes_unfiltered[key])))
    for key in point_source_keys], ignore_index=True)
points_clean, point_exact_mask, point_missing, point_conflict_rows = audit_duplicate_key_frame(
```

```
    points_with_origin, ["match_id", "Pt"], exact_columns=point_columns)
point_conflict_keys = set(point_conflict_rows[["match_id", "Pt"]].itertuples(index=False, name=None))
point_conflict_match_ids = set(point_conflict_rows["match_id"].dropna().astype(str))
point_missing_ids = set(point_missing["match_id"].dropna().astype(str))
point_numbers = pd.to_numeric(points_clean["Pt"], errors="coerce")
invalid_point_rows = (point_numbers.isna() | point_numbers.lt(1) | point_numbers.mod(1).ne(0)
                      | ~pd.to_numeric(points_clean["Svr"], errors="coerce").isin([1, 2])
                      | ~pd.to_numeric(points_clean["PtWinner"], errors="coerce").isin([1, 2]))
invalid_point_match_ids = set(points_clean.loc[invalid_point_rows, "match_id"].astype(str))
excluded_match_ids = (conflicted_match_ids | invalid_metadata_ids | point_conflict_match_ids
                      | point_missing_ids | invalid_point_match_ids)

all_player_match_conflicts = set()
for key in STAT_SOURCE_CONFIG:
    flagged = audit_quarantined_rows[key]
    if "player" in flagged:
        all_player_match_conflicts.update(flagged[["match_id", "player"]].dropna().itertuples(index=False, name=None))
    else:
        for role in ["server", "returner"]:
            all_player_match_conflicts.update(flagged[["match_id", role]].dropna().itertuples(index=False, name=None))

metadata_eligible = metadata.loc[~metadata["match_id"].astype(str).isin(excluded_match_ids)].copy()
valid_match_ids = set(metadata_eligible["match_id"])
participant_pairs = set(metadata_eligible[["match_id", "Player 1"]].itertuples(index=False, name=None)) | set(
    metadata_eligible[["match_id", "Player 2"]].itertuples(index=False, name=None))
for key, clean in clean_sources.items():
    reasons = pd.Series("", index=clean.index, dtype="string")
    reasons.loc[~clean["match_id"].isin(valid_match_ids)] = "shared match exclusion or orphan metadata"
    if "player" in clean:
        pairs = list(clean[["match_id", "player"]].itertuples(index=False, name=None))
        reasons.loc[[pair not in participant_pairs for pair in pairs]] = "player absent from eligible match metadata"
        reasons.loc[[pair in all_player_match_conflicts for pair in pairs]] = "shared player-match exclusion"
    elif key == "charting_m_stats_Rally":
        for role in ["server", "returner"]:
            pairs = list(clean[["match_id", role]].itertuples(index=False, name=None))
            reasons.loc[[pair not in participant_pairs for pair in pairs]] = "participant absent from eligible match metadata"
            reasons.loc[[pair in all_player_match_conflicts for pair in pairs]] = "shared player-match exclusion"
    if key == "charting_m_matches":
        reasons.loc[clean["match_id"].astype(str).isin(invalid_metadata_ids)] = "invalid match metadata"
    rejected = reasons.ne("")
    audit_quarantined_rows[key] = pd.concat([audit_quarantined_rows[key],
        clean.loc[rejected].assign(_audit_reason=reasons.loc[rejected])], ignore_index=True)
    analysis_dataframes[key] = clean.loc[~rejected].copy()

for key in point_source_keys:
    original = raw_dataframes_unfiltered[key]
    clean = points_clean.loc[points_clean["_audit_source"].eq(key)]
    rejected = ~clean["match_id"].isin(valid_match_ids)
    early = pd.concat([point_missing.assign(_audit_reason="missing point key"),
                       point_conflict_rows.assign(_audit_reason="conflicting point key")], ignore_index=True)
    audit_quarantined_rows[key] = pd.concat([
        early.loc[early["_audit_source"].eq(key)],
        clean.loc[rejected].assign(_audit_reason="shared match exclusion or orphan metadata")], ignore_index=True)
    exact_rows = points_with_origin.loc[point_exact_mask & points_with_origin["_audit_source"].eq(key)]
    audit_exact_duplicate_rows[key] = exact_rows.copy()
    analysis_dataframes[key] = clean.loc[~rejected, point_columns].copy()
    audit_records[key] = {"dataset": key, "key": "match_id, Pt (across all era files)",
```

```
        "raw_rows": len(original), "exact_duplicates_removed": len(exact_rows),
        "missing_key_rows": int(point_missing["_audit_source"].eq(key).sum()),
        "conflicting_key_rows": int(point_conflict_rows["_audit_source"].eq(key).sum()),
        "conflicting_keys": len(point_conflict_rows.loc[point_conflict_rows["_audit_source"].eq(key), ["match_id", "Pt"]].drop_duplicates())}

for key, record in audit_records.items():
    record["quarantined_rows"] = len(audit_quarantined_rows[key])
    record["eligible_rows"] = len(analysis_dataframes[key])
    record["rows_after_shared_quarantine"] = record["eligible_rows"]
    if record["raw_rows"] != record["exact_duplicates_removed"] + record["quarantined_rows"] + record["eligible_rows"]:
        raise AssertionError(f"Row accounting failed for {key}")
audit_summary = pd.DataFrame(audit_records.values()).sort_values("dataset").reset_index(drop=True)
reference_overview_before = filter_stat_slice(raw_dataframes_unfiltered["charting_m_stats_Overview"], None, "total")
reference_overview_after = filter_stat_slice(analysis_dataframes["charting_m_stats_Overview"], None, "total")
conflicted_big_three_player_matches = {pair for pair in all_player_match_conflicts if pair[1] in BIG_THREE}
duplicate_audit_exclusions = pd.DataFrame([
    {"measure": "conflicting match IDs", "count": len(conflicted_match_ids)},
    {"measure": "invalid metadata match IDs", "count": len(invalid_metadata_ids)},
    {"measure": "conflicting point keys (match_id, Pt)", "count": len(point_conflict_keys)},
    {"measure": "matches excluded due to point-key conflicts", "count": len(point_conflict_match_ids)},
    {"measure": "invalid point-code match IDs", "count": len(invalid_point_match_ids)},
    {"measure": "player-match exclusions across all statistical slices", "count": len(all_player_match_conflicts)},
    {"measure": "conflicting Big Three player-match keys", "count": len(conflicted_big_three_player_matches)},
    {"measure": "Overview total rows before shared quarantine", "count": len(reference_overview_before)},
    {"measure": "Overview player-match rows after shared quarantine", "count": len(reference_overview_after)},
])
duplicate_audit_state = {"conflicted_match_ids": conflicted_match_ids, "point_conflict_keys": point_conflict_keys,
    "point_conflict_match_ids": point_conflict_match_ids, "excluded_match_ids": excluded_match_ids,
    "player_match_conflicts": all_player_match_conflicts, "big_three_player_match_conflicts": conflicted_big_three_player_matches,
    "audit_summary": audit_summary, "exclusions": duplicate_audit_exclusions,
    "malformed_match_records": malformed_match_records, "overview_rows_before_quarantine": len(reference_overview_before),
    "overview_rows_after_quarantine": len(reference_overview_after)}
print("Complete source accounting: raw = exact duplicates removed + quarantined + eligible")
report_table(audit_summary)
print("Shared exclusions (categories can overlap; do not sum counts):")
report_table(duplicate_audit_exclusions)
print("Conflicting raw match metadata, retained unchanged:")
report_table(malformed_match_records)
```

```
Complete source accounting: raw = exact duplicates removed + quarantined + eligible
```

dataset	key	raw_rows	exact_duplicates_removed	missing_key_rows	conflicting_key_rows	conflicting_keys	quarantined_rows	eligible_rows	rows_after_shared_quarantine
charting_m_matches	match_id	7566	0	0	2	1	10	7556	7556
charting_m_points_2010s	match_id, Pt (across all era files)	357229	299	0	0	0	0	356930	356930
charting_m_points_2020s	match_id, Pt (across all era files)	547478	83	0	394	197	593	546802	546802
charting_m_points_to_2009	match_id, Pt (across all era files)	379569	1243	0	578	289	1842	376484	376484
charting_m_stats_KeyPointsReturn	match_id, player, row	60464	84	0	40	20	92	60288	60288
charting_m_stats_KeyPointsServe	match_id, player, row	60464	84	0	40	20	92	60288	60288
charting_m_stats_NetPoints	match_id, player, row	59396	74	0	60	30	102	59220	59220
charting_m_stats_Overview	match_id, player, set	56850	56	0	92	46	126	56668	56668
charting_m_stats_Rally	match_id, server, returner, row	96706	107	0	110	55	167	96432	96432
charting_m_stats_ReturnDepth	match_id, player, row	270002	323	0	276	138	457	269222	269222
charting_m_stats_ReturnOutcomes	match_id, player, row	310496	378	0	308	154	527	309591	309591
charting_m_stats_ServeBasics	match_id, player, row	45341	47	0	62	31	85	45209	45209
charting_m_stats_ServeDirection	match_id, player, row	45341	47	0	62	31	85	45209	45209
charting_m_stats_ServeInfluence	match_id, player, row	30235	34	0	36	18	54	30147	30147
charting_m_stats_ShotDirOutcomes	match_id, player, row	174687	227	0	154	77	290	174170	174170
charting_m_stats_ShotDirection	match_id, player, row	59997	66	0	76	38	110	59821	59821
charting_m_stats_SnV	match_id, player, row	51300	78	0	66	33	116	51106	51106
charting_m_stats_SvBreakSplit	match_id, player, row	164871	196	0	168	84	277	164398	164398

dataset	key	raw_rows	exact_duplicates_removed	missing_key_rows	conflicting_key_rows	conflicting_keys	quarantined_rows	eligible_rows	rows_after_shared_quarantine
charting_m_stats_SvBreakTotal	match_id, player, row	164871	196	0	168	84	277	164398	164398

Shared exclusions (categories can overlap; do not sum counts):

measure	count
conflicting match IDs	1
invalid metadata match IDs	0
conflicting point keys (match_id, Pt)	486
matches excluded due to point-key conflicts	8
invalid point-code match IDs	0
player-match exclusions across all statistical slices	16
conflicting Big Three player-match keys	0
Overview total rows before shared quarantine	15116
Overview player-match rows after shared quarantine	15072

Conflicting raw match metadata, retained unchanged:

match_id	Player 1	Player 2	Pl 1 hand	Pl 2 hand	Date	Tournament
20240915-M-Davis_Cup_World_Group-RR-Botic_Van_De_Zandschulp-Matteo_Berrettini	R	R	20240915	Davis Cup World Group	RR	15:15
20240915-M-Davis_Cup_World_Group-RR-Botic_Van_De_Zandschulp-Matteo_Berrettini	Botic Van De Zandschulp	Matteo Berrettini	Unavailable	Unavailable	20240915	Davis_Cup_World_Group

Round	Time	Court	Surface	Umpire	Best of	Final TB?
Unipol Arena	Hard	Arnaud Gabas	3	1	Zindaras	Unavailable
RR	Unavailable	Unavailable	Hard	Unavailable	Unavailable	Unavailable

Charted by
Unavailable
Zindaras

The audit preserves every row/set slice, including NetPoints and Approach, and includes Rally in duplicate-key checks. The tables above are computed from the current originals; exclusion counts refer to distinct scopes and are not additive. The next checks verify the retained grains and flag numerical anomalies for review without silently changing source values.

3.2 Integrity and numerical consistency checks

In [6]:
```
integrity_checks = []
def record_integrity(check, failures):
    integrity_checks.append({"check": check, "failures": int(failures)})

record_integrity("All sources reconcile raw/duplicate/quarantine/eligible row counts", sum(
    r["raw_rows"] != r["exact_duplicates_removed"] + r["quarantined_rows"] + r["eligible_rows"]
    for r in audit_records.values()))
record_integrity("Match IDs unique", analysis_dataframes["charting_m_matches"].duplicated("match_id").sum())
for key in STAT_SOURCE_CONFIG:
    grain = (["match_id", "server", "returner", "row"] if key.endswith("_Rally") else stat_key_columns[key])
    record_integrity(f"Unique full grain: {key}", analysis_dataframes[key].duplicated(grain).sum())
retained_points = pd.concat([analysis_dataframes[k] for k in point_source_keys], ignore_index=True)
record_integrity("Point keys unique across era files", retained_points.duplicated(["match_id", "Pt"]).sum())
for key, frame in analysis_dataframes.items():
    record_integrity(f"No orphan match IDs: {key}", (~frame["match_id"].isin(valid_match_ids)).sum())

# Flags remain inspectable at source-row grain; review before using affected metrics.
numeric_quality_parts = []
for key in STAT_SOURCE_CONFIG:
    frame = analysis_dataframes[key]
    labels = {"match_id", "player", "row", "set", "server", "returner"}
    for column in frame.columns.difference(list(labels)):
        is_percentage = key.endswith("_ServeInfluence") and column.startswith("won_")
        values = pd.to_numeric(frame[column].astype("string").str.rstrip("%") if is_percentage else frame[column], errors="coerce")
        flag = frame[column].isna() | values.isna() | ~np.isfinite(values) | values.lt(0)
        flag |= values.gt(100) if is_percentage else values.mod(1).ne(0)
        if flag.any():
            numeric_quality_parts.append(frame.loc[flag, [c for c in ["match_id", "player", "server", "returner", "row", "set"] if c in frame]]
                .assign(dataset=key, check=(f"missing/invalid percentage: {column}" if is_percentage else f"missing/nonfinite/negative/noninteger count:
{column}")))
    for numerator, denominator in [("pts_won", "pts"), ("bp_saved", "bk_pts"), ("first_won", "first_in"),
                                   ("second_won", "second_in"), ("return_pts_won", "return_pts"),
                                   ("aces", "serve_pts"), ("dfs", "second_in")]:
        if numerator in frame and denominator in frame:
            flag = pd.to_numeric(frame[numerator], errors="coerce").gt(pd.to_numeric(frame[denominator], errors="coerce"))
            if flag.any():
                numeric_quality_parts.append(frame.loc[flag, [c for c in ["match_id", "player", "row", "set"] if c in frame]]
                    .assign(dataset=key, check=f"{numerator} exceeds {denominator}"))
    if key.endswith("_Overview"):
        flag = frame["first_in"] + frame["second_in"] != frame["serve_pts"]
        if flag.any():
            numeric_quality_parts.append(frame.loc[flag, ["match_id", "player", "set"]]
                .assign(dataset=key, check="first_in + second_in differs from serve_pts"))
numeric_quality_flags = (pd.concat(numeric_quality_parts, ignore_index=True) if numeric_quality_parts else
                         pd.DataFrame(columns=["dataset", "match_id", "player", "check"]))
```

```python
numeric_quality_summary = numeric_quality_flags.groupby(["dataset", "check"], as_index=False).size().rename(columns={"size": "flagged_rows"})
integrity_check_results = pd.DataFrame(integrity_checks)
if integrity_check_results["failures"].sum():
    raise AssertionError("Shared integrity checks failed; inspect integrity_check_results")

# Source-code/state diagnostics are review flags, not automatic match deletions.
point_state_quality = pd.DataFrame([
    {"field": column, "missing_or_invalid_rows": int((pd.to_numeric(retained_points[column], errors="coerce").isna()
        | pd.to_numeric(retained_points[column], errors="coerce").lt(0)
        | pd.to_numeric(retained_points[column], errors="coerce").mod(1).ne(0)).sum())}
    for column in ["Set1", "Set2", "Gm1", "Gm2"]])
sequence_coverage_checks = retained_points.groupby("match_id")["Pt"].agg(["min", "max", "count"])
sequence_coverage_checks["gap_or_nonunit_start"] = (sequence_coverage_checks["min"].ne(1)
    | sequence_coverage_checks["max"].ne(sequence_coverage_checks["count"]))
point_sequence_flags = sequence_coverage_checks.loc[sequence_coverage_checks["gap_or_nonunit_start"]].reset_index()
net_slice_coverage = analysis_dataframes["charting_m_stats_NetPoints"].groupby("row", as_index=False).size().rename(columns={"size": "retained_rows"})
if not {"Approach", "NetPoints"}.issubset(set(net_slice_coverage["row"])):
    raise AssertionError("Required net-point slices were lost")
point_key_sizes = points_with_origin.groupby(["match_id", "Pt"], dropna=False).size()
all_point_duplicate_excess = int(point_key_sizes.sub(1).clip(lower=0).sum())
big_three_match_ids = set(matches_source.loc[matches_source["Player 1"].isin(BIG_THREE)
    | matches_source["Player 2"].isin(BIG_THREE), "match_id"])
big_three_point_key_sizes = points_with_origin.loc[points_with_origin["match_id"].isin(big_three_match_ids)].groupby(["match_id", "Pt"], dropna=False).size()
big_three_point_duplicate_excess = int(big_three_point_key_sizes.sub(1).clip(lower=0).sum())
point_duplicate_summary = pd.DataFrame([
    {"scope": "All point sources", "duplicate_key_excess_rows": all_point_duplicate_excess},
    {"scope": "Big Three matches", "duplicate_key_excess_rows": big_three_point_duplicate_excess}])
print("Hard integrity checks (all must pass):")
report_table(integrity_check_results)
print("Numerical review flags; retained unchanged pending metric-specific validation:")
report_table(numeric_quality_summary)
metadata_field_quality = pd.DataFrame([
    {"field": c, "missing_or_unrecognized_matches": int((~analysis_dataframes["charting_m_matches"][c].isin(["R", "L"])).sum())}
    for c in ["Pl 1 hand", "Pl 2 hand"]] + [
    {"field": "Best of (3/5 required for current deciding-set model)", "missing_or_unrecognized_matches": int((~pd.to_numeric(
        analysis_dataframes["charting_m_matches"]["Best of"], errors="coerce").isin([3, 5])).sum())}])
print("Metadata fields requiring metric-specific handling (matches retained):")
report_table(metadata_field_quality)
print("Point-state review flags:")
report_table(point_state_quality)
print(f"Point sequences with gaps or a start other than 1: {len(point_sequence_flags)}")
report_table(point_sequence_flags)
print("Net-point slices preserved for later analyses:")
report_table(net_slice_coverage)
report_table(point_duplicate_summary)
```

```
Hard integrity checks (all must pass):
```

check	failures
All sources reconcile raw/duplicate/quarantine/eligible row counts	0
Match IDs unique	0
Unique full grain: charting_m_stats_KeyPointsReturn	0
Unique full grain: charting_m_stats_KeyPointsServe	0
Unique full grain: charting_m_stats_NetPoints	0
Unique full grain: charting_m_stats_Overview	0
Unique full grain: charting_m_stats_Rally	0
Unique full grain: charting_m_stats_ReturnDepth	0
Unique full grain: charting_m_stats_ReturnOutcomes	0
Unique full grain: charting_m_stats_ServeBasics	0
Unique full grain: charting_m_stats_ServeDirection	0
Unique full grain: charting_m_stats_ServeInfluence	0
Unique full grain: charting_m_stats_ShotDirection	0
Unique full grain: charting_m_stats_ShotDirOutcomes	0
Unique full grain: charting_m_stats_SnV	0
Unique full grain: charting_m_stats_SvBreakSplit	0
Unique full grain: charting_m_stats_SvBreakTotal	0
Point keys unique across era files	0
No orphan match IDs: charting_m_matches	0
No orphan match IDs: charting_m_stats_KeyPointsReturn	0
No orphan match IDs: charting_m_stats_KeyPointsServe	0
No orphan match IDs: charting_m_stats_NetPoints	0
No orphan match IDs: charting_m_stats_Overview	0
No orphan match IDs: charting_m_stats_Rally	0
No orphan match IDs: charting_m_stats_ReturnDepth	0
No orphan match IDs: charting_m_stats_ReturnOutcomes	0
No orphan match IDs: charting_m_stats_ServeBasics	0
No orphan match IDs: charting_m_stats_ServeDirection	0
No orphan match IDs: charting_m_stats_ServeInfluence	0
No orphan match IDs: charting_m_stats_ShotDirection	0
No orphan match IDs: charting_m_stats_ShotDirOutcomes	0
No orphan match IDs: charting_m_stats_SnV	0

check	failures
No orphan match IDs: charting_m_stats_SvBreakSplit	0
No orphan match IDs: charting_m_stats_SvBreakTotal	0
No orphan match IDs: charting_m_points_2010s	0
No orphan match IDs: charting_m_points_2020s	0
No orphan match IDs: charting_m_points_to_2009	0

Numerical review flags; retained unchanged pending metric-specific validation:

dataset	check	flagged_rows
charting_m_stats_ServeInfluence	missing/invalid percentage: won_10+	4688
charting_m_stats_ServeInfluence	missing/invalid percentage: won_2+	1
charting_m_stats_ServeInfluence	missing/invalid percentage: won_3+	1
charting_m_stats_ServeInfluence	missing/invalid percentage: won_4+	23
charting_m_stats_ServeInfluence	missing/invalid percentage: won_5+	81
charting_m_stats_ServeInfluence	missing/invalid percentage: won_6+	641
charting_m_stats_ServeInfluence	missing/invalid percentage: won_7+	1205
charting_m_stats_ServeInfluence	missing/invalid percentage: won_8+	2243
charting_m_stats_ServeInfluence	missing/invalid percentage: won_9+	3255

Metadata fields requiring metric-specific handling (matches retained):

field	missing_or_unrecognized_matches
Pl 1 hand	32
Pl 2 hand	32
Best of (3/5 required for current deciding-set model)	33

Point-state review flags:

field	missing_or_invalid_rows
Set1	0
Set2	0
Gm1	1
Gm2	2

Point sequences with gaps or a start other than 1: 0

No records in this review table.

Net-point slices preserved for later analyses:

row	retained_rows
Approach	14654
ApproachRallies	14506
NetPoints	15044
NetPointsRallies	15016

scope	duplicate_key_excess_rows
All point sources	2111
Big Three matches	0

3.3 Retained analytical coverage and review register

Compare original and audited player-match membership in each source using unambiguous metadata. Statistical source presence means any retained slice, not guaranteed eligibility for every KPI. Detailed quarantined and numerically flagged records remain available in notebook variables. Hard key and accounting checks must pass; numerical/state flags remain unresolved until the relevant metric is validated. No inferred outcomes or performance rankings are computed in this step.

In [7]:
```
audit_player_coverage_rows = []
for key, frame in analysis_dataframes.items():
    original = raw_dataframes_unfiltered[key]
    for player in BIG_THREE:
        metadata_ids = set(coverage_participation.loc[coverage_participation["player"].eq(player), "match_id"])
        def source_ids(table):
            if "player" in table:
                return set(table.loc[table["player"].eq(player), "match_id"]) & metadata_ids
            if "server" in table and "returner" in table:
                return set(table.loc[table["server"].eq(player) | table["returner"].eq(player), "match_id"]) & metadata_ids
            return set(table["match_id"]) & metadata_ids
        before, after = source_ids(original), source_ids(frame)
        audit_player_coverage_rows.append({"player": player, "dataset": key,
            "source_matches_before": len(before), "source_matches_after": len(after),
            "matches_removed": len(before - after)})
audit_player_coverage = pd.DataFrame(audit_player_coverage_rows).sort_values(["player", "dataset"]).reset_index(drop=True)
quarantine_reason_summary = pd.concat([
    rows.assign(dataset=key)[["dataset", "_audit_reason"]]
    for key, rows in audit_quarantined_rows.items()], ignore_index=True).groupby(
        ["dataset", "_audit_reason"], as_index=False).size().rename(columns={"size": "rows"})
audit_review_register = pd.DataFrame([
    {"issue": "Numerical statistical field flags (not distinct rows)", "count": len(numeric_quality_flags), "next_action": "Unavailable percentages retained;
count/rate eligibility applied in Sections IV–VIII"},
    {"issue": "Metadata field flags", "count": int(metadata_field_quality["missing_or_unrecognized_matches"].sum()), "next_action": "Handle handedness/format
unknowns only in metrics requiring these fields"},
    {"issue": "Point-state flags", "count": int(point_state_quality["missing_or_invalid_rows"].sum()), "next_action": "Invalid score/state points excluded from
```

```
validated event cohorts"},
    {"issue": "Gapped/nonunit-start point sequences", "count": len(point_sequence_flags), "next_action": "Terminal-score and event-completeness rules applied in
Section IV"},
])
print("Quarantine rows by source and reason (each nonduplicate excluded row counted once):")
report_table(quarantine_reason_summary)
print("Big Three source coverage before and after shared audit:")
report_table(audit_player_coverage)
print("Remaining review work, not certified as metric-ready:")
report_table(audit_review_register)
# Re-run invariant: originals stay untouched and the audited inputs remain independent.
for key, original in raw_dataframes_unfiltered.items():
    if analysis_dataframes[key] is original:
        raise AssertionError(f"Original and analysis frame alias: {key}")
raw_dataframes = analysis_dataframes  # Legacy consumers use audited inputs; originals remain separate.
print("Step 3 complete: original inputs preserved, all 19 sources accounted for, full slices retained.")
```

```
Quarantine rows by source and reason (each nonduplicate excluded row counted once):
```

▼ **Complete evidence table: 51 rows × 3 columns**

dataset	_audit_reason	rows
charting_m_matches	conflicting key	2
charting_m_matches	shared match exclusion or orphan metadata	8
charting_m_points_2020s	conflicting point key	394
charting_m_points_2020s	shared match exclusion or orphan metadata	199
charting_m_points_to_2009	conflicting point key	578
charting_m_points_to_2009	shared match exclusion or orphan metadata	1264
charting_m_stats_KeyPointsReturn	conflicting key	40
charting_m_stats_KeyPointsReturn	player absent from eligible match metadata	8
charting_m_stats_KeyPointsReturn	shared player-match exclusion	44
charting_m_stats_KeyPointsServe	conflicting key	40
charting_m_stats_KeyPointsServe	player absent from eligible match metadata	8
charting_m_stats_KeyPointsServe	shared player-match exclusion	44
charting_m_stats_NetPoints	conflicting key	60
charting_m_stats_NetPoints	player absent from eligible match metadata	8
charting_m_stats_NetPoints	shared player-match exclusion	34
charting_m_stats_Overview	conflicting key	92
charting_m_stats_Overview	player absent from eligible match metadata	8
charting_m_stats_Overview	shared player-match exclusion	26
charting_m_stats_Rally	conflicting key	110
charting_m_stats_Rally	participant absent from eligible match metadata	13
charting_m_stats_Rally	shared player-match exclusion	44
charting_m_stats_ReturnDepth	conflicting key	276
charting_m_stats_ReturnDepth	player absent from eligible match metadata	36
charting_m_stats_ReturnDepth	shared player-match exclusion	145
charting_m_stats_ReturnOutcomes	conflicting key	308
charting_m_stats_ReturnOutcomes	player absent from eligible match metadata	42
charting_m_stats_ReturnOutcomes	shared player-match exclusion	177
charting_m_stats_ServeBasics	conflicting key	62
charting_m_stats_ServeBasics	player absent from eligible match metadata	6
charting_m_stats_ServeBasics	shared player-match exclusion	17

dataset	_audit_reason	rows
charting_m_stats_ServeDirection	conflicting key	62
charting_m_stats_ServeDirection	player absent from eligible match metadata	6
charting_m_stats_ServeDirection	shared player-match exclusion	17
charting_m_stats_ServeInfluence	conflicting key	36
charting_m_stats_ServeInfluence	player absent from eligible match metadata	4
charting_m_stats_ServeInfluence	shared player-match exclusion	14
charting_m_stats_ShotDirOutcomes	conflicting key	154
charting_m_stats_ShotDirOutcomes	player absent from eligible match metadata	23
charting_m_stats_ShotDirOutcomes	shared player-match exclusion	113
charting_m_stats_ShotDirection	conflicting key	76
charting_m_stats_ShotDirection	player absent from eligible match metadata	8
charting_m_stats_ShotDirection	shared player-match exclusion	26
charting_m_stats_SnV	conflicting key	66
charting_m_stats_SnV	player absent from eligible match metadata	10
charting_m_stats_SnV	shared player-match exclusion	40
charting_m_stats_SvBreakSplit	conflicting key	168
charting_m_stats_SvBreakSplit	player absent from eligible match metadata	22
charting_m_stats_SvBreakSplit	shared player-match exclusion	87
charting_m_stats_SvBreakTotal	conflicting key	168
charting_m_stats_SvBreakTotal	player absent from eligible match metadata	22
charting_m_stats_SvBreakTotal	shared player-match exclusion	87

```
Big Three source coverage before and after shared audit:
```

▼ Complete evidence table: 57 rows × 5 columns

player	dataset	source_matches_before	source_matches_after	matches_removed
Novak Djokovic	charting_m_matches	553	553	0
Novak Djokovic	charting_m_points_2010s	283	283	0
Novak Djokovic	charting_m_points_2020s	197	197	0
Novak Djokovic	charting_m_points_to_2009	72	72	0
Novak Djokovic	charting_m_stats_KeyPointsReturn	550	550	0
Novak Djokovic	charting_m_stats_KeyPointsServe	550	550	0
Novak Djokovic	charting_m_stats_NetPoints	551	551	0
Novak Djokovic	charting_m_stats_Overview	550	550	0
Novak Djokovic	charting_m_stats_Rally	550	550	0
Novak Djokovic	charting_m_stats_ReturnDepth	550	550	0
Novak Djokovic	charting_m_stats_ReturnOutcomes	550	550	0
Novak Djokovic	charting_m_stats_ServeBasics	550	550	0
Novak Djokovic	charting_m_stats_ServeDirection	550	550	0
Novak Djokovic	charting_m_stats_ServeInfluence	551	551	0
Novak Djokovic	charting_m_stats_ShotDirOutcomes	548	548	0
Novak Djokovic	charting_m_stats_ShotDirection	552	552	0
Novak Djokovic	charting_m_stats_SnV	328	328	0
Novak Djokovic	charting_m_stats_SvBreakSplit	550	550	0
Novak Djokovic	charting_m_stats_SvBreakTotal	550	550	0
Rafael Nadal	charting_m_matches	425	425	0
Rafael Nadal	charting_m_points_2010s	262	262	0
Rafael Nadal	charting_m_points_2020s	51	51	0
Rafael Nadal	charting_m_points_to_2009	110	110	0
Rafael Nadal	charting_m_stats_KeyPointsReturn	424	424	0
Rafael Nadal	charting_m_stats_KeyPointsServe	424	424	0
Rafael Nadal	charting_m_stats_NetPoints	425	425	0
Rafael Nadal	charting_m_stats_Overview	424	424	0
Rafael Nadal	charting_m_stats_Rally	424	424	0
Rafael Nadal	charting_m_stats_ReturnDepth	424	424	0
Rafael Nadal	charting_m_stats_ReturnOutcomes	423	423	0

player	dataset	source_matches_before	source_matches_after	matches_removed
Rafael Nadal	charting_m_stats_ServeBasics	423	423	0
Rafael Nadal	charting_m_stats_ServeDirection	423	423	0
Rafael Nadal	charting_m_stats_ServeInfluence	425	425	0
Rafael Nadal	charting_m_stats_ShotDirOutcomes	422	422	0
Rafael Nadal	charting_m_stats_ShotDirection	425	425	0
Rafael Nadal	charting_m_stats_SnV	174	174	0
Rafael Nadal	charting_m_stats_SvBreakSplit	423	423	0
Rafael Nadal	charting_m_stats_SvBreakTotal	423	423	0
Roger Federer	charting_m_matches	723	723	0
Roger Federer	charting_m_points_2010s	344	344	0
Roger Federer	charting_m_points_2020s	14	14	0
Roger Federer	charting_m_points_to_2009	364	364	0
Roger Federer	charting_m_stats_KeyPointsReturn	722	722	0
Roger Federer	charting_m_stats_KeyPointsServe	722	722	0
Roger Federer	charting_m_stats_NetPoints	722	722	0
Roger Federer	charting_m_stats_Overview	722	722	0
Roger Federer	charting_m_stats_Rally	722	722	0
Roger Federer	charting_m_stats_ReturnDepth	722	722	0
Roger Federer	charting_m_stats_ReturnOutcomes	722	722	0
Roger Federer	charting_m_stats_ServeBasics	722	722	0
Roger Federer	charting_m_stats_ServeDirection	722	722	0
Roger Federer	charting_m_stats_ServeInfluence	722	722	0
Roger Federer	charting_m_stats_ShotDirOutcomes	719	719	0
Roger Federer	charting_m_stats_ShotDirection	722	722	0
Roger Federer	charting_m_stats_SnV	669	669	0
Roger Federer	charting_m_stats_SvBreakSplit	722	722	0
Roger Federer	charting_m_stats_SvBreakTotal	722	722	0

Remaining review work, not certified as metric-ready:

issue	count	next_action
Numerical statistical field flags (not distinct rows)	12138	Unavailable percentages retained; count/rate eligibility applied in Sections IV–VIII
Metadata field flags	97	Handle handedness/format unknowns only in metrics requiring these fields
Point-state flags	3	Invalid score/state points excluded from validated event cohorts
Gapped/nonunit-start point sequences	0	Terminal-score and event-completeness rules applied in Section IV

```
Step 3 complete: original inputs preserved, all 19 sources accounted for, full slices retained.
```

The audit outputs provide key checks, exact row accounting, retained-slice coverage, and an explicit review register. A passed shared audit establishes unique joins and transparent exclusions; it does not certify every numeric field or reconstructed event. All statistical slices now survive when eligible, so downstream consumers must explicitly choose Total, set, stroke, or tactical slices before aggregating.

Current audit findings: 1 conflicting match ID, 486 conflicting point keys across 8 matches, 1,625 exact duplicate point rows, and 16 affected statistical player-match keys. None of the conflicting statistical player-match keys or duplicate point-key excess rows involve the Big Three. Overview Total rows decrease from 15,116 to 15,072; full Overview set slices remain available separately. The 12,138 statistical field flags are unavailable ServeInfluence percentage entries represented as `–`, not 12,138 corrupt matches; these values must remain unavailable rather than becoming zero. Three point-state field flags are handled by the event-eligibility rules in Section IV. No retained sequence has a point-number gap or a nonunit start, but contiguous numbering alone does not prove match completion.

IV. Canonical match, point, and shot tables

Reconstruction question: Which recorded points establish a completed game, tiebreak, set, or match? A last recorded point can belong to an unfinished chart or retirement; treating it as a match winner would change the central comparison.

4.1 A complete player-match spine

Start with audited match participation, not Overview presence. Retain every eligible Big Three player-match even when Overview or point records are missing; preserve missing values and expose coverage flags. The local workbook defines `Pts` relative to the **server and returner**, unlike `Set1/Set2` and `Gm1/Gm2`, which refer to Player 1/Player 2. Point ordering and all metadata joins are validated before event reconstruction.

In [8]:

```python
def prepare_base_tables(raw_dataframes, big_three):
    matches = raw_dataframes["charting_m_matches"].rename(columns={
        "Player 1": "player1", "Player 2": "player2", "Date": "match_date", "Surface": "surface"}).copy()
    matches["match_date"] = pd.to_datetime(matches["match_date"].astype("string"), format="%Y%m%d", errors="coerce")
    matches["surface"] = matches["surface"].astype("string").str.strip().str.title().replace(
        {"Hard Court": "Hard", "Grass Court": "Grass", "Clay Court": "Clay"}).fillna("Unknown")
    matches["Final TB?"] = matches["Final TB?"].astype("string").str.strip().str.upper()
    matches["Best of"] = pd.to_numeric(matches["Best of"], errors="coerce")
    if matches["match_id"].duplicated().any():
        raise AssertionError("Match metadata must be unique")
    points = pd.concat([frame for key, frame in raw_dataframes.items() if key.startswith("charting_m_points")], ignore_index=True)
    for column in ["Pt", "Svr", "PtWinner", "Set1", "Set2", "Gm1", "Gm2"]:
        points[column] = pd.to_numeric(points[column], errors="coerce")
    if points.duplicated(["match_id", "Pt"]).any():
        raise AssertionError("Point keys must be unique across all source files")
    big = matches[matches["player1"].isin(big_three) | matches["player2"].isin(big_three)].copy()
    return matches, points, big


def build_master_from_overview(raw_dataframes, matches, big_three):
    parts = []
    for own, other, hand in [("player1", "player2", "Pl 1 hand"), ("player2", "player1", "Pl 2 hand")]:
        part = matches.loc[matches[own].isin(big_three)].copy()
        part["player"], part["opponent"], part["handedness"] = part[own], part[other], part[hand]
        parts.append(part)
    spine = pd.concat(parts, ignore_index=True)
    exclusions = globals().get("all_player_match_conflicts", set())
    spine = spine.loc[[pair not in exclusions for pair in spine[["match_id", "player"]].itertuples(index=False, name=None)]].copy()
    overview = filter_stat_slice(raw_dataframes["charting_m_stats_Overview"], None, "total").drop(columns="set")
    if overview.duplicated(["match_id", "player"]).any():
        raise AssertionError("Overview Total must be unique")
    master = spine.merge(overview.assign(has_overview=True), on=["match_id", "player"], how="left", validate="one_to_one")
    master["has_overview"] = master["has_overview"].astype("boolean").fillna(False).astype(bool)
    if master.duplicated(["match_id", "player"]).any() or len(master) != len(spine):
        raise AssertionError("Player-match spine changed during enrichment")
    return master


# Shared token grammar: preserve undirected and modified strokes before assigning hitters.
SHOT_TOKEN_PATTERN = re.compile(r"([fbrsvzopuylmhijktq])([+\-=;^]*)([0-3])?([789])?")
SHOT_FAMILIES = {"f": "groundstroke", "b": "groundstroke", "r": "slice", "s": "slice",
    "v": "volley", "z": "volley", "o": "overhead", "p": "overhead", "u": "drop shot", "y": "drop shot",
    "l": "lob", "m": "lob", "h": "half-volley", "i": "half-volley", "j": "swinging volley", "k": "swinging volley", "t": "other", "q": "unknown"}
FOREHAND_CODES = set("frvoulhj")
BACKHAND_CODES = set("bszpymik")
NET_SHOT_CODES = set("vzophijk")


def select_point_sequence(first, second):
    if pd.notna(second) and str(second).strip():
        return str(second).strip(), 2
    return (str(first).strip(), 1) if pd.notna(first) and str(first).strip() else ("", None)


def parse_point_sequence(first, second):
    sequence, serve_number = select_point_sequence(first, second)
```

```
    tokens = list(SHOT_TOKEN_PATTERN.finditer(sequence))
    prefix = sequence[:tokens[0].start()] if tokens else sequence
    serve_codes = [int(char) for char in prefix if char in "456"]
    unknown_point = not serve_codes and "0" not in prefix  # Includes S/R/P/Q placeholders; never fabricate rally length.
    double_fault = serve_number == 2 and not tokens and bool(re.search(r"[nwdxe!]", prefix))
    end_error = bool(tokens and re.search(r"[@#]|[nwdxe!]", sequence[tokens[-1].end():]))
    rally_length = (np.nan if unknown_point else 0 if double_fault else 1 + len(tokens) - int(end_error))
    return {"sequence": sequence, "serve_number": serve_number,
        "serve_direction_code": serve_codes[-1] if serve_codes else 0 if "0" in prefix else pd.NA,
        "serve_attempts": sum(str(value).count(char) for value in [first, second] if pd.notna(value) for char in "456"),
        "double_fault_from_code": double_fault, "rally_length": rally_length,
        "sequence_known": not unknown_point, "tokens": tokens, "last_stroke_error": end_error}
```

4.2 Reconstruct pressure events and verify match completion

Tiebreak games are identified as groups using numeric nonstandard scores and tied game counts, with match-tiebreak formats handled separately. Preserve all points in each identified tiebreak, including its 0–0 start. Break points use receiver-leading server-relative scores; no-ad deciding points are explicitly recognized. A match winner is accepted only when the final point ends a valid game/tiebreak, a set, and the required best-of match. Otherwise the outcome remains unknown. Tiebreak service court alternates with **total points played**, not the two-point service-turn pattern. Special or unsupported states remain visible in validation tables.

In [9]:
```
def build_canonical_events(matches, points_all, include_shots=True):
    points = points_all.loc[points_all["match_id"].isin(matches["match_id"])].copy()
    points = points.merge(matches, on="match_id", how="inner", validate="many_to_one").sort_values(["match_id", "Pt"]).reset_index(drop=True)
    points["game_number"] = points["Gm#"].astype("string").str.split("(", regex=False).str[0].str.strip()
    scores = points["Pts"].astype("string").str.upper().str.strip().str.extract(r"^(AD|A|\d+)-(AD|A|\d+)$")
    left, right = pd.to_numeric(scores[0], errors="coerce"), pd.to_numeric(scores[1], errors="coerce")
    nonstandard = left.notna() & right.notna() & (~left.isin([0, 15, 30, 40]) | ~right.isin([0, 15, 30, 40]))
    tied_games = points["Gm1"].eq(points["Gm2"])
    possible_tb = tied_games & (points["Gm1"].ge(3) | (points["Final TB?"].eq("S") & points["Gm1"].eq(0)))
    keys = ["match_id", "Set1", "Set2", "game_number"]
    candidate_keys = points.loc[nonstandard & possible_tb, keys].drop_duplicates()
    points["is_tiebreak_point"] = pd.MultiIndex.from_frame(points[keys]).isin(pd.MultiIndex.from_frame(candidate_keys))
    rank = {"0": 0, "15": 1, "30": 2, "40": 3, "AD": 4, "A": 4}
    server_rank, receiver_rank = scores[0].map(rank), scores[1].map(rank)
    regular_valid = (server_rank.notna() & receiver_rank.notna()
        & (~server_rank.eq(4) | receiver_rank.eq(3)) & (~receiver_rank.eq(4) | server_rank.eq(3)))
    tb_valid = left.notna() & right.notna() & left.ge(0) & right.ge(0)
    state_valid = points[["Set1", "Set2", "Gm1", "Gm2"]].notna().all(axis=1)
    state_valid &= points[["Set1", "Set2", "Gm1", "Gm2"]].ge(0).all(axis=1)
    points["score_state_valid"] = state_valid & np.where(points["is_tiebreak_point"], tb_valid, regular_valid)
    points["server_name"] = np.where(points["Svr"].eq(1), points["player1"], points["player2"])
    points["returner_name"] = np.where(points["Svr"].eq(1), points["player2"], points["player1"])
    points["point_winner"] = np.where(points["PtWinner"].eq(1), points["player1"], points["player2"])
    server_won = points["Svr"].eq(points["PtWinner"])
    no_ad = points["Final TB?"].eq("N") & server_rank.eq(3) & receiver_rank.eq(3)
    points["is_break_point"] = ~points["is_tiebreak_point"] & ((receiver_rank.ge(3) & receiver_rank.gt(server_rank)) | no_ad)
    points["break_point_saved"] = points["is_break_point"] & server_won
    game_end_regular = ((np.where(server_won, server_rank, receiver_rank) >= 3)
        & (np.where(server_won, server_rank - receiver_rank, receiver_rank - server_rank) >= 1)) | no_ad
```

```
    # Match/set tiebreak formats: 10 in S, or A deciding sets; otherwise standard seven-point TB.
    deciding = points["Set1"].add(points["Set2"]).add(1).eq(points["Best of"])
    tb_target = np.where(points["Final TB?"].eq("S") | (points["Final TB?"].eq("A") & deciding), 10, 7)
    winner_pre = np.where(server_won, left, right)
    loser_pre = np.where(server_won, right, left)
    game_end_tb = (winner_pre + 1 >= tb_target) & (winner_pre + 1 - loser_pre >= 2)
    points["game_ended"] = points["score_state_valid"] & np.where(points["is_tiebreak_point"], game_end_tb, game_end_regular)
    points["set_number"] = points["Set1"] + points["Set2"] + 1
    points["is_deciding_set"] = deciding
    score_sum = np.where(points["is_tiebreak_point"], left + right, server_rank + receiver_rank)
    points["service_court"] = pd.Series(np.where(score_sum % 2 == 0, "deuce", "ad"), index=points.index).where(points["score_state_valid"])
    points.loc[no_ad, "service_court"] = pd.NA  # Receiver chooses; cannot infer from parity.
    points["score_server_numeric"], points["score_receiver_numeric"] = left, right
    parsed = [parse_point_sequence(a, b) for a, b in points[["1st", "2nd"]].itertuples(index=False, name=None)]
    for field in ["sequence", "serve_number", "serve_direction_code", "serve_attempts", "double_fault_from_code", "rally_length", "sequence_known",
"last_stroke_error"]:
        points[field] = [item[field] for item in parsed]
    points["previous_rally_length"] = points.groupby("match_id")["rally_length"].shift(1)
    shot_rows = []
    for row, item in (zip(points.itertuples(index=False), parsed) if include_shots else []):
        for index, token in enumerate(item["tokens"]):
            kind, modifiers, direction, depth = token.groups()
            hitter = row.returner_name if index % 2 == 0 else row.server_name
            if hitter not in BIG_THREE:
                continue
            shot_rows.append({"match_id": row.match_id, "Pt": row.Pt, "shot_number": index + 1,
                "player": hitter, "surface": row.surface, "match_date": row.match_date,
                "shot_code": kind, "shot_family": SHOT_FAMILIES[kind],
                "wing": "forehand" if kind in FOREHAND_CODES else "backhand" if kind in BACKHAND_CODES else "unknown",
                "direction_code": int(direction) if direction else pd.NA, "depth_code": int(depth) if depth else pd.NA,
                "at_net_inferred": True if "-" in modifiers else False if "=" in modifiers else kind in NET_SHOT_CODES,
                "approach_marker": "+" in modifiers, "is_return": index == 0,
                "is_error_attempt": index == len(item["tokens"]) - 1 and item["last_stroke_error"]})
    shots = pd.DataFrame(shot_rows)
    return points, shots


def validated_match_results(matches, points):
    last = points.groupby("match_id", sort=False).tail(1).copy()
    win1 = last["PtWinner"].eq(1)
    winner_games = np.where(win1, last["Gm1"], last["Gm2"]) + last["game_ended"].astype(int)
    loser_games = np.where(win1, last["Gm2"], last["Gm1"])
    needed_games = np.where(last["Final TB?"].eq("N"), 4, 6)
    set_end = last["game_ended"] & (((winner_games >= needed_games) & (winner_games - loser_games >= 2))
        | last["is_tiebreak_point"])
    sets_won = np.where(win1, last["Set1"], last["Set2"]) + set_end.astype(int)
    supported = last["Best of"].isin([3, 5])
    required_sets = last["Best of"] // 2 + 1
    format_valid = last["Set1"].lt(required_sets) & last["Set2"].lt(required_sets)
    complete = set_end & supported & format_valid & (sets_won == required_sets)
    last["winner"] = last["point_winner"].where(complete)
    last["outcome_status"] = np.select([complete, ~supported, ~format_valid],
        ["validated terminal score", "unsupported match format", "format metadata contradicts set state"], default="terminal score not verified")
    result = matches.merge(last[["match_id", "winner", "outcome_status"]], on="match_id", how="left", validate="one_to_one")
    result["outcome_status"] = result["outcome_status"].fillna("no charted points")
    return result
```

```
def infer_match_results(matches_big, points_all):
    # Later report consumers inherit the corrected winner policy.
    events, _ = build_canonical_events(matches_big, points_all, include_shots=False)
    return validated_match_results(matches_big, events)
```

4.3 Construct tables and reconcile source counts

The reconciliation uses matched player-match rows, with exact counts for service/return points and pressure events. Any discrepancy is retained in a review table; source values are not overwritten to force agreement. Point-derived pressure components are eligible only where their required score states and source comparisons agree. Match outcomes remain independent: a verified terminal score does not automatically certify a whole point sequence or every derived event.

The workbook uses S/R for entirely uncharted points with known server/returner outcomes, and P/Q for point penalties. Keep those valid point outcomes in the point table, but distinguish them from charted serve/rally statistics. Reconciliation of Overview counts uses sequence-known points; overall point-win counts retain valid recorded outcomes. Double faults can end with net, wide, deep, or other error codes, not only `d`.

In [10]:
```
canonical_matches_all, canonical_points_all, canonical_matches = prepare_base_tables(analysis_dataframes, BIG_THREE)
canonical_points, canonical_shots = build_canonical_events(canonical_matches, canonical_points_all)
canonical_match_results = validated_match_results(canonical_matches, canonical_points)
canonical_master = build_master_from_overview(analysis_dataframes, canonical_matches, BIG_THREE)
canonical_master = canonical_master.merge(canonical_match_results[["match_id", "winner", "outcome_status"]],
    on="match_id", how="left", validate="many_to_one")
canonical_master["is_win"] = canonical_master["player"].eq(canonical_master["winner"]).astype("Int64").where(canonical_master["winner"].notna())
player_point_parts = []
for number, column in [(1, "player1"), (2, "player2")]:
    part = canonical_points.loc[canonical_points[column].isin(BIG_THREE), ["match_id", "Pt", "surface", "match_date", "Svr", "PtWinner",
        "is_break_point", "score_state_valid", "is_tiebreak_point", "game_ended", "game_number", "Set1", "Set2", "set_number", "is_deciding_set",
"rally_length", "previous_rally_length", "double_fault_from_code", "sequence_known"]].copy()
    part["player"] = canonical_points.loc[part.index, column]
    part["is_serving"] = part["Svr"].eq(number)
    part["point_won"] = part["PtWinner"].eq(number)
    part["bp_faced_from_score"] = part["is_break_point"] & part["is_serving"]
    part["bp_faced"] = part["bp_faced_from_score"] & part["sequence_known"]
    part["bp_saved"] = part["bp_faced"] & part["point_won"]
    part["bp_opportunity"] = part["is_break_point"] & ~part["is_serving"] & part["sequence_known"]
    part["bp_converted"] = part["bp_opportunity"] & part["point_won"]
    part["service_point"] = (part["is_serving"] & part["sequence_known"]).astype(int)
    part["return_point"] = (~part["is_serving"] & part["sequence_known"]).astype(int)
    part["service_point_won"] = part["is_serving"] & part["point_won"] & part["sequence_known"]
    part["return_point_won"] = ~part["is_serving"] & part["point_won"] & part["sequence_known"]
    part["dfs_parsed"] = part["is_serving"] & part["double_fault_from_code"]
    player_point_parts.append(part)
canonical_player_points = pd.concat(player_point_parts, ignore_index=True)
point_totals = canonical_player_points.groupby(["match_id", "player"], as_index=False).agg(
    points_played=("Pt", "size"), points_won=("point_won", "sum"),
```

```
    service_points_parsed=("service_point", "sum"), service_wins_parsed=("service_point_won", "sum"),
    return_points_parsed=("return_point", "sum"), return_wins_parsed=("return_point_won", "sum"),
    bp_faced_parsed=("bp_faced", "sum"), bp_saved_parsed=("bp_saved", "sum"),
    bp_opportunities_parsed=("bp_opportunity", "sum"), bp_converted_parsed=("bp_converted", "sum"),
    dfs_parsed=("dfs_parsed", "sum"), sequence_known_points=("sequence_known", "sum"), valid_score_points=("score_state_valid", "sum"))
canonical_master = canonical_master.merge(point_totals, on=["match_id", "player"], how="left", validate="one_to_one")
canonical_master["has_points"] = canonical_master["points_played"].notna()
canonical_master["source_service_wins"] = canonical_master["first_won"] + canonical_master["second_won"]
comparisons = {"service_points_parsed": "serve_pts", "service_wins_parsed": "source_service_wins",
    "return_points_parsed": "return_pts", "return_wins_parsed": "return_pts_won", "bp_faced_parsed": "bk_pts",
    "bp_saved_parsed": "bp_saved", "dfs_parsed": "dfs"}
reconciliation_parts = []
for parsed_field, source_field in comparisons.items():
    rows = canonical_master.loc[canonical_master["has_points"] & canonical_master["has_overview"],
        ["match_id", "player", "surface", parsed_field, source_field]].copy()
    rows["metric"] = parsed_field
    rows["parsed_count"], rows["source_count"] = rows[parsed_field], rows[source_field]
    rows["difference"] = rows["parsed_count"] - rows["source_count"]
    reconciliation_parts.append(rows[["match_id", "player", "surface", "metric", "parsed_count", "source_count", "difference"]])
canonical_reconciliation = pd.concat(reconciliation_parts, ignore_index=True)
canonical_reconciliation_summary = canonical_reconciliation.groupby("metric", as_index=False).agg(
    compared_player_matches=("difference", "size"), mismatching_player_matches=("difference", lambda x: int(x.ne(0).sum())),
    total_parsed=("parsed_count", "sum"), total_source=("source_count", "sum"))
canonical_reconciliation_flags = canonical_reconciliation.loc[canonical_reconciliation["difference"].ne(0)].copy()
pressure_disagreements = set(canonical_reconciliation_flags.loc[canonical_reconciliation_flags["metric"].isin([
    "bp_faced_parsed", "bp_saved_parsed", "service_points_parsed", "return_points_parsed"]), ["match_id", "player"]].itertuples(index=False, name=None))
canonical_master["bp_events_validated"] = (canonical_master["has_overview"] & canonical_master["has_points"]
    & canonical_master["valid_score_points"].eq(canonical_master["points_played"])
    & pd.Series([pair not in pressure_disagreements for pair in canonical_master[["match_id", "player"]].itertuples(index=False, name=None)],
index=canonical_master.index))

# Completed tiebreaks are outcomes, not a count of every numeric-score candidate.
tb_keys = ["match_id", "Set1", "Set2", "game_number"]
canonical_tiebreaks = canonical_points.loc[canonical_points["is_tiebreak_point"]].groupby(tb_keys, sort=False).tail(1).copy()
canonical_tiebreaks["tiebreak_winner"] = canonical_tiebreaks["point_winner"].where(canonical_tiebreaks["game_ended"])
canonical_games = canonical_points.groupby(["match_id", "game_number"], sort=False).tail(1).copy()
canonical_games["game_winner"] = canonical_games["point_winner"].where(canonical_games["game_ended"])
canonical_sets = canonical_points.groupby(["match_id", "Set1", "Set2"], sort=False).tail(1).copy()
set_win1 = canonical_sets["PtWinner"].eq(1)
set_winner_games = np.where(set_win1, canonical_sets["Gm1"], canonical_sets["Gm2"]) + canonical_sets["game_ended"].astype(int)
set_loser_games = np.where(set_win1, canonical_sets["Gm2"], canonical_sets["Gm1"])
set_target = np.where(canonical_sets["Final TB?"].eq("N"), 4, 6)
canonical_sets["set_ended"] = canonical_sets["game_ended"] & (
    ((set_winner_games >= set_target) & (set_winner_games - set_loser_games >= 2)) | canonical_sets["is_tiebreak_point"])
canonical_sets["set_winner"] = canonical_sets["point_winner"].where(canonical_sets["set_ended"])
# Reject deciding-set labels when observed set counts contradict declared format.
max_observed_sets = canonical_points.groupby("match_id")[["Set1", "Set2"]].max().max(axis=1)
format_contradiction_ids = set(max_observed_sets.index[max_observed_sets.ge(2)]) & set(
    canonical_matches.loc[canonical_matches["Best of"].eq(3), "match_id"])
canonical_sets["format_consistent"] = ~canonical_sets["match_id"].isin(format_contradiction_ids)
canonical_sets["deciding_set_validated"] = canonical_sets["is_deciding_set"] & canonical_sets["set_ended"] & canonical_sets["format_consistent"]
canonical_format_review = canonical_matches.loc[canonical_matches["match_id"].isin(format_contradiction_ids), ["match_id", "Best of"]].copy()
canonical_format_review["maximum_prepoint_sets_won"] = canonical_format_review["match_id"].map(max_observed_sets)

# Assertions check real source identities and parser edge cases rather than duplicated formulas.
assert select_point_sequence("6f27b3*", np.nan) == ("6f27b3*", 1)
assert select_point_sequence("5d", "4b+2f*") == ("4b+2f*", 2)
```

```
assert len(parse_point_sequence("6fb+2z-3*", None)["tokens"]) == 3
assert parse_point_sequence("6f27b3*", None)["rally_length"] == 3
assert parse_point_sequence("6f27b3n@", None)["rally_length"] == 2
assert parse_point_sequence("5d", "6d")["rally_length"] == 0
assert parse_point_sequence("5n", "4n")["double_fault_from_code"]
assert parse_point_sequence("0f1*", None)["sequence_known"]
assert pd.isna(parse_point_sequence("S", None)["rally_length"])
fixture_matches = canonical_matches.head(1).copy()
fixture_id = fixture_matches.iloc[0]["match_id"]
fixture_points = pd.DataFrame([
    {"match_id": fixture_id, "Pt": 1, "Svr": 2, "PtWinner": 2, "Set1": 0, "Set2": 0,
     "Gm1": 0, "Gm2": 0, "Pts": "30-40", "Gm#": "1", "TbSet": True, "1st": "6f27*", "2nd": np.nan},
    {"match_id": fixture_id, "Pt": 2, "Svr": 2, "PtWinner": 1, "Set1": 0, "Set2": 0,
     "Gm1": 0, "Gm2": 0, "Pts": "40-AD", "Gm#": "1", "TbSet": True, "1st": "5n", "2nd": "4n"}])
fixture_events, _ = build_canonical_events(fixture_matches, fixture_points, include_shots=False)
assert fixture_events["is_break_point"].all()  # Server is Player 2; receiver is still the right score.
assert fixture_events["break_point_saved"].tolist() == [True, False]
assert fixture_events["game_ended"].tolist() == [False, True]
assert "z" in NET_SHOT_CODES
assert not canonical_master.duplicated(["match_id", "player"]).any()
assert not canonical_player_points.duplicated(["match_id", "Pt", "player"]).any()
assert not canonical_shots.duplicated(["match_id", "Pt", "shot_number"]).any()
assert canonical_master.loc[~canonical_master["has_points"], "points_played"].isna().all()
assert canonical_master.loc[canonical_master["winner"].isna(), "is_win"].isna().all()
canonical_coverage_summary = canonical_master.groupby("player", as_index=False).agg(
    charted_matches=("match_id", "size"), matches_with_overview=("has_overview", "sum"),
    matches_with_points=("has_points", "sum"), verified_outcomes=("winner", "count"),
    verified_wins=("is_win", "sum"), validated_bp_player_matches=("bp_events_validated", "sum"),
    earliest_charted_date=("match_date", "min"), latest_charted_date=("match_date", "max"))
canonical_outcome_review = canonical_master.loc[canonical_master["winner"].isna(),
    ["match_id", "player", "has_overview", "has_points", "outcome_status"]]
print("Canonical coverage and verified outcomes:")
report_table(canonical_coverage_summary)
print("Source reconciliation on matched coverage (source statistics exclude S/R uncharted-rally placeholders):")
report_table(canonical_reconciliation_summary)
print("Declared match formats contradicted by observed set counts (not silently overridden):")
report_table(canonical_format_review)
print("Outcome review (unknown outcomes remain missing):")
report_table(canonical_outcome_review)
print(f"Reconciliation discrepancies: {len(canonical_reconciliation_flags)} field-level comparisons")
report_table(canonical_reconciliation_flags)
canonical_event_quality = pd.DataFrame([
    {"check": "invalid score/state point rows", "count": int((~canonical_points["score_state_valid"]).sum())},
    {"check": "unavailable rally lengths", "count": int(canonical_points["rally_length"].isna().sum())},
    {"check": "completed reconstructed tiebreaks", "count": int(canonical_tiebreaks["tiebreak_winner"].notna().sum())},
    {"check": "incomplete reconstructed tiebreak groups", "count": int(canonical_tiebreaks["tiebreak_winner"].isna().sum())},
    {"check": "declared formats contradicted by recorded sets", "count": len(canonical_format_review)},
])
print("Event coverage and review checks:")
report_table(canonical_event_quality)
print("Canonical player-match preview:")
report_table(canonical_master[["match_id", "player", "surface", "winner", "is_win", "has_overview", "has_points", "bp_events_validated"]].head(10))
print(f"Built {len(canonical_points):,} match-point rows, {len(canonical_player_points):,} Big Three player-point rows, and {len(canonical_shots):,} Big Three
attempted rally strokes.")
print("Parser edge cases, unique grains, missing-outcome handling, and join checks passed.")
```

```
Canonical coverage and verified outcomes:
```

player	charted_matches	matches_with_overview	matches_with_points	verified_outcomes	verified_wins	validated_bp_player_matches	earliest_charted_date	latest_charted_date
Novak Djokovic	553	550	552	526	404	546	2005-01-17	2026-03-11
Rafael Nadal	425	424	423	412	298	404	2003-05-15	2024-11-19
Roger Federer	723	722	722	709	559	702	1998-07-04	2021-07-05

Source reconciliation on matched coverage (source statistics exclude S/R uncharted-rally placeholders):

metric	compared_player_matches	mismatching_player_matches	total_parsed	total_source
bp_faced_parsed	1694	4.000	9,137.000	9,139.000
bp_saved_parsed	1694	2.000	5,909.000	5,909.000
dfs_parsed	1694	10.000	3,404.000	3,406.000
return_points_parsed	1694	27.000	147,573.000	147,576.000
return_wins_parsed	1694	19.000	58,812.000	58,822.000
service_points_parsed	1694	19.000	139,906.000	139,903.000
service_wins_parsed	1694	10.000	94,683.000	94,675.000

Declared match formats contradicted by observed set counts (not silently overridden):

	match_id	Best of	maximum_prepoint_sets_won
1662	20230903-M-US_Open-R16-Borna_Gojo-Novak_Djokovic	3.000	2
1669	20230901-M-US_Open-R32-Novak_Djokovic-Laslo_Djere	3.000	2
1684	20230829-M-US_Open-R128-Alexandre_Muller-Novak_Djokovic	3.000	2
1841	20230529-M-Roland_Garros-R128-Novak_Djokovic-Aleksandar_Kovacevic	3.000	2
5223	20130125-M-Australian_Open-SF-David_Ferrer-Novak_Djokovic	3.000	2
5293	20120804-M-Olympics-F-Andy_Murray-Roger_Federer	3.000	2
5421	20110911-M-US_Open-F-Rafael_Nadal-Novak_Djokovic	3.000	2
5483	20110130-M-Australian_Open-F-Novak_Djokovic-Andy_Murray	3.000	2
5535	20100904-M-US_Open-R32-Novak_Djokovic-James_Blake	3.000	2
5770	20080603-M-Roland_Garros-QF-Novak_Djokovic-Ernests_Gulbis	3.000	2
5797	20080121-M-Australian_Open-R16-Tomas_Berdych-Roger_Federer	3.000	2
5930	20060922-M-Davis_Cup_World_Group_PO-RR-Roger_Federer-Novak_Djokovic	3.000	2
6191	20040627-M-Wimbledon-R32-Roger_Federer-Thomas_Johansson	3.000	2

Outcome review (unknown outcomes remain missing):

▼ Complete evidence table: 54 rows × 5 columns

	match_id	player	has_overview	has_points	outcome_status
2	20260128-M-Australian_Open-QF-Novak_Djokovic-Lorenzo_Musetti	Novak Djokovic	True	True	terminal score not verified
26	20250124-M-Australian_Open-SF-Novak_Djokovic-Alexander_Zverev	Novak Djokovic	True	True	terminal score not verified
40	20240828-M-US_Open-R64-Novak_Djokovic-Laslo_Djere	Novak Djokovic	True	True	terminal score not verified
82	20230901-M-US_Open-R32-Novak_Djokovic-Laslo_Djere	Novak Djokovic	True	True	format metadata contradicts set state
93	20230529-M-Roland_Garros-R128-Novak_Djokovic-Aleksandar_Kovacevic	Novak Djokovic	True	True	format metadata contradicts set state
120	20221008-M-Astana-SF-Novak_Djokovic-Daniil_Medvedev	Novak Djokovic	True	True	terminal score not verified
170	20210629-M-Wimbledon-R128-Roger_Federer-Adrian_Mannarino	Roger Federer	True	True	terminal score not verified
186	20210210-M-Australian_Open-R64-Novak_Djokovic-Francis_Tiafoe	Novak Djokovic	False	False	no charted points
192	20200906-M-US_Open-R16-Novak_Djokovic-Pablo_Carreno_Busta	Novak Djokovic	True	True	terminal score not verified
227	20190902-M-US_Open-R16-Novak_Djokovic-Stan_Wawrinka	Novak Djokovic	True	True	terminal score not verified
269	20190123-M-Australian_Open-QF-Novak_Djokovic-Kei_Nishikori	Novak Djokovic	True	True	terminal score not verified
270	20190122-M-Australian_Open-QF-Rafael_Nadal-Francis_Tiafoe	Rafael Nadal	True	False	no charted points
283	20180907-M-US_Open-SF-Rafael_Nadal-Juan_Martin_Del_Potro	Rafael Nadal	True	True	terminal score not verified
291	20180818-M-Cincinnati_Masters-SF-Roger_Federer-David_Goffin	Roger Federer	True	True	terminal score not verified
324	20180123-M-Australian_Open-QF-Rafael_Nadal-Marin_Cilic	Rafael Nadal	True	True	terminal score not verified
348	20170712-M-Wimbledon-QF-Novak_Djokovic-Tomas_Berdych	Novak Djokovic	True	True	terminal score not verified
372	20170327-M-Miami_Masters-R32-Roger_Federer-Juan_Martin_Del_Potro	Roger Federer	False	False	no charted points
526	20140624-M-Wimbledon-R128-Rafael_Nadal-Martin_Klizan	Rafael Nadal	True	False	no charted points
535	20140511-M-Madrid_Masters-F-Rafael_Nadal-Kei_Nishikori	Rafael Nadal	True	True	terminal score not verified

	match_id	player	has_overview	has_points	outcome_status
667	20110911-M-US_Open-F-Rafael_Nadal-Novak_Djokovic	Rafael Nadal	True	True	format metadata contradicts set state
673	20110821-M-Cincinnati_Masters-F-Novak_Djokovic-Andy_Murray	Novak Djokovic	True	True	terminal score not verified
674	20110820-M-Cincinnati_Masters-SF-Novak_Djokovic-Tomas_Berdych	Novak Djokovic	True	True	terminal score not verified
676	20110813-M-Canada_Masters-SF-Novak_Djokovic-Jo_Wilfried_Tsonga	Novak Djokovic	True	True	terminal score not verified
700	20110130-M-Australian_Open-F-Novak_Djokovic-Andy_Murray	Novak Djokovic	True	True	format metadata contradicts set state
723	20100904-M-US_Open-R32-Novak_Djokovic-James_Blake	Novak Djokovic	True	True	format metadata contradicts set state
828	20080603-M-Roland_Garros-QF-Novak_Djokovic-Ernests_Gulbis	Novak Djokovic	True	True	format metadata contradicts set state
835	20080510-M-Rome_Masters-SF-Novak_Djokovic-Radek_Stepanek	Novak Djokovic	True	True	terminal score not verified
839	20080419-M-Monte_Carlo_Masters-SF-Roger_Federer-Novak_Djokovic	Roger Federer	True	True	terminal score not verified
872	20070707-M-Wimbledon-SF-Novak_Djokovic-Rafael_Nadal	Novak Djokovic	True	True	terminal score not verified
912	20060922-M-Davis_Cup_World_Group_PO-RR-Roger_Federer-Novak_Djokovic	Roger Federer	True	True	format metadata contradicts set state
924	20060616-M-Queens_Club-QF-Rafael_Nadal-Lleyton_Hewitt	Rafael Nadal	True	True	terminal score not verified
926	20060609-M-Roland_Garros-SF-Roger_Federer-David_Nalbandian	Roger Federer	True	True	terminal score not verified
928	20060607-M-Roland_Garros-QF-Novak_Djokovic-Rafael_Nadal	Novak Djokovic	False	True	terminal score not verified
1017	20040627-M-Wimbledon-R32-Roger_Federer-Thomas_Johansson	Roger Federer	True	True	format metadata contradicts set state
1069	20030510-M-Rome_Masters-SF-Roger_Federer-Juan_Carlos_Ferrero	Roger Federer	True	True	terminal score not verified
1097	20011214-M-St_Anton_EXH-RR-Roger_Federer-Markus_Hipfl	Roger Federer	True	True	terminal score not verified
1136	20230903-M-US_Open-R16-Borna_Gojo-Novak_Djokovic	Novak Djokovic	True	True	format metadata contradicts set state

	match_id	player	has_overview	has_points	outcome_status
1138	20230829-M-US_Open-R128-Alexandre_Muller-Novak_Djokovic	Novak Djokovic	True	True	format metadata contradicts set state
1171	20190922-M-Laver_Cup-RR-John_Isner-Roger_Federer	Roger Federer	True	True	terminal score not verified
1172	20190921-M-Laver_Cup-RR-Milos_Raonic-Rafael_Nadal	Rafael Nadal	True	True	terminal score not verified
1222	20180126-M-Australian_Open-SF-Hyeon_Chung-Roger_Federer	Roger Federer	True	True	terminal score not verified
1374	20140113-M-Australian_Open-R128-Bernard_Tomic-Rafael_Nadal	Rafael Nadal	True	True	terminal score not verified
1409	20130125-M-Australian_Open-SF-David_Ferrer-Novak_Djokovic	Novak Djokovic	True	True	format metadata contradicts set state
1418	20120804-M-Olympics-F-Andy_Murray-Roger_Federer	Roger Federer	True	True	format metadata contradicts set state
1452	20110911-M-US_Open-F-Rafael_Nadal-Novak_Djokovic	Novak Djokovic	True	True	format metadata contradicts set state
1513	20100126-M-Australian_Open-QF-Andy_Murray-Rafael_Nadal	Rafael Nadal	True	True	terminal score not verified
1521	20091017-M-Shanghai_Masters-SF-Feliciano_Lopez-Rafael_Nadal	Rafael Nadal	True	True	terminal score not verified
1545	20090127-M-Australian_Open-QF-Andy_Roddick-Novak_Djokovic	Novak Djokovic	True	True	terminal score not verified
1566	20080420-M-Estoril-F-Nikolay_Davydenko-Roger_Federer	Roger Federer	True	True	terminal score not verified
1567	20080419-M-Monte_Carlo_Masters-SF-Roger_Federer-Novak_Djokovic	Novak Djokovic	True	True	terminal score not verified
1574	20080121-M-Australian_Open-R16-Tomas_Berdych-Roger_Federer	Roger Federer	True	True	format metadata contradicts set state
1589	20070707-M-Wimbledon-SF-Novak_Djokovic-Rafael_Nadal	Rafael Nadal	True	True	terminal score not verified
1615	20060922-M-Davis_Cup_World_Group_PO-RR-Roger_Federer-Novak_Djokovic	Novak Djokovic	True	True	format metadata contradicts set state
1629	20060607-M-Roland_Garros-QF-Novak_Djokovic-Rafael_Nadal	Rafael Nadal	False	True	terminal score not verified

```
Reconciliation discrepancies: 91 field-level comparisons
```

▼ Complete evidence table: 91 rows × 7 columns

	match_id	player	surface	metric	parsed_count	source_count	difference
146	20220130-M-Australian_Open-F-Rafael_Nadal-Daniil_Medvedev	Rafael Nadal	Hard	service_points_parsed	188.000	189.000	-1.000
430	20160108-M-Brisbane-QF-Roger_Federer-Grigor_Dimitrov	Roger Federer	Hard	service_points_parsed	82.000	83.000	-1.000
447	20151101-M-Basel-F-Rafael_Nadal-Roger_Federer	Rafael Nadal	Hard	service_points_parsed	90.000	83.000	7.000
516	20140810-M-Cincinnati_Masters-F-Roger_Federer-David_Ferrer	Roger Federer	Hard	service_points_parsed	73.000	74.000	-1.000
521	20140626-M-Wimbledon-R64-Rafael_Nadal-Lukas_Rosol	Rafael Nadal	Grass	service_points_parsed	121.000	122.000	-1.000
539	20140223-M-Rio_de_Janeiro-F-Rafael_Nadal-Alexandr_Dolgopolov	Rafael Nadal	Clay	service_points_parsed	67.000	68.000	-1.000
556	20131010-M-Shanghai_Masters-R16-Roger_Federer-Gael_Monfils	Roger Federer	Hard	service_points_parsed	101.000	102.000	-1.000
595	20121106-M-Tour_Finals-RR-Roger_Federer-Janko_Tipsarevic	Roger Federer	Hard	service_points_parsed	40.000	41.000	-1.000
628	20120605-M-Roland_Garros-QF-Novak_Djokovic-Jo_Wilfried_Tsonga	Novak Djokovic	Clay	service_points_parsed	167.000	168.000	-1.000
789	20090123-M-Australian_Open-R32-Roger_Federer-Marat_Safin	Roger Federer	Hard	service_points_parsed	88.000	89.000	-1.000
833	20080423-M-Monte_Carlo_Masters-R32-Roger_Federer-Ruben_Ramirez_Hidalgo	Roger Federer	Clay	service_points_parsed	91.000	92.000	-1.000
875	20070520-M-Hamburg_Masters-F-Roger_Federer-Rafael_Nadal	Roger Federer	Clay	service_points_parsed	73.000	72.000	1.000
880	20070415-M-Monte_Carlo_Masters-F-Roger_Federer-Rafael_Nadal	Roger Federer	Clay	service_points_parsed	66.000	64.000	2.000
1049	20030919-M-Davis_Cup_World_Group_SF-RR-Roger_Federer-Mark_Philippoussis	Roger Federer	Hard	service_points_parsed	112.000	113.000	-1.000
1270	20151101-M-Basel-F-Rafael_Nadal-Roger_Federer	Roger Federer	Hard	service_points_parsed	90.000	88.000	2.000
1299	20150418-M-Monte_Carlo_Masters-SF-Novak_Djokovic-Rafael_Nadal	Rafael Nadal	Clay	service_points_parsed	65.000	66.000	-1.000
1334	20140531-M-Roland_Garros-R32-Leonardo_Mayer-Rafael_Nadal	Rafael Nadal	Clay	service_points_parsed	81.000	82.000	-1.000

	match_id	player	surface	metric	parsed_count	source_count	difference
1358	20140308-M-Indian_Wells_Masters-R64-Radek_Stepanek-Rafael_Nadal	Rafael Nadal	Hard	service_points_parsed	80.000	81.000	-1.000
1503	20100502-M-Rome_Masters-F-David_Ferrer-Rafael_Nadal	Rafael Nadal	Clay	service_points_parsed	55.000	50.000	5.000
2141	20151101-M-Basel-F-Rafael_Nadal-Roger_Federer	Rafael Nadal	Hard	service_wins_parsed	58.000	51.000	7.000
2210	20140810-M-Cincinnati_Masters-F-Roger_Federer-David_Ferrer	Roger Federer	Hard	service_wins_parsed	48.000	49.000	-1.000
2233	20140223-M-Rio_de_Janeiro-F-Rafael_Nadal-Alexandr_Dolgopolov	Rafael Nadal	Clay	service_wins_parsed	47.000	48.000	-1.000
2483	20090123-M-Australian_Open-R32-Roger_Federer-Marat_Safin	Roger Federer	Hard	service_wins_parsed	67.000	68.000	-1.000
2527	20080423-M-Monte_Carlo_Masters-R32-Roger_Federer-Ruben_Ramirez_Hidalgo	Roger Federer	Clay	service_wins_parsed	57.000	58.000	-1.000
2574	20070415-M-Monte_Carlo_Masters-F-Roger_Federer-Rafael_Nadal	Roger Federer	Clay	service_wins_parsed	39.000	37.000	2.000
2964	20151101-M-Basel-F-Rafael_Nadal-Roger_Federer	Roger Federer	Hard	service_wins_parsed	63.000	62.000	1.000
3028	20140531-M-Roland_Garros-R32-Leonardo_Mayer-Rafael_Nadal	Rafael Nadal	Clay	service_wins_parsed	56.000	57.000	-1.000
3052	20140308-M-Indian_Wells_Masters-R64-Radek_Stepanek-Rafael_Nadal	Rafael Nadal	Hard	service_wins_parsed	54.000	55.000	-1.000
3197	20100502-M-Rome_Masters-F-David_Ferrer-Rafael_Nadal	Rafael Nadal	Clay	service_wins_parsed	41.000	37.000	4.000
3564	20210605-M-Roland_Garros-R32-Rafael_Nadal-Cameron_Norrie	Rafael Nadal	Clay	return_points_parsed	80.000	81.000	-1.000
3835	20151101-M-Basel-F-Rafael_Nadal-Roger_Federer	Rafael Nadal	Hard	return_points_parsed	90.000	88.000	2.000
3878	20150418-M-Monte_Carlo_Masters-SF-Novak_Djokovic-Rafael_Nadal	Novak Djokovic	Clay	return_points_parsed	65.000	66.000	-1.000
3914	20140606-M-Roland_Garros-SF-Rafael_Nadal-Andy_Murray	Rafael Nadal	Clay	return_points_parsed	66.000	67.000	-1.000
4177	20090123-M-Australian_Open-R32-Roger_Federer-Marat_Safin	Roger Federer	Hard	return_points_parsed	87.000	88.000	-1.000
4179	20081116-M-Tour_Finals-F-Novak_Djokovic-Nikolay_Davydenko	Novak Djokovic	Hard	return_points_parsed	66.000	64.000	2.000

	match_id	player	surface	metric	parsed_count	source_count	difference
4192	20080831-M-US_Open-R32-Roger_Federer-Radek_Stepanek	Roger Federer	Hard	return_points_parsed	85.000	86.000	-1.000
4271	20070325-M-Miami_Masters-F-Novak_Djokovic-Guillermo_Canas	Novak Djokovic	Hard	return_points_parsed	68.000	69.000	-1.000
4282	20061116-M-Masters_Cup-RR-Roger_Federer-Ivan_Ljubicic	Roger Federer	Hard	return_points_parsed	66.000	67.000	-1.000
4410	20040326-M-Miami_Masters-R64-Roger_Federer-Nikolay_Davydenko	Roger Federer	Hard	return_points_parsed	78.000	79.000	-1.000
4658	20151101-M-Basel-F-Rafael_Nadal-Roger_Federer	Roger Federer	Hard	return_points_parsed	90.000	83.000	7.000
4701	20150121-M-Australian_Open-R64-Simone_Bolelli-Roger_Federer	Roger Federer	Hard	return_points_parsed	108.000	109.000	-1.000
4722	20140531-M-Roland_Garros-R32-Leonardo_Mayer-Rafael_Nadal	Rafael Nadal	Clay	return_points_parsed	91.000	92.000	-1.000
4730	20140510-M-Madrid_Masters-SF-Roberto_Bautista_Agut-Rafael_Nadal	Rafael Nadal	Clay	return_points_parsed	76.000	77.000	-1.000
4749	20140126-M-Australian_Open-F-Stan_Wawrinka-Rafael_Nadal	Rafael Nadal	Hard	return_points_parsed	94.000	95.000	-1.000
4750	20140122-M-Australian_Open-QF-Grigor_Dimitrov-Rafael_Nadal	Rafael Nadal	Hard	return_points_parsed	154.000	155.000	-1.000
4817	20120303-M-Dubai-F-Andy_Murray-Roger_Federer	Roger Federer	Hard	return_points_parsed	65.000	66.000	-1.000
4822	20120122-M-Australian_Open-R16-Bernard_Tomic-Roger_Federer	Roger Federer	Hard	return_points_parsed	94.000	95.000	-1.000
4891	20100502-M-Rome_Masters-F-David_Ferrer-Rafael_Nadal	Rafael Nadal	Clay	return_points_parsed	81.000	77.000	4.000
4913	20090531-M-Roland_Garros-R16-Robin_Soderling-Rafael_Nadal	Rafael Nadal	Clay	return_points_parsed	142.000	143.000	-1.000
4979	20070520-M-Hamburg_Masters-F-Roger_Federer-Rafael_Nadal	Rafael Nadal	Clay	return_points_parsed	73.000	72.000	1.000
4985	20070415-M-Monte_Carlo_Masters-F-Roger_Federer-Rafael_Nadal	Rafael Nadal	Clay	return_points_parsed	66.000	64.000	2.000
4994	20061117-M-Masters_Cup-RR-Nikolay_Davydenko-Rafael_Nadal	Rafael Nadal	Hard	return_points_parsed	81.000	82.000	-1.000
4999	20060906-M-US_Open-QF-Mikhail_Youzhny-Rafael_Nadal	Rafael Nadal	Hard	return_points_parsed	127.000	128.000	-1.000

	match_id	player	surface	metric	parsed_count	source_count	difference
5000	20060901-M-US_Open-R32-Vincent_Spadea-Roger_Federer	Roger Federer	Hard	return_points_parsed	74.000	75.000	-1.000
5063	20020208-M-Davis_Cup_World_Group_R1-RR-Marat_Safin-Roger_Federer	Roger Federer	Clay	return_points_parsed	81.000	82.000	-1.000
5079	19990215-M-Rotterdam-QF-Yevgeny_Kafelnikov-Roger_Federer	Roger Federer	Hard	return_points_parsed	92.000	93.000	-1.000
5258	20210605-M-Roland_Garros-R32-Rafael_Nadal-Cameron_Norrie	Rafael Nadal	Clay	return_wins_parsed	38.000	39.000	-1.000
5529	20151101-M-Basel-F-Rafael_Nadal-Roger_Federer	Rafael Nadal	Hard	return_wins_parsed	27.000	26.000	1.000
5572	20150418-M-Monte_Carlo_Masters-SF-Novak_Djokovic-Rafael_Nadal	Novak Djokovic	Clay	return_wins_parsed	29.000	30.000	-1.000
5608	20140606-M-Roland_Garros-SF-Rafael_Nadal-Andy_Murray	Rafael Nadal	Clay	return_wins_parsed	33.000	34.000	-1.000
5871	20090123-M-Australian_Open-R32-Roger_Federer-Marat_Safin	Roger Federer	Hard	return_wins_parsed	33.000	34.000	-1.000
5873	20081116-M-Tour_Finals-F-Novak_Djokovic-Nikolay_Davydenko	Novak Djokovic	Hard	return_wins_parsed	33.000	31.000	2.000
5965	20070325-M-Miami_Masters-F-Novak_Djokovic-Guillermo_Canas	Novak Djokovic	Hard	return_wins_parsed	28.000	29.000	-1.000
6104	20040326-M-Miami_Masters-R64-Roger_Federer-Nikolay_Davydenko	Roger Federer	Hard	return_wins_parsed	29.000	30.000	-1.000
6395	20150121-M-Australian_Open-R64-Simone_Bolelli-Roger_Federer	Roger Federer	Hard	return_wins_parsed	44.000	45.000	-1.000
6424	20140510-M-Madrid_Masters-SF-Roberto_Bautista_Agut-Rafael_Nadal	Rafael Nadal	Clay	return_wins_parsed	37.000	38.000	-1.000
6511	20120303-M-Dubai-F-Andy_Murray-Roger_Federer	Roger Federer	Hard	return_wins_parsed	25.000	26.000	-1.000
6516	20120122-M-Australian_Open-R16-Bernard_Tomic-Roger_Federer	Roger Federer	Hard	return_wins_parsed	45.000	46.000	-1.000
6585	20100502-M-Rome_Masters-F-David_Ferrer-Rafael_Nadal	Rafael Nadal	Clay	return_wins_parsed	36.000	35.000	1.000
6607	20090531-M-Roland_Garros-R16-Robin_Soderling-Rafael_Nadal	Rafael Nadal	Clay	return_wins_parsed	48.000	49.000	-1.000
6673	20070520-M-Hamburg_Masters-F-Roger_Federer-Rafael_Nadal	Rafael Nadal	Clay	return_wins_parsed	27.000	26.000	1.000

	match_id	player	surface	metric	parsed_count	source_count	difference
6693	20060906-M-US_Open-QF-Mikhail_Youzhny-Rafael_Nadal	Rafael Nadal	Hard	return_wins_parsed	37.000	38.000	-1.000
6694	20060901-M-US_Open-R32-Vincent_Spadea-Roger_Federer	Roger Federer	Hard	return_wins_parsed	35.000	36.000	-1.000
6757	20020208-M-Davis_Cup_World_Group_R1-RR-Marat_Safin-Roger_Federer	Roger Federer	Clay	return_wins_parsed	37.000	38.000	-1.000
6773	19990215-M-Rotterdam-QF-Yevgeny_Kafelnikov-Roger_Federer	Roger Federer	Hard	return_wins_parsed	36.000	37.000	-1.000
7223	20151101-M-Basel-F-Rafael_Nadal-Roger_Federer	Rafael Nadal	Hard	bp_faced_parsed	7.000	6.000	1.000
7292	20140810-M-Cincinnati_Masters-F-Roger_Federer-David_Ferrer	Roger Federer	Hard	bp_faced_parsed	10.000	11.000	-1.000
7404	20120605-M-Roland_Garros-QF-Novak_Djokovic-Jo_Wilfried_Tsonga	Novak Djokovic	Clay	bp_faced_parsed	10.000	11.000	-1.000
7825	20030919-M-Davis_Cup_World_Group_SF-RR-Roger_Federer-Mark_Philippoussis	Roger Federer	Hard	bp_faced_parsed	14.000	15.000	-1.000
8917	20151101-M-Basel-F-Rafael_Nadal-Roger_Federer	Rafael Nadal	Hard	bp_saved_parsed	4.000	3.000	1.000
8986	20140810-M-Cincinnati_Masters-F-Roger_Federer-David_Ferrer	Roger Federer	Hard	bp_saved_parsed	8.000	9.000	-1.000
10253	20230606-M-Roland_Garros-QF-Novak_Djokovic-Karen_Khachanov	Novak Djokovic	Clay	dfs_parsed	5.000	6.000	-1.000
10256	20230531-M-Roland_Garros-R64-Novak_Djokovic-Marton_Fucsovics	Novak Djokovic	Clay	dfs_parsed	3.000	4.000	-1.000
10389	20190904-M-US_Open-QF-Rafael_Nadal-Diego_Schwartzman	Rafael Nadal	Hard	dfs_parsed	2.000	1.000	1.000
10554	20170117-M-Australian_Open-R128-Rafael_Nadal-Florian_Mayer	Rafael Nadal	Hard	dfs_parsed	0.000	1.000	-1.000
10685	20140626-M-Wimbledon-R64-Rafael_Nadal-Lukas_Rosol	Rafael Nadal	Grass	dfs_parsed	2.000	3.000	-1.000
10903	20100131-M-Australian_Open-F-Roger_Federer-Andy_Murray	Roger Federer	Hard	dfs_parsed	2.000	1.000	1.000
11054	20070122-M-Australian_Open-R16-Rafael_Nadal-Andy_Murray	Rafael Nadal	Hard	dfs_parsed	0.000	1.000	-1.000
11691	20090515-M-Madrid_Masters-SF-Novak_Djokovic-Rafael_Nadal	Rafael Nadal	Clay	dfs_parsed	4.000	3.000	1.000
11795	20060220-M-Dubai-R32-Stan_Wawrinka-Roger_Federer	Roger Federer	Hard	dfs_parsed	1.000	0.000	1.000

	match_id	player	surface	metric	parsed_count	source_count	difference
11827	20040409-M-Davis_Cup_World_Group_QF-RR-Nicolas_Escude-Roger_Federer	Roger Federer	Hard	dfs_parsed	4.000	5.000	-1.000

Event coverage and review checks:

check	count
invalid score/state point rows	16
unavailable rally lengths	1840
completed reconstructed tiebreaks	745
incomplete reconstructed tiebreak groups	7
declared formats contradicted by recorded sets	13

Canonical player-match preview:

match_id	player	surface	winner	is_win	has_overview	has_points	bp_events_validated
20260311-M-Indian_Wells_Masters-R16-Novak_Djokovic-Jack_Draper	Novak Djokovic	Hard	Jack Draper	0	True	True	True
20260201-M-Australian_Open-F-Novak_Djokovic-Carlos_Alcaraz	Novak Djokovic	Hard	Carlos Alcaraz	0	True	True	True
20260128-M-Australian_Open-QF-Novak_Djokovic-Lorenzo_Musetti	Novak Djokovic	Hard	Unavailable	<NA>	True	True	True
20260124-M-Australian_Open-R32-Novak_Djokovic-Botic_Van_De_Zandschulp	Novak Djokovic	Hard	Novak Djokovic	1	True	True	True
20251108-M-Athens-F-Novak_Djokovic-Lorenzo_Musetti	Novak Djokovic	Hard	Novak Djokovic	1	True	True	True
20251107-M-Athens-SF-Novak_Djokovic-Yannick_Hanfmann	Novak Djokovic	Hard	Novak Djokovic	1	True	True	True
20251003-M-Shanghai_Masters-R64-Novak_Djokovic-Marin_Cilic	Novak Djokovic	Hard	Novak Djokovic	1	True	True	True
20250905-M-US_Open-SF-Novak_Djokovic-Carlos_Alcaraz	Novak Djokovic	Hard	Carlos Alcaraz	0	True	True	True
20250902-M-US_Open-QF-Novak_Djokovic-Taylor_Fritz	Novak Djokovic	Hard	Novak Djokovic	1	True	True	True
20250831-M-US_Open-R16-Novak_Djokovic-Jan_Lennard_Struff	Novak Djokovic	Hard	Novak Djokovic	1	True	True	True

Built 265,339 match-point rows, 289,837 Big Three player-point rows, and 574,281 Big Three attempted rally strokes.
Parser edge cases, unique grains, missing-outcome handling, and join checks passed.

4.4 Interpretation and downstream handoff

The canonical tables above are the inputs for the performance and pressure analyses. Unknown outcomes and missing point totals remain missing; presence of a terminal score is not proof against every transcription error. Reconciliation discrepancies remain visible and are excluded from validated break-point components where relevant. Shot rows represent attempted strokes, including flagged error attempts; serves are separate point attributes. Missing preceding-point rally length remains missing, not a fabricated zero.

V. Core performance KPIs

Performance question: Is observed success driven more by serving, returning, or the surface schedule? Service and return rates separate two routes to winning; DR combines them on a common eligible sample.

5.1 Eligibility and coverage

Use each player's complete available charted coverage. **Wins are verified wins in this sample**, not complete career wins: terminal-score uncertainty remains unknown rather than a loss. Individual coverage dates, source availability, and unknown outcomes accompany the results. Weeks at No. 1 are unavailable in the supplied files.

Performance rates pool valid numerators and denominators. Overview statistics do not require a verified match outcome. A source/point discrepancy excludes only the affected KPI fields, not the whole player-match; source rows without points can still contribute valid statistics, with their unverified reconciliation coverage disclosed. DR requires a common eligible sample for both serving and returning. No radar-speed estimates or arbitrary Serve Impact composite are used.

Break-point rates here use source-compatible charted events on score-valid player-matches with matching Overview break-point and point-count checks. Conversion additionally reconciles against the opponent’s Overview break points faced and saved. Uncharted S/R placeholders have outcomes but no charted serve/rally detail; total point-win percentage includes their recorded outcomes. Completed regular-game hold/break rates exclude tiebreaks, missing 0–0 starts, changing servers, invalid score states or within-game score transitions, and incomplete games. These remain descriptive outcomes, not pressure-adjusted ability.

In [11]:
```
# Metric records retain counts, match coverage, source reconciliation, and uncertainty.
kpi_master = canonical_master.copy()
kpi_master["first_service_wins"] = kpi_master["first_won"]
```

```
kpi_master["second_service_wins"] = kpi_master["second_won"]
kpi_master["total_service_wins"] = kpi_master["first_won"] + kpi_master["second_won"]
opponent_pressure = filter_stat_slice(analysis_dataframes["charting_m_stats_Overview"], None, "total")[["match_id", "player", "bk_pts", "bp_saved"]].rename(
    columns={"player": "opponent", "bk_pts": "opponent_bp_faced", "bp_saved": "opponent_bp_saved"})
kpi_master = kpi_master.merge(opponent_pressure, on=["match_id", "opponent"], how="left", validate="many_to_one")
kpi_master["source_bp_converted"] = kpi_master["opponent_bp_faced"] - kpi_master["opponent_bp_saved"]
kpi_master["bp_conversion_reconciled"] = (kpi_master["opponent_bp_faced"].notna()
    & kpi_master["bp_opportunities_parsed"].eq(kpi_master["opponent_bp_faced"])
    & kpi_master["bp_converted_parsed"].eq(kpi_master["source_bp_converted"]))
bp_conversion_source_review = kpi_master.loc[kpi_master["has_points"] & kpi_master["opponent_bp_faced"].notna()
    & ~kpi_master["bp_conversion_reconciled"], ["match_id", "player", "bp_opportunities_parsed", "opponent_bp_faced", "bp_converted_parsed",
"source_bp_converted"]].copy()
kpi_master["known_outcome"] = kpi_master["winner"].notna().astype(int)
kpi_master["verified_win_count"] = kpi_master["is_win"].fillna(0).astype(int)

# Completed service games: check the entire game, not only its final point.
game_keys = ["match_id", "game_number"]
game_starts = canonical_points.groupby(game_keys, sort=False).head(1)[game_keys + ["Pts"]].rename(columns={"Pts": "starting_score"})
# Validate every within-game score transition in the regular-game denominator.
score_parts = canonical_points["Pts"].astype("string").str.upper().str.extract(r"^(AD|A|\d+)-(AD|A|\d+)$")
score_rank_map = {"0": 0, "15": 1, "30": 2, "40": 3, "AD": 4, "A": 4}
server_rank = score_parts[0].map(score_rank_map)
receiver_rank = score_parts[1].map(score_rank_map)
server_won = canonical_points["Svr"].eq(canonical_points["PtWinner"])
expected_server = server_rank + server_won.astype(int)
expected_receiver = receiver_rank + (~server_won).astype(int)
back_to_deuce = expected_server.eq(4) & expected_receiver.eq(4)
expected_server = expected_server.mask(back_to_deuce, 3)
expected_receiver = expected_receiver.mask(back_to_deuce, 3)
next_server = server_rank.groupby([canonical_points[k] for k in game_keys]).shift(-1)
next_receiver = receiver_rank.groupby([canonical_points[k] for k in game_keys]).shift(-1)
is_game_last = ~canonical_points.duplicated(game_keys, keep="last")
within_game_valid = (is_game_last | (~canonical_points["game_ended"]
    & expected_server.eq(next_server) & expected_receiver.eq(next_receiver)))
game_transition_quality = canonical_points[game_keys].assign(valid_transition=within_game_valid).groupby(game_keys, as_index=False).agg(
    all_transitions_valid=("valid_transition", "all"))
game_quality = canonical_points.groupby(game_keys, as_index=False).agg(
    all_score_states_valid=("score_state_valid", "all"), server_count=("Svr", "nunique"))
kpi_games = canonical_games.merge(game_starts, on=game_keys, validate="one_to_one").merge(game_quality, on=game_keys, validate="one_to_one")
kpi_games = kpi_games.merge(game_transition_quality, on=game_keys, validate="one_to_one")
kpi_games["eligible_regular_game"] = (~kpi_games["is_tiebreak_point"] & kpi_games["game_ended"]
    & kpi_games["starting_score"].eq("0-0") & kpi_games["all_score_states_valid"] & kpi_games["all_transitions_valid"] & kpi_games["server_count"].eq(1))
game_parts = []
for role in ["server_name", "returner_name"]:
    part = kpi_games.loc[kpi_games["eligible_regular_game"] & kpi_games[role].isin(BIG_THREE)].copy()
    part["player"] = part[role]
    part["service_games"] = int(role == "server_name")
    part["return_games"] = int(role == "returner_name")
    part["service_holds"] = part["game_winner"].eq(part["player"]) & part["service_games"].eq(1)
    part["return_breaks"] = part["game_winner"].eq(part["player"]) & part["return_games"].eq(1)
    game_parts.append(part)
game_counts = pd.concat(game_parts, ignore_index=True).groupby(["match_id", "player"], as_index=False)[
    ["service_games", "return_games", "service_holds", "return_breaks"]].sum()
kpi_master = kpi_master.merge(game_counts, on=["match_id", "player"], how="left", validate="one_to_one")

# Exclusion sets apply only to the corresponding metric, never to unknown match outcomes.
reconciliation_bad_pairs = {
```

```
    metric: set(group[["match_id", "player"]].itertuples(index=False, name=None))
    for metric, group in canonical_reconciliation_flags.groupby("metric")
}
kpi_pairs = list(kpi_master[["match_id", "player"]].itertuples(index=False, name=None))
def valid_count_ratio(numerator, denominator):
    a, b = pd.to_numeric(kpi_master[numerator], errors="coerce"), pd.to_numeric(kpi_master[denominator], errors="coerce")
    return a.notna() & b.notna() & np.isfinite(a) & np.isfinite(b) & a.ge(0) & b.gt(0) & a.le(b)

def reconciled_mask(fields):
    bad = set().union(*(reconciliation_bad_pairs.get(field, set()) for field in fields))
    return pd.Series([pair not in bad for pair in kpi_pairs], index=kpi_master.index)

source_valid = kpi_master["has_overview"]
rate_specs = {
    "match_win_pct": ("verified_win_count", "known_outcome", "matches", []),
    "point_win_pct": ("points_won", "points_played", "points", []),
    "first_serve_in_pct": ("first_in", "serve_pts", "points", ["service_points_parsed"]),
    "first_serve_win_pct": ("first_service_wins", "first_in", "points", ["service_wins_parsed", "service_points_parsed"]),
    "second_serve_win_pct": ("second_service_wins", "second_in", "points", ["service_wins_parsed", "service_points_parsed"]),
    "service_point_win_pct": ("total_service_wins", "serve_pts", "points", ["service_wins_parsed", "service_points_parsed"]),
    "return_point_win_pct": ("return_pts_won", "return_pts", "points", ["return_wins_parsed", "return_points_parsed"]),
    "ace_pct": ("aces", "serve_pts", "points", ["service_points_parsed"]),
    "double_fault_pct": ("dfs", "serve_pts", "points", ["dfs_parsed", "service_points_parsed"]),
    "break_point_save_pct": ("bp_saved_parsed", "bp_faced_parsed", "events", []),
    "break_point_conversion_pct": ("bp_converted_parsed", "bp_opportunities_parsed", "events", []),
    "service_hold_pct": ("service_holds", "service_games", "events", []),
    "return_break_pct": ("return_breaks", "return_games", "events", []),
}
source_metric_names = set(rate_specs) - {"match_win_pct", "point_win_pct", "break_point_save_pct", "break_point_conversion_pct", "service_hold_pct",
"return_break_pct"}
kpi_masks = {}
for metric, (num, den, basis, fields) in rate_specs.items():
    mask = valid_count_ratio(num, den)
    if metric in source_metric_names:
        mask &= source_valid & reconciled_mask(fields)
    if metric.startswith("break_point_"):
        mask &= kpi_master["bp_events_validated"]
    if metric == "break_point_conversion_pct":
        mask &= kpi_master["bp_conversion_reconciled"]
    kpi_masks[metric] = mask
# DR uses the same observations for service and return terms.
kpi_masks["dominance_ratio"] = (valid_count_ratio("total_service_wins", "serve_pts")
    & valid_count_ratio("return_pts_won", "return_pts") & source_valid
    & reconciled_mask(["service_wins_parsed", "service_points_parsed", "return_wins_parsed", "return_points_parsed"]))

kpi_scope_groups = [(player, surface) for player in BIG_THREE for surface in ["Overall"] + PROJECT_SURFACES]
kpi_rows, bootstrap_inputs, bootstrap_descriptors = [], [], []
bootstrap_match_ids = sorted(kpi_master["match_id"].unique())
bootstrap_match_index = {match_id: index for index, match_id in enumerate(bootstrap_match_ids)}
for player, surface in kpi_scope_groups:
    group_mask = kpi_master["player"].eq(player)
    if surface != "Overall":
        group_mask &= kpi_master["surface"].eq(surface)
    cohort = kpi_master.loc[group_mask]
    for metric in list(rate_specs) + ["dominance_ratio"]:
        eligible = group_mask & kpi_masks[metric]
        sample = kpi_master.loc[eligible]
```

```
        if metric == "dominance_ratio":
            fields = ["return_pts_won", "return_pts", "total_service_wins", "serve_pts"]
            totals = sample[fields].sum(min_count=1)
            service_loss_rate = 1 - totals["total_service_wins"] / totals["serve_pts"] if totals["serve_pts"] > 0 else np.nan
            estimate = (totals["return_pts_won"] / totals["return_pts"] / service_loss_rate
                        if totals["return_pts"] > 0 and service_loss_rate > 0 else np.nan)
            numerator, denominator = np.nan, np.nan
            sample_size = min(totals["return_pts"], totals["serve_pts"])
            small = pd.isna(sample_size) or sample_size < SMALL_SAMPLE_THRESHOLDS["points"]
            near_zero = pd.notna(service_loss_rate) and service_loss_rate < .01
        else:
            num, den, basis, _ = rate_specs[metric]
            fields = [num, den]
            numerator, denominator = sample[num].sum(min_count=1), sample[den].sum(min_count=1)
            estimate = 100 * numerator / denominator if denominator > 0 else np.nan
            small = pd.isna(denominator) or denominator < SMALL_SAMPLE_THRESHOLDS[basis]
            near_zero = False
        row_index = len(kpi_rows)
        kpi_rows.append({"player": player, "surface": surface, "metric": metric,
            "estimate": estimate, "unit": "ratio" if metric == "dominance_ratio" else "%",
            "numerator": numerator, "denominator": denominator, "eligible_matches": len(sample),
            "cohort_matches": len(cohort), "coverage_pct": 100 * len(sample) / len(cohort) if len(cohort) else np.nan,
            "matches_with_point_reconciliation": int(sample["has_points"].sum()) if metric in source_metric_names or metric == "dominance_ratio" else pd.NA,
            "small_sample": bool(small), "near_zero_dr_denominator": bool(near_zero),
            "return_points_won_dr": totals["return_pts_won"] if metric == "dominance_ratio" else np.nan,
            "return_points_dr": totals["return_pts"] if metric == "dominance_ratio" else np.nan,
            "service_points_won_dr": totals["total_service_wins"] if metric == "dominance_ratio" else np.nan,
            "service_points_dr": totals["serve_pts"] if metric == "dominance_ratio" else np.nan})
        columns = []
        for field in fields:
            vector = np.zeros(len(bootstrap_match_ids))
            for match_id, value in sample[["match_id", field]].itertuples(index=False, name=None):
                vector[bootstrap_match_index[match_id]] = value
            columns.append(len(bootstrap_inputs))
            bootstrap_inputs.append(vector)
        bootstrap_descriptors.append((row_index, metric, columns))

core_kpi_long = pd.DataFrame(kpi_rows)
# One joint match bootstrap retains both observations of a rivalry match.
bootstrap_rng = np.random.default_rng(RANDOM_SEED)
bootstrap_weights = bootstrap_rng.multinomial(len(bootstrap_match_ids),
    np.full(len(bootstrap_match_ids), 1 / len(bootstrap_match_ids)), size=BOOTSTRAP_REPLICATES)
bootstrap_totals = bootstrap_weights @ np.column_stack(bootstrap_inputs)
for row_index, metric, columns in bootstrap_descriptors:
    with np.errstate(divide="ignore", invalid="ignore"):
        if metric == "dominance_ratio":
            rw, rp, sw, sp = [bootstrap_totals[:, column] for column in columns]
            loss = 1 - sw / sp
            values = np.where((rp > 0) & (sp > 0) & (loss > 0), (rw / rp) / loss, np.nan)
        else:
            numerator, denominator = [bootstrap_totals[:, column] for column in columns]
            values = np.where(denominator > 0, 100 * numerator / denominator, np.nan)
    values = values[np.isfinite(values)]
    core_kpi_long.loc[row_index, "valid_bootstrap_replicates"] = len(values)
    core_kpi_long.loc[row_index, "ci95_lower"] = np.percentile(values, 2.5) if len(values) >= .95 * BOOTSTRAP_REPLICATES else np.nan
    core_kpi_long.loc[row_index, "ci95_upper"] = np.percentile(values, 97.5) if len(values) >= .95 * BOOTSTRAP_REPLICATES else np.nan
core_kpi_long["valid_bootstrap_replicates"] = core_kpi_long["valid_bootstrap_replicates"].astype(int)
```

```
core_kpi_long["status"] = np.select([core_kpi_long["estimate"].isna(), core_kpi_long["near_zero_dr_denominator"], core_kpi_long["small_sample"]],
    ["unavailable", "unstable DR denominator", "small sample"], default="available")

coverage_rows = []
for player, surface in kpi_scope_groups:
    group = kpi_master.loc[kpi_master["player"].eq(player)]
    if surface != "Overall":
        group = group.loc[group["surface"].eq(surface)]
    metadata_row = player_metadata.set_index("player").loc[player]
    hand = metadata_row["handedness"]
    coverage_rows.append({"player": player, "surface": surface, "charted_matches": len(group),
        "verified_wins": int(group["is_win"].sum()), "known_outcome_matches": int(group["winner"].notna().sum()),
        "unknown_outcome_matches": int(group["winner"].isna().sum()),
        "matches_with_overview": int(group["has_overview"].sum()), "matches_with_points": int(group["has_points"].sum()),
        "earliest_charted_date": group["match_date"].min(), "latest_charted_date": group["match_date"].max(),
        "handedness": hand, "handedness_resolution": metadata_row["resolution_status"],
        "handedness_conflicting_records": metadata_row["conflicting_records"], "weeks_at_no_1": "Unavailable in supplied files"})
core_kpi_coverage = pd.DataFrame(coverage_rows)
core_kpi_wide = core_kpi_coverage.merge(core_kpi_long.pivot(index=["player", "surface"], columns="metric", values="estimate").reset_index(),
    on=["player", "surface"], validate="one_to_one")
core_kpi_overall = core_kpi_wide.loc[core_kpi_wide["surface"].eq("Overall")].reset_index(drop=True)
core_kpi_by_surface = core_kpi_wide.loc[core_kpi_wide["surface"].ne("Overall")].reset_index(drop=True)

# Data and output invariants; confidence intervals need not contain the point estimate.
assert not core_kpi_long.duplicated(["player", "surface", "metric"]).any()
assert len(core_kpi_long) == 3 * 4 * 14
assert core_kpi_long.loc[core_kpi_long["unit"].eq("%"), "estimate"].dropna().between(0, 100).all()
assert core_kpi_long["eligible_matches"].le(core_kpi_long["cohort_matches"]).all()
assert core_kpi_long["valid_bootstrap_replicates"].between(0, BOOTSTRAP_REPLICATES).all()
assert (core_kpi_coverage["known_outcome_matches"] + core_kpi_coverage["unknown_outcome_matches"]).eq(core_kpi_coverage["charted_matches"]).all()
for player in BIG_THREE:
    overall = core_kpi_overall.loc[core_kpi_overall["player"].eq(player)].iloc[0]
    by_surface = core_kpi_by_surface.loc[core_kpi_by_surface["player"].eq(player)]
    other_surfaces = kpi_master.loc[kpi_master["player"].eq(player) & ~kpi_master["surface"].isin(PROJECT_SURFACES)]
    assert by_surface["verified_wins"].sum() + other_surfaces["is_win"].sum() == overall["verified_wins"]
print(f"Core KPIs validated: {len(core_kpi_long)} player/scope/metric rows; {BOOTSTRAP_REPLICATES:,} shared match-bootstrap replicates (seed {RANDOM_SEED}).")
print("Intervals condition on supplied charted matches, fixed eligibility, and the recorded outcomes; they do not repair unknown outcomes or selection bias.")
```

```
Core KPIs validated: 168 player/scope/metric rows; 2,000 shared match-bootstrap replicates (seed 42).
Intervals condition on supplied charted matches, fixed eligibility, and the recorded outcomes; they do not repair unknown outcomes or selection bias.
```

5.2 Overall and surface scorecards

Percentages are displayed on a 0–100 scale; DR is a ratio. The long-form evidence table contains every numerator, denominator, eligible-match count, small-sample flag, and 95% interval. Different KPIs can use different samples; compare their coverage before interpreting differences. First-serve win percentage is conditional on the first serve landing in; ace and double-fault percentages use all eligible service points. No rankings are inferred from overlapping or nonoverlapping marginal intervals alone.

In [12]:

```
print("Coverage and player context:")
report_table(core_kpi_coverage)
scorecard_columns = ["player", "surface", "verified_wins", "known_outcome_matches", "unknown_outcome_matches",
    "match_win_pct", "point_win_pct", "dominance_ratio", "first_serve_in_pct", "first_serve_win_pct", "second_serve_win_pct",
```

```
    "service_point_win_pct", "return_point_win_pct", "ace_pct", "double_fault_pct", "break_point_save_pct",
    "break_point_conversion_pct", "service_hold_pct", "return_break_pct"]
print("Overall performance (eligible samples vary by KPI):")
report_table(core_kpi_overall[scorecard_columns].round(2))
print("Surface-specific performance:")
report_table(core_kpi_by_surface[scorecard_columns].round(2))
print("Complete metric evidence and uncertainty:")
with pd.option_context("display.max_rows", None, "display.max_columns", None):
    report_table(core_kpi_long.round(3))
print("Unavailable or small-sample estimates:")
report_table(core_kpi_long.loc[core_kpi_long["status"].ne("available"), ["player", "surface", "metric", "status", "denominator", "eligible_matches"]])
print("Regular-game eligibility:")
report_table(kpi_games.groupby(["is_tiebreak_point", "eligible_regular_game"], as_index=False).size().rename(columns={"size": "games"}))

print("Break-point conversion cross-check against opponent Overview counts (flagged matches excluded from conversion KPI):")
report_table(bp_conversion_source_review)
```

```
Coverage and player context:
```

player	surface	charted_matches	verified_wins	known_outcome_matches	unknown_outcome_matches	matches_with_overview	matches_with_points	earliest_charted_date
Roger Federer	Overall	723	559	709	14	722	722	1998-07-04
Roger Federer	Clay	130	87	126	4	130	130	1999-07-16
Roger Federer	Grass	119	97	116	3	119	119	1998-07-04
Roger Federer	Hard	474	375	467	7	473	473	1998-10-05
Rafael Nadal	Overall	425	298	412	13	424	423	2003-05-15
Rafael Nadal	Clay	166	139	164	2	165	166	2003-05-15
Rafael Nadal	Grass	40	22	37	3	40	39	2003-06-27
Rafael Nadal	Hard	219	137	211	8	219	218	2004-03-28
Novak Djokovic	Overall	553	404	526	27	550	552	2005-01-17
Novak Djokovic	Clay	129	87	124	5	128	129	2006-04-19
Novak Djokovic	Grass	58	44	56	2	58	58	2007-07-05
Novak Djokovic	Hard	366	273	346	20	364	365	2005-01-17

player	surface	latest_charted_date	handedness	handedness_resolution	handedness_conflicting_records	weeks_at_no_1
Roger Federer	Overall	2021-07-05	Right-handed	consistent recognized source records	0	Unavailable in supplied files
Roger Federer	Clay	2021-06-05	Right-handed	consistent recognized source records	0	Unavailable in supplied files
Roger Federer	Grass	2021-07-05	Right-handed	consistent recognized source records	0	Unavailable in supplied files
Roger Federer	Hard	2021-03-10	Right-handed	consistent recognized source records	0	Unavailable in supplied files
Rafael Nadal	Overall	2024-11-19	Left-handed	documented source consensus; contradictions retained	3	Unavailable in supplied files
Rafael Nadal	Clay	2024-07-29	Left-handed	documented source consensus; contradictions retained	3	Unavailable in supplied files
Rafael Nadal	Grass	2022-07-04	Left-handed	documented source consensus; contradictions retained	3	Unavailable in supplied files
Rafael Nadal	Hard	2024-11-19	Left-handed	documented source consensus; contradictions retained	3	Unavailable in supplied files
Novak Djokovic	Overall	2026-03-11	Right-handed	consistent recognized source records	0	Unavailable in supplied files
Novak Djokovic	Clay	2025-06-06	Right-handed	consistent recognized source records	0	Unavailable in supplied files
Novak Djokovic	Grass	2025-07-11	Right-handed	consistent recognized source records	0	Unavailable in supplied files
Novak Djokovic	Hard	2026-03-11	Right-handed	consistent recognized source records	0	Unavailable in supplied files

Overall performance (eligible samples vary by KPI):

player	surface	verified_wins	known_outcome_matches	unknown_outcome_matches	match_win_pct	point_win_pct	dominance_ratio	first_serve_in_pct
Roger Federer	Overall	559	709	14	78.840	53.760	1.280	62.450
Rafael Nadal	Overall	298	412	13	72.330	52.750	1.180	68.880
Novak Djokovic	Overall	404	526	27	76.810	53.420	1.220	65.250

player	surface	first_serve_win_pct	second_serve_win_pct	service_point_win_pct	return_point_win_pct	ace_pct	double_fault_pct	break_point_save_pct
Roger Federer	Overall	76.990	56.460	69.280	39.230	10.180	2.200	65.790
Rafael Nadal	Overall	70.550	55.790	65.960	40.170	4.280	2.320	63.860
Novak Djokovic	Overall	73.270	54.930	66.900	40.450	7.190	2.820	63.920

player	surface	break_point_conversion_pct	service_hold_pct	return_break_pct
Roger Federer	Overall	40.640	88.210	25.680
Rafael Nadal	Overall	42.850	83.230	28.310
Novak Djokovic	Overall	42.730	84.720	28.140

Surface-specific performance:

player	surface	verified_wins	known_outcome_matches	unknown_outcome_matches	match_win_pct	point_win_pct	dominance_ratio	first_serve_in_pct
Roger Federer	Clay	87	126	4	69.050	52.580	1.160	61.090
Roger Federer	Grass	97	116	3	83.620	54.100	1.360	65.400
Roger Federer	Hard	375	467	7	80.300	54.000	1.290	61.970
Rafael Nadal	Clay	139	164	2	84.760	54.650	1.280	70.500
Rafael Nadal	Grass	22	37	3	59.460	51.650	1.130	69.110
Rafael Nadal	Hard	137	211	8	64.930	51.670	1.120	67.710
Novak Djokovic	Clay	87	124	5	70.160	52.540	1.140	65.630
Novak Djokovic	Grass	44	56	2	78.570	52.850	1.210	66.990
Novak Djokovic	Hard	273	346	20	78.900	53.870	1.260	64.750

player	surface	first_serve_win_pct	second_serve_win_pct	service_point_win_pct	return_point_win_pct	ace_pct	double_fault_pct	break_point_save_pct
Roger Federer	Clay	72.980	54.400	65.750	39.910	7.290	1.930	62.500
Roger Federer	Grass	79.740	59.660	72.790	37.000	11.890	1.580	71.820
Roger Federer	Hard	77.300	56.210	69.280	39.710	10.520	2.470	65.710
Rafael Nadal	Clay	68.970	55.620	65.030	44.820	2.610	2.140	65.310
Rafael Nadal	Grass	73.650	57.540	68.670	35.280	7.010	1.940	67.760
Rafael Nadal	Hard	70.950	55.500	65.960	38.150	4.790	2.530	61.930
Novak Djokovic	Clay	69.640	53.460	64.080	41.070	4.890	2.500	60.960
Novak Djokovic	Grass	76.480	53.320	68.830	37.680	9.270	2.820	68.450
Novak Djokovic	Hard	73.940	55.790	67.540	40.810	7.610	2.930	64.410

player	surface	break_point_conversion_pct	service_hold_pct	return_break_pct
Roger Federer	Clay	40.560	82.450	27.010
Roger Federer	Grass	39.720	92.850	21.320
Roger Federer	Hard	40.900	88.460	26.630
Rafael Nadal	Clay	45.840	81.860	37.950
Rafael Nadal	Grass	38.230	88.050	18.870
Rafael Nadal	Hard	40.930	83.040	23.920
Novak Djokovic	Clay	42.420	79.630	29.590
Novak Djokovic	Grass	37.630	87.960	22.000
Novak Djokovic	Hard	43.810	85.890	28.930

Complete metric evidence and uncertainty:

▼ Complete evidence table: 168 rows × 21 columns

player	surface	metric	estimate	unit	numerator	denominator	eligible_matches	cohort_matches	coverage_pct
Roger Federer	Overall	match_win_pct	78.843	%	559.000	709.000	709	723	98.064
Roger Federer	Overall	point_win_pct	53.755	%	67,376.000	125,338.000	722	723	99.862
Roger Federer	Overall	first_serve_in_pct	62.452	%	36,873.000	59,042.000	712	723	98.479
Roger Federer	Overall	first_serve_win_pct	76.994	%	28,390.000	36,873.000	712	723	98.479
Roger Federer	Overall	second_serve_win_pct	56.457	%	12,516.000	22,169.000	712	723	98.479
Roger Federer	Overall	service_point_win_pct	69.283	%	40,906.000	59,042.000	712	723	98.479
Roger Federer	Overall	return_point_win_pct	39.228	%	24,853.000	63,355.000	711	723	98.340
Roger Federer	Overall	ace_pct	10.178	%	6,009.000	59,042.000	712	723	98.479
Roger Federer	Overall	double_fault_pct	2.198	%	1,292.000	58,775.000	709	723	98.064
Roger Federer	Overall	break_point_save_pct	65.794	%	2,187.000	3,324.000	612	723	84.647
Roger Federer	Overall	break_point_conversion_pct	40.639	%	2,431.000	5,982.000	697	723	96.404
Roger Federer	Overall	service_hold_pct	88.207	%	8,811.000	9,989.000	722	723	99.862
Roger Federer	Overall	return_break_pct	25.683	%	2,518.000	9,804.000	722	723	99.862
Roger Federer	Overall	dominance_ratio	1.277	ratio	Unavailable	Unavailable	703	723	97.234
Roger Federer	Clay	match_win_pct	69.048	%	87.000	126.000	126	130	96.923
Roger Federer	Clay	point_win_pct	52.577	%	12,252.000	23,303.000	130	130	100.000
Roger Federer	Clay	first_serve_in_pct	61.092	%	6,725.000	11,008.000	127	130	97.692
Roger Federer	Clay	first_serve_win_pct	72.981	%	4,908.000	6,725.000	127	130	97.692
Roger Federer	Clay	second_serve_win_pct	54.401	%	2,330.000	4,283.000	127	130	97.692
Roger Federer	Clay	service_point_win_pct	65.752	%	7,238.000	11,008.000	127	130	97.692
Roger Federer	Clay	return_point_win_pct	39.906	%	4,679.000	11,725.000	129	130	99.231
Roger Federer	Clay	ace_pct	7.295	%	803.000	11,008.000	127	130	97.692
Roger Federer	Clay	double_fault_pct	1.926	%	212.000	11,008.000	127	130	97.692
Roger Federer	Clay	break_point_save_pct	62.500	%	515.000	824.000	121	130	93.077
Roger Federer	Clay	break_point_conversion_pct	40.557	%	466.000	1,149.000	126	130	96.923
Roger Federer	Clay	service_hold_pct	82.450	%	1,494.000	1,812.000	130	130	100.000
Roger Federer	Clay	return_break_pct	27.009	%	484.000	1,792.000	130	130	100.000
Roger Federer	Clay	dominance_ratio	1.162	ratio	Unavailable	Unavailable	126	130	96.923
Roger Federer	Grass	match_win_pct	83.621	%	97.000	116.000	116	119	97.479
Roger Federer	Grass	point_win_pct	54.096	%	12,679.000	23,438.000	119	119	100.000

player	surface	metric	estimate	unit	numerator	denominator	eligible_matches	cohort_matches	coverage_pct
Roger Federer	Grass	first_serve_in_pct	65.404	%	7,269.000	11,114.000	119	119	100.000
Roger Federer	Grass	first_serve_win_pct	79.736	%	5,796.000	7,269.000	119	119	100.000
Roger Federer	Grass	second_serve_win_pct	59.662	%	2,294.000	3,845.000	119	119	100.000
Roger Federer	Grass	service_point_win_pct	72.791	%	8,090.000	11,114.000	119	119	100.000
Roger Federer	Grass	return_point_win_pct	36.998	%	4,493.000	12,144.000	119	119	100.000
Roger Federer	Grass	ace_pct	11.886	%	1,321.000	11,114.000	119	119	100.000
Roger Federer	Grass	double_fault_pct	1.584	%	176.000	11,114.000	119	119	100.000
Roger Federer	Grass	break_point_save_pct	71.816	%	344.000	479.000	99	119	83.193
Roger Federer	Grass	break_point_conversion_pct	39.719	%	396.000	997.000	118	119	99.160
Roger Federer	Grass	service_hold_pct	92.850	%	1,766.000	1,902.000	119	119	100.000
Roger Federer	Grass	return_break_pct	21.322	%	400.000	1,876.000	119	119	100.000
Roger Federer	Grass	dominance_ratio	1.360	ratio	Unavailable	Unavailable	119	119	100.000
Roger Federer	Hard	match_win_pct	80.300	%	375.000	467.000	467	474	98.523
Roger Federer	Hard	point_win_pct	54.003	%	42,445.000	78,597.000	473	474	99.789
Roger Federer	Hard	first_serve_in_pct	61.969	%	22,879.000	36,920.000	466	474	98.312
Roger Federer	Hard	first_serve_win_pct	77.302	%	17,686.000	22,879.000	466	474	98.312
Roger Federer	Hard	second_serve_win_pct	56.207	%	7,892.000	14,041.000	466	474	98.312
Roger Federer	Hard	service_point_win_pct	69.280	%	25,578.000	36,920.000	466	474	98.312
Roger Federer	Hard	return_point_win_pct	39.713	%	15,681.000	39,486.000	463	474	97.679
Roger Federer	Hard	ace_pct	10.523	%	3,885.000	36,920.000	466	474	98.312
Roger Federer	Hard	double_fault_pct	2.466	%	904.000	36,653.000	463	474	97.679
Roger Federer	Hard	break_point_save_pct	65.710	%	1,328.000	2,021.000	392	474	82.700
Roger Federer	Hard	break_point_conversion_pct	40.902	%	1,569.000	3,836.000	453	474	95.570
Roger Federer	Hard	service_hold_pct	88.462	%	5,551.000	6,275.000	473	474	99.789
Roger Federer	Hard	return_break_pct	26.630	%	1,634.000	6,136.000	473	474	99.789
Roger Federer	Hard	dominance_ratio	1.294	ratio	Unavailable	Unavailable	458	474	96.624
Rafael Nadal	Overall	match_win_pct	72.330	%	298.000	412.000	412	425	96.941
Rafael Nadal	Overall	point_win_pct	52.751	%	37,398.000	70,896.000	423	425	99.529
Rafael Nadal	Overall	first_serve_in_pct	68.877	%	23,385.000	33,952.000	416	425	97.882
Rafael Nadal	Overall	first_serve_win_pct	70.554	%	16,499.000	23,385.000	416	425	97.882
Rafael Nadal	Overall	second_serve_win_pct	55.787	%	5,895.000	10,567.000	416	425	97.882

player	surface	metric	estimate	unit	numerator	denominator	eligible_matches	cohort_matches	coverage_pct
Rafael Nadal	Overall	service_point_win_pct	65.958	%	22,394.000	33,952.000	416	425	97.882
Rafael Nadal	Overall	return_point_win_pct	40.174	%	14,059.000	34,995.000	411	425	96.706
Rafael Nadal	Overall	ace_pct	4.277	%	1,452.000	33,952.000	416	425	97.882
Rafael Nadal	Overall	double_fault_pct	2.317	%	776.000	33,497.000	412	425	96.941
Rafael Nadal	Overall	break_point_save_pct	63.858	%	1,569.000	2,457.000	361	425	84.941
Rafael Nadal	Overall	break_point_conversion_pct	42.853	%	1,499.000	3,498.000	394	425	92.706
Rafael Nadal	Overall	service_hold_pct	83.232	%	4,636.000	5,570.000	422	425	99.294
Rafael Nadal	Overall	return_break_pct	28.314	%	1,570.000	5,545.000	423	425	99.529
Rafael Nadal	Overall	dominance_ratio	1.183	ratio	Unavailable	Unavailable	406	425	95.529
Rafael Nadal	Clay	match_win_pct	84.756	%	139.000	164.000	164	166	98.795
Rafael Nadal	Clay	point_win_pct	54.650	%	14,092.000	25,786.000	166	166	100.000
Rafael Nadal	Clay	first_serve_in_pct	70.496	%	8,566.000	12,151.000	161	166	96.988
Rafael Nadal	Clay	first_serve_win_pct	68.970	%	5,908.000	8,566.000	161	166	96.988
Rafael Nadal	Clay	second_serve_win_pct	55.621	%	1,994.000	3,585.000	161	166	96.988
Rafael Nadal	Clay	service_point_win_pct	65.032	%	7,902.000	12,151.000	161	166	96.988
Rafael Nadal	Clay	return_point_win_pct	44.823	%	5,606.000	12,507.000	157	166	94.578
Rafael Nadal	Clay	ace_pct	2.609	%	317.000	12,151.000	161	166	96.988
Rafael Nadal	Clay	double_fault_pct	2.138	%	257.000	12,018.000	160	166	96.386
Rafael Nadal	Clay	break_point_save_pct	65.306	%	640.000	980.000	139	166	83.735
Rafael Nadal	Clay	break_point_conversion_pct	45.841	%	711.000	1,551.000	154	166	92.771
Rafael Nadal	Clay	service_hold_pct	81.859	%	1,638.000	2,001.000	166	166	100.000
Rafael Nadal	Clay	return_break_pct	37.954	%	753.000	1,984.000	166	166	100.000
Rafael Nadal	Clay	dominance_ratio	1.283	ratio	Unavailable	Unavailable	155	166	93.373
Rafael Nadal	Grass	match_win_pct	59.459	%	22.000	37.000	37	40	92.500
Rafael Nadal	Grass	point_win_pct	51.647	%	4,391.000	8,502.000	39	40	97.500
Rafael Nadal	Grass	first_serve_in_pct	69.108	%	2,850.000	4,124.000	39	40	97.500
Rafael Nadal	Grass	first_serve_win_pct	73.649	%	2,099.000	2,850.000	39	40	97.500
Rafael Nadal	Grass	second_serve_win_pct	57.535	%	733.000	1,274.000	39	40	97.500
Rafael Nadal	Grass	service_point_win_pct	68.671	%	2,832.000	4,124.000	39	40	97.500
Rafael Nadal	Grass	return_point_win_pct	35.279	%	1,553.000	4,402.000	40	40	100.000
Rafael Nadal	Grass	ace_pct	7.008	%	289.000	4,124.000	39	40	97.500

player	surface	metric	estimate	unit	numerator	denominator	eligible_matches	cohort_matches	coverage_pct
Rafael Nadal	Grass	double_fault_pct	1.940	%	80.000	4,124.000	39	40	97.500
Rafael Nadal	Grass	break_point_save_pct	67.755	%	166.000	245.000	33	40	82.500
Rafael Nadal	Grass	break_point_conversion_pct	38.226	%	125.000	327.000	38	40	95.000
Rafael Nadal	Grass	service_hold_pct	88.047	%	604.000	686.000	39	40	97.500
Rafael Nadal	Grass	return_break_pct	18.868	%	130.000	689.000	39	40	97.500
Rafael Nadal	Grass	dominance_ratio	1.129	ratio	Unavailable	Unavailable	39	40	97.500
Rafael Nadal	Hard	match_win_pct	64.929	%	137.000	211.000	211	219	96.347
Rafael Nadal	Hard	point_win_pct	51.669	%	18,915.000	36,608.000	218	219	99.543
Rafael Nadal	Hard	first_serve_in_pct	67.709	%	11,969.000	17,677.000	216	219	98.630
Rafael Nadal	Hard	first_serve_win_pct	70.950	%	8,492.000	11,969.000	216	219	98.630
Rafael Nadal	Hard	second_serve_win_pct	55.501	%	3,168.000	5,708.000	216	219	98.630
Rafael Nadal	Hard	service_point_win_pct	65.961	%	11,660.000	17,677.000	216	219	98.630
Rafael Nadal	Hard	return_point_win_pct	38.151	%	6,900.000	18,086.000	214	219	97.717
Rafael Nadal	Hard	ace_pct	4.786	%	846.000	17,677.000	216	219	98.630
Rafael Nadal	Hard	double_fault_pct	2.530	%	439.000	17,355.000	213	219	97.260
Rafael Nadal	Hard	break_point_save_pct	61.932	%	763.000	1,232.000	189	219	86.301
Rafael Nadal	Hard	break_point_conversion_pct	40.926	%	663.000	1,620.000	202	219	92.237
Rafael Nadal	Hard	service_hold_pct	83.039	%	2,394.000	2,883.000	217	219	99.087
Rafael Nadal	Hard	return_break_pct	23.921	%	687.000	2,872.000	218	219	99.543
Rafael Nadal	Hard	dominance_ratio	1.123	ratio	Unavailable	Unavailable	212	219	96.804
Novak Djokovic	Overall	match_win_pct	76.806	%	404.000	526.000	526	553	95.118
Novak Djokovic	Overall	point_win_pct	53.422	%	50,005.000	93,603.000	552	553	99.819
Novak Djokovic	Overall	first_serve_in_pct	65.253	%	29,612.000	45,380.000	549	553	99.277
Novak Djokovic	Overall	first_serve_win_pct	73.271	%	21,697.000	29,612.000	549	553	99.277
Novak Djokovic	Overall	second_serve_win_pct	54.934	%	8,662.000	15,768.000	549	553	99.277
Novak Djokovic	Overall	service_point_win_pct	66.900	%	30,359.000	45,380.000	549	553	99.277
Novak Djokovic	Overall	return_point_win_pct	40.448	%	19,038.000	47,068.000	547	553	98.915
Novak Djokovic	Overall	ace_pct	7.195	%	3,265.000	45,380.000	549	553	99.277
Novak Djokovic	Overall	double_fault_pct	2.817	%	1,272.000	45,150.000	547	553	98.915
Novak Djokovic	Overall	break_point_save_pct	63.921	%	1,979.000	3,096.000	501	553	90.597
Novak Djokovic	Overall	break_point_conversion_pct	42.726	%	1,997.000	4,674.000	537	553	97.107

player	surface	metric	estimate	unit	numerator	denominator	eligible_matches	cohort_matches	coverage_pct
Novak Djokovic	Overall	service_hold_pct	84.716	%	6,280.000	7,413.000	552	553	99.819
Novak Djokovic	Overall	return_break_pct	28.138	%	2,029.000	7,211.000	552	553	99.819
Novak Djokovic	Overall	dominance_ratio	1.221	ratio	Unavailable	Unavailable	546	553	98.734
Novak Djokovic	Clay	match_win_pct	70.161	%	87.000	124.000	124	129	96.124
Novak Djokovic	Clay	point_win_pct	52.543	%	11,538.000	21,959.000	129	129	100.000
Novak Djokovic	Clay	first_serve_in_pct	65.634	%	6,992.000	10,653.000	127	129	98.450
Novak Djokovic	Clay	first_serve_win_pct	69.637	%	4,869.000	6,992.000	127	129	98.450
Novak Djokovic	Clay	second_serve_win_pct	53.455	%	1,957.000	3,661.000	127	129	98.450
Novak Djokovic	Clay	service_point_win_pct	64.076	%	6,826.000	10,653.000	127	129	98.450
Novak Djokovic	Clay	return_point_win_pct	41.074	%	4,468.000	10,878.000	127	129	98.450
Novak Djokovic	Clay	ace_pct	4.891	%	521.000	10,653.000	127	129	98.450
Novak Djokovic	Clay	double_fault_pct	2.504	%	261.000	10,423.000	125	129	96.899
Novak Djokovic	Clay	break_point_save_pct	60.959	%	534.000	876.000	121	129	93.798
Novak Djokovic	Clay	break_point_conversion_pct	42.416	%	481.000	1,134.000	124	129	96.124
Novak Djokovic	Clay	service_hold_pct	79.629	%	1,372.000	1,723.000	129	129	100.000
Novak Djokovic	Clay	return_break_pct	29.588	%	495.000	1,673.000	129	129	100.000
Novak Djokovic	Clay	dominance_ratio	1.142	ratio	Unavailable	Unavailable	126	129	97.674
Novak Djokovic	Grass	match_win_pct	78.571	%	44.000	56.000	56	58	96.552
Novak Djokovic	Grass	point_win_pct	52.845	%	6,677.000	12,635.000	58	58	100.000
Novak Djokovic	Grass	first_serve_in_pct	66.986	%	4,060.000	6,061.000	58	58	100.000
Novak Djokovic	Grass	first_serve_win_pct	76.478	%	3,105.000	4,060.000	58	58	100.000
Novak Djokovic	Grass	second_serve_win_pct	53.323	%	1,067.000	2,001.000	58	58	100.000
Novak Djokovic	Grass	service_point_win_pct	68.834	%	4,172.000	6,061.000	58	58	100.000
Novak Djokovic	Grass	return_point_win_pct	37.680	%	2,404.000	6,380.000	58	58	100.000
Novak Djokovic	Grass	ace_pct	9.272	%	562.000	6,061.000	58	58	100.000
Novak Djokovic	Grass	double_fault_pct	2.821	%	171.000	6,061.000	58	58	100.000
Novak Djokovic	Grass	break_point_save_pct	68.449	%	256.000	374.000	55	58	94.828
Novak Djokovic	Grass	break_point_conversion_pct	37.633	%	213.000	566.000	56	58	96.552
Novak Djokovic	Grass	service_hold_pct	87.964	%	877.000	997.000	58	58	100.000
Novak Djokovic	Grass	return_break_pct	21.996	%	216.000	982.000	58	58	100.000
Novak Djokovic	Grass	dominance_ratio	1.209	ratio	Unavailable	Unavailable	58	58	100.000
Novak Djokovic	Hard	match_win_pct	78.902	%	273.000	346.000	346	366	94.536

player	surface	metric	estimate	unit	numerator	denominator	eligible_matches	cohort_matches	coverage_pct
Novak Djokovic	Hard	point_win_pct	53.873	%	31,790.000	59,009.000	365	366	99.727
Novak Djokovic	Hard	first_serve_in_pct	64.746	%	18,560.000	28,666.000	364	366	99.454
Novak Djokovic	Hard	first_serve_win_pct	73.939	%	13,723.000	18,560.000	364	366	99.454
Novak Djokovic	Hard	second_serve_win_pct	55.789	%	5,638.000	10,106.000	364	366	99.454
Novak Djokovic	Hard	service_point_win_pct	67.540	%	19,361.000	28,666.000	364	366	99.454
Novak Djokovic	Hard	return_point_win_pct	40.812	%	12,166.000	29,810.000	362	366	98.907
Novak Djokovic	Hard	ace_pct	7.612	%	2,182.000	28,666.000	364	366	99.454
Novak Djokovic	Hard	double_fault_pct	2.930	%	840.000	28,666.000	364	366	99.454
Novak Djokovic	Hard	break_point_save_pct	64.410	%	1,189.000	1,846.000	325	366	88.798
Novak Djokovic	Hard	break_point_conversion_pct	43.813	%	1,303.000	2,974.000	357	366	97.541
Novak Djokovic	Hard	service_hold_pct	85.894	%	4,031.000	4,693.000	365	366	99.727
Novak Djokovic	Hard	return_break_pct	28.929	%	1,318.000	4,556.000	365	366	99.727
Novak Djokovic	Hard	dominance_ratio	1.257	ratio	Unavailable	Unavailable	362	366	98.907

player	surface	metric	matches_with_point_reconciliation	small_sample	near_zero_dr_denominator	return_points_won_dr	return_points_dr	service_points_won_dr	service_points_dr
Roger Federer	Overall	match_win_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Roger Federer	Overall	point_win_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Roger Federer	Overall	first_serve_in_pct	712	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Roger Federer	Overall	first_serve_win_pct	712	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Roger Federer	Overall	second_serve_win_pct	712	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Roger Federer	Overall	service_point_win_pct	712	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Roger Federer	Overall	return_point_win_pct	711	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Roger Federer	Overall	ace_pct	712	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Roger Federer	Overall	double_fault_pct	709	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Roger Federer	Overall	break_point_save_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Roger Federer	Overall	break_point_conversion_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Roger Federer	Overall	service_hold_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Roger Federer	Overall	return_break_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Roger Federer	Overall	dominance_ratio	703	False	False	24,576.000	62,654.000	40,413.000	58,327.000
Roger Federer	Clay	match_win_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable

player	surface	metric	matches_with_point_reconciliation	small_sample	near_zero_dr_denominator	return_points_won_dr	return_points_dr	service_points_won_dr	service_points_dr
Roger Federer	Clay	point_win_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Roger Federer	Clay	first_serve_in_pct	127	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Roger Federer	Clay	first_serve_win_pct	127	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Roger Federer	Clay	second_serve_win_pct	127	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Roger Federer	Clay	service_point_win_pct	127	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Roger Federer	Clay	return_point_win_pct	129	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Roger Federer	Clay	ace_pct	127	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Roger Federer	Clay	double_fault_pct	127	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Roger Federer	Clay	break_point_save_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Roger Federer	Clay	break_point_conversion_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Roger Federer	Clay	service_hold_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Roger Federer	Clay	return_break_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Roger Federer	Clay	dominance_ratio	126	False	False	4,575.000	11,479.000	7,183.000	10,931.000
Roger Federer	Grass	match_win_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Roger Federer	Grass	point_win_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Roger Federer	Grass	first_serve_in_pct	119	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Roger Federer	Grass	first_serve_win_pct	119	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Roger Federer	Grass	second_serve_win_pct	119	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Roger Federer	Grass	service_point_win_pct	119	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Roger Federer	Grass	return_point_win_pct	119	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Roger Federer	Grass	ace_pct	119	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Roger Federer	Grass	double_fault_pct	119	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Roger Federer	Grass	break_point_save_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Roger Federer	Grass	break_point_conversion_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Roger Federer	Grass	service_hold_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Roger Federer	Grass	return_break_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Roger Federer	Grass	dominance_ratio	119	False	False	4,493.000	12,144.000	8,090.000	11,114.000
Roger Federer	Hard	match_win_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Roger Federer	Hard	point_win_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Roger Federer	Hard	first_serve_in_pct	466	False	False	Unavailable	Unavailable	Unavailable	Unavailable

player	surface	metric	matches_with_point_reconciliation	small_sample	near_zero_dr_denominator	return_points_won_dr	return_points_dr	service_points_won_dr	service_points_dr
Roger Federer	Hard	first_serve_win_pct	466	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Roger Federer	Hard	second_serve_win_pct	466	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Roger Federer	Hard	service_point_win_pct	466	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Roger Federer	Hard	return_point_win_pct	463	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Roger Federer	Hard	ace_pct	466	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Roger Federer	Hard	double_fault_pct	463	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Roger Federer	Hard	break_point_save_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Roger Federer	Hard	break_point_conversion_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Roger Federer	Hard	service_hold_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Roger Federer	Hard	return_break_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Roger Federer	Hard	dominance_ratio	458	False	False	15,508.000	39,031.000	25,140.000	36,282.000
Rafael Nadal	Overall	match_win_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Rafael Nadal	Overall	point_win_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Rafael Nadal	Overall	first_serve_in_pct	414	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Rafael Nadal	Overall	first_serve_win_pct	414	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Rafael Nadal	Overall	second_serve_win_pct	414	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Rafael Nadal	Overall	service_point_win_pct	414	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Rafael Nadal	Overall	return_point_win_pct	409	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Rafael Nadal	Overall	ace_pct	414	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Rafael Nadal	Overall	double_fault_pct	410	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Rafael Nadal	Overall	break_point_save_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Rafael Nadal	Overall	break_point_conversion_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Rafael Nadal	Overall	service_hold_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Rafael Nadal	Overall	return_break_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Rafael Nadal	Overall	dominance_ratio	404	False	False	13,863.000	34,464.000	21,797.000	33,023.000
Rafael Nadal	Clay	match_win_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Rafael Nadal	Clay	point_win_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Rafael Nadal	Clay	first_serve_in_pct	161	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Rafael Nadal	Clay	first_serve_win_pct	161	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Rafael Nadal	Clay	second_serve_win_pct	161	False	False	Unavailable	Unavailable	Unavailable	Unavailable

player	surface	metric	matches_with_point_reconciliation	small_sample	near_zero_dr_denominator	return_points_won_dr	return_points_dr	service_points_won_dr	service_points_dr
Rafael Nadal	Clay	service_point_win_pct	161	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Rafael Nadal	Clay	return_point_win_pct	157	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Rafael Nadal	Clay	ace_pct	161	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Rafael Nadal	Clay	double_fault_pct	160	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Rafael Nadal	Clay	break_point_save_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Rafael Nadal	Clay	break_point_conversion_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Rafael Nadal	Clay	service_hold_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Rafael Nadal	Clay	return_break_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Rafael Nadal	Clay	dominance_ratio	155	False	False	5,560.000	12,383.000	7,601.000	11,692.000
Rafael Nadal	Grass	match_win_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Rafael Nadal	Grass	point_win_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Rafael Nadal	Grass	first_serve_in_pct	38	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Rafael Nadal	Grass	first_serve_win_pct	38	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Rafael Nadal	Grass	second_serve_win_pct	38	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Rafael Nadal	Grass	service_point_win_pct	38	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Rafael Nadal	Grass	return_point_win_pct	39	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Rafael Nadal	Grass	ace_pct	38	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Rafael Nadal	Grass	double_fault_pct	38	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Rafael Nadal	Grass	break_point_save_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Rafael Nadal	Grass	break_point_conversion_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Rafael Nadal	Grass	service_hold_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Rafael Nadal	Grass	return_break_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Rafael Nadal	Grass	dominance_ratio	38	False	False	1,513.000	4,279.000	2,832.000	4,124.000
Rafael Nadal	Hard	match_win_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Rafael Nadal	Hard	point_win_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Rafael Nadal	Hard	first_serve_in_pct	215	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Rafael Nadal	Hard	first_serve_win_pct	215	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Rafael Nadal	Hard	second_serve_win_pct	215	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Rafael Nadal	Hard	service_point_win_pct	215	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Rafael Nadal	Hard	return_point_win_pct	213	False	False	Unavailable	Unavailable	Unavailable	Unavailable

player	surface	metric	matches_with_point_reconciliation	small_sample	near_zero_dr_denominator	return_points_won_dr	return_points_dr	service_points_won_dr	service_points_dr
Rafael Nadal	Hard	ace_pct	215	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Rafael Nadal	Hard	double_fault_pct	212	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Rafael Nadal	Hard	break_point_save_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Rafael Nadal	Hard	break_point_conversion_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Rafael Nadal	Hard	service_hold_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Rafael Nadal	Hard	return_break_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Rafael Nadal	Hard	dominance_ratio	211	False	False	6,790.000	17,802.000	11,364.000	17,207.000
Novak Djokovic	Overall	match_win_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Novak Djokovic	Overall	point_win_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Novak Djokovic	Overall	first_serve_in_pct	549	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Novak Djokovic	Overall	first_serve_win_pct	549	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Novak Djokovic	Overall	second_serve_win_pct	549	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Novak Djokovic	Overall	service_point_win_pct	549	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Novak Djokovic	Overall	return_point_win_pct	547	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Novak Djokovic	Overall	ace_pct	549	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Novak Djokovic	Overall	double_fault_pct	547	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Novak Djokovic	Overall	break_point_save_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Novak Djokovic	Overall	break_point_conversion_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Novak Djokovic	Overall	service_hold_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Novak Djokovic	Overall	return_break_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Novak Djokovic	Overall	dominance_ratio	546	False	False	18,973.000	46,913.000	30,214.000	45,172.000
Novak Djokovic	Clay	match_win_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Novak Djokovic	Clay	point_win_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Novak Djokovic	Clay	first_serve_in_pct	127	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Novak Djokovic	Clay	first_serve_win_pct	127	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Novak Djokovic	Clay	second_serve_win_pct	127	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Novak Djokovic	Clay	service_point_win_pct	127	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Novak Djokovic	Clay	return_point_win_pct	127	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Novak Djokovic	Clay	ace_pct	127	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Novak Djokovic	Clay	double_fault_pct	125	False	False	Unavailable	Unavailable	Unavailable	Unavailable

player	surface	metric	matches_with_point_reconciliation	small_sample	near_zero_dr_denominator	return_points_won_dr	return_points_dr	service_points_won_dr	service_points_dr
Novak Djokovic	Clay	break_point_save_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Novak Djokovic	Clay	break_point_conversion_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Novak Djokovic	Clay	service_hold_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Novak Djokovic	Clay	return_break_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Novak Djokovic	Clay	dominance_ratio	126	False	False	4,403.000	10,723.000	6,787.000	10,596.000
Novak Djokovic	Grass	match_win_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Novak Djokovic	Grass	point_win_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Novak Djokovic	Grass	first_serve_in_pct	58	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Novak Djokovic	Grass	first_serve_win_pct	58	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Novak Djokovic	Grass	second_serve_win_pct	58	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Novak Djokovic	Grass	service_point_win_pct	58	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Novak Djokovic	Grass	return_point_win_pct	58	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Novak Djokovic	Grass	ace_pct	58	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Novak Djokovic	Grass	double_fault_pct	58	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Novak Djokovic	Grass	break_point_save_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Novak Djokovic	Grass	break_point_conversion_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Novak Djokovic	Grass	service_hold_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Novak Djokovic	Grass	return_break_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Novak Djokovic	Grass	dominance_ratio	58	False	False	2,404.000	6,380.000	4,172.000	6,061.000
Novak Djokovic	Hard	match_win_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Novak Djokovic	Hard	point_win_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Novak Djokovic	Hard	first_serve_in_pct	364	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Novak Djokovic	Hard	first_serve_win_pct	364	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Novak Djokovic	Hard	second_serve_win_pct	364	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Novak Djokovic	Hard	service_point_win_pct	364	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Novak Djokovic	Hard	return_point_win_pct	362	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Novak Djokovic	Hard	ace_pct	364	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Novak Djokovic	Hard	double_fault_pct	364	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Novak Djokovic	Hard	break_point_save_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Novak Djokovic	Hard	break_point_conversion_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable

player	surface	metric	matches_with_point_reconciliation	small_sample	near_zero_dr_denominator	return_points_won_dr	return_points_dr	service_points_won_dr	service_points_dr
Novak Djokovic	Hard	service_hold_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Novak Djokovic	Hard	return_break_pct	<NA>	False	False	Unavailable	Unavailable	Unavailable	Unavailable
Novak Djokovic	Hard	dominance_ratio	362	False	False	12,166.000	29,810.000	19,255.000	28,515.000

player	surface	metric	valid_bootstrap_replicates	ci95_lower	ci95_upper	status
Roger Federer	Overall	match_win_pct	2000	75.921	81.818	available
Roger Federer	Overall	point_win_pct	2000	53.415	54.122	available
Roger Federer	Overall	first_serve_in_pct	2000	61.971	62.909	available
Roger Federer	Overall	first_serve_win_pct	2000	76.460	77.530	available
Roger Federer	Overall	second_serve_win_pct	2000	55.686	57.229	available
Roger Federer	Overall	service_point_win_pct	2000	68.743	69.800	available
Roger Federer	Overall	return_point_win_pct	2000	38.679	39.763	available
Roger Federer	Overall	ace_pct	2000	9.763	10.573	available
Roger Federer	Overall	double_fault_pct	2000	2.060	2.342	available
Roger Federer	Overall	break_point_save_pct	2000	64.236	67.388	available
Roger Federer	Overall	break_point_conversion_pct	2000	39.327	42.017	available
Roger Federer	Overall	service_hold_pct	2000	87.395	88.989	available
Roger Federer	Overall	return_break_pct	2000	24.539	26.841	available
Roger Federer	Overall	dominance_ratio	2000	1.250	1.306	available
Roger Federer	Clay	match_win_pct	2000	60.869	76.925	available
Roger Federer	Clay	point_win_pct	2000	51.752	53.444	available
Roger Federer	Clay	first_serve_in_pct	2000	59.991	62.167	available
Roger Federer	Clay	first_serve_win_pct	2000	71.647	74.310	available
Roger Federer	Clay	second_serve_win_pct	2000	52.557	56.218	available
Roger Federer	Clay	service_point_win_pct	2000	64.415	67.065	available
Roger Federer	Clay	return_point_win_pct	2000	38.759	41.054	available
Roger Federer	Clay	ace_pct	2000	6.511	8.130	available
Roger Federer	Clay	double_fault_pct	2000	1.602	2.251	available
Roger Federer	Clay	break_point_save_pct	2000	59.148	65.644	available
Roger Federer	Clay	break_point_conversion_pct	2000	37.310	43.886	available
Roger Federer	Clay	service_hold_pct	2000	80.053	84.724	available
Roger Federer	Clay	return_break_pct	2000	24.746	29.405	available
Roger Federer	Clay	dominance_ratio	2000	1.107	1.227	available
Roger Federer	Grass	match_win_pct	2000	76.423	90.000	available
Roger Federer	Grass	point_win_pct	2000	53.306	54.874	available

player	surface	metric	valid_bootstrap_replicates	ci95_lower	ci95_upper	status
Roger Federer	Grass	first_serve_in_pct	2000	64.366	66.333	available
Roger Federer	Grass	first_serve_win_pct	2000	78.568	80.869	available
Roger Federer	Grass	second_serve_win_pct	2000	57.860	61.400	available
Roger Federer	Grass	service_point_win_pct	2000	71.591	73.921	available
Roger Federer	Grass	return_point_win_pct	2000	35.838	38.219	available
Roger Federer	Grass	ace_pct	2000	10.965	12.847	available
Roger Federer	Grass	double_fault_pct	2000	1.325	1.853	available
Roger Federer	Grass	break_point_save_pct	2000	68.110	75.294	available
Roger Federer	Grass	break_point_conversion_pct	2000	36.737	42.980	available
Roger Federer	Grass	service_hold_pct	2000	91.467	94.090	available
Roger Federer	Grass	return_break_pct	2000	18.886	23.865	available
Roger Federer	Grass	dominance_ratio	2000	1.285	1.438	available
Roger Federer	Hard	match_win_pct	2000	76.652	84.081	available
Roger Federer	Hard	point_win_pct	2000	53.571	54.463	available
Roger Federer	Hard	first_serve_in_pct	2000	61.402	62.549	available
Roger Federer	Hard	first_serve_win_pct	2000	76.610	77.947	available
Roger Federer	Hard	second_serve_win_pct	2000	55.270	57.169	available
Roger Federer	Hard	service_point_win_pct	2000	68.655	69.928	available
Roger Federer	Hard	return_point_win_pct	2000	39.032	40.400	available
Roger Federer	Hard	ace_pct	2000	10.044	11.000	available
Roger Federer	Hard	double_fault_pct	2000	2.284	2.650	available
Roger Federer	Hard	break_point_save_pct	2000	63.615	67.824	available
Roger Federer	Hard	break_point_conversion_pct	2000	39.250	42.721	available
Roger Federer	Hard	service_hold_pct	2000	87.477	89.456	available
Roger Federer	Hard	return_break_pct	2000	25.184	28.087	available
Roger Federer	Hard	dominance_ratio	2000	1.257	1.334	available
Rafael Nadal	Overall	match_win_pct	2000	68.000	76.538	available
Rafael Nadal	Overall	point_win_pct	2000	52.262	53.274	available
Rafael Nadal	Overall	first_serve_in_pct	2000	68.282	69.477	available
Rafael Nadal	Overall	first_serve_win_pct	2000	69.733	71.368	available
Rafael Nadal	Overall	second_serve_win_pct	2000	54.757	56.862	available
Rafael Nadal	Overall	service_point_win_pct	2000	65.216	66.661	available
Rafael Nadal	Overall	return_point_win_pct	2000	39.417	40.958	available
Rafael Nadal	Overall	ace_pct	2000	3.927	4.639	available
Rafael Nadal	Overall	double_fault_pct	2000	2.138	2.494	available
Rafael Nadal	Overall	break_point_save_pct	2000	61.974	65.762	available
Rafael Nadal	Overall	break_point_conversion_pct	2000	41.244	44.584	available

player	surface	metric	valid_bootstrap_replicates	ci95_lower	ci95_upper	status
Rafael Nadal	Overall	service_hold_pct	2000	81.894	84.459	available
Rafael Nadal	Overall	return_break_pct	2000	26.742	29.987	available
Rafael Nadal	Overall	dominance_ratio	2000	1.149	1.222	available
Rafael Nadal	Clay	match_win_pct	2000	79.041	89.832	available
Rafael Nadal	Clay	point_win_pct	2000	53.833	55.532	available
Rafael Nadal	Clay	first_serve_in_pct	2000	69.405	71.550	available
Rafael Nadal	Clay	first_serve_win_pct	2000	67.737	70.240	available
Rafael Nadal	Clay	second_serve_win_pct	2000	53.826	57.538	available
Rafael Nadal	Clay	service_point_win_pct	2000	63.865	66.242	available
Rafael Nadal	Clay	return_point_win_pct	2000	43.628	46.027	available
Rafael Nadal	Clay	ace_pct	2000	2.283	2.958	available
Rafael Nadal	Clay	double_fault_pct	2000	1.872	2.405	available
Rafael Nadal	Clay	break_point_save_pct	2000	62.116	68.488	available
Rafael Nadal	Clay	break_point_conversion_pct	2000	43.478	48.278	available
Rafael Nadal	Clay	service_hold_pct	2000	79.753	83.994	available
Rafael Nadal	Clay	return_break_pct	2000	35.267	40.629	available
Rafael Nadal	Clay	dominance_ratio	2000	1.226	1.345	available
Rafael Nadal	Grass	match_win_pct	2000	43.583	75.000	available
Rafael Nadal	Grass	point_win_pct	2000	50.417	53.102	available
Rafael Nadal	Grass	first_serve_in_pct	2000	67.539	70.458	available
Rafael Nadal	Grass	first_serve_win_pct	2000	71.614	75.932	available
Rafael Nadal	Grass	second_serve_win_pct	2000	54.186	60.848	available
Rafael Nadal	Grass	service_point_win_pct	2000	66.614	70.880	available
Rafael Nadal	Grass	return_point_win_pct	2000	33.326	37.481	available
Rafael Nadal	Grass	ace_pct	2000	5.727	8.522	available
Rafael Nadal	Grass	double_fault_pct	2000	1.588	2.321	available
Rafael Nadal	Grass	break_point_save_pct	2000	60.949	74.163	available
Rafael Nadal	Grass	break_point_conversion_pct	2000	32.848	44.022	available
Rafael Nadal	Grass	service_hold_pct	2000	84.684	91.290	available
Rafael Nadal	Grass	return_break_pct	2000	15.253	22.939	available
Rafael Nadal	Grass	dominance_ratio	2000	1.033	1.257	available
Rafael Nadal	Hard	match_win_pct	2000	58.410	71.491	available
Rafael Nadal	Hard	point_win_pct	2000	50.985	52.395	available
Rafael Nadal	Hard	first_serve_in_pct	2000	66.866	68.522	available
Rafael Nadal	Hard	first_serve_win_pct	2000	69.739	72.121	available
Rafael Nadal	Hard	second_serve_win_pct	2000	54.059	56.898	available
Rafael Nadal	Hard	service_point_win_pct	2000	64.905	66.998	available
Rafael Nadal	Hard	return_point_win_pct	2000	37.286	39.033	available

player	surface	metric	valid_bootstrap_replicates	ci95_lower	ci95_upper	status
Rafael Nadal	Hard	ace_pct	2000	4.312	5.287	available
Rafael Nadal	Hard	double_fault_pct	2000	2.266	2.794	available
Rafael Nadal	Hard	break_point_save_pct	2000	59.251	64.665	available
Rafael Nadal	Hard	break_point_conversion_pct	2000	38.434	43.597	available
Rafael Nadal	Hard	service_hold_pct	2000	81.206	84.726	available
Rafael Nadal	Hard	return_break_pct	2000	22.139	25.858	available
Rafael Nadal	Hard	dominance_ratio	2000	1.075	1.176	available
Novak Djokovic	Overall	match_win_pct	2000	73.176	80.274	available
Novak Djokovic	Overall	point_win_pct	2000	53.011	53.823	available
Novak Djokovic	Overall	first_serve_in_pct	2000	64.789	65.707	available
Novak Djokovic	Overall	first_serve_win_pct	2000	72.526	73.984	available
Novak Djokovic	Overall	second_serve_win_pct	2000	54.113	55.858	available
Novak Djokovic	Overall	service_point_win_pct	2000	66.295	67.517	available
Novak Djokovic	Overall	return_point_win_pct	2000	39.861	41.041	available
Novak Djokovic	Overall	ace_pct	2000	6.813	7.590	available
Novak Djokovic	Overall	double_fault_pct	2000	2.637	3.001	available
Novak Djokovic	Overall	break_point_save_pct	2000	62.134	65.695	available
Novak Djokovic	Overall	break_point_conversion_pct	2000	41.327	44.174	available
Novak Djokovic	Overall	service_hold_pct	2000	83.714	85.738	available
Novak Djokovic	Overall	return_break_pct	2000	26.851	29.466	available
Novak Djokovic	Overall	dominance_ratio	2000	1.190	1.252	available
Novak Djokovic	Clay	match_win_pct	2000	61.947	78.152	available
Novak Djokovic	Clay	point_win_pct	2000	51.739	53.428	available
Novak Djokovic	Clay	first_serve_in_pct	2000	64.667	66.619	available
Novak Djokovic	Clay	first_serve_win_pct	2000	68.135	71.128	available
Novak Djokovic	Clay	second_serve_win_pct	2000	51.845	55.249	available
Novak Djokovic	Clay	service_point_win_pct	2000	62.781	65.415	available
Novak Djokovic	Clay	return_point_win_pct	2000	39.924	42.227	available
Novak Djokovic	Clay	ace_pct	2000	4.405	5.408	available
Novak Djokovic	Clay	double_fault_pct	2000	2.180	2.841	available
Novak Djokovic	Clay	break_point_save_pct	2000	57.385	64.305	available
Novak Djokovic	Clay	break_point_conversion_pct	2000	39.732	45.438	available
Novak Djokovic	Clay	service_hold_pct	2000	77.218	82.028	available
Novak Djokovic	Clay	return_break_pct	2000	27.172	32.127	available
Novak Djokovic	Clay	dominance_ratio	2000	1.090	1.202	available
Novak Djokovic	Grass	match_win_pct	2000	66.667	89.093	available
Novak Djokovic	Grass	point_win_pct	2000	51.818	53.903	available

player	surface	metric	valid_bootstrap_replicates	ci95_lower	ci95_upper	status
Novak Djokovic	Grass	first_serve_in_pct	2000	65.763	68.299	available
Novak Djokovic	Grass	first_serve_win_pct	2000	74.624	78.212	available
Novak Djokovic	Grass	second_serve_win_pct	2000	50.644	56.036	available
Novak Djokovic	Grass	service_point_win_pct	2000	67.294	70.304	available
Novak Djokovic	Grass	return_point_win_pct	2000	35.970	39.470	available
Novak Djokovic	Grass	ace_pct	2000	8.015	10.579	available
Novak Djokovic	Grass	double_fault_pct	2000	2.277	3.367	available
Novak Djokovic	Grass	break_point_save_pct	2000	64.052	72.694	available
Novak Djokovic	Grass	break_point_conversion_pct	2000	33.813	42.052	available
Novak Djokovic	Grass	service_hold_pct	2000	85.397	90.410	available
Novak Djokovic	Grass	return_break_pct	2000	18.811	25.593	available
Novak Djokovic	Grass	dominance_ratio	2000	1.126	1.298	available
Novak Djokovic	Hard	match_win_pct	2000	74.474	83.003	available
Novak Djokovic	Hard	point_win_pct	2000	53.344	54.404	available
Novak Djokovic	Hard	first_serve_in_pct	2000	64.176	65.333	available
Novak Djokovic	Hard	first_serve_win_pct	2000	73.090	74.806	available
Novak Djokovic	Hard	second_serve_win_pct	2000	54.799	56.929	available
Novak Djokovic	Hard	service_point_win_pct	2000	66.768	68.291	available
Novak Djokovic	Hard	return_point_win_pct	2000	40.034	41.575	available
Novak Djokovic	Hard	ace_pct	2000	7.172	8.103	available
Novak Djokovic	Hard	double_fault_pct	2000	2.699	3.172	available
Novak Djokovic	Hard	break_point_save_pct	2000	62.085	66.789	available
Novak Djokovic	Hard	break_point_conversion_pct	2000	42.075	45.651	available
Novak Djokovic	Hard	service_hold_pct	2000	84.700	87.071	available
Novak Djokovic	Hard	return_break_pct	2000	27.251	30.717	available
Novak Djokovic	Hard	dominance_ratio	2000	1.217	1.299	available

Unavailable or small-sample estimates:

No records in this review table.

Regular-game eligibility:

is_tiebreak_point	eligible_regular_game	games
False	False	24
False	True	41734
True	False	752

Break-point conversion cross-check against opponent Overview counts (flagged matches excluded from conversion KPI):

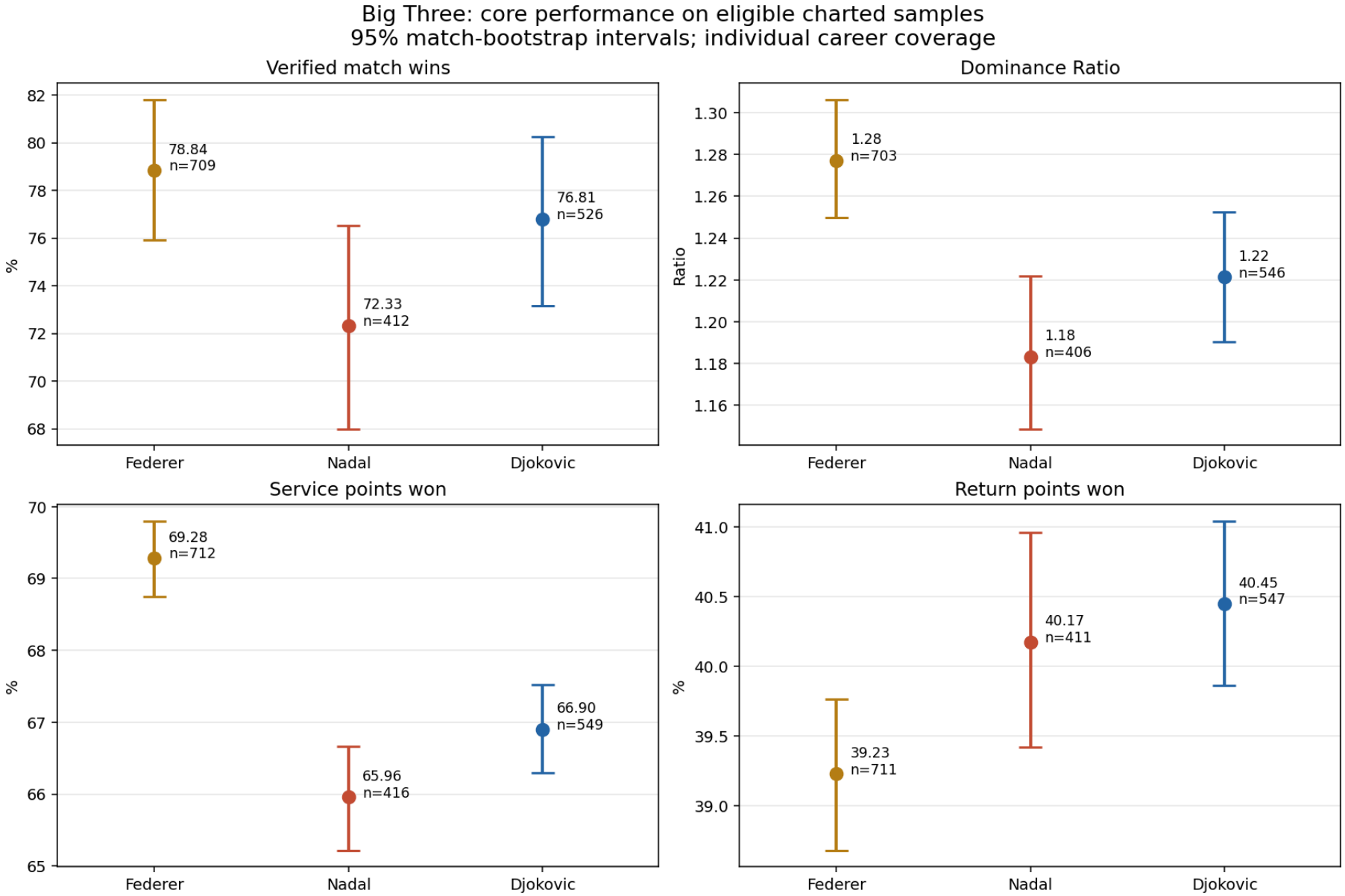
	match_id	player	bp_opportunities_parsed	opponent_bp_faced	bp_converted_parsed	source_bp_converted
1275	20151101-M-Basel-F-Rafael_Nadal-Roger_Federer	Roger Federer	7	6.000	3	3.000
1682	20020208-M-Davis_Cup_World_Group_R1-RR-Marat_Safin-Roger_Federer	Roger Federer	10	11.000	5	6.000

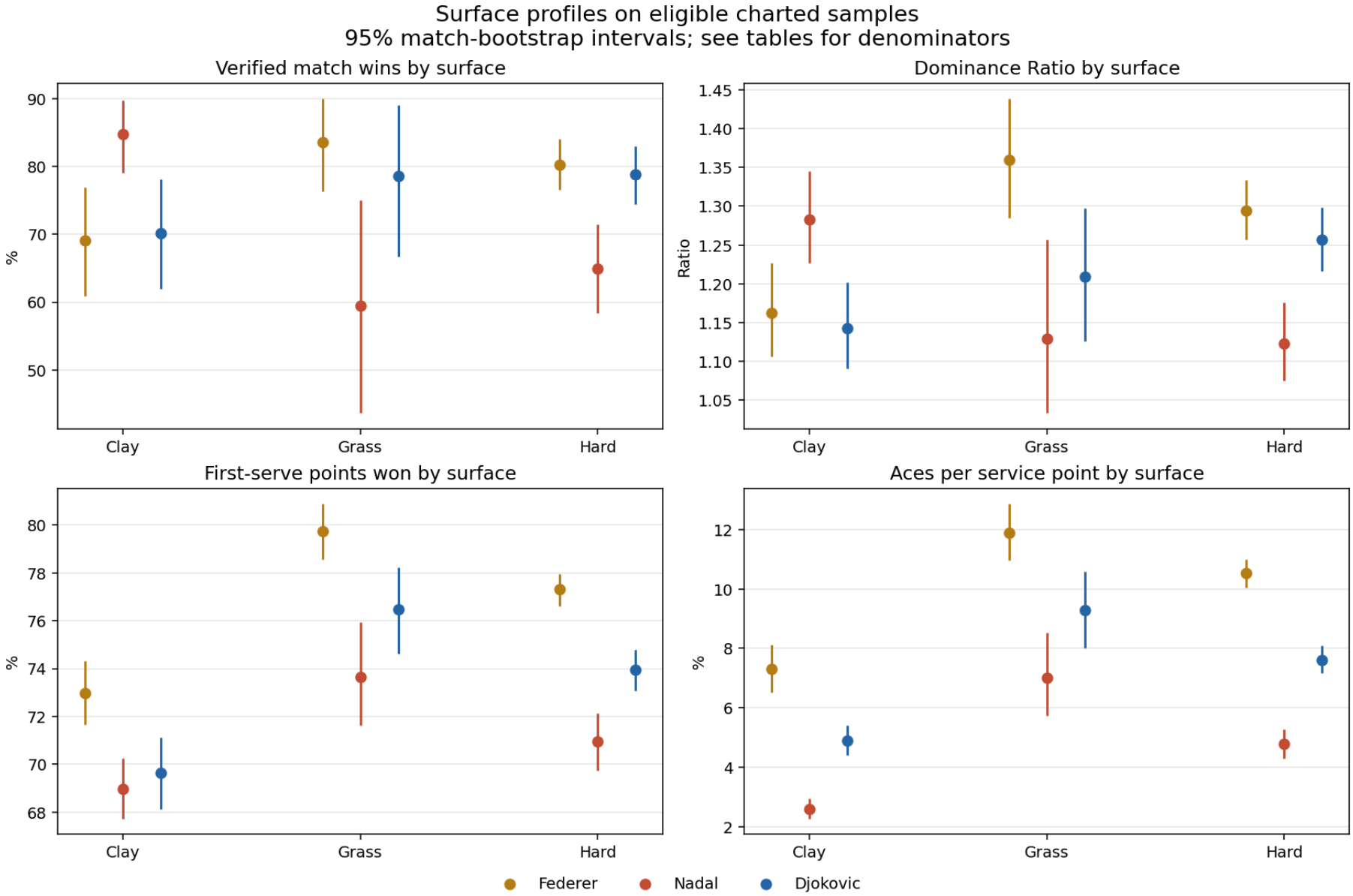
5.3 Visual comparison with uncertainty

Points show pooled estimates; whiskers show 95% percentile intervals from the same joint match bootstrap. Verified win rates exclude unknown outcomes. Small samples are labeled in the evidence table; intervals are conditional on recorded data and eligibility. A lower double-fault percentage is generally preferable; a higher percentage is not universally better for every style measure.

In [13]:
```
import io
from IPython.display import Image

KPI_PLAYER_COLORS = {"Roger Federer": "#B78016", "Rafael Nadal": "#C44E36", "Novak Djokovic": "#2766A5"}
KPI_SHORT_NAMES = {"Roger Federer": "Federer", "Rafael Nadal": "Nadal", "Novak Djokovic": "Djokovic"}
def display_kpi_figure(figure):
    buffer = io.BytesIO()
    figure.savefig(buffer, format="png", dpi=140, bbox_inches="tight")
    display(Image(data=buffer.getvalue()))
    plt.close(figure)

with plt.style.context("default"):
    fig, axes = plt.subplots(2, 2, figsize=(12, 8), constrained_layout=True)
    for ax, metric, title in zip(axes.flat,
        ["match_win_pct", "dominance_ratio", "service_point_win_pct", "return_point_win_pct"],
        ["Verified match wins", "Dominance Ratio", "Service points won", "Return points won"]):
        for index, player in enumerate(BIG_THREE):
            row = core_kpi_long.loc[core_kpi_long["player"].eq(player) & core_kpi_long["surface"].eq("Overall") & core_kpi_long["metric"].eq(metric)].iloc[0]
            if pd.isna(row["estimate"]):
                continue
            ax.scatter(index, row["estimate"], color=KPI_PLAYER_COLORS[player], s=65, zorder=3)
            if pd.notna(row["ci95_lower"]):
                ax.vlines(index, row["ci95_lower"], row["ci95_upper"], color=KPI_PLAYER_COLORS[player], linewidth=2)
                ax.hlines([row["ci95_lower"], row["ci95_upper"]], index-.06, index+.06, color=KPI_PLAYER_COLORS[player])
            ax.annotate(f"{row['estimate']:.2f}\nn={int(row['eligible_matches'])}", (index, row["estimate"]), xytext=(9, 0), textcoords="offset points",
fontsize=9)
        ax.set_xticks(range(3), [KPI_SHORT_NAMES[p] for p in BIG_THREE])
        ax.set_xlim(-.5, 2.6)
        ax.set_title(title)
        ax.set_ylabel("Ratio" if metric == "dominance_ratio" else "%")
        ax.grid(axis="y", alpha=.25)
    fig.suptitle("Big Three: core performance on eligible charted samples\n95% match-bootstrap intervals; individual career coverage", fontsize=14)
    display_kpi_figure(fig)

    fig, axes = plt.subplots(2, 2, figsize=(12, 8), constrained_layout=True)
```

```
    for ax, metric, title in zip(axes.flat,
        ["match_win_pct", "dominance_ratio", "first_serve_win_pct", "ace_pct"],
        ["Verified match wins by surface", "Dominance Ratio by surface", "First-serve points won by surface", "Aces per service point by surface"]):
        for player_index, player in enumerate(BIG_THREE):
            group = core_kpi_long.loc[core_kpi_long["player"].eq(player) & core_kpi_long["metric"].eq(metric)].set_index("surface").reindex(PROJECT_SURFACES)
            x = np.arange(3) + (player_index - 1) * .16
            ax.scatter(x, group["estimate"], color=KPI_PLAYER_COLORS[player], label=KPI_SHORT_NAMES[player], s=40, zorder=3)
            for xpos, (_, row) in zip(x, group.iterrows()):
                if pd.notna(row["ci95_lower"]):
                    ax.vlines(xpos, row["ci95_lower"], row["ci95_upper"], color=KPI_PLAYER_COLORS[player], linewidth=1.5)
        ax.set_xticks(range(3), PROJECT_SURFACES)
        ax.set_title(title)
        ax.set_ylabel("Ratio" if metric == "dominance_ratio" else "%")
        ax.grid(axis="y", alpha=.25)
    fig.legend(*axes.flat[0].get_legend_handles_labels(), loc="outside lower center", ncol=3, frameon=False)
    fig.suptitle("Surface profiles on eligible charted samples\n95% match-bootstrap intervals; see tables for denominators", fontsize=14)
    display_kpi_figure(fig)
```


Big Three: core performance on eligible charted samples
95% match-bootstrap intervals; individual career coverage
Verified match wins
%
78.84
n=709
72.33
n=412
76.81
n=526
Federer
Nadal
Djokovic
Dominance Ratio
Ratio
1.28
n=703
1.18
n=406
1.22
n=546
Federer
Nadal
Djokovic
Service points won
%
69.28
n=712
65.96
n=416
66.90
n=549
Federer
Nadal
Djokovic
Return points won
%
39.23
n=711
40.17
n=411
40.45
n=547
Federer
Nadal
Djokovic


Surface profiles on eligible charted samples
95% match-bootstrap intervals; see tables for denominators
Verified match wins by surface
%
90
80
70
60
50
Clay
Grass
Hard
Dominance Ratio by surface
Ratio
1.45
1.40
1.35
1.30
1.25
1.20
1.15
1.10
1.05
Clay
Grass
Hard
First-serve points won by surface
%
80
78
76
74
72
70
68
Clay
Grass
Hard
Aces per service point by surface
%
12
10
8
6
4
2
Clay
Grass
Hard
Federer
Nadal
Djokovic

5.4 Reading the performance results

These scorecards summarize verified match outcomes and eligible source/point statistics. Unknown endings, source discrepancies, and different charted schedules affect who contributes to each estimate. The source-compatible service/return rates deliberately use different coverage from the all-recorded-outcome point-win rate. A high ace rate describes one serving weapon, not complete serving strength; return-point success and DR add complementary context. Hold and break rates are reconstructed from fully eligible regular games, not inferred from overall match wins.

The tables keep wins and unknown outcomes beside the rates, and the figures separate serving from returning before looking across surfaces. That distinction matters: frequent aces can coexist with a less imposing return game, while a strong returner can win through sustained pressure rather than service dominance. The computed findings below establish the baseline for the pressure analysis.

5.5 Evidence: serving, returning, and surface specialization

In [14]:
```
def evidence_metric(player,metric,surface="Overall"):
    return core_kpi_long.loc[core_kpi_long["player"].eq(player)&core_kpi_long["surface"].eq(surface)&core_kpi_long["metric"].eq(metric)].iloc[0]
def interval_text(low,high,digits=2):
    return f"{low:.{digits}f}–{high:.{digits}f}" if pd.notna(low) and pd.notna(high) else "unavailable"
performance_focus=core_kpi_long.loc[core_kpi_long["surface"].eq("Overall") & core_kpi_long["metric"].isin(
    ["match_win_pct","dominance_ratio","service_point_win_pct","return_point_win_pct","service_hold_pct","return_break_pct"]),
    ["player","metric","estimate","unit","eligible_matches","numerator","denominator","ci95_lower","ci95_upper"]]
report_table(performance_focus,caption="Headline performance: pooled counts, eligible matches, and 95% intervals",expanded=True)
lines=["**Observed result.**"]
for player in BIG_THREE:

win=evidence_metric(player,"match_win_pct");serve=evidence_metric(player,"service_point_win_pct");ret=evidence_metric(player,"return_point_win_pct");dr=evidence
_metric(player,"dominance_ratio")
    lines.append(f"- {player}: {int(win.numerator)}/{int(win.denominator)} verified wins ({win.estimate:.2f}%, 95% interval
{interval_text(win.ci95_lower,win.ci95_upper)}); service points {serve.estimate:.2f}%, return points {ret.estimate:.2f}%; DR {dr.estimate:.3f} on
{int(dr.eligible_matches)} eligible matches.")
surface_evidence=core_kpi_long.loc[core_kpi_long["surface"].ne("Overall")&core_kpi_long["metric"].eq("match_win_pct"),
    ["player","surface","estimate","numerator","denominator","ci95_lower","ci95_upper","status"]]
lines.append("\n**Tennis interpretation.** Higher service-point success can shorten exposure to return games; return-point success creates a different route to
winning. DR asks how often a player wins return points relative to losing service points, rather than treating aces as complete serving strength. A player can
lead one component without leading the other.")
lines.append("\n**Alternative explanations.** These are pooled player samples with different opponents, career windows, and surface schedules. The intervals
describe variation under match resampling; they do not justify a causal player comparison or repair chart selection. Head-to-head results later test a narrower
matchup question. Next, we examine whether pressure outcomes add a distinct perspective.")
display(Markdown("\n\n".join(lines)))
report_table(surface_evidence,caption="Surface match-win evidence: eligible known outcomes, not complete career results",expanded=True)
```

Headline performance: pooled counts, eligible matches, and 95% intervals

	player	metric	estimate	unit	eligible_matches	numerator	denominator	ci95_lower	ci95_upper
0	Roger Federer	match_win_pct	78.843	%	709	559.000	709.000	75.921	81.818
5	Roger Federer	service_point_win_pct	69.283	%	712	40,906.000	59,042.000	68.743	69.800
6	Roger Federer	return_point_win_pct	39.228	%	711	24,853.000	63,355.000	38.679	39.763
11	Roger Federer	service_hold_pct	88.207	%	722	8,811.000	9,989.000	87.395	88.989
12	Roger Federer	return_break_pct	25.683	%	722	2,518.000	9,804.000	24.539	26.841
13	Roger Federer	dominance_ratio	1.277	ratio	703	Unavailable	Unavailable	1.250	1.306
56	Rafael Nadal	match_win_pct	72.330	%	412	298.000	412.000	68.000	76.538
61	Rafael Nadal	service_point_win_pct	65.958	%	416	22,394.000	33,952.000	65.216	66.661
62	Rafael Nadal	return_point_win_pct	40.174	%	411	14,059.000	34,995.000	39.417	40.958
67	Rafael Nadal	service_hold_pct	83.232	%	422	4,636.000	5,570.000	81.894	84.459
68	Rafael Nadal	return_break_pct	28.314	%	423	1,570.000	5,545.000	26.742	29.987
69	Rafael Nadal	dominance_ratio	1.183	ratio	406	Unavailable	Unavailable	1.149	1.222
112	Novak Djokovic	match_win_pct	76.806	%	526	404.000	526.000	73.176	80.274
117	Novak Djokovic	service_point_win_pct	66.900	%	549	30,359.000	45,380.000	66.295	67.517
118	Novak Djokovic	return_point_win_pct	40.448	%	547	19,038.000	47,068.000	39.861	41.041
123	Novak Djokovic	service_hold_pct	84.716	%	552	6,280.000	7,413.000	83.714	85.738
124	Novak Djokovic	return_break_pct	28.138	%	552	2,029.000	7,211.000	26.851	29.466
125	Novak Djokovic	dominance_ratio	1.221	ratio	546	Unavailable	Unavailable	1.190	1.252

Observed result.

- Roger Federer: 559/709 verified wins (78.84%, 95% interval 75.92–81.82); service points 69.28%, return points 39.23%; DR 1.277 on 703 eligible matches.
- Rafael Nadal: 298/412 verified wins (72.33%, 95% interval 68.00–76.54); service points 65.96%, return points 40.17%; DR 1.183 on 406 eligible matches.
- Novak Djokovic: 404/526 verified wins (76.81%, 95% interval 73.18–80.27); service points 66.90%, return points 40.45%; DR 1.221 on 546 eligible matches.

Tennis interpretation. Higher service-point success can shorten exposure to return games; return-point success creates a different route to winning. DR asks how often a player wins return points relative to losing service points, rather than treating aces as complete serving strength. A player can lead one component without leading the other.

Alternative explanations. These are pooled player samples with different opponents, career windows, and surface schedules. The intervals describe variation under match resampling; they do not justify a causal player comparison or repair chart selection. Head-to-head results later test a narrower matchup question. Next, we examine whether pressure outcomes add a distinct perspective.

Surface match-win evidence: eligible known outcomes, not complete career results

	player	surface	estimate	numerator	denominator	ci95_lower	ci95_upper	status
14	Roger Federer	Clay	69.048	87.000	126.000	60.869	76.925	available
28	Roger Federer	Grass	83.621	97.000	116.000	76.423	90.000	available
42	Roger Federer	Hard	80.300	375.000	467.000	76.652	84.081	available
70	Rafael Nadal	Clay	84.756	139.000	164.000	79.041	89.832	available
84	Rafael Nadal	Grass	59.459	22.000	37.000	43.583	75.000	available
98	Rafael Nadal	Hard	64.929	137.000	211.000	58.410	71.491	available
126	Novak Djokovic	Clay	70.161	87.000	124.000	61.947	78.152	available
140	Novak Djokovic	Grass	78.571	44.000	56.000	66.667	89.093	available
154	Novak Djokovic	Hard	78.902	273.000	346.000	74.474	83.003	available

VI. Pressure outcomes and Clutch Index

Pressure question: Do raw pressure outcomes and context-adjusted point associations tell the same story? Break points, tiebreaks, and deciding sets represent different tennis situations; their declared composite is a preference, not a psychological diagnosis.

6.1 Completed events and a common comparison sample

CI = **0.50 × break-point save rate + 0.30 × tiebreak win rate + 0.20 × deciding-set win rate**. It is a descriptive 0–1 composite with declared preference weights, not a measurement of mental toughness. Each full CI uses one common eligible player-match cohort for all three components. Require validated source-compatible break-point counts, valid score states, fully validated tiebreak groups (if present), a verified match outcome, and consistent supported match format. No-tiebreak and no-decider matches contribute genuine zero event counts, not invented wins or losses. Zero component denominators make full CI unavailable; an optional available-component result is explicitly labeled partial and never substitutes for full CI.

Tiebreak validation checks 0–0 starts, numerical score progression through serving changes, and a valid terminal score. Deciding-set outcomes require completed sets, verified match outcomes, and supported best-of metadata; a match tiebreak is identified separately and is not also counted as a conventional deciding set. Component-specific evidence is shown alongside the stricter common-cohort CI. Surface CIs use the same formula but may be small-sample. Uncertainty uses the 2,000 joint match draws from Section V. A resampled match carries all its points and both Big Three observations together; we do not pretend that thousands of points are thousands of independent matches.

In [15]:
```
# Validate complete tiebreak groups, including score rotation when the server changes.
tb_keys = ["match_id", "Set1", "Set2", "game_number"]
tb_points = canonical_points.loc[canonical_points["is_tiebreak_point"]].copy()
tb_groupers = [tb_points[k] for k in tb_keys]
tb_first = ~tb_points.duplicated(tb_keys, keep="first")
tb_last = ~tb_points.duplicated(tb_keys, keep="last")
tb_server_won = tb_points["Svr"].eq(tb_points["PtWinner"])
tb_next_svr = tb_points["Svr"].groupby(tb_groupers).shift(-1)
tb_next_left = tb_points["score_server_numeric"].groupby(tb_groupers).shift(-1)
tb_next_right = tb_points["score_receiver_numeric"].groupby(tb_groupers).shift(-1)
tb_new_left = tb_points["score_server_numeric"] + tb_server_won.astype(int)
tb_new_right = tb_points["score_receiver_numeric"] + (~tb_server_won).astype(int)
tb_switched = tb_points["Svr"].ne(tb_next_svr)
tb_transition_valid = tb_last | (~tb_points["game_ended"]
    & pd.Series(np.where(tb_switched, tb_new_right, tb_new_left), index=tb_points.index).eq(tb_next_left)
    & pd.Series(np.where(tb_switched, tb_new_left, tb_new_right), index=tb_points.index).eq(tb_next_right))
tb_points["tb_row_valid"] = (tb_points["score_state_valid"] & tb_transition_valid
    & (~tb_first | (tb_points["score_server_numeric"].eq(0) & tb_points["score_receiver_numeric"].eq(0))))
tb_group_quality = tb_points.groupby(tb_keys, as_index=False).agg(all_tb_rows_valid=("tb_row_valid", "all"), tiebreak_points=("Pt", "size"))
validated_tiebreaks = canonical_tiebreaks.merge(tb_group_quality, on=tb_keys, validate="one_to_one")
```

```
validated_tiebreaks["tb_complete_validated"] = validated_tiebreaks["all_tb_rows_valid"] & validated_tiebreaks["tiebreak_winner"].notna()
# A deciding-set A rule cannot be trusted when recorded set counts contradict Best of.
validated_tiebreaks.loc[validated_tiebreaks["Final TB?"].eq("A") & validated_tiebreaks["match_id"].isin(format_contradiction_ids), "tb_complete_validated"] =
False
bad_tb_match_ids = set(validated_tiebreaks.loc[~validated_tiebreaks["tb_complete_validated"], "match_id"])
valid_tb_keys = pd.MultiIndex.from_frame(validated_tiebreaks.loc[validated_tiebreaks["tb_complete_validated"], tb_keys])

clutch_match_events = kpi_master[["match_id", "player", "surface", "match_date", "opponent", "Best of", "Final TB?", "has_points",
    "has_overview", "winner", "outcome_status", "bp_events_validated", "bp_faced_parsed", "bp_saved_parsed", "points_played", "valid_score_points"]].copy()
clutch_match_events["bp_faced"] = clutch_match_events["bp_faced_parsed"].where(clutch_match_events["bp_events_validated"])
clutch_match_events["bp_saved"] = clutch_match_events["bp_saved_parsed"].where(clutch_match_events["bp_events_validated"])
clutch_match_events["tb_match_eligible"] = (clutch_match_events["has_points"] & ~clutch_match_events["match_id"].isin(bad_tb_match_ids)
    & clutch_match_events["valid_score_points"].eq(clutch_match_events["points_played"]))
clutch_match_events["deciding_match_eligible"] = (clutch_match_events["winner"].notna()
    & clutch_match_events["Best of"].isin([3, 5]) & ~clutch_match_events["match_id"].isin(format_contradiction_ids)
    & ~clutch_match_events["Final TB?"].eq("S"))
tb_player_rows, deciding_player_rows = [], []
for role in ["player1", "player2"]:
    a = validated_tiebreaks.loc[validated_tiebreaks["tb_complete_validated"] & validated_tiebreaks[role].isin(BIG_THREE)].copy()
    a["player"] = a[role]
    a["tiebreaks_played"] = 1
    a["tiebreaks_won"] = a["player"].eq(a["tiebreak_winner"]).astype(int)
    tb_player_rows.append(a)
    b = canonical_sets.loc[canonical_sets["deciding_set_validated"] & canonical_sets[role].isin(BIG_THREE)
        & ~canonical_sets["Final TB?"].eq("S")].copy()
    b["player"] = b[role]
    b["deciding_sets_played"] = 1
    b["deciding_sets_won"] = b["player"].eq(b["set_winner"]).astype(int)
    deciding_player_rows.append(b)
tb_match_counts = pd.concat(tb_player_rows, ignore_index=True).groupby(["match_id", "player"], as_index=False)[["tiebreaks_played", "tiebreaks_won"]].sum()
deciding_match_counts = pd.concat(deciding_player_rows, ignore_index=True).groupby(["match_id", "player"], as_index=False)[["deciding_sets_played",
"deciding_sets_won"]].sum()
clutch_match_events = clutch_match_events.merge(tb_match_counts, on=["match_id", "player"], how="left", validate="one_to_one").merge(
    deciding_match_counts, on=["match_id", "player"], how="left", validate="one_to_one")
for fields, eligible_field in [(["tiebreaks_played", "tiebreaks_won"], "tb_match_eligible"),
                               (["deciding_sets_played", "deciding_sets_won"], "deciding_match_eligible")]:
    for field in fields:
        clutch_match_events[field] = clutch_match_events[field].fillna(0).where(clutch_match_events[eligible_field])
clutch_match_events["common_ci_eligible"] = (clutch_match_events["bp_events_validated"]
    & clutch_match_events["tb_match_eligible"] & clutch_match_events["deciding_match_eligible"])

CI_COMPONENTS = {"bp": ("bp_saved", "bp_faced", .5), "tb": ("tiebreaks_won", "tiebreaks_played", .3),
                 "deciding": ("deciding_sets_won", "deciding_sets_played", .2)}
clutch_rows, clutch_boot_arrays, clutch_boot_descriptors, component_evidence = [], [], [], []
for player, surface in kpi_scope_groups:
    group = clutch_match_events.loc[clutch_match_events["player"].eq(player)]
    if surface != "Overall":
        group = group.loc[group["surface"].eq(surface)]
    common = group.loc[group["common_ci_eligible"]]
    row = {"player": player, "surface": surface, "cohort_matches": len(group), "common_ci_matches": len(common),
           "common_ci_coverage_pct": 100 * len(common) / len(group) if len(group) else np.nan}
    available_weight, available_score = 0., 0.
    columns = []
    for component, (num, den, weight) in CI_COMPONENTS.items():
        numerator, denominator = common[num].sum(min_count=1), common[den].sum(min_count=1)
        rate = numerator / denominator if denominator > 0 else np.nan
        row.update({num: numerator, den: denominator, component + "_component": rate})
```

```
        broad = group.loc[group[num].notna() & group[den].notna()]
        broad_den = broad[den].sum(min_count=1)
        component_evidence.append({"player": player, "surface": surface, "component": component,
            "component_eligible_matches": len(broad), "component_wins": broad[num].sum(min_count=1), "component_events": broad_den,
            "component_only_rate": broad[num].sum(min_count=1) / broad_den if broad_den > 0 else np.nan,
            "common_ci_wins": numerator, "common_ci_events": denominator, "common_ci_rate": rate})
        if pd.notna(rate):
            available_weight += weight
            available_score += weight * rate
        for field in [num, den]:
            vector = np.zeros(len(bootstrap_match_ids))
            for match_id, value in common[["match_id", field]].itertuples(index=False, name=None):
                vector[bootstrap_match_index[match_id]] = value
            columns.append(len(clutch_boot_arrays))
            clutch_boot_arrays.append(vector)
    rates = [row[c + "_component"] for c in CI_COMPONENTS]
    row["clutch_index_full"] = sum(rate * spec[2] for rate, spec in zip(rates, CI_COMPONENTS.values())) if all(pd.notna(r) for r in rates) else np.nan
    row["clutch_index_partial_renorm"] = available_score / available_weight if available_weight else np.nan
    row["partial_index_label"] = "Available-component index; never substituted for full CI"
    row["small_sample"] = any(pd.isna(row[spec[1]]) or row[spec[1]] < SMALL_SAMPLE_THRESHOLDS["events"] for spec in CI_COMPONENTS.values())
    clutch_boot_descriptors.append((len(clutch_rows), columns))
    clutch_rows.append(row)
clutch_summary = pd.DataFrame(clutch_rows)
clutch_component_evidence = pd.DataFrame(component_evidence)
clutch_boot_totals = bootstrap_weights @ np.column_stack(clutch_boot_arrays)
clutch_bootstrap_distributions = {}
for index, columns in clutch_boot_descriptors:
    with np.errstate(divide="ignore", invalid="ignore"):
        rates = [np.where(clutch_boot_totals[:, columns[j+1]] > 0,
            clutch_boot_totals[:, columns[j]] / clutch_boot_totals[:, columns[j+1]], np.nan) for j in [0, 2, 4]]
        full = .5 * rates[0] + .3 * rates[1] + .2 * rates[2]
    for metric, values in zip(["bp_component", "tb_component", "deciding_component", "clutch_index_full"], rates + [full]):
        valid = values[np.isfinite(values)]
        clutch_summary.loc[index, metric + "_valid_bootstraps"] = len(valid)
        clutch_summary.loc[index, metric + "_ci95_lower"] = np.percentile(valid, 2.5) if len(valid) >= .95 * BOOTSTRAP_REPLICATES else np.nan
        clutch_summary.loc[index, metric + "_ci95_upper"] = np.percentile(valid, 97.5) if len(valid) >= .95 * BOOTSTRAP_REPLICATES else np.nan
    clutch_bootstrap_distributions[(clutch_summary.loc[index, "player"], clutch_summary.loc[index, "surface"])] = full
clutch_summary["status"] = np.where(clutch_summary["clutch_index_full"].isna(), "unavailable", np.where(clutch_summary["small_sample"], "small sample",
"available"))
clutch_overall = clutch_summary.loc[clutch_summary["surface"].eq("Overall")].reset_index(drop=True)
clutch_by_surface = clutch_summary.loc[clutch_summary["surface"].ne("Overall")].reset_index(drop=True)
assert not clutch_match_events.duplicated(["match_id", "player"]).any()
assert not clutch_summary.duplicated(["player", "surface"]).any()
assert clutch_summary["clutch_index_full"].dropna().between(0, 1).all()
for num, den, _ in CI_COMPONENTS.values():
    assert clutch_summary[num].dropna().le(clutch_summary[den].dropna()).all()
assert clutch_summary["common_ci_matches"].le(clutch_summary["cohort_matches"]).all()
print("Clutch Index on a common cohort: full counts, components, and uncertainty")
with pd.option_context("display.max_columns", None):
    report_table(clutch_summary.round(4))
print("Component-specific evidence versus the stricter common CI cohort:")
report_table(clutch_component_evidence.round(4))
print("Tiebreak groups requiring review (excluded, not assigned a winner):")
report_table(validated_tiebreaks.loc[~validated_tiebreaks["tb_complete_validated"], tb_keys + ["tiebreak_points", "all_tb_rows_valid", "game_ended"]])
print("Common-cohort eligibility counts:")
report_table(clutch_match_events.groupby("player", as_index=False)[["bp_events_validated", "tb_match_eligible", "deciding_match_eligible",
"common_ci_eligible"]].sum())
```

Clutch Index on a common cohort: full counts, components, and uncertainty

▼ Complete evidence table: 12 rows × 31 columns

player	surface	cohort_matches	common_ci_matches	common_ci_coverage_pct	bp_saved	bp_faced	bp_component	tiebreaks_won
Roger Federer	Overall	723	687	95.021	2154	3271	0.658	229.000
Roger Federer	Clay	130	122	93.846	505	809	0.624	36.000
Roger Federer	Grass	119	116	97.479	331	459	0.721	60.000
Roger Federer	Hard	474	449	94.726	1318	2003	0.658	133.000
Rafael Nadal	Overall	425	392	92.235	1489	2338	0.637	89.000
Rafael Nadal	Clay	166	153	92.169	635	970	0.655	27.000
Rafael Nadal	Grass	40	36	90.000	155	232	0.668	14.000
Rafael Nadal	Hard	219	203	92.694	699	1136	0.615	48.000
Novak Djokovic	Overall	553	519	93.852	1901	2957	0.643	146.000
Novak Djokovic	Clay	129	122	94.574	525	860	0.611	29.000
Novak Djokovic	Grass	58	56	96.552	253	367	0.689	25.000
Novak Djokovic	Hard	366	341	93.169	1123	1730	0.649	92.000

player	surface	tiebreaks_played	tb_component	deciding_sets_won	deciding_sets_played	deciding_component	clutch_index_full	clutch_index_partial_renorm
Roger Federer	Overall	366.000	0.626	109.000	179.000	0.609	0.639	0.639
Roger Federer	Clay	64.000	0.562	19.000	32.000	0.594	0.600	0.600
Roger Federer	Grass	91.000	0.659	16.000	27.000	0.593	0.677	0.677
Roger Federer	Hard	211.000	0.630	74.000	120.000	0.617	0.641	0.641
Rafael Nadal	Overall	172.000	0.517	65.000	97.000	0.670	0.608	0.608
Rafael Nadal	Clay	48.000	0.562	28.000	32.000	0.875	0.671	0.671
Rafael Nadal	Grass	29.000	0.483	4.000	9.000	0.444	0.568	0.568
Rafael Nadal	Hard	95.000	0.505	33.000	56.000	0.589	0.577	0.577
Novak Djokovic	Overall	237.000	0.616	90.000	132.000	0.682	0.643	0.643
Novak Djokovic	Clay	48.000	0.604	24.000	37.000	0.649	0.616	0.616
Novak Djokovic	Grass	44.000	0.568	8.000	10.000	0.800	0.675	0.675
Novak Djokovic	Hard	145.000	0.634	58.000	85.000	0.682	0.651	0.651

player	surface	partial_index_label	small_sample	bp_component_valid_bootstraps	bp_component_ci95_lower	bp_component_ci95_upper	tb_component_valid_bootstraps	tb_component_ci95_lower
Roger Federer	Overall	Available-component index; never substituted for full CI	False	2,000.000	0.643	0.675	2,000.000	0.573

player	surface	partial_index_label	small_sample	bp_component_valid_bootstraps	bp_component_ci95_lower	bp_component_ci95_upper	tb_component_valid_bootstraps	tb_component_ci95_lower
Roger Federer	Clay	Available-component index; never substituted for full CI	False	2,000.000	0.591	0.656	2,000.000	0.443
Roger Federer	Grass	Available-component index; never substituted for full CI	True	2,000.000	0.684	0.756	2,000.000	0.553
Roger Federer	Hard	Available-component index; never substituted for full CI	False	2,000.000	0.637	0.679	2,000.000	0.562
Rafael Nadal	Overall	Available-component index; never substituted for full CI	False	2,000.000	0.618	0.656	2,000.000	0.440
Rafael Nadal	Clay	Available-component index; never substituted for full CI	False	2,000.000	0.623	0.687	2,000.000	0.432
Rafael Nadal	Grass	Available-component index; never substituted for full CI	True	2,000.000	0.598	0.735	2,000.000	0.281
Rafael Nadal	Hard	Available-component index; never substituted for full CI	False	2,000.000	0.587	0.644	2,000.000	0.398
Novak Djokovic	Overall	Available-component index; never substituted for full CI	False	2,000.000	0.625	0.661	2,000.000	0.550
Novak Djokovic	Clay	Available-component index; never substituted for full CI	False	2,000.000	0.575	0.645	2,000.000	0.447
Novak Djokovic	Grass	Available-component index; never substituted for full CI	True	2,000.000	0.645	0.734	2,000.000	0.405
Novak Djokovic	Hard	Available-component index; never substituted for full CI	False	2,000.000	0.626	0.673	2,000.000	0.553

player	surface	tb_component_ci95_upper	deciding_component_valid_bootstraps	deciding_component_ci95_lower	deciding_component_ci95_upper	clutch_index_full_valid_bootstraps	clutch_index_full_ci95_lower	clutch_index_full_ci95_upper
Roger Federer	Overall	0.676	2,000.000	0.537	0.681	2,000.000	0.614	0.664
Roger Federer	Clay	0.677	2,000.000	0.424	0.765	2,000.000	0.541	0.659

player	surface	tb_component_ci95_upper	deciding_component_valid_bootstraps	deciding_component_ci95_lower	deciding_component_ci95_upper	clutch_index_full_valid_bootstraps	clutch_index_full_ci95_lower	clutch_index_full_ci95_upper
Roger Federer	Grass	0.759	2,000.000	0.393	0.781	2,000.000	0.614	0.736
Roger Federer	Hard	0.698	2,000.000	0.529	0.704	2,000.000	0.609	0.674
Rafael Nadal	Overall	0.595	2,000.000	0.571	0.762	2,000.000	0.572	0.640
Rafael Nadal	Clay	0.692	2,000.000	0.741	0.971	2,000.000	0.614	0.721
Rafael Nadal	Grass	0.707	2,000.000	0.100	0.800	2,000.000	0.477	0.661
Rafael Nadal	Hard	0.613	2,000.000	0.451	0.717	2,000.000	0.529	0.625
Novak Djokovic	Overall	0.679	2,000.000	0.605	0.764	2,000.000	0.614	0.671
Novak Djokovic	Clay	0.761	2,000.000	0.488	0.800	2,000.000	0.564	0.677
Novak Djokovic	Grass	0.720	2,000.000	0.500	1.000	2,000.000	0.596	0.745
Novak Djokovic	Hard	0.710	2,000.000	0.582	0.781	2,000.000	0.614	0.688

player	surface	status
Roger Federer	Overall	available
Roger Federer	Clay	available
Roger Federer	Grass	small sample
Roger Federer	Hard	available
Rafael Nadal	Overall	available
Rafael Nadal	Clay	available
Rafael Nadal	Grass	small sample
Rafael Nadal	Hard	available
Novak Djokovic	Overall	available
Novak Djokovic	Clay	available
Novak Djokovic	Grass	small sample
Novak Djokovic	Hard	available

Component-specific evidence versus the stricter common CI cohort:

player	surface	component	component_eligible_matches	component_wins	component_events	component_only_rate	common_ci_wins	common_ci_events	common_ci_rate
Roger Federer	Overall	bp	702	2,187.000	3,324.000	0.658	2,154.000	3,271.000	0.658
Roger Federer	Overall	tb	717	237.000	376.000	0.630	229.000	366.000	0.626
Roger Federer	Overall	deciding	707	115.000	187.000	0.615	109.000	179.000	0.609
Roger Federer	Clay	bp	126	515.000	824.000	0.625	505.000	809.000	0.624
Roger Federer	Clay	tb	130	38.000	66.000	0.576	36.000	64.000	0.562
Roger Federer	Clay	deciding	126	21.000	34.000	0.618	19.000	32.000	0.594
Roger Federer	Grass	bp	119	344.000	479.000	0.718	331.000	459.000	0.721
Roger Federer	Grass	tb	119	60.000	92.000	0.652	60.000	91.000	0.659
Roger Federer	Grass	deciding	116	16.000	27.000	0.593	16.000	27.000	0.593
Roger Federer	Hard	bp	457	1,328.000	2,021.000	0.657	1,318.000	2,003.000	0.658
Roger Federer	Hard	tb	468	139.000	218.000	0.638	133.000	211.000	0.630
Roger Federer	Hard	deciding	465	78.000	126.000	0.619	74.000	120.000	0.617
Rafael Nadal	Overall	bp	404	1,569.000	2,457.000	0.639	1,489.000	2,338.000	0.637
Rafael Nadal	Overall	tb	420	96.000	184.000	0.522	89.000	172.000	0.517
Rafael Nadal	Overall	deciding	412	68.000	103.000	0.660	65.000	97.000	0.670
Rafael Nadal	Clay	bp	155	640.000	980.000	0.653	635.000	970.000	0.655
Rafael Nadal	Clay	tb	165	29.000	51.000	0.569	27.000	48.000	0.562
Rafael Nadal	Clay	deciding	164	28.000	33.000	0.849	28.000	32.000	0.875

player	surface	component	component_eligible_matches	component_wins	component_events	component_only_rate	common_ci_wins	common_ci_events	common_ci_rate
Rafael Nadal	Grass	bp	38	166.000	245.000	0.678	155.000	232.000	0.668
Rafael Nadal	Grass	tb	39	15.000	30.000	0.500	14.000	29.000	0.483
Rafael Nadal	Grass	deciding	37	4.000	9.000	0.444	4.000	9.000	0.444
Rafael Nadal	Hard	bp	211	763.000	1,232.000	0.619	699.000	1,136.000	0.615
Rafael Nadal	Hard	tb	216	52.000	103.000	0.505	48.000	95.000	0.505
Rafael Nadal	Hard	deciding	211	36.000	61.000	0.590	33.000	56.000	0.589
Novak Djokovic	Overall	bp	546	1,979.000	3,096.000	0.639	1,901.000	2,957.000	0.643
Novak Djokovic	Overall	tb	550	153.000	247.000	0.619	146.000	237.000	0.616
Novak Djokovic	Overall	deciding	526	92.000	134.000	0.687	90.000	132.000	0.682
Novak Djokovic	Clay	bp	126	534.000	876.000	0.610	525.000	860.000	0.611
Novak Djokovic	Clay	tb	129	32.000	51.000	0.627	29.000	48.000	0.604
Novak Djokovic	Clay	deciding	124	25.000	38.000	0.658	24.000	37.000	0.649
Novak Djokovic	Grass	bp	58	256.000	374.000	0.684	253.000	367.000	0.689
Novak Djokovic	Grass	tb	58	25.000	45.000	0.556	25.000	44.000	0.568
Novak Djokovic	Grass	deciding	56	8.000	10.000	0.800	8.000	10.000	0.800
Novak Djokovic	Hard	bp	362	1,189.000	1,846.000	0.644	1,123.000	1,730.000	0.649
Novak Djokovic	Hard	tb	363	96.000	151.000	0.636	92.000	145.000	0.634
Novak Djokovic	Hard	deciding	346	59.000	86.000	0.686	58.000	85.000	0.682

Tiebreak groups requiring review (excluded, not assigned a winner):

	match_id	Set1	Set2	game_number	tiebreak_points	all_tb_rows_valid	game_ended
315	20120129-M-Australian_Open-F-Novak_Djokovic-Rafael_Nadal	2	1	43.0	12	False	False
344	20121112-M-Tour_Finals-F-Roger_Federer-Novak_Djokovic	0	0	13.0	14	False	False
390	20140423-M-Barcelona-R32-Albert_Ramos-Rafael_Nadal	0	0	13.0	9	False	False
523	20170924-M-Laver_Cup-RR-Roger_Federer-Nick_Kyrgios	0	1	23.0	14	True	False
611	20190921-M-Laver_Cup-RR-Milos_Raonic-Rafael_Nadal	0	1	22.0	8	True	False
612	20190921-M-Laver_Cup-RR-Roger_Federer-Nick_Kyrgios	0	0	13.0	12	True	False
614	20190922-M-Laver_Cup-RR-John_Isner-Roger_Federer	0	1	23.0	10	True	False

Common-cohort eligibility counts:

player	bp_events_validated	tb_match_eligible	deciding_match_eligible	common_ci_eligible
Novak Djokovic	546	550	526	519
Rafael Nadal	404	420	412	392
Roger Federer	702	717	707	687

6.2 Separate point-outcome models

Fit **two inferential logistic regressions**: saving a break point while serving, and winning a point in a validated completed tiebreak. The tiebreak model includes whether the player is serving. Both use player effects, surface, set number, deciding-set state, preceding-point rally length, and a descriptive opponent-strength proxy. Unsupported or contradicted match formats are excluded from model context. Break-point observations follow the source-compatible eligibility used in Section V; match completion is not required for the point model.

Opponent context is pooled return-point success for the BP opponent and pooled total-point success for the TB opponent, calculated from the local Overview tables **excluding the current match**. Require at least ten other Overview matches; otherwise retain a missing-context indicator and impute the observed model-sample mean. Prior-rally missingness is also explicitly indicated, with mean imputation for that covariate. These are retrospective contextual associations, not leakage-free forecasts or causal adjustments. Age, injury, career phase, and unmeasured opponent differences remain limitations.

Player indicators are fitted without a global intercept. Their exponentials are **reference-profile odds**, not player-to-player odds ratios. Direct pairwise contrasts use coefficient differences and the full match-clustered covariance. Wald intervals are unadjusted 95%; Holm-adjusted p-values cover all six planned player comparisons across both models. Point observations within a match, including

complementary rivalry tiebreak rows, share one uncertainty cluster. No model result measures psychology or establishes an overall GOAT.

For a binary outcome, the model is $\text{logit}(p_i) = x_i^T\beta$. A player contrast is $\exp(\beta_a - \beta_b)$; an odds ratio of 1 is equality of conditional odds, not equality of observed win percentages. Hard is the surface reference, first set and non-deciding are the state reference, and centered continuous covariates are evaluated at their model-sample means. The opponent coefficient refers to a full 0–1 change; the added diagnostic table reports the more interpretable **10-percentage-point** change. Robust covariance aggregates score contributions by match, allowing point dependence within a match. Independence across match clusters and the additive logit form remain assumptions; career/era dependence is not eliminated.

The six planned contrasts form one Holm family. An unadjusted interval can exclude 1 even when its Holm-adjusted test does not reject. Failure to reject is not proof of equivalence. In-sample Brier scores summarize fit only; the model is not evaluated or advertised as a forecast.

In [16]:
```
# Assemble eligible pressure observations without re-parsing point scores or rally codes.
context_master = kpi_master[["match_id", "player", "opponent", "Best of", "Final TB?", "bp_events_validated"]]
pressure_player_points = canonical_player_points.merge(context_master, on=["match_id", "player"], how="left", validate="many_to_one")
valid_context = (pressure_player_points["Best of"].isin([3, 5])
    & ~pressure_player_points["match_id"].isin(format_contradiction_ids)
    & pressure_player_points["surface"].isin(PROJECT_SURFACES))
bp_model_events = pressure_player_points.loc[valid_context & pressure_player_points["bp_events_validated"] & pressure_player_points["bp_faced"]].copy()
bp_model_events["success"] = bp_model_events["point_won"].astype(int)
tb_point_keys = canonical_points.loc[canonical_points["is_tiebreak_point"]].copy()
tb_point_keys = tb_point_keys.loc[pd.MultiIndex.from_frame(tb_point_keys[tb_keys]).isin(valid_tb_keys), ["match_id", "Pt"]]
tb_model_events = pressure_player_points.loc[valid_context & pressure_player_points["is_tiebreak_point"]].merge(tb_point_keys,
    on=["match_id", "Pt"], how="inner", validate="many_to_one")
tb_model_events["success"] = tb_model_events["point_won"].astype(int)

opponent_overview = filter_stat_slice(analysis_dataframes["charting_m_stats_Overview"], None, "total").copy()
opponent_overview["context_point_wins"] = opponent_overview["first_won"] + opponent_overview["second_won"] + opponent_overview["return_pts_won"]
opponent_overview["context_points"] = opponent_overview["serve_pts"] + opponent_overview["return_pts"]
context_fields = ["return_pts_won", "return_pts", "context_point_wins", "context_points"]
context_valid = opponent_overview[context_fields].notna().all(axis=1) & opponent_overview[context_fields].ge(0).all(axis=1)
context_valid &= opponent_overview["return_pts_won"].le(opponent_overview["return_pts"]) &
opponent_overview["context_point_wins"].le(opponent_overview["context_points"])
opponent_overview = opponent_overview.loc[context_valid]
opponent_totals = opponent_overview.groupby("player", as_index=False)[context_fields].sum().rename(columns={"player": "opponent"})
opponent_match_counts = opponent_overview.groupby("player")["match_id"].nunique()
opponent_totals["opponent_matches"] = opponent_totals["opponent"].map(opponent_match_counts)
current_opponent_rows = opponent_overview[["match_id", "player"] + context_fields].rename(columns={"player": "opponent", **{c: "current_" + c for c in
context_fields}})
PLAYER_PRESSURE_TERMS = {p: "player_" + p.split()[-1].lower() for p in BIG_THREE}

def fit_pressure_outcome_model(events, kind):
    events = events.merge(opponent_totals, on="opponent", how="left", validate="many_to_one").merge(
        current_opponent_rows, on=["match_id", "opponent"], how="left", validate="many_to_one")
    current_present = events["current_context_points"].notna().astype(int)
```

```
    other_matches = events["opponent_matches"] - current_present
    wins_field, points_field = ("return_pts_won", "return_pts") if kind == "BP save" else ("context_point_wins", "context_points")
    other_wins = events[wins_field] - events["current_" + wins_field].fillna(0)
    other_points = events[points_field] - events["current_" + points_field].fillna(0)
    events["opponent_strength"] = (other_wins / other_points).where(other_matches.ge(10) & other_points.gt(0))
    events = events.reset_index(drop=True)
    design = pd.DataFrame({term: events["player"].eq(player).astype(float) for player, term in PLAYER_PRESSURE_TERMS.items()})
    design["surface_clay"] = events["surface"].eq("Clay").astype(float)
    design["surface_grass"] = events["surface"].eq("Grass").astype(float)
    design["set_after_first"] = events["set_number"] - 1
    design["deciding_set"] = events["is_deciding_set"].astype(float)
    references = {}
    for source, name in [("previous_rally_length", "prior_rally"), ("opponent_strength", "opponent_strength")]:
        values = pd.to_numeric(events[source], errors="coerce")
        reference = float(values.mean())
        if not np.isfinite(reference):
            raise ValueError(f"No supported {name} covariate for {kind}")
        references[name] = reference
        design[name + "_centered"] = values.fillna(reference) - reference
        if values.isna().any():
            design[name + "_missing"] = values.isna().astype(float)
    if kind == "TB point":
        design["is_serving"] = events["is_serving"].astype(float)
    design = design.astype(float)
    X, y = design.to_numpy(), events["success"].to_numpy(dtype=float)
    rank = np.linalg.matrix_rank(X)
    if rank != X.shape[1] or len(np.unique(y)) != 2:
        raise ValueError(f"Unsupported design or outcome variation for {kind}")
    model = LogisticRegression(penalty=None, fit_intercept=False, solver="lbfgs", max_iter=3000, tol=1e-10)
    model.fit(design, y)
    if int(model.n_iter_[0]) >= model.max_iter:
        raise RuntimeError(f"{kind} model did not converge")
    beta = model.coef_[0]
    fitted = model.predict_proba(design)[:, 1]
    information = X.T @ (X * (fitted * (1 - fitted))[:, None])
    if np.linalg.matrix_rank(information) != X.shape[1]:
        raise RuntimeError(f"{kind} information matrix is singular")
    bread = np.linalg.inv(information)
    scores = pd.DataFrame(X * (y - fitted)[:, None])
    scores["match_id"] = events["match_id"].to_numpy()
    clustered_scores = scores.groupby("match_id").sum().to_numpy()
    G, N, K = len(clustered_scores), len(y), X.shape[1]
    if G <= K or N <= K:
        raise ValueError(f"Too few independent match clusters for {kind}")
    correction = G / (G - 1) * (N - 1) / (N - K)
    covariance = correction * bread @ (clustered_scores.T @ clustered_scores) @ bread
    se = np.sqrt(np.diag(covariance))
    if not np.isfinite(se).all() or (se <= 0).any():
        raise RuntimeError(f"{kind}: invalid match-clustered uncertainty")
    critical = stats.norm.ppf(.975)
    coefficients = pd.DataFrame({"model": kind, "term": design.columns, "coefficient": beta, "clustered_se": se,
        "exp_coefficient": np.exp(beta), "ci95_lower": np.exp(beta-critical*se), "ci95_upper": np.exp(beta+critical*se),
        "p_value_unadjusted": 2 * stats.norm.sf(np.abs(beta / se))})
    diagnostics = {"model": kind, "observations": N, "match_clusters": G, "parameters": K, "design_rank": rank,
        "iterations": int(model.n_iter_[0]), "successes": int(y.sum()), "raw_success_pct": 100*y.mean(),
        "brier_in_sample": float(np.mean((y-fitted)**2)), "prior_rally_missing": int(events["previous_rally_length"].isna().sum()),
        "opponent_context_missing": int(events["opponent_strength"].isna().sum()), "information_condition_number": float(np.linalg.cond(information))}
```

```
    return {"kind": kind, "events": events, "design": design, "model": model, "beta": beta, "covariance": covariance,
        "coefficients": coefficients, "diagnostics": diagnostics, "references": references}

pressure_models = {kind: fit_pressure_outcome_model(events, kind) for kind, events in [("BP save", bp_model_events), ("TB point", tb_model_events)]}
pressure_model_diagnostics = pd.DataFrame([r["diagnostics"] for r in pressure_models.values()])
pressure_model_coefficients = pd.concat([r["coefficients"] for r in pressure_models.values()], ignore_index=True)
contrast_rows, profile_rows = [], []
for kind, result in pressure_models.items():
    columns = list(result["design"].columns)
    for a, b in [(BIG_THREE[0], BIG_THREE[1]), (BIG_THREE[0], BIG_THREE[2]), (BIG_THREE[1], BIG_THREE[2])]:
        contrast = np.zeros(len(columns))
        contrast[columns.index(PLAYER_PRESSURE_TERMS[a])] = 1
        contrast[columns.index(PLAYER_PRESSURE_TERMS[b])] = -1
        delta = contrast @ result["beta"]
        se = float(np.sqrt(contrast @ result["covariance"] @ contrast))
        contrast_rows.append({"model": kind, "comparison": a + " / " + b, "odds_ratio": np.exp(delta),
            "ci95_lower": np.exp(delta - 1.96*se), "ci95_upper": np.exp(delta + 1.96*se),
            "p_value_unadjusted": 2*stats.norm.sf(abs(delta/se))})
    for player in BIG_THREE:
        for serving in ([True] if kind == "BP save" else [False, True]):
            x = np.zeros(len(columns))
            x[columns.index(PLAYER_PRESSURE_TERMS[player])] = 1
            if "is_serving" in columns:
                x[columns.index("is_serving")] = int(serving)
            logit = x @ result["beta"]
            se = float(np.sqrt(x @ result["covariance"] @ x))
            logistic = lambda z: 1 / (1 + np.exp(-z))
            profile_rows.append({"model": kind, "player": player, "is_serving": serving,
                "reference_probability_pct": 100*logistic(logit), "ci95_lower": 100*logistic(logit-1.96*se),
                "ci95_upper": 100*logistic(logit+1.96*se), "reference_surface": "Hard", "reference_set": 1,
                "prior_rally_reference": result["references"]["prior_rally"], "opponent_strength_reference": result["references"]["opponent_strength"]})
pressure_pairwise = pd.DataFrame(contrast_rows)
order = np.argsort(pressure_pairwise["p_value_unadjusted"].to_numpy())
adjusted = np.minimum(1, np.maximum.accumulate((len(order)-np.arange(len(order))) * pressure_pairwise.iloc[order]["p_value_unadjusted"].to_numpy()))
pressure_pairwise.loc[order, "p_value_holm_six_tests"] = adjusted
pressure_reference_profiles = pd.DataFrame(profile_rows)
pressure_event_coverage = pd.concat([result["events"].assign(model=kind) for kind, result in pressure_models.items()], ignore_index=True).groupby(
    ["model", "player", "surface"], as_index=False).agg(observations=("success", "size"), successes=("success", "sum"), match_clusters=("match_id", "nunique"))
pressure_event_coverage["observed_success_pct"] = 100*pressure_event_coverage["successes"] / pressure_event_coverage["observations"]
assert pressure_pairwise["p_value_holm_six_tests"].between(0, 1).all()
assert pressure_pairwise["odds_ratio"].gt(0).all()
assert pressure_reference_profiles["reference_probability_pct"].between(0, 100).all()
assert pressure_pairwise["ci95_lower"].le(pressure_pairwise["ci95_upper"]).all()
assert not bp_model_events.duplicated(["match_id", "Pt", "player"]).any()
assert not tb_model_events.duplicated(["match_id", "Pt", "player"]).any()
print("Pressure model diagnostics (in-sample fit describes these observations, not predictive performance):")
report_table(pressure_model_diagnostics.round(5))
print("Observed event coverage by player and surface:")
report_table(pressure_event_coverage.round(3))
print("Model coefficients: player exponentials are reference odds; other exponentials are conditional odds ratios:")
with pd.option_context("display.max_rows", None):
    report_table(pressure_model_coefficients)
print("Direct player comparisons: unadjusted 95% intervals and Holm p-values across all six planned tests:")
with pd.option_context("display.max_colwidth", None):
    report_table(pressure_pairwise)
print("Conditional probabilities: Hard, set 1, non-deciding, model-average prior rally and opponent context, neither covariate missing:")
report_table(pressure_reference_profiles.round(4))
```

Pressure model diagnostics (in-sample fit describes these observations, not predictive performance):

model	observations	match_clusters	parameters	design_rank	iterations	successes
BP save	8773	1342	11	11	100	5673
TB point	9586	595	12	12	96	5101

raw_success_pct	brier_in_sample	prior_rally_missing	opponent_context_missing	information_condition_number
64.664	0.227	8	688	29,522.560
53.213	0.229	18	860	61,970.064

Observed event coverage by player and surface:

model	player	surface	observations	successes	match_clusters	observed_success_pct
BP save	Novak Djokovic	Clay	865	527	119	60.925
BP save	Novak Djokovic	Grass	374	256	55	68.449
BP save	Novak Djokovic	Hard	1797	1161	318	64.608
BP save	Rafael Nadal	Clay	980	640	139	65.306
BP save	Rafael Nadal	Grass	245	166	33	67.755
BP save	Rafael Nadal	Hard	1206	748	188	62.023
BP save	Roger Federer	Clay	824	515	121	62.500
BP save	Roger Federer	Grass	467	337	97	72.163
BP save	Roger Federer	Hard	2015	1323	390	65.658
TB point	Novak Djokovic	Clay	551	309	42	56.080
TB point	Novak Djokovic	Grass	569	303	33	53.251
TB point	Novak Djokovic	Hard	1716	931	126	54.254
TB point	Rafael Nadal	Clay	590	305	42	51.695
TB point	Rafael Nadal	Grass	381	192	22	50.394
TB point	Rafael Nadal	Hard	1202	612	83	50.915
TB point	Roger Federer	Clay	826	433	51	52.421
TB point	Roger Federer	Grass	1081	585	69	54.117
TB point	Roger Federer	Hard	2670	1431	174	53.596

Model coefficients: player exponentials are reference odds; other exponentials are conditional odds ratios:

model	term	coefficient	clustered_se	exp_coefficient	ci95_lower	ci95_upper	p_value_unadjusted
BP save	player_federer	0.575	0.048	1.777	1.618	1.952	3.1e-33
BP save	player_nadal	0.533	0.056	1.704	1.525	1.903	3.78e-21
BP save	player_djokovic	0.529	0.050	1.697	1.539	1.872	2.71e-26
BP save	surface_clay	-0.028	0.052	0.973	0.879	1.076	0.591
BP save	surface_grass	0.214	0.073	1.238	1.074	1.429	0.00332
BP save	set_after_first	0.030	0.026	1.030	0.979	1.084	0.248
BP save	deciding_set	-0.042	0.084	0.958	0.814	1.129	0.611
BP save	prior_rally_centered	0.001	0.005	1.001	0.992	1.010	0.82
BP save	prior_rally_missing	-0.709	0.645	0.492	0.139	1.743	0.272
BP save	opponent_strength_centered	-4.257	0.794	0.014	0.003	0.067	8.21e-08
BP save	opponent_strength_missing	0.208	0.094	1.232	1.025	1.480	0.026
TB point	player_federer	-0.420	0.050	0.657	0.596	0.724	2.47e-17
TB point	player_nadal	-0.507	0.061	0.602	0.535	0.678	5.79e-17
TB point	player_djokovic	-0.359	0.053	0.698	0.630	0.774	9.99e-12
TB point	surface_clay	0.031	0.052	1.031	0.932	1.141	0.555
TB point	surface_grass	0.00085	0.051	1.001	0.905	1.106	0.987
TB point	set_after_first	-0.035	0.021	0.966	0.926	1.006	0.0981
TB point	deciding_set	0.005	0.065	1.005	0.884	1.142	0.941
TB point	prior_rally_centered	-0.006	0.004	0.994	0.985	1.002	0.135
TB point	prior_rally_missing	-0.242	0.382	0.785	0.371	1.661	0.527
TB point	opponent_strength_centered	-2.229	0.931	0.108	0.017	0.668	0.0167
TB point	opponent_strength_missing	0.091	0.081	1.095	0.935	1.282	0.261
TB point	is_serving	1.168	0.048	3.216	2.928	3.532	1.21e-131

Direct player comparisons: unadjusted 95% intervals and Holm p-values across all six planned tests:

model	comparison	odds_ratio	ci95_lower	ci95_upper	p_value_unadjusted	p_value_holm_six_tests
BP save	Roger Federer / Rafael Nadal	1.043	0.930	1.170	0.473	1
BP save	Roger Federer / Novak Djokovic	1.047	0.942	1.163	0.393	1
BP save	Rafael Nadal / Novak Djokovic	1.004	0.890	1.132	0.951	1
TB point	Roger Federer / Rafael Nadal	1.091	0.974	1.221	0.132	0.659
TB point	Roger Federer / Novak Djokovic	0.941	0.841	1.052	0.286	1
TB point	Rafael Nadal / Novak Djokovic	0.863	0.758	0.982	0.0257	0.154

Conditional probabilities: Hard, set 1, non-deciding, model-average prior rally and opponent context, neither covariate missing:

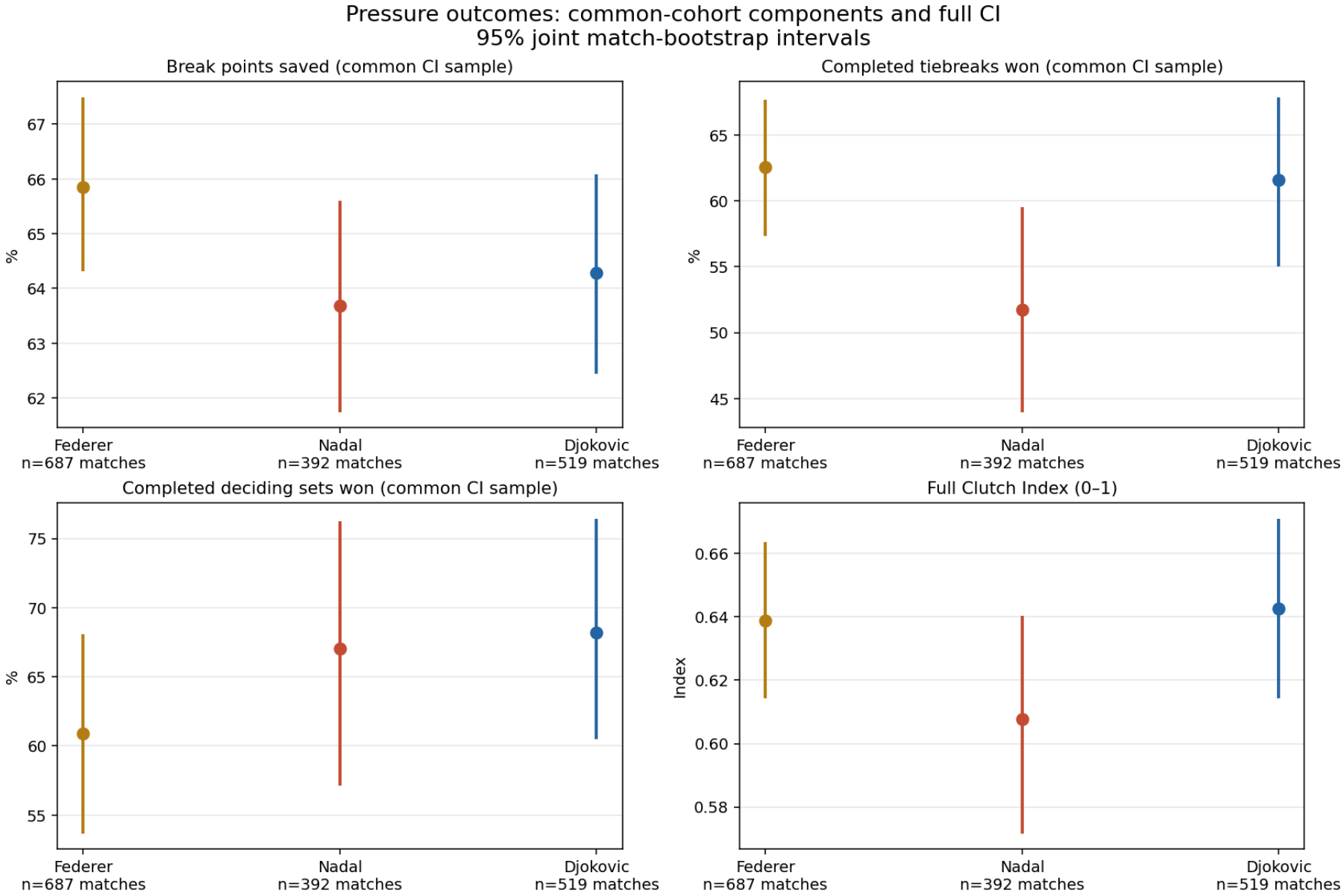
model	player	is_serving	reference_probability_pct	ci95_lower	ci95_upper	reference_surface	reference_set	prior_rally_reference	opponent_strength_reference
BP save	Roger Federer	True	63.989	61.800	66.121	Hard	1	5.661	0.367
BP save	Rafael Nadal	True	63.015	60.401	65.555	Hard	1	5.661	0.367
BP save	Novak Djokovic	True	62.927	60.620	65.178	Hard	1	5.661	0.367
TB point	Roger Federer	False	39.652	37.352	42.000	Hard	1	4.762	0.502
TB point	Roger Federer	True	67.875	65.832	69.852	Hard	1	4.762	0.502
TB point	Rafael Nadal	False	37.591	34.850	40.415	Hard	1	4.762	0.502
TB point	Rafael Nadal	True	65.950	63.102	68.688	Hard	1	4.762	0.502
TB point	Novak Djokovic	False	41.118	38.639	43.642	Hard	1	4.762	0.502
TB point	Novak Djokovic	True	69.188	66.856	71.425	Hard	1	4.762	0.502

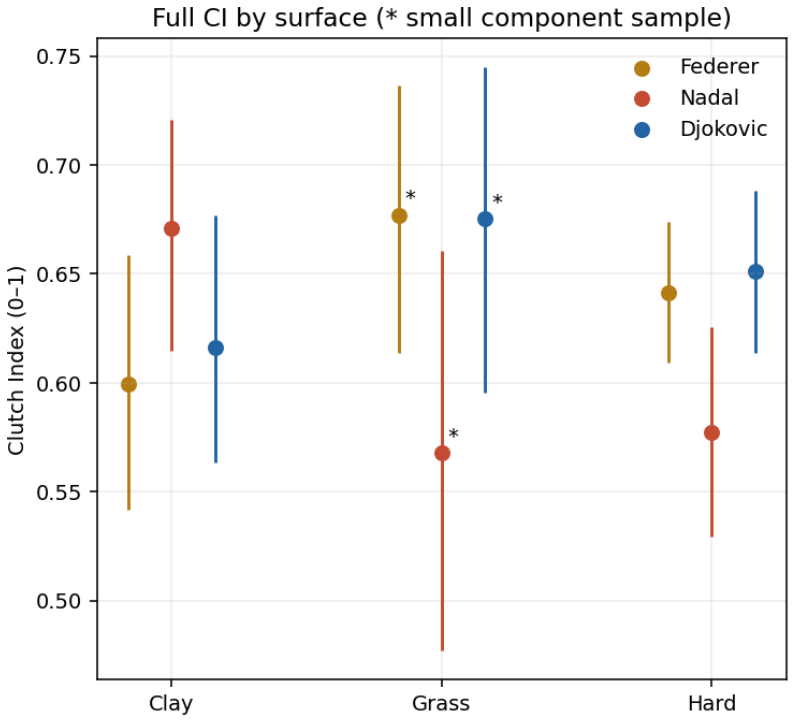
6.3 Component, surface, and model comparisons

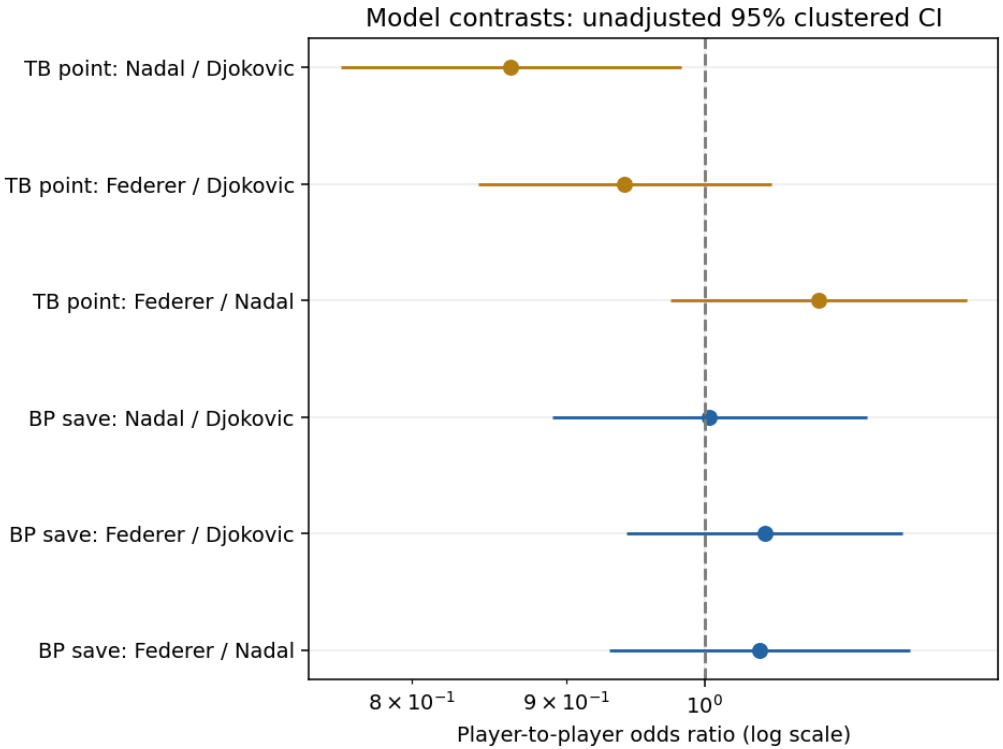
The common-sample component plots and CI describe pooled event outcomes; the model profiles condition on a shared reference scenario. Their samples and questions differ. A tiebreak point win probability is not a tiebreak win percentage. Model probabilities are not unconditional player abilities. Pairwise odds-ratio whiskers use the clustered covariance; use the Holm-adjusted table for the six planned comparisons rather than inferring differences from overlapping separate player intervals.

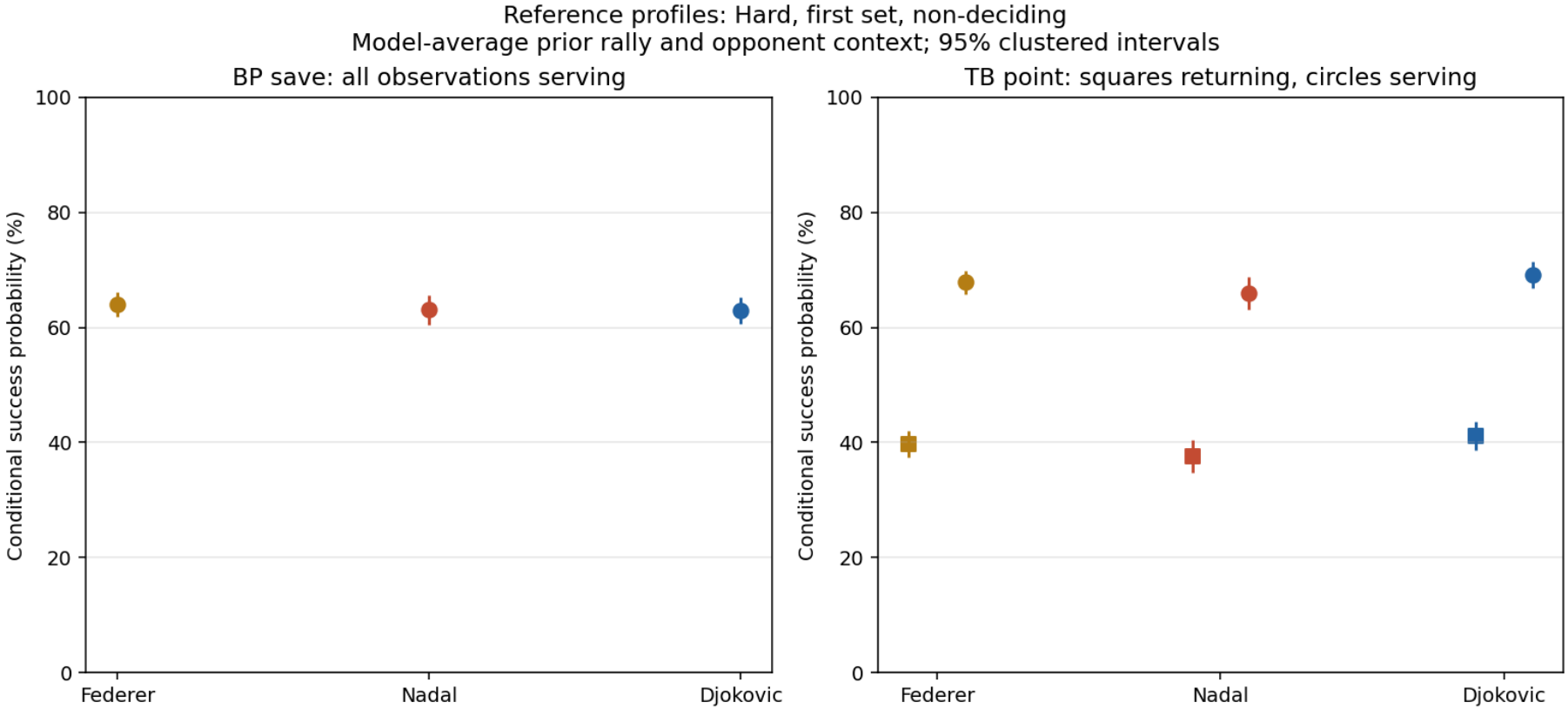
In [17]:
```
with plt.style.context("default"):
    fig, axes = plt.subplots(2, 2, figsize=(12, 8), constrained_layout=True)
    for ax, metric, title in zip(axes.flat, ["bp_component", "tb_component", "deciding_component", "clutch_index_full"],
        ["Break points saved (common CI sample)", "Completed tiebreaks won (common CI sample)", "Completed deciding sets won (common CI sample)", "Full Clutch Index (0–1)"]):
        scale = 1 if metric == "clutch_index_full" else 100
        for index, player in enumerate(BIG_THREE):
            row = clutch_overall.loc[clutch_overall["player"].eq(player)].iloc[0]
            ax.scatter(index, row[metric]*scale, color=KPI_PLAYER_COLORS[player], s=55, zorder=3)
            ax.vlines(index, row[metric+"_ci95_lower"]*scale, row[metric+"_ci95_upper"]*scale, color=KPI_PLAYER_COLORS[player], linewidth=2)
        ax.set_xticks(range(3), [KPI_SHORT_NAMES[p]+f"\nn={int(clutch_overall.set_index('player').loc[p,'common_ci_matches'])} matches" for p in BIG_THREE])
```

```
        ax.set_ylabel("Index" if scale == 1 else "%")
        ax.set_title(title, fontsize=11)
        ax.grid(axis="y", alpha=.25)
    fig.suptitle("Pressure outcomes: common-cohort components and full CI\n95% joint match-bootstrap intervals", fontsize=14)
    display_kpi_figure(fig)

    fig, axes = plt.subplots(1, 2, figsize=(12, 5), constrained_layout=True)
    for index, player in enumerate(BIG_THREE):
        rows = clutch_by_surface.loc[clutch_by_surface["player"].eq(player)].set_index("surface").reindex(PROJECT_SURFACES)
        x = np.arange(3) + (index-1)*.16
        axes[0].scatter(x, rows["clutch_index_full"], color=KPI_PLAYER_COLORS[player], label=KPI_SHORT_NAMES[player], s=45)
        for xpos, (_, row) in zip(x, rows.iterrows()):
            axes[0].vlines(xpos, row["clutch_index_full_ci95_lower"], row["clutch_index_full_ci95_upper"], color=KPI_PLAYER_COLORS[player])
            if row["small_sample"]:
                axes[0].annotate("*", (xpos, row["clutch_index_full"]), xytext=(3, 5), textcoords="offset points")
    axes[0].set_xticks(range(3), PROJECT_SURFACES)
    axes[0].set_title("Full CI by surface (* small component sample)")
    axes[0].set_ylabel("Clutch Index (0-1)")
    axes[0].legend(frameon=False)
    for index, row in pressure_pairwise.iterrows():
        color = "#2766A5" if row["model"] == "BP save" else "#B78016"
        axes[1].scatter(row["odds_ratio"], index, color=color, s=45)
        axes[1].hlines(index, row["ci95_lower"], row["ci95_upper"], color=color)
    axes[1].set_yticks(range(len(pressure_pairwise)), [r.model + ": " + r.comparison.replace("Roger ", "").replace("Rafael ", "").replace("Novak ", "") for r in
pressure_pairwise.itertuples()])
    axes[1].axvline(1, color="gray", linestyle="--")
    axes[1].set_xscale("log")
    axes[1].set_xlabel("Player-to-player odds ratio (log scale)")
    axes[1].set_title("Model contrasts: unadjusted 95% clustered CI")
    for ax in axes:
        ax.grid(alpha=.2)
    display_kpi_figure(fig)

    fig, axes = plt.subplots(1, 2, figsize=(11, 5), constrained_layout=True)
    for ax, kind in zip(axes, ["BP save", "TB point"]):
        for index, player in enumerate(BIG_THREE):
            for _, row in pressure_reference_profiles.loc[pressure_reference_profiles["model"].eq(kind) &
pressure_reference_profiles["player"].eq(player)].iterrows():
                xpos = index if kind == "BP save" else index + (.1 if row["is_serving"] else -.1)
                ax.scatter(xpos, row["reference_probability_pct"], marker="o" if row["is_serving"] else "s", color=KPI_PLAYER_COLORS[player], s=55)
                ax.vlines(xpos, row["ci95_lower"], row["ci95_upper"], color=KPI_PLAYER_COLORS[player])
        ax.set_xticks(range(3), [KPI_SHORT_NAMES[p] for p in BIG_THREE])
        ax.set_ylabel("Conditional success probability (%)")
        ax.set_ylim(0, 100)
        ax.set_title("BP save: all observations serving" if kind == "BP save" else "TB point: squares returning, circles serving")
        ax.grid(axis="y", alpha=.25)
    fig.suptitle("Reference profiles: Hard, first set, non-deciding\nModel-average prior rally and opponent context; 95% clustered intervals", fontsize=12)
    display_kpi_figure(fig)
```


Pressure outcomes: common-cohort components and full CI
95% joint match-bootstrap intervals
Break points saved (common CI sample)
%
67
66
65
64
63
62
Federer
n=687 matches
Nadal
n=392 matches
Djokovic
n=519 matches
Completed tiebreaks won (common CI sample)
%
65
60
55
50
45
Federer
n=687 matches
Nadal
n=392 matches
Djokovic
n=519 matches
Completed deciding sets won (common CI sample)
%
75
70
65
60
55
Federer
n=687 matches
Nadal
n=392 matches
Djokovic
n=519 matches
Full Clutch Index (0–1)
Index
0.66
0.64
0.62
0.60
0.58
Federer
n=687 matches
Nadal
n=392 matches
Djokovic
n=519 matches


Full CI by surface (* small component sample)
Clutch Index (0–1)
0.75
0.70
0.65
0.60
0.55
0.50
Federer
Nadal
Djokovic
*
*
*
Clay
Grass
Hard


Model contrasts: unadjusted 95% clustered CI
TB point: Nadal / Djokovic
TB point: Federer / Djokovic
TB point: Federer / Nadal
BP save: Nadal / Djokovic
BP save: Federer / Djokovic
BP save: Federer / Nadal
8×10^{-1}
9×10^{-1}
10^{0}
Player-to-player odds ratio (log scale)


Reference profiles: Hard, first set, non-deciding
Model-average prior rally and opponent context; 95% clustered intervals
BP save: all observations serving
TB point: squares returning, circles serving
Conditional success probability (%)
100
80
60
40
20
0
Federer
Nadal
Djokovic

6.4 Interpretation and validation limits

The full CI rewards the declared blend of pressure outcomes on a common sample. It is sensitive to the weights, selection of charted matches, opponent mix, and component event counts; surface estimates with fewer than 30 events in any component are flagged. Ordinary serving/returning strength can explain part of pressure success. The point models provide context-adjusted associations, not a direct measure of performing above a player's normal expectation under pressure. An above-baseline pressure effect is a different estimand and is not claimed here.

The models deliberately exclude unsupported contexts and retain missing-covariate indicators. In-sample Brier scores and convergence checks are diagnostics, not out-of-sample forecasting claims. Descriptive opponent proxies and preceding-point rally length do not exhaust confounding. The preference comparison in Section IX consumes full common-cohort CI and its paired uncertainty, not the logistic coefficients or fitted probabilities.

6.5 Evidence: raw pressure success versus conditional associations

In [18]:
```
pressure_headline=clutch_overall[["player","common_ci_matches","bp_saved","bp_faced","tb_component","tiebreaks_played",
    "deciding_component","deciding_sets_played","clutch_index_full","clutch_index_full_ci95_lower","clutch_index_full_ci95_upper"]]
report_table(pressure_headline,caption="Pressure evidence on each player's common CI cohort",expanded=True)
lines=["**Observed result.**"]
for row in clutch_overall.itertuples(index=False):
    lines.append(f"- {row.player}: CI {row.clutch_index_full:.3f} ({interval_text(row.clutch_index_full_ci95_lower,row.clutch_index_full_ci95_upper,3)}), on
{row.common_ci_matches} common-cohort matches; {int(row.bp_saved)}/{int(row.bp_faced)} break points saved, {int(row.tiebreaks_won)}/{int(row.tiebreaks_played)}
completed tiebreaks won, {int(row.deciding_sets_won)}/{int(row.deciding_sets_played)} deciding sets won.")
rejections=int(pressure_pairwise["p_value_holm_six_tests"].lt(.05).sum())
lines.append(f"\nThe models use {len(bp_model_events):,} BP-save observations and {len(tb_model_events):,} TB player-point observations (rivalry points can
contribute complementary player rows). {rejections} of six planned contrasts reject at Holm-adjusted 5%. This is not an equivalence test. The full contrast
table, including unadjusted intervals and adjusted p-values, remains visible in Section 6.2.")
lines.append("\n**Tennis interpretation.** A strong serve can save break points without implying an extra psychological effect. Winning a tiebreak involves both
serving and returning, which is why its point model includes serving status. Deciding-set win rates mix match formats and opponents; CI does not isolate
performance above a player's usual baseline.")
lines.append("\n**Alternative explanations.** Common-cohort filtering removes uncertain outcomes and events, and may select a different schedule for each
player. The retrospective opponent proxy, preceding-point rally, and additive surface terms leave age, injury, era, matchup interactions, and time dependence
unmodeled. Next we examine the shot patterns that may accompany these outcomes, without interpreting them as causal mechanisms.")
display(Markdown("\n\n".join(lines)))
model_diagnostic_rows=[]
for kind,result in pressure_models.items():
    X=result["design"].to_numpy();y=result["events"]["success"].to_numpy(dtype=float)
    fitted=result["model"].predict_proba(result["design"])[:,1]
    model_diagnostic_rows.append({"model":kind,"observations":len(y),"match_clusters":result["diagnostics"]["match_clusters"],
        "max_absolute_score_per_observation":float(np.max(np.abs(X.T@(y-fitted)))/len(y)),
        "minimum_fitted_probability":fitted.min(),"maximum_fitted_probability":fitted.max(),
        "minimum_covariance_eigenvalue":float(np.linalg.eigvalsh(result["covariance"]).min())})
pressure_numerical_diagnostics=pd.DataFrame(model_diagnostic_rows)
report_table(pressure_numerical_diagnostics,caption="Numerical diagnostics: score residual, fitted range and clustered covariance",expanded=True)
```

```
context_effect_rows=[]
for kind,result in pressure_models.items():
    j=list(result["design"].columns).index("opponent_strength_centered")
    effect=.1*result["beta"][j];se=.1*np.sqrt(result["covariance"][j,j])
    context_effect_rows.append({"model":kind,"opponent_proxy_change":"+10 percentage points", "conditional_odds_ratio":np.exp(effect),
        "ci95_lower":np.exp(effect-1.96*se),"ci95_upper":np.exp(effect+1.96*se),
        "interpretation":"retrospective association; unadjusted interval, not a player contrast"})
pressure_context_effects=pd.DataFrame(context_effect_rows)
report_table(pressure_context_effects,expanded=True)
```

Pressure evidence on each player's common CI cohort

player	common_ci_matches	bp_saved	bp_faced	tb_component	tiebreaks_played	deciding_component	deciding_sets_played
Roger Federer	687	2154	3271	0.626	366.000	0.609	179.000
Rafael Nadal	392	1489	2338	0.517	172.000	0.670	97.000
Novak Djokovic	519	1901	2957	0.616	237.000	0.682	132.000

player	clutch_index_full	clutch_index_full_ci95_lower	clutch_index_full_ci95_upper
Roger Federer	0.639	0.614	0.664
Rafael Nadal	0.608	0.572	0.640
Novak Djokovic	0.643	0.614	0.671

Observed result.

- Roger Federer: CI 0.639 (0.614–0.664), on 687 common-cohort matches; 2154/3271 break points saved, 229/366 completed tiebreaks won, 109/179 deciding sets won.
- Rafael Nadal: CI 0.608 (0.572–0.640), on 392 common-cohort matches; 1489/2338 break points saved, 89/172 completed tiebreaks won, 65/97 deciding sets won.
- Novak Djokovic: CI 0.643 (0.614–0.671), on 519 common-cohort matches; 1901/2957 break points saved, 146/237 completed tiebreaks won, 90/132 deciding sets won.

The models use 8,773 BP-save observations and 9,586 TB player-point observations (rivalry points can contribute complementary player rows). 0 of six planned contrasts reject at Holm-adjusted 5%. This is not an equivalence test. The full contrast table, including unadjusted intervals and adjusted p-values, remains visible in Section 6.2.

Tennis interpretation. A strong serve can save break points without implying an extra psychological effect. Winning a tiebreak involves both serving and returning, which is why its point model includes serving status. Deciding-set win rates mix match formats and opponents; CI does not isolate performance above a player's usual baseline.

Alternative explanations. Common-cohort filtering removes uncertain outcomes and events, and may select a different schedule for each player. The retrospective opponent proxy, preceding-point rally, and additive surface terms leave age, injury, era, matchup interactions, and time dependence unmodeled. Next we examine the shot patterns that may accompany these outcomes, without interpreting them as causal mechanisms.

Numerical diagnostics: score residual, fitted range and clustered covariance

model	observations	match_clusters	max_absolute_score_per_observation	minimum_fitted_probability	maximum_fitted_probability	minimum_covariance_eigenvalue
BP save	8773	1342	1.93e-07	0.410	0.783	2.11e-05
TB point	9586	595	1.59e-07	0.302	0.727	1.69e-05

model	opponent_proxy_change	conditional_odds_ratio	ci95_lower	ci95_upper	interpretation
BP save	+10 percentage points	0.653	0.559	0.763	retrospective association; unadjusted interval, not a player contrast
TB point	+10 percentage points	0.800	0.667	0.960	retrospective association; unadjusted interval, not a player contrast

VII. Shot mix, diversity, and coded placement

Tactical question: Do the players build points with different stroke families and directional patterns? Repeated patterns can be effective; variety describes options used, rather than the quality of those choices.

7.1 Observed categories and denominators

Use every available eligible charted career period. The shared parser assigns hitters **before** dropping unknown types or directions. Rally strokes exclude serves and include attempted error strokes, which are flagged separately; a stroke ending in `#` or `@` is an error attempt even without an explicit net/wide/deep suffix. Entirely uncharted S/R points contain no recoverable shot mix. Point-sequence coverage and unknown-category shares accompany all comparisons.

Shot-type shares include nine fixed known families plus unknown. Known-family entropy excludes unknown types and uses fixed K=9; zero-count families remain in the normalization. Source directional mix uses the five ShotDirection categories (crosscourt, middle, line, inside-out, inside-in), separately for Total and source F/B/S stroke classes. These source classes are not an exhaustive taxonomy of every parsed stroke; their denominator is the sum of the five classified source counts, not all rally shots. Source direction availability is shown as matches, not an invented unknown-shot fraction.

Parsed target codes 1/2/3 answer a different question: the target's right/center/left, defined relative to a right-hander's forehand/backhand reference in the workbook. In a hitter-facing-opponent court view, code 1 is on the left of the page and code 3 on the right, irrespective of the hitter's handedness. These codes alone do not establish crosscourt versus down-the-line trajectories. Return-only target diversity is reported separately. No diversity index contributes automatically to performance or greatness.

In [19]:
```
SHOT_KNOWN_FAMILIES = ["groundstroke", "slice", "volley", "overhead", "drop shot", "lob", "half-volley", "swinging volley", "other"]
SHOT_ALL_FAMILIES = SHOT_KNOWN_FAMILIES + ["unknown"]
SOURCE_DIRECTION_CATEGORIES = ["crosscourt", "down_middle", "down_the_line", "inside_out", "inside_in"]
TARGET_CATEGORIES = ["1", "2", "3"]
SERVE_ZONE_CATEGORIES = ["deuce_wide", "deuce_body", "deuce_t", "ad_wide", "ad_body", "ad_t"]
RETURN_JOINT_CATEGORIES = [str(depth) + "|" + str(direction) for depth in [7, 8, 9] for direction in [1, 2, 3]]
shot_analysis = canonical_shots.copy()
```

```
shot_analysis["target_category"] = shot_analysis["direction_code"].map({1: "1", 2: "2", 3: "3"})
shot_analysis["return_joint_category"] = [str(int(depth)) + "|" + str(int(direction))
    if ret and pd.notna(depth) and depth in [7, 8, 9] and pd.notna(direction) and direction in [1, 2, 3] else None
    for ret, depth, direction in shot_analysis[["is_return", "depth_code", "direction_code"]].itertuples(index=False, name=None)]
source_direction = analysis_dataframes["charting_m_stats_ShotDirection"].copy()
source_direction = source_direction.loc[source_direction["player"].isin(BIG_THREE)].merge(
    canonical_master[["match_id", "player", "surface"]], on=["match_id", "player"], how="inner", validate="many_to_one")
for field in SOURCE_DIRECTION_CATEGORIES:
    source_direction[field] = pd.to_numeric(source_direction[field], errors="coerce")
source_direction["direction_counts_valid"] = (source_direction[SOURCE_DIRECTION_CATEGORIES].notna().all(axis=1)
    & source_direction[SOURCE_DIRECTION_CATEGORIES].ge(0).all(axis=1)
    & source_direction[SOURCE_DIRECTION_CATEGORIES].mod(1).eq(0).all(axis=1))
source_direction_invalid = source_direction.loc[~source_direction["direction_counts_valid"]].copy()
source_direction = source_direction.loc[source_direction["direction_counts_valid"]].copy()

# Active point serve = first serve when it was played, otherwise the second serve.
# Includes coded double-fault attempts; first-serve faults preceding a second serve are not counted again.
serve_analysis = canonical_points.loc[canonical_points["server_name"].isin(BIG_THREE)].copy()
serve_analysis["player"] = serve_analysis["server_name"]
serve_analysis["target_category"] = [str(court) + "_" + {4: "wide", 5: "body", 6: "t"}[int(code)]
    if pd.notna(court) and court in ["deuce", "ad"] and pd.notna(code) and code in [4, 5, 6] else None
    for court, code in serve_analysis[["service_court", "serve_direction_code"]].itertuples(index=False, name=None)]

# Category counts at player-match grain allow the same joint match bootstrap as Sections V–VI.
def parsed_category_counts(frame, category):
    return frame.dropna(subset=[category]).groupby(["match_id", "player", "surface", category], as_index=False).size().rename(columns={category: "category",
"size": "count"})

distribution_inputs = {
    "shot_type": (parsed_category_counts(shot_analysis, "shot_family"), SHOT_ALL_FAMILIES, SHOT_KNOWN_FAMILIES),
    "target_all_attempts": (parsed_category_counts(shot_analysis, "target_category"), TARGET_CATEGORIES, TARGET_CATEGORIES),
    "target_non_error_strokes": (parsed_category_counts(shot_analysis.loc[~shot_analysis["is_error_attempt"]], "target_category"), TARGET_CATEGORIES,
TARGET_CATEGORIES),
    "target_returns": (parsed_category_counts(shot_analysis.loc[shot_analysis["is_return"]], "target_category"), TARGET_CATEGORIES, TARGET_CATEGORIES),
    "active_serve_target": (parsed_category_counts(serve_analysis, "target_category"), SERVE_ZONE_CATEGORIES, None),
    "return_depth_target": (parsed_category_counts(shot_analysis, "return_joint_category"), RETURN_JOINT_CATEGORIES, None),
}
for row_value in ["Total", "F", "B", "S"]:
    selected = source_direction.loc[source_direction["row"].eq(row_value)]
    counts = selected.melt(id_vars=["match_id", "player", "surface"], value_vars=SOURCE_DIRECTION_CATEGORIES,
        var_name="category", value_name="count")
    distribution_inputs["source_direction_" + row_value] = (counts, SOURCE_DIRECTION_CATEGORIES, SOURCE_DIRECTION_CATEGORIES)

def normalized_entropy(counts, known_indices):
    known = counts[..., known_indices]
    denominator = known.sum(axis=-1, keepdims=True)
    with np.errstate(divide="ignore", invalid="ignore"):
        probabilities = np.divide(known, denominator, out=np.zeros_like(known, dtype=float), where=denominator > 0)
        terms = np.where(probabilities > 0, probabilities * np.log(probabilities), 0)
        entropy = -terms.sum(axis=-1) / np.log(len(known_indices))
    return np.where(denominator[..., 0] > 0, entropy, np.nan)

shot_distribution_rows, shot_diversity_rows = [], []
for distribution, (counts, categories, entropy_categories) in distribution_inputs.items():
    for player, surface in kpi_scope_groups:
        data = counts.loc[counts["player"].eq(player)]
        if surface != "Overall":
```

```
            data = data.loc[data["surface"].eq(surface)]
        # One vector per category, including absent categories as zeros.
        category_matrix = np.zeros((len(bootstrap_match_ids), len(categories)))
        for match_id, category, count in data[["match_id", "category", "count"]].itertuples(index=False, name=None):
            if category in categories and match_id in bootstrap_match_index:
                category_matrix[bootstrap_match_index[match_id], categories.index(category)] += count
        totals = category_matrix.sum(axis=0)
        total = totals.sum()
        contributing_matches = int((category_matrix.sum(axis=1) > 0).sum())
        boot = bootstrap_weights @ category_matrix
        boot_den = boot.sum(axis=1)
        for index, category in enumerate(categories):
            with np.errstate(divide="ignore", invalid="ignore"):
                values = np.where(boot_den > 0, 100 * boot[:, index] / boot_den, np.nan)
            valid = values[np.isfinite(values)]
            shot_distribution_rows.append({"player": player, "surface": surface, "distribution": distribution, "category": category,
                "count": int(totals[index]), "classified_denominator": int(total), "share_pct": 100*totals[index]/total if total else np.nan,
                "contributing_matches": contributing_matches, "valid_bootstraps": len(valid),
                "ci95_lower": np.percentile(valid, 2.5) if len(valid) >= .95*BOOTSTRAP_REPLICATES else np.nan,
                "ci95_upper": np.percentile(valid, 97.5) if len(valid) >= .95*BOOTSTRAP_REPLICATES else np.nan})
        if entropy_categories is not None:
            indices = [categories.index(category) for category in entropy_categories]
            point_entropy = float(normalized_entropy(totals, indices))
            values = normalized_entropy(boot, indices)
            valid = values[np.isfinite(values)]
            known_count = int(totals[indices].sum())
            shot_diversity_rows.append({"player": player, "surface": surface, "distribution": distribution,
                "normalized_entropy": point_entropy, "fixed_categories_K": len(indices), "classified_strokes": known_count,
                "contributing_matches": contributing_matches, "small_sample": known_count < SMALL_SAMPLE_THRESHOLDS["shots"],
                "valid_bootstraps": len(valid), "ci95_lower": np.percentile(valid, 2.5) if len(valid) >= .95*BOOTSTRAP_REPLICATES else np.nan,
                "ci95_upper": np.percentile(valid, 97.5) if len(valid) >= .95*BOOTSTRAP_REPLICATES else np.nan})
shot_distribution_long = pd.DataFrame(shot_distribution_rows)
shot_diversity_summary = pd.DataFrame(shot_diversity_rows)

shot_coverage_rows = []
for player, surface in kpi_scope_groups:
    shots = shot_analysis.loc[shot_analysis["player"].eq(player)]
    serves = serve_analysis.loc[serve_analysis["player"].eq(player)]
    player_points = canonical_player_points.loc[canonical_player_points["player"].eq(player)]
    metadata = canonical_master.loc[canonical_master["player"].eq(player)]
    direction_rows = source_direction.loc[source_direction["player"].eq(player) & source_direction["row"].eq("Total")]
    if surface != "Overall":
        shots, serves, player_points, metadata, direction_rows = [frame.loc[frame["surface"].eq(surface)] for frame in [shots, serves, player_points, metadata,
direction_rows]]
    returns = shots.loc[shots["is_return"]]
    known_targets = int(shots["target_category"].notna().sum())
    known_wings = shots["wing"].isin(["forehand", "backhand"])
    shot_coverage_rows.append({"player": player, "surface": surface, "cohort_matches": len(metadata),
        "matches_with_parsed_strokes": shots["match_id"].nunique(), "recorded_player_points": len(player_points),
        "points_with_sequence_codes": int(player_points["sequence_known"].sum()),
        "sequence_code_coverage_pct": 100*player_points["sequence_known"].mean() if len(player_points) else np.nan,
        "attempted_rally_strokes": len(shots), "error_attempts": int(shots["is_error_attempt"].sum()),
        "unknown_type_strokes": int(shots["shot_family"].eq("unknown").sum()),
        "unknown_type_pct": 100*shots["shot_family"].eq("unknown").mean() if len(shots) else np.nan,
        "known_wing_strokes": int(known_wings.sum()),
        "forehand_share_known_wing_pct": 100*shots["wing"].eq("forehand").sum()/known_wings.sum() if known_wings.sum() else np.nan,
        "classified_targets": known_targets, "target_direction_coverage_pct": 100*known_targets/len(shots) if len(shots) else np.nan,
```

```
        "parsed_return_attempts": len(returns), "returns_with_known_depth": int(returns["depth_code"].isin([7, 8, 9]).sum()),
        "return_depth_coverage_pct": 100*returns["depth_code"].isin([7, 8, 9]).mean() if len(returns) else np.nan,
        "returns_with_depth_and_direction": int(returns["return_joint_category"].notna().sum()),
        "recorded_service_points": len(serves), "classified_active_serve_targets": int(serves["target_category"].notna().sum()),
        "active_serve_target_coverage_pct": 100*serves["target_category"].notna().mean() if len(serves) else np.nan,
        "source_Total_direction_matches": direction_rows["match_id"].nunique()})
shot_coverage_summary = pd.DataFrame(shot_coverage_rows)
# Classification denominator checks and identities, including the entropy boundary cases.
assert np.isclose(normalized_entropy(np.array([1., 0, 0]), [0, 1, 2]), 0)
assert np.isclose(normalized_entropy(np.ones(3), [0, 1, 2]), 1)
assert pd.isna(normalized_entropy(np.zeros(3), [0, 1, 2]))
assert parse_point_sequence("6f#", None)["last_stroke_error"]
assert parse_point_sequence("6f#", None)["rally_length"] == 1
assert not shot_distribution_long.duplicated(["player", "surface", "distribution", "category"]).any()
assert shot_distribution_long["share_pct"].dropna().between(0, 100).all()
assert shot_diversity_summary["normalized_entropy"].dropna().between(-1e-12, 1+1e-12).all()
for _, group in shot_distribution_long.groupby(["player", "surface", "distribution"]):
    if group["classified_denominator"].iloc[0] > 0:
        assert np.isclose(group["share_pct"].sum(), 100)
        assert group["count"].sum() == group["classified_denominator"].iloc[0]
for player in BIG_THREE:
    family_rows = shot_distribution_long.loc[shot_distribution_long["player"].eq(player) & shot_distribution_long["surface"].eq("Overall") &
shot_distribution_long["distribution"].eq("shot_type")]
    assert family_rows["count"].sum() == len(shot_analysis.loc[shot_analysis["player"].eq(player)])
print("Coverage and missing-category evidence:")
with pd.option_context("display.max_columns", None):
    report_table(shot_coverage_summary.round(3))
print("Shot diversity: different distributions describe different concepts; do not rank them as performance:")
with pd.option_context("display.max_rows", None):
    report_table(shot_diversity_summary.round(4))
print("Complete category counts, percentages, and match-bootstrap intervals:")
with pd.option_context("display.max_rows", None):
    report_table(shot_distribution_long.round(3))
print(f"Invalid source-direction count rows excluded: {len(source_direction_invalid)}")
report_table(source_direction_invalid)
```

```
Coverage and missing-category evidence:
```

player	surface	cohort_matches	matches_with_parsed_strokes	recorded_player_points	points_with_sequence_codes	sequence_code_coverage_pct	attempted_rally_strokes	error_attempts
Roger Federer	Overall	723	722	125338	124133	99.039	220280	39628
Roger Federer	Clay	130	130	23303	23044	98.889	45793	7809
Roger Federer	Grass	119	119	23438	23258	99.232	35126	6933
Roger Federer	Hard	474	473	78597	77831	99.025	139361	24886
Rafael Nadal	Overall	425	423	70896	70639	99.637	151792	20050
Rafael Nadal	Clay	166	166	25786	25708	99.698	62935	7309
Rafael Nadal	Grass	40	39	8502	8413	98.953	14056	2323
Rafael Nadal	Hard	219	218	36608	36518	99.754	74801	10418
Novak Djokovic	Overall	553	552	93603	93110	99.473	202209	27378
Novak Djokovic	Clay	129	129	21959	21870	99.595	52098	6617
Novak Djokovic	Grass	58	58	12635	12441	98.465	23538	3578
Novak Djokovic	Hard	366	365	59009	58799	99.644	126573	17183

player	surface	unknown_type_strokes	unknown_type_pct	known_wing_strokes	forehand_share_known_wing_pct	classified_targets	target_direction_coverage_pct	parsed_return_attempts
Roger Federer	Overall	32	0.015	220184	49.187	215848	97.988	57377
Roger Federer	Clay	12	0.026	45777	49.197	45132	98.557	10889
Roger Federer	Grass	4	0.011	35108	49.220	33574	95.582	10535
Roger Federer	Hard	16	0.011	139299	49.176	137142	98.408	35953
Rafael Nadal	Overall	11	0.007	151738	55.412	145635	95.944	31719
Rafael Nadal	Clay	6	0.010	62917	55.821	59868	95.127	12228
Rafael Nadal	Grass	3	0.021	14043	54.226	13581	96.621	3542
Rafael Nadal	Hard	2	0.003	74778	55.290	72186	96.504	15949
Novak Djokovic	Overall	25	0.012	202154	50.100	199028	98.427	41780
Novak Djokovic	Clay	14	0.027	52080	50.340	51435	98.727	10022
Novak Djokovic	Grass	2	0.008	23527	49.156	23272	98.870	5538
Novak Djokovic	Hard	9	0.007	126547	50.177	124321	98.221	26220

player	surface	returns_with_known_depth	return_depth_coverage_pct	returns_with_depth_and_direction	recorded_service_points	classified_active_serve_targets	active_serve_target_coverage_pct	source_Total_direction_matches
Roger Federer	Overall	42924	74.810	42882	60460	59819	98.940	722
Roger Federer	Clay	8429	77.408	8427	11371	11226	98.725	130
Roger Federer	Grass	7228	68.609	7221	11202	11109	99.170	119
Roger Federer	Hard	27267	75.841	27234	37887	37484	98.936	473
Rafael Nadal	Overall	25627	80.794	22262	34660	34542	99.660	425
Rafael Nadal	Clay	10428	85.280	8508	12513	12464	99.608	166
Rafael Nadal	Grass	2722	76.849	2431	4162	4121	99.015	40

player	surface	returns_with_known_depth	return_depth_coverage_pct	returns_with_depth_and_direction	recorded_service_points	classified_active_serve_targets	active_serve_target_coverage_pct	source_Total_direction_matches
Rafael Nadal	Hard	12477	78.231	11323	17985	17957	99.844	219
Novak Djokovic	Overall	32139	76.924	31898	45935	45681	99.447	552
Novak Djokovic	Clay	7946	79.286	7756	10927	10874	99.515	129
Novak Djokovic	Grass	4062	73.348	4062	6159	6057	98.344	58
Novak Djokovic	Hard	20131	76.777	20080	28849	28750	99.657	365

Shot diversity: different distributions describe different concepts; do not rank them as performance:

▼ Complete evidence table: 96 rows × 11 columns

player	surface	distribution	normalized_entropy	fixed_categories_K	classified_strokes	contributing_matches	small_sample	valid_bootstraps	ci95_lower
Roger Federer	Overall	shot_type	0.410	9	220248	722	False	2000	0.406
Roger Federer	Clay	shot_type	0.373	9	45781	130	False	2000	0.362
Roger Federer	Grass	shot_type	0.451	9	35122	119	False	2000	0.439
Roger Federer	Hard	shot_type	0.410	9	139345	473	False	2000	0.405
Rafael Nadal	Overall	shot_type	0.257	9	151781	423	False	2000	0.250
Rafael Nadal	Clay	shot_type	0.217	9	62929	166	False	2000	0.209
Rafael Nadal	Grass	shot_type	0.335	9	14053	39	False	2000	0.314
Rafael Nadal	Hard	shot_type	0.270	9	74799	218	False	2000	0.262
Novak Djokovic	Overall	shot_type	0.280	9	202184	552	False	2000	0.274
Novak Djokovic	Clay	shot_type	0.258	9	52084	129	False	2000	0.246
Novak Djokovic	Grass	shot_type	0.328	9	23536	58	False	2000	0.309
Novak Djokovic	Hard	shot_type	0.277	9	126564	365	False	2000	0.270
Roger Federer	Overall	target_all_attempts	0.991	3	215848	719	False	2000	0.990
Roger Federer	Clay	target_all_attempts	0.995	3	45132	130	False	2000	0.991
Roger Federer	Grass	target_all_attempts	0.993	3	33574	117	False	2000	0.990
Roger Federer	Hard	target_all_attempts	0.990	3	137142	472	False	2000	0.988
Rafael Nadal	Overall	target_all_attempts	0.988	3	145635	421	False	2000	0.986
Rafael Nadal	Clay	target_all_attempts	0.982	3	59868	166	False	2000	0.977
Rafael Nadal	Grass	target_all_attempts	0.983	3	13581	39	False	2000	0.974
Rafael Nadal	Hard	target_all_attempts	0.993	3	72186	216	False	2000	0.990
Novak Djokovic	Overall	target_all_attempts	0.994	3	199028	549	False	2000	0.992
Novak Djokovic	Clay	target_all_attempts	0.993	3	51435	129	False	2000	0.989
Novak Djokovic	Grass	target_all_attempts	0.993	3	23272	58	False	2000	0.988
Novak Djokovic	Hard	target_all_attempts	0.994	3	124321	362	False	2000	0.992
Roger Federer	Overall	target_non_error_strokes	0.990	3	179579	719	False	2000	0.989
Roger Federer	Clay	target_non_error_strokes	0.994	3	37959	130	False	2000	0.990
Roger Federer	Grass	target_non_error_strokes	0.991	3	27464	117	False	2000	0.988

player	surface	distribution	normalized_entropy	fixed_categories_K	classified_strokes	contributing_matches	small_sample	valid_bootstraps	ci95_lower
Roger Federer	Hard	target_non_error_strokes	0.988	3	114156	472	False	2000	0.986
Rafael Nadal	Overall	target_non_error_strokes	0.987	3	127703	421	False	2000	0.984
Rafael Nadal	Clay	target_non_error_strokes	0.981	3	53459	166	False	2000	0.975
Rafael Nadal	Grass	target_non_error_strokes	0.983	3	11475	39	False	2000	0.973
Rafael Nadal	Hard	target_non_error_strokes	0.992	3	62769	216	False	2000	0.989
Novak Djokovic	Overall	target_non_error_strokes	0.993	3	173861	549	False	2000	0.991
Novak Djokovic	Clay	target_non_error_strokes	0.993	3	45299	129	False	2000	0.988
Novak Djokovic	Grass	target_non_error_strokes	0.992	3	19954	58	False	2000	0.986
Novak Djokovic	Hard	target_non_error_strokes	0.992	3	108608	362	False	2000	0.991
Roger Federer	Overall	target_returns	0.909	3	55456	719	False	2000	0.904
Roger Federer	Clay	target_returns	0.911	3	10615	130	False	2000	0.896
Roger Federer	Grass	target_returns	0.916	3	9913	117	False	2000	0.903
Roger Federer	Hard	target_returns	0.907	3	34928	472	False	2000	0.900
Rafael Nadal	Overall	target_returns	0.944	3	27493	415	False	2000	0.937
Rafael Nadal	Clay	target_returns	0.946	3	10051	163	False	2000	0.934
Rafael Nadal	Grass	target_returns	0.954	3	3178	39	False	2000	0.935
Rafael Nadal	Hard	target_returns	0.939	3	14264	213	False	2000	0.929
Novak Djokovic	Overall	target_returns	0.897	3	40270	548	False	2000	0.890
Novak Djokovic	Clay	target_returns	0.908	3	9622	129	False	2000	0.893
Novak Djokovic	Grass	target_returns	0.898	3	5392	58	False	2000	0.878
Novak Djokovic	Hard	target_returns	0.892	3	25256	361	False	2000	0.883
Roger Federer	Overall	source_direction_Total	0.860	5	132560	719	False	2000	0.857
Roger Federer	Clay	source_direction_Total	0.868	5	29093	130	False	2000	0.861
Roger Federer	Grass	source_direction_Total	0.866	5	18665	117	False	2000	0.858
Roger Federer	Hard	source_direction_Total	0.856	5	84802	472	False	2000	0.852
Rafael Nadal	Overall	source_direction_Total	0.859	5	105412	422	False	2000	0.853
Rafael Nadal	Clay	source_direction_Total	0.861	5	44589	165	False	2000	0.849
Rafael Nadal	Grass	source_direction_Total	0.859	5	9114	40	False	2000	0.845

player	surface	distribution	normalized_entropy	fixed_categories_K	classified_strokes	contributing_matches	small_sample	valid_bootstraps	ci95_lower
Rafael Nadal	Hard	source_direction_Total	0.856	5	51709	217	False	2000	0.849
Novak Djokovic	Overall	source_direction_Total	0.830	5	138901	549	False	2000	0.827
Novak Djokovic	Clay	source_direction_Total	0.839	5	36480	129	False	2000	0.831
Novak Djokovic	Grass	source_direction_Total	0.823	5	15240	58	False	2000	0.813
Novak Djokovic	Hard	source_direction_Total	0.827	5	87181	362	False	2000	0.823
Roger Federer	Overall	source_direction_F	0.908	5	70862	719	False	2000	0.905
Roger Federer	Clay	source_direction_F	0.907	5	15922	130	False	2000	0.898
Roger Federer	Grass	source_direction_F	0.910	5	10097	117	False	2000	0.902
Roger Federer	Hard	source_direction_F	0.908	5	44843	472	False	2000	0.904
Rafael Nadal	Overall	source_direction_F	0.853	5	62069	421	False	2000	0.846
Rafael Nadal	Clay	source_direction_F	0.848	5	26013	164	False	2000	0.836
Rafael Nadal	Grass	source_direction_F	0.857	5	5449	40	False	2000	0.835
Rafael Nadal	Hard	source_direction_F	0.855	5	30607	217	False	2000	0.846
Novak Djokovic	Overall	source_direction_F	0.874	5	70632	549	False	2000	0.870
Novak Djokovic	Clay	source_direction_F	0.878	5	19246	129	False	2000	0.868
Novak Djokovic	Grass	source_direction_F	0.866	5	7410	58	False	2000	0.855
Novak Djokovic	Hard	source_direction_F	0.873	5	43976	362	False	2000	0.868
Roger Federer	Overall	source_direction_B	0.714	5	44026	719	False	2000	0.709
Roger Federer	Clay	source_direction_B	0.713	5	10186	130	False	2000	0.702
Roger Federer	Grass	source_direction_B	0.716	5	5667	117	False	2000	0.706
Roger Federer	Hard	source_direction_B	0.714	5	28173	472	False	2000	0.706
Rafael Nadal	Overall	source_direction_B	0.769	5	35305	422	False	2000	0.753
Rafael Nadal	Clay	source_direction_B	0.788	5	16480	165	False	2000	0.759
Rafael Nadal	Grass	source_direction_B	0.722	5	2564	40	False	2000	0.698

player	surface	distribution	normalized_entropy	fixed_categories_K	classified_strokes	contributing_matches	small_sample	valid_bootstraps	ci95_lower
Rafael Nadal	Hard	source_direction_B	0.754	5	16261	217	False	2000	0.740
Novak Djokovic	Overall	source_direction_B	0.739	5	59630	548	False	2000	0.730
Novak Djokovic	Clay	source_direction_B	0.741	5	16003	129	False	2000	0.713
Novak Djokovic	Grass	source_direction_B	0.748	5	6385	58	False	2000	0.728
Novak Djokovic	Hard	source_direction_B	0.736	5	37242	361	False	2000	0.729
Roger Federer	Overall	source_direction_S	0.662	5	17672	718	False	2000	0.651
Roger Federer	Clay	source_direction_S	0.667	5	2985	130	False	2000	0.639
Roger Federer	Grass	source_direction_S	0.687	5	2901	117	False	2000	0.667
Roger Federer	Hard	source_direction_S	0.654	5	11786	471	False	2000	0.639
Rafael Nadal	Overall	source_direction_S	0.789	5	8038	418	False	2000	0.779
Rafael Nadal	Clay	source_direction_S	0.786	5	2096	162	False	2000	0.759
Rafael Nadal	Grass	source_direction_S	0.782	5	1101	40	False	2000	0.765
Rafael Nadal	Hard	source_direction_S	0.789	5	4841	216	False	2000	0.777
Novak Djokovic	Overall	source_direction_S	0.685	5	8441	546	False	2000	0.671
Novak Djokovic	Clay	source_direction_S	0.725	5	1231	127	False	2000	0.693
Novak Djokovic	Grass	source_direction_S	0.705	5	1445	58	False	2000	0.660
Novak Djokovic	Hard	source_direction_S	0.669	5	5765	361	False	2000	0.652

player	surface	distribution	ci95_upper
Roger Federer	Overall	shot_type	0.415
Roger Federer	Clay	shot_type	0.384
Roger Federer	Grass	shot_type	0.464
Roger Federer	Hard	shot_type	0.416
Rafael Nadal	Overall	shot_type	0.264
Rafael Nadal	Clay	shot_type	0.226
Rafael Nadal	Grass	shot_type	0.356
Rafael Nadal	Hard	shot_type	0.278

player	surface	distribution	ci95_upper
Novak Djokovic	Overall	shot_type	0.286
Novak Djokovic	Clay	shot_type	0.269
Novak Djokovic	Grass	shot_type	0.345
Novak Djokovic	Hard	shot_type	0.284
Roger Federer	Overall	target_all_attempts	0.993
Roger Federer	Clay	target_all_attempts	0.997
Roger Federer	Grass	target_all_attempts	0.995
Roger Federer	Hard	target_all_attempts	0.991
Rafael Nadal	Overall	target_all_attempts	0.990
Rafael Nadal	Clay	target_all_attempts	0.986
Rafael Nadal	Grass	target_all_attempts	0.991
Rafael Nadal	Hard	target_all_attempts	0.995
Novak Djokovic	Overall	target_all_attempts	0.995
Novak Djokovic	Clay	target_all_attempts	0.996
Novak Djokovic	Grass	target_all_attempts	0.997
Novak Djokovic	Hard	target_all_attempts	0.995
Roger Federer	Overall	target_non_error_strokes	0.991
Roger Federer	Clay	target_non_error_strokes	0.997
Roger Federer	Grass	target_non_error_strokes	0.994
Roger Federer	Hard	target_non_error_strokes	0.990
Rafael Nadal	Overall	target_non_error_strokes	0.989
Rafael Nadal	Clay	target_non_error_strokes	0.985
Rafael Nadal	Grass	target_non_error_strokes	0.991
Rafael Nadal	Hard	target_non_error_strokes	0.994
Novak Djokovic	Overall	target_non_error_strokes	0.994
Novak Djokovic	Clay	target_non_error_strokes	0.996
Novak Djokovic	Grass	target_non_error_strokes	0.996
Novak Djokovic	Hard	target_non_error_strokes	0.994
Roger Federer	Overall	target_returns	0.914
Roger Federer	Clay	target_returns	0.923
Roger Federer	Grass	target_returns	0.928
Roger Federer	Hard	target_returns	0.913
Rafael Nadal	Overall	target_returns	0.951
Rafael Nadal	Clay	target_returns	0.958
Rafael Nadal	Grass	target_returns	0.968
Rafael Nadal	Hard	target_returns	0.949
Novak Djokovic	Overall	target_returns	0.903
Novak Djokovic	Clay	target_returns	0.922
Novak Djokovic	Grass	target_returns	0.913
Novak Djokovic	Hard	target_returns	0.899

player	surface	distribution	ci95_upper
Roger Federer	Overall	source_direction_Total	0.863
Roger Federer	Clay	source_direction_Total	0.875
Roger Federer	Grass	source_direction_Total	0.873
Roger Federer	Hard	source_direction_Total	0.859
Rafael Nadal	Overall	source_direction_Total	0.866
Rafael Nadal	Clay	source_direction_Total	0.874
Rafael Nadal	Grass	source_direction_Total	0.872
Rafael Nadal	Hard	source_direction_Total	0.864
Novak Djokovic	Overall	source_direction_Total	0.834
Novak Djokovic	Clay	source_direction_Total	0.849
Novak Djokovic	Grass	source_direction_Total	0.831
Novak Djokovic	Hard	source_direction_Total	0.831
Roger Federer	Overall	source_direction_F	0.911
Roger Federer	Clay	source_direction_F	0.914
Roger Federer	Grass	source_direction_F	0.916
Roger Federer	Hard	source_direction_F	0.912
Rafael Nadal	Overall	source_direction_F	0.860
Rafael Nadal	Clay	source_direction_F	0.859
Rafael Nadal	Grass	source_direction_F	0.877
Rafael Nadal	Hard	source_direction_F	0.863
Novak Djokovic	Overall	source_direction_F	0.878
Novak Djokovic	Clay	source_direction_F	0.886
Novak Djokovic	Grass	source_direction_F	0.877
Novak Djokovic	Hard	source_direction_F	0.878
Roger Federer	Overall	source_direction_B	0.721
Roger Federer	Clay	source_direction_B	0.723
Roger Federer	Grass	source_direction_B	0.725
Roger Federer	Hard	source_direction_B	0.723
Rafael Nadal	Overall	source_direction_B	0.787
Rafael Nadal	Clay	source_direction_B	0.818
Rafael Nadal	Grass	source_direction_B	0.743
Rafael Nadal	Hard	source_direction_B	0.771
Novak Djokovic	Overall	source_direction_B	0.749
Novak Djokovic	Clay	source_direction_B	0.769
Novak Djokovic	Grass	source_direction_B	0.765
Novak Djokovic	Hard	source_direction_B	0.742
Roger Federer	Overall	source_direction_S	0.675
Roger Federer	Clay	source_direction_S	0.691
Roger Federer	Grass	source_direction_S	0.705
Roger Federer	Hard	source_direction_S	0.670

player	surface	distribution	ci95_upper
Rafael Nadal	Overall	source_direction_S	0.800
Rafael Nadal	Clay	source_direction_S	0.812
Rafael Nadal	Grass	source_direction_S	0.797
Rafael Nadal	Hard	source_direction_S	0.801
Novak Djokovic	Overall	source_direction_S	0.701
Novak Djokovic	Clay	source_direction_S	0.755
Novak Djokovic	Grass	source_direction_S	0.745
Novak Djokovic	Hard	source_direction_S	0.685

Complete category counts, percentages, and match-bootstrap intervals:

▼ Complete evidence table: 648 rows × 11 columns

player	surface	distribution	category	count	classified_denominator	share_pct	contributing_matches	valid_bootstraps	ci95_lower
Roger Federer	Overall	shot_type	groundstroke	158256	220280	71.843	722	2000	71.354
Roger Federer	Overall	shot_type	slice	42402	220280	19.249	722	2000	18.805
Roger Federer	Overall	shot_type	volley	9847	220280	4.470	722	2000	4.285
Roger Federer	Overall	shot_type	overhead	2687	220280	1.220	722	2000	1.159
Roger Federer	Overall	shot_type	drop shot	2135	220280	0.969	722	2000	0.906
Roger Federer	Overall	shot_type	lob	2226	220280	1.011	722	2000	0.951
Roger Federer	Overall	shot_type	half-volley	1652	220280	0.750	722	2000	0.685
Roger Federer	Overall	shot_type	swinging volley	979	220280	0.444	722	2000	0.411
Roger Federer	Overall	shot_type	other	64	220280	0.029	722	2000	0.021
Roger Federer	Overall	shot_type	unknown	32	220280	0.015	722	2000	0.009
Roger Federer	Clay	shot_type	groundstroke	35034	45793	76.505	130	2000	75.408
Roger Federer	Clay	shot_type	slice	7016	45793	15.321	130	2000	14.372
Roger Federer	Clay	shot_type	volley	1704	45793	3.721	130	2000	3.376
Roger Federer	Clay	shot_type	overhead	524	45793	1.144	130	2000	1.013
Roger Federer	Clay	shot_type	drop shot	696	45793	1.520	130	2000	1.342
Roger Federer	Clay	shot_type	lob	405	45793	0.884	130	2000	0.760
Roger Federer	Clay	shot_type	half-volley	221	45793	0.483	130	2000	0.400
Roger Federer	Clay	shot_type	swinging volley	177	45793	0.387	130	2000	0.313
Roger Federer	Clay	shot_type	other	4	45793	0.009	130	2000	0.002
Roger Federer	Clay	shot_type	unknown	12	45793	0.026	130	2000	0.007
Roger Federer	Grass	shot_type	groundstroke	23774	35126	67.682	119	2000	66.490
Roger Federer	Grass	shot_type	slice	7618	35126	21.688	119	2000	20.597
Roger Federer	Grass	shot_type	volley	1891	35126	5.383	119	2000	4.813
Roger Federer	Grass	shot_type	overhead	454	35126	1.292	119	2000	1.120
Roger Federer	Grass	shot_type	drop shot	309	35126	0.880	119	2000	0.751
Roger Federer	Grass	shot_type	lob	428	35126	1.218	119	2000	1.075
Roger Federer	Grass	shot_type	half-volley	435	35126	1.238	119	2000	0.987
Roger Federer	Grass	shot_type	swinging volley	199	35126	0.567	119	2000	0.476
Roger Federer	Grass	shot_type	other	14	35126	0.040	119	2000	0.020
Roger Federer	Grass	shot_type	unknown	4	35126	0.011	119	2000	0.003
Roger Federer	Hard	shot_type	groundstroke	99448	139361	71.360	473	2000	70.816
Roger Federer	Hard	shot_type	slice	27768	139361	19.925	473	2000	19.445
Roger Federer	Hard	shot_type	volley	6252	139361	4.486	473	2000	4.268
Roger Federer	Hard	shot_type	overhead	1709	139361	1.226	473	2000	1.150
Roger Federer	Hard	shot_type	drop shot	1130	139361	0.811	473	2000	0.739
Roger Federer	Hard	shot_type	lob	1393	139361	1.000	473	2000	0.925
Roger Federer	Hard	shot_type	half-volley	996	139361	0.715	473	2000	0.648

player	surface	distribution	category	count	classified_denominator	share_pct	contributing_matches	valid_bootstraps	ci95_lower
Roger Federer	Hard	shot_type	swinging volley	603	139361	0.433	473	2000	0.394
Roger Federer	Hard	shot_type	other	46	139361	0.033	473	2000	0.023
Roger Federer	Hard	shot_type	unknown	16	139361	0.011	473	2000	0.005
Rafael Nadal	Overall	shot_type	groundstroke	130511	151792	85.980	423	2000	85.401
Rafael Nadal	Overall	shot_type	slice	14443	151792	9.515	423	2000	8.998
Rafael Nadal	Overall	shot_type	volley	2245	151792	1.479	423	2000	1.381
Rafael Nadal	Overall	shot_type	overhead	1127	151792	0.742	423	2000	0.689
Rafael Nadal	Overall	shot_type	drop shot	1234	151792	0.813	423	2000	0.744
Rafael Nadal	Overall	shot_type	lob	1653	151792	1.089	423	2000	1.005
Rafael Nadal	Overall	shot_type	half-volley	381	151792	0.251	423	2000	0.212
Rafael Nadal	Overall	shot_type	swinging volley	144	151792	0.095	423	2000	0.077
Rafael Nadal	Overall	shot_type	other	43	151792	0.028	423	2000	0.020
Rafael Nadal	Overall	shot_type	unknown	11	151792	0.007	423	2000	0.002
Rafael Nadal	Clay	shot_type	groundstroke	56272	62935	89.413	166	2000	88.889
Rafael Nadal	Clay	shot_type	slice	3959	62935	6.291	166	2000	5.888
Rafael Nadal	Clay	shot_type	volley	702	62935	1.115	166	2000	0.999
Rafael Nadal	Clay	shot_type	overhead	413	62935	0.656	166	2000	0.580
Rafael Nadal	Clay	shot_type	drop shot	662	62935	1.052	166	2000	0.938
Rafael Nadal	Clay	shot_type	lob	720	62935	1.144	166	2000	1.005
Rafael Nadal	Clay	shot_type	half-volley	123	62935	0.195	166	2000	0.141
Rafael Nadal	Clay	shot_type	swinging volley	66	62935	0.105	166	2000	0.075
Rafael Nadal	Clay	shot_type	other	12	62935	0.019	166	2000	0.009
Rafael Nadal	Clay	shot_type	unknown	6	62935	0.010	166	2000	0.001
Rafael Nadal	Grass	shot_type	groundstroke	11108	14056	79.027	39	2000	76.971
Rafael Nadal	Grass	shot_type	slice	2109	14056	15.004	39	2000	13.197
Rafael Nadal	Grass	shot_type	volley	267	14056	1.900	39	2000	1.594
Rafael Nadal	Grass	shot_type	overhead	137	14056	0.975	39	2000	0.783
Rafael Nadal	Grass	shot_type	drop shot	158	14056	1.124	39	2000	0.854
Rafael Nadal	Grass	shot_type	lob	172	14056	1.224	39	2000	1.052
Rafael Nadal	Grass	shot_type	half-volley	75	14056	0.534	39	2000	0.339
Rafael Nadal	Grass	shot_type	swinging volley	17	14056	0.121	39	2000	0.058
Rafael Nadal	Grass	shot_type	other	10	14056	0.071	39	2000	0.021
Rafael Nadal	Grass	shot_type	unknown	3	14056	0.021	39	2000	0.000
Rafael Nadal	Hard	shot_type	groundstroke	63131	74801	84.399	218	2000	83.649
Rafael Nadal	Hard	shot_type	slice	8375	74801	11.196	218	2000	10.434
Rafael Nadal	Hard	shot_type	volley	1276	74801	1.706	218	2000	1.559
Rafael Nadal	Hard	shot_type	overhead	577	74801	0.771	218	2000	0.702
Rafael Nadal	Hard	shot_type	drop shot	414	74801	0.553	218	2000	0.484
Rafael Nadal	Hard	shot_type	lob	761	74801	1.017	218	2000	0.907

player	surface	distribution	category	count	classified_denominator	share_pct	contributing_matches	valid_bootstraps	ci95_lower
Rafael Nadal	Hard	shot_type	half-volley	183	74801	0.245	218	2000	0.193
Rafael Nadal	Hard	shot_type	swinging volley	61	74801	0.082	218	2000	0.058
Rafael Nadal	Hard	shot_type	other	21	74801	0.028	218	2000	0.017
Rafael Nadal	Hard	shot_type	unknown	2	74801	0.003	218	2000	0.000
Novak Djokovic	Overall	shot_type	groundstroke	172474	202209	85.295	552	2000	84.851
Novak Djokovic	Overall	shot_type	slice	17121	202209	8.467	552	2000	8.071
Novak Djokovic	Overall	shot_type	volley	4326	202209	2.139	552	2000	2.030
Novak Djokovic	Overall	shot_type	overhead	1632	202209	0.807	552	2000	0.749
Novak Djokovic	Overall	shot_type	drop shot	3127	202209	1.546	552	2000	1.454
Novak Djokovic	Overall	shot_type	lob	2518	202209	1.245	552	2000	1.163
Novak Djokovic	Overall	shot_type	half-volley	528	202209	0.261	552	2000	0.224
Novak Djokovic	Overall	shot_type	swinging volley	428	202209	0.212	552	2000	0.188
Novak Djokovic	Overall	shot_type	other	30	202209	0.015	552	2000	0.010
Novak Djokovic	Overall	shot_type	unknown	25	202209	0.012	552	2000	0.007
Novak Djokovic	Clay	shot_type	groundstroke	45665	52098	87.652	129	2000	86.947
Novak Djokovic	Clay	shot_type	slice	2801	52098	5.376	129	2000	4.991
Novak Djokovic	Clay	shot_type	volley	966	52098	1.854	129	2000	1.671
Novak Djokovic	Clay	shot_type	overhead	453	52098	0.870	129	2000	0.730
Novak Djokovic	Clay	shot_type	drop shot	1237	52098	2.374	129	2000	2.144
Novak Djokovic	Clay	shot_type	lob	671	52098	1.288	129	2000	1.113
Novak Djokovic	Clay	shot_type	half-volley	164	52098	0.315	129	2000	0.219
Novak Djokovic	Clay	shot_type	swinging volley	123	52098	0.236	129	2000	0.182
Novak Djokovic	Clay	shot_type	other	4	52098	0.008	129	2000	0.002
Novak Djokovic	Clay	shot_type	unknown	14	52098	0.027	129	2000	0.011
Novak Djokovic	Grass	shot_type	groundstroke	19019	23538	80.801	58	2000	79.116
Novak Djokovic	Grass	shot_type	slice	2905	23538	12.342	58	2000	11.001
Novak Djokovic	Grass	shot_type	volley	625	23538	2.655	58	2000	2.306
Novak Djokovic	Grass	shot_type	overhead	172	23538	0.731	58	2000	0.589
Novak Djokovic	Grass	shot_type	drop shot	338	23538	1.436	58	2000	1.210
Novak Djokovic	Grass	shot_type	lob	344	23538	1.461	58	2000	1.242
Novak Djokovic	Grass	shot_type	half-volley	74	23538	0.314	58	2000	0.207
Novak Djokovic	Grass	shot_type	swinging volley	50	23538	0.212	58	2000	0.147
Novak Djokovic	Grass	shot_type	other	9	23538	0.038	58	2000	0.015
Novak Djokovic	Grass	shot_type	unknown	2	23538	0.008	58	2000	0.000
Novak Djokovic	Hard	shot_type	groundstroke	107790	126573	85.160	365	2000	84.660
Novak Djokovic	Hard	shot_type	slice	11415	126573	9.019	365	2000	8.584
Novak Djokovic	Hard	shot_type	volley	2735	126573	2.161	365	2000	2.023
Novak Djokovic	Hard	shot_type	overhead	1007	126573	0.796	365	2000	0.729
Novak Djokovic	Hard	shot_type	drop shot	1552	126573	1.226	365	2000	1.134

player	surface	distribution	category	count	classified_denominator	share_pct	contributing_matches	valid_bootstraps	ci95_lower
Novak Djokovic	Hard	shot_type	lob	1503	126573	1.187	365	2000	1.094
Novak Djokovic	Hard	shot_type	half-volley	290	126573	0.229	365	2000	0.197
Novak Djokovic	Hard	shot_type	swinging volley	255	126573	0.201	365	2000	0.174
Novak Djokovic	Hard	shot_type	other	17	126573	0.013	365	2000	0.007
Novak Djokovic	Hard	shot_type	unknown	9	126573	0.007	365	2000	0.002
Roger Federer	Overall	target_all_attempts	1	61501	215848	28.493	719	2000	28.074
Roger Federer	Overall	target_all_attempts	2	68982	215848	31.959	719	2000	31.373
Roger Federer	Overall	target_all_attempts	3	85365	215848	39.549	719	2000	38.985
Roger Federer	Clay	target_all_attempts	1	13470	45132	29.846	130	2000	28.665
Roger Federer	Clay	target_all_attempts	2	14296	45132	31.676	130	2000	30.178
Roger Federer	Clay	target_all_attempts	3	17366	45132	38.478	130	2000	36.897
Roger Federer	Grass	target_all_attempts	1	9569	33574	28.501	117	2000	27.458
Roger Federer	Grass	target_all_attempts	2	10964	33574	32.656	117	2000	31.372
Roger Federer	Grass	target_all_attempts	3	13041	33574	38.843	117	2000	37.662
Roger Federer	Hard	target_all_attempts	1	38462	137142	28.045	472	2000	27.570
Roger Federer	Hard	target_all_attempts	2	43722	137142	31.881	472	2000	31.128
Roger Federer	Hard	target_all_attempts	3	54958	137142	40.074	472	2000	39.433
Rafael Nadal	Overall	target_all_attempts	1	46958	145635	32.244	421	2000	31.697
Rafael Nadal	Overall	target_all_attempts	2	39771	145635	27.309	421	2000	26.495
Rafael Nadal	Overall	target_all_attempts	3	58906	145635	40.448	421	2000	39.765
Rafael Nadal	Clay	target_all_attempts	1	18973	59868	31.691	166	2000	30.705
Rafael Nadal	Clay	target_all_attempts	2	15645	59868	26.132	166	2000	24.728
Rafael Nadal	Clay	target_all_attempts	3	25250	59868	42.176	166	2000	41.148
Rafael Nadal	Grass	target_all_attempts	1	4622	13581	34.033	39	2000	32.290
Rafael Nadal	Grass	target_all_attempts	2	3425	13581	25.219	39	2000	23.366
Rafael Nadal	Grass	target_all_attempts	3	5534	13581	40.748	39	2000	38.297

player	surface	distribution	category	count	classified_denominator	share_pct	contributing_matches	valid_bootstraps	ci95_lower
Rafael Nadal	Hard	target_all_attempts	1	23363	72186	32.365	216	2000	31.718
Rafael Nadal	Hard	target_all_attempts	2	20701	72186	28.677	216	2000	27.733
Rafael Nadal	Hard	target_all_attempts	3	28122	72186	38.958	216	2000	38.029
Novak Djokovic	Overall	target_all_attempts	1	60357	199028	30.326	549	2000	29.830
Novak Djokovic	Overall	target_all_attempts	2	60977	199028	30.637	549	2000	29.982
Novak Djokovic	Overall	target_all_attempts	3	77694	199028	39.037	549	2000	38.433
Novak Djokovic	Clay	target_all_attempts	1	16265	51435	31.622	129	2000	30.475
Novak Djokovic	Clay	target_all_attempts	2	15006	51435	29.175	129	2000	27.688
Novak Djokovic	Clay	target_all_attempts	3	20164	51435	39.203	129	2000	37.764
Novak Djokovic	Grass	target_all_attempts	1	7105	23272	30.530	58	2000	29.157
Novak Djokovic	Grass	target_all_attempts	2	7072	23272	30.388	58	2000	28.379
Novak Djokovic	Grass	target_all_attempts	3	9095	23272	39.081	58	2000	37.233
Novak Djokovic	Hard	target_all_attempts	1	36987	124321	29.751	362	2000	29.161
Novak Djokovic	Hard	target_all_attempts	2	38899	124321	31.289	362	2000	30.533
Novak Djokovic	Hard	target_all_attempts	3	48435	124321	38.960	362	2000	38.306
Roger Federer	Overall	target_non_error_strokes	1	49636	179579	27.640	719	2000	27.189
Roger Federer	Overall	target_non_error_strokes	2	58567	179579	32.614	719	2000	32.005
Roger Federer	Overall	target_non_error_strokes	3	71376	179579	39.746	719	2000	39.128
Roger Federer	Clay	target_non_error_strokes	1	11099	37959	29.239	130	2000	27.934
Roger Federer	Clay	target_non_error_strokes	2	12279	37959	32.348	130	2000	30.752
Roger Federer	Clay	target_non_error_strokes	3	14581	37959	38.412	130	2000	36.643
Roger Federer	Grass	target_non_error_strokes	1	7611	27464	27.713	117	2000	26.701
Roger Federer	Grass	target_non_error_strokes	2	9175	27464	33.407	117	2000	32.113
Roger Federer	Grass	target_non_error_strokes	3	10678	27464	38.880	117	2000	37.628

player	surface	distribution	category	count	classified_denominator	share_pct	contributing_matches	valid_bootstraps	ci95_lower
Roger Federer	Hard	target_non_error_strokes	1	30926	114156	27.091	472	2000	26.592
Roger Federer	Hard	target_non_error_strokes	2	37113	114156	32.511	472	2000	31.751
Roger Federer	Hard	target_non_error_strokes	3	46117	114156	40.398	472	2000	39.721
Rafael Nadal	Overall	target_non_error_strokes	1	39642	127703	31.042	421	2000	30.475
Rafael Nadal	Overall	target_non_error_strokes	2	35504	127703	27.802	421	2000	26.954
Rafael Nadal	Overall	target_non_error_strokes	3	52557	127703	41.156	421	2000	40.430
Rafael Nadal	Clay	target_non_error_strokes	1	16351	53459	30.586	166	2000	29.574
Rafael Nadal	Clay	target_non_error_strokes	2	14227	53459	26.613	166	2000	25.139
Rafael Nadal	Clay	target_non_error_strokes	3	22881	53459	42.801	166	2000	41.713
Rafael Nadal	Grass	target_non_error_strokes	1	3748	11475	32.662	39	2000	30.686
Rafael Nadal	Grass	target_non_error_strokes	2	2955	11475	25.752	39	2000	23.750
Rafael Nadal	Grass	target_non_error_strokes	3	4772	11475	41.586	39	2000	38.947
Rafael Nadal	Hard	target_non_error_strokes	1	19543	62769	31.135	216	2000	30.483
Rafael Nadal	Hard	target_non_error_strokes	2	18322	62769	29.190	216	2000	28.202
Rafael Nadal	Hard	target_non_error_strokes	3	24904	62769	39.676	216	2000	38.653
Novak Djokovic	Overall	target_non_error_strokes	1	51648	173861	29.706	549	2000	29.164
Novak Djokovic	Overall	target_non_error_strokes	2	53782	173861	30.934	549	2000	30.245
Novak Djokovic	Overall	target_non_error_strokes	3	68431	173861	39.360	549	2000	38.701
Novak Djokovic	Clay	target_non_error_strokes	1	14120	45299	31.171	129	2000	29.949
Novak Djokovic	Clay	target_non_error_strokes	2	13343	45299	29.455	129	2000	27.912
Novak Djokovic	Clay	target_non_error_strokes	3	17836	45299	39.374	129	2000	37.806
Novak Djokovic	Grass	target_non_error_strokes	1	5951	19954	29.824	58	2000	28.310
Novak Djokovic	Grass	target_non_error_strokes	2	6111	19954	30.625	58	2000	28.498
Novak Djokovic	Grass	target_non_error_strokes	3	7892	19954	39.551	58	2000	37.505

player	surface	distribution	category	count	classified_denominator	share_pct	contributing_matches	valid_bootstraps	ci95_lower
Novak Djokovic	Hard	target_non_error_strokes	1	31577	108608	29.074	362	2000	28.451
Novak Djokovic	Hard	target_non_error_strokes	2	34328	108608	31.607	362	2000	30.826
Novak Djokovic	Hard	target_non_error_strokes	3	42703	108608	39.318	362	2000	38.634
Roger Federer	Overall	target_returns	1	8531	55456	15.383	719	2000	14.847
Roger Federer	Overall	target_returns	2	28137	55456	50.738	719	2000	50.002
Roger Federer	Overall	target_returns	3	18788	55456	33.879	719	2000	33.067
Roger Federer	Clay	target_returns	1	1655	10615	15.591	130	2000	14.052
Roger Federer	Clay	target_returns	2	5384	10615	50.721	130	2000	48.832
Roger Federer	Clay	target_returns	3	3576	10615	33.688	130	2000	31.281
Roger Federer	Grass	target_returns	1	1644	9913	16.584	117	2000	15.190
Roger Federer	Grass	target_returns	2	5062	9913	51.064	117	2000	49.415
Roger Federer	Grass	target_returns	3	3207	9913	32.351	117	2000	30.504
Roger Federer	Hard	target_returns	1	5232	34928	14.979	472	2000	14.372
Roger Federer	Hard	target_returns	2	17691	34928	50.650	472	2000	49.736
Roger Federer	Hard	target_returns	3	12005	34928	34.371	472	2000	33.443
Rafael Nadal	Overall	target_returns	1	5891	27493	21.427	415	2000	20.606
Rafael Nadal	Overall	target_returns	2	13605	27493	49.485	415	2000	48.337
Rafael Nadal	Overall	target_returns	3	7997	27493	29.087	415	2000	28.162
Rafael Nadal	Clay	target_returns	1	2056	10051	20.456	163	2000	19.190
Rafael Nadal	Clay	target_returns	2	4852	10051	48.274	163	2000	46.227
Rafael Nadal	Clay	target_returns	3	3143	10051	31.271	163	2000	29.714
Rafael Nadal	Grass	target_returns	1	709	3178	22.310	39	2000	19.705
Rafael Nadal	Grass	target_returns	2	1525	3178	47.986	39	2000	45.288
Rafael Nadal	Grass	target_returns	3	944	3178	29.704	39	2000	26.606
Rafael Nadal	Hard	target_returns	1	3126	14264	21.915	213	2000	20.780
Rafael Nadal	Hard	target_returns	2	7228	14264	50.673	213	2000	49.218
Rafael Nadal	Hard	target_returns	3	3910	14264	27.412	213	2000	26.244
Novak Djokovic	Overall	target_returns	1	6395	40270	15.880	548	2000	15.323
Novak Djokovic	Overall	target_returns	2	21825	40270	54.197	548	2000	53.345
Novak Djokovic	Overall	target_returns	3	12050	40270	29.923	548	2000	29.124
Novak Djokovic	Clay	target_returns	1	1597	9622	16.597	129	2000	15.357
Novak Djokovic	Clay	target_returns	2	5078	9622	52.775	129	2000	50.651
Novak Djokovic	Clay	target_returns	3	2947	9622	30.628	129	2000	28.468
Novak Djokovic	Grass	target_returns	1	845	5392	15.671	58	2000	14.208
Novak Djokovic	Grass	target_returns	2	2898	5392	53.746	58	2000	51.353
Novak Djokovic	Grass	target_returns	3	1649	5392	30.582	58	2000	28.172
Novak Djokovic	Hard	target_returns	1	3953	25256	15.652	361	2000	14.992

player	surface	distribution	category	count	classified_denominator	share_pct	contributing_matches	valid_bootstraps	ci95_lower
Novak Djokovic	Hard	target_returns	2	13849	25256	54.834	361	2000	53.890
Novak Djokovic	Hard	target_returns	3	7454	25256	29.514	361	2000	28.616
Roger Federer	Overall	active_serve_target	deuce_wide	12989	59819	21.714	722	2000	21.319
Roger Federer	Overall	active_serve_target	deuce_body	5346	59819	8.937	722	2000	8.563
Roger Federer	Overall	active_serve_target	deuce_t	13037	59819	21.794	722	2000	21.361
Roger Federer	Overall	active_serve_target	ad_wide	14875	59819	24.867	722	2000	24.490
Roger Federer	Overall	active_serve_target	ad_body	3877	59819	6.481	722	2000	6.182
Roger Federer	Overall	active_serve_target	ad_t	9695	59819	16.207	722	2000	15.851
Roger Federer	Clay	active_serve_target	deuce_wide	2278	11226	20.292	130	2000	19.125
Roger Federer	Clay	active_serve_target	deuce_body	1024	11226	9.122	130	2000	8.145
Roger Federer	Clay	active_serve_target	deuce_t	2553	11226	22.742	130	2000	21.401
Roger Federer	Clay	active_serve_target	ad_wide	2976	11226	26.510	130	2000	25.331
Roger Federer	Clay	active_serve_target	ad_body	739	11226	6.583	130	2000	5.670
Roger Federer	Clay	active_serve_target	ad_t	1656	11226	14.751	130	2000	13.879
Roger Federer	Grass	active_serve_target	deuce_wide	2525	11109	22.729	119	2000	21.891
Roger Federer	Grass	active_serve_target	deuce_body	911	11109	8.201	119	2000	7.426
Roger Federer	Grass	active_serve_target	deuce_t	2402	11109	21.622	119	2000	20.826
Roger Federer	Grass	active_serve_target	ad_wide	2668	11109	24.017	119	2000	23.198
Roger Federer	Grass	active_serve_target	ad_body	652	11109	5.869	119	2000	5.387
Roger Federer	Grass	active_serve_target	ad_t	1951	11109	17.562	119	2000	16.875
Roger Federer	Hard	active_serve_target	deuce_wide	8186	37484	21.839	473	2000	21.362
Roger Federer	Hard	active_serve_target	deuce_body	3411	37484	9.100	473	2000	8.626
Roger Federer	Hard	active_serve_target	deuce_t	8082	37484	21.561	473	2000	21.011
Roger Federer	Hard	active_serve_target	ad_wide	9231	37484	24.627	473	2000	24.213

player	surface	distribution	category	count	classified_denominator	share_pct	contributing_matches	valid_bootstraps	ci95_lower
Roger Federer	Hard	active_serve_target	ad_body	2486	37484	6.632	473	2000	6.253
Roger Federer	Hard	active_serve_target	ad_t	6088	37484	16.242	473	2000	15.835
Rafael Nadal	Overall	active_serve_target	deuce_wide	4194	34542	12.142	423	2000	11.456
Rafael Nadal	Overall	active_serve_target	deuce_body	4933	34542	14.281	423	2000	13.558
Rafael Nadal	Overall	active_serve_target	deuce_t	8969	34542	25.965	423	2000	25.071
Rafael Nadal	Overall	active_serve_target	ad_wide	8781	34542	25.421	423	2000	24.476
Rafael Nadal	Overall	active_serve_target	ad_body	4364	34542	12.634	423	2000	11.973
Rafael Nadal	Overall	active_serve_target	ad_t	3301	34542	9.556	423	2000	8.964
Rafael Nadal	Clay	active_serve_target	deuce_wide	1254	12464	10.061	166	2000	8.972
Rafael Nadal	Clay	active_serve_target	deuce_body	2015	12464	16.167	166	2000	14.842
Rafael Nadal	Clay	active_serve_target	deuce_t	3252	12464	26.091	166	2000	24.405
Rafael Nadal	Clay	active_serve_target	ad_wide	3639	12464	29.196	166	2000	27.635
Rafael Nadal	Clay	active_serve_target	ad_body	1353	12464	10.855	166	2000	9.857
Rafael Nadal	Clay	active_serve_target	ad_t	951	12464	7.630	166	2000	6.712
Rafael Nadal	Grass	active_serve_target	deuce_wide	526	4121	12.764	39	2000	10.499
Rafael Nadal	Grass	active_serve_target	deuce_body	528	4121	12.812	39	2000	10.919
Rafael Nadal	Grass	active_serve_target	deuce_t	1106	4121	26.838	39	2000	23.802
Rafael Nadal	Grass	active_serve_target	ad_wide	872	4121	21.160	39	2000	18.618
Rafael Nadal	Grass	active_serve_target	ad_body	527	4121	12.788	39	2000	11.066
Rafael Nadal	Grass	active_serve_target	ad_t	562	4121	13.637	39	2000	11.892
Rafael Nadal	Hard	active_serve_target	deuce_wide	2414	17957	13.443	218	2000	12.523
Rafael Nadal	Hard	active_serve_target	deuce_body	2390	17957	13.310	218	2000	12.419
Rafael Nadal	Hard	active_serve_target	deuce_t	4611	17957	25.678	218	2000	24.476
Rafael Nadal	Hard	active_serve_target	ad_wide	4270	17957	23.779	218	2000	22.647

player	surface	distribution	category	count	classified_denominator	share_pct	contributing_matches	valid_bootstraps	ci95_lower
Rafael Nadal	Hard	active_serve_target	ad_body	2484	17957	13.833	218	2000	12.907
Rafael Nadal	Hard	active_serve_target	ad_t	1788	17957	9.957	218	2000	9.185
Novak Djokovic	Overall	active_serve_target	deuce_wide	11034	45681	24.154	552	2000	23.709
Novak Djokovic	Overall	active_serve_target	deuce_body	4496	45681	9.842	552	2000	9.362
Novak Djokovic	Overall	active_serve_target	deuce_t	8429	45681	18.452	552	2000	17.931
Novak Djokovic	Overall	active_serve_target	ad_wide	9207	45681	20.155	552	2000	19.679
Novak Djokovic	Overall	active_serve_target	ad_body	3786	45681	8.288	552	2000	7.871
Novak Djokovic	Overall	active_serve_target	ad_t	8729	45681	19.109	552	2000	18.633
Novak Djokovic	Clay	active_serve_target	deuce_wide	2673	10874	24.582	129	2000	23.550
Novak Djokovic	Clay	active_serve_target	deuce_body	972	10874	8.939	129	2000	7.993
Novak Djokovic	Clay	active_serve_target	deuce_t	2037	10874	18.733	129	2000	17.559
Novak Djokovic	Clay	active_serve_target	ad_wide	2384	10874	21.924	129	2000	20.787
Novak Djokovic	Clay	active_serve_target	ad_body	730	10874	6.713	129	2000	5.948
Novak Djokovic	Clay	active_serve_target	ad_t	2078	10874	19.110	129	2000	18.100
Novak Djokovic	Grass	active_serve_target	deuce_wide	1411	6057	23.295	58	2000	22.181
Novak Djokovic	Grass	active_serve_target	deuce_body	555	6057	9.163	58	2000	8.077
Novak Djokovic	Grass	active_serve_target	deuce_t	1216	6057	20.076	58	2000	18.683
Novak Djokovic	Grass	active_serve_target	ad_wide	1151	6057	19.003	58	2000	17.859
Novak Djokovic	Grass	active_serve_target	ad_body	495	6057	8.172	58	2000	7.162
Novak Djokovic	Grass	active_serve_target	ad_t	1229	6057	20.291	58	2000	19.178
Novak Djokovic	Hard	active_serve_target	deuce_wide	6950	28750	24.174	365	2000	23.601
Novak Djokovic	Hard	active_serve_target	deuce_body	2969	28750	10.327	365	2000	9.699
Novak Djokovic	Hard	active_serve_target	deuce_t	5176	28750	18.003	365	2000	17.415
Novak Djokovic	Hard	active_serve_target	ad_wide	5672	28750	19.729	365	2000	19.189

player	surface	distribution	category	count	classified_denominator	share_pct	contributing_matches	valid_bootstraps	ci95_lower
Novak Djokovic	Hard	active_serve_target	ad_body	2561	28750	8.908	365	2000	8.348
Novak Djokovic	Hard	active_serve_target	ad_t	5422	28750	18.859	365	2000	18.279
Roger Federer	Overall	return_depth_target	7|1	1517	42882	3.538	698	2000	3.328
Roger Federer	Overall	return_depth_target	7|2	5273	42882	12.297	698	2000	11.818
Roger Federer	Overall	return_depth_target	7|3	3825	42882	8.920	698	2000	8.547
Roger Federer	Overall	return_depth_target	8|1	3162	42882	7.374	698	2000	7.037
Roger Federer	Overall	return_depth_target	8|2	11919	42882	27.795	698	2000	27.057
Roger Federer	Overall	return_depth_target	8|3	7859	42882	18.327	698	2000	17.644
Roger Federer	Overall	return_depth_target	9|1	1338	42882	3.120	698	2000	2.873
Roger Federer	Overall	return_depth_target	9|2	5091	42882	11.872	698	2000	11.343
Roger Federer	Overall	return_depth_target	9|3	2898	42882	6.758	698	2000	6.326
Roger Federer	Clay	return_depth_target	7|1	278	8427	3.299	125	2000	2.791
Roger Federer	Clay	return_depth_target	7|2	1013	8427	12.021	125	2000	10.942
Roger Federer	Clay	return_depth_target	7|3	739	8427	8.769	125	2000	7.872
Roger Federer	Clay	return_depth_target	8|1	685	8427	8.129	125	2000	7.161
Roger Federer	Clay	return_depth_target	8|2	2516	8427	29.856	125	2000	28.111
Roger Federer	Clay	return_depth_target	8|3	1648	8427	19.556	125	2000	17.641
Roger Federer	Clay	return_depth_target	9|1	256	8427	3.038	125	2000	2.351
Roger Federer	Clay	return_depth_target	9|2	858	8427	10.182	125	2000	9.063
Roger Federer	Clay	return_depth_target	9|3	434	8427	5.150	125	2000	4.302
Roger Federer	Grass	return_depth_target	7|1	315	7221	4.362	113	2000	3.854
Roger Federer	Grass	return_depth_target	7|2	1070	7221	14.818	113	2000	13.441
Roger Federer	Grass	return_depth_target	7|3	687	7221	9.514	113	2000	8.506
Roger Federer	Grass	return_depth_target	8|1	523	7221	7.243	113	2000	6.415

player	surface	distribution	category	count	classified_denominator	share_pct	contributing_matches	valid_bootstraps	ci95_lower
Roger Federer	Grass	return_depth_target	8|2	2007	7221	27.794	113	2000	26.227
Roger Federer	Grass	return_depth_target	8|3	1209	7221	16.743	113	2000	15.427
Roger Federer	Grass	return_depth_target	9|1	225	7221	3.116	113	2000	2.543
Roger Federer	Grass	return_depth_target	9|2	763	7221	10.566	113	2000	9.296
Roger Federer	Grass	return_depth_target	9|3	422	7221	5.844	113	2000	4.862
Roger Federer	Hard	return_depth_target	7|1	924	27234	3.393	460	2000	3.128
Roger Federer	Hard	return_depth_target	7|2	3190	27234	11.713	460	2000	11.188
Roger Federer	Hard	return_depth_target	7|3	2399	27234	8.809	460	2000	8.368
Roger Federer	Hard	return_depth_target	8|1	1954	27234	7.175	460	2000	6.767
Roger Federer	Hard	return_depth_target	8|2	7396	27234	27.157	460	2000	26.187
Roger Federer	Hard	return_depth_target	8|3	5002	27234	18.367	460	2000	17.634
Roger Federer	Hard	return_depth_target	9|1	857	27234	3.147	460	2000	2.854
Roger Federer	Hard	return_depth_target	9|2	3470	27234	12.741	460	2000	12.079
Roger Federer	Hard	return_depth_target	9|3	2042	27234	7.498	460	2000	6.968
Rafael Nadal	Overall	return_depth_target	7|1	1470	22262	6.603	368	2000	6.183
Rafael Nadal	Overall	return_depth_target	7|2	3309	22262	14.864	368	2000	14.103
Rafael Nadal	Overall	return_depth_target	7|3	2218	22262	9.963	368	2000	9.468
Rafael Nadal	Overall	return_depth_target	8|1	2098	22262	9.424	368	2000	8.884
Rafael Nadal	Overall	return_depth_target	8|2	5174	22262	23.241	368	2000	22.385
Rafael Nadal	Overall	return_depth_target	8|3	3104	22262	13.943	368	2000	13.278
Rafael Nadal	Overall	return_depth_target	9|1	959	22262	4.308	368	2000	3.918
Rafael Nadal	Overall	return_depth_target	9|2	2693	22262	12.097	368	2000	11.486
Rafael Nadal	Overall	return_depth_target	9|3	1237	22262	5.557	368	2000	5.121
Rafael Nadal	Clay	return_depth_target	7|1	535	8508	6.288	135	2000	5.696

player	surface	distribution	category	count	classified_denominator	share_pct	contributing_matches	valid_bootstraps	ci95_lower
Rafael Nadal	Clay	return_depth_target	7|2	1170	8508	13.752	135	2000	12.440
Rafael Nadal	Clay	return_depth_target	7|3	960	8508	11.283	135	2000	10.475
Rafael Nadal	Clay	return_depth_target	8|1	779	8508	9.156	135	2000	8.310
Rafael Nadal	Clay	return_depth_target	8|2	1989	8508	23.378	135	2000	21.892
Rafael Nadal	Clay	return_depth_target	8|3	1296	8508	15.233	135	2000	14.135
Rafael Nadal	Clay	return_depth_target	9|1	351	8508	4.126	135	2000	3.524
Rafael Nadal	Clay	return_depth_target	9|2	971	8508	11.413	135	2000	10.305
Rafael Nadal	Clay	return_depth_target	9|3	457	8508	5.371	135	2000	4.699
Rafael Nadal	Grass	return_depth_target	7|1	188	2431	7.733	36	2000	6.143
Rafael Nadal	Grass	return_depth_target	7|2	336	2431	13.821	36	2000	12.134
Rafael Nadal	Grass	return_depth_target	7|3	241	2431	9.914	36	2000	8.391
Rafael Nadal	Grass	return_depth_target	8|1	240	2431	9.872	36	2000	8.330
Rafael Nadal	Grass	return_depth_target	8|2	557	2431	22.912	36	2000	20.788
Rafael Nadal	Grass	return_depth_target	8|3	337	2431	13.863	36	2000	11.929
Rafael Nadal	Grass	return_depth_target	9|1	83	2431	3.414	36	2000	2.543
Rafael Nadal	Grass	return_depth_target	9|2	293	2431	12.053	36	2000	10.064
Rafael Nadal	Grass	return_depth_target	9|3	156	2431	6.417	36	2000	4.886
Rafael Nadal	Hard	return_depth_target	7|1	747	11323	6.597	197	2000	6.070
Rafael Nadal	Hard	return_depth_target	7|2	1803	11323	15.923	197	2000	14.906
Rafael Nadal	Hard	return_depth_target	7|3	1017	11323	8.982	197	2000	8.308
Rafael Nadal	Hard	return_depth_target	8|1	1079	11323	9.529	197	2000	8.802
Rafael Nadal	Hard	return_depth_target	8|2	2628	11323	23.209	197	2000	21.977
Rafael Nadal	Hard	return_depth_target	8|3	1471	11323	12.991	197	2000	12.008
Rafael Nadal	Hard	return_depth_target	9|1	525	11323	4.637	197	2000	4.043

player	surface	distribution	category	count	classified_denominator	share_pct	contributing_matches	valid_bootstraps	ci95_lower
Rafael Nadal	Hard	return_depth_target	9|2	1429	11323	12.620	197	2000	11.646
Rafael Nadal	Hard	return_depth_target	9|3	624	11323	5.511	197	2000	4.932
Novak Djokovic	Overall	return_depth_target	7|1	958	31898	3.003	524	2000	2.795
Novak Djokovic	Overall	return_depth_target	7|2	3197	31898	10.023	524	2000	9.571
Novak Djokovic	Overall	return_depth_target	7|3	2397	31898	7.515	524	2000	7.167
Novak Djokovic	Overall	return_depth_target	8|1	2366	31898	7.417	524	2000	7.027
Novak Djokovic	Overall	return_depth_target	8|2	8670	31898	27.180	524	2000	26.396
Novak Djokovic	Overall	return_depth_target	8|3	4940	31898	15.487	524	2000	14.844
Novak Djokovic	Overall	return_depth_target	9|1	1414	31898	4.433	524	2000	4.125
Novak Djokovic	Overall	return_depth_target	9|2	5607	31898	17.578	524	2000	16.836
Novak Djokovic	Overall	return_depth_target	9|3	2349	31898	7.364	524	2000	6.889
Novak Djokovic	Clay	return_depth_target	7|1	205	7756	2.643	120	2000	2.256
Novak Djokovic	Clay	return_depth_target	7|2	696	7756	8.974	120	2000	8.079
Novak Djokovic	Clay	return_depth_target	7|3	522	7756	6.730	120	2000	5.941
Novak Djokovic	Clay	return_depth_target	8|1	646	7756	8.329	120	2000	7.503
Novak Djokovic	Clay	return_depth_target	8|2	2119	7756	27.321	120	2000	25.693
Novak Djokovic	Clay	return_depth_target	8|3	1278	7756	16.478	120	2000	15.092
Novak Djokovic	Clay	return_depth_target	9|1	387	7756	4.990	120	2000	4.369
Novak Djokovic	Clay	return_depth_target	9|2	1257	7756	16.207	120	2000	14.817
Novak Djokovic	Clay	return_depth_target	9|3	646	7756	8.329	120	2000	7.193
Novak Djokovic	Grass	return_depth_target	7|1	142	4062	3.496	54	2000	2.923
Novak Djokovic	Grass	return_depth_target	7|2	417	4062	10.266	54	2000	9.265
Novak Djokovic	Grass	return_depth_target	7|3	334	4062	8.223	54	2000	7.097
Novak Djokovic	Grass	return_depth_target	8|1	286	4062	7.041	54	2000	5.979

player	surface	distribution	category	count	classified_denominator	share_pct	contributing_matches	valid_bootstraps	ci95_lower
Novak Djokovic	Grass	return_depth_target	8|2	1130	4062	27.819	54	2000	25.775
Novak Djokovic	Grass	return_depth_target	8|3	656	4062	16.150	54	2000	14.150
Novak Djokovic	Grass	return_depth_target	9|1	149	4062	3.668	54	2000	2.916
Novak Djokovic	Grass	return_depth_target	9|2	686	4062	16.888	54	2000	14.533
Novak Djokovic	Grass	return_depth_target	9|3	262	4062	6.450	54	2000	5.271
Novak Djokovic	Hard	return_depth_target	7|1	611	20080	3.043	350	2000	2.783
Novak Djokovic	Hard	return_depth_target	7|2	2084	20080	10.378	350	2000	9.817
Novak Djokovic	Hard	return_depth_target	7|3	1541	20080	7.674	350	2000	7.252
Novak Djokovic	Hard	return_depth_target	8|1	1434	20080	7.141	350	2000	6.675
Novak Djokovic	Hard	return_depth_target	8|2	5421	20080	26.997	350	2000	26.009
Novak Djokovic	Hard	return_depth_target	8|3	3006	20080	14.970	350	2000	14.260
Novak Djokovic	Hard	return_depth_target	9|1	878	20080	4.373	350	2000	4.016
Novak Djokovic	Hard	return_depth_target	9|2	3664	20080	18.247	350	2000	17.333
Novak Djokovic	Hard	return_depth_target	9|3	1441	20080	7.176	350	2000	6.641
Roger Federer	Overall	source_direction_Total	crosscourt	55964	132560	42.218	719	2000	41.746
Roger Federer	Overall	source_direction_Total	down_middle	34016	132560	25.661	719	2000	25.019
Roger Federer	Overall	source_direction_Total	down_the_line	19099	132560	14.408	719	2000	14.041
Roger Federer	Overall	source_direction_Total	inside_out	19106	132560	14.413	719	2000	14.102
Roger Federer	Overall	source_direction_Total	inside_in	4375	132560	3.300	719	2000	3.146
Roger Federer	Clay	source_direction_Total	crosscourt	12090	29093	41.556	130	2000	40.559
Roger Federer	Clay	source_direction_Total	down_middle	7567	29093	26.010	130	2000	24.322
Roger Federer	Clay	source_direction_Total	down_the_line	4043	29093	13.897	130	2000	13.079
Roger Federer	Clay	source_direction_Total	inside_out	4271	29093	14.681	130	2000	14.026
Roger Federer	Clay	source_direction_Total	inside_in	1122	29093	3.857	130	2000	3.449

player	surface	distribution	category	count	classified_denominator	share_pct	contributing_matches	valid_bootstraps	ci95_lower
Roger Federer	Grass	source_direction_Total	crosscourt	7680	18665	41.147	117	2000	40.251
Roger Federer	Grass	source_direction_Total	down_middle	4725	18665	25.315	117	2000	23.950
Roger Federer	Grass	source_direction_Total	down_the_line	2975	18665	15.939	117	2000	15.120
Roger Federer	Grass	source_direction_Total	inside_out	2708	18665	14.508	117	2000	13.773
Roger Federer	Grass	source_direction_Total	inside_in	577	18665	3.091	117	2000	2.754
Roger Federer	Hard	source_direction_Total	crosscourt	36194	84802	42.681	472	2000	42.073
Roger Federer	Hard	source_direction_Total	down_middle	21724	84802	25.617	472	2000	24.842
Roger Federer	Hard	source_direction_Total	down_the_line	12081	84802	14.246	472	2000	13.784
Roger Federer	Hard	source_direction_Total	inside_out	12127	84802	14.300	472	2000	13.930
Roger Federer	Hard	source_direction_Total	inside_in	2676	84802	3.156	472	2000	2.983
Rafael Nadal	Overall	source_direction_Total	crosscourt	47009	105412	44.595	422	2000	43.750
Rafael Nadal	Overall	source_direction_Total	down_middle	23075	105412	21.890	422	2000	21.043
Rafael Nadal	Overall	source_direction_Total	down_the_line	16605	105412	15.752	422	2000	15.249
Rafael Nadal	Overall	source_direction_Total	inside_out	14796	105412	14.036	422	2000	13.519
Rafael Nadal	Overall	source_direction_Total	inside_in	3927	105412	3.725	422	2000	3.457
Rafael Nadal	Clay	source_direction_Total	crosscourt	20078	44589	45.029	165	2000	43.576
Rafael Nadal	Clay	source_direction_Total	down_middle	9503	44589	21.312	165	2000	19.874
Rafael Nadal	Clay	source_direction_Total	down_the_line	6913	44589	15.504	165	2000	14.569
Rafael Nadal	Clay	source_direction_Total	inside_out	6240	44589	13.994	165	2000	13.002
Rafael Nadal	Clay	source_direction_Total	inside_in	1855	44589	4.160	165	2000	3.685
Rafael Nadal	Grass	source_direction_Total	crosscourt	4151	9114	45.545	40	2000	43.654
Rafael Nadal	Grass	source_direction_Total	down_middle	1667	9114	18.291	40	2000	16.444
Rafael Nadal	Grass	source_direction_Total	down_the_line	1630	9114	17.885	40	2000	16.281
Rafael Nadal	Grass	source_direction_Total	inside_out	1312	9114	14.395	40	2000	13.271

player	surface	distribution	category	count	classified_denominator	share_pct	contributing_matches	valid_bootstraps	ci95_lower
Rafael Nadal	Grass	source_direction_Total	inside_in	354	9114	3.884	40	2000	3.357
Rafael Nadal	Hard	source_direction_Total	crosscourt	22780	51709	44.054	217	2000	43.036
Rafael Nadal	Hard	source_direction_Total	down_middle	11905	51709	23.023	217	2000	22.021
Rafael Nadal	Hard	source_direction_Total	down_the_line	8062	51709	15.591	217	2000	15.014
Rafael Nadal	Hard	source_direction_Total	inside_out	7244	51709	14.009	217	2000	13.471
Rafael Nadal	Hard	source_direction_Total	inside_in	1718	51709	3.322	217	2000	2.998
Novak Djokovic	Overall	source_direction_Total	crosscourt	62058	138901	44.678	549	2000	44.089
Novak Djokovic	Overall	source_direction_Total	down_middle	33771	138901	24.313	549	2000	23.619
Novak Djokovic	Overall	source_direction_Total	down_the_line	20009	138901	14.405	549	2000	13.962
Novak Djokovic	Overall	source_direction_Total	inside_out	20664	138901	14.877	549	2000	14.528
Novak Djokovic	Overall	source_direction_Total	inside_in	2399	138901	1.727	549	2000	1.607
Novak Djokovic	Clay	source_direction_Total	crosscourt	16211	36480	44.438	129	2000	42.869
Novak Djokovic	Clay	source_direction_Total	down_middle	8618	36480	23.624	129	2000	22.119
Novak Djokovic	Clay	source_direction_Total	down_the_line	5202	36480	14.260	129	2000	13.368
Novak Djokovic	Clay	source_direction_Total	inside_out	5662	36480	15.521	129	2000	14.770
Novak Djokovic	Clay	source_direction_Total	inside_in	787	36480	2.157	129	2000	1.831
Novak Djokovic	Grass	source_direction_Total	crosscourt	6921	15240	45.413	58	2000	43.603
Novak Djokovic	Grass	source_direction_Total	down_middle	3469	15240	22.762	58	2000	20.641
Novak Djokovic	Grass	source_direction_Total	down_the_line	2482	15240	16.286	58	2000	14.781
Novak Djokovic	Grass	source_direction_Total	inside_out	2175	15240	14.272	58	2000	13.355
Novak Djokovic	Grass	source_direction_Total	inside_in	193	15240	1.266	58	2000	0.998
Novak Djokovic	Hard	source_direction_Total	crosscourt	38926	87181	44.650	362	2000	43.953
Novak Djokovic	Hard	source_direction_Total	down_middle	21684	87181	24.872	362	2000	24.013
Novak Djokovic	Hard	source_direction_Total	down_the_line	12325	87181	14.137	362	2000	13.649

player	surface	distribution	category	count	classified_denominator	share_pct	contributing_matches	valid_bootstraps	ci95_lower
Novak Djokovic	Hard	source_direction_Total	inside_out	12827	87181	14.713	362	2000	14.307
Novak Djokovic	Hard	source_direction_Total	inside_in	1419	87181	1.628	362	2000	1.505
Roger Federer	Overall	source_direction_F	crosscourt	26333	70862	37.161	719	2000	36.625
Roger Federer	Overall	source_direction_F	down_middle	14399	70862	20.320	719	2000	19.738
Roger Federer	Overall	source_direction_F	down_the_line	8570	70862	12.094	719	2000	11.692
Roger Federer	Overall	source_direction_F	inside_out	17237	70862	24.325	719	2000	23.850
Roger Federer	Overall	source_direction_F	inside_in	4323	70862	6.101	719	2000	5.813
Roger Federer	Clay	source_direction_F	crosscourt	6039	15922	37.929	130	2000	36.628
Roger Federer	Clay	source_direction_F	down_middle	3146	15922	19.759	130	2000	18.217
Roger Federer	Clay	source_direction_F	down_the_line	1719	15922	10.796	130	2000	10.003
Roger Federer	Clay	source_direction_F	inside_out	3903	15922	24.513	130	2000	23.350
Roger Federer	Clay	source_direction_F	inside_in	1115	15922	7.003	130	2000	6.260
Roger Federer	Grass	source_direction_F	crosscourt	3696	10097	36.605	117	2000	35.336
Roger Federer	Grass	source_direction_F	down_middle	2022	10097	20.026	117	2000	18.627
Roger Federer	Grass	source_direction_F	down_the_line	1328	10097	13.152	117	2000	12.319
Roger Federer	Grass	source_direction_F	inside_out	2480	10097	24.562	117	2000	23.402
Roger Federer	Grass	source_direction_F	inside_in	571	10097	5.655	117	2000	5.087
Roger Federer	Hard	source_direction_F	crosscourt	16598	44843	37.014	472	2000	36.398
Roger Federer	Hard	source_direction_F	down_middle	9231	44843	20.585	472	2000	19.834
Roger Federer	Hard	source_direction_F	down_the_line	5523	44843	12.316	472	2000	11.832
Roger Federer	Hard	source_direction_F	inside_out	10854	44843	24.204	472	2000	23.653
Roger Federer	Hard	source_direction_F	inside_in	2637	44843	5.881	472	2000	5.560
Rafael Nadal	Overall	source_direction_F	crosscourt	29672	62069	47.805	421	2000	46.794
Rafael Nadal	Overall	source_direction_F	down_middle	9535	62069	15.362	421	2000	14.621

player	surface	distribution	category	count	classified_denominator	share_pct	contributing_matches	valid_bootstraps	ci95_lower
Rafael Nadal	Overall	source_direction_F	down_the_line	6926	62069	11.159	421	2000	10.730
Rafael Nadal	Overall	source_direction_F	inside_out	12281	62069	19.786	421	2000	19.151
Rafael Nadal	Overall	source_direction_F	inside_in	3655	62069	5.889	421	2000	5.556
Rafael Nadal	Clay	source_direction_F	crosscourt	12687	26013	48.772	164	2000	47.025
Rafael Nadal	Clay	source_direction_F	down_middle	3846	26013	14.785	164	2000	13.490
Rafael Nadal	Clay	source_direction_F	down_the_line	2724	26013	10.472	164	2000	9.723
Rafael Nadal	Clay	source_direction_F	inside_out	5075	26013	19.509	164	2000	18.299
Rafael Nadal	Clay	source_direction_F	inside_in	1681	26013	6.462	164	2000	5.930
Rafael Nadal	Grass	source_direction_F	crosscourt	2586	5449	47.458	40	2000	44.661
Rafael Nadal	Grass	source_direction_F	down_middle	662	5449	12.149	40	2000	10.625
Rafael Nadal	Grass	source_direction_F	down_the_line	721	5449	13.232	40	2000	11.956
Rafael Nadal	Grass	source_direction_F	inside_out	1132	5449	20.774	40	2000	19.073
Rafael Nadal	Grass	source_direction_F	inside_in	348	5449	6.386	40	2000	5.498
Rafael Nadal	Hard	source_direction_F	crosscourt	14399	30607	47.045	217	2000	45.812
Rafael Nadal	Hard	source_direction_F	down_middle	5027	30607	16.424	217	2000	15.529
Rafael Nadal	Hard	source_direction_F	down_the_line	3481	30607	11.373	217	2000	10.842
Rafael Nadal	Hard	source_direction_F	inside_out	6074	30607	19.845	217	2000	19.094
Rafael Nadal	Hard	source_direction_F	inside_in	1626	30607	5.313	217	2000	4.841
Novak Djokovic	Overall	source_direction_F	crosscourt	28443	70632	40.269	549	2000	39.613
Novak Djokovic	Overall	source_direction_F	down_middle	14225	70632	20.140	549	2000	19.480
Novak Djokovic	Overall	source_direction_F	down_the_line	9292	70632	13.156	549	2000	12.675
Novak Djokovic	Overall	source_direction_F	inside_out	16352	70632	23.151	549	2000	22.670
Novak Djokovic	Overall	source_direction_F	inside_in	2320	70632	3.285	549	2000	3.075
Novak Djokovic	Clay	source_direction_F	crosscourt	7710	19246	40.060	129	2000	38.528

player	surface	distribution	category	count	classified_denominator	share_pct	contributing_matches	valid_bootstraps	ci95_lower
Novak Djokovic	Clay	source_direction_F	down_middle	3838	19246	19.942	129	2000	18.408
Novak Djokovic	Clay	source_direction_F	down_the_line	2366	19246	12.293	129	2000	11.474
Novak Djokovic	Clay	source_direction_F	inside_out	4597	19246	23.885	129	2000	22.898
Novak Djokovic	Clay	source_direction_F	inside_in	735	19246	3.819	129	2000	3.348
Novak Djokovic	Grass	source_direction_F	crosscourt	3054	7410	41.215	58	2000	39.430
Novak Djokovic	Grass	source_direction_F	down_middle	1430	7410	19.298	58	2000	17.198
Novak Djokovic	Grass	source_direction_F	down_the_line	1110	7410	14.980	58	2000	13.309
Novak Djokovic	Grass	source_direction_F	inside_out	1626	7410	21.943	58	2000	20.705
Novak Djokovic	Grass	source_direction_F	inside_in	190	7410	2.564	58	2000	2.010
Novak Djokovic	Hard	source_direction_F	crosscourt	17679	43976	40.201	362	2000	39.502
Novak Djokovic	Hard	source_direction_F	down_middle	8957	43976	20.368	362	2000	19.568
Novak Djokovic	Hard	source_direction_F	down_the_line	5816	43976	13.225	362	2000	12.652
Novak Djokovic	Hard	source_direction_F	inside_out	10129	43976	23.033	362	2000	22.424
Novak Djokovic	Hard	source_direction_F	inside_in	1395	43976	3.172	362	2000	2.938
Roger Federer	Overall	source_direction_B	crosscourt	19848	44026	45.082	719	2000	44.244
Roger Federer	Overall	source_direction_B	down_middle	14913	44026	33.873	719	2000	33.035
Roger Federer	Overall	source_direction_B	down_the_line	7791	44026	17.696	719	2000	17.154
Roger Federer	Overall	source_direction_B	inside_out	1442	44026	3.275	719	2000	3.032
Roger Federer	Overall	source_direction_B	inside_in	32	44026	0.073	719	2000	0.031
Roger Federer	Clay	source_direction_B	crosscourt	4478	10186	43.962	130	2000	41.972
Roger Federer	Clay	source_direction_B	down_middle	3561	10186	34.960	130	2000	32.883
Roger Federer	Clay	source_direction_B	down_the_line	1827	10186	17.936	130	2000	16.724
Roger Federer	Clay	source_direction_B	inside_out	317	10186	3.112	130	2000	2.619
Roger Federer	Clay	source_direction_B	inside_in	3	10186	0.029	130	2000	0.000

player	surface	distribution	category	count	classified_denominator	share_pct	contributing_matches	valid_bootstraps	ci95_lower
Roger Federer	Grass	source_direction_B	crosscourt	2508	5667	44.256	117	2000	42.786
Roger Federer	Grass	source_direction_B	down_middle	1868	5667	32.963	117	2000	31.192
Roger Federer	Grass	source_direction_B	down_the_line	1128	5667	19.905	117	2000	18.354
Roger Federer	Grass	source_direction_B	inside_out	160	5667	2.823	117	2000	2.357
Roger Federer	Grass	source_direction_B	inside_in	3	5667	0.053	117	2000	0.000
Roger Federer	Hard	source_direction_B	crosscourt	12862	28173	45.654	472	2000	44.670
Roger Federer	Hard	source_direction_B	down_middle	9484	28173	33.663	472	2000	32.626
Roger Federer	Hard	source_direction_B	down_the_line	4836	28173	17.165	472	2000	16.551
Roger Federer	Hard	source_direction_B	inside_out	965	28173	3.425	472	2000	3.104
Roger Federer	Hard	source_direction_B	inside_in	26	28173	0.092	472	2000	0.030
Rafael Nadal	Overall	source_direction_B	crosscourt	15615	35305	44.229	422	2000	43.094
Rafael Nadal	Overall	source_direction_B	down_middle	10398	35305	29.452	422	2000	28.440
Rafael Nadal	Overall	source_direction_B	down_the_line	7090	35305	20.082	422	2000	19.245
Rafael Nadal	Overall	source_direction_B	inside_out	1967	35305	5.571	422	2000	5.017
Rafael Nadal	Overall	source_direction_B	inside_in	235	35305	0.666	422	2000	0.294
Rafael Nadal	Clay	source_direction_B	crosscourt	7012	16480	42.549	165	2000	40.555
Rafael Nadal	Clay	source_direction_B	down_middle	4810	16480	29.187	165	2000	27.354
Rafael Nadal	Clay	source_direction_B	down_the_line	3478	16480	21.104	165	2000	19.586
Rafael Nadal	Clay	source_direction_B	inside_out	1024	16480	6.214	165	2000	5.216
Rafael Nadal	Clay	source_direction_B	inside_in	156	16480	0.947	165	2000	0.242
Rafael Nadal	Grass	source_direction_B	crosscourt	1294	2564	50.468	40	2000	47.529
Rafael Nadal	Grass	source_direction_B	down_middle	629	2564	24.532	40	2000	21.971
Rafael Nadal	Grass	source_direction_B	down_the_line	528	2564	20.593	40	2000	17.612
Rafael Nadal	Grass	source_direction_B	inside_out	107	2564	4.173	40	2000	3.325

player	surface	distribution	category	count	classified_denominator	share_pct	contributing_matches	valid_bootstraps	ci95_lower
Rafael Nadal	Grass	source_direction_B	inside_in	6	2564	0.234	40	2000	0.045
Rafael Nadal	Hard	source_direction_B	crosscourt	7309	16261	44.948	217	2000	43.674
Rafael Nadal	Hard	source_direction_B	down_middle	4959	16261	30.496	217	2000	29.323
Rafael Nadal	Hard	source_direction_B	down_the_line	3084	16261	18.966	217	2000	18.017
Rafael Nadal	Hard	source_direction_B	inside_out	836	16261	5.141	217	2000	4.601
Rafael Nadal	Hard	source_direction_B	inside_in	73	16261	0.449	217	2000	0.164
Novak Djokovic	Overall	source_direction_B	crosscourt	29052	59630	48.720	548	2000	47.712
Novak Djokovic	Overall	source_direction_B	down_middle	16961	59630	28.444	548	2000	27.554
Novak Djokovic	Overall	source_direction_B	down_the_line	9590	59630	16.083	548	2000	15.461
Novak Djokovic	Overall	source_direction_B	inside_out	3959	59630	6.639	548	2000	6.283
Novak Djokovic	Overall	source_direction_B	inside_in	68	59630	0.114	548	2000	0.028
Novak Djokovic	Clay	source_direction_B	crosscourt	7918	16003	49.478	129	2000	46.779
Novak Djokovic	Clay	source_direction_B	down_middle	4399	16003	27.489	129	2000	25.613
Novak Djokovic	Clay	source_direction_B	down_the_line	2621	16003	16.378	129	2000	14.967
Novak Djokovic	Clay	source_direction_B	inside_out	1017	16003	6.355	129	2000	5.434
Novak Djokovic	Clay	source_direction_B	inside_in	48	16003	0.300	129	2000	0.013
Novak Djokovic	Grass	source_direction_B	crosscourt	3115	6385	48.786	58	2000	45.956
Novak Djokovic	Grass	source_direction_B	down_middle	1618	6385	25.341	58	2000	22.768
Novak Djokovic	Grass	source_direction_B	down_the_line	1186	6385	18.575	58	2000	16.642
Novak Djokovic	Grass	source_direction_B	inside_out	464	6385	7.267	58	2000	6.205
Novak Djokovic	Grass	source_direction_B	inside_in	2	6385	0.031	58	2000	0.000
Novak Djokovic	Hard	source_direction_B	crosscourt	18019	37242	48.384	361	2000	47.317
Novak Djokovic	Hard	source_direction_B	down_middle	10944	37242	29.386	361	2000	28.374
Novak Djokovic	Hard	source_direction_B	down_the_line	5783	37242	15.528	361	2000	14.859

player	surface	distribution	category	count	classified_denominator	share_pct	contributing_matches	valid_bootstraps	ci95_lower
Novak Djokovic	Hard	source_direction_B	inside_out	2478	37242	6.654	361	2000	6.319
Novak Djokovic	Hard	source_direction_B	inside_in	18	37242	0.048	361	2000	0.023
Roger Federer	Overall	source_direction_S	crosscourt	9783	17672	55.359	718	2000	54.003
Roger Federer	Overall	source_direction_S	down_middle	4704	17672	26.618	718	2000	25.581
Roger Federer	Overall	source_direction_S	down_the_line	2738	17672	15.493	718	2000	14.715
Roger Federer	Overall	source_direction_S	inside_out	427	17672	2.416	718	2000	2.082
Roger Federer	Overall	source_direction_S	inside_in	20	17672	0.113	718	2000	0.052
Roger Federer	Clay	source_direction_S	crosscourt	1573	2985	52.697	130	2000	49.078
Roger Federer	Clay	source_direction_S	down_middle	860	2985	28.811	130	2000	25.993
Roger Federer	Clay	source_direction_S	down_the_line	497	2985	16.650	130	2000	14.413
Roger Federer	Clay	source_direction_S	inside_out	51	2985	1.709	130	2000	1.134
Roger Federer	Clay	source_direction_S	inside_in	4	2985	0.134	130	2000	0.030
Roger Federer	Grass	source_direction_S	crosscourt	1476	2901	50.879	117	2000	47.785
Roger Federer	Grass	source_direction_S	down_middle	835	2901	28.783	117	2000	26.536
Roger Federer	Grass	source_direction_S	down_the_line	519	2901	17.890	117	2000	16.020
Roger Federer	Grass	source_direction_S	inside_out	68	2901	2.344	117	2000	1.802
Roger Federer	Grass	source_direction_S	inside_in	3	2901	0.103	117	2000	0.000
Roger Federer	Hard	source_direction_S	crosscourt	6734	11786	57.136	471	2000	55.608
Roger Federer	Hard	source_direction_S	down_middle	3009	11786	25.530	471	2000	24.304
Roger Federer	Hard	source_direction_S	down_the_line	1722	11786	14.611	471	2000	13.684
Roger Federer	Hard	source_direction_S	inside_out	308	11786	2.613	471	2000	2.172
Roger Federer	Hard	source_direction_S	inside_in	13	11786	0.110	471	2000	0.033
Rafael Nadal	Overall	source_direction_S	crosscourt	1722	8038	21.423	418	2000	19.776
Rafael Nadal	Overall	source_direction_S	down_middle	3142	8038	39.089	418	2000	37.344

player	surface	distribution	category	count	classified_denominator	share_pct	contributing_matches	valid_bootstraps	ci95_lower
Rafael Nadal	Overall	source_direction_s	down_the_line	2589	8038	32.210	418	2000	30.225
Rafael Nadal	Overall	source_direction_s	inside_out	548	8038	6.818	418	2000	6.172
Rafael Nadal	Overall	source_direction_s	inside_in	37	8038	0.460	418	2000	0.229
Rafael Nadal	Clay	source_direction_s	crosscourt	379	2096	18.082	162	2000	15.851
Rafael Nadal	Clay	source_direction_s	down_middle	847	2096	40.410	162	2000	37.357
Rafael Nadal	Clay	source_direction_s	down_the_line	711	2096	33.922	162	2000	30.523
Rafael Nadal	Clay	source_direction_s	inside_out	141	2096	6.727	162	2000	5.559
Rafael Nadal	Clay	source_direction_s	inside_in	18	2096	0.859	162	2000	0.217
Rafael Nadal	Grass	source_direction_s	crosscourt	271	1101	24.614	40	2000	20.961
Rafael Nadal	Grass	source_direction_s	down_middle	376	1101	34.151	40	2000	29.403
Rafael Nadal	Grass	source_direction_s	down_the_line	381	1101	34.605	40	2000	29.412
Rafael Nadal	Grass	source_direction_s	inside_out	73	1101	6.630	40	2000	5.150
Rafael Nadal	Grass	source_direction_s	inside_in	0	1101	0.000	40	2000	0.000
Rafael Nadal	Hard	source_direction_s	crosscourt	1072	4841	22.144	216	2000	19.833
Rafael Nadal	Hard	source_direction_s	down_middle	1919	4841	39.641	216	2000	37.374
Rafael Nadal	Hard	source_direction_s	down_the_line	1497	4841	30.923	216	2000	28.508
Rafael Nadal	Hard	source_direction_s	inside_out	334	4841	6.899	216	2000	6.023
Rafael Nadal	Hard	source_direction_s	inside_in	19	4841	0.392	216	2000	0.150
Novak Djokovic	Overall	source_direction_s	crosscourt	4456	8441	52.790	546	2000	51.065
Novak Djokovic	Overall	source_direction_s	down_middle	2532	8441	29.996	546	2000	28.563
Novak Djokovic	Overall	source_direction_s	down_the_line	1096	8441	12.984	546	2000	12.053
Novak Djokovic	Overall	source_direction_s	inside_out	346	8441	4.099	546	2000	3.497
Novak Djokovic	Overall	source_direction_s	inside_in	11	8441	0.130	546	2000	0.049
Novak Djokovic	Clay	source_direction_s	crosscourt	583	1231	47.360	127	2000	43.243

player	surface	distribution	category	count	classified_denominator	share_pct	contributing_matches	valid_bootstraps	ci95_lower
Novak Djokovic	Clay	source_direction_S	down_middle	381	1231	30.950	127	2000	27.542
Novak Djokovic	Clay	source_direction_S	down_the_line	215	1231	17.465	127	2000	14.800
Novak Djokovic	Clay	source_direction_S	inside_out	48	1231	3.899	127	2000	2.595
Novak Djokovic	Clay	source_direction_S	inside_in	4	1231	0.325	127	2000	0.000
Novak Djokovic	Grass	source_direction_S	crosscourt	752	1445	52.042	58	2000	47.156
Novak Djokovic	Grass	source_direction_S	down_middle	421	1445	29.135	58	2000	25.776
Novak Djokovic	Grass	source_direction_S	down_the_line	186	1445	12.872	58	2000	10.289
Novak Djokovic	Grass	source_direction_S	inside_out	85	1445	5.882	58	2000	3.854
Novak Djokovic	Grass	source_direction_S	inside_in	1	1445	0.069	58	2000	0.000
Novak Djokovic	Hard	source_direction_S	crosscourt	3121	5765	54.137	361	2000	52.219
Novak Djokovic	Hard	source_direction_S	down_middle	1730	5765	30.009	361	2000	28.176
Novak Djokovic	Hard	source_direction_S	down_the_line	695	5765	12.056	361	2000	10.995
Novak Djokovic	Hard	source_direction_S	inside_out	213	5765	3.695	361	2000	3.089
Novak Djokovic	Hard	source_direction_S	inside_in	6	5765	0.104	361	2000	0.019

player	surface	distribution	ci95_upper
Roger Federer	Overall	shot_type	72.331
Roger Federer	Overall	shot_type	19.694
Roger Federer	Overall	shot_type	4.653
Roger Federer	Overall	shot_type	1.282
Roger Federer	Overall	shot_type	1.039
Roger Federer	Overall	shot_type	1.067
Roger Federer	Overall	shot_type	0.825
Roger Federer	Overall	shot_type	0.480
Roger Federer	Overall	shot_type	0.037
Roger Federer	Overall	shot_type	0.022
Roger Federer	Clay	shot_type	77.592
Roger Federer	Clay	shot_type	16.336
Roger Federer	Clay	shot_type	4.084

player	surface	distribution	ci95_upper
Roger Federer	Clay	shot_type	1.275
Roger Federer	Clay	shot_type	1.704
Roger Federer	Clay	shot_type	1.019
Roger Federer	Clay	shot_type	0.576
Roger Federer	Clay	shot_type	0.469
Roger Federer	Clay	shot_type	0.018
Roger Federer	Clay	shot_type	0.050
Roger Federer	Grass	shot_type	68.866
Roger Federer	Grass	shot_type	22.862
Roger Federer	Grass	shot_type	6.020
Roger Federer	Grass	shot_type	1.478
Roger Federer	Grass	shot_type	1.012
Roger Federer	Grass	shot_type	1.368
Roger Federer	Grass	shot_type	1.563
Roger Federer	Grass	shot_type	0.665
Roger Federer	Grass	shot_type	0.062
Roger Federer	Grass	shot_type	0.024
Roger Federer	Hard	shot_type	71.922
Roger Federer	Hard	shot_type	20.382
Roger Federer	Hard	shot_type	4.715
Roger Federer	Hard	shot_type	1.303
Roger Federer	Hard	shot_type	0.883
Roger Federer	Hard	shot_type	1.072
Roger Federer	Hard	shot_type	0.792
Roger Federer	Hard	shot_type	0.474
Roger Federer	Hard	shot_type	0.043
Roger Federer	Hard	shot_type	0.019
Rafael Nadal	Overall	shot_type	86.547
Rafael Nadal	Overall	shot_type	10.070
Rafael Nadal	Overall	shot_type	1.580
Rafael Nadal	Overall	shot_type	0.800
Rafael Nadal	Overall	shot_type	0.887
Rafael Nadal	Overall	shot_type	1.168
Rafael Nadal	Overall	shot_type	0.293
Rafael Nadal	Overall	shot_type	0.116
Rafael Nadal	Overall	shot_type	0.038
Rafael Nadal	Overall	shot_type	0.013
Rafael Nadal	Clay	shot_type	89.929
Rafael Nadal	Clay	shot_type	6.731
Rafael Nadal	Clay	shot_type	1.240

player	surface	distribution	ci95_upper
Rafael Nadal	Clay	shot_type	0.738
Rafael Nadal	Clay	shot_type	1.183
Rafael Nadal	Clay	shot_type	1.289
Rafael Nadal	Clay	shot_type	0.255
Rafael Nadal	Clay	shot_type	0.139
Rafael Nadal	Clay	shot_type	0.031
Rafael Nadal	Clay	shot_type	0.023
Rafael Nadal	Grass	shot_type	80.934
Rafael Nadal	Grass	shot_type	16.971
Rafael Nadal	Grass	shot_type	2.248
Rafael Nadal	Grass	shot_type	1.148
Rafael Nadal	Grass	shot_type	1.401
Rafael Nadal	Grass	shot_type	1.378
Rafael Nadal	Grass	shot_type	0.763
Rafael Nadal	Grass	shot_type	0.188
Rafael Nadal	Grass	shot_type	0.138
Rafael Nadal	Grass	shot_type	0.059
Rafael Nadal	Hard	shot_type	85.162
Rafael Nadal	Hard	shot_type	11.944
Rafael Nadal	Hard	shot_type	1.868
Rafael Nadal	Hard	shot_type	0.853
Rafael Nadal	Hard	shot_type	0.628
Rafael Nadal	Hard	shot_type	1.129
Rafael Nadal	Hard	shot_type	0.297
Rafael Nadal	Hard	shot_type	0.108
Rafael Nadal	Hard	shot_type	0.040
Rafael Nadal	Hard	shot_type	0.007
Novak Djokovic	Overall	shot_type	85.733
Novak Djokovic	Overall	shot_type	8.855
Novak Djokovic	Overall	shot_type	2.248
Novak Djokovic	Overall	shot_type	0.864
Novak Djokovic	Overall	shot_type	1.645
Novak Djokovic	Overall	shot_type	1.331
Novak Djokovic	Overall	shot_type	0.299
Novak Djokovic	Overall	shot_type	0.235
Novak Djokovic	Overall	shot_type	0.021
Novak Djokovic	Overall	shot_type	0.019
Novak Djokovic	Clay	shot_type	88.285
Novak Djokovic	Clay	shot_type	5.818
Novak Djokovic	Clay	shot_type	2.061

player	surface	distribution	ci95_upper
Novak Djokovic	Clay	shot_type	1.017
Novak Djokovic	Clay	shot_type	2.621
Novak Djokovic	Clay	shot_type	1.470
Novak Djokovic	Clay	shot_type	0.436
Novak Djokovic	Clay	shot_type	0.292
Novak Djokovic	Clay	shot_type	0.016
Novak Djokovic	Clay	shot_type	0.047
Novak Djokovic	Grass	shot_type	82.384
Novak Djokovic	Grass	shot_type	13.794
Novak Djokovic	Grass	shot_type	3.028
Novak Djokovic	Grass	shot_type	0.880
Novak Djokovic	Grass	shot_type	1.655
Novak Djokovic	Grass	shot_type	1.681
Novak Djokovic	Grass	shot_type	0.437
Novak Djokovic	Grass	shot_type	0.285
Novak Djokovic	Grass	shot_type	0.067
Novak Djokovic	Grass	shot_type	0.023
Novak Djokovic	Hard	shot_type	85.686
Novak Djokovic	Hard	shot_type	9.455
Novak Djokovic	Hard	shot_type	2.301
Novak Djokovic	Hard	shot_type	0.864
Novak Djokovic	Hard	shot_type	1.316
Novak Djokovic	Hard	shot_type	1.292
Novak Djokovic	Hard	shot_type	0.262
Novak Djokovic	Hard	shot_type	0.229
Novak Djokovic	Hard	shot_type	0.021
Novak Djokovic	Hard	shot_type	0.013
Roger Federer	Overall	target_all_attempts	28.942
Roger Federer	Overall	target_all_attempts	32.577
Roger Federer	Overall	target_all_attempts	40.100
Roger Federer	Clay	target_all_attempts	31.168
Roger Federer	Clay	target_all_attempts	33.230
Roger Federer	Clay	target_all_attempts	40.009
Roger Federer	Grass	target_all_attempts	29.576
Roger Federer	Grass	target_all_attempts	34.069
Roger Federer	Grass	target_all_attempts	39.984
Roger Federer	Hard	target_all_attempts	28.543
Roger Federer	Hard	target_all_attempts	32.638
Roger Federer	Hard	target_all_attempts	40.684
Rafael Nadal	Overall	target_all_attempts	32.784

player	surface	distribution	ci95_upper
Rafael Nadal	Overall	target_all_attempts	28.086
Rafael Nadal	Overall	target_all_attempts	41.144
Rafael Nadal	Clay	target_all_attempts	32.731
Rafael Nadal	Clay	target_all_attempts	27.503
Rafael Nadal	Clay	target_all_attempts	43.281
Rafael Nadal	Grass	target_all_attempts	35.878
Rafael Nadal	Grass	target_all_attempts	27.554
Rafael Nadal	Grass	target_all_attempts	42.927
Rafael Nadal	Hard	target_all_attempts	33.021
Rafael Nadal	Hard	target_all_attempts	29.654
Rafael Nadal	Hard	target_all_attempts	39.855
Novak Djokovic	Overall	target_all_attempts	30.844
Novak Djokovic	Overall	target_all_attempts	31.339
Novak Djokovic	Overall	target_all_attempts	39.577
Novak Djokovic	Clay	target_all_attempts	32.722
Novak Djokovic	Clay	target_all_attempts	30.804
Novak Djokovic	Clay	target_all_attempts	40.595
Novak Djokovic	Grass	target_all_attempts	32.055
Novak Djokovic	Grass	target_all_attempts	32.592
Novak Djokovic	Grass	target_all_attempts	40.929
Novak Djokovic	Hard	target_all_attempts	30.346
Novak Djokovic	Hard	target_all_attempts	32.108
Novak Djokovic	Hard	target_all_attempts	39.582
Roger Federer	Overall	target_non_error_strokes	28.136
Roger Federer	Overall	target_non_error_strokes	33.252
Roger Federer	Overall	target_non_error_strokes	40.352
Roger Federer	Clay	target_non_error_strokes	30.726
Roger Federer	Clay	target_non_error_strokes	34.008
Roger Federer	Clay	target_non_error_strokes	40.106
Roger Federer	Grass	target_non_error_strokes	28.799
Roger Federer	Grass	target_non_error_strokes	34.794
Roger Federer	Grass	target_non_error_strokes	40.068
Roger Federer	Hard	target_non_error_strokes	27.613
Roger Federer	Hard	target_non_error_strokes	33.278
Roger Federer	Hard	target_non_error_strokes	41.041
Rafael Nadal	Overall	target_non_error_strokes	31.625
Rafael Nadal	Overall	target_non_error_strokes	28.607
Rafael Nadal	Overall	target_non_error_strokes	41.910
Rafael Nadal	Clay	target_non_error_strokes	31.632
Rafael Nadal	Clay	target_non_error_strokes	28.038

player	surface	distribution	ci95_upper
Rafael Nadal	Clay	target_non_error_strokes	43.969
Rafael Nadal	Grass	target_non_error_strokes	34.727
Rafael Nadal	Grass	target_non_error_strokes	28.185
Rafael Nadal	Grass	target_non_error_strokes	43.942
Rafael Nadal	Hard	target_non_error_strokes	31.799
Rafael Nadal	Hard	target_non_error_strokes	30.232
Rafael Nadal	Hard	target_non_error_strokes	40.652
Novak Djokovic	Overall	target_non_error_strokes	30.273
Novak Djokovic	Overall	target_non_error_strokes	31.657
Novak Djokovic	Overall	target_non_error_strokes	39.962
Novak Djokovic	Clay	target_non_error_strokes	32.377
Novak Djokovic	Clay	target_non_error_strokes	31.150
Novak Djokovic	Clay	target_non_error_strokes	40.898
Novak Djokovic	Grass	target_non_error_strokes	31.557
Novak Djokovic	Grass	target_non_error_strokes	32.994
Novak Djokovic	Grass	target_non_error_strokes	41.584
Novak Djokovic	Hard	target_non_error_strokes	29.717
Novak Djokovic	Hard	target_non_error_strokes	32.445
Novak Djokovic	Hard	target_non_error_strokes	39.989
Roger Federer	Overall	target_returns	15.895
Roger Federer	Overall	target_returns	51.492
Roger Federer	Overall	target_returns	34.711
Roger Federer	Clay	target_returns	17.204
Roger Federer	Clay	target_returns	52.654
Roger Federer	Clay	target_returns	36.013
Roger Federer	Grass	target_returns	17.960
Roger Federer	Grass	target_returns	52.778
Roger Federer	Grass	target_returns	34.286
Roger Federer	Hard	target_returns	15.598
Roger Federer	Hard	target_returns	51.547
Roger Federer	Hard	target_returns	35.306
Rafael Nadal	Overall	target_returns	22.282
Rafael Nadal	Overall	target_returns	50.533
Rafael Nadal	Overall	target_returns	30.016
Rafael Nadal	Clay	target_returns	21.819
Rafael Nadal	Clay	target_returns	50.289
Rafael Nadal	Clay	target_returns	32.955
Rafael Nadal	Grass	target_returns	25.052
Rafael Nadal	Grass	target_returns	50.915
Rafael Nadal	Grass	target_returns	32.740

player	surface	distribution	ci95_upper
Rafael Nadal	Hard	target_returns	23.069
Rafael Nadal	Hard	target_returns	52.103
Rafael Nadal	Hard	target_returns	28.611
Novak Djokovic	Overall	target_returns	16.397
Novak Djokovic	Overall	target_returns	55.070
Novak Djokovic	Overall	target_returns	30.715
Novak Djokovic	Clay	target_returns	17.814
Novak Djokovic	Clay	target_returns	54.946
Novak Djokovic	Clay	target_returns	32.728
Novak Djokovic	Grass	target_returns	17.052
Novak Djokovic	Grass	target_returns	56.322
Novak Djokovic	Grass	target_returns	33.010
Novak Djokovic	Hard	target_returns	16.279
Novak Djokovic	Hard	target_returns	55.813
Novak Djokovic	Hard	target_returns	30.371
Roger Federer	Overall	active_serve_target	22.121
Roger Federer	Overall	active_serve_target	9.337
Roger Federer	Overall	active_serve_target	22.241
Roger Federer	Overall	active_serve_target	25.244
Roger Federer	Overall	active_serve_target	6.800
Roger Federer	Overall	active_serve_target	16.558
Roger Federer	Clay	active_serve_target	21.524
Roger Federer	Clay	active_serve_target	10.121
Roger Federer	Clay	active_serve_target	24.076
Roger Federer	Clay	active_serve_target	27.577
Roger Federer	Clay	active_serve_target	7.638
Roger Federer	Clay	active_serve_target	15.728
Roger Federer	Grass	active_serve_target	23.492
Roger Federer	Grass	active_serve_target	9.014
Roger Federer	Grass	active_serve_target	22.433
Roger Federer	Grass	active_serve_target	24.849
Roger Federer	Grass	active_serve_target	6.378
Roger Federer	Grass	active_serve_target	18.266
Roger Federer	Hard	active_serve_target	22.311
Roger Federer	Hard	active_serve_target	9.604
Roger Federer	Hard	active_serve_target	22.118
Roger Federer	Hard	active_serve_target	25.059
Roger Federer	Hard	active_serve_target	7.015
Roger Federer	Hard	active_serve_target	16.674
Rafael Nadal	Overall	active_serve_target	12.844

player	surface	distribution	ci95_upper
Rafael Nadal	Overall	active_serve_target	15.037
Rafael Nadal	Overall	active_serve_target	26.859
Rafael Nadal	Overall	active_serve_target	26.367
Rafael Nadal	Overall	active_serve_target	13.287
Rafael Nadal	Overall	active_serve_target	10.202
Rafael Nadal	Clay	active_serve_target	11.217
Rafael Nadal	Clay	active_serve_target	17.481
Rafael Nadal	Clay	active_serve_target	27.703
Rafael Nadal	Clay	active_serve_target	30.673
Rafael Nadal	Clay	active_serve_target	11.870
Rafael Nadal	Clay	active_serve_target	8.559
Rafael Nadal	Grass	active_serve_target	15.222
Rafael Nadal	Grass	active_serve_target	14.940
Rafael Nadal	Grass	active_serve_target	29.900
Rafael Nadal	Grass	active_serve_target	23.599
Rafael Nadal	Grass	active_serve_target	14.474
Rafael Nadal	Grass	active_serve_target	15.469
Rafael Nadal	Hard	active_serve_target	14.388
Rafael Nadal	Hard	active_serve_target	14.247
Rafael Nadal	Hard	active_serve_target	26.909
Rafael Nadal	Hard	active_serve_target	24.973
Rafael Nadal	Hard	active_serve_target	14.765
Rafael Nadal	Hard	active_serve_target	10.738
Novak Djokovic	Overall	active_serve_target	24.609
Novak Djokovic	Overall	active_serve_target	10.347
Novak Djokovic	Overall	active_serve_target	18.972
Novak Djokovic	Overall	active_serve_target	20.646
Novak Djokovic	Overall	active_serve_target	8.708
Novak Djokovic	Overall	active_serve_target	19.573
Novak Djokovic	Clay	active_serve_target	25.669
Novak Djokovic	Clay	active_serve_target	9.974
Novak Djokovic	Clay	active_serve_target	19.932
Novak Djokovic	Clay	active_serve_target	23.044
Novak Djokovic	Clay	active_serve_target	7.545
Novak Djokovic	Clay	active_serve_target	20.138
Novak Djokovic	Grass	active_serve_target	24.393
Novak Djokovic	Grass	active_serve_target	10.262
Novak Djokovic	Grass	active_serve_target	21.533
Novak Djokovic	Grass	active_serve_target	20.229
Novak Djokovic	Grass	active_serve_target	9.187

player	surface	distribution	ci95_upper
Novak Djokovic	Grass	active_serve_target	21.401
Novak Djokovic	Hard	active_serve_target	24.741
Novak Djokovic	Hard	active_serve_target	10.998
Novak Djokovic	Hard	active_serve_target	18.614
Novak Djokovic	Hard	active_serve_target	20.270
Novak Djokovic	Hard	active_serve_target	9.463
Novak Djokovic	Hard	active_serve_target	19.426
Roger Federer	Overall	return_depth_target	3.749
Roger Federer	Overall	return_depth_target	12.778
Roger Federer	Overall	return_depth_target	9.284
Roger Federer	Overall	return_depth_target	7.712
Roger Federer	Overall	return_depth_target	28.538
Roger Federer	Overall	return_depth_target	18.971
Roger Federer	Overall	return_depth_target	3.374
Roger Federer	Overall	return_depth_target	12.408
Roger Federer	Overall	return_depth_target	7.183
Roger Federer	Clay	return_depth_target	3.837
Roger Federer	Clay	return_depth_target	13.123
Roger Federer	Clay	return_depth_target	9.677
Roger Federer	Clay	return_depth_target	9.111
Roger Federer	Clay	return_depth_target	31.712
Roger Federer	Clay	return_depth_target	21.471
Roger Federer	Clay	return_depth_target	3.735
Roger Federer	Clay	return_depth_target	11.354
Roger Federer	Clay	return_depth_target	6.130
Roger Federer	Grass	return_depth_target	4.913
Roger Federer	Grass	return_depth_target	16.240
Roger Federer	Grass	return_depth_target	10.574
Roger Federer	Grass	return_depth_target	8.085
Roger Federer	Grass	return_depth_target	29.369
Roger Federer	Grass	return_depth_target	18.168
Roger Federer	Grass	return_depth_target	3.689
Roger Federer	Grass	return_depth_target	11.826
Roger Federer	Grass	return_depth_target	6.881
Roger Federer	Hard	return_depth_target	3.645
Roger Federer	Hard	return_depth_target	12.289
Roger Federer	Hard	return_depth_target	9.224
Roger Federer	Hard	return_depth_target	7.589
Roger Federer	Hard	return_depth_target	28.115
Roger Federer	Hard	return_depth_target	19.119

player	surface	distribution	ci95_upper
Roger Federer	Hard	return_depth_target	3.449
Roger Federer	Hard	return_depth_target	13.434
Roger Federer	Hard	return_depth_target	8.067
Rafael Nadal	Overall	return_depth_target	7.013
Rafael Nadal	Overall	return_depth_target	15.599
Rafael Nadal	Overall	return_depth_target	10.469
Rafael Nadal	Overall	return_depth_target	9.940
Rafael Nadal	Overall	return_depth_target	24.150
Rafael Nadal	Overall	return_depth_target	14.626
Rafael Nadal	Overall	return_depth_target	4.722
Rafael Nadal	Overall	return_depth_target	12.757
Rafael Nadal	Overall	return_depth_target	6.010
Rafael Nadal	Clay	return_depth_target	6.916
Rafael Nadal	Clay	return_depth_target	14.999
Rafael Nadal	Clay	return_depth_target	12.115
Rafael Nadal	Clay	return_depth_target	10.048
Rafael Nadal	Clay	return_depth_target	24.989
Rafael Nadal	Clay	return_depth_target	16.343
Rafael Nadal	Clay	return_depth_target	4.788
Rafael Nadal	Clay	return_depth_target	12.443
Rafael Nadal	Clay	return_depth_target	6.107
Rafael Nadal	Grass	return_depth_target	9.429
Rafael Nadal	Grass	return_depth_target	15.517
Rafael Nadal	Grass	return_depth_target	11.417
Rafael Nadal	Grass	return_depth_target	11.384
Rafael Nadal	Grass	return_depth_target	25.271
Rafael Nadal	Grass	return_depth_target	15.658
Rafael Nadal	Grass	return_depth_target	4.320
Rafael Nadal	Grass	return_depth_target	14.161
Rafael Nadal	Grass	return_depth_target	8.159
Rafael Nadal	Hard	return_depth_target	7.150
Rafael Nadal	Hard	return_depth_target	16.988
Rafael Nadal	Hard	return_depth_target	9.650
Rafael Nadal	Hard	return_depth_target	10.297
Rafael Nadal	Hard	return_depth_target	24.474
Rafael Nadal	Hard	return_depth_target	13.926
Rafael Nadal	Hard	return_depth_target	5.240
Rafael Nadal	Hard	return_depth_target	13.558
Rafael Nadal	Hard	return_depth_target	6.119
Novak Djokovic	Overall	return_depth_target	3.221

player	surface	distribution	ci95_upper
Novak Djokovic	Overall	return_depth_target	10.444
Novak Djokovic	Overall	return_depth_target	7.865
Novak Djokovic	Overall	return_depth_target	7.783
Novak Djokovic	Overall	return_depth_target	27.981
Novak Djokovic	Overall	return_depth_target	16.097
Novak Djokovic	Overall	return_depth_target	4.743
Novak Djokovic	Overall	return_depth_target	18.314
Novak Djokovic	Overall	return_depth_target	7.830
Novak Djokovic	Clay	return_depth_target	3.075
Novak Djokovic	Clay	return_depth_target	9.903
Novak Djokovic	Clay	return_depth_target	7.503
Novak Djokovic	Clay	return_depth_target	9.193
Novak Djokovic	Clay	return_depth_target	28.999
Novak Djokovic	Clay	return_depth_target	17.802
Novak Djokovic	Clay	return_depth_target	5.692
Novak Djokovic	Clay	return_depth_target	17.652
Novak Djokovic	Clay	return_depth_target	9.494
Novak Djokovic	Grass	return_depth_target	4.155
Novak Djokovic	Grass	return_depth_target	11.359
Novak Djokovic	Grass	return_depth_target	9.359
Novak Djokovic	Grass	return_depth_target	8.077
Novak Djokovic	Grass	return_depth_target	29.938
Novak Djokovic	Grass	return_depth_target	18.275
Novak Djokovic	Grass	return_depth_target	4.506
Novak Djokovic	Grass	return_depth_target	19.171
Novak Djokovic	Grass	return_depth_target	7.722
Novak Djokovic	Hard	return_depth_target	3.313
Novak Djokovic	Hard	return_depth_target	10.934
Novak Djokovic	Hard	return_depth_target	8.119
Novak Djokovic	Hard	return_depth_target	7.593
Novak Djokovic	Hard	return_depth_target	27.981
Novak Djokovic	Hard	return_depth_target	15.744
Novak Djokovic	Hard	return_depth_target	4.736
Novak Djokovic	Hard	return_depth_target	19.198
Novak Djokovic	Hard	return_depth_target	7.723
Roger Federer	Overall	source_direction_Total	42.661
Roger Federer	Overall	source_direction_Total	26.328
Roger Federer	Overall	source_direction_Total	14.770
Roger Federer	Overall	source_direction_Total	14.741
Roger Federer	Overall	source_direction_Total	3.448

player	surface	distribution	ci95_upper
Roger Federer	Clay	source_direction_Total	42.498
Roger Federer	Clay	source_direction_Total	27.720
Roger Federer	Clay	source_direction_Total	14.795
Roger Federer	Clay	source_direction_Total	15.387
Roger Federer	Clay	source_direction_Total	4.304
Roger Federer	Grass	source_direction_Total	41.962
Roger Federer	Grass	source_direction_Total	26.791
Roger Federer	Grass	source_direction_Total	16.826
Roger Federer	Grass	source_direction_Total	15.243
Roger Federer	Grass	source_direction_Total	3.445
Roger Federer	Hard	source_direction_Total	43.244
Roger Federer	Hard	source_direction_Total	26.441
Roger Federer	Hard	source_direction_Total	14.695
Roger Federer	Hard	source_direction_Total	14.728
Roger Federer	Hard	source_direction_Total	3.341
Rafael Nadal	Overall	source_direction_Total	45.360
Rafael Nadal	Overall	source_direction_Total	22.691
Rafael Nadal	Overall	source_direction_Total	16.270
Rafael Nadal	Overall	source_direction_Total	14.662
Rafael Nadal	Overall	source_direction_Total	4.032
Rafael Nadal	Clay	source_direction_Total	46.394
Rafael Nadal	Clay	source_direction_Total	22.667
Rafael Nadal	Clay	source_direction_Total	16.404
Rafael Nadal	Clay	source_direction_Total	15.226
Rafael Nadal	Clay	source_direction_Total	4.717
Rafael Nadal	Grass	source_direction_Total	47.318
Rafael Nadal	Grass	source_direction_Total	20.490
Rafael Nadal	Grass	source_direction_Total	19.383
Rafael Nadal	Grass	source_direction_Total	15.547
Rafael Nadal	Grass	source_direction_Total	4.382
Rafael Nadal	Hard	source_direction_Total	44.993
Rafael Nadal	Hard	source_direction_Total	24.080
Rafael Nadal	Hard	source_direction_Total	16.161
Rafael Nadal	Hard	source_direction_Total	14.641
Rafael Nadal	Hard	source_direction_Total	3.700
Novak Djokovic	Overall	source_direction_Total	45.275
Novak Djokovic	Overall	source_direction_Total	25.037
Novak Djokovic	Overall	source_direction_Total	14.851
Novak Djokovic	Overall	source_direction_Total	15.231
Novak Djokovic	Overall	source_direction_Total	1.865

player	surface	distribution	ci95_upper
Novak Djokovic	Clay	source_direction_Total	45.793
Novak Djokovic	Clay	source_direction_Total	25.298
Novak Djokovic	Clay	source_direction_Total	15.214
Novak Djokovic	Clay	source_direction_Total	16.470
Novak Djokovic	Clay	source_direction_Total	2.510
Novak Djokovic	Grass	source_direction_Total	47.064
Novak Djokovic	Grass	source_direction_Total	25.128
Novak Djokovic	Grass	source_direction_Total	17.843
Novak Djokovic	Grass	source_direction_Total	15.230
Novak Djokovic	Grass	source_direction_Total	1.591
Novak Djokovic	Hard	source_direction_Total	45.305
Novak Djokovic	Hard	source_direction_Total	25.744
Novak Djokovic	Hard	source_direction_Total	14.625
Novak Djokovic	Hard	source_direction_Total	15.121
Novak Djokovic	Hard	source_direction_Total	1.758
Roger Federer	Overall	source_direction_F	37.685
Roger Federer	Overall	source_direction_F	20.953
Roger Federer	Overall	source_direction_F	12.457
Roger Federer	Overall	source_direction_F	24.783
Roger Federer	Overall	source_direction_F	6.375
Roger Federer	Clay	source_direction_F	39.300
Roger Federer	Clay	source_direction_F	21.297
Roger Federer	Clay	source_direction_F	11.657
Roger Federer	Clay	source_direction_F	25.802
Roger Federer	Clay	source_direction_F	7.810
Roger Federer	Grass	source_direction_F	37.766
Roger Federer	Grass	source_direction_F	21.514
Roger Federer	Grass	source_direction_F	13.986
Roger Federer	Grass	source_direction_F	25.723
Roger Federer	Grass	source_direction_F	6.293
Roger Federer	Hard	source_direction_F	37.641
Roger Federer	Hard	source_direction_F	21.354
Roger Federer	Hard	source_direction_F	12.774
Roger Federer	Hard	source_direction_F	24.720
Roger Federer	Hard	source_direction_F	6.198
Rafael Nadal	Overall	source_direction_F	48.730
Rafael Nadal	Overall	source_direction_F	16.105
Rafael Nadal	Overall	source_direction_F	11.584
Rafael Nadal	Overall	source_direction_F	20.533
Rafael Nadal	Overall	source_direction_F	6.238

player	surface	distribution	ci95_upper
Rafael Nadal	Clay	source_direction_F	50.375
Rafael Nadal	Clay	source_direction_F	16.079
Rafael Nadal	Clay	source_direction_F	11.205
Rafael Nadal	Clay	source_direction_F	21.014
Rafael Nadal	Clay	source_direction_F	7.019
Rafael Nadal	Grass	source_direction_F	50.190
Rafael Nadal	Grass	source_direction_F	13.990
Rafael Nadal	Grass	source_direction_F	14.507
Rafael Nadal	Grass	source_direction_F	22.465
Rafael Nadal	Grass	source_direction_F	7.240
Rafael Nadal	Hard	source_direction_F	48.221
Rafael Nadal	Hard	source_direction_F	17.388
Rafael Nadal	Hard	source_direction_F	11.922
Rafael Nadal	Hard	source_direction_F	20.614
Rafael Nadal	Hard	source_direction_F	5.799
Novak Djokovic	Overall	source_direction_F	40.934
Novak Djokovic	Overall	source_direction_F	20.871
Novak Djokovic	Overall	source_direction_F	13.640
Novak Djokovic	Overall	source_direction_F	23.633
Novak Djokovic	Overall	source_direction_F	3.502
Novak Djokovic	Clay	source_direction_F	41.545
Novak Djokovic	Clay	source_direction_F	21.612
Novak Djokovic	Clay	source_direction_F	13.179
Novak Djokovic	Clay	source_direction_F	24.985
Novak Djokovic	Clay	source_direction_F	4.273
Novak Djokovic	Grass	source_direction_F	42.954
Novak Djokovic	Grass	source_direction_F	21.542
Novak Djokovic	Grass	source_direction_F	16.770
Novak Djokovic	Grass	source_direction_F	23.173
Novak Djokovic	Grass	source_direction_F	3.210
Novak Djokovic	Hard	source_direction_F	40.967
Novak Djokovic	Hard	source_direction_F	21.244
Novak Djokovic	Hard	source_direction_F	13.775
Novak Djokovic	Hard	source_direction_F	23.594
Novak Djokovic	Hard	source_direction_F	3.429
Roger Federer	Overall	source_direction_B	45.915
Roger Federer	Overall	source_direction_B	34.732
Roger Federer	Overall	source_direction_B	18.238
Roger Federer	Overall	source_direction_B	3.545
Roger Federer	Overall	source_direction_B	0.141

player	surface	distribution	ci95_upper
Roger Federer	Clay	source_direction_B	45.979
Roger Federer	Clay	source_direction_B	37.166
Roger Federer	Clay	source_direction_B	19.348
Roger Federer	Clay	source_direction_B	3.610
Roger Federer	Clay	source_direction_B	0.070
Roger Federer	Grass	source_direction_B	45.716
Roger Federer	Grass	source_direction_B	34.797
Roger Federer	Grass	source_direction_B	21.517
Roger Federer	Grass	source_direction_B	3.321
Roger Federer	Grass	source_direction_B	0.115
Roger Federer	Hard	source_direction_B	46.655
Roger Federer	Hard	source_direction_B	34.711
Roger Federer	Hard	source_direction_B	17.772
Roger Federer	Hard	source_direction_B	3.819
Roger Federer	Hard	source_direction_B	0.193
Rafael Nadal	Overall	source_direction_B	45.292
Rafael Nadal	Overall	source_direction_B	30.480
Rafael Nadal	Overall	source_direction_B	20.956
Rafael Nadal	Overall	source_direction_B	6.247
Rafael Nadal	Overall	source_direction_B	1.174
Rafael Nadal	Clay	source_direction_B	44.386
Rafael Nadal	Clay	source_direction_B	30.936
Rafael Nadal	Clay	source_direction_B	22.640
Rafael Nadal	Clay	source_direction_B	7.464
Rafael Nadal	Clay	source_direction_B	1.932
Rafael Nadal	Grass	source_direction_B	53.357
Rafael Nadal	Grass	source_direction_B	27.584
Rafael Nadal	Grass	source_direction_B	23.050
Rafael Nadal	Grass	source_direction_B	5.174
Rafael Nadal	Grass	source_direction_B	0.481
Rafael Nadal	Hard	source_direction_B	46.189
Rafael Nadal	Hard	source_direction_B	31.734
Rafael Nadal	Hard	source_direction_B	19.875
Rafael Nadal	Hard	source_direction_B	5.784
Rafael Nadal	Hard	source_direction_B	0.912
Novak Djokovic	Overall	source_direction_B	49.734
Novak Djokovic	Overall	source_direction_B	29.323
Novak Djokovic	Overall	source_direction_B	16.707
Novak Djokovic	Overall	source_direction_B	7.006
Novak Djokovic	Overall	source_direction_B	0.263

player	surface	distribution	ci95_upper
Novak Djokovic	Clay	source_direction_B	52.006
Novak Djokovic	Clay	source_direction_B	29.502
Novak Djokovic	Clay	source_direction_B	17.836
Novak Djokovic	Clay	source_direction_B	7.479
Novak Djokovic	Clay	source_direction_B	0.869
Novak Djokovic	Grass	source_direction_B	51.531
Novak Djokovic	Grass	source_direction_B	28.276
Novak Djokovic	Grass	source_direction_B	20.511
Novak Djokovic	Grass	source_direction_B	8.383
Novak Djokovic	Grass	source_direction_B	0.081
Novak Djokovic	Hard	source_direction_B	49.460
Novak Djokovic	Hard	source_direction_B	30.415
Novak Djokovic	Hard	source_direction_B	16.204
Novak Djokovic	Hard	source_direction_B	7.001
Novak Djokovic	Hard	source_direction_B	0.081
Roger Federer	Overall	source_direction_S	56.691
Roger Federer	Overall	source_direction_S	27.717
Roger Federer	Overall	source_direction_S	16.303
Roger Federer	Overall	source_direction_S	2.820
Roger Federer	Overall	source_direction_S	0.200
Roger Federer	Clay	source_direction_S	56.212
Roger Federer	Clay	source_direction_S	31.645
Roger Federer	Clay	source_direction_S	19.320
Roger Federer	Clay	source_direction_S	2.316
Roger Federer	Clay	source_direction_S	0.276
Roger Federer	Grass	source_direction_S	53.800
Roger Federer	Grass	source_direction_S	31.191
Roger Federer	Grass	source_direction_S	19.896
Roger Federer	Grass	source_direction_S	2.955
Roger Federer	Grass	source_direction_S	0.239
Roger Federer	Hard	source_direction_S	58.650
Roger Federer	Hard	source_direction_S	26.897
Roger Federer	Hard	source_direction_S	15.527
Roger Federer	Hard	source_direction_S	3.190
Roger Federer	Hard	source_direction_S	0.232
Rafael Nadal	Overall	source_direction_S	23.105
Rafael Nadal	Overall	source_direction_S	40.839
Rafael Nadal	Overall	source_direction_S	34.194
Rafael Nadal	Overall	source_direction_S	7.532
Rafael Nadal	Overall	source_direction_S	0.753

player	surface	distribution	ci95_upper
Rafael Nadal	Clay	source_direction_S	20.695
Rafael Nadal	Clay	source_direction_S	43.586
Rafael Nadal	Clay	source_direction_S	37.335
Rafael Nadal	Clay	source_direction_S	7.992
Rafael Nadal	Clay	source_direction_S	1.743
Rafael Nadal	Grass	source_direction_S	29.461
Rafael Nadal	Grass	source_direction_S	38.983
Rafael Nadal	Grass	source_direction_S	39.325
Rafael Nadal	Grass	source_direction_S	8.360
Rafael Nadal	Grass	source_direction_S	0.000
Rafael Nadal	Hard	source_direction_S	24.412
Rafael Nadal	Hard	source_direction_S	41.832
Rafael Nadal	Hard	source_direction_S	33.425
Rafael Nadal	Hard	source_direction_S	7.893
Rafael Nadal	Hard	source_direction_S	0.676
Novak Djokovic	Overall	source_direction_S	54.424
Novak Djokovic	Overall	source_direction_S	31.429
Novak Djokovic	Overall	source_direction_S	14.046
Novak Djokovic	Overall	source_direction_S	4.741
Novak Djokovic	Overall	source_direction_S	0.228
Novak Djokovic	Clay	source_direction_S	50.857
Novak Djokovic	Clay	source_direction_S	34.406
Novak Djokovic	Clay	source_direction_S	20.588
Novak Djokovic	Clay	source_direction_S	5.356
Novak Djokovic	Clay	source_direction_S	0.744
Novak Djokovic	Grass	source_direction_S	57.121
Novak Djokovic	Grass	source_direction_S	32.414
Novak Djokovic	Grass	source_direction_S	15.831
Novak Djokovic	Grass	source_direction_S	8.202
Novak Djokovic	Grass	source_direction_S	0.252
Novak Djokovic	Hard	source_direction_S	56.025
Novak Djokovic	Hard	source_direction_S	31.721
Novak Djokovic	Hard	source_direction_S	13.162
Novak Djokovic	Hard	source_direction_S	4.369
Novak Djokovic	Hard	source_direction_S	0.205

`Invalid source-direction count rows excluded: 0`

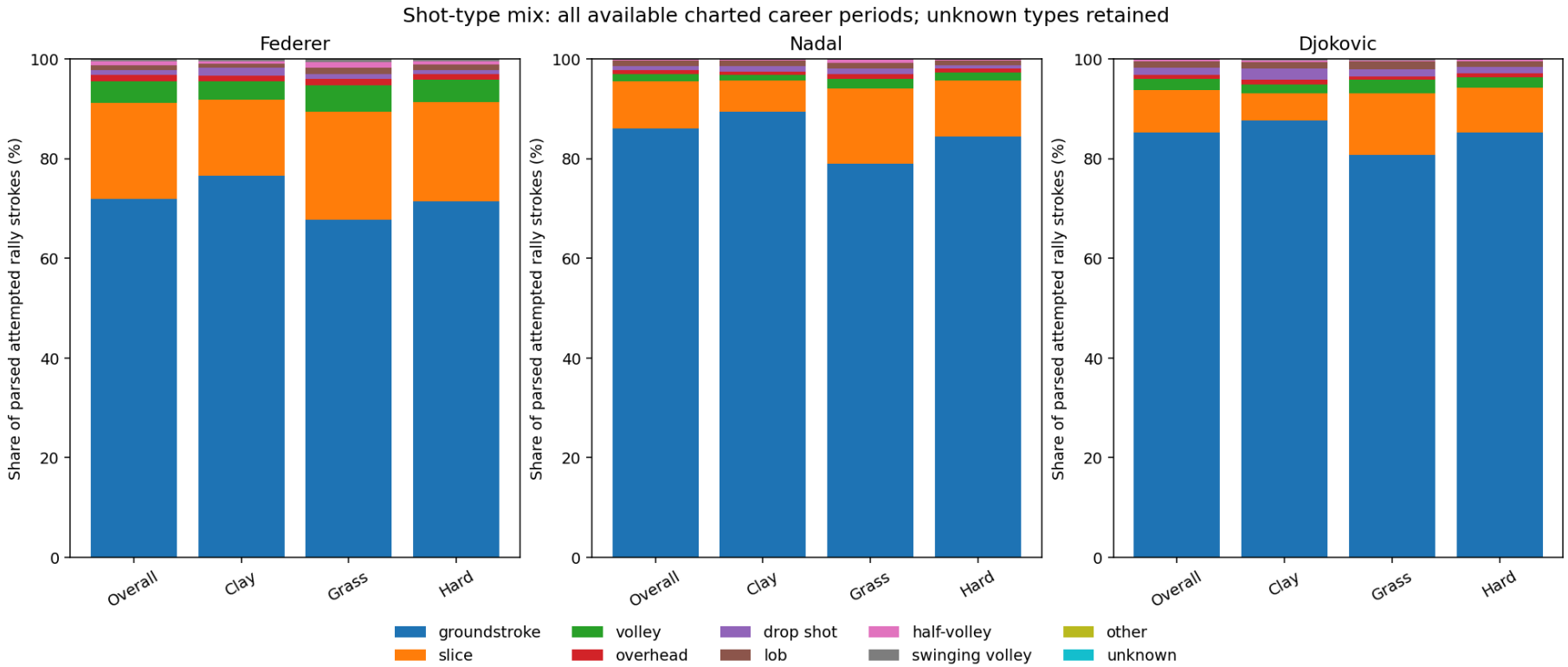
No records in this review table.

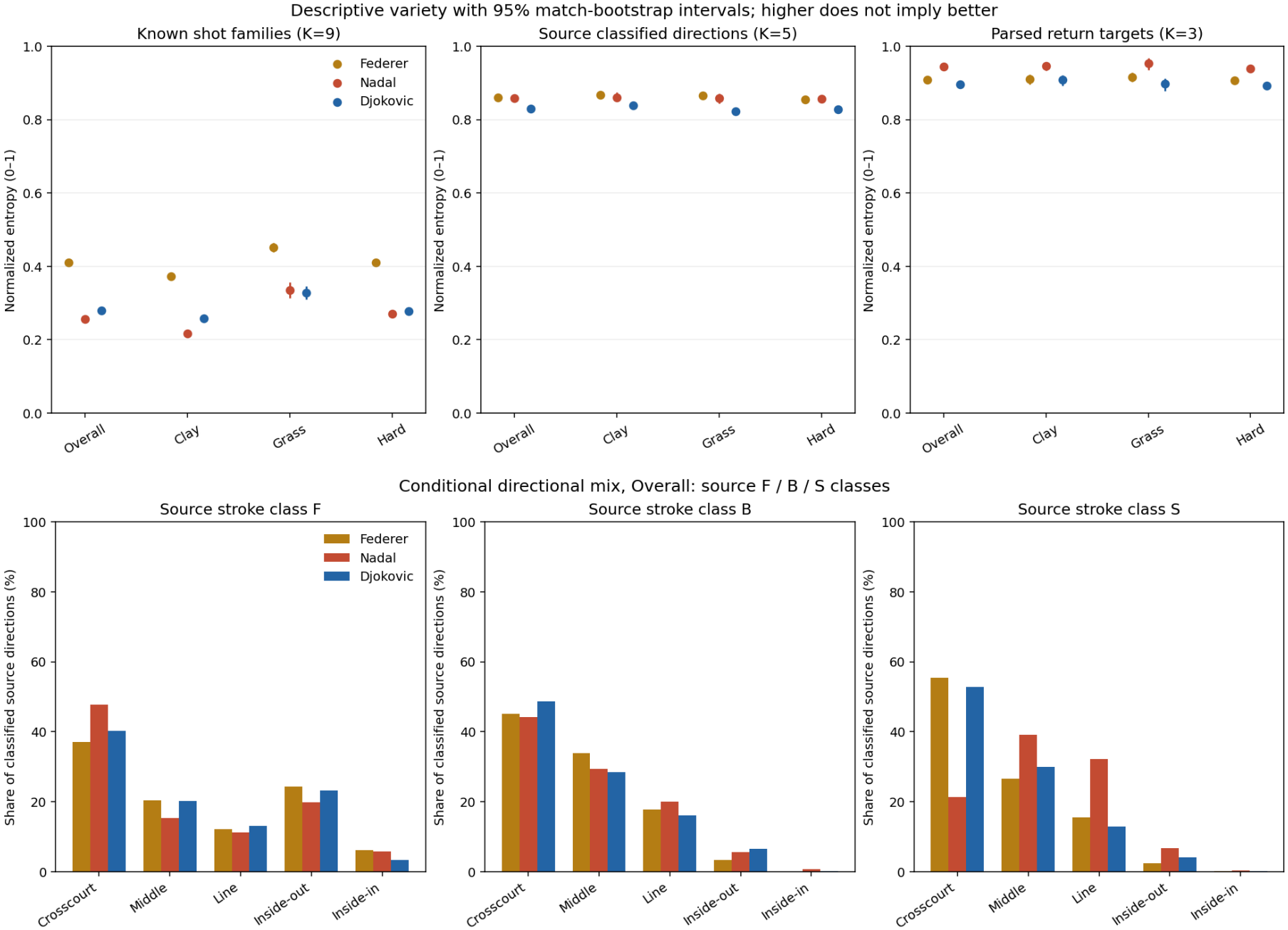
7.2 Shot mix and conditional directional variety

Shot-type mix uses all parsed attempted rally strokes, including unknown types. Entropy uses known types only; classifications and K are fixed across players and surfaces. Source directional shares and their F/B/S breakdown use the source's classified counts; they are not interchangeable with the parser's three target-side codes. Returns have a separate three-target distribution. All rates pool counts before division; bootstrap draws resample matches jointly rather than treating strokes as independent observations. No uncertainty interval addresses uncharted or misclassified strokes.

In [20]:
```
def get_shot_distribution(player, surface, distribution, categories):
    return shot_distribution_long.loc[shot_distribution_long["player"].eq(player)
        & shot_distribution_long["surface"].eq(surface) & shot_distribution_long["distribution"].eq(distribution)].set_index("category").reindex(categories)

with plt.style.context("default"):
    family_colors = plt.get_cmap("tab20")(np.linspace(0, .9, len(SHOT_ALL_FAMILIES)))
    fig, axes = plt.subplots(1, 3, figsize=(14, 6), constrained_layout=True)
    for ax, player in zip(axes, BIG_THREE):
        bottom = np.zeros(4)
        for family, color in zip(SHOT_ALL_FAMILIES, family_colors):
            values = [get_shot_distribution(player, scope, "shot_type", SHOT_ALL_FAMILIES).loc[family, "share_pct"] for scope in ["Overall"] + PROJECT_SURFACES]
            ax.bar(range(4), values, bottom=bottom, label=family, color=color)
            bottom += np.nan_to_num(values)
        ax.set_xticks(range(4), ["Overall"] + PROJECT_SURFACES, rotation=30)
        ax.set_ylim(0, 100)
        ax.set_title(KPI_SHORT_NAMES[player])
        ax.set_ylabel("Share of parsed attempted rally strokes (%)")
    handles, labels = axes[0].get_legend_handles_labels()
    fig.legend(handles, labels, loc="outside lower center", ncol=5, frameon=False)
    fig.suptitle("Shot-type mix: all available charted career periods; unknown types retained", fontsize=13)
    display_kpi_figure(fig)

    fig, axes = plt.subplots(1, 3, figsize=(14, 5), constrained_layout=True)
    for ax, distribution, title in zip(axes,
        ["shot_type", "source_direction_Total", "target_returns"],
        ["Known shot families (K=9)", "Source classified directions (K=5)", "Parsed return targets (K=3)"]):
        for index, player in enumerate(BIG_THREE):
            rows = shot_diversity_summary.loc[shot_diversity_summary["player"].eq(player)
                & shot_diversity_summary["distribution"].eq(distribution)].set_index("surface").reindex(["Overall"]+PROJECT_SURFACES)
            x = np.arange(4) + (index-1)*.16
            ax.scatter(x, rows["normalized_entropy"], color=KPI_PLAYER_COLORS[player], label=KPI_SHORT_NAMES[player], s=35)
            for xpos, (_, row) in zip(x, rows.iterrows()):
                ax.vlines(xpos, row["ci95_lower"], row["ci95_upper"], color=KPI_PLAYER_COLORS[player])
        ax.set_xticks(range(4), ["Overall"]+PROJECT_SURFACES, rotation=30)
        ax.set_ylim(0, 1)
        ax.set_title(title)
        ax.set_ylabel("Normalized entropy (0–1)")
        ax.grid(axis="y", alpha=.2)
    axes[0].legend(frameon=False)
    fig.suptitle("Descriptive variety with 95% match-bootstrap intervals; higher does not imply better", fontsize=13)
    display_kpi_figure(fig)
```

```
fig, axes = plt.subplots(1, 3, figsize=(14, 5), constrained_layout=True)
for ax, stroke_class in zip(axes, ["F", "B", "S"]):
    for index, player in enumerate(BIG_THREE):
        data = get_shot_distribution(player, "Overall", "source_direction_"+stroke_class, SOURCE_DIRECTION_CATEGORIES)
        ax.bar(np.arange(5)+(index-1)*.25, data["share_pct"], width=.25, color=KPI_PLAYER_COLORS[player], label=KPI_SHORT_NAMES[player])
    ax.set_xticks(range(5), ["Crosscourt", "Middle", "Line", "Inside-out", "Inside-in"], rotation=35, ha="right")
    ax.set_ylim(0, 100)
    ax.set_title("Source stroke class " + stroke_class)
    ax.set_ylabel("Share of classified source directions (%)")
axes[0].legend(frameon=False)
fig.suptitle("Conditional directional mix, Overall: source F / B / S classes", fontsize=13)
display_kpi_figure(fig)
```




Descriptive variety with 95% match-bootstrap intervals; higher does not imply better
Known shot families (K=9)
Source classified directions (K=5)
Parsed return targets (K=3)
Normalized entropy (0–1)
Federer
Nadal
Djokovic
Overall
Clay
Grass
Hard
Conditional directional mix, Overall: source F / B / S classes
Source stroke class F
Source stroke class B
Source stroke class S
Share of classified source directions (%)
Crosscourt
Middle
Line
Inside-out
Inside-in

7.3 Court-zone illustrations, not ball tracking

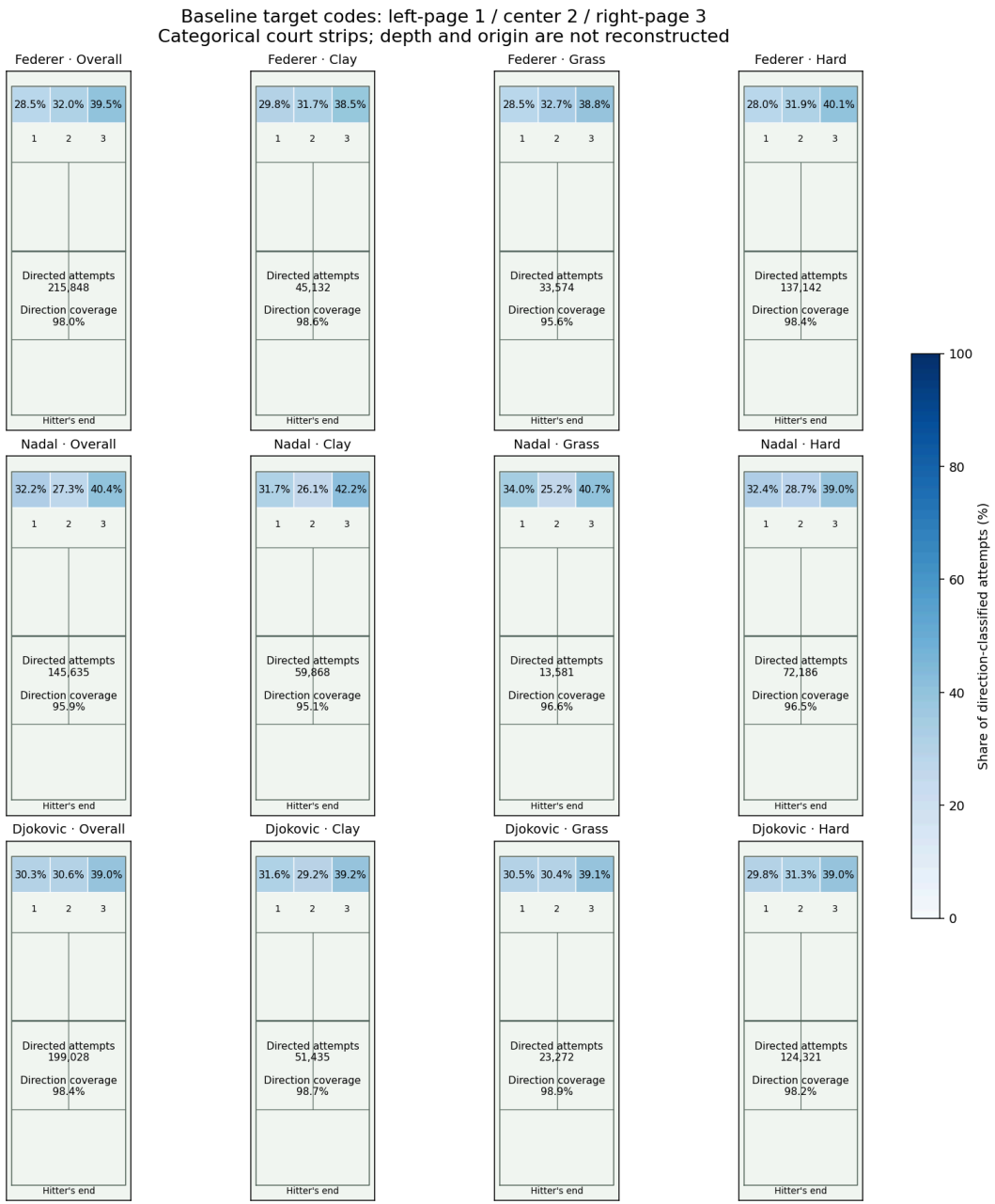
The first court grid displays **baseline-direction shares** for all parsed attempted rally strokes. It places categorical color strips at the opponent baseline to emphasize the workbook's baseline-crossing definition; it does not estimate bounce depth, shot origins, player movement, or tracking coordinates. Code 1 is the opponent's right side (left of the page), code 2 center, and code 3 the opponent's left. Nadal's left-handedness does not reverse these target labels. Shot-type and target-side shares cannot identify crosscourt forehands without the appropriate directional classification.

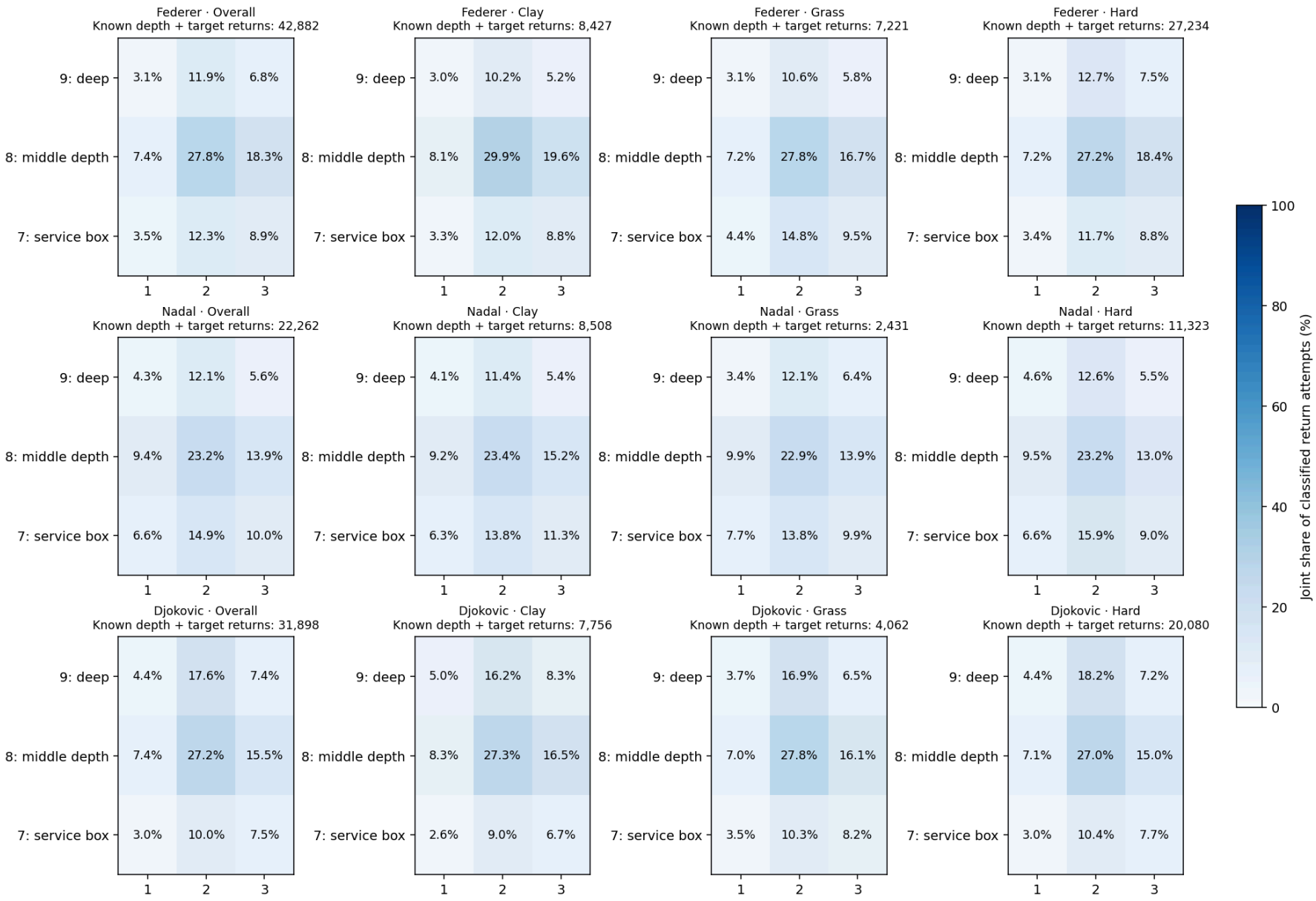
The second heatmap shows the joint categories for **return attempts with both recorded depth and direction**. Depth 7 means service-box depth, 8 the nearer half between service line and baseline, and 9 the deeper half. Unknown depth is excluded with its coverage reported; it is never projected to the baseline. Joint cells sum to 100% within each panel. These are categorical code combinations, not precise landing coordinates. Error attempts may indicate intended/coded targets rather than successful landings. No KDE, synthetic jitter, or inferred origin positions are used.

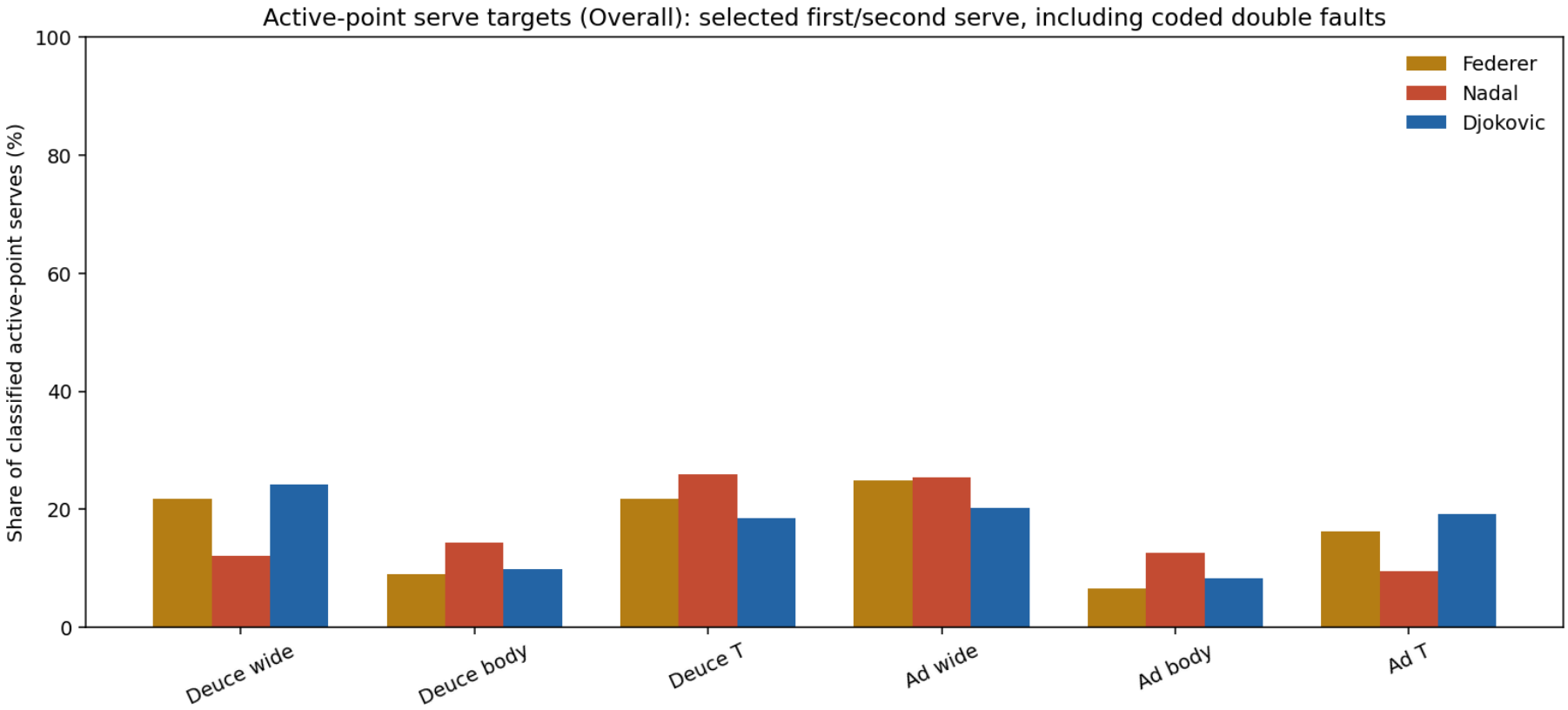
Active-point serve placement uses the selected first or second serve, including coded second-serve faults, and classifies deuce/ad × wide/body/T where score parity and direction are known. It excludes first-serve faults that precede a second serve from this particular denominator; it is not the distribution of every serve attempt. Unknown direction, missing sequence, and no-ad court choice reduce classification coverage.

In [21]:
```
def draw_reference_court(ax):
    # Regulation singles reference geometry, in metres; no measured positions are implied.
    half_width, baseline, service = 4.115, 11.885, 6.4
    ax.set_facecolor("#f1f5f2")
    ax.plot([-half_width, half_width, half_width, -half_width, -half_width],
            [-baseline, -baseline, baseline, baseline, -baseline], color="#53645a", linewidth=.8)
    ax.hlines([-service, 0, service], -half_width, half_width, color="#53645a", linewidth=[.6, 1.3, .6])
    ax.vlines(0, -service, service, color="#53645a", linewidth=.6)
    ax.set_xlim(-4.5, 4.5)
    ax.set_ylim(-13, 13)
    ax.set_aspect("equal")
    ax.set_xticks([])
    ax.set_yticks([])
    ax.text(0, -12.5, "Hitter's end", ha="center", fontsize=7)

with plt.style.context("default"):
    cmap = plt.get_cmap("Blues")
    norm = plt.Normalize(0, 100)
    fig, axes = plt.subplots(3, 4, figsize=(12, 13), constrained_layout=True)
    for row_index, player in enumerate(BIG_THREE):
        for column, scope in enumerate(["Overall"]+PROJECT_SURFACES):
            ax = axes[row_index, column]
            draw_reference_court(ax)
```

```
            data = get_shot_distribution(player, scope, "target_all_attempts", TARGET_CATEGORIES)
            for index, category in enumerate(TARGET_CATEGORIES):
                value = data.loc[category, "share_pct"]
                left = -4.115 + index*(8.23/3)
                ax.add_patch(plt.Rectangle((left, 9.3), 8.23/3, 2.585, facecolor=cmap(norm(value)) if pd.notna(value) else "lightgray", edgecolor="white",
linewidth=.7))
                ax.text(left+8.23/6, 10.55, f"{value:.1f}%" if pd.notna(value) else "n/a", ha="center", va="center", fontsize=8, color="white" if value > 55
else "black")
            coverage = shot_coverage_summary.loc[shot_coverage_summary["player"].eq(player) & shot_coverage_summary["surface"].eq(scope)].iloc[0]
            ax.text(0, 7.9, "1          2          3", ha="center", fontsize=7)
            ax.text(0, -3.5, f"Directed attempts\n{int(coverage['classified_targets']):,}\n\nDirection
coverage\n{coverage['target_direction_coverage_pct']:.1f}%", ha="center", va="center", fontsize=8)
            ax.set_title(KPI_SHORT_NAMES[player]+" · "+scope, fontsize=10)
    fig.colorbar(plt.cm.ScalarMappable(norm=norm, cmap=cmap), ax=axes, shrink=.5, label="Share of direction-classified attempts (%)")
    fig.suptitle("Baseline target codes: left-page 1 / center 2 / right-page 3\nCategorical court strips; depth and origin are not reconstructed", fontsize=14)
    display_kpi_figure(fig)

    fig, axes = plt.subplots(3, 4, figsize=(14, 10), constrained_layout=True)
    for row_index, player in enumerate(BIG_THREE):
        for column, scope in enumerate(["Overall"]+PROJECT_SURFACES):
            ax = axes[row_index, column]
            data = get_shot_distribution(player, scope, "return_depth_target", RETURN_JOINT_CATEGORIES)
            matrix = data["share_pct"].to_numpy().reshape(3, 3)
            ax.imshow(matrix, origin="lower", vmin=0, vmax=100, cmap="Blues", aspect="auto")
            for y in range(3):
                for x in range(3):
                    value = matrix[y, x]
                    ax.text(x, y, f"{value:.1f}%" if pd.notna(value) else "n/a", ha="center", va="center", fontsize=9, color="white" if value > 55 else "black")
            ax.set_xticks(range(3), ["1", "2", "3"])
            ax.set_yticks(range(3), ["7: service box", "8: middle depth", "9: deep"])
            n = int(data["classified_denominator"].iloc[0])
            ax.set_title(KPI_SHORT_NAMES[player]+" · "+scope+f"\nKnown depth + target returns: {n:,}", fontsize=9)
    fig.colorbar(plt.cm.ScalarMappable(norm=norm, cmap=cmap), ax=axes, shrink=.6, label="Joint share of classified return attempts (%)")
    fig.suptitle("Recorded return depth × baseline target code\nCategorical heatmap; missing depth excluded, no fabricated coordinates", fontsize=14)
    display_kpi_figure(fig)

    fig, ax = plt.subplots(figsize=(11, 5), constrained_layout=True)
    for index, player in enumerate(BIG_THREE):
        data = get_shot_distribution(player, "Overall", "active_serve_target", SERVE_ZONE_CATEGORIES)
        ax.bar(np.arange(6)+(index-1)*.25, data["share_pct"], width=.25, color=KPI_PLAYER_COLORS[player], label=KPI_SHORT_NAMES[player])
    ax.set_xticks(range(6), ["Deuce wide", "Deuce body", "Deuce T", "Ad wide", "Ad body", "Ad T"], rotation=25)
    ax.set_ylabel("Share of classified active-point serves (%)")
    ax.set_ylim(0, 100)
    ax.legend(frameon=False)
    ax.set_title("Active-point serve targets (Overall): selected first/second serve, including coded double faults")
    display_kpi_figure(fig)
```


Baseline target codes: left-page 1 / center 2 / right-page 3
Categorical court strips; depth and origin are not reconstructed
Federer · Overall
28.5%
32.0%
39.5%
1
2
3
Directed attempts
215,848
Direction coverage
98.0%
Hitter's end
Federer · Clay
29.8%
31.7%
38.5%
1
2
3
Directed attempts
45,132
Direction coverage
98.6%
Hitter's end
Federer · Grass
28.5%
32.7%
38.8%
1
2
3
Directed attempts
33,574
Direction coverage
95.6%
Hitter's end
Federer · Hard
28.0%
31.9%
40.1%
1
2
3
Directed attempts
137,142
Direction coverage
98.4%
Hitter's end
Nadal · Overall
32.2%
27.3%
40.4%
1
2
3
Directed attempts
145,635
Direction coverage
95.9%
Hitter's end
Nadal · Clay
31.7%
26.1%
42.2%
1
2
3
Directed attempts
59,868
Direction coverage
95.1%
Hitter's end
Nadal · Grass
34.0%
25.2%
40.7%
1
2
3
Directed attempts
13,581
Direction coverage
96.6%
Hitter's end
Nadal · Hard
32.4%
28.7%
39.0%
1
2
3
Directed attempts
72,186
Direction coverage
96.5%
Hitter's end
Djokovic · Overall
30.3%
30.6%
39.0%
1
2
3
Directed attempts
199,028
Direction coverage
98.4%
Hitter's end
Djokovic · Clay
31.6%
29.2%
39.2%
1
2
3
Directed attempts
51,435
Direction coverage
98.7%
Hitter's end
Djokovic · Grass
30.5%
30.4%
39.1%
1
2
3
Directed attempts
23,272
Direction coverage
98.9%
Hitter's end
Djokovic · Hard
29.8%
31.3%
39.0%
1
2
3
Directed attempts
124,321
Direction coverage
98.2%
Hitter's end
Share of direction-classified attempts (%)
100
80
60
40
20
0


Recorded return depth × baseline target code
Categorical heatmap; missing depth excluded, no fabricated coordinates
Federer · Overall
Known depth + target returns: 42,882
9: deep
8: middle depth
7: service box
3.1%
11.9%
6.8%
7.4%
27.8%
18.3%
3.5%
12.3%
8.9%
1
2
3
Federer · Clay
Known depth + target returns: 8,427
3.0%
10.2%
5.2%
8.1%
29.9%
19.6%
3.3%
12.0%
8.8%
Federer · Grass
Known depth + target returns: 7,221
3.1%
10.6%
5.8%
7.2%
27.8%
16.7%
4.4%
14.8%
9.5%
Federer · Hard
Known depth + target returns: 27,234
3.1%
12.7%
7.5%
7.2%
27.2%
18.4%
3.4%
11.7%
8.8%
Nadal · Overall
Known depth + target returns: 22,262
4.3%
12.1%
5.6%
9.4%
23.2%
13.9%
6.6%
14.9%
10.0%
Nadal · Clay
Known depth + target returns: 8,508
4.1%
11.4%
5.4%
9.2%
23.4%
15.2%
6.3%
13.8%
11.3%
Nadal · Grass
Known depth + target returns: 2,431
3.4%
12.1%
6.4%
9.9%
22.9%
13.9%
7.7%
13.8%
9.9%
Nadal · Hard
Known depth + target returns: 11,323
4.6%
12.6%
5.5%
9.5%
23.2%
13.0%
6.6%
15.9%
9.0%
Djokovic · Overall
Known depth + target returns: 31,898
4.4%
17.6%
7.4%
7.4%
27.2%
15.5%
3.0%
10.0%
7.5%
Djokovic · Clay
Known depth + target returns: 7,756
5.0%
16.2%
8.3%
8.3%
27.3%
16.5%
2.6%
9.0%
6.7%
Djokovic · Grass
Known depth + target returns: 4,062
3.7%
16.9%
6.5%
7.0%
27.8%
16.1%
3.5%
10.3%
8.2%
Djokovic · Hard
Known depth + target returns: 20,080
4.4%
18.2%
7.2%
7.1%
27.0%
15.0%
3.0%
10.4%
7.7%
Joint share of classified return attempts (%)
0
20
40
60
80
100



7.4 Interpretation and limitations

The tables distinguish stroke family, source-defined direction, return target, return depth, and active serve placement. Entropy is descriptive: a reliable repeated crosscourt pattern may be more effective than a varied but unsuccessful pattern. Stroke-type diversity does not measure athleticism, invention, or match-winning ability. Differences may reflect opponents, career periods, surfaces, error rates, and charting coverage. Conditional F/B/S and return tables help explain aggregates without manufacturing shot combinations absent from the sources.

The court panels keep a common color scale and expose how much direction and depth information was actually recorded. A repeated crosscourt pattern and a varied menu of strokes can each have a tactical purpose; neither becomes effective merely because its frequency looks distinctive. The next question is whether these frequencies changed within each observed career.

7.5 Evidence: variety and repeated patterns

In [22]:
```
shot_headline_rows=[]
for player in BIG_THREE:
    coverage=shot_coverage_summary.loc[shot_coverage_summary["player"].eq(player)&shot_coverage_summary["surface"].eq("Overall")].iloc[0]
    families=get_shot_distribution(player,"Overall","shot_type",SHOT_ALL_FAMILIES)

entropy=shot_diversity_summary.loc[shot_diversity_summary["player"].eq(player)&shot_diversity_summary["surface"].eq("Overall")&shot_diversity_summary["distribution"].eq("shot_type")].iloc[0]
    shot_headline_rows.append({"player":player,"attempted_rally_strokes":coverage.attempted_rally_strokes,
        "groundstroke_share_pct":families.loc["groundstroke","share_pct"],"slice_share_pct":families.loc["slice","share_pct"],
        "volley_share_pct":families.loc["volley","share_pct"],"known_family_entropy":entropy.normalized_entropy,
        "entropy_ci95_lower":entropy.ci95_lower,"entropy_ci95_upper":entropy.ci95_upper,
        "target_coverage_pct":coverage.target_direction_coverage_pct,"return_depth_coverage_pct":coverage.return_depth_coverage_pct})
shot_headline=pd.DataFrame(shot_headline_rows)
report_table(shot_headline,caption="Stroke-family shares use all attempted rally strokes; entropy uses known families",expanded=True)
most_varied=shot_headline.sort_values("known_family_entropy",ascending=False).iloc[0]
lines=[f"**Observed result.** {most_varied.player} has the highest observed known-family entropy ({most_varied.known_family_entropy:.3f}, 95% interval {interval_text(most_varied.entropy_ci95_lower,most_varied.entropy_ci95_upper,3)}). The table shows the groundstroke/slice/volley mix behind the index and the separate direction/depth coverage. This is a descriptive ordering, not a tested superiority claim.",
    "**Tennis interpretation.** A slice can change pace or trajectory, a volley can end a point at net, and a repeated groundstroke pattern can constrain an opponent. These are plausible uses of the coded strokes; the present distributions do not measure their causal effectiveness. Crosscourt versus line shares use source directional categories, while target strips show baseline-crossing sides. The heatmap does not establish where players stood or where each ball bounced.",
    "**Alternative explanations.** Stroke frequencies depend on surface, opponent, rally opportunities, era, error coding, and which matches were charted. Missing return depth is substantial and need not be random; the heatmap is conditional on recorded depth and direction. Source handedness anomalies can affect source-derived classifications, so coded target sides and source directions remain separate. Next we ask whether tactical frequencies change within the observed careers."]
display(Markdown("\n\n".join(lines)))
```

Stroke-family shares use all attempted rally strokes; entropy uses known families

player	attempted_rally_strokes	groundstroke_share_pct	slice_share_pct	volley_share_pct	known_family_entropy	entropy_ci95_lower	entropy_ci95_upper	target_coverage_pct	return_depth_coverage_pct
Roger Federer	220280	71.843	19.249	4.470	0.410	0.406	0.415	97.988	74.810
Rafael Nadal	151792	85.980	9.515	1.479	0.257	0.250	0.264	95.944	80.794
Novak Djokovic	202209	85.295	8.467	2.139	0.280	0.274	0.286	98.427	76.924

Observed result. Roger Federer has the highest observed known-family entropy (0.410, 95% interval 0.406–0.415). The table shows the groundstroke/slice/volley mix behind the index and the separate direction/depth coverage. This is a descriptive ordering, not a tested superiority claim.

Tennis interpretation. A slice can change pace or trajectory, a volley can end a point at net, and a repeated groundstroke pattern can constrain an opponent. These are plausible uses of the coded strokes; the present distributions do not measure their causal effectiveness. Crosscourt versus line shares use source directional categories, while target strips show baseline-crossing sides. The heatmap does not establish where players stood or where each ball bounced.

Alternative explanations. Stroke frequencies depend on surface, opponent, rally opportunities, era, error coding, and which matches were charted. Missing return depth is substantial and need not be random; the heatmap is conditional on recorded depth and direction. Source handedness anomalies can affect source-derived classifications, so coded target sides and source directions remain separate. Next we ask whether tactical frequencies change within the observed careers.

VIII. Surface adaptability and tactical evolution

Evolution question: How balanced are observed results across surfaces, and which tactical frequencies change over supported career periods? Cross-surface success and within-career tactical change answer separate questions.

8.1 Primary adaptability definition and sensitivity

Use known match outcomes from each player's entire available charted career on Clay, Grass, and Hard. Let p be the pooled win rate in this three-surface universe and p_s the rate on surface s. **Parity = 100 × [1 – sqrt(mean_s((p_s – p)²))]**. **SAS = 0.60 × parity + 0.40 × [100 × min_s(p_s)]**. Require evidence on all three surfaces; a score for an individual surface would not measure cross-surface adaptability. Unknown outcomes remain excluded and counted separately.

Parity rewards consistency, including consistency at modest win rates. The weakest-surface component adds winning strength. The 60/40 weights are a declared preference, not a learned truth. Report sensitivity over parity weights from 0 to 1 and an alternative equal-surface anchor for p to expose dependence on surface scheduling. The primary definition remains unchanged. Joint match bootstraps recompute all surface rates, parity, weakest rate, and SAS; the weakest surface can change in a replicate. Small-sample flags apply when any surface has fewer than 30 known-outcome matches. Intervals condition on charted selection and verified outcomes.

In [23]:

```
adaptability_rows, adaptability_bootstraps, adaptability_weight_rows = [], {}, []
for player in BIG_THREE:
    group = kpi_master.loc[kpi_master["player"].eq(player) & kpi_master["surface"].isin(PROJECT_SURFACES)]
    point_counts = []
    vectors = []
    for surface in PROJECT_SURFACES:
        sample = group.loc[group["surface"].eq(surface) & group["winner"].notna()]
        point_counts.append([int(sample["is_win"].sum()), len(sample)])
        for value_field in ["verified_win_count", "known_outcome"]:
            vector = np.zeros(len(bootstrap_match_ids))
            for mid, value in sample[["match_id", value_field]].itertuples(index=False, name=None):
                vector[bootstrap_match_index[mid]] = value
            vectors.append(vector)
    counts = np.array(point_counts, dtype=float)
    totals = bootstrap_weights @ np.column_stack(vectors)
    with np.errstate(divide="ignore", invalid="ignore"):
        surface_rates = counts[:, 0] / counts[:, 1]
        boot_rates = totals[:, [0, 2, 4]] / totals[:, [1, 3, 5]]
        overall_rate = counts[:, 0].sum() / counts[:, 1].sum()
        boot_overall = totals[:, [0, 2, 4]].sum(axis=1) / totals[:, [1, 3, 5]].sum(axis=1)
        parity = 100 * (1 - np.sqrt(np.mean((surface_rates - overall_rate)**2)))
        boot_parity = 100 * (1 - np.sqrt(np.mean((boot_rates - boot_overall[:, None])**2, axis=1)))
        weakest_rate_pct = 100 * np.min(surface_rates)
        boot_weakest = 100 * np.min(boot_rates, axis=1)
        score = .6 * parity + .4 * weakest_rate_pct
        boot_score = .6 * boot_parity + .4 * boot_weakest
        alternate_parity = 100 * (1 - np.sqrt(np.mean((surface_rates - surface_rates.mean())**2)))
    row = {"player": player, "known_outcome_matches": int(counts[:, 1].sum()),
        "unknown_outcome_matches": int(group["winner"].isna().sum()), "win_rate_overall_pct": 100*overall_rate,
        "parity_score": parity, "weakest_surface_win_pct": weakest_rate_pct,
        "weakest_surface": ", ".join(s for s, rate in zip(PROJECT_SURFACES, surface_rates) if np.isclose(rate, surface_rates.min())) if
np.isfinite(surface_rates).all() else "Unavailable",
        "surface_adaptability_score": score, "equal_surface_anchor_sas_sensitivity": .6*alternate_parity + .4*weakest_rate_pct,
        "small_surface_sample": bool(np.any(counts[:, 1] < SMALL_SAMPLE_THRESHOLDS["matches"]))}
    for surface, count, rate in zip(PROJECT_SURFACES, counts, surface_rates):
        row[surface + "_known_matches"] = int(count[1])
        row[surface + "_wins"] = int(count[0])
        row[surface + "_win_pct"] = 100*rate
    for name, values in [("parity_score", boot_parity), ("weakest_surface_win_pct", boot_weakest), ("surface_adaptability_score", boot_score)]:
        valid = values[np.isfinite(values)]
        row[name + "_valid_bootstraps"] = len(valid)
        row[name + "_ci95_lower"] = np.percentile(valid, 2.5) if len(valid) >= .95*BOOTSTRAP_REPLICATES else np.nan
        row[name + "_ci95_upper"] = np.percentile(valid, 97.5) if len(valid) >= .95*BOOTSTRAP_REPLICATES else np.nan
    adaptability_rows.append(row)
    adaptability_bootstraps[player] = {"surface_rates": boot_rates, "parity": boot_parity, "weakest_rate_pct": boot_weakest, "sas": boot_score}
    for parity_weight in np.linspace(0, 1, 11):
        adaptability_weight_rows.append({"player": player, "parity_weight": parity_weight,
            "weakest_surface_weight": 1-parity_weight, "score": parity_weight*parity+(1-parity_weight)*weakest_rate_pct})
surface_adaptability_summary_v2 = pd.DataFrame(adaptability_rows)
surface_adaptability_weight_sensitivity = pd.DataFrame(adaptability_weight_rows)
surface_adaptability_weight_sensitivity["rank_within_weight"] = surface_adaptability_weight_sensitivity.groupby("parity_weight")["score"].rank(ascending=False,
method="min")
assert surface_adaptability_summary_v2["surface_adaptability_score"].dropna().between(0, 100).all()
assert surface_adaptability_summary_v2["parity_score"].dropna().between(0, 100).all()
for player in BIG_THREE:
    row = surface_adaptability_summary_v2.loc[surface_adaptability_summary_v2["player"].eq(player)].iloc[0]
```

```
    source = core_kpi_by_surface.loc[core_kpi_by_surface["player"].eq(player)].set_index("surface")
    for surface in PROJECT_SURFACES:
        assert row[surface+"_known_matches"] == source.loc[surface, "known_outcome_matches"]
        assert np.isclose(row[surface+"_win_pct"], source.loc[surface, "match_win_pct"])
    assert np.isclose(row["surface_adaptability_score"], .6*row["parity_score"]+.4*row["weakest_surface_win_pct"])
print("Primary surface adaptability, evidence rates, and uncertainty:")
with pd.option_context("display.max_columns", None):
    report_table(surface_adaptability_summary_v2.round(4))
print("Weight sensitivity; descriptive score ranks are conditional on the definition:")
with pd.option_context("display.max_rows", None):
    report_table(surface_adaptability_weight_sensitivity.round(4))
```

```
Primary surface adaptability, evidence rates, and uncertainty:
```

▼ Complete evidence table: 3 rows × 28 columns

player	known_outcome_matches	unknown_outcome_matches	win_rate_overall_pct	parity_score	weakest_surface_win_pct	weakest_surface	surface_adaptability_score
Roger Federer	709	14	78.843	93.652	69.048	Clay	83.810
Rafael Nadal	412	13	72.330	88.822	59.459	Grass	77.077
Novak Djokovic	526	27	76.806	95.850	70.161	Clay	85.575

player	equal_surface_anchor_sas_sensitivity	small_surface_sample	Clay_known_matches	Clay_wins	Clay_win_pct	Grass_known_matches	Grass_wins
Roger Federer	83.877	False	126	87	69.048	116	97
Rafael Nadal	77.263	False	164	139	84.756	37	22
Novak Djokovic	85.638	False	124	87	70.161	56	44

player	Grass_win_pct	Hard_known_matches	Hard_wins	Hard_win_pct	parity_score_valid_bootstraps	parity_score_ci95_lower	parity_score_ci95_upper
Roger Federer	83.621	467	375	80.300	2000	89.161	97.433
Rafael Nadal	59.459	211	137	64.929	2000	81.609	93.377
Novak Djokovic	78.571	346	273	78.902	2000	90.474	98.699

player	weakest_surface_win_pct_valid_bootstraps	weakest_surface_win_pct_ci95_lower	weakest_surface_win_pct_ci95_upper	surface_adaptability_score_valid_bootstraps	surface_adaptability_score_ci95_lower	surface_adaptability_score_ci95_upper
Roger Federer	2000	60.869	76.787	2000	77.811	88.900
Rafael Nadal	2000	43.583	68.185	2000	66.502	83.046
Novak Djokovic	2000	61.817	76.745	2000	79.299	89.564

```
Weight sensitivity; descriptive score ranks are conditional on the definition:
```

player	parity_weight	weakest_surface_weight	score	rank_within_weight
Roger Federer	0.000	1.000	69.048	2.000
Roger Federer	0.100	0.900	71.508	2.000
Roger Federer	0.200	0.800	73.968	2.000
Roger Federer	0.300	0.700	76.429	2.000
Roger Federer	0.400	0.600	78.889	2.000
Roger Federer	0.500	0.500	81.350	2.000
Roger Federer	0.600	0.400	83.810	2.000
Roger Federer	0.700	0.300	86.270	2.000
Roger Federer	0.800	0.200	88.731	2.000
Roger Federer	0.900	0.100	91.191	2.000
Roger Federer	1.000	0.000	93.652	2.000
Rafael Nadal	0.000	1.000	59.459	3.000
Rafael Nadal	0.100	0.900	62.396	3.000
Rafael Nadal	0.200	0.800	65.332	3.000
Rafael Nadal	0.300	0.700	68.268	3.000
Rafael Nadal	0.400	0.600	71.204	3.000
Rafael Nadal	0.500	0.500	74.141	3.000
Rafael Nadal	0.600	0.400	77.077	3.000
Rafael Nadal	0.700	0.300	80.013	3.000
Rafael Nadal	0.800	0.200	82.950	3.000
Rafael Nadal	0.900	0.100	85.886	3.000
Rafael Nadal	1.000	0.000	88.822	3.000
Novak Djokovic	0.000	1.000	70.161	1.000
Novak Djokovic	0.100	0.900	72.730	1.000
Novak Djokovic	0.200	0.800	75.299	1.000
Novak Djokovic	0.300	0.700	77.868	1.000
Novak Djokovic	0.400	0.600	80.437	1.000
Novak Djokovic	0.500	0.500	83.006	1.000
Novak Djokovic	0.600	0.400	85.575	1.000
Novak Djokovic	0.700	0.300	88.144	1.000
Novak Djokovic	0.800	0.200	90.713	1.000
Novak Djokovic	0.900	0.100	93.281	1.000

player	parity_weight	weakest_surface_weight	score	rank_within_weight
Novak Djokovic	1.000	0.000	95.850	1.000

8.2 Prespecified career windows and tactical inputs

Use **four-calendar-year bins anchored to each player's first charted year**, with the final bin truncated to the last available year. Boundaries use coverage only, not observed tactical changes. The first and last periods are coverage edges, not evidence of a professional debut, retirement date, or a complete calendar period. Show every period, including small samples and unavailable surfaces; do not carry forward missing values or force common career dates.

Approach and NetPoints frequencies use their distinct source slices divided by source-matched service plus return points. Serve-and-volley frequency uses the SnV numerator and service points: a missing SnV row becomes zero **only when the nonSnV complement equals all source service points**. Missing counts can also be recovered from explicit serve-prefix `+` codes when the coded service-point total reconciles to Overview; an absent `+` on a fully coded serve is a genuine non-SnV observation. Source/code disagreements are flagged and excluded. Otherwise missing counts stay unavailable; present rows must reconcile complements or the independently coded serve points. Parsed drop-shot and slice shares use known stroke families, with attempted error strokes included. Rally shares use known rally lengths of at least one shot (serve included, ending error excluded), excluding double faults and entirely uncharted sequences. Serve target shares are conditional on a known court and direction and are calculated separately within deuce and ad courts.

For each metric, show the raw pooled period rate and within-surface rates. A separate **equal-surface standardized** series averages the three surface rates with fixed 1/3 weights, only when all three are available. It does not extrapolate missing grass data. Percentage-point contrasts compare the first and last periods with a supported standardized estimate, identifying the endpoints explicitly; they may exclude career-edge periods without three-surface support. Counts and small-sample flags remain visible, and wide intervals are not evidence of stability. Evolution measures change, not invention, improvement, or influence on tennis.

Endpoint-change intervals are exploratory, unadjusted 95% intervals across multiple tactical measures; they are not a family of confirmatory significance tests.

In [24]:
```
TACTICAL_PERIOD_YEARS = 4
period_plan_rows = []
for player in BIG_THREE:
    dates = canonical_master.loc[canonical_master["player"].eq(player), "match_date"]
    first_year, last_year = int(dates.min().year), int(dates.max().year)
    for index, start in enumerate(range(first_year, last_year+1, TACTICAL_PERIOD_YEARS)):
        end = min(start+TACTICAL_PERIOD_YEARS-1, last_year)
```

```
        sample = canonical_master.loc[canonical_master["player"].eq(player) & canonical_master["match_date"].dt.year.between(start, end)]
        period_plan_rows.append({"player": player, "period_index": index, "period": f"{start}–{end}", "start_year": start,
            "end_year": end, "nominal_years": TACTICAL_PERIOD_YEARS, "coverage_edge": start == first_year or end == last_year,
            "first_charted_date": sample["match_date"].min(), "last_charted_date": sample["match_date"].max(), "charted_matches": len(sample)})
tactical_period_plan = pd.DataFrame(period_plan_rows)
tactical_match_table = kpi_master.copy()
tactical_match_table["period_index"] = [int((date.year - int(tactical_period_plan.loc[tactical_period_plan["player"].eq(player), "start_year"].min())) //
TACTICAL_PERIOD_YEARS)
    for player, date in tactical_match_table[["player", "match_date"]].itertuples(index=False, name=None)]
tactical_match_table = tactical_match_table.merge(tactical_period_plan[["player", "period_index", "period", "coverage_edge"]], on=["player", "period_index"],
how="left", validate="many_to_one")
tactical_match_table["source_player_points"] = tactical_match_table["serve_pts"] + tactical_match_table["return_pts"]
for slice_name, field in [("Approach", "approach_points"), ("NetPoints", "net_points_all")]:
    source = analysis_dataframes["charting_m_stats_NetPoints"]
    selected = source.loc[source["row"].eq(slice_name), ["match_id", "player", "net_pts"]].rename(columns={"net_pts": field})
    tactical_match_table = tactical_match_table.merge(selected, on=["match_id", "player"], how="left", validate="one_to_one")
for slice_name, field in [("SnV", "snv_points"), ("nonSnV", "non_snv_points")]:
    source = analysis_dataframes["charting_m_stats_SnV"]
    selected = source.loc[source["row"].eq(slice_name), ["match_id", "player", "snv_pts"]].rename(columns={"snv_pts": field})
    tactical_match_table = tactical_match_table.merge(selected, on=["match_id", "player"], how="left", validate="one_to_one")
proven_no_snv = tactical_match_table["snv_points"].isna() & tactical_match_table["non_snv_points"].eq(tactical_match_table["serve_pts"]) &
tactical_match_table["serve_pts"].notna()
tactical_match_table["snv_zero_proven_by_complement"] = proven_no_snv
tactical_match_table.loc[proven_no_snv, "snv_points"] = 0
snv_complete = (tactical_match_table["snv_points"].add(tactical_match_table["non_snv_points"]).eq(tactical_match_table["serve_pts"])
    | (tactical_match_table["non_snv_points"].isna() & tactical_match_table["snv_points"].eq(tactical_match_table["serve_pts"])))

# Independently recover SnV from explicit serve-prefix '+' markers, not from missing rows.
def serve_prefix_snv(sequence):
    token = SHOT_TOKEN_PATTERN.search(sequence)
    prefix = sequence[:token.start()] if token else sequence
    return "+" in prefix
snv_point_codes = serve_analysis.loc[serve_analysis["sequence_known"]].copy()
snv_point_codes["snv_from_code"] = snv_point_codes["sequence"].map(serve_prefix_snv).astype(int)
snv_point_codes["coded_service_points"] = 1
snv_code_counts = snv_point_codes.groupby(["match_id", "player"], as_index=False)[["snv_from_code", "coded_service_points"]].sum()
tactical_match_table = tactical_match_table.merge(snv_code_counts, on=["match_id", "player"], how="left", validate="one_to_one")
complete_snv_codes = (tactical_match_table["coded_service_points"].eq(tactical_match_table["serve_pts"])
    & tactical_match_table["serve_pts"].gt(0) & reconciled_mask(["service_points_parsed"]))
source_snv_present = tactical_match_table["snv_points"].notna()
source_snv_code_disagreement = source_snv_present & complete_snv_codes & tactical_match_table["snv_points"].ne(tactical_match_table["snv_from_code"])
snv_code_source_review = tactical_match_table.loc[source_snv_code_disagreement,
    ["match_id", "player", "snv_points", "snv_from_code", "coded_service_points", "serve_pts"]].copy()
recover_snv = ~source_snv_present & complete_snv_codes
tactical_match_table["snv_recovered_from_explicit_codes"] = recover_snv
tactical_match_table.loc[recover_snv, "snv_points"] = tactical_match_table.loc[recover_snv, "snv_from_code"]
snv_complete = ((snv_complete | complete_snv_codes) & ~source_snv_code_disagreement)

shot_counts = shot_analysis.assign(known_family=shot_analysis["shot_family"].ne("unknown").astype(int),
    drop_count=shot_analysis["shot_family"].eq("drop shot").astype(int),
    slice_count=shot_analysis["shot_family"].eq("slice").astype(int)).groupby(["match_id", "player"], as_index=False)[["known_family", "drop_count",
"slice_count"]].sum()
tactical_match_table = tactical_match_table.merge(shot_counts, on=["match_id", "player"], how="left", validate="one_to_one")
rallies = canonical_player_points.loc[canonical_player_points["rally_length"].notna() & canonical_player_points["rally_length"].ge(1)].copy()
rallies["known_rallies"] = 1
rallies["short_rallies"] = rallies["rally_length"].le(4).astype(int)
rallies["medium_rallies"] = rallies["rally_length"].between(5, 8).astype(int)
```

```
rallies["long_rallies"] = rallies["rally_length"].ge(9).astype(int)
rally_counts = rallies.groupby(["match_id", "player"], as_index=False)[["known_rallies", "short_rallies", "medium_rallies", "long_rallies"]].sum()
tactical_match_table = tactical_match_table.merge(rally_counts, on=["match_id", "player"], how="left", validate="one_to_one")
for court in ["deuce", "ad"]:
    serves = serve_analysis.loc[serve_analysis["service_court"].eq(court) & serve_analysis["target_category"].notna()].copy()
    serves[court+"_classified_serves"] = 1
    for code, name in [(4, "wide"), (5, "body"), (6, "t")]:
        serves[court+"_"+name+"_serves"] = serves["serve_direction_code"].eq(code).astype(int)
    fields = [court+"_classified_serves"] + [court+"_"+name+"_serves" for name in ["wide", "body", "t"]]
    tactical_match_table = tactical_match_table.merge(serves.groupby(["match_id", "player"], as_index=False)[fields].sum(), on=["match_id", "player"],
how="left", validate="one_to_one")
assert list(tactical_match_table[["match_id", "player"]].itertuples(index=False, name=None)) == kpi_pairs

tactical_metric_specs = {
    "approach_frequency_pct": ("approach_points", "source_player_points", "points"),
    "net_point_frequency_pct": ("net_points_all", "source_player_points", "points"),
    "serve_volley_frequency_pct": ("snv_points", "serve_pts", "points"),
    "drop_shot_share_pct": ("drop_count", "known_family", "shots"),
    "slice_share_pct": ("slice_count", "known_family", "shots"),
    "short_rally_share_pct": ("short_rallies", "known_rallies", "points"),
    "medium_rally_share_pct": ("medium_rallies", "known_rallies", "points"),
    "long_rally_share_pct": ("long_rallies", "known_rallies", "points"),
}
for court in ["deuce", "ad"]:
    for name in ["wide", "body", "t"]:
        tactical_metric_specs[court+"_"+name+"_serve_share_pct"] = (court+"_"+name+"_serves", court+"_classified_serves", "points")
tactical_masks = {}
for metric, (num, den, basis) in tactical_metric_specs.items():
    a, b = tactical_match_table[num], tactical_match_table[den]
    mask = a.notna() & b.notna() & np.isfinite(a) & np.isfinite(b) & a.ge(0) & b.gt(0) & a.le(b)
    if metric in ["approach_frequency_pct", "net_point_frequency_pct"]:
        mask &= reconciled_mask(["service_points_parsed", "return_points_parsed"])
    if metric == "serve_volley_frequency_pct":
        mask &= snv_complete & reconciled_mask(["service_points_parsed"])
    tactical_masks[metric] = mask

tactical_rows, tactical_rate_bootstraps = [], {}
for plan in tactical_period_plan.itertuples(index=False):
    period_mask = tactical_match_table["player"].eq(plan.player) & tactical_match_table["period_index"].eq(plan.period_index)
    for scope in ["Overall"] + PROJECT_SURFACES:
        group_mask = period_mask if scope == "Overall" else period_mask & tactical_match_table["surface"].eq(scope)
        for metric, (num, den, basis) in tactical_metric_specs.items():
            sample = tactical_match_table.loc[group_mask & tactical_masks[metric]]
            numerator, denominator = sample[num].sum(min_count=1), sample[den].sum(min_count=1)
            vectors = np.zeros((len(bootstrap_match_ids), 2))
            for mid, a, b in sample[["match_id", num, den]].itertuples(index=False, name=None):
                vectors[bootstrap_match_index[mid]] = [a, b]
            totals = bootstrap_weights @ vectors
            with np.errstate(divide="ignore", invalid="ignore"):
                values = np.where(totals[:, 1] > 0, 100*totals[:, 0]/totals[:, 1], np.nan)
            valid = values[np.isfinite(values)]
            tactical_rows.append({"player": plan.player, "period_index": plan.period_index, "period": plan.period,
                "surface": scope, "metric": metric, "numerator": numerator, "denominator": denominator,
                "estimate_pct": 100*numerator/denominator if denominator > 0 else np.nan,
                "contributing_matches": len(sample), "cohort_matches": int(group_mask.sum()),
                "coverage_pct": 100*len(sample)/group_mask.sum() if group_mask.sum() else np.nan,
                "small_sample": len(sample) < SMALL_SAMPLE_THRESHOLDS["matches"] or pd.isna(denominator) or denominator < SMALL_SAMPLE_THRESHOLDS[basis],
```

```
                "coverage_edge_period": plan.coverage_edge, "valid_bootstraps": len(valid),
                "ci95_lower": np.percentile(valid, 2.5) if len(valid) >= .95*BOOTSTRAP_REPLICATES else np.nan,
                "ci95_upper": np.percentile(valid, 97.5) if len(valid) >= .95*BOOTSTRAP_REPLICATES else np.nan})
            tactical_rate_bootstraps[(plan.player, plan.period_index, scope, metric)] = values
tactical_evolution_long = pd.DataFrame(tactical_rows)
standardized_rows, standardized_bootstraps = [], {}
for plan in tactical_period_plan.itertuples(index=False):
    for metric in tactical_metric_specs:
        rates = tactical_evolution_long.loc[tactical_evolution_long["player"].eq(plan.player)
            & tactical_evolution_long["period_index"].eq(plan.period_index) &
tactical_evolution_long["metric"].eq(metric)].set_index("surface").reindex(PROJECT_SURFACES)
        all_surfaces = rates["estimate_pct"].notna().all()
        boot = np.column_stack([tactical_rate_bootstraps[(plan.player, plan.period_index, s, metric)] for s in PROJECT_SURFACES])
        values = np.mean(boot, axis=1) if all_surfaces else np.full(BOOTSTRAP_REPLICATES, np.nan)
        valid = values[np.isfinite(values)]
        standardized_rows.append({"player": plan.player, "period_index": plan.period_index, "period": plan.period, "metric": metric,
            "standardized_pct": rates["estimate_pct"].mean() if all_surfaces else np.nan,
            "surfaces_available": int(rates["estimate_pct"].notna().sum()), "small_sample": bool(rates["small_sample"].any()),
            "ci95_lower": np.percentile(valid, 2.5) if len(valid) >= .95*BOOTSTRAP_REPLICATES else np.nan,
            "ci95_upper": np.percentile(valid, 97.5) if len(valid) >= .95*BOOTSTRAP_REPLICATES else np.nan,
            "valid_bootstraps": len(valid)})
        standardized_bootstraps[(plan.player, plan.period_index, metric)] = values
tactical_standardized_trends = pd.DataFrame(standardized_rows)
contrast_rows = []
for player in BIG_THREE:
    for metric in tactical_metric_specs:
        available = tactical_standardized_trends.loc[tactical_standardized_trends["player"].eq(player)
            & tactical_standardized_trends["metric"].eq(metric) & tactical_standardized_trends["standardized_pct"].notna()].sort_values("period_index")
        if len(available) < 2:
            contrast_rows.append({"player": player, "metric": metric, "status": "fewer than two three-surface periods"})
            continue
        first, last = available.iloc[0], available.iloc[-1]
        delta = standardized_bootstraps[(player, last["period_index"], metric)] - standardized_bootstraps[(player, first["period_index"], metric)]
        valid = delta[np.isfinite(delta)]
        contrast_rows.append({"player": player, "metric": metric, "first_supported_period": first["period"], "last_supported_period": last["period"],
            "change_percentage_points": last["standardized_pct"]-first["standardized_pct"],
            "ci95_lower": np.percentile(valid, 2.5) if len(valid) >= .95*BOOTSTRAP_REPLICATES else np.nan,
            "ci95_upper": np.percentile(valid, 97.5) if len(valid) >= .95*BOOTSTRAP_REPLICATES else np.nan,
            "valid_bootstraps": len(valid), "status": "small endpoint sample" if first["small_sample"] or last["small_sample"] else "available"})
tactical_change_summary = pd.DataFrame(contrast_rows)
snv_source_coverage = tactical_match_table.groupby("player", as_index=False).agg(
    cohort_matches=("match_id", "size"), proven_zero_snv_matches=("snv_zero_proven_by_complement", "sum"),
    matches_with_known_snv_count=("snv_points", "count"),
    matches_recovered_from_explicit_snv_codes=("snv_recovered_from_explicit_codes", "sum"))
assert not tactical_evolution_long.duplicated(["player", "period_index", "surface", "metric"]).any()
assert tactical_evolution_long["estimate_pct"].dropna().between(0, 100).all()
assert tactical_standardized_trends.loc[tactical_standardized_trends["surfaces_available"].lt(3), "standardized_pct"].isna().all()
assert tactical_period_plan.groupby("player")["charted_matches"].sum().reindex(BIG_THREE).eq(canonical_master.groupby("player").size().reindex(BIG_THREE)).all()
assert
(tactical_match_table["short_rallies"]+tactical_match_table["medium_rallies"]+tactical_match_table["long_rallies"]).dropna().eq(tactical_match_table["known_rall
ies"].dropna()).all()
print("Career periods, fixed before inspecting tactical values:")
report_table(tactical_period_plan)
print("Serve-and-volley source/complement coverage (missing does not automatically mean zero):")
report_table(snv_source_coverage)
print("SnV source/code discrepancies (excluded, not forced to agree):")
report_table(snv_code_source_review)
```

```
print("All tactical rates: pooled and within-surface, with eligibility, counts, and uncertainty:")
for player in BIG_THREE:
    print(player)
    with pd.option_context("display.max_rows", None):
        report_table(tactical_evolution_long.loc[tactical_evolution_long["player"].eq(player)].round(3))
print("Equal-surface standardized trends; no estimate without all three surfaces:")
with pd.option_context("display.max_rows", None):
    report_table(tactical_standardized_trends.round(3))
print("First/last supported standardized-period contrasts, in percentage points (not performance gains):")
with pd.option_context("display.max_rows", None):
    report_table(tactical_change_summary.round(3))
```

```
Career periods, fixed before inspecting tactical values:
```

player	period_index	period	start_year	end_year	nominal_years	coverage_edge	first_charted_date	last_charted_date	charted_matches
Roger Federer	0	1998–2001	1998	2001	4	True	1998-07-04	2001-12-14	32
Roger Federer	1	2002–2005	2002	2005	4	False	2002-01-12	2005-11-19	162
Roger Federer	2	2006–2009	2006	2009	4	False	2006-01-06	2009-11-28	170
Roger Federer	3	2010–2013	2010	2013	4	False	2010-01-08	2013-11-05	134
Roger Federer	4	2014–2017	2014	2017	4	False	2014-01-01	2017-10-29	126
Roger Federer	5	2018–2021	2018	2021	4	True	2018-01-16	2021-07-05	99
Rafael Nadal	0	2003–2006	2003	2006	4	True	2003-05-15	2006-11-17	38
Rafael Nadal	1	2007–2010	2007	2010	4	False	2007-01-16	2010-11-28	96
Rafael Nadal	2	2011–2014	2011	2014	4	False	2011-01-07	2014-07-01	111
Rafael Nadal	3	2015–2018	2015	2018	4	False	2015-01-05	2018-09-07	104
Rafael Nadal	4	2019–2022	2019	2022	4	False	2019-01-22	2022-11-17	62
Rafael Nadal	5	2023–2024	2023	2024	4	True	2023-01-16	2024-11-19	14
Novak Djokovic	0	2005–2008	2005	2008	4	True	2005-01-17	2008-11-16	50
Novak Djokovic	1	2009–2012	2009	2012	4	False	2009-01-27	2012-11-12	106
Novak Djokovic	2	2013–2016	2013	2016	4	False	2013-01-13	2016-11-20	132
Novak Djokovic	3	2017–2020	2017	2020	4	False	2017-01-07	2020-11-21	79
Novak Djokovic	4	2021–2024	2021	2024	4	False	2021-02-10	2024-12-31	143
Novak Djokovic	5	2025–2026	2025	2026	4	True	2025-01-02	2026-03-11	43

Serve-and-volley source/complement coverage (missing does not automatically mean zero):

player	cohort_matches	proven_zero_snv_matches	matches_with_known_snv_count	matches_recovered_from_explicit_snv_codes
Novak Djokovic	553	0	550	222
Rafael Nadal	425	0	415	241
Roger Federer	723	0	720	51

SnV source/code discrepancies (excluded, not forced to agree):

	match_id	player	snv_points	snv_from_code	coded_service_points	serve_pts
198	20200130-M-Australian_Open-SF-Roger_Federer-Novak_Djokovic	Roger Federer	7.000	8.000	104.000	104.000
231	20190714-M-Wimbledon-F-Roger_Federer-Novak_Djokovic	Roger Federer	11.000	12.000	203.000	203.000
244	20190620-M-Halle-R16-Roger_Federer-Jo_Wilfried_Tsonga	Roger Federer	12.000	13.000	100.000	100.000
253	20190510-M-Madrid_Masters-QF-Roger_Federer-Dominic_Thiem	Roger Federer	20.000	21.000	106.000	106.000
258	20190317-M-Indian_Wells_Masters-F-Roger_Federer-Dominic_Thiem	Roger Federer	14.000	15.000	84.000	84.000
443	20151117-M-Tour_Finals-RR-Roger_Federer-Novak_Djokovic	Roger Federer	5.000	6.000	54.000	54.000
1046	20031110-M-Masters_Cup-RR-Roger_Federer-Andre_Agassi	Roger Federer	41.000	42.000	124.000	124.000
1274	20151104-M-Paris_Masters-R32-Andreas_Seppi-Roger_Federer	Roger Federer	4.000	5.000	40.000	40.000
1641	20051117-M-Masters_Cup-RR-Guillermo_Coria-Roger_Federer	Roger Federer	1.000	2.000	64.000	64.000
1682	20020208-M-Davis_Cup_World_Group_R1-RR-Marat_Safin-Roger_Federer	Roger Federer	6.000	7.000	77.000	77.000

All tactical rates: pooled and within-surface, with eligibility, counts, and uncertainty:
Roger Federer

▼ Complete evidence table: 336 rows × 16 columns

	player	surface	period	period_index	metric	numerator	denominator	estimate_pct	contributing_matches	cohort_matches
0	Roger Federer	Overall	1998–2001	0	approach_frequency_pct	1,209.000	5,693.000	21.237	30	32
1	Roger Federer	Overall	1998–2001	0	net_point_frequency_pct	1,292.000	5,810.000	22.238	31	32
2	Roger Federer	Overall	1998–2001	0	serve_volley_frequency_pct	684.000	2,950.000	23.186	32	32
3	Roger Federer	Overall	1998–2001	0	drop_shot_share_pct	32.000	9,562.000	0.335	32	32
4	Roger Federer	Overall	1998–2001	0	slice_share_pct	1,717.000	9,562.000	17.956	32	32
5	Roger Federer	Overall	1998–2001	0	short_rally_share_pct	4,151.000	5,749.000	72.204	32	32
6	Roger Federer	Overall	1998–2001	0	medium_rally_share_pct	1,150.000	5,749.000	20.003	32	32
7	Roger Federer	Overall	1998–2001	0	long_rally_share_pct	448.000	5,749.000	7.793	32	32
8	Roger Federer	Overall	1998–2001	0	deuce_wide_serve_share_pct	564.000	1,537.000	36.695	32	32
9	Roger Federer	Overall	1998–2001	0	deuce_body_serve_share_pct	299.000	1,537.000	19.453	32	32
10	Roger Federer	Overall	1998–2001	0	deuce_t_serve_share_pct	674.000	1,537.000	43.852	32	32
11	Roger Federer	Overall	1998–2001	0	ad_wide_serve_share_pct	812.000	1,405.000	57.794	32	32
12	Roger Federer	Overall	1998–2001	0	ad_body_serve_share_pct	192.000	1,405.000	13.665	32	32
13	Roger Federer	Overall	1998–2001	0	ad_t_serve_share_pct	401.000	1,405.000	28.541	32	32
14	Roger Federer	Clay	1998–2001	0	approach_frequency_pct	168.000	1,233.000	13.625	5	5
15	Roger Federer	Clay	1998–2001	0	net_point_frequency_pct	193.000	1,233.000	15.653	5	5
16	Roger Federer	Clay	1998–2001	0	serve_volley_frequency_pct	23.000	615.000	3.740	5	5
17	Roger Federer	Clay	1998–2001	0	drop_shot_share_pct	19.000	2,599.000	0.731	5	5
18	Roger Federer	Clay	1998–2001	0	slice_share_pct	451.000	2,599.000	17.353	5	5
19	Roger Federer	Clay	1998–2001	0	short_rally_share_pct	709.000	1,182.000	59.983	5	5
20	Roger Federer	Clay	1998–2001	0	medium_rally_share_pct	312.000	1,182.000	26.396	5	5
21	Roger Federer	Clay	1998–2001	0	long_rally_share_pct	161.000	1,182.000	13.621	5	5
22	Roger Federer	Clay	1998–2001	0	deuce_wide_serve_share_pct	102.000	313.000	32.588	5	5

	player	surface	period	period_index	metric	numerator	denominator	estimate_pct	contributing_matches	cohort_matches
23	Roger Federer	Clay	1998–2001	0	deuce_body_serve_share_pct	53.000	313.000	16.933	5	5
24	Roger Federer	Clay	1998–2001	0	deuce_t_serve_share_pct	158.000	313.000	50.479	5	5
25	Roger Federer	Clay	1998–2001	0	ad_wide_serve_share_pct	205.000	302.000	67.881	5	5
26	Roger Federer	Clay	1998–2001	0	ad_body_serve_share_pct	29.000	302.000	9.603	5	5
27	Roger Federer	Clay	1998–2001	0	ad_t_serve_share_pct	68.000	302.000	22.517	5	5
28	Roger Federer	Grass	1998–2001	0	approach_frequency_pct	389.000	1,227.000	31.703	5	6
29	Roger Federer	Grass	1998–2001	0	net_point_frequency_pct	403.000	1,344.000	29.985	6	6
30	Roger Federer	Grass	1998–2001	0	serve_volley_frequency_pct	374.000	655.000	57.099	6	6
31	Roger Federer	Grass	1998–2001	0	drop_shot_share_pct	2.000	1,348.000	0.148	6	6
32	Roger Federer	Grass	1998–2001	0	slice_share_pct	161.000	1,348.000	11.944	6	6
33	Roger Federer	Grass	1998–2001	0	short_rally_share_pct	1,121.000	1,287.000	87.102	6	6
34	Roger Federer	Grass	1998–2001	0	medium_rally_share_pct	148.000	1,287.000	11.500	6	6
35	Roger Federer	Grass	1998–2001	0	long_rally_share_pct	18.000	1,287.000	1.399	6	6
36	Roger Federer	Grass	1998–2001	0	deuce_wide_serve_share_pct	125.000	342.000	36.550	6	6
37	Roger Federer	Grass	1998–2001	0	deuce_body_serve_share_pct	70.000	342.000	20.468	6	6
38	Roger Federer	Grass	1998–2001	0	deuce_t_serve_share_pct	147.000	342.000	42.982	6	6
39	Roger Federer	Grass	1998–2001	0	ad_wide_serve_share_pct	162.000	313.000	51.757	6	6
40	Roger Federer	Grass	1998–2001	0	ad_body_serve_share_pct	45.000	313.000	14.377	6	6
41	Roger Federer	Grass	1998–2001	0	ad_t_serve_share_pct	106.000	313.000	33.866	6	6
42	Roger Federer	Hard	1998–2001	0	approach_frequency_pct	652.000	3,233.000	20.167	20	21
43	Roger Federer	Hard	1998–2001	0	net_point_frequency_pct	696.000	3,233.000	21.528	20	21
44	Roger Federer	Hard	1998–2001	0	serve_volley_frequency_pct	287.000	1,680.000	17.083	21	21
45	Roger Federer	Hard	1998–2001	0	drop_shot_share_pct	11.000	5,615.000	0.196	21	21
46	Roger Federer	Hard	1998–2001	0	slice_share_pct	1,105.000	5,615.000	19.679	21	21

	player	surface	period	period_index	metric	numerator	denominator	estimate_pct	contributing_matches	cohort_matches
47	Roger Federer	Hard	1998–2001	0	short_rally_share_pct	2,321.000	3,280.000	70.762	21	21
48	Roger Federer	Hard	1998–2001	0	medium_rally_share_pct	690.000	3,280.000	21.037	21	21
49	Roger Federer	Hard	1998–2001	0	long_rally_share_pct	269.000	3,280.000	8.201	21	21
50	Roger Federer	Hard	1998–2001	0	deuce_wide_serve_share_pct	337.000	882.000	38.209	21	21
51	Roger Federer	Hard	1998–2001	0	deuce_body_serve_share_pct	176.000	882.000	19.955	21	21
52	Roger Federer	Hard	1998–2001	0	deuce_t_serve_share_pct	369.000	882.000	41.837	21	21
53	Roger Federer	Hard	1998–2001	0	ad_wide_serve_share_pct	445.000	790.000	56.329	21	21
54	Roger Federer	Hard	1998–2001	0	ad_body_serve_share_pct	118.000	790.000	14.937	21	21
55	Roger Federer	Hard	1998–2001	0	ad_t_serve_share_pct	227.000	790.000	28.734	21	21
56	Roger Federer	Overall	2002–2005	1	approach_frequency_pct	4,529.000	25,828.000	17.535	159	162
57	Roger Federer	Overall	2002–2005	1	net_point_frequency_pct	4,953.000	25,828.000	19.177	159	162
58	Roger Federer	Overall	2002–2005	1	serve_volley_frequency_pct	2,072.000	12,390.000	16.723	158	162
59	Roger Federer	Overall	2002–2005	1	drop_shot_share_pct	301.000	46,472.000	0.648	162	162
60	Roger Federer	Overall	2002–2005	1	slice_share_pct	9,203.000	46,472.000	19.803	162	162
61	Roger Federer	Overall	2002–2005	1	short_rally_share_pct	17,493.000	25,507.000	68.581	162	162
62	Roger Federer	Overall	2002–2005	1	medium_rally_share_pct	5,552.000	25,507.000	21.767	162	162
63	Roger Federer	Overall	2002–2005	1	long_rally_share_pct	2,462.000	25,507.000	9.652	162	162
64	Roger Federer	Overall	2002–2005	1	deuce_wide_serve_share_pct	2,642.000	6,700.000	39.433	162	162
65	Roger Federer	Overall	2002–2005	1	deuce_body_serve_share_pct	1,230.000	6,700.000	18.358	162	162
66	Roger Federer	Overall	2002–2005	1	deuce_t_serve_share_pct	2,828.000	6,700.000	42.209	162	162
67	Roger Federer	Overall	2002–2005	1	ad_wide_serve_share_pct	3,379.000	6,062.000	55.741	162	162
68	Roger Federer	Overall	2002–2005	1	ad_body_serve_share_pct	830.000	6,062.000	13.692	162	162
69	Roger Federer	Overall	2002–2005	1	ad_t_serve_share_pct	1,853.000	6,062.000	30.567	162	162
70	Roger Federer	Clay	2002–2005	1	approach_frequency_pct	814.000	4,673.000	17.419	30	31

	player	surface	period	period_index	metric	numerator	denominator	estimate_pct	contributing_matches	cohort_matches
71	Roger Federer	Clay	2002–2005	1	net_point_frequency_pct	910.000	4,673.000	19.474	30	31
72	Roger Federer	Clay	2002–2005	1	serve_volley_frequency_pct	323.000	2,251.000	14.349	30	31
73	Roger Federer	Clay	2002–2005	1	drop_shot_share_pct	99.000	9,040.000	1.095	31	31
74	Roger Federer	Clay	2002–2005	1	slice_share_pct	1,564.000	9,040.000	17.301	31	31
75	Roger Federer	Clay	2002–2005	1	short_rally_share_pct	3,114.000	4,672.000	66.652	31	31
76	Roger Federer	Clay	2002–2005	1	medium_rally_share_pct	1,046.000	4,672.000	22.389	31	31
77	Roger Federer	Clay	2002–2005	1	long_rally_share_pct	512.000	4,672.000	10.959	31	31
78	Roger Federer	Clay	2002–2005	1	deuce_wide_serve_share_pct	433.000	1,216.000	35.609	31	31
79	Roger Federer	Clay	2002–2005	1	deuce_body_serve_share_pct	238.000	1,216.000	19.572	31	31
80	Roger Federer	Clay	2002–2005	1	deuce_t_serve_share_pct	545.000	1,216.000	44.819	31	31
81	Roger Federer	Clay	2002–2005	1	ad_wide_serve_share_pct	670.000	1,110.000	60.360	31	31
82	Roger Federer	Clay	2002–2005	1	ad_body_serve_share_pct	125.000	1,110.000	11.261	31	31
83	Roger Federer	Clay	2002–2005	1	ad_t_serve_share_pct	315.000	1,110.000	28.378	31	31
84	Roger Federer	Grass	2002–2005	1	approach_frequency_pct	717.000	3,564.000	20.118	21	21
85	Roger Federer	Grass	2002–2005	1	net_point_frequency_pct	776.000	3,564.000	21.773	21	21
86	Roger Federer	Grass	2002–2005	1	serve_volley_frequency_pct	460.000	1,718.000	26.775	21	21
87	Roger Federer	Grass	2002–2005	1	drop_shot_share_pct	24.000	5,435.000	0.442	21	21
88	Roger Federer	Grass	2002–2005	1	slice_share_pct	1,039.000	5,435.000	19.117	21	21
89	Roger Federer	Grass	2002–2005	1	short_rally_share_pct	2,550.000	3,472.000	73.445	21	21
90	Roger Federer	Grass	2002–2005	1	medium_rally_share_pct	687.000	3,472.000	19.787	21	21
91	Roger Federer	Grass	2002–2005	1	long_rally_share_pct	235.000	3,472.000	6.768	21	21
92	Roger Federer	Grass	2002–2005	1	deuce_wide_serve_share_pct	370.000	911.000	40.615	21	21
93	Roger Federer	Grass	2002–2005	1	deuce_body_serve_share_pct	156.000	911.000	17.124	21	21
94	Roger Federer	Grass	2002–2005	1	deuce_t_serve_share_pct	385.000	911.000	42.261	21	21

	player	surface	period	period_index	metric	numerator	denominator	estimate_pct	contributing_matches	cohort_matches
95	Roger Federer	Grass	2002–2005	1	ad_wide_serve_share_pct	438.000	807.000	54.275	21	21
96	Roger Federer	Grass	2002–2005	1	ad_body_serve_share_pct	112.000	807.000	13.879	21	21
97	Roger Federer	Grass	2002–2005	1	ad_t_serve_share_pct	257.000	807.000	31.846	21	21
98	Roger Federer	Hard	2002–2005	1	approach_frequency_pct	2,998.000	17,591.000	17.043	108	110
99	Roger Federer	Hard	2002–2005	1	net_point_frequency_pct	3,267.000	17,591.000	18.572	108	110
100	Roger Federer	Hard	2002–2005	1	serve_volley_frequency_pct	1,289.000	8,421.000	15.307	107	110
101	Roger Federer	Hard	2002–2005	1	drop_shot_share_pct	178.000	31,997.000	0.556	110	110
102	Roger Federer	Hard	2002–2005	1	slice_share_pct	6,600.000	31,997.000	20.627	110	110
103	Roger Federer	Hard	2002–2005	1	short_rally_share_pct	11,829.000	17,363.000	68.128	110	110
104	Roger Federer	Hard	2002–2005	1	medium_rally_share_pct	3,819.000	17,363.000	21.995	110	110
105	Roger Federer	Hard	2002–2005	1	long_rally_share_pct	1,715.000	17,363.000	9.877	110	110
106	Roger Federer	Hard	2002–2005	1	deuce_wide_serve_share_pct	1,839.000	4,573.000	40.214	110	110
107	Roger Federer	Hard	2002–2005	1	deuce_body_serve_share_pct	836.000	4,573.000	18.281	110	110
108	Roger Federer	Hard	2002–2005	1	deuce_t_serve_share_pct	1,898.000	4,573.000	41.504	110	110
109	Roger Federer	Hard	2002–2005	1	ad_wide_serve_share_pct	2,271.000	4,145.000	54.789	110	110
110	Roger Federer	Hard	2002–2005	1	ad_body_serve_share_pct	593.000	4,145.000	14.306	110	110
111	Roger Federer	Hard	2002–2005	1	ad_t_serve_share_pct	1,281.000	4,145.000	30.905	110	110
112	Roger Federer	Overall	2006–2009	2	approach_frequency_pct	3,926.000	28,017.000	14.013	161	170
113	Roger Federer	Overall	2006–2009	2	net_point_frequency_pct	4,441.000	28,763.000	15.440	163	170
114	Roger Federer	Overall	2006–2009	2	serve_volley_frequency_pct	717.000	14,075.000	5.094	166	170
115	Roger Federer	Overall	2006–2009	2	drop_shot_share_pct	480.000	56,924.000	0.843	170	170
116	Roger Federer	Overall	2006–2009	2	slice_share_pct	10,839.000	56,924.000	19.041	170	170
117	Roger Federer	Overall	2006–2009	2	short_rally_share_pct	19,058.000	29,133.000	65.417	170	170
118	Roger Federer	Overall	2006–2009	2	medium_rally_share_pct	6,692.000	29,133.000	22.971	170	170

	player	surface	period	period_index	metric	numerator	denominator	estimate_pct	contributing_matches	cohort_matches
119	Roger Federer	Overall	2006–2009	2	long_rally_share_pct	3,383.000	29,133.000	11.612	170	170
120	Roger Federer	Overall	2006–2009	2	deuce_wide_serve_share_pct	2,947.000	7,551.000	39.028	170	170
121	Roger Federer	Overall	2006–2009	2	deuce_body_serve_share_pct	1,294.000	7,551.000	17.137	170	170
122	Roger Federer	Overall	2006–2009	2	deuce_t_serve_share_pct	3,310.000	7,551.000	43.835	170	170
123	Roger Federer	Overall	2006–2009	2	ad_wide_serve_share_pct	3,542.000	6,832.000	51.844	170	170
124	Roger Federer	Overall	2006–2009	2	ad_body_serve_share_pct	916.000	6,832.000	13.407	170	170
125	Roger Federer	Overall	2006–2009	2	ad_t_serve_share_pct	2,374.000	6,832.000	34.748	170	170
126	Roger Federer	Clay	2006–2009	2	approach_frequency_pct	1,023.000	6,851.000	14.932	39	43
127	Roger Federer	Clay	2006–2009	2	net_point_frequency_pct	1,179.000	7,161.000	16.464	40	43
128	Roger Federer	Clay	2006–2009	2	serve_volley_frequency_pct	153.000	3,475.000	4.403	40	43
129	Roger Federer	Clay	2006–2009	2	drop_shot_share_pct	242.000	16,800.000	1.440	43	43
130	Roger Federer	Clay	2006–2009	2	slice_share_pct	2,247.000	16,800.000	13.375	43	43
131	Roger Federer	Clay	2006–2009	2	short_rally_share_pct	4,439.000	7,476.000	59.377	43	43
132	Roger Federer	Clay	2006–2009	2	medium_rally_share_pct	1,922.000	7,476.000	25.709	43	43
133	Roger Federer	Clay	2006–2009	2	long_rally_share_pct	1,115.000	7,476.000	14.914	43	43
134	Roger Federer	Clay	2006–2009	2	deuce_wide_serve_share_pct	779.000	1,932.000	40.321	43	43
135	Roger Federer	Clay	2006–2009	2	deuce_body_serve_share_pct	295.000	1,932.000	15.269	43	43
136	Roger Federer	Clay	2006–2009	2	deuce_t_serve_share_pct	858.000	1,932.000	44.410	43	43
137	Roger Federer	Clay	2006–2009	2	ad_wide_serve_share_pct	931.000	1,768.000	52.658	43	43
138	Roger Federer	Clay	2006–2009	2	ad_body_serve_share_pct	283.000	1,768.000	16.007	43	43
139	Roger Federer	Clay	2006–2009	2	ad_t_serve_share_pct	554.000	1,768.000	31.335	43	43
140	Roger Federer	Grass	2006–2009	2	approach_frequency_pct	449.000	3,589.000	12.510	19	20
141	Roger Federer	Grass	2006–2009	2	net_point_frequency_pct	530.000	4,025.000	13.168	20	20
142	Roger Federer	Grass	2006–2009	2	serve_volley_frequency_pct	104.000	1,919.000	5.419	20	20

	player	surface	period	period_index	metric	numerator	denominator	estimate_pct	contributing_matches	cohort_matches
143	Roger Federer	Grass	2006–2009	2	drop_shot_share_pct	38.000	6,219.000	0.611	20	20
144	Roger Federer	Grass	2006–2009	2	slice_share_pct	1,448.000	6,219.000	23.283	20	20
145	Roger Federer	Grass	2006–2009	2	short_rally_share_pct	2,878.000	3,949.000	72.879	20	20
146	Roger Federer	Grass	2006–2009	2	medium_rally_share_pct	777.000	3,949.000	19.676	20	20
147	Roger Federer	Grass	2006–2009	2	long_rally_share_pct	294.000	3,949.000	7.445	20	20
148	Roger Federer	Grass	2006–2009	2	deuce_wide_serve_share_pct	399.000	1,006.000	39.662	20	20
149	Roger Federer	Grass	2006–2009	2	deuce_body_serve_share_pct	171.000	1,006.000	16.998	20	20
150	Roger Federer	Grass	2006–2009	2	deuce_t_serve_share_pct	436.000	1,006.000	43.340	20	20
151	Roger Federer	Grass	2006–2009	2	ad_wide_serve_share_pct	447.000	913.000	48.959	20	20
152	Roger Federer	Grass	2006–2009	2	ad_body_serve_share_pct	113.000	913.000	12.377	20	20
153	Roger Federer	Grass	2006–2009	2	ad_t_serve_share_pct	353.000	913.000	38.664	20	20
154	Roger Federer	Hard	2006–2009	2	approach_frequency_pct	2,454.000	17,577.000	13.961	103	107
155	Roger Federer	Hard	2006–2009	2	net_point_frequency_pct	2,732.000	17,577.000	15.543	103	107
156	Roger Federer	Hard	2006–2009	2	serve_volley_frequency_pct	460.000	8,681.000	5.299	106	107
157	Roger Federer	Hard	2006–2009	2	drop_shot_share_pct	200.000	33,905.000	0.590	107	107
158	Roger Federer	Hard	2006–2009	2	slice_share_pct	7,144.000	33,905.000	21.071	107	107
159	Roger Federer	Hard	2006–2009	2	short_rally_share_pct	11,741.000	17,708.000	66.303	107	107
160	Roger Federer	Hard	2006–2009	2	medium_rally_share_pct	3,993.000	17,708.000	22.549	107	107
161	Roger Federer	Hard	2006–2009	2	long_rally_share_pct	1,974.000	17,708.000	11.148	107	107
162	Roger Federer	Hard	2006–2009	2	deuce_wide_serve_share_pct	1,769.000	4,613.000	38.348	107	107
163	Roger Federer	Hard	2006–2009	2	deuce_body_serve_share_pct	828.000	4,613.000	17.949	107	107
164	Roger Federer	Hard	2006–2009	2	deuce_t_serve_share_pct	2,016.000	4,613.000	43.703	107	107
165	Roger Federer	Hard	2006–2009	2	ad_wide_serve_share_pct	2,164.000	4,151.000	52.132	107	107
166	Roger Federer	Hard	2006–2009	2	ad_body_serve_share_pct	520.000	4,151.000	12.527	107	107

	player	surface	period	period_index	metric	numerator	denominator	estimate_pct	contributing_matches	cohort_matches
167	Roger Federer	Hard	2006–2009	2	ad_t_serve_share_pct	1,467.000	4,151.000	35.341	107	107
168	Roger Federer	Overall	2010–2013	3	approach_frequency_pct	2,632.000	21,155.000	12.442	122	134
169	Roger Federer	Overall	2010–2013	3	net_point_frequency_pct	3,197.000	22,433.000	14.251	130	134
170	Roger Federer	Overall	2010–2013	3	serve_volley_frequency_pct	479.000	11,070.000	4.327	132	134
171	Roger Federer	Overall	2010–2013	3	drop_shot_share_pct	571.000	43,329.000	1.318	134	134
172	Roger Federer	Overall	2010–2013	3	slice_share_pct	7,966.000	43,329.000	18.385	134	134
173	Roger Federer	Overall	2010–2013	3	short_rally_share_pct	14,798.000	22,464.000	65.874	134	134
174	Roger Federer	Overall	2010–2013	3	medium_rally_share_pct	5,168.000	22,464.000	23.006	134	134
175	Roger Federer	Overall	2010–2013	3	long_rally_share_pct	2,498.000	22,464.000	11.120	134	134
176	Roger Federer	Overall	2010–2013	3	deuce_wide_serve_share_pct	2,556.000	5,859.000	43.625	134	134
177	Roger Federer	Overall	2010–2013	3	deuce_body_serve_share_pct	1,077.000	5,859.000	18.382	134	134
178	Roger Federer	Overall	2010–2013	3	deuce_t_serve_share_pct	2,226.000	5,859.000	37.993	134	134
179	Roger Federer	Overall	2010–2013	3	ad_wide_serve_share_pct	2,636.000	5,344.000	49.326	134	134
180	Roger Federer	Overall	2010–2013	3	ad_body_serve_share_pct	891.000	5,344.000	16.673	134	134
181	Roger Federer	Overall	2010–2013	3	ad_t_serve_share_pct	1,817.000	5,344.000	34.001	134	134
182	Roger Federer	Clay	2010–2013	3	approach_frequency_pct	484.000	4,029.000	12.013	22	23
183	Roger Federer	Clay	2010–2013	3	net_point_frequency_pct	536.000	4,126.000	12.991	23	23
184	Roger Federer	Clay	2010–2013	3	serve_volley_frequency_pct	86.000	2,024.000	4.249	23	23
185	Roger Federer	Clay	2010–2013	3	drop_shot_share_pct	145.000	8,026.000	1.807	23	23
186	Roger Federer	Clay	2010–2013	3	slice_share_pct	1,191.000	8,026.000	14.839	23	23
187	Roger Federer	Clay	2010–2013	3	short_rally_share_pct	2,571.000	4,027.000	63.844	23	23
188	Roger Federer	Clay	2010–2013	3	medium_rally_share_pct	980.000	4,027.000	24.336	23	23
189	Roger Federer	Clay	2010–2013	3	long_rally_share_pct	476.000	4,027.000	11.820	23	23
190	Roger Federer	Clay	2010–2013	3	deuce_wide_serve_share_pct	455.000	1,055.000	43.128	23	23

	player	surface	period	period_index	metric	numerator	denominator	estimate_pct	contributing_matches	cohort_matches
191	Roger Federer	Clay	2010–2013	3	deuce_body_serve_share_pct	175.000	1,055.000	16.588	23	23
192	Roger Federer	Clay	2010–2013	3	deuce_t_serve_share_pct	425.000	1,055.000	40.284	23	23
193	Roger Federer	Clay	2010–2013	3	ad_wide_serve_share_pct	494.000	968.000	51.033	23	23
194	Roger Federer	Clay	2010–2013	3	ad_body_serve_share_pct	147.000	968.000	15.186	23	23
195	Roger Federer	Clay	2010–2013	3	ad_t_serve_share_pct	327.000	968.000	33.781	23	23
196	Roger Federer	Grass	2010–2013	3	approach_frequency_pct	508.000	3,818.000	13.305	19	21
197	Roger Federer	Grass	2010–2013	3	net_point_frequency_pct	659.000	4,322.000	15.248	21	21
198	Roger Federer	Grass	2010–2013	3	serve_volley_frequency_pct	98.000	2,093.000	4.682	21	21
199	Roger Federer	Grass	2010–2013	3	drop_shot_share_pct	85.000	7,159.000	1.187	21	21
200	Roger Federer	Grass	2010–2013	3	slice_share_pct	1,544.000	7,159.000	21.567	21	21
201	Roger Federer	Grass	2010–2013	3	short_rally_share_pct	2,970.000	4,235.000	70.130	21	21
202	Roger Federer	Grass	2010–2013	3	medium_rally_share_pct	914.000	4,235.000	21.582	21	21
203	Roger Federer	Grass	2010–2013	3	long_rally_share_pct	351.000	4,235.000	8.288	21	21
204	Roger Federer	Grass	2010–2013	3	deuce_wide_serve_share_pct	483.000	1,094.000	44.150	21	21
205	Roger Federer	Grass	2010–2013	3	deuce_body_serve_share_pct	196.000	1,094.000	17.916	21	21
206	Roger Federer	Grass	2010–2013	3	deuce_t_serve_share_pct	415.000	1,094.000	37.934	21	21
207	Roger Federer	Grass	2010–2013	3	ad_wide_serve_share_pct	505.000	996.000	50.703	21	21
208	Roger Federer	Grass	2010–2013	3	ad_body_serve_share_pct	134.000	996.000	13.454	21	21
209	Roger Federer	Grass	2010–2013	3	ad_t_serve_share_pct	357.000	996.000	35.843	21	21
210	Roger Federer	Hard	2010–2013	3	approach_frequency_pct	1,640.000	13,308.000	12.323	81	90
211	Roger Federer	Hard	2010–2013	3	net_point_frequency_pct	2,002.000	13,985.000	14.315	86	90
212	Roger Federer	Hard	2010–2013	3	serve_volley_frequency_pct	295.000	6,953.000	4.243	88	90
213	Roger Federer	Hard	2010–2013	3	drop_shot_share_pct	341.000	28,144.000	1.212	90	90
214	Roger Federer	Hard	2010–2013	3	slice_share_pct	5,231.000	28,144.000	18.587	90	90

	player	surface	period	period_index	metric	numerator	denominator	estimate_pct	contributing_matches	cohort_matches
215	Roger Federer	Hard	2010–2013	3	short_rally_share_pct	9,257.000	14,202.000	65.181	90	90
216	Roger Federer	Hard	2010–2013	3	medium_rally_share_pct	3,274.000	14,202.000	23.053	90	90
217	Roger Federer	Hard	2010–2013	3	long_rally_share_pct	1,671.000	14,202.000	11.766	90	90
218	Roger Federer	Hard	2010–2013	3	deuce_wide_serve_share_pct	1,618.000	3,710.000	43.612	90	90
219	Roger Federer	Hard	2010–2013	3	deuce_body_serve_share_pct	706.000	3,710.000	19.030	90	90
220	Roger Federer	Hard	2010–2013	3	deuce_t_serve_share_pct	1,386.000	3,710.000	37.358	90	90
221	Roger Federer	Hard	2010–2013	3	ad_wide_serve_share_pct	1,637.000	3,380.000	48.432	90	90
222	Roger Federer	Hard	2010–2013	3	ad_body_serve_share_pct	610.000	3,380.000	18.047	90	90
223	Roger Federer	Hard	2010–2013	3	ad_t_serve_share_pct	1,133.000	3,380.000	33.521	90	90
224	Roger Federer	Overall	2014–2017	4	approach_frequency_pct	2,862.000	20,003.000	14.308	116	126
225	Roger Federer	Overall	2014–2017	4	net_point_frequency_pct	3,375.000	20,605.000	16.380	121	126
226	Roger Federer	Overall	2014–2017	4	serve_volley_frequency_pct	1,051.000	9,851.000	10.669	120	126
227	Roger Federer	Overall	2014–2017	4	drop_shot_share_pct	336.000	34,766.000	0.966	125	126
228	Roger Federer	Overall	2014–2017	4	slice_share_pct	6,726.000	34,766.000	19.346	125	126
229	Roger Federer	Overall	2014–2017	4	short_rally_share_pct	14,768.000	20,744.000	71.192	125	126
230	Roger Federer	Overall	2014–2017	4	medium_rally_share_pct	4,356.000	20,744.000	20.999	125	126
231	Roger Federer	Overall	2014–2017	4	long_rally_share_pct	1,620.000	20,744.000	7.809	125	126
232	Roger Federer	Overall	2014–2017	4	deuce_wide_serve_share_pct	2,310.000	5,345.000	43.218	125	126
233	Roger Federer	Overall	2014–2017	4	deuce_body_serve_share_pct	863.000	5,345.000	16.146	125	126
234	Roger Federer	Overall	2014–2017	4	deuce_t_serve_share_pct	2,172.000	5,345.000	40.636	125	126
235	Roger Federer	Overall	2014–2017	4	ad_wide_serve_share_pct	2,518.000	4,838.000	52.046	125	126
236	Roger Federer	Overall	2014–2017	4	ad_body_serve_share_pct	573.000	4,838.000	11.844	125	126
237	Roger Federer	Overall	2014–2017	4	ad_t_serve_share_pct	1,747.000	4,838.000	36.110	125	126
238	Roger Federer	Clay	2014–2017	4	approach_frequency_pct	370.000	2,981.000	12.412	17	19

	player	surface	period	period_index	metric	numerator	denominator	estimate_pct	contributing_matches	cohort_matches
239	Roger Federer	Clay	2014–2017	4	net_point_frequency_pct	472.000	3,212.000	14.695	19	19
240	Roger Federer	Clay	2014–2017	4	serve_volley_frequency_pct	123.000	1,589.000	7.741	19	19
241	Roger Federer	Clay	2014–2017	4	drop_shot_share_pct	96.000	5,876.000	1.634	19	19
242	Roger Federer	Clay	2014–2017	4	slice_share_pct	996.000	5,876.000	16.950	19	19
243	Roger Federer	Clay	2014–2017	4	short_rally_share_pct	2,112.000	3,153.000	66.984	19	19
244	Roger Federer	Clay	2014–2017	4	medium_rally_share_pct	731.000	3,153.000	23.184	19	19
245	Roger Federer	Clay	2014–2017	4	long_rally_share_pct	310.000	3,153.000	9.832	19	19
246	Roger Federer	Clay	2014–2017	4	deuce_wide_serve_share_pct	307.000	829.000	37.033	19	19
247	Roger Federer	Clay	2014–2017	4	deuce_body_serve_share_pct	190.000	829.000	22.919	19	19
248	Roger Federer	Clay	2014–2017	4	deuce_t_serve_share_pct	332.000	829.000	40.048	19	19
249	Roger Federer	Clay	2014–2017	4	ad_wide_serve_share_pct	434.000	757.000	57.332	19	19
250	Roger Federer	Clay	2014–2017	4	ad_body_serve_share_pct	101.000	757.000	13.342	19	19
251	Roger Federer	Clay	2014–2017	4	ad_t_serve_share_pct	222.000	757.000	29.326	19	19
252	Roger Federer	Grass	2014–2017	4	approach_frequency_pct	711.000	4,672.000	15.218	23	24
253	Roger Federer	Grass	2014–2017	4	net_point_frequency_pct	815.000	4,763.000	17.111	24	24
254	Roger Federer	Grass	2014–2017	4	serve_volley_frequency_pct	317.000	2,294.000	13.819	24	24
255	Roger Federer	Grass	2014–2017	4	drop_shot_share_pct	67.000	6,895.000	0.972	24	24
256	Roger Federer	Grass	2014–2017	4	slice_share_pct	1,577.000	6,895.000	22.872	24	24
257	Roger Federer	Grass	2014–2017	4	short_rally_share_pct	3,476.000	4,652.000	74.721	24	24
258	Roger Federer	Grass	2014–2017	4	medium_rally_share_pct	912.000	4,652.000	19.604	24	24
259	Roger Federer	Grass	2014–2017	4	long_rally_share_pct	264.000	4,652.000	5.675	24	24
260	Roger Federer	Grass	2014–2017	4	deuce_wide_serve_share_pct	545.000	1,199.000	45.455	24	24
261	Roger Federer	Grass	2014–2017	4	deuce_body_serve_share_pct	178.000	1,199.000	14.846	24	24
262	Roger Federer	Grass	2014–2017	4	deuce_t_serve_share_pct	476.000	1,199.000	39.700	24	24

	player	surface	period	period_index	metric	numerator	denominator	estimate_pct	contributing_matches	cohort_matches
263	Roger Federer	Grass	2014–2017	4	ad_wide_serve_share_pct	567.000	1,093.000	51.876	24	24
264	Roger Federer	Grass	2014–2017	4	ad_body_serve_share_pct	117.000	1,093.000	10.704	24	24
265	Roger Federer	Grass	2014–2017	4	ad_t_serve_share_pct	409.000	1,093.000	37.420	24	24
266	Roger Federer	Hard	2014–2017	4	approach_frequency_pct	1,781.000	12,350.000	14.421	76	83
267	Roger Federer	Hard	2014–2017	4	net_point_frequency_pct	2,088.000	12,630.000	16.532	78	83
268	Roger Federer	Hard	2014–2017	4	serve_volley_frequency_pct	611.000	5,968.000	10.238	77	83
269	Roger Federer	Hard	2014–2017	4	drop_shot_share_pct	173.000	21,995.000	0.787	82	83
270	Roger Federer	Hard	2014–2017	4	slice_share_pct	4,153.000	21,995.000	18.882	82	83
271	Roger Federer	Hard	2014–2017	4	short_rally_share_pct	9,180.000	12,939.000	70.948	82	83
272	Roger Federer	Hard	2014–2017	4	medium_rally_share_pct	2,713.000	12,939.000	20.968	82	83
273	Roger Federer	Hard	2014–2017	4	long_rally_share_pct	1,046.000	12,939.000	8.084	82	83
274	Roger Federer	Hard	2014–2017	4	deuce_wide_serve_share_pct	1,458.000	3,317.000	43.955	82	83
275	Roger Federer	Hard	2014–2017	4	deuce_body_serve_share_pct	495.000	3,317.000	14.923	82	83
276	Roger Federer	Hard	2014–2017	4	deuce_t_serve_share_pct	1,364.000	3,317.000	41.121	82	83
277	Roger Federer	Hard	2014–2017	4	ad_wide_serve_share_pct	1,517.000	2,988.000	50.770	82	83
278	Roger Federer	Hard	2014–2017	4	ad_body_serve_share_pct	355.000	2,988.000	11.881	82	83
279	Roger Federer	Hard	2014–2017	4	ad_t_serve_share_pct	1,116.000	2,988.000	37.349	82	83
280	Roger Federer	Overall	2018–2021	5	approach_frequency_pct	2,871.000	17,542.000	16.366	99	99
281	Roger Federer	Overall	2018–2021	5	net_point_frequency_pct	3,234.000	17,542.000	18.436	99	99
282	Roger Federer	Overall	2018–2021	5	serve_volley_frequency_pct	928.000	7,750.000	11.974	94	99
283	Roger Federer	Overall	2018–2021	5	drop_shot_share_pct	415.000	29,195.000	1.421	99	99
284	Roger Federer	Overall	2018–2021	5	slice_share_pct	5,951.000	29,195.000	20.384	99	99
285	Roger Federer	Overall	2018–2021	5	short_rally_share_pct	12,140.000	17,073.000	71.106	99	99
286	Roger Federer	Overall	2018–2021	5	medium_rally_share_pct	3,490.000	17,073.000	20.442	99	99

	player	surface	period	period_index	metric	numerator	denominator	estimate_pct	contributing_matches	cohort_matches
287	Roger Federer	Overall	2018–2021	5	long_rally_share_pct	1,443.000	17,073.000	8.452	99	99
288	Roger Federer	Overall	2018–2021	5	deuce_wide_serve_share_pct	1,970.000	4,380.000	44.977	99	99
289	Roger Federer	Overall	2018–2021	5	deuce_body_serve_share_pct	583.000	4,380.000	13.311	99	99
290	Roger Federer	Overall	2018–2021	5	deuce_t_serve_share_pct	1,827.000	4,380.000	41.712	99	99
291	Roger Federer	Overall	2018–2021	5	ad_wide_serve_share_pct	1,988.000	3,966.000	50.126	99	99
292	Roger Federer	Overall	2018–2021	5	ad_body_serve_share_pct	475.000	3,966.000	11.977	99	99
293	Roger Federer	Overall	2018–2021	5	ad_t_serve_share_pct	1,503.000	3,966.000	37.897	99	99
294	Roger Federer	Clay	2018–2021	5	approach_frequency_pct	319.000	2,005.000	15.910	9	9
295	Roger Federer	Clay	2018–2021	5	net_point_frequency_pct	356.000	2,005.000	17.756	9	9
296	Roger Federer	Clay	2018–2021	5	serve_volley_frequency_pct	117.000	871.000	13.433	8	9
297	Roger Federer	Clay	2018–2021	5	drop_shot_share_pct	95.000	3,440.000	2.762	9	9
298	Roger Federer	Clay	2018–2021	5	slice_share_pct	567.000	3,440.000	16.483	9	9
299	Roger Federer	Clay	2018–2021	5	short_rally_share_pct	1,353.000	1,951.000	69.349	9	9
300	Roger Federer	Clay	2018–2021	5	medium_rally_share_pct	422.000	1,951.000	21.630	9	9
301	Roger Federer	Clay	2018–2021	5	long_rally_share_pct	176.000	1,951.000	9.021	9	9
302	Roger Federer	Clay	2018–2021	5	deuce_wide_serve_share_pct	202.000	510.000	39.608	9	9
303	Roger Federer	Clay	2018–2021	5	deuce_body_serve_share_pct	73.000	510.000	14.314	9	9
304	Roger Federer	Clay	2018–2021	5	deuce_t_serve_share_pct	235.000	510.000	46.078	9	9
305	Roger Federer	Clay	2018–2021	5	ad_wide_serve_share_pct	242.000	466.000	51.931	9	9
306	Roger Federer	Clay	2018–2021	5	ad_body_serve_share_pct	54.000	466.000	11.588	9	9
307	Roger Federer	Clay	2018–2021	5	ad_t_serve_share_pct	170.000	466.000	36.481	9	9
308	Roger Federer	Grass	2018–2021	5	approach_frequency_pct	807.000	5,240.000	15.401	27	27
309	Roger Federer	Grass	2018–2021	5	net_point_frequency_pct	929.000	5,240.000	17.729	27	27
310	Roger Federer	Grass	2018–2021	5	serve_volley_frequency_pct	271.000	2,132.000	12.711	25	27

	player	surface	period	period_index	metric	numerator	denominator	estimate_pct	contributing_matches	cohort_matches
311	Roger Federer	Grass	2018–2021	5	drop_shot_share_pct	93.000	8,066.000	1.153	27	27
312	Roger Federer	Grass	2018–2021	5	slice_share_pct	1,849.000	8,066.000	22.923	27	27
313	Roger Federer	Grass	2018–2021	5	short_rally_share_pct	3,809.000	5,113.000	74.496	27	27
314	Roger Federer	Grass	2018–2021	5	medium_rally_share_pct	960.000	5,113.000	18.776	27	27
315	Roger Federer	Grass	2018–2021	5	long_rally_share_pct	344.000	5,113.000	6.728	27	27
316	Roger Federer	Grass	2018–2021	5	deuce_wide_serve_share_pct	603.000	1,286.000	46.890	27	27
317	Roger Federer	Grass	2018–2021	5	deuce_body_serve_share_pct	140.000	1,286.000	10.886	27	27
318	Roger Federer	Grass	2018–2021	5	deuce_t_serve_share_pct	543.000	1,286.000	42.224	27	27
319	Roger Federer	Grass	2018–2021	5	ad_wide_serve_share_pct	549.000	1,149.000	47.781	27	27
320	Roger Federer	Grass	2018–2021	5	ad_body_serve_share_pct	131.000	1,149.000	11.401	27	27
321	Roger Federer	Grass	2018–2021	5	ad_t_serve_share_pct	469.000	1,149.000	40.818	27	27
322	Roger Federer	Hard	2018–2021	5	approach_frequency_pct	1,745.000	10,297.000	16.947	63	63
323	Roger Federer	Hard	2018–2021	5	net_point_frequency_pct	1,949.000	10,297.000	18.928	63	63
324	Roger Federer	Hard	2018–2021	5	serve_volley_frequency_pct	540.000	4,747.000	11.376	61	63
325	Roger Federer	Hard	2018–2021	5	drop_shot_share_pct	227.000	17,689.000	1.283	63	63
326	Roger Federer	Hard	2018–2021	5	slice_share_pct	3,535.000	17,689.000	19.984	63	63
327	Roger Federer	Hard	2018–2021	5	short_rally_share_pct	6,978.000	10,009.000	69.717	63	63
328	Roger Federer	Hard	2018–2021	5	medium_rally_share_pct	2,108.000	10,009.000	21.061	63	63
329	Roger Federer	Hard	2018–2021	5	long_rally_share_pct	923.000	10,009.000	9.222	63	63
330	Roger Federer	Hard	2018–2021	5	deuce_wide_serve_share_pct	1,165.000	2,584.000	45.085	63	63
331	Roger Federer	Hard	2018–2021	5	deuce_body_serve_share_pct	370.000	2,584.000	14.319	63	63
332	Roger Federer	Hard	2018–2021	5	deuce_t_serve_share_pct	1,049.000	2,584.000	40.596	63	63
333	Roger Federer	Hard	2018–2021	5	ad_wide_serve_share_pct	1,197.000	2,351.000	50.915	63	63
334	Roger Federer	Hard	2018–2021	5	ad_body_serve_share_pct	290.000	2,351.000	12.335	63	63

	player	surface	period	period_index	metric	numerator	denominator	estimate_pct	contributing_matches	cohort_matches
335	Roger Federer	Hard	2018–2021	5	ad_t_serve_share_pct	864.000	2,351.000	36.750	63	63

	player	surface	period	coverage_pct	small_sample	coverage_edge_period	valid_bootstraps	ci95_lower	ci95_upper
0	Roger Federer	Overall	1998–2001	93.750	False	True	2000	17.623	25.119
1	Roger Federer	Overall	1998–2001	96.875	False	True	2000	18.599	26.016
2	Roger Federer	Overall	1998–2001	100.000	False	True	2000	13.746	33.295
3	Roger Federer	Overall	1998–2001	100.000	False	True	2000	0.182	0.512
4	Roger Federer	Overall	1998–2001	100.000	False	True	2000	15.620	20.101
5	Roger Federer	Overall	1998–2001	100.000	False	True	2000	67.333	77.098
6	Roger Federer	Overall	1998–2001	100.000	False	True	2000	17.138	22.824
7	Roger Federer	Overall	1998–2001	100.000	False	True	2000	5.534	10.183
8	Roger Federer	Overall	1998–2001	100.000	False	True	2000	33.786	39.865
9	Roger Federer	Overall	1998–2001	100.000	False	True	2000	15.834	22.885
10	Roger Federer	Overall	1998–2001	100.000	False	True	2000	41.052	46.769
11	Roger Federer	Overall	1998–2001	100.000	False	True	2000	53.799	61.693
12	Roger Federer	Overall	1998–2001	100.000	False	True	2000	11.063	16.425
13	Roger Federer	Overall	1998–2001	100.000	False	True	2000	25.361	31.709
14	Roger Federer	Clay	1998–2001	100.000	True	True	1986	10.115	16.495
15	Roger Federer	Clay	1998–2001	100.000	True	True	1986	10.345	20.619
16	Roger Federer	Clay	1998–2001	100.000	True	True	1986	1.691	6.417
17	Roger Federer	Clay	1998–2001	100.000	True	True	1986	0.367	1.238
18	Roger Federer	Clay	1998–2001	100.000	True	True	1986	13.503	21.145
19	Roger Federer	Clay	1998–2001	100.000	True	True	1986	51.310	74.013
20	Roger Federer	Clay	1998–2001	100.000	True	True	1986	18.538	32.715
21	Roger Federer	Clay	1998–2001	100.000	True	True	1986	7.566	16.275
22	Roger Federer	Clay	1998–2001	100.000	True	True	1986	21.488	47.905
23	Roger Federer	Clay	1998–2001	100.000	True	True	1986	5.556	26.471
24	Roger Federer	Clay	1998–2001	100.000	True	True	1986	41.406	57.025
25	Roger Federer	Clay	1998–2001	100.000	True	True	1986	57.407	75.692
26	Roger Federer	Clay	1998–2001	100.000	True	True	1986	5.594	14.904
27	Roger Federer	Clay	1998–2001	100.000	True	True	1986	16.667	29.071
28	Roger Federer	Grass	1998–2001	83.333	True	True	1989	22.145	44.011
29	Roger Federer	Grass	1998–2001	100.000	True	True	1997	17.597	41.078
30	Roger Federer	Grass	1998–2001	100.000	True	True	1997	33.735	79.327
31	Roger Federer	Grass	1998–2001	100.000	True	True	1997	0.000	0.322
32	Roger Federer	Grass	1998–2001	100.000	True	True	1997	8.103	18.295
33	Roger Federer	Grass	1998–2001	100.000	True	True	1997	83.121	90.439

	player	surface	period	coverage_pct	small_sample	coverage_edge_period	valid_bootstraps	ci95_lower	ci95_upper
34	Roger Federer	Grass	1998–2001	100.000	True	True	1997	8.629	14.787
35	Roger Federer	Grass	1998–2001	100.000	True	True	1997	0.265	2.700
36	Roger Federer	Grass	1998–2001	100.000	True	True	1997	29.545	42.857
37	Roger Federer	Grass	1998–2001	100.000	True	True	1997	13.907	25.806
38	Roger Federer	Grass	1998–2001	100.000	True	True	1997	35.526	51.351
39	Roger Federer	Grass	1998–2001	100.000	True	True	1997	40.000	58.377
40	Roger Federer	Grass	1998–2001	100.000	True	True	1997	8.770	24.510
41	Roger Federer	Grass	1998–2001	100.000	True	True	1997	30.234	38.168
42	Roger Federer	Hard	1998–2001	95.238	True	True	2000	16.512	24.330
43	Roger Federer	Hard	1998–2001	95.238	True	True	2000	17.866	25.783
44	Roger Federer	Hard	1998–2001	100.000	True	True	2000	10.538	24.904
45	Roger Federer	Hard	1998–2001	100.000	True	True	2000	0.084	0.337
46	Roger Federer	Hard	1998–2001	100.000	True	True	2000	16.699	22.226
47	Roger Federer	Hard	1998–2001	100.000	True	True	2000	67.116	74.688
48	Roger Federer	Hard	1998–2001	100.000	True	True	2000	18.832	22.869
49	Roger Federer	Hard	1998–2001	100.000	True	True	2000	5.991	10.641
50	Roger Federer	Hard	1998–2001	100.000	True	True	2000	35.016	41.815
51	Roger Federer	Hard	1998–2001	100.000	True	True	2000	15.189	24.732
52	Roger Federer	Hard	1998–2001	100.000	True	True	2000	38.836	44.633
53	Roger Federer	Hard	1998–2001	100.000	True	True	2000	51.799	60.976
54	Roger Federer	Hard	1998–2001	100.000	True	True	2000	11.428	18.280
55	Roger Federer	Hard	1998–2001	100.000	True	True	2000	24.597	33.065
56	Roger Federer	Overall	2002–2005	98.148	False	False	2000	16.408	18.633
57	Roger Federer	Overall	2002–2005	98.148	False	False	2000	18.073	20.278
58	Roger Federer	Overall	2002–2005	97.531	False	False	2000	14.481	19.014
59	Roger Federer	Overall	2002–2005	100.000	False	False	2000	0.542	0.750
60	Roger Federer	Overall	2002–2005	100.000	False	False	2000	18.932	20.637
61	Roger Federer	Overall	2002–2005	100.000	False	False	2000	67.271	69.783
62	Roger Federer	Overall	2002–2005	100.000	False	False	2000	21.025	22.517
63	Roger Federer	Overall	2002–2005	100.000	False	False	2000	8.933	10.416
64	Roger Federer	Overall	2002–2005	100.000	False	False	2000	38.008	40.969
65	Roger Federer	Overall	2002–2005	100.000	False	False	2000	16.857	19.983
66	Roger Federer	Overall	2002–2005	100.000	False	False	2000	40.345	44.085
67	Roger Federer	Overall	2002–2005	100.000	False	False	2000	54.321	57.234
68	Roger Federer	Overall	2002–2005	100.000	False	False	2000	12.504	14.948
69	Roger Federer	Overall	2002–2005	100.000	False	False	2000	29.221	31.817
70	Roger Federer	Clay	2002–2005	96.774	False	False	2000	14.523	20.272
71	Roger Federer	Clay	2002–2005	96.774	False	False	2000	16.649	22.242
72	Roger Federer	Clay	2002–2005	96.774	False	False	2000	9.206	20.405

	player	surface	period	coverage_pct	small_sample	coverage_edge_period	valid_bootstraps	ci95_lower	ci95_upper
73	Roger Federer	Clay	2002–2005	100.000	False	False	2000	0.831	1.425
74	Roger Federer	Clay	2002–2005	100.000	False	False	2000	15.210	19.616
75	Roger Federer	Clay	2002–2005	100.000	False	False	2000	63.611	69.773
76	Roger Federer	Clay	2002–2005	100.000	False	False	2000	20.576	24.037
77	Roger Federer	Clay	2002–2005	100.000	False	False	2000	9.334	12.712
78	Roger Federer	Clay	2002–2005	100.000	False	False	2000	32.870	38.377
79	Roger Federer	Clay	2002–2005	100.000	False	False	2000	14.764	24.494
80	Roger Federer	Clay	2002–2005	100.000	False	False	2000	38.818	50.375
81	Roger Federer	Clay	2002–2005	100.000	False	False	2000	56.763	64.018
82	Roger Federer	Clay	2002–2005	100.000	False	False	2000	8.617	14.006
83	Roger Federer	Clay	2002–2005	100.000	False	False	2000	25.126	31.736
84	Roger Federer	Grass	2002–2005	100.000	True	False	2000	16.963	23.495
85	Roger Federer	Grass	2002–2005	100.000	True	False	2000	18.652	25.232
86	Roger Federer	Grass	2002–2005	100.000	True	False	2000	18.721	35.443
87	Roger Federer	Grass	2002–2005	100.000	True	False	2000	0.252	0.684
88	Roger Federer	Grass	2002–2005	100.000	True	False	2000	16.800	21.666
89	Roger Federer	Grass	2002–2005	100.000	True	False	2000	69.880	76.851
90	Roger Federer	Grass	2002–2005	100.000	True	False	2000	17.698	21.781
91	Roger Federer	Grass	2002–2005	100.000	True	False	2000	5.085	8.626
92	Roger Federer	Grass	2002–2005	100.000	True	False	2000	37.555	43.820
93	Roger Federer	Grass	2002–2005	100.000	True	False	2000	12.410	21.908
94	Roger Federer	Grass	2002–2005	100.000	True	False	2000	38.095	46.366
95	Roger Federer	Grass	2002–2005	100.000	True	False	2000	49.367	59.781
96	Roger Federer	Grass	2002–2005	100.000	True	False	2000	9.966	17.523
97	Roger Federer	Grass	2002–2005	100.000	True	False	2000	28.200	35.664
98	Roger Federer	Hard	2002–2005	98.182	False	False	2000	15.768	18.360
99	Roger Federer	Hard	2002–2005	98.182	False	False	2000	17.317	19.870
100	Roger Federer	Hard	2002–2005	97.273	False	False	2000	12.968	17.924
101	Roger Federer	Hard	2002–2005	100.000	False	False	2000	0.442	0.670
102	Roger Federer	Hard	2002–2005	100.000	False	False	2000	19.684	21.488
103	Roger Federer	Hard	2002–2005	100.000	False	False	2000	66.652	69.517
104	Roger Federer	Hard	2002–2005	100.000	False	False	2000	21.136	22.869
105	Roger Federer	Hard	2002–2005	100.000	False	False	2000	8.984	10.735
106	Roger Federer	Hard	2002–2005	100.000	False	False	2000	38.487	42.228
107	Roger Federer	Hard	2002–2005	100.000	False	False	2000	16.620	20.029
108	Roger Federer	Hard	2002–2005	100.000	False	False	2000	39.316	43.716
109	Roger Federer	Hard	2002–2005	100.000	False	False	2000	53.268	56.342
110	Roger Federer	Hard	2002–2005	100.000	False	False	2000	12.852	15.728
111	Roger Federer	Hard	2002–2005	100.000	False	False	2000	29.423	32.322

	player	surface	period	coverage_pct	small_sample	coverage_edge_period	valid_bootstraps	ci95_lower	ci95_upper
112	Roger Federer	Overall	2006–2009	94.706	False	False	2000	13.024	15.042
113	Roger Federer	Overall	2006–2009	95.882	False	False	2000	14.444	16.470
114	Roger Federer	Overall	2006–2009	97.647	False	False	2000	4.346	5.878
115	Roger Federer	Overall	2006–2009	100.000	False	False	2000	0.728	0.962
116	Roger Federer	Overall	2006–2009	100.000	False	False	2000	18.060	20.086
117	Roger Federer	Overall	2006–2009	100.000	False	False	2000	64.179	66.665
118	Roger Federer	Overall	2006–2009	100.000	False	False	2000	22.281	23.670
119	Roger Federer	Overall	2006–2009	100.000	False	False	2000	10.848	12.346
120	Roger Federer	Overall	2006–2009	100.000	False	False	2000	37.173	40.867
121	Roger Federer	Overall	2006–2009	100.000	False	False	2000	15.432	18.891
122	Roger Federer	Overall	2006–2009	100.000	False	False	2000	41.780	46.092
123	Roger Federer	Overall	2006–2009	100.000	False	False	2000	50.232	53.474
124	Roger Federer	Overall	2006–2009	100.000	False	False	2000	11.684	15.138
125	Roger Federer	Overall	2006–2009	100.000	False	False	2000	33.051	36.530
126	Roger Federer	Clay	2006–2009	90.698	False	False	2000	12.705	17.227
127	Roger Federer	Clay	2006–2009	93.023	False	False	2000	14.168	18.725
128	Roger Federer	Clay	2006–2009	93.023	False	False	2000	3.213	5.719
129	Roger Federer	Clay	2006–2009	100.000	False	False	2000	1.175	1.718
130	Roger Federer	Clay	2006–2009	100.000	False	False	2000	11.810	15.114
131	Roger Federer	Clay	2006–2009	100.000	False	False	2000	57.188	61.739
132	Roger Federer	Clay	2006–2009	100.000	False	False	2000	24.214	27.269
133	Roger Federer	Clay	2006–2009	100.000	False	False	2000	13.572	16.302
134	Roger Federer	Clay	2006–2009	100.000	False	False	2000	35.282	45.242
135	Roger Federer	Clay	2006–2009	100.000	False	False	2000	12.180	18.347
136	Roger Federer	Clay	2006–2009	100.000	False	False	2000	38.605	50.154
137	Roger Federer	Clay	2006–2009	100.000	False	False	2000	48.585	56.426
138	Roger Federer	Clay	2006–2009	100.000	False	False	2000	11.194	21.045
139	Roger Federer	Clay	2006–2009	100.000	False	False	2000	28.164	34.780
140	Roger Federer	Grass	2006–2009	95.000	True	False	2000	9.505	15.160
141	Roger Federer	Grass	2006–2009	100.000	True	False	2000	9.968	16.002
142	Roger Federer	Grass	2006–2009	100.000	True	False	2000	3.736	7.045
143	Roger Federer	Grass	2006–2009	100.000	True	False	2000	0.416	0.780
144	Roger Federer	Grass	2006–2009	100.000	True	False	2000	19.821	27.300
145	Roger Federer	Grass	2006–2009	100.000	True	False	2000	70.586	75.624
146	Roger Federer	Grass	2006–2009	100.000	True	False	2000	17.819	21.153
147	Roger Federer	Grass	2006–2009	100.000	True	False	2000	6.165	8.549
148	Roger Federer	Grass	2006–2009	100.000	True	False	2000	34.707	44.315
149	Roger Federer	Grass	2006–2009	100.000	True	False	2000	12.948	21.518
150	Roger Federer	Grass	2006–2009	100.000	True	False	2000	39.594	47.226

	player	surface	period	coverage_pct	small_sample	coverage_edge_period	valid_bootstraps	ci95_lower	ci95_upper
151	Roger Federer	Grass	2006–2009	100.000	True	False	2000	45.850	52.448
152	Roger Federer	Grass	2006–2009	100.000	True	False	2000	10.121	15.007
153	Roger Federer	Grass	2006–2009	100.000	True	False	2000	35.145	41.657
154	Roger Federer	Hard	2006–2009	96.262	False	False	2000	12.754	15.352
155	Roger Federer	Hard	2006–2009	96.262	False	False	2000	14.346	16.882
156	Roger Federer	Hard	2006–2009	99.065	False	False	2000	4.338	6.398
157	Roger Federer	Hard	2006–2009	100.000	False	False	2000	0.487	0.694
158	Roger Federer	Hard	2006–2009	100.000	False	False	2000	20.137	21.927
159	Roger Federer	Hard	2006–2009	100.000	False	False	2000	64.913	67.610
160	Roger Federer	Hard	2006–2009	100.000	False	False	2000	21.780	23.331
161	Roger Federer	Hard	2006–2009	100.000	False	False	2000	10.286	12.027
162	Roger Federer	Hard	2006–2009	100.000	False	False	2000	36.355	40.505
163	Roger Federer	Hard	2006–2009	100.000	False	False	2000	15.682	20.285
164	Roger Federer	Hard	2006–2009	100.000	False	False	2000	41.221	46.359
165	Roger Federer	Hard	2006–2009	100.000	False	False	2000	50.171	54.141
166	Roger Federer	Hard	2006–2009	100.000	False	False	2000	10.818	14.217
167	Roger Federer	Hard	2006–2009	100.000	False	False	2000	33.080	37.708
168	Roger Federer	Overall	2010–2013	91.045	False	False	2000	11.546	13.371
169	Roger Federer	Overall	2010–2013	97.015	False	False	2000	13.349	15.175
170	Roger Federer	Overall	2010–2013	98.507	False	False	2000	3.684	4.997
171	Roger Federer	Overall	2010–2013	100.000	False	False	2000	1.186	1.450
172	Roger Federer	Overall	2010–2013	100.000	False	False	2000	17.420	19.409
173	Roger Federer	Overall	2010–2013	100.000	False	False	2000	64.591	67.082
174	Roger Federer	Overall	2010–2013	100.000	False	False	2000	22.338	23.711
175	Roger Federer	Overall	2010–2013	100.000	False	False	2000	10.313	11.999
176	Roger Federer	Overall	2010–2013	100.000	False	False	2000	41.759	45.402
177	Roger Federer	Overall	2010–2013	100.000	False	False	2000	16.845	20.037
178	Roger Federer	Overall	2010–2013	100.000	False	False	2000	36.164	39.830
179	Roger Federer	Overall	2010–2013	100.000	False	False	2000	47.428	51.065
180	Roger Federer	Overall	2010–2013	100.000	False	False	2000	14.980	18.440
181	Roger Federer	Overall	2010–2013	100.000	False	False	2000	32.229	35.810
182	Roger Federer	Clay	2010–2013	95.652	True	False	2000	9.518	14.703
183	Roger Federer	Clay	2010–2013	100.000	True	False	2000	10.416	15.725
184	Roger Federer	Clay	2010–2013	100.000	True	False	2000	2.736	5.906
185	Roger Federer	Clay	2010–2013	100.000	True	False	2000	1.395	2.167
186	Roger Federer	Clay	2010–2013	100.000	True	False	2000	12.409	17.268
187	Roger Federer	Clay	2010–2013	100.000	True	False	2000	61.457	66.318
188	Roger Federer	Clay	2010–2013	100.000	True	False	2000	22.496	26.313
189	Roger Federer	Clay	2010–2013	100.000	True	False	2000	10.032	13.415

	player	surface	period	coverage_pct	small_sample	coverage_edge_period	valid_bootstraps	ci95_lower	ci95_upper
190	Roger Federer	Clay	2010–2013	100.000	True	False	2000	37.549	48.760
191	Roger Federer	Clay	2010–2013	100.000	True	False	2000	12.994	20.144
192	Roger Federer	Clay	2010–2013	100.000	True	False	2000	35.455	44.841
193	Roger Federer	Clay	2010–2013	100.000	True	False	2000	46.091	56.235
194	Roger Federer	Clay	2010–2013	100.000	True	False	2000	10.810	20.589
195	Roger Federer	Clay	2010–2013	100.000	True	False	2000	28.172	39.103
196	Roger Federer	Grass	2010–2013	90.476	True	False	2000	11.989	14.716
197	Roger Federer	Grass	2010–2013	100.000	True	False	2000	13.512	16.812
198	Roger Federer	Grass	2010–2013	100.000	True	False	2000	3.238	6.211
199	Roger Federer	Grass	2010–2013	100.000	True	False	2000	0.908	1.477
200	Roger Federer	Grass	2010–2013	100.000	True	False	2000	19.920	23.233
201	Roger Federer	Grass	2010–2013	100.000	True	False	2000	67.294	72.999
202	Roger Federer	Grass	2010–2013	100.000	True	False	2000	19.951	23.183
203	Roger Federer	Grass	2010–2013	100.000	True	False	2000	6.733	9.874
204	Roger Federer	Grass	2010–2013	100.000	True	False	2000	41.061	46.970
205	Roger Federer	Grass	2010–2013	100.000	True	False	2000	14.601	21.237
206	Roger Federer	Grass	2010–2013	100.000	True	False	2000	35.444	40.786
207	Roger Federer	Grass	2010–2013	100.000	True	False	2000	46.604	54.665
208	Roger Federer	Grass	2010–2013	100.000	True	False	2000	11.154	15.944
209	Roger Federer	Grass	2010–2013	100.000	True	False	2000	33.192	38.819
210	Roger Federer	Hard	2010–2013	90.000	False	False	2000	11.188	13.501
211	Roger Federer	Hard	2010–2013	95.556	False	False	2000	13.203	15.497
212	Roger Federer	Hard	2010–2013	97.778	False	False	2000	3.424	5.141
213	Roger Federer	Hard	2010–2013	100.000	False	False	2000	1.071	1.355
214	Roger Federer	Hard	2010–2013	100.000	False	False	2000	17.471	19.703
215	Roger Federer	Hard	2010–2013	100.000	False	False	2000	63.712	66.645
216	Roger Federer	Hard	2010–2013	100.000	False	False	2000	22.256	23.861
217	Roger Federer	Hard	2010–2013	100.000	False	False	2000	10.739	12.809
218	Roger Federer	Hard	2010–2013	100.000	False	False	2000	41.247	45.700
219	Roger Federer	Hard	2010–2013	100.000	False	False	2000	17.064	21.189
220	Roger Federer	Hard	2010–2013	100.000	False	False	2000	35.007	39.790
221	Roger Federer	Hard	2010–2013	100.000	False	False	2000	46.109	50.577
222	Roger Federer	Hard	2010–2013	100.000	False	False	2000	15.826	20.292
223	Roger Federer	Hard	2010–2013	100.000	False	False	2000	31.372	35.820
224	Roger Federer	Overall	2014–2017	92.063	False	False	2000	13.303	15.320
225	Roger Federer	Overall	2014–2017	96.032	False	False	2000	15.353	17.371
226	Roger Federer	Overall	2014–2017	95.238	False	False	2000	9.593	11.824
227	Roger Federer	Overall	2014–2017	99.206	False	False	2000	0.823	1.133
228	Roger Federer	Overall	2014–2017	99.206	False	False	2000	18.418	20.222

	player	surface	period	coverage_pct	small_sample	coverage_edge_period	valid_bootstraps	ci95_lower	ci95_upper
229	Roger Federer	Overall	2014–2017	99.206	False	False	2000	69.986	72.471
230	Roger Federer	Overall	2014–2017	99.206	False	False	2000	20.245	21.736
231	Roger Federer	Overall	2014–2017	99.206	False	False	2000	7.082	8.522
232	Roger Federer	Overall	2014–2017	99.206	False	False	2000	41.705	44.661
233	Roger Federer	Overall	2014–2017	99.206	False	False	2000	14.502	17.824
234	Roger Federer	Overall	2014–2017	99.206	False	False	2000	38.909	42.358
235	Roger Federer	Overall	2014–2017	99.206	False	False	2000	50.363	53.781
236	Roger Federer	Overall	2014–2017	99.206	False	False	2000	10.583	13.078
237	Roger Federer	Overall	2014–2017	99.206	False	False	2000	34.623	37.525
238	Roger Federer	Clay	2014–2017	89.474	True	False	2000	11.111	13.866
239	Roger Federer	Clay	2014–2017	100.000	True	False	2000	13.355	16.098
240	Roger Federer	Clay	2014–2017	100.000	True	False	2000	5.899	9.678
241	Roger Federer	Clay	2014–2017	100.000	True	False	2000	1.164	2.196
242	Roger Federer	Clay	2014–2017	100.000	True	False	2000	14.990	18.929
243	Roger Federer	Clay	2014–2017	100.000	True	False	2000	64.110	70.080
244	Roger Federer	Clay	2014–2017	100.000	True	False	2000	21.051	25.382
245	Roger Federer	Clay	2014–2017	100.000	True	False	2000	8.243	11.523
246	Roger Federer	Clay	2014–2017	100.000	True	False	2000	33.085	40.759
247	Roger Federer	Clay	2014–2017	100.000	True	False	2000	18.003	27.837
248	Roger Federer	Clay	2014–2017	100.000	True	False	2000	35.153	45.279
249	Roger Federer	Clay	2014–2017	100.000	True	False	2000	53.234	60.621
250	Roger Federer	Clay	2014–2017	100.000	True	False	2000	10.584	16.476
251	Roger Federer	Clay	2014–2017	100.000	True	False	2000	26.377	32.952
252	Roger Federer	Grass	2014–2017	95.833	True	False	2000	12.464	18.015
253	Roger Federer	Grass	2014–2017	100.000	True	False	2000	14.423	19.826
254	Roger Federer	Grass	2014–2017	100.000	True	False	2000	10.860	17.110
255	Roger Federer	Grass	2014–2017	100.000	True	False	2000	0.707	1.292
256	Roger Federer	Grass	2014–2017	100.000	True	False	2000	21.126	24.647
257	Roger Federer	Grass	2014–2017	100.000	True	False	2000	72.368	77.154
258	Roger Federer	Grass	2014–2017	100.000	True	False	2000	18.137	20.941
259	Roger Federer	Grass	2014–2017	100.000	True	False	2000	4.150	7.206
260	Roger Federer	Grass	2014–2017	100.000	True	False	2000	42.327	48.174
261	Roger Federer	Grass	2014–2017	100.000	True	False	2000	12.300	17.745
262	Roger Federer	Grass	2014–2017	100.000	True	False	2000	35.739	43.296
263	Roger Federer	Grass	2014–2017	100.000	True	False	2000	48.799	55.237
264	Roger Federer	Grass	2014–2017	100.000	True	False	2000	8.812	12.533
265	Roger Federer	Grass	2014–2017	100.000	True	False	2000	34.391	40.132
266	Roger Federer	Hard	2014–2017	91.566	False	False	2000	13.203	15.596
267	Roger Federer	Hard	2014–2017	93.976	False	False	2000	15.256	17.759

	player	surface	period	coverage_pct	small_sample	coverage_edge_period	valid_bootstraps	ci95_lower	ci95_upper
268	Roger Federer	Hard	2014–2017	92.771	False	False	2000	8.997	11.549
269	Roger Federer	Hard	2014–2017	98.795	False	False	2000	0.631	0.957
270	Roger Federer	Hard	2014–2017	98.795	False	False	2000	17.707	19.978
271	Roger Federer	Hard	2014–2017	98.795	False	False	2000	69.530	72.352
272	Roger Federer	Hard	2014–2017	98.795	False	False	2000	20.105	21.850
273	Roger Federer	Hard	2014–2017	98.795	False	False	2000	7.197	8.945
274	Roger Federer	Hard	2014–2017	98.795	False	False	2000	42.191	45.775
275	Roger Federer	Hard	2014–2017	98.795	False	False	2000	12.983	16.971
276	Roger Federer	Hard	2014–2017	98.795	False	False	2000	38.948	43.323
277	Roger Federer	Hard	2014–2017	98.795	False	False	2000	48.585	52.959
278	Roger Federer	Hard	2014–2017	98.795	False	False	2000	10.077	13.591
279	Roger Federer	Hard	2014–2017	98.795	False	False	2000	35.709	38.831
280	Roger Federer	Overall	2018–2021	100.000	False	True	2000	15.192	17.461
281	Roger Federer	Overall	2018–2021	100.000	False	True	2000	17.141	19.594
282	Roger Federer	Overall	2018–2021	94.949	False	True	2000	10.597	13.409
283	Roger Federer	Overall	2018–2021	100.000	False	True	2000	1.194	1.659
284	Roger Federer	Overall	2018–2021	100.000	False	True	2000	19.132	21.684
285	Roger Federer	Overall	2018–2021	100.000	False	True	2000	69.790	72.494
286	Roger Federer	Overall	2018–2021	100.000	False	True	2000	19.584	21.244
287	Roger Federer	Overall	2018–2021	100.000	False	True	2000	7.660	9.253
288	Roger Federer	Overall	2018–2021	100.000	False	True	2000	43.236	46.831
289	Roger Federer	Overall	2018–2021	100.000	False	True	2000	11.933	14.756
290	Roger Federer	Overall	2018–2021	100.000	False	True	2000	39.869	43.520
291	Roger Federer	Overall	2018–2021	100.000	False	True	2000	48.212	52.053
292	Roger Federer	Overall	2018–2021	100.000	False	True	2000	10.764	13.318
293	Roger Federer	Overall	2018–2021	100.000	False	True	2000	36.161	39.680
294	Roger Federer	Clay	2018–2021	100.000	True	True	2000	11.508	18.947
295	Roger Federer	Clay	2018–2021	100.000	True	True	2000	12.721	21.518
296	Roger Federer	Clay	2018–2021	88.889	True	True	2000	8.050	18.167
297	Roger Federer	Clay	2018–2021	100.000	True	True	2000	1.934	3.367
298	Roger Federer	Clay	2018–2021	100.000	True	True	2000	13.987	19.836
299	Roger Federer	Clay	2018–2021	100.000	True	True	2000	63.307	74.534
300	Roger Federer	Clay	2018–2021	100.000	True	True	2000	18.512	25.246
301	Roger Federer	Clay	2018–2021	100.000	True	True	2000	6.769	12.025
302	Roger Federer	Clay	2018–2021	100.000	True	True	2000	33.526	46.307
303	Roger Federer	Clay	2018–2021	100.000	True	True	2000	8.247	19.964
304	Roger Federer	Clay	2018–2021	100.000	True	True	2000	39.155	53.062
305	Roger Federer	Clay	2018–2021	100.000	True	True	2000	41.970	61.328
306	Roger Federer	Clay	2018–2021	100.000	True	True	2000	5.818	19.544

	player	surface	period	coverage_pct	small_sample	coverage_edge_period	valid_bootstraps	ci95_lower	ci95_upper
307	Roger Federer	Clay	2018–2021	100.000	True	True	2000	27.673	44.918
308	Roger Federer	Grass	2018–2021	100.000	True	True	2000	13.644	17.233
309	Roger Federer	Grass	2018–2021	100.000	True	True	2000	15.752	19.673
310	Roger Federer	Grass	2018–2021	92.593	True	True	2000	9.919	15.803
311	Roger Federer	Grass	2018–2021	100.000	True	True	2000	0.852	1.448
312	Roger Federer	Grass	2018–2021	100.000	True	True	2000	20.324	25.131
313	Roger Federer	Grass	2018–2021	100.000	True	True	2000	71.747	77.219
314	Roger Federer	Grass	2018–2021	100.000	True	True	2000	17.246	20.201
315	Roger Federer	Grass	2018–2021	100.000	True	True	2000	5.213	8.416
316	Roger Federer	Grass	2018–2021	100.000	True	True	2000	43.886	49.734
317	Roger Federer	Grass	2018–2021	100.000	True	True	2000	9.204	12.643
318	Roger Federer	Grass	2018–2021	100.000	True	True	2000	39.680	44.857
319	Roger Federer	Grass	2018–2021	100.000	True	True	2000	44.425	51.000
320	Roger Federer	Grass	2018–2021	100.000	True	True	2000	9.970	12.913
321	Roger Federer	Grass	2018–2021	100.000	True	True	2000	37.852	44.039
322	Roger Federer	Hard	2018–2021	100.000	False	True	2000	15.378	18.422
323	Roger Federer	Hard	2018–2021	100.000	False	True	2000	17.275	20.434
324	Roger Federer	Hard	2018–2021	96.825	False	True	2000	9.726	13.124
325	Roger Federer	Hard	2018–2021	100.000	False	True	2000	1.004	1.589
326	Roger Federer	Hard	2018–2021	100.000	False	True	2000	18.615	21.556
327	Roger Federer	Hard	2018–2021	100.000	False	True	2000	68.330	71.001
328	Roger Federer	Hard	2018–2021	100.000	False	True	2000	20.157	22.006
329	Roger Federer	Hard	2018–2021	100.000	False	True	2000	8.359	10.128
330	Roger Federer	Hard	2018–2021	100.000	False	True	2000	42.819	47.554
331	Roger Federer	Hard	2018–2021	100.000	False	True	2000	12.566	16.235
332	Roger Federer	Hard	2018–2021	100.000	False	True	2000	38.112	43.188
333	Roger Federer	Hard	2018–2021	100.000	False	True	2000	48.789	52.873
334	Roger Federer	Hard	2018–2021	100.000	False	True	2000	10.780	13.981
335	Roger Federer	Hard	2018–2021	100.000	False	True	2000	34.984	38.625

Rafael Nadal

▼ Complete evidence table: 336 rows × 16 columns

	player	surface	period	period_index	metric	numerator	denominator	estimate_pct	contributing_matches	cohort_matches
336	Rafael Nadal	Overall	2003–2006	0	approach_frequency_pct	541.000	6,744.000	8.022	35	38
337	Rafael Nadal	Overall	2003–2006	0	net_point_frequency_pct	703.000	6,744.000	10.424	35	38
338	Rafael Nadal	Overall	2003–2006	0	serve_volley_frequency_pct	46.000	3,599.000	1.278	37	38
339	Rafael Nadal	Overall	2003–2006	0	drop_shot_share_pct	198.000	16,821.000	1.177	38	38
340	Rafael Nadal	Overall	2003–2006	0	slice_share_pct	1,304.000	16,821.000	7.752	38	38
341	Rafael Nadal	Overall	2003–2006	0	short_rally_share_pct	3,945.000	7,093.000	55.618	38	38
342	Rafael Nadal	Overall	2003–2006	0	medium_rally_share_pct	1,863.000	7,093.000	26.265	38	38
343	Rafael Nadal	Overall	2003–2006	0	long_rally_share_pct	1,285.000	7,093.000	18.116	38	38
344	Rafael Nadal	Overall	2003–2006	0	deuce_wide_serve_share_pct	380.000	1,899.000	20.011	38	38
345	Rafael Nadal	Overall	2003–2006	0	deuce_body_serve_share_pct	434.000	1,899.000	22.854	38	38
346	Rafael Nadal	Overall	2003–2006	0	deuce_t_serve_share_pct	1,085.000	1,899.000	57.135	38	38
347	Rafael Nadal	Overall	2003–2006	0	ad_wide_serve_share_pct	1,132.000	1,748.000	64.760	38	38
348	Rafael Nadal	Overall	2003–2006	0	ad_body_serve_share_pct	350.000	1,748.000	20.023	38	38
349	Rafael Nadal	Overall	2003–2006	0	ad_t_serve_share_pct	266.000	1,748.000	15.217	38	38
350	Rafael Nadal	Clay	2003–2006	0	approach_frequency_pct	200.000	2,650.000	7.547	12	13
351	Rafael Nadal	Clay	2003–2006	0	net_point_frequency_pct	286.000	2,650.000	10.792	12	13
352	Rafael Nadal	Clay	2003–2006	0	serve_volley_frequency_pct	9.000	1,301.000	0.692	12	13
353	Rafael Nadal	Clay	2003–2006	0	drop_shot_share_pct	112.000	7,453.000	1.503	13	13
354	Rafael Nadal	Clay	2003–2006	0	slice_share_pct	449.000	7,453.000	6.024	13	13
355	Rafael Nadal	Clay	2003–2006	0	short_rally_share_pct	1,290.000	2,688.000	47.991	13	13
356	Rafael Nadal	Clay	2003–2006	0	medium_rally_share_pct	761.000	2,688.000	28.311	13	13
357	Rafael Nadal	Clay	2003–2006	0	long_rally_share_pct	637.000	2,688.000	23.698	13	13
358	Rafael Nadal	Clay	2003–2006	0	deuce_wide_serve_share_pct	95.000	704.000	13.494	13	13

	player	surface	period	period_index	metric	numerator	denominator	estimate_pct	contributing_matches	cohort_matches
359	Rafael Nadal	Clay	2003–2006	0	deuce_body_serve_share_pct	186.000	704.000	26.420	13	13
360	Rafael Nadal	Clay	2003–2006	0	deuce_t_serve_share_pct	423.000	704.000	60.085	13	13
361	Rafael Nadal	Clay	2003–2006	0	ad_wide_serve_share_pct	491.000	645.000	76.124	13	13
362	Rafael Nadal	Clay	2003–2006	0	ad_body_serve_share_pct	95.000	645.000	14.729	13	13
363	Rafael Nadal	Clay	2003–2006	0	ad_t_serve_share_pct	59.000	645.000	9.147	13	13
364	Rafael Nadal	Grass	2003–2006	0	approach_frequency_pct	160.000	1,393.000	11.486	7	7
365	Rafael Nadal	Grass	2003–2006	0	net_point_frequency_pct	187.000	1,393.000	13.424	7	7
366	Rafael Nadal	Grass	2003–2006	0	serve_volley_frequency_pct	18.000	696.000	2.586	7	7
367	Rafael Nadal	Grass	2003–2006	0	drop_shot_share_pct	29.000	2,468.000	1.175	7	7
368	Rafael Nadal	Grass	2003–2006	0	slice_share_pct	323.000	2,468.000	13.088	7	7
369	Rafael Nadal	Grass	2003–2006	0	short_rally_share_pct	884.000	1,353.000	65.336	7	7
370	Rafael Nadal	Grass	2003–2006	0	medium_rally_share_pct	329.000	1,353.000	24.316	7	7
371	Rafael Nadal	Grass	2003–2006	0	long_rally_share_pct	140.000	1,353.000	10.347	7	7
372	Rafael Nadal	Grass	2003–2006	0	deuce_wide_serve_share_pct	80.000	359.000	22.284	7	7
373	Rafael Nadal	Grass	2003–2006	0	deuce_body_serve_share_pct	56.000	359.000	15.599	7	7
374	Rafael Nadal	Grass	2003–2006	0	deuce_t_serve_share_pct	223.000	359.000	62.117	7	7
375	Rafael Nadal	Grass	2003–2006	0	ad_wide_serve_share_pct	173.000	337.000	51.335	7	7
376	Rafael Nadal	Grass	2003–2006	0	ad_body_serve_share_pct	75.000	337.000	22.255	7	7
377	Rafael Nadal	Grass	2003–2006	0	ad_t_serve_share_pct	89.000	337.000	26.409	7	7
378	Rafael Nadal	Hard	2003–2006	0	approach_frequency_pct	181.000	2,701.000	6.701	16	18
379	Rafael Nadal	Hard	2003–2006	0	net_point_frequency_pct	230.000	2,701.000	8.515	16	18
380	Rafael Nadal	Hard	2003–2006	0	serve_volley_frequency_pct	19.000	1,602.000	1.186	18	18
381	Rafael Nadal	Hard	2003–2006	0	drop_shot_share_pct	57.000	6,900.000	0.826	18	18
382	Rafael Nadal	Hard	2003–2006	0	slice_share_pct	532.000	6,900.000	7.710	18	18

	player	surface	period	period_index	metric	numerator	denominator	estimate_pct	contributing_matches	cohort_matches
383	Rafael Nadal	Hard	2003–2006	0	short_rally_share_pct	1,771.000	3,052.000	58.028	18	18
384	Rafael Nadal	Hard	2003–2006	0	medium_rally_share_pct	773.000	3,052.000	25.328	18	18
385	Rafael Nadal	Hard	2003–2006	0	long_rally_share_pct	508.000	3,052.000	16.645	18	18
386	Rafael Nadal	Hard	2003–2006	0	deuce_wide_serve_share_pct	205.000	836.000	24.522	18	18
387	Rafael Nadal	Hard	2003–2006	0	deuce_body_serve_share_pct	192.000	836.000	22.967	18	18
388	Rafael Nadal	Hard	2003–2006	0	deuce_t_serve_share_pct	439.000	836.000	52.512	18	18
389	Rafael Nadal	Hard	2003–2006	0	ad_wide_serve_share_pct	468.000	766.000	61.097	18	18
390	Rafael Nadal	Hard	2003–2006	0	ad_body_serve_share_pct	180.000	766.000	23.499	18	18
391	Rafael Nadal	Hard	2003–2006	0	ad_t_serve_share_pct	118.000	766.000	15.405	18	18
392	Rafael Nadal	Overall	2007–2010	1	approach_frequency_pct	944.000	14,722.000	6.412	90	96
393	Rafael Nadal	Overall	2007–2010	1	net_point_frequency_pct	1,287.000	15,119.000	8.512	92	96
394	Rafael Nadal	Overall	2007–2010	1	serve_volley_frequency_pct	77.000	7,621.000	1.010	95	96
395	Rafael Nadal	Overall	2007–2010	1	drop_shot_share_pct	270.000	35,826.000	0.754	96	96
396	Rafael Nadal	Overall	2007–2010	1	slice_share_pct	4,024.000	35,826.000	11.232	96	96
397	Rafael Nadal	Overall	2007–2010	1	short_rally_share_pct	8,948.000	15,453.000	57.905	96	96
398	Rafael Nadal	Overall	2007–2010	1	medium_rally_share_pct	3,880.000	15,453.000	25.108	96	96
399	Rafael Nadal	Overall	2007–2010	1	long_rally_share_pct	2,625.000	15,453.000	16.987	96	96
400	Rafael Nadal	Overall	2007–2010	1	deuce_wide_serve_share_pct	791.000	4,024.000	19.657	96	96
401	Rafael Nadal	Overall	2007–2010	1	deuce_body_serve_share_pct	1,139.000	4,024.000	28.305	96	96
402	Rafael Nadal	Overall	2007–2010	1	deuce_t_serve_share_pct	2,094.000	4,024.000	52.038	96	96
403	Rafael Nadal	Overall	2007–2010	1	ad_wide_serve_share_pct	1,975.000	3,652.000	54.080	96	96
404	Rafael Nadal	Overall	2007–2010	1	ad_body_serve_share_pct	893.000	3,652.000	24.452	96	96
405	Rafael Nadal	Overall	2007–2010	1	ad_t_serve_share_pct	784.000	3,652.000	21.468	96	96
406	Rafael Nadal	Clay	2007–2010	1	approach_frequency_pct	206.000	4,473.000	4.605	29	34

	player	surface	period	period_index	metric	numerator	denominator	estimate_pct	contributing_matches	cohort_matches
407	Rafael Nadal	Clay	2007–2010	1	net_point_frequency_pct	328.000	4,647.000	7.058	30	34
408	Rafael Nadal	Clay	2007–2010	1	serve_volley_frequency_pct	9.000	2,529.000	0.356	33	34
409	Rafael Nadal	Clay	2007–2010	1	drop_shot_share_pct	131.000	13,705.000	0.956	34	34
410	Rafael Nadal	Clay	2007–2010	1	slice_share_pct	1,009.000	13,705.000	7.362	34	34
411	Rafael Nadal	Clay	2007–2010	1	short_rally_share_pct	2,756.000	5,229.000	52.706	34	34
412	Rafael Nadal	Clay	2007–2010	1	medium_rally_share_pct	1,385.000	5,229.000	26.487	34	34
413	Rafael Nadal	Clay	2007–2010	1	long_rally_share_pct	1,088.000	5,229.000	20.807	34	34
414	Rafael Nadal	Clay	2007–2010	1	deuce_wide_serve_share_pct	239.000	1,346.000	17.756	34	34
415	Rafael Nadal	Clay	2007–2010	1	deuce_body_serve_share_pct	414.000	1,346.000	30.758	34	34
416	Rafael Nadal	Clay	2007–2010	1	deuce_t_serve_share_pct	693.000	1,346.000	51.486	34	34
417	Rafael Nadal	Clay	2007–2010	1	ad_wide_serve_share_pct	779.000	1,238.000	62.924	34	34
418	Rafael Nadal	Clay	2007–2010	1	ad_body_serve_share_pct	248.000	1,238.000	20.032	34	34
419	Rafael Nadal	Clay	2007–2010	1	ad_t_serve_share_pct	211.000	1,238.000	17.044	34	34
420	Rafael Nadal	Grass	2007–2010	1	approach_frequency_pct	193.000	2,341.000	8.244	12	12
421	Rafael Nadal	Grass	2007–2010	1	net_point_frequency_pct	246.000	2,341.000	10.508	12	12
422	Rafael Nadal	Grass	2007–2010	1	serve_volley_frequency_pct	15.000	1,148.000	1.307	12	12
423	Rafael Nadal	Grass	2007–2010	1	drop_shot_share_pct	35.000	4,016.000	0.872	12	12
424	Rafael Nadal	Grass	2007–2010	1	slice_share_pct	546.000	4,016.000	13.596	12	12
425	Rafael Nadal	Grass	2007–2010	1	short_rally_share_pct	1,569.000	2,298.000	68.277	12	12
426	Rafael Nadal	Grass	2007–2010	1	medium_rally_share_pct	509.000	2,298.000	22.150	12	12
427	Rafael Nadal	Grass	2007–2010	1	long_rally_share_pct	220.000	2,298.000	9.574	12	12
428	Rafael Nadal	Grass	2007–2010	1	deuce_wide_serve_share_pct	98.000	603.000	16.252	12	12
429	Rafael Nadal	Grass	2007–2010	1	deuce_body_serve_share_pct	160.000	603.000	26.534	12	12
430	Rafael Nadal	Grass	2007–2010	1	deuce_t_serve_share_pct	345.000	603.000	57.214	12	12

	player	surface	period	period_index	metric	numerator	denominator	estimate_pct	contributing_matches	cohort_matches
431	Rafael Nadal	Grass	2007–2010	1	ad_wide_serve_share_pct	277.000	545.000	50.826	12	12
432	Rafael Nadal	Grass	2007–2010	1	ad_body_serve_share_pct	134.000	545.000	24.587	12	12
433	Rafael Nadal	Grass	2007–2010	1	ad_t_serve_share_pct	134.000	545.000	24.587	12	12
434	Rafael Nadal	Hard	2007–2010	1	approach_frequency_pct	545.000	7,908.000	6.892	49	50
435	Rafael Nadal	Hard	2007–2010	1	net_point_frequency_pct	713.000	8,131.000	8.769	50	50
436	Rafael Nadal	Hard	2007–2010	1	serve_volley_frequency_pct	53.000	3,944.000	1.344	50	50
437	Rafael Nadal	Hard	2007–2010	1	drop_shot_share_pct	104.000	18,105.000	0.574	50	50
438	Rafael Nadal	Hard	2007–2010	1	slice_share_pct	2,469.000	18,105.000	13.637	50	50
439	Rafael Nadal	Hard	2007–2010	1	short_rally_share_pct	4,623.000	7,926.000	58.327	50	50
440	Rafael Nadal	Hard	2007–2010	1	medium_rally_share_pct	1,986.000	7,926.000	25.057	50	50
441	Rafael Nadal	Hard	2007–2010	1	long_rally_share_pct	1,317.000	7,926.000	16.616	50	50
442	Rafael Nadal	Hard	2007–2010	1	deuce_wide_serve_share_pct	454.000	2,075.000	21.880	50	50
443	Rafael Nadal	Hard	2007–2010	1	deuce_body_serve_share_pct	565.000	2,075.000	27.229	50	50
444	Rafael Nadal	Hard	2007–2010	1	deuce_t_serve_share_pct	1,056.000	2,075.000	50.892	50	50
445	Rafael Nadal	Hard	2007–2010	1	ad_wide_serve_share_pct	919.000	1,869.000	49.171	50	50
446	Rafael Nadal	Hard	2007–2010	1	ad_body_serve_share_pct	511.000	1,869.000	27.341	50	50
447	Rafael Nadal	Hard	2007–2010	1	ad_t_serve_share_pct	439.000	1,869.000	23.488	50	50
448	Rafael Nadal	Overall	2011–2014	2	approach_frequency_pct	807.000	16,100.000	5.012	100	111
449	Rafael Nadal	Overall	2011–2014	2	net_point_frequency_pct	1,264.000	16,678.000	7.579	103	111
450	Rafael Nadal	Overall	2011–2014	2	serve_volley_frequency_pct	47.000	8,347.000	0.563	106	111
451	Rafael Nadal	Overall	2011–2014	2	drop_shot_share_pct	214.000	39,726.000	0.539	110	111
452	Rafael Nadal	Overall	2011–2014	2	slice_share_pct	3,775.000	39,726.000	9.503	110	111
453	Rafael Nadal	Overall	2011–2014	2	short_rally_share_pct	10,254.000	17,463.000	58.718	110	111
454	Rafael Nadal	Overall	2011–2014	2	medium_rally_share_pct	4,331.000	17,463.000	24.801	110	111

	player	surface	period	period_index	metric	numerator	denominator	estimate_pct	contributing_matches	cohort_matches
455	Rafael Nadal	Overall	2011–2014	2	long_rally_share_pct	2,878.000	17,463.000	16.481	110	111
456	Rafael Nadal	Overall	2011–2014	2	deuce_wide_serve_share_pct	858.000	4,558.000	18.824	110	111
457	Rafael Nadal	Overall	2011–2014	2	deuce_body_serve_share_pct	1,700.000	4,558.000	37.297	110	111
458	Rafael Nadal	Overall	2011–2014	2	deuce_t_serve_share_pct	2,000.000	4,558.000	43.879	110	111
459	Rafael Nadal	Overall	2011–2014	2	ad_wide_serve_share_pct	2,240.000	4,136.000	54.159	110	111
460	Rafael Nadal	Overall	2011–2014	2	ad_body_serve_share_pct	1,164.000	4,136.000	28.143	110	111
461	Rafael Nadal	Overall	2011–2014	2	ad_t_serve_share_pct	732.000	4,136.000	17.698	110	111
462	Rafael Nadal	Clay	2011–2014	2	approach_frequency_pct	369.000	7,676.000	4.807	50	54
463	Rafael Nadal	Clay	2011–2014	2	net_point_frequency_pct	584.000	7,676.000	7.608	50	54
464	Rafael Nadal	Clay	2011–2014	2	serve_volley_frequency_pct	13.000	3,832.000	0.339	52	54
465	Rafael Nadal	Clay	2011–2014	2	drop_shot_share_pct	131.000	19,940.000	0.657	54	54
466	Rafael Nadal	Clay	2011–2014	2	slice_share_pct	1,188.000	19,940.000	5.958	54	54
467	Rafael Nadal	Clay	2011–2014	2	short_rally_share_pct	4,364.000	8,034.000	54.319	54	54
468	Rafael Nadal	Clay	2011–2014	2	medium_rally_share_pct	2,175.000	8,034.000	27.072	54	54
469	Rafael Nadal	Clay	2011–2014	2	long_rally_share_pct	1,495.000	8,034.000	18.608	54	54
470	Rafael Nadal	Clay	2011–2014	2	deuce_wide_serve_share_pct	379.000	2,088.000	18.151	54	54
471	Rafael Nadal	Clay	2011–2014	2	deuce_body_serve_share_pct	828.000	2,088.000	39.655	54	54
472	Rafael Nadal	Clay	2011–2014	2	deuce_t_serve_share_pct	881.000	2,088.000	42.193	54	54
473	Rafael Nadal	Clay	2011–2014	2	ad_wide_serve_share_pct	1,141.000	1,890.000	60.370	54	54
474	Rafael Nadal	Clay	2011–2014	2	ad_body_serve_share_pct	467.000	1,890.000	24.709	54	54
475	Rafael Nadal	Clay	2011–2014	2	ad_t_serve_share_pct	282.000	1,890.000	14.921	54	54
476	Rafael Nadal	Grass	2011–2014	2	approach_frequency_pct	136.000	2,193.000	6.202	10	11
477	Rafael Nadal	Grass	2011–2014	2	net_point_frequency_pct	188.000	2,193.000	8.573	10	11
478	Rafael Nadal	Grass	2011–2014	2	serve_volley_frequency_pct	13.000	941.000	1.382	9	11

	player	surface	period	period_index	metric	numerator	denominator	estimate_pct	contributing_matches	cohort_matches
479	Rafael Nadal	Grass	2011–2014	2	drop_shot_share_pct	30.000	3,532.000	0.849	10	11
480	Rafael Nadal	Grass	2011–2014	2	slice_share_pct	714.000	3,532.000	20.215	10	11
481	Rafael Nadal	Grass	2011–2014	2	short_rally_share_pct	1,516.000	2,158.000	70.250	10	11
482	Rafael Nadal	Grass	2011–2014	2	medium_rally_share_pct	448.000	2,158.000	20.760	10	11
483	Rafael Nadal	Grass	2011–2014	2	long_rally_share_pct	194.000	2,158.000	8.990	10	11
484	Rafael Nadal	Grass	2011–2014	2	deuce_wide_serve_share_pct	114.000	564.000	20.213	10	11
485	Rafael Nadal	Grass	2011–2014	2	deuce_body_serve_share_pct	183.000	564.000	32.447	10	11
486	Rafael Nadal	Grass	2011–2014	2	deuce_t_serve_share_pct	267.000	564.000	47.340	10	11
487	Rafael Nadal	Grass	2011–2014	2	ad_wide_serve_share_pct	198.000	498.000	39.759	10	11
488	Rafael Nadal	Grass	2011–2014	2	ad_body_serve_share_pct	153.000	498.000	30.723	10	11
489	Rafael Nadal	Grass	2011–2014	2	ad_t_serve_share_pct	147.000	498.000	29.518	10	11
490	Rafael Nadal	Hard	2011–2014	2	approach_frequency_pct	302.000	6,231.000	4.847	40	46
491	Rafael Nadal	Hard	2011–2014	2	net_point_frequency_pct	492.000	6,809.000	7.226	43	46
492	Rafael Nadal	Hard	2011–2014	2	serve_volley_frequency_pct	21.000	3,574.000	0.588	45	46
493	Rafael Nadal	Hard	2011–2014	2	drop_shot_share_pct	53.000	16,254.000	0.326	46	46
494	Rafael Nadal	Hard	2011–2014	2	slice_share_pct	1,873.000	16,254.000	11.523	46	46
495	Rafael Nadal	Hard	2011–2014	2	short_rally_share_pct	4,374.000	7,271.000	60.157	46	46
496	Rafael Nadal	Hard	2011–2014	2	medium_rally_share_pct	1,708.000	7,271.000	23.491	46	46
497	Rafael Nadal	Hard	2011–2014	2	long_rally_share_pct	1,189.000	7,271.000	16.353	46	46
498	Rafael Nadal	Hard	2011–2014	2	deuce_wide_serve_share_pct	365.000	1,906.000	19.150	46	46
499	Rafael Nadal	Hard	2011–2014	2	deuce_body_serve_share_pct	689.000	1,906.000	36.149	46	46
500	Rafael Nadal	Hard	2011–2014	2	deuce_t_serve_share_pct	852.000	1,906.000	44.701	46	46
501	Rafael Nadal	Hard	2011–2014	2	ad_wide_serve_share_pct	901.000	1,748.000	51.545	46	46
502	Rafael Nadal	Hard	2011–2014	2	ad_body_serve_share_pct	544.000	1,748.000	31.121	46	46

	player	surface	period	period_index	metric	numerator	denominator	estimate_pct	contributing_matches	cohort_matches
503	Rafael Nadal	Hard	2011–2014	2	ad_t_serve_share_pct	303.000	1,748.000	17.334	46	46
504	Rafael Nadal	Overall	2015–2018	3	approach_frequency_pct	1,284.000	16,595.000	7.737	101	104
505	Rafael Nadal	Overall	2015–2018	3	net_point_frequency_pct	1,623.000	16,731.000	9.701	102	104
506	Rafael Nadal	Overall	2015–2018	3	serve_volley_frequency_pct	83.000	8,125.000	1.022	102	104
507	Rafael Nadal	Overall	2015–2018	3	drop_shot_share_pct	286.000	34,287.000	0.834	104	104
508	Rafael Nadal	Overall	2015–2018	3	slice_share_pct	2,952.000	34,287.000	8.610	104	104
509	Rafael Nadal	Overall	2015–2018	3	short_rally_share_pct	10,146.000	16,527.000	61.390	104	104
510	Rafael Nadal	Overall	2015–2018	3	medium_rally_share_pct	4,046.000	16,527.000	24.481	104	104
511	Rafael Nadal	Overall	2015–2018	3	long_rally_share_pct	2,335.000	16,527.000	14.128	104	104
512	Rafael Nadal	Overall	2015–2018	3	deuce_wide_serve_share_pct	1,271.000	4,341.000	29.279	104	104
513	Rafael Nadal	Overall	2015–2018	3	deuce_body_serve_share_pct	1,018.000	4,341.000	23.451	104	104
514	Rafael Nadal	Overall	2015–2018	3	deuce_t_serve_share_pct	2,052.000	4,341.000	47.270	104	104
515	Rafael Nadal	Overall	2015–2018	3	ad_wide_serve_share_pct	1,979.000	3,936.000	50.279	104	104
516	Rafael Nadal	Overall	2015–2018	3	ad_body_serve_share_pct	1,136.000	3,936.000	28.862	104	104
517	Rafael Nadal	Overall	2015–2018	3	ad_t_serve_share_pct	821.000	3,936.000	20.859	104	104
518	Rafael Nadal	Clay	2015–2018	3	approach_frequency_pct	393.000	5,303.000	7.411	39	40
519	Rafael Nadal	Clay	2015–2018	3	net_point_frequency_pct	508.000	5,303.000	9.579	39	40
520	Rafael Nadal	Clay	2015–2018	3	serve_volley_frequency_pct	22.000	2,530.000	0.870	39	40
521	Rafael Nadal	Clay	2015–2018	3	drop_shot_share_pct	146.000	12,651.000	1.154	40	40
522	Rafael Nadal	Clay	2015–2018	3	slice_share_pct	776.000	12,651.000	6.134	40	40
523	Rafael Nadal	Clay	2015–2018	3	short_rally_share_pct	2,902.000	5,282.000	54.941	40	40
524	Rafael Nadal	Clay	2015–2018	3	medium_rally_share_pct	1,435.000	5,282.000	27.168	40	40
525	Rafael Nadal	Clay	2015–2018	3	long_rally_share_pct	945.000	5,282.000	17.891	40	40
526	Rafael Nadal	Clay	2015–2018	3	deuce_wide_serve_share_pct	358.000	1,359.000	26.343	40	40

	player	surface	period	period_index	metric	numerator	denominator	estimate_pct	contributing_matches	cohort_matches
527	Rafael Nadal	Clay	2015–2018	3	deuce_body_serve_share_pct	392.000	1,359.000	28.845	40	40
528	Rafael Nadal	Clay	2015–2018	3	deuce_t_serve_share_pct	609.000	1,359.000	44.812	40	40
529	Rafael Nadal	Clay	2015–2018	3	ad_wide_serve_share_pct	713.000	1,236.000	57.686	40	40
530	Rafael Nadal	Clay	2015–2018	3	ad_body_serve_share_pct	320.000	1,236.000	25.890	40	40
531	Rafael Nadal	Clay	2015–2018	3	ad_t_serve_share_pct	203.000	1,236.000	16.424	40	40
532	Rafael Nadal	Grass	2015–2018	3	approach_frequency_pct	148.000	1,365.000	10.842	4	4
533	Rafael Nadal	Grass	2015–2018	3	net_point_frequency_pct	185.000	1,365.000	13.553	4	4
534	Rafael Nadal	Grass	2015–2018	3	serve_volley_frequency_pct	10.000	683.000	1.464	4	4
535	Rafael Nadal	Grass	2015–2018	3	drop_shot_share_pct	44.000	2,226.000	1.977	4	4
536	Rafael Nadal	Grass	2015–2018	3	slice_share_pct	294.000	2,226.000	13.208	4	4
537	Rafael Nadal	Grass	2015–2018	3	short_rally_share_pct	923.000	1,325.000	69.660	4	4
538	Rafael Nadal	Grass	2015–2018	3	medium_rally_share_pct	267.000	1,325.000	20.151	4	4
539	Rafael Nadal	Grass	2015–2018	3	long_rally_share_pct	135.000	1,325.000	10.189	4	4
540	Rafael Nadal	Grass	2015–2018	3	deuce_wide_serve_share_pct	125.000	353.000	35.411	4	4
541	Rafael Nadal	Grass	2015–2018	3	deuce_body_serve_share_pct	69.000	353.000	19.547	4	4
542	Rafael Nadal	Grass	2015–2018	3	deuce_t_serve_share_pct	159.000	353.000	45.042	4	4
543	Rafael Nadal	Grass	2015–2018	3	ad_wide_serve_share_pct	125.000	327.000	38.226	4	4
544	Rafael Nadal	Grass	2015–2018	3	ad_body_serve_share_pct	103.000	327.000	31.498	4	4
545	Rafael Nadal	Grass	2015–2018	3	ad_t_serve_share_pct	99.000	327.000	30.275	4	4
546	Rafael Nadal	Hard	2015–2018	3	approach_frequency_pct	743.000	9,927.000	7.485	58	60
547	Rafael Nadal	Hard	2015–2018	3	net_point_frequency_pct	930.000	10,063.000	9.242	59	60
548	Rafael Nadal	Hard	2015–2018	3	serve_volley_frequency_pct	51.000	4,912.000	1.038	59	60
549	Rafael Nadal	Hard	2015–2018	3	drop_shot_share_pct	96.000	19,410.000	0.495	60	60
550	Rafael Nadal	Hard	2015–2018	3	slice_share_pct	1,882.000	19,410.000	9.696	60	60

	player	surface	period	period_index	metric	numerator	denominator	estimate_pct	contributing_matches	cohort_matches
551	Rafael Nadal	Hard	2015–2018	3	short_rally_share_pct	6,321.000	9,920.000	63.720	60	60
552	Rafael Nadal	Hard	2015–2018	3	medium_rally_share_pct	2,344.000	9,920.000	23.629	60	60
553	Rafael Nadal	Hard	2015–2018	3	long_rally_share_pct	1,255.000	9,920.000	12.651	60	60
554	Rafael Nadal	Hard	2015–2018	3	deuce_wide_serve_share_pct	788.000	2,629.000	29.973	60	60
555	Rafael Nadal	Hard	2015–2018	3	deuce_body_serve_share_pct	557.000	2,629.000	21.187	60	60
556	Rafael Nadal	Hard	2015–2018	3	deuce_t_serve_share_pct	1,284.000	2,629.000	48.840	60	60
557	Rafael Nadal	Hard	2015–2018	3	ad_wide_serve_share_pct	1,141.000	2,373.000	48.083	60	60
558	Rafael Nadal	Hard	2015–2018	3	ad_body_serve_share_pct	713.000	2,373.000	30.046	60	60
559	Rafael Nadal	Hard	2015–2018	3	ad_t_serve_share_pct	519.000	2,373.000	21.871	60	60
560	Rafael Nadal	Overall	2019–2022	4	approach_frequency_pct	707.000	9,971.000	7.091	58	62
561	Rafael Nadal	Overall	2019–2022	4	net_point_frequency_pct	1,036.000	10,164.000	10.193	60	62
562	Rafael Nadal	Overall	2019–2022	4	serve_volley_frequency_pct	158.000	5,025.000	3.144	60	62
563	Rafael Nadal	Overall	2019–2022	4	drop_shot_share_pct	212.000	20,729.000	1.023	61	62
564	Rafael Nadal	Overall	2019–2022	4	slice_share_pct	2,069.000	20,729.000	9.981	61	62
565	Rafael Nadal	Overall	2019–2022	4	short_rally_share_pct	6,353.000	10,190.000	62.345	61	62
566	Rafael Nadal	Overall	2019–2022	4	medium_rally_share_pct	2,391.000	10,190.000	23.464	61	62
567	Rafael Nadal	Overall	2019–2022	4	long_rally_share_pct	1,446.000	10,190.000	14.190	61	62
568	Rafael Nadal	Overall	2019–2022	4	deuce_wide_serve_share_pct	796.000	2,735.000	29.104	61	62
569	Rafael Nadal	Overall	2019–2022	4	deuce_body_serve_share_pct	551.000	2,735.000	20.146	61	62
570	Rafael Nadal	Overall	2019–2022	4	deuce_t_serve_share_pct	1,388.000	2,735.000	50.750	61	62
571	Rafael Nadal	Overall	2019–2022	4	ad_wide_serve_share_pct	1,194.000	2,476.000	48.223	61	62
572	Rafael Nadal	Overall	2019–2022	4	ad_body_serve_share_pct	668.000	2,476.000	26.979	61	62
573	Rafael Nadal	Overall	2019–2022	4	ad_t_serve_share_pct	614.000	2,476.000	24.798	61	62
574	Rafael Nadal	Clay	2019–2022	4	approach_frequency_pct	167.000	2,782.000	6.003	16	18

	player	surface	period	period_index	metric	numerator	denominator	estimate_pct	contributing_matches	cohort_matches
575	Rafael Nadal	Clay	2019–2022	4	net_point_frequency_pct	282.000	2,863.000	9.850	17	18
576	Rafael Nadal	Clay	2019–2022	4	serve_volley_frequency_pct	14.000	1,481.000	0.945	18	18
577	Rafael Nadal	Clay	2019–2022	4	drop_shot_share_pct	102.000	7,049.000	1.447	18	18
578	Rafael Nadal	Clay	2019–2022	4	slice_share_pct	424.000	7,049.000	6.015	18	18
579	Rafael Nadal	Clay	2019–2022	4	short_rally_share_pct	1,551.000	2,913.000	53.244	18	18
580	Rafael Nadal	Clay	2019–2022	4	medium_rally_share_pct	818.000	2,913.000	28.081	18	18
581	Rafael Nadal	Clay	2019–2022	4	long_rally_share_pct	544.000	2,913.000	18.675	18	18
582	Rafael Nadal	Clay	2019–2022	4	deuce_wide_serve_share_pct	152.000	775.000	19.613	18	18
583	Rafael Nadal	Clay	2019–2022	4	deuce_body_serve_share_pct	148.000	775.000	19.097	18	18
584	Rafael Nadal	Clay	2019–2022	4	deuce_t_serve_share_pct	475.000	775.000	61.290	18	18
585	Rafael Nadal	Clay	2019–2022	4	ad_wide_serve_share_pct	385.000	705.000	54.610	18	18
586	Rafael Nadal	Clay	2019–2022	4	ad_body_serve_share_pct	165.000	705.000	23.404	18	18
587	Rafael Nadal	Clay	2019–2022	4	ad_t_serve_share_pct	155.000	705.000	21.986	18	18
588	Rafael Nadal	Grass	2019–2022	4	approach_frequency_pct	68.000	1,111.000	6.121	6	6
589	Rafael Nadal	Grass	2019–2022	4	net_point_frequency_pct	110.000	1,111.000	9.901	6	6
590	Rafael Nadal	Grass	2019–2022	4	serve_volley_frequency_pct	11.000	535.000	2.056	6	6
591	Rafael Nadal	Grass	2019–2022	4	drop_shot_share_pct	20.000	1,811.000	1.104	6	6
592	Rafael Nadal	Grass	2019–2022	4	slice_share_pct	232.000	1,811.000	12.811	6	6
593	Rafael Nadal	Grass	2019–2022	4	short_rally_share_pct	760.000	1,080.000	70.370	6	6
594	Rafael Nadal	Grass	2019–2022	4	medium_rally_share_pct	230.000	1,080.000	21.296	6	6
595	Rafael Nadal	Grass	2019–2022	4	long_rally_share_pct	90.000	1,080.000	8.333	6	6
596	Rafael Nadal	Grass	2019–2022	4	deuce_wide_serve_share_pct	109.000	281.000	38.790	6	6
597	Rafael Nadal	Grass	2019–2022	4	deuce_body_serve_share_pct	60.000	281.000	21.352	6	6
598	Rafael Nadal	Grass	2019–2022	4	deuce_t_serve_share_pct	112.000	281.000	39.858	6	6

	player	surface	period	period_index	metric	numerator	denominator	estimate_pct	contributing_matches	cohort_matches
599	Rafael Nadal	Grass	2019–2022	4	ad_wide_serve_share_pct	99.000	254.000	38.976	6	6
600	Rafael Nadal	Grass	2019–2022	4	ad_body_serve_share_pct	62.000	254.000	24.409	6	6
601	Rafael Nadal	Grass	2019–2022	4	ad_t_serve_share_pct	93.000	254.000	36.614	6	6
602	Rafael Nadal	Hard	2019–2022	4	approach_frequency_pct	472.000	6,078.000	7.766	36	38
603	Rafael Nadal	Hard	2019–2022	4	net_point_frequency_pct	644.000	6,190.000	10.404	37	38
604	Rafael Nadal	Hard	2019–2022	4	serve_volley_frequency_pct	133.000	3,009.000	4.420	36	38
605	Rafael Nadal	Hard	2019–2022	4	drop_shot_share_pct	90.000	11,869.000	0.758	37	38
606	Rafael Nadal	Hard	2019–2022	4	slice_share_pct	1,413.000	11,869.000	11.905	37	38
607	Rafael Nadal	Hard	2019–2022	4	short_rally_share_pct	4,042.000	6,197.000	65.225	37	38
608	Rafael Nadal	Hard	2019–2022	4	medium_rally_share_pct	1,343.000	6,197.000	21.672	37	38
609	Rafael Nadal	Hard	2019–2022	4	long_rally_share_pct	812.000	6,197.000	13.103	37	38
610	Rafael Nadal	Hard	2019–2022	4	deuce_wide_serve_share_pct	535.000	1,679.000	31.864	37	38
611	Rafael Nadal	Hard	2019–2022	4	deuce_body_serve_share_pct	343.000	1,679.000	20.429	37	38
612	Rafael Nadal	Hard	2019–2022	4	deuce_t_serve_share_pct	801.000	1,679.000	47.707	37	38
613	Rafael Nadal	Hard	2019–2022	4	ad_wide_serve_share_pct	710.000	1,517.000	46.803	37	38
614	Rafael Nadal	Hard	2019–2022	4	ad_body_serve_share_pct	441.000	1,517.000	29.071	37	38
615	Rafael Nadal	Hard	2019–2022	4	ad_t_serve_share_pct	366.000	1,517.000	24.127	37	38
616	Rafael Nadal	Overall	2023–2024	5	approach_frequency_pct	170.000	2,051.000	8.289	14	14
617	Rafael Nadal	Overall	2023–2024	5	net_point_frequency_pct	211.000	2,051.000	10.288	14	14
618	Rafael Nadal	Overall	2023–2024	5	serve_volley_frequency_pct	53.000	1,037.000	5.111	14	14
619	Rafael Nadal	Overall	2023–2024	5	drop_shot_share_pct	54.000	4,392.000	1.230	14	14
620	Rafael Nadal	Overall	2023–2024	5	slice_share_pct	319.000	4,392.000	7.263	14	14
621	Rafael Nadal	Overall	2023–2024	5	short_rally_share_pct	1,154.000	1,980.000	58.283	14	14
622	Rafael Nadal	Overall	2023–2024	5	medium_rally_share_pct	518.000	1,980.000	26.162	14	14

	player	surface	period	period_index	metric	numerator	denominator	estimate_pct	contributing_matches	cohort_matches
623	Rafael Nadal	Overall	2023–2024	5	long_rally_share_pct	308.000	1,980.000	15.556	14	14
624	Rafael Nadal	Overall	2023–2024	5	deuce_wide_serve_share_pct	98.000	539.000	18.182	14	14
625	Rafael Nadal	Overall	2023–2024	5	deuce_body_serve_share_pct	91.000	539.000	16.883	14	14
626	Rafael Nadal	Overall	2023–2024	5	deuce_t_serve_share_pct	350.000	539.000	64.935	14	14
627	Rafael Nadal	Overall	2023–2024	5	ad_wide_serve_share_pct	261.000	498.000	52.410	14	14
628	Rafael Nadal	Overall	2023–2024	5	ad_body_serve_share_pct	153.000	498.000	30.723	14	14
629	Rafael Nadal	Overall	2023–2024	5	ad_t_serve_share_pct	84.000	498.000	16.867	14	14
630	Rafael Nadal	Clay	2023–2024	5	approach_frequency_pct	63.000	936.000	6.731	7	7
631	Rafael Nadal	Clay	2023–2024	5	net_point_frequency_pct	85.000	936.000	9.081	7	7
632	Rafael Nadal	Clay	2023–2024	5	serve_volley_frequency_pct	14.000	478.000	2.929	7	7
633	Rafael Nadal	Clay	2023–2024	5	drop_shot_share_pct	40.000	2,131.000	1.877	7	7
634	Rafael Nadal	Clay	2023–2024	5	slice_share_pct	113.000	2,131.000	5.303	7	7
635	Rafael Nadal	Clay	2023–2024	5	short_rally_share_pct	488.000	908.000	53.744	7	7
636	Rafael Nadal	Clay	2023–2024	5	medium_rally_share_pct	268.000	908.000	29.515	7	7
637	Rafael Nadal	Clay	2023–2024	5	long_rally_share_pct	152.000	908.000	16.740	7	7
638	Rafael Nadal	Clay	2023–2024	5	deuce_wide_serve_share_pct	31.000	249.000	12.450	7	7
639	Rafael Nadal	Clay	2023–2024	5	deuce_body_serve_share_pct	47.000	249.000	18.876	7	7
640	Rafael Nadal	Clay	2023–2024	5	deuce_t_serve_share_pct	171.000	249.000	68.675	7	7
641	Rafael Nadal	Clay	2023–2024	5	ad_wide_serve_share_pct	130.000	229.000	56.769	7	7
642	Rafael Nadal	Clay	2023–2024	5	ad_body_serve_share_pct	58.000	229.000	25.328	7	7
643	Rafael Nadal	Clay	2023–2024	5	ad_t_serve_share_pct	41.000	229.000	17.904	7	7
644	Rafael Nadal	Grass	2023–2024	5	approach_frequency_pct	Unavailable	Unavailable	Unavailable	0	0
645	Rafael Nadal	Grass	2023–2024	5	net_point_frequency_pct	Unavailable	Unavailable	Unavailable	0	0
646	Rafael Nadal	Grass	2023–2024	5	serve_volley_frequency_pct	Unavailable	Unavailable	Unavailable	0	0

	player	surface	period	period_index	metric	numerator	denominator	estimate_pct	contributing_matches	cohort_matches
647	Rafael Nadal	Grass	2023–2024	5	drop_shot_share_pct	Unavailable	Unavailable	Unavailable	0	0
648	Rafael Nadal	Grass	2023–2024	5	slice_share_pct	Unavailable	Unavailable	Unavailable	0	0
649	Rafael Nadal	Grass	2023–2024	5	short_rally_share_pct	Unavailable	Unavailable	Unavailable	0	0
650	Rafael Nadal	Grass	2023–2024	5	medium_rally_share_pct	Unavailable	Unavailable	Unavailable	0	0
651	Rafael Nadal	Grass	2023–2024	5	long_rally_share_pct	Unavailable	Unavailable	Unavailable	0	0
652	Rafael Nadal	Grass	2023–2024	5	deuce_wide_serve_share_pct	Unavailable	Unavailable	Unavailable	0	0
653	Rafael Nadal	Grass	2023–2024	5	deuce_body_serve_share_pct	Unavailable	Unavailable	Unavailable	0	0
654	Rafael Nadal	Grass	2023–2024	5	deuce_t_serve_share_pct	Unavailable	Unavailable	Unavailable	0	0
655	Rafael Nadal	Grass	2023–2024	5	ad_wide_serve_share_pct	Unavailable	Unavailable	Unavailable	0	0
656	Rafael Nadal	Grass	2023–2024	5	ad_body_serve_share_pct	Unavailable	Unavailable	Unavailable	0	0
657	Rafael Nadal	Grass	2023–2024	5	ad_t_serve_share_pct	Unavailable	Unavailable	Unavailable	0	0
658	Rafael Nadal	Hard	2023–2024	5	approach_frequency_pct	107.000	1,115.000	9.596	7	7
659	Rafael Nadal	Hard	2023–2024	5	net_point_frequency_pct	126.000	1,115.000	11.300	7	7
660	Rafael Nadal	Hard	2023–2024	5	serve_volley_frequency_pct	39.000	559.000	6.977	7	7
661	Rafael Nadal	Hard	2023–2024	5	drop_shot_share_pct	14.000	2,261.000	0.619	7	7
662	Rafael Nadal	Hard	2023–2024	5	slice_share_pct	206.000	2,261.000	9.111	7	7
663	Rafael Nadal	Hard	2023–2024	5	short_rally_share_pct	666.000	1,072.000	62.127	7	7
664	Rafael Nadal	Hard	2023–2024	5	medium_rally_share_pct	250.000	1,072.000	23.321	7	7
665	Rafael Nadal	Hard	2023–2024	5	long_rally_share_pct	156.000	1,072.000	14.552	7	7
666	Rafael Nadal	Hard	2023–2024	5	deuce_wide_serve_share_pct	67.000	290.000	23.103	7	7
667	Rafael Nadal	Hard	2023–2024	5	deuce_body_serve_share_pct	44.000	290.000	15.172	7	7
668	Rafael Nadal	Hard	2023–2024	5	deuce_t_serve_share_pct	179.000	290.000	61.724	7	7
669	Rafael Nadal	Hard	2023–2024	5	ad_wide_serve_share_pct	131.000	269.000	48.699	7	7
670	Rafael Nadal	Hard	2023–2024	5	ad_body_serve_share_pct	95.000	269.000	35.316	7	7

	player	surface	period	period_index	metric	numerator	denominator	estimate_pct	contributing_matches	cohort_matches
671	Rafael Nadal	Hard	2023–2024	5	ad_t_serve_share_pct	43.000	269.000	15.985	7	7

	player	surface	period	coverage_pct	small_sample	coverage_edge_period	valid_bootstraps	ci95_lower	ci95_upper
336	Rafael Nadal	Overall	2003–2006	92.105	False	True	2000	6.720	9.392
337	Rafael Nadal	Overall	2003–2006	92.105	False	True	2000	8.921	12.016
338	Rafael Nadal	Overall	2003–2006	97.368	False	True	2000	0.836	1.748
339	Rafael Nadal	Overall	2003–2006	100.000	False	True	2000	0.929	1.448
340	Rafael Nadal	Overall	2003–2006	100.000	False	True	2000	6.656	9.127
341	Rafael Nadal	Overall	2003–2006	100.000	False	True	2000	52.225	58.691
342	Rafael Nadal	Overall	2003–2006	100.000	False	True	2000	24.837	27.616
343	Rafael Nadal	Overall	2003–2006	100.000	False	True	2000	15.534	21.132
344	Rafael Nadal	Overall	2003–2006	100.000	False	True	2000	15.100	25.208
345	Rafael Nadal	Overall	2003–2006	100.000	False	True	2000	18.319	27.639
346	Rafael Nadal	Overall	2003–2006	100.000	False	True	2000	50.266	64.101
347	Rafael Nadal	Overall	2003–2006	100.000	False	True	2000	58.741	70.350
348	Rafael Nadal	Overall	2003–2006	100.000	False	True	2000	16.572	23.587
349	Rafael Nadal	Overall	2003–2006	100.000	False	True	2000	11.870	19.472
350	Rafael Nadal	Clay	2003–2006	92.308	True	True	2000	5.937	9.584
351	Rafael Nadal	Clay	2003–2006	92.308	True	True	2000	8.725	13.413
352	Rafael Nadal	Clay	2003–2006	92.308	True	True	2000	0.234	1.290
353	Rafael Nadal	Clay	2003–2006	100.000	True	True	2000	1.102	1.980
354	Rafael Nadal	Clay	2003–2006	100.000	True	True	2000	5.369	7.034
355	Rafael Nadal	Clay	2003–2006	100.000	True	True	2000	43.190	52.388
356	Rafael Nadal	Clay	2003–2006	100.000	True	True	2000	26.155	30.970
357	Rafael Nadal	Clay	2003–2006	100.000	True	True	2000	19.372	28.809
358	Rafael Nadal	Clay	2003–2006	100.000	True	True	2000	6.559	22.182
359	Rafael Nadal	Clay	2003–2006	100.000	True	True	2000	18.051	37.007
360	Rafael Nadal	Clay	2003–2006	100.000	True	True	2000	47.368	71.762
361	Rafael Nadal	Clay	2003–2006	100.000	True	True	2000	68.565	83.886
362	Rafael Nadal	Clay	2003–2006	100.000	True	True	2000	9.170	20.555
363	Rafael Nadal	Clay	2003–2006	100.000	True	True	2000	5.618	13.092
364	Rafael Nadal	Grass	2003–2006	100.000	True	True	1999	7.806	15.447
365	Rafael Nadal	Grass	2003–2006	100.000	True	True	1999	9.246	18.102
366	Rafael Nadal	Grass	2003–2006	100.000	True	True	1999	1.374	3.820
367	Rafael Nadal	Grass	2003–2006	100.000	True	True	1999	0.547	1.919
368	Rafael Nadal	Grass	2003–2006	100.000	True	True	1999	9.329	17.345
369	Rafael Nadal	Grass	2003–2006	100.000	True	True	1999	58.345	70.739

	player	surface	period	coverage_pct	small_sample	coverage_edge_period	valid_bootstraps	ci95_lower	ci95_upper
370	Rafael Nadal	Grass	2003–2006	100.000	True	True	1999	20.428	29.020
371	Rafael Nadal	Grass	2003–2006	100.000	True	True	1999	8.207	12.637
372	Rafael Nadal	Grass	2003–2006	100.000	True	True	1999	10.848	35.145
373	Rafael Nadal	Grass	2003–2006	100.000	True	True	1999	8.271	27.029
374	Rafael Nadal	Grass	2003–2006	100.000	True	True	1999	40.404	79.889
375	Rafael Nadal	Grass	2003–2006	100.000	True	True	1999	37.010	65.036
376	Rafael Nadal	Grass	2003–2006	100.000	True	True	1999	12.413	32.444
377	Rafael Nadal	Grass	2003–2006	100.000	True	True	1999	14.813	37.870
378	Rafael Nadal	Hard	2003–2006	88.889	True	True	2000	5.129	8.112
379	Rafael Nadal	Hard	2003–2006	88.889	True	True	2000	6.667	10.398
380	Rafael Nadal	Hard	2003–2006	100.000	True	True	2000	0.612	1.689
381	Rafael Nadal	Hard	2003–2006	100.000	True	True	2000	0.573	1.107
382	Rafael Nadal	Hard	2003–2006	100.000	True	True	2000	6.186	9.361
383	Rafael Nadal	Hard	2003–2006	100.000	True	True	2000	55.532	60.576
384	Rafael Nadal	Hard	2003–2006	100.000	True	True	2000	23.659	26.818
385	Rafael Nadal	Hard	2003–2006	100.000	True	True	2000	14.275	18.960
386	Rafael Nadal	Hard	2003–2006	100.000	True	True	2000	17.228	31.782
387	Rafael Nadal	Hard	2003–2006	100.000	True	True	2000	17.339	28.699
388	Rafael Nadal	Hard	2003–2006	100.000	True	True	2000	43.684	61.807
389	Rafael Nadal	Hard	2003–2006	100.000	True	True	2000	53.493	67.966
390	Rafael Nadal	Hard	2003–2006	100.000	True	True	2000	19.360	28.390
391	Rafael Nadal	Hard	2003–2006	100.000	True	True	2000	11.509	19.815
392	Rafael Nadal	Overall	2007–2010	93.750	False	False	2000	5.687	7.244
393	Rafael Nadal	Overall	2007–2010	95.833	False	False	2000	7.736	9.359
394	Rafael Nadal	Overall	2007–2010	98.958	False	False	2000	0.676	1.396
395	Rafael Nadal	Overall	2007–2010	100.000	False	False	2000	0.631	0.889
396	Rafael Nadal	Overall	2007–2010	100.000	False	False	2000	10.147	12.421
397	Rafael Nadal	Overall	2007–2010	100.000	False	False	2000	56.052	59.717
398	Rafael Nadal	Overall	2007–2010	100.000	False	False	2000	24.214	26.063
399	Rafael Nadal	Overall	2007–2010	100.000	False	False	2000	15.598	18.447
400	Rafael Nadal	Overall	2007–2010	100.000	False	False	2000	17.255	22.249
401	Rafael Nadal	Overall	2007–2010	100.000	False	False	2000	25.445	31.454
402	Rafael Nadal	Overall	2007–2010	100.000	False	False	2000	47.868	55.859
403	Rafael Nadal	Overall	2007–2010	100.000	False	False	2000	49.634	58.653
404	Rafael Nadal	Overall	2007–2010	100.000	False	False	2000	21.298	27.850
405	Rafael Nadal	Overall	2007–2010	100.000	False	False	2000	18.624	24.719
406	Rafael Nadal	Clay	2007–2010	85.294	True	False	2000	3.687	5.666
407	Rafael Nadal	Clay	2007–2010	88.235	False	False	2000	6.020	8.137
408	Rafael Nadal	Clay	2007–2010	97.059	False	False	2000	0.130	0.643

	player	surface	period	coverage_pct	small_sample	coverage_edge_period	valid_bootstraps	ci95_lower	ci95_upper
409	Rafael Nadal	Clay	2007–2010	100.000	False	False	2000	0.722	1.220
410	Rafael Nadal	Clay	2007–2010	100.000	False	False	2000	6.166	8.499
411	Rafael Nadal	Clay	2007–2010	100.000	False	False	2000	50.010	55.463
412	Rafael Nadal	Clay	2007–2010	100.000	False	False	2000	24.752	28.351
413	Rafael Nadal	Clay	2007–2010	100.000	False	False	2000	18.390	23.121
414	Rafael Nadal	Clay	2007–2010	100.000	False	False	2000	13.358	22.256
415	Rafael Nadal	Clay	2007–2010	100.000	False	False	2000	24.742	37.327
416	Rafael Nadal	Clay	2007–2010	100.000	False	False	2000	43.765	59.138
417	Rafael Nadal	Clay	2007–2010	100.000	False	False	2000	54.967	71.144
418	Rafael Nadal	Clay	2007–2010	100.000	False	False	2000	14.593	25.805
419	Rafael Nadal	Clay	2007–2010	100.000	False	False	2000	12.174	22.045
420	Rafael Nadal	Grass	2007–2010	100.000	True	False	2000	6.628	10.107
421	Rafael Nadal	Grass	2007–2010	100.000	True	False	2000	8.942	12.866
422	Rafael Nadal	Grass	2007–2010	100.000	True	False	2000	0.656	2.276
423	Rafael Nadal	Grass	2007–2010	100.000	True	False	2000	0.501	1.276
424	Rafael Nadal	Grass	2007–2010	100.000	True	False	2000	10.559	17.703
425	Rafael Nadal	Grass	2007–2010	100.000	True	False	2000	65.930	71.384
426	Rafael Nadal	Grass	2007–2010	100.000	True	False	2000	20.511	23.161
427	Rafael Nadal	Grass	2007–2010	100.000	True	False	2000	7.543	11.519
428	Rafael Nadal	Grass	2007–2010	100.000	True	False	2000	11.268	20.743
429	Rafael Nadal	Grass	2007–2010	100.000	True	False	2000	18.980	37.235
430	Rafael Nadal	Grass	2007–2010	100.000	True	False	2000	45.640	67.143
431	Rafael Nadal	Grass	2007–2010	100.000	True	False	2000	35.252	62.430
432	Rafael Nadal	Grass	2007–2010	100.000	True	False	2000	15.478	35.080
433	Rafael Nadal	Grass	2007–2010	100.000	True	False	2000	18.970	33.166
434	Rafael Nadal	Hard	2007–2010	98.000	False	False	2000	5.816	8.114
435	Rafael Nadal	Hard	2007–2010	100.000	False	False	2000	7.671	9.990
436	Rafael Nadal	Hard	2007–2010	100.000	False	False	2000	0.745	2.025
437	Rafael Nadal	Hard	2007–2010	100.000	False	False	2000	0.450	0.704
438	Rafael Nadal	Hard	2007–2010	100.000	False	False	2000	12.116	15.112
439	Rafael Nadal	Hard	2007–2010	100.000	False	False	2000	56.167	60.307
440	Rafael Nadal	Hard	2007–2010	100.000	False	False	2000	23.903	26.241
441	Rafael Nadal	Hard	2007–2010	100.000	False	False	2000	15.208	18.234
442	Rafael Nadal	Hard	2007–2010	100.000	False	False	2000	18.250	25.472
443	Rafael Nadal	Hard	2007–2010	100.000	False	False	2000	23.839	30.498
444	Rafael Nadal	Hard	2007–2010	100.000	False	False	2000	45.454	56.291
445	Rafael Nadal	Hard	2007–2010	100.000	False	False	2000	43.732	54.684
446	Rafael Nadal	Hard	2007–2010	100.000	False	False	2000	22.927	31.997
447	Rafael Nadal	Hard	2007–2010	100.000	False	False	2000	19.408	28.425

	player	surface	period	coverage_pct	small_sample	coverage_edge_period	valid_bootstraps	ci95_lower	ci95_upper
448	Rafael Nadal	Overall	2011–2014	90.090	False	False	2000	4.424	5.647
449	Rafael Nadal	Overall	2011–2014	92.793	False	False	2000	6.926	8.249
450	Rafael Nadal	Overall	2011–2014	95.495	False	False	2000	0.370	0.759
451	Rafael Nadal	Overall	2011–2014	99.099	False	False	2000	0.447	0.639
452	Rafael Nadal	Overall	2011–2014	99.099	False	False	2000	8.266	10.817
453	Rafael Nadal	Overall	2011–2014	99.099	False	False	2000	56.765	60.610
454	Rafael Nadal	Overall	2011–2014	99.099	False	False	2000	23.887	25.718
455	Rafael Nadal	Overall	2011–2014	99.099	False	False	2000	15.160	17.925
456	Rafael Nadal	Overall	2011–2014	99.099	False	False	2000	16.801	20.884
457	Rafael Nadal	Overall	2011–2014	99.099	False	False	2000	34.559	39.978
458	Rafael Nadal	Overall	2011–2014	99.099	False	False	2000	40.774	47.133
459	Rafael Nadal	Overall	2011–2014	99.099	False	False	2000	49.924	58.280
460	Rafael Nadal	Overall	2011–2014	99.099	False	False	2000	25.287	31.195
461	Rafael Nadal	Overall	2011–2014	99.099	False	False	2000	15.296	20.204
462	Rafael Nadal	Clay	2011–2014	92.593	False	False	2000	4.014	5.660
463	Rafael Nadal	Clay	2011–2014	92.593	False	False	2000	6.566	8.687
464	Rafael Nadal	Clay	2011–2014	96.296	False	False	2000	0.138	0.578
465	Rafael Nadal	Clay	2011–2014	100.000	False	False	2000	0.516	0.810
466	Rafael Nadal	Clay	2011–2014	100.000	False	False	2000	5.362	6.563
467	Rafael Nadal	Clay	2011–2014	100.000	False	False	2000	51.866	56.754
468	Rafael Nadal	Clay	2011–2014	100.000	False	False	2000	25.914	28.263
469	Rafael Nadal	Clay	2011–2014	100.000	False	False	2000	16.654	20.709
470	Rafael Nadal	Clay	2011–2014	100.000	False	False	2000	14.716	21.938
471	Rafael Nadal	Clay	2011–2014	100.000	False	False	2000	35.915	43.434
472	Rafael Nadal	Clay	2011–2014	100.000	False	False	2000	37.759	46.709
473	Rafael Nadal	Clay	2011–2014	100.000	False	False	2000	54.327	66.049
474	Rafael Nadal	Clay	2011–2014	100.000	False	False	2000	20.765	28.934
475	Rafael Nadal	Clay	2011–2014	100.000	False	False	2000	11.806	18.570
476	Rafael Nadal	Grass	2011–2014	90.909	True	False	2000	3.791	8.832
477	Rafael Nadal	Grass	2011–2014	90.909	True	False	2000	6.083	11.310
478	Rafael Nadal	Grass	2011–2014	81.818	True	False	2000	0.397	2.400
479	Rafael Nadal	Grass	2011–2014	90.909	True	False	2000	0.385	1.415
480	Rafael Nadal	Grass	2011–2014	90.909	True	False	2000	16.399	23.563
481	Rafael Nadal	Grass	2011–2014	90.909	True	False	2000	64.842	76.210
482	Rafael Nadal	Grass	2011–2014	90.909	True	False	2000	17.540	23.733
483	Rafael Nadal	Grass	2011–2014	90.909	True	False	2000	6.079	11.779
484	Rafael Nadal	Grass	2011–2014	90.909	True	False	2000	15.077	25.765
485	Rafael Nadal	Grass	2011–2014	90.909	True	False	2000	26.790	38.558
486	Rafael Nadal	Grass	2011–2014	90.909	True	False	2000	39.927	55.468

	player	surface	period	coverage_pct	small_sample	coverage_edge_period	valid_bootstraps	ci95_lower	ci95_upper
487	Rafael Nadal	Grass	2011–2014	90.909	True	False	2000	32.857	46.080
488	Rafael Nadal	Grass	2011–2014	90.909	True	False	2000	24.123	38.158
489	Rafael Nadal	Grass	2011–2014	90.909	True	False	2000	22.195	36.957
490	Rafael Nadal	Hard	2011–2014	86.957	False	False	2000	4.086	5.680
491	Rafael Nadal	Hard	2011–2014	93.478	False	False	2000	6.370	8.193
492	Rafael Nadal	Hard	2011–2014	97.826	False	False	2000	0.337	0.863
493	Rafael Nadal	Hard	2011–2014	100.000	False	False	2000	0.250	0.410
494	Rafael Nadal	Hard	2011–2014	100.000	False	False	2000	9.806	13.187
495	Rafael Nadal	Hard	2011–2014	100.000	False	False	2000	57.991	62.335
496	Rafael Nadal	Hard	2011–2014	100.000	False	False	2000	22.430	24.639
497	Rafael Nadal	Hard	2011–2014	100.000	False	False	2000	14.644	18.070
498	Rafael Nadal	Hard	2011–2014	100.000	False	False	2000	16.528	21.807
499	Rafael Nadal	Hard	2011–2014	100.000	False	False	2000	31.331	40.933
500	Rafael Nadal	Hard	2011–2014	100.000	False	False	2000	39.027	50.191
501	Rafael Nadal	Hard	2011–2014	100.000	False	False	2000	45.542	57.719
502	Rafael Nadal	Hard	2011–2014	100.000	False	False	2000	26.523	36.018
503	Rafael Nadal	Hard	2011–2014	100.000	False	False	2000	13.937	20.778
504	Rafael Nadal	Overall	2015–2018	97.115	False	False	2000	6.844	8.683
505	Rafael Nadal	Overall	2015–2018	98.077	False	False	2000	8.861	10.567
506	Rafael Nadal	Overall	2015–2018	98.077	False	False	2000	0.649	1.445
507	Rafael Nadal	Overall	2015–2018	100.000	False	False	2000	0.687	0.991
508	Rafael Nadal	Overall	2015–2018	100.000	False	False	2000	7.708	9.590
509	Rafael Nadal	Overall	2015–2018	100.000	False	False	2000	59.620	63.139
510	Rafael Nadal	Overall	2015–2018	100.000	False	False	2000	23.626	25.369
511	Rafael Nadal	Overall	2015–2018	100.000	False	False	2000	13.004	15.294
512	Rafael Nadal	Overall	2015–2018	100.000	False	False	2000	26.602	32.066
513	Rafael Nadal	Overall	2015–2018	100.000	False	False	2000	21.268	25.654
514	Rafael Nadal	Overall	2015–2018	100.000	False	False	2000	44.374	50.012
515	Rafael Nadal	Overall	2015–2018	100.000	False	False	2000	46.812	53.958
516	Rafael Nadal	Overall	2015–2018	100.000	False	False	2000	26.264	31.573
517	Rafael Nadal	Overall	2015–2018	100.000	False	False	2000	18.464	23.207
518	Rafael Nadal	Clay	2015–2018	97.500	False	False	2000	6.416	8.536
519	Rafael Nadal	Clay	2015–2018	97.500	False	False	2000	8.566	10.687
520	Rafael Nadal	Clay	2015–2018	97.500	False	False	2000	0.439	1.332
521	Rafael Nadal	Clay	2015–2018	100.000	False	False	2000	0.917	1.422
522	Rafael Nadal	Clay	2015–2018	100.000	False	False	2000	5.284	7.094
523	Rafael Nadal	Clay	2015–2018	100.000	False	False	2000	52.491	57.416
524	Rafael Nadal	Clay	2015–2018	100.000	False	False	2000	25.712	28.682
525	Rafael Nadal	Clay	2015–2018	100.000	False	False	2000	16.215	19.623

	player	surface	period	coverage_pct	small_sample	coverage_edge_period	valid_bootstraps	ci95_lower	ci95_upper
526	Rafael Nadal	Clay	2015–2018	100.000	False	False	2000	21.970	30.765
527	Rafael Nadal	Clay	2015–2018	100.000	False	False	2000	24.980	32.722
528	Rafael Nadal	Clay	2015–2018	100.000	False	False	2000	40.107	49.094
529	Rafael Nadal	Clay	2015–2018	100.000	False	False	2000	51.840	63.148
530	Rafael Nadal	Clay	2015–2018	100.000	False	False	2000	22.210	29.843
531	Rafael Nadal	Clay	2015–2018	100.000	False	False	2000	12.908	20.273
532	Rafael Nadal	Grass	2015–2018	100.000	True	False	1969	3.213	14.076
533	Rafael Nadal	Grass	2015–2018	100.000	True	False	1969	10.442	14.663
534	Rafael Nadal	Grass	2015–2018	100.000	True	False	1969	0.000	4.819
535	Rafael Nadal	Grass	2015–2018	100.000	True	False	1969	1.633	2.443
536	Rafael Nadal	Grass	2015–2018	100.000	True	False	1969	10.537	15.083
537	Rafael Nadal	Grass	2015–2018	100.000	True	False	1969	58.995	82.988
538	Rafael Nadal	Grass	2015–2018	100.000	True	False	1969	15.353	23.545
539	Rafael Nadal	Grass	2015–2018	100.000	True	False	1969	1.660	17.460
540	Rafael Nadal	Grass	2015–2018	100.000	True	False	1969	20.312	43.617
541	Rafael Nadal	Grass	2015–2018	100.000	True	False	1969	15.957	28.125
542	Rafael Nadal	Grass	2015–2018	100.000	True	False	1969	40.426	51.562
543	Rafael Nadal	Grass	2015–2018	100.000	True	False	1969	27.451	48.101
544	Rafael Nadal	Grass	2015–2018	100.000	True	False	1969	26.506	34.314
545	Rafael Nadal	Grass	2015–2018	100.000	True	False	1969	17.722	38.235
546	Rafael Nadal	Hard	2015–2018	96.667	False	False	2000	6.373	8.762
547	Rafael Nadal	Hard	2015–2018	98.333	False	False	2000	8.158	10.372
548	Rafael Nadal	Hard	2015–2018	98.333	False	False	2000	0.548	1.580
549	Rafael Nadal	Hard	2015–2018	100.000	False	False	2000	0.380	0.611
550	Rafael Nadal	Hard	2015–2018	100.000	False	False	2000	8.405	11.022
551	Rafael Nadal	Hard	2015–2018	100.000	False	False	2000	61.927	65.338
552	Rafael Nadal	Hard	2015–2018	100.000	False	False	2000	22.668	24.587
553	Rafael Nadal	Hard	2015–2018	100.000	False	False	2000	11.483	13.855
554	Rafael Nadal	Hard	2015–2018	100.000	False	False	2000	26.313	33.891
555	Rafael Nadal	Hard	2015–2018	100.000	False	False	2000	18.383	23.808
556	Rafael Nadal	Hard	2015–2018	100.000	False	False	2000	44.612	52.788
557	Rafael Nadal	Hard	2015–2018	100.000	False	False	2000	43.585	52.510
558	Rafael Nadal	Hard	2015–2018	100.000	False	False	2000	26.230	33.877
559	Rafael Nadal	Hard	2015–2018	100.000	False	False	2000	19.049	24.975
560	Rafael Nadal	Overall	2019–2022	93.548	False	False	2000	6.172	8.049
561	Rafael Nadal	Overall	2019–2022	96.774	False	False	2000	9.157	11.190
562	Rafael Nadal	Overall	2019–2022	96.774	False	False	2000	2.283	4.173
563	Rafael Nadal	Overall	2019–2022	98.387	False	False	2000	0.800	1.241
564	Rafael Nadal	Overall	2019–2022	98.387	False	False	2000	8.708	11.446

	player	surface	period	coverage_pct	small_sample	coverage_edge_period	valid_bootstraps	ci95_lower	ci95_upper
565	Rafael Nadal	Overall	2019–2022	98.387	False	False	2000	59.976	64.828
566	Rafael Nadal	Overall	2019–2022	98.387	False	False	2000	22.153	24.767
567	Rafael Nadal	Overall	2019–2022	98.387	False	False	2000	12.646	15.682
568	Rafael Nadal	Overall	2019–2022	98.387	False	False	2000	25.848	32.242
569	Rafael Nadal	Overall	2019–2022	98.387	False	False	2000	18.038	22.205
570	Rafael Nadal	Overall	2019–2022	98.387	False	False	2000	47.336	54.214
571	Rafael Nadal	Overall	2019–2022	98.387	False	False	2000	44.309	52.288
572	Rafael Nadal	Overall	2019–2022	98.387	False	False	2000	23.995	30.186
573	Rafael Nadal	Overall	2019–2022	98.387	False	False	2000	22.080	27.623
574	Rafael Nadal	Clay	2019–2022	88.889	True	False	2000	4.659	7.256
575	Rafael Nadal	Clay	2019–2022	94.444	True	False	2000	8.088	11.470
576	Rafael Nadal	Clay	2019–2022	100.000	True	False	2000	0.265	1.845
577	Rafael Nadal	Clay	2019–2022	100.000	True	False	2000	1.101	1.783
578	Rafael Nadal	Clay	2019–2022	100.000	True	False	2000	4.714	7.264
579	Rafael Nadal	Clay	2019–2022	100.000	True	False	2000	49.747	56.869
580	Rafael Nadal	Clay	2019–2022	100.000	True	False	2000	25.974	29.921
581	Rafael Nadal	Clay	2019–2022	100.000	True	False	2000	16.659	20.759
582	Rafael Nadal	Clay	2019–2022	100.000	True	False	2000	15.398	24.251
583	Rafael Nadal	Clay	2019–2022	100.000	True	False	2000	16.141	22.193
584	Rafael Nadal	Clay	2019–2022	100.000	True	False	2000	55.981	65.679
585	Rafael Nadal	Clay	2019–2022	100.000	True	False	2000	45.746	63.255
586	Rafael Nadal	Clay	2019–2022	100.000	True	False	2000	18.757	27.912
587	Rafael Nadal	Clay	2019–2022	100.000	True	False	2000	16.208	28.960
588	Rafael Nadal	Grass	2019–2022	100.000	True	False	1994	4.024	9.312
589	Rafael Nadal	Grass	2019–2022	100.000	True	False	1994	6.770	13.678
590	Rafael Nadal	Grass	2019–2022	100.000	True	False	1994	0.292	5.024
591	Rafael Nadal	Grass	2019–2022	100.000	True	False	1994	0.579	1.550
592	Rafael Nadal	Grass	2019–2022	100.000	True	False	1994	9.669	15.808
593	Rafael Nadal	Grass	2019–2022	100.000	True	False	1994	67.675	72.958
594	Rafael Nadal	Grass	2019–2022	100.000	True	False	1994	18.549	23.956
595	Rafael Nadal	Grass	2019–2022	100.000	True	False	1994	6.621	9.792
596	Rafael Nadal	Grass	2019–2022	100.000	True	False	1994	27.517	49.675
597	Rafael Nadal	Grass	2019–2022	100.000	True	False	1994	16.286	26.596
598	Rafael Nadal	Grass	2019–2022	100.000	True	False	1994	24.409	54.717
599	Rafael Nadal	Grass	2019–2022	100.000	True	False	1994	32.386	44.560
600	Rafael Nadal	Grass	2019–2022	100.000	True	False	1994	20.502	28.667
601	Rafael Nadal	Grass	2019–2022	100.000	True	False	1994	32.727	42.449
602	Rafael Nadal	Hard	2019–2022	94.737	False	False	2000	6.442	9.031
603	Rafael Nadal	Hard	2019–2022	97.368	False	False	2000	8.969	11.844

	player	surface	period	coverage_pct	small_sample	coverage_edge_period	valid_bootstraps	ci95_lower	ci95_upper
604	Rafael Nadal	Hard	2019–2022	94.737	False	False	2000	3.117	5.887
605	Rafael Nadal	Hard	2019–2022	97.368	False	False	2000	0.490	1.056
606	Rafael Nadal	Hard	2019–2022	97.368	False	False	2000	10.267	13.595
607	Rafael Nadal	Hard	2019–2022	97.368	False	False	2000	62.501	67.941
608	Rafael Nadal	Hard	2019–2022	97.368	False	False	2000	20.308	23.054
609	Rafael Nadal	Hard	2019–2022	97.368	False	False	2000	11.202	15.041
610	Rafael Nadal	Hard	2019–2022	97.368	False	False	2000	28.586	35.533
611	Rafael Nadal	Hard	2019–2022	97.368	False	False	2000	17.455	23.580
612	Rafael Nadal	Hard	2019–2022	97.368	False	False	2000	44.175	51.400
613	Rafael Nadal	Hard	2019–2022	97.368	False	False	2000	42.512	51.401
614	Rafael Nadal	Hard	2019–2022	97.368	False	False	2000	24.780	33.581
615	Rafael Nadal	Hard	2019–2022	97.368	False	False	2000	21.360	27.178
616	Rafael Nadal	Overall	2023–2024	100.000	True	True	2000	6.011	10.813
617	Rafael Nadal	Overall	2023–2024	100.000	True	True	2000	8.097	12.766
618	Rafael Nadal	Overall	2023–2024	100.000	True	True	2000	2.553	8.151
619	Rafael Nadal	Overall	2023–2024	100.000	True	True	2000	0.777	1.674
620	Rafael Nadal	Overall	2023–2024	100.000	True	True	2000	5.524	9.187
621	Rafael Nadal	Overall	2023–2024	100.000	True	True	2000	55.019	61.468
622	Rafael Nadal	Overall	2023–2024	100.000	True	True	2000	23.630	28.504
623	Rafael Nadal	Overall	2023–2024	100.000	True	True	2000	13.434	17.791
624	Rafael Nadal	Overall	2023–2024	100.000	True	True	2000	13.527	22.687
625	Rafael Nadal	Overall	2023–2024	100.000	True	True	2000	11.751	21.925
626	Rafael Nadal	Overall	2023–2024	100.000	True	True	2000	59.644	70.757
627	Rafael Nadal	Overall	2023–2024	100.000	True	True	2000	46.711	57.959
628	Rafael Nadal	Overall	2023–2024	100.000	True	True	2000	23.617	38.424
629	Rafael Nadal	Overall	2023–2024	100.000	True	True	2000	13.226	19.902
630	Rafael Nadal	Clay	2023–2024	100.000	True	True	1998	3.333	9.314
631	Rafael Nadal	Clay	2023–2024	100.000	True	True	1998	6.114	11.612
632	Rafael Nadal	Clay	2023–2024	100.000	True	True	1998	1.156	4.684
633	Rafael Nadal	Clay	2023–2024	100.000	True	True	1998	1.375	2.320
634	Rafael Nadal	Clay	2023–2024	100.000	True	True	1998	3.681	7.654
635	Rafael Nadal	Clay	2023–2024	100.000	True	True	1998	49.512	58.814
636	Rafael Nadal	Clay	2023–2024	100.000	True	True	1998	26.158	32.826
637	Rafael Nadal	Clay	2023–2024	100.000	True	True	1998	12.267	20.867
638	Rafael Nadal	Clay	2023–2024	100.000	True	True	1998	8.772	16.031
639	Rafael Nadal	Clay	2023–2024	100.000	True	True	1998	9.673	27.632
640	Rafael Nadal	Clay	2023–2024	100.000	True	True	1998	60.471	77.153
641	Rafael Nadal	Clay	2023–2024	100.000	True	True	1998	47.602	66.901
642	Rafael Nadal	Clay	2023–2024	100.000	True	True	1998	14.622	39.506

	player	surface	period	coverage_pct	small_sample	coverage_edge_period	valid_bootstraps	ci95_lower	ci95_upper
643	Rafael Nadal	Clay	2023–2024	100.000	True	True	1998	10.000	22.690
644	Rafael Nadal	Grass	2023–2024	Unavailable	True	True	0	Unavailable	Unavailable
645	Rafael Nadal	Grass	2023–2024	Unavailable	True	True	0	Unavailable	Unavailable
646	Rafael Nadal	Grass	2023–2024	Unavailable	True	True	0	Unavailable	Unavailable
647	Rafael Nadal	Grass	2023–2024	Unavailable	True	True	0	Unavailable	Unavailable
648	Rafael Nadal	Grass	2023–2024	Unavailable	True	True	0	Unavailable	Unavailable
649	Rafael Nadal	Grass	2023–2024	Unavailable	True	True	0	Unavailable	Unavailable
650	Rafael Nadal	Grass	2023–2024	Unavailable	True	True	0	Unavailable	Unavailable
651	Rafael Nadal	Grass	2023–2024	Unavailable	True	True	0	Unavailable	Unavailable
652	Rafael Nadal	Grass	2023–2024	Unavailable	True	True	0	Unavailable	Unavailable
653	Rafael Nadal	Grass	2023–2024	Unavailable	True	True	0	Unavailable	Unavailable
654	Rafael Nadal	Grass	2023–2024	Unavailable	True	True	0	Unavailable	Unavailable
655	Rafael Nadal	Grass	2023–2024	Unavailable	True	True	0	Unavailable	Unavailable
656	Rafael Nadal	Grass	2023–2024	Unavailable	True	True	0	Unavailable	Unavailable
657	Rafael Nadal	Grass	2023–2024	Unavailable	True	True	0	Unavailable	Unavailable
658	Rafael Nadal	Hard	2023–2024	100.000	True	True	1997	6.378	14.291
659	Rafael Nadal	Hard	2023–2024	100.000	True	True	1997	7.847	15.925
660	Rafael Nadal	Hard	2023–2024	100.000	True	True	1997	2.644	12.977
661	Rafael Nadal	Hard	2023–2024	100.000	True	True	1997	0.299	0.977
662	Rafael Nadal	Hard	2023–2024	100.000	True	True	1997	6.860	11.840
663	Rafael Nadal	Hard	2023–2024	100.000	True	True	1997	59.659	65.337
664	Rafael Nadal	Hard	2023–2024	100.000	True	True	1997	20.271	25.303
665	Rafael Nadal	Hard	2023–2024	100.000	True	True	1997	12.710	16.279
666	Rafael Nadal	Hard	2023–2024	100.000	True	True	1997	16.803	28.110
667	Rafael Nadal	Hard	2023–2024	100.000	True	True	1997	8.771	20.804
668	Rafael Nadal	Hard	2023–2024	100.000	True	True	1997	55.725	70.576
669	Rafael Nadal	Hard	2023–2024	100.000	True	True	1997	42.103	55.469
670	Rafael Nadal	Hard	2023–2024	100.000	True	True	1997	27.238	43.191
671	Rafael Nadal	Hard	2023–2024	100.000	True	True	1997	12.500	19.334

Novak Djokovic

▼ Complete evidence table: 336 rows × 16 columns

	player	surface	period	period_index	metric	numerator	denominator	estimate_pct	contributing_matches	cohort_matches
672	Novak Djokovic	Overall	2005–2008	0	approach_frequency_pct	907.000	7,419.000	12.225	47	50
673	Novak Djokovic	Overall	2005–2008	0	net_point_frequency_pct	1,056.000	7,419.000	14.234	47	50
674	Novak Djokovic	Overall	2005–2008	0	serve_volley_frequency_pct	86.000	3,965.000	2.169	49	50
675	Novak Djokovic	Overall	2005–2008	0	drop_shot_share_pct	312.000	16,497.000	1.891	50	50
676	Novak Djokovic	Overall	2005–2008	0	slice_share_pct	966.000	16,497.000	5.856	50	50
677	Novak Djokovic	Overall	2005–2008	0	short_rally_share_pct	4,630.000	7,585.000	61.042	50	50
678	Novak Djokovic	Overall	2005–2008	0	medium_rally_share_pct	1,840.000	7,585.000	24.258	50	50
679	Novak Djokovic	Overall	2005–2008	0	long_rally_share_pct	1,115.000	7,585.000	14.700	50	50
680	Novak Djokovic	Overall	2005–2008	0	deuce_wide_serve_share_pct	1,004.000	2,096.000	47.901	50	50
681	Novak Djokovic	Overall	2005–2008	0	deuce_body_serve_share_pct	453.000	2,096.000	21.613	50	50
682	Novak Djokovic	Overall	2005–2008	0	deuce_t_serve_share_pct	639.000	2,096.000	30.487	50	50
683	Novak Djokovic	Overall	2005–2008	0	ad_wide_serve_share_pct	901.000	1,925.000	46.805	50	50
684	Novak Djokovic	Overall	2005–2008	0	ad_body_serve_share_pct	338.000	1,925.000	17.558	50	50
685	Novak Djokovic	Overall	2005–2008	0	ad_t_serve_share_pct	686.000	1,925.000	35.636	50	50
686	Novak Djokovic	Clay	2005–2008	0	approach_frequency_pct	255.000	1,713.000	14.886	11	12
687	Novak Djokovic	Clay	2005–2008	0	net_point_frequency_pct	298.000	1,713.000	17.396	11	12
688	Novak Djokovic	Clay	2005–2008	0	serve_volley_frequency_pct	10.000	877.000	1.140	11	12
689	Novak Djokovic	Clay	2005–2008	0	drop_shot_share_pct	121.000	4,487.000	2.697	12	12
690	Novak Djokovic	Clay	2005–2008	0	slice_share_pct	192.000	4,487.000	4.279	12	12
691	Novak Djokovic	Clay	2005–2008	0	short_rally_share_pct	945.000	1,774.000	53.269	12	12
692	Novak Djokovic	Clay	2005–2008	0	medium_rally_share_pct	492.000	1,774.000	27.734	12	12
693	Novak Djokovic	Clay	2005–2008	0	long_rally_share_pct	337.000	1,774.000	18.997	12	12
694	Novak Djokovic	Clay	2005–2008	0	deuce_wide_serve_share_pct	253.000	484.000	52.273	12	12

	player	surface	period	period_index	metric	numerator	denominator	estimate_pct	contributing_matches	cohort_matches
695	Novak Djokovic	Clay	2005–2008	0	deuce_body_serve_share_pct	92.000	484.000	19.008	12	12
696	Novak Djokovic	Clay	2005–2008	0	deuce_t_serve_share_pct	139.000	484.000	28.719	12	12
697	Novak Djokovic	Clay	2005–2008	0	ad_wide_serve_share_pct	208.000	449.000	46.325	12	12
698	Novak Djokovic	Clay	2005–2008	0	ad_body_serve_share_pct	52.000	449.000	11.581	12	12
699	Novak Djokovic	Clay	2005–2008	0	ad_t_serve_share_pct	189.000	449.000	42.094	12	12
700	Novak Djokovic	Grass	2005–2008	0	approach_frequency_pct	75.000	732.000	10.246	4	4
701	Novak Djokovic	Grass	2005–2008	0	net_point_frequency_pct	87.000	732.000	11.885	4	4
702	Novak Djokovic	Grass	2005–2008	0	serve_volley_frequency_pct	8.000	386.000	2.073	4	4
703	Novak Djokovic	Grass	2005–2008	0	drop_shot_share_pct	14.000	1,442.000	0.971	4	4
704	Novak Djokovic	Grass	2005–2008	0	slice_share_pct	103.000	1,442.000	7.143	4	4
705	Novak Djokovic	Grass	2005–2008	0	short_rally_share_pct	474.000	711.000	66.667	4	4
706	Novak Djokovic	Grass	2005–2008	0	medium_rally_share_pct	144.000	711.000	20.253	4	4
707	Novak Djokovic	Grass	2005–2008	0	long_rally_share_pct	93.000	711.000	13.080	4	4
708	Novak Djokovic	Grass	2005–2008	0	deuce_wide_serve_share_pct	86.000	199.000	43.216	4	4
709	Novak Djokovic	Grass	2005–2008	0	deuce_body_serve_share_pct	51.000	199.000	25.628	4	4
710	Novak Djokovic	Grass	2005–2008	0	deuce_t_serve_share_pct	62.000	199.000	31.156	4	4
711	Novak Djokovic	Grass	2005–2008	0	ad_wide_serve_share_pct	87.000	187.000	46.524	4	4
712	Novak Djokovic	Grass	2005–2008	0	ad_body_serve_share_pct	33.000	187.000	17.647	4	4
713	Novak Djokovic	Grass	2005–2008	0	ad_t_serve_share_pct	67.000	187.000	35.829	4	4
714	Novak Djokovic	Hard	2005–2008	0	approach_frequency_pct	577.000	4,974.000	11.600	32	34
715	Novak Djokovic	Hard	2005–2008	0	net_point_frequency_pct	671.000	4,974.000	13.490	32	34
716	Novak Djokovic	Hard	2005–2008	0	serve_volley_frequency_pct	68.000	2,702.000	2.517	34	34
717	Novak Djokovic	Hard	2005–2008	0	drop_shot_share_pct	177.000	10,568.000	1.675	34	34
718	Novak Djokovic	Hard	2005–2008	0	slice_share_pct	671.000	10,568.000	6.349	34	34

	player	surface	period	period_index	metric	numerator	denominator	estimate_pct	contributing_matches	cohort_matches
719	Novak Djokovic	Hard	2005–2008	0	short_rally_share_pct	3,211.000	5,100.000	62.961	34	34
720	Novak Djokovic	Hard	2005–2008	0	medium_rally_share_pct	1,204.000	5,100.000	23.608	34	34
721	Novak Djokovic	Hard	2005–2008	0	long_rally_share_pct	685.000	5,100.000	13.431	34	34
722	Novak Djokovic	Hard	2005–2008	0	deuce_wide_serve_share_pct	665.000	1,413.000	47.063	34	34
723	Novak Djokovic	Hard	2005–2008	0	deuce_body_serve_share_pct	310.000	1,413.000	21.939	34	34
724	Novak Djokovic	Hard	2005–2008	0	deuce_t_serve_share_pct	438.000	1,413.000	30.998	34	34
725	Novak Djokovic	Hard	2005–2008	0	ad_wide_serve_share_pct	606.000	1,289.000	47.013	34	34
726	Novak Djokovic	Hard	2005–2008	0	ad_body_serve_share_pct	253.000	1,289.000	19.628	34	34
727	Novak Djokovic	Hard	2005–2008	0	ad_t_serve_share_pct	430.000	1,289.000	33.359	34	34
728	Novak Djokovic	Overall	2009–2012	1	approach_frequency_pct	1,370.000	16,430.000	8.338	99	106
729	Novak Djokovic	Overall	2009–2012	1	net_point_frequency_pct	1,863.000	17,100.000	10.895	103	106
730	Novak Djokovic	Overall	2009–2012	1	serve_volley_frequency_pct	74.000	8,635.000	0.857	104	106
731	Novak Djokovic	Overall	2009–2012	1	drop_shot_share_pct	490.000	42,165.000	1.162	106	106
732	Novak Djokovic	Overall	2009–2012	1	slice_share_pct	3,065.000	42,165.000	7.269	106	106
733	Novak Djokovic	Overall	2009–2012	1	short_rally_share_pct	10,023.000	17,322.000	57.863	106	106
734	Novak Djokovic	Overall	2009–2012	1	medium_rally_share_pct	4,100.000	17,322.000	23.669	106	106
735	Novak Djokovic	Overall	2009–2012	1	long_rally_share_pct	3,199.000	17,322.000	18.468	106	106
736	Novak Djokovic	Overall	2009–2012	1	deuce_wide_serve_share_pct	2,014.000	4,653.000	43.284	106	106
737	Novak Djokovic	Overall	2009–2012	1	deuce_body_serve_share_pct	1,083.000	4,653.000	23.275	106	106
738	Novak Djokovic	Overall	2009–2012	1	deuce_t_serve_share_pct	1,556.000	4,653.000	33.441	106	106
739	Novak Djokovic	Overall	2009–2012	1	ad_wide_serve_share_pct	1,702.000	4,237.000	40.170	106	106
740	Novak Djokovic	Overall	2009–2012	1	ad_body_serve_share_pct	1,075.000	4,237.000	25.372	106	106
741	Novak Djokovic	Overall	2009–2012	1	ad_t_serve_share_pct	1,460.000	4,237.000	34.458	106	106
742	Novak Djokovic	Clay	2009–2012	1	approach_frequency_pct	259.000	3,742.000	6.921	24	26

	player	surface	period	period_index	metric	numerator	denominator	estimate_pct	contributing_matches	cohort_matches
743	Novak Djokovic	Clay	2009–2012	1	net_point_frequency_pct	376.000	3,916.000	9.602	25	26
744	Novak Djokovic	Clay	2009–2012	1	serve_volley_frequency_pct	15.000	1,971.000	0.761	25	26
745	Novak Djokovic	Clay	2009–2012	1	drop_shot_share_pct	135.000	10,985.000	1.229	26	26
746	Novak Djokovic	Clay	2009–2012	1	slice_share_pct	540.000	10,985.000	4.916	26	26
747	Novak Djokovic	Clay	2009–2012	1	short_rally_share_pct	2,228.000	4,119.000	54.091	26	26
748	Novak Djokovic	Clay	2009–2012	1	medium_rally_share_pct	996.000	4,119.000	24.181	26	26
749	Novak Djokovic	Clay	2009–2012	1	long_rally_share_pct	895.000	4,119.000	21.729	26	26
750	Novak Djokovic	Clay	2009–2012	1	deuce_wide_serve_share_pct	531.000	1,112.000	47.752	26	26
751	Novak Djokovic	Clay	2009–2012	1	deuce_body_serve_share_pct	203.000	1,112.000	18.255	26	26
752	Novak Djokovic	Clay	2009–2012	1	deuce_t_serve_share_pct	378.000	1,112.000	33.993	26	26
753	Novak Djokovic	Clay	2009–2012	1	ad_wide_serve_share_pct	441.000	1,025.000	43.024	26	26
754	Novak Djokovic	Clay	2009–2012	1	ad_body_serve_share_pct	192.000	1,025.000	18.732	26	26
755	Novak Djokovic	Clay	2009–2012	1	ad_t_serve_share_pct	392.000	1,025.000	38.244	26	26
756	Novak Djokovic	Grass	2009–2012	1	approach_frequency_pct	142.000	1,545.000	9.191	9	11
757	Novak Djokovic	Grass	2009–2012	1	net_point_frequency_pct	201.000	1,924.000	10.447	11	11
758	Novak Djokovic	Grass	2009–2012	1	serve_volley_frequency_pct	10.000	935.000	1.070	11	11
759	Novak Djokovic	Grass	2009–2012	1	drop_shot_share_pct	55.000	3,488.000	1.577	11	11
760	Novak Djokovic	Grass	2009–2012	1	slice_share_pct	346.000	3,488.000	9.920	11	11
761	Novak Djokovic	Grass	2009–2012	1	short_rally_share_pct	1,288.000	1,883.000	68.401	11	11
762	Novak Djokovic	Grass	2009–2012	1	medium_rally_share_pct	378.000	1,883.000	20.074	11	11
763	Novak Djokovic	Grass	2009–2012	1	long_rally_share_pct	217.000	1,883.000	11.524	11	11
764	Novak Djokovic	Grass	2009–2012	1	deuce_wide_serve_share_pct	199.000	493.000	40.365	11	11
765	Novak Djokovic	Grass	2009–2012	1	deuce_body_serve_share_pct	98.000	493.000	19.878	11	11
766	Novak Djokovic	Grass	2009–2012	1	deuce_t_serve_share_pct	196.000	493.000	39.757	11	11

	player	surface	period	period_index	metric	numerator	denominator	estimate_pct	contributing_matches	cohort_matches
767	Novak Djokovic	Grass	2009–2012	1	ad_wide_serve_share_pct	177.000	441.000	40.136	11	11
768	Novak Djokovic	Grass	2009–2012	1	ad_body_serve_share_pct	102.000	441.000	23.129	11	11
769	Novak Djokovic	Grass	2009–2012	1	ad_t_serve_share_pct	162.000	441.000	36.735	11	11
770	Novak Djokovic	Hard	2009–2012	1	approach_frequency_pct	969.000	11,143.000	8.696	66	69
771	Novak Djokovic	Hard	2009–2012	1	net_point_frequency_pct	1,286.000	11,260.000	11.421	67	69
772	Novak Djokovic	Hard	2009–2012	1	serve_volley_frequency_pct	49.000	5,729.000	0.855	68	69
773	Novak Djokovic	Hard	2009–2012	1	drop_shot_share_pct	300.000	27,692.000	1.083	69	69
774	Novak Djokovic	Hard	2009–2012	1	slice_share_pct	2,179.000	27,692.000	7.869	69	69
775	Novak Djokovic	Hard	2009–2012	1	short_rally_share_pct	6,507.000	11,320.000	57.482	69	69
776	Novak Djokovic	Hard	2009–2012	1	medium_rally_share_pct	2,726.000	11,320.000	24.081	69	69
777	Novak Djokovic	Hard	2009–2012	1	long_rally_share_pct	2,087.000	11,320.000	18.436	69	69
778	Novak Djokovic	Hard	2009–2012	1	deuce_wide_serve_share_pct	1,284.000	3,048.000	42.126	69	69
779	Novak Djokovic	Hard	2009–2012	1	deuce_body_serve_share_pct	782.000	3,048.000	25.656	69	69
780	Novak Djokovic	Hard	2009–2012	1	deuce_t_serve_share_pct	982.000	3,048.000	32.218	69	69
781	Novak Djokovic	Hard	2009–2012	1	ad_wide_serve_share_pct	1,084.000	2,771.000	39.119	69	69
782	Novak Djokovic	Hard	2009–2012	1	ad_body_serve_share_pct	781.000	2,771.000	28.185	69	69
783	Novak Djokovic	Hard	2009–2012	1	ad_t_serve_share_pct	906.000	2,771.000	32.696	69	69
784	Novak Djokovic	Overall	2013–2016	2	approach_frequency_pct	1,637.000	20,944.000	7.816	126	132
785	Novak Djokovic	Overall	2013–2016	2	net_point_frequency_pct	2,326.000	21,817.000	10.661	131	132
786	Novak Djokovic	Overall	2013–2016	2	serve_volley_frequency_pct	114.000	10,637.000	1.072	132	132
787	Novak Djokovic	Overall	2013–2016	2	drop_shot_share_pct	581.000	47,193.000	1.231	132	132
788	Novak Djokovic	Overall	2013–2016	2	slice_share_pct	3,860.000	47,193.000	8.179	132	132
789	Novak Djokovic	Overall	2013–2016	2	short_rally_share_pct	12,901.000	21,307.000	60.548	132	132
790	Novak Djokovic	Overall	2013–2016	2	medium_rally_share_pct	5,070.000	21,307.000	23.795	132	132

	player	surface	period	period_index	metric	numerator	denominator	estimate_pct	contributing_matches	cohort_matches
791	Novak Djokovic	Overall	2013–2016	2	long_rally_share_pct	3,336.000	21,307.000	15.657	132	132
792	Novak Djokovic	Overall	2013–2016	2	deuce_wide_serve_share_pct	2,456.000	5,591.000	43.928	132	132
793	Novak Djokovic	Overall	2013–2016	2	deuce_body_serve_share_pct	1,099.000	5,591.000	19.657	132	132
794	Novak Djokovic	Overall	2013–2016	2	deuce_t_serve_share_pct	2,036.000	5,591.000	36.416	132	132
795	Novak Djokovic	Overall	2013–2016	2	ad_wide_serve_share_pct	2,207.000	5,043.000	43.764	132	132
796	Novak Djokovic	Overall	2013–2016	2	ad_body_serve_share_pct	819.000	5,043.000	16.240	132	132
797	Novak Djokovic	Overall	2013–2016	2	ad_t_serve_share_pct	2,017.000	5,043.000	39.996	132	132
798	Novak Djokovic	Clay	2013–2016	2	approach_frequency_pct	195.000	2,984.000	6.535	17	21
799	Novak Djokovic	Clay	2013–2016	2	net_point_frequency_pct	324.000	3,549.000	9.129	20	21
800	Novak Djokovic	Clay	2013–2016	2	serve_volley_frequency_pct	16.000	1,801.000	0.888	21	21
801	Novak Djokovic	Clay	2013–2016	2	drop_shot_share_pct	181.000	8,546.000	2.118	21	21
802	Novak Djokovic	Clay	2013–2016	2	slice_share_pct	486.000	8,546.000	5.687	21	21
803	Novak Djokovic	Clay	2013–2016	2	short_rally_share_pct	2,032.000	3,594.000	56.539	21	21
804	Novak Djokovic	Clay	2013–2016	2	medium_rally_share_pct	939.000	3,594.000	26.127	21	21
805	Novak Djokovic	Clay	2013–2016	2	long_rally_share_pct	623.000	3,594.000	17.334	21	21
806	Novak Djokovic	Clay	2013–2016	2	deuce_wide_serve_share_pct	441.000	949.000	46.470	21	21
807	Novak Djokovic	Clay	2013–2016	2	deuce_body_serve_share_pct	174.000	949.000	18.335	21	21
808	Novak Djokovic	Clay	2013–2016	2	deuce_t_serve_share_pct	334.000	949.000	35.195	21	21
809	Novak Djokovic	Clay	2013–2016	2	ad_wide_serve_share_pct	356.000	851.000	41.833	21	21
810	Novak Djokovic	Clay	2013–2016	2	ad_body_serve_share_pct	165.000	851.000	19.389	21	21
811	Novak Djokovic	Clay	2013–2016	2	ad_t_serve_share_pct	330.000	851.000	38.778	21	21
812	Novak Djokovic	Grass	2013–2016	2	approach_frequency_pct	264.000	2,794.000	9.449	11	12
813	Novak Djokovic	Grass	2013–2016	2	net_point_frequency_pct	349.000	3,004.000	11.618	12	12
814	Novak Djokovic	Grass	2013–2016	2	serve_volley_frequency_pct	22.000	1,469.000	1.498	12	12

	player	surface	period	period_index	metric	numerator	denominator	estimate_pct	contributing_matches	cohort_matches
815	Novak Djokovic	Grass	2013–2016	2	drop_shot_share_pct	59.000	5,735.000	1.029	12	12
816	Novak Djokovic	Grass	2013–2016	2	slice_share_pct	617.000	5,735.000	10.759	12	12
817	Novak Djokovic	Grass	2013–2016	2	short_rally_share_pct	1,907.000	2,930.000	65.085	12	12
818	Novak Djokovic	Grass	2013–2016	2	medium_rally_share_pct	666.000	2,930.000	22.730	12	12
819	Novak Djokovic	Grass	2013–2016	2	long_rally_share_pct	357.000	2,930.000	12.184	12	12
820	Novak Djokovic	Grass	2013–2016	2	deuce_wide_serve_share_pct	313.000	773.000	40.492	12	12
821	Novak Djokovic	Grass	2013–2016	2	deuce_body_serve_share_pct	156.000	773.000	20.181	12	12
822	Novak Djokovic	Grass	2013–2016	2	deuce_t_serve_share_pct	304.000	773.000	39.327	12	12
823	Novak Djokovic	Grass	2013–2016	2	ad_wide_serve_share_pct	308.000	696.000	44.253	12	12
824	Novak Djokovic	Grass	2013–2016	2	ad_body_serve_share_pct	106.000	696.000	15.230	12	12
825	Novak Djokovic	Grass	2013–2016	2	ad_t_serve_share_pct	282.000	696.000	40.517	12	12
826	Novak Djokovic	Hard	2013–2016	2	approach_frequency_pct	1,178.000	15,166.000	7.767	98	99
827	Novak Djokovic	Hard	2013–2016	2	net_point_frequency_pct	1,653.000	15,264.000	10.829	99	99
828	Novak Djokovic	Hard	2013–2016	2	serve_volley_frequency_pct	76.000	7,367.000	1.032	99	99
829	Novak Djokovic	Hard	2013–2016	2	drop_shot_share_pct	341.000	32,912.000	1.036	99	99
830	Novak Djokovic	Hard	2013–2016	2	slice_share_pct	2,757.000	32,912.000	8.377	99	99
831	Novak Djokovic	Hard	2013–2016	2	short_rally_share_pct	8,962.000	14,783.000	60.624	99	99
832	Novak Djokovic	Hard	2013–2016	2	medium_rally_share_pct	3,465.000	14,783.000	23.439	99	99
833	Novak Djokovic	Hard	2013–2016	2	long_rally_share_pct	2,356.000	14,783.000	15.937	99	99
834	Novak Djokovic	Hard	2013–2016	2	deuce_wide_serve_share_pct	1,702.000	3,869.000	43.991	99	99
835	Novak Djokovic	Hard	2013–2016	2	deuce_body_serve_share_pct	769.000	3,869.000	19.876	99	99
836	Novak Djokovic	Hard	2013–2016	2	deuce_t_serve_share_pct	1,398.000	3,869.000	36.133	99	99
837	Novak Djokovic	Hard	2013–2016	2	ad_wide_serve_share_pct	1,543.000	3,496.000	44.136	99	99
838	Novak Djokovic	Hard	2013–2016	2	ad_body_serve_share_pct	548.000	3,496.000	15.675	99	99

	player	surface	period	period_index	metric	numerator	denominator	estimate_pct	contributing_matches	cohort_matches
839	Novak Djokovic	Hard	2013–2016	2	ad_t_serve_share_pct	1,405.000	3,496.000	40.189	99	99
840	Novak Djokovic	Overall	2017–2020	3	approach_frequency_pct	1,208.000	13,742.000	8.791	77	79
841	Novak Djokovic	Overall	2017–2020	3	net_point_frequency_pct	1,582.000	13,983.000	11.314	79	79
842	Novak Djokovic	Overall	2017–2020	3	serve_volley_frequency_pct	93.000	6,853.000	1.357	79	79
843	Novak Djokovic	Overall	2017–2020	3	drop_shot_share_pct	469.000	30,849.000	1.520	79	79
844	Novak Djokovic	Overall	2017–2020	3	slice_share_pct	2,873.000	30,849.000	9.313	79	79
845	Novak Djokovic	Overall	2017–2020	3	short_rally_share_pct	8,023.000	13,537.000	59.267	79	79
846	Novak Djokovic	Overall	2017–2020	3	medium_rally_share_pct	3,195.000	13,537.000	23.602	79	79
847	Novak Djokovic	Overall	2017–2020	3	long_rally_share_pct	2,319.000	13,537.000	17.131	79	79
848	Novak Djokovic	Overall	2017–2020	3	deuce_wide_serve_share_pct	1,693.000	3,596.000	47.080	79	79
849	Novak Djokovic	Overall	2017–2020	3	deuce_body_serve_share_pct	579.000	3,596.000	16.101	79	79
850	Novak Djokovic	Overall	2017–2020	3	deuce_t_serve_share_pct	1,324.000	3,596.000	36.819	79	79
851	Novak Djokovic	Overall	2017–2020	3	ad_wide_serve_share_pct	1,405.000	3,256.000	43.151	79	79
852	Novak Djokovic	Overall	2017–2020	3	ad_body_serve_share_pct	519.000	3,256.000	15.940	79	79
853	Novak Djokovic	Overall	2017–2020	3	ad_t_serve_share_pct	1,332.000	3,256.000	40.909	79	79
854	Novak Djokovic	Clay	2017–2020	3	approach_frequency_pct	302.000	3,768.000	8.015	21	21
855	Novak Djokovic	Clay	2017–2020	3	net_point_frequency_pct	405.000	3,768.000	10.748	21	21
856	Novak Djokovic	Clay	2017–2020	3	serve_volley_frequency_pct	27.000	1,918.000	1.408	21	21
857	Novak Djokovic	Clay	2017–2020	3	drop_shot_share_pct	197.000	8,931.000	2.206	21	21
858	Novak Djokovic	Clay	2017–2020	3	slice_share_pct	495.000	8,931.000	5.542	21	21
859	Novak Djokovic	Clay	2017–2020	3	short_rally_share_pct	2,066.000	3,673.000	56.248	21	21
860	Novak Djokovic	Clay	2017–2020	3	medium_rally_share_pct	914.000	3,673.000	24.884	21	21
861	Novak Djokovic	Clay	2017–2020	3	long_rally_share_pct	693.000	3,673.000	18.867	21	21
862	Novak Djokovic	Clay	2017–2020	3	deuce_wide_serve_share_pct	472.000	999.000	47.247	21	21

	player	surface	period	period_index	metric	numerator	denominator	estimate_pct	contributing_matches	cohort_matches
863	Novak Djokovic	Clay	2017–2020	3	deuce_body_serve_share_pct	147.000	999.000	14.715	21	21
864	Novak Djokovic	Clay	2017–2020	3	deuce_t_serve_share_pct	380.000	999.000	38.038	21	21
865	Novak Djokovic	Clay	2017–2020	3	ad_wide_serve_share_pct	493.000	919.000	53.645	21	21
866	Novak Djokovic	Clay	2017–2020	3	ad_body_serve_share_pct	104.000	919.000	11.317	21	21
867	Novak Djokovic	Clay	2017–2020	3	ad_t_serve_share_pct	322.000	919.000	35.038	21	21
868	Novak Djokovic	Grass	2017–2020	3	approach_frequency_pct	267.000	2,687.000	9.937	13	13
869	Novak Djokovic	Grass	2017–2020	3	net_point_frequency_pct	338.000	2,687.000	12.579	13	13
870	Novak Djokovic	Grass	2017–2020	3	serve_volley_frequency_pct	19.000	1,318.000	1.442	13	13
871	Novak Djokovic	Grass	2017–2020	3	drop_shot_share_pct	62.000	5,396.000	1.149	13	13
872	Novak Djokovic	Grass	2017–2020	3	slice_share_pct	734.000	5,396.000	13.603	13	13
873	Novak Djokovic	Grass	2017–2020	3	short_rally_share_pct	1,621.000	2,586.000	62.684	13	13
874	Novak Djokovic	Grass	2017–2020	3	medium_rally_share_pct	570.000	2,586.000	22.042	13	13
875	Novak Djokovic	Grass	2017–2020	3	long_rally_share_pct	395.000	2,586.000	15.275	13	13
876	Novak Djokovic	Grass	2017–2020	3	deuce_wide_serve_share_pct	320.000	695.000	46.043	13	13
877	Novak Djokovic	Grass	2017–2020	3	deuce_body_serve_share_pct	115.000	695.000	16.547	13	13
878	Novak Djokovic	Grass	2017–2020	3	deuce_t_serve_share_pct	260.000	695.000	37.410	13	13
879	Novak Djokovic	Grass	2017–2020	3	ad_wide_serve_share_pct	235.000	623.000	37.721	13	13
880	Novak Djokovic	Grass	2017–2020	3	ad_body_serve_share_pct	102.000	623.000	16.372	13	13
881	Novak Djokovic	Grass	2017–2020	3	ad_t_serve_share_pct	286.000	623.000	45.907	13	13
882	Novak Djokovic	Hard	2017–2020	3	approach_frequency_pct	639.000	7,287.000	8.769	43	45
883	Novak Djokovic	Hard	2017–2020	3	net_point_frequency_pct	839.000	7,528.000	11.145	45	45
884	Novak Djokovic	Hard	2017–2020	3	serve_volley_frequency_pct	47.000	3,617.000	1.299	45	45
885	Novak Djokovic	Hard	2017–2020	3	drop_shot_share_pct	210.000	16,522.000	1.271	45	45
886	Novak Djokovic	Hard	2017–2020	3	slice_share_pct	1,644.000	16,522.000	9.950	45	45

	player	surface	period	period_index	metric	numerator	denominator	estimate_pct	contributing_matches	cohort_matches
887	Novak Djokovic	Hard	2017–2020	3	short_rally_share_pct	4,336.000	7,278.000	59.577	45	45
888	Novak Djokovic	Hard	2017–2020	3	medium_rally_share_pct	1,711.000	7,278.000	23.509	45	45
889	Novak Djokovic	Hard	2017–2020	3	long_rally_share_pct	1,231.000	7,278.000	16.914	45	45
890	Novak Djokovic	Hard	2017–2020	3	deuce_wide_serve_share_pct	901.000	1,902.000	47.371	45	45
891	Novak Djokovic	Hard	2017–2020	3	deuce_body_serve_share_pct	317.000	1,902.000	16.667	45	45
892	Novak Djokovic	Hard	2017–2020	3	deuce_t_serve_share_pct	684.000	1,902.000	35.962	45	45
893	Novak Djokovic	Hard	2017–2020	3	ad_wide_serve_share_pct	677.000	1,714.000	39.498	45	45
894	Novak Djokovic	Hard	2017–2020	3	ad_body_serve_share_pct	313.000	1,714.000	18.261	45	45
895	Novak Djokovic	Hard	2017–2020	3	ad_t_serve_share_pct	724.000	1,714.000	42.240	45	45
896	Novak Djokovic	Overall	2021–2024	4	approach_frequency_pct	2,319.000	24,126.000	9.612	142	143
897	Novak Djokovic	Overall	2021–2024	4	net_point_frequency_pct	3,172.000	24,126.000	13.148	142	143
898	Novak Djokovic	Overall	2021–2024	4	serve_volley_frequency_pct	585.000	11,618.000	5.035	142	143
899	Novak Djokovic	Overall	2021–2024	4	drop_shot_share_pct	961.000	50,493.000	1.903	142	143
900	Novak Djokovic	Overall	2021–2024	4	slice_share_pct	4,902.000	50,493.000	9.708	142	143
901	Novak Djokovic	Overall	2021–2024	4	short_rally_share_pct	14,375.000	23,349.000	61.566	142	143
902	Novak Djokovic	Overall	2021–2024	4	medium_rally_share_pct	5,313.000	23,349.000	22.755	142	143
903	Novak Djokovic	Overall	2021–2024	4	long_rally_share_pct	3,661.000	23,349.000	15.679	142	143
904	Novak Djokovic	Overall	2021–2024	4	deuce_wide_serve_share_pct	2,913.000	6,091.000	47.825	142	143
905	Novak Djokovic	Overall	2021–2024	4	deuce_body_serve_share_pct	1,012.000	6,091.000	16.615	142	143
906	Novak Djokovic	Overall	2021–2024	4	deuce_t_serve_share_pct	2,166.000	6,091.000	35.561	142	143
907	Novak Djokovic	Overall	2021–2024	4	ad_wide_serve_share_pct	2,286.000	5,521.000	41.406	142	143
908	Novak Djokovic	Overall	2021–2024	4	ad_body_serve_share_pct	797.000	5,521.000	14.436	142	143
909	Novak Djokovic	Overall	2021–2024	4	ad_t_serve_share_pct	2,438.000	5,521.000	44.159	142	143
910	Novak Djokovic	Clay	2021–2024	4	approach_frequency_pct	664.000	6,977.000	9.517	41	41

	player	surface	period	period_index	metric	numerator	denominator	estimate_pct	contributing_matches	cohort_matches
911	Novak Djokovic	Clay	2021–2024	4	net_point_frequency_pct	1,014.000	6,977.000	14.533	41	41
912	Novak Djokovic	Clay	2021–2024	4	serve_volley_frequency_pct	107.000	3,398.000	3.149	41	41
913	Novak Djokovic	Clay	2021–2024	4	drop_shot_share_pct	492.000	16,016.000	3.072	41	41
914	Novak Djokovic	Clay	2021–2024	4	slice_share_pct	855.000	16,016.000	5.338	41	41
915	Novak Djokovic	Clay	2021–2024	4	short_rally_share_pct	3,806.000	6,778.000	56.152	41	41
916	Novak Djokovic	Clay	2021–2024	4	medium_rally_share_pct	1,778.000	6,778.000	26.232	41	41
917	Novak Djokovic	Clay	2021–2024	4	long_rally_share_pct	1,194.000	6,778.000	17.616	41	41
918	Novak Djokovic	Clay	2021–2024	4	deuce_wide_serve_share_pct	798.000	1,775.000	44.958	41	41
919	Novak Djokovic	Clay	2021–2024	4	deuce_body_serve_share_pct	305.000	1,775.000	17.183	41	41
920	Novak Djokovic	Clay	2021–2024	4	deuce_t_serve_share_pct	672.000	1,775.000	37.859	41	41
921	Novak Djokovic	Clay	2021–2024	4	ad_wide_serve_share_pct	747.000	1,623.000	46.026	41	41
922	Novak Djokovic	Clay	2021–2024	4	ad_body_serve_share_pct	180.000	1,623.000	11.091	41	41
923	Novak Djokovic	Clay	2021–2024	4	ad_t_serve_share_pct	696.000	1,623.000	42.884	41	41
924	Novak Djokovic	Grass	2021–2024	4	approach_frequency_pct	336.000	2,996.000	11.215	13	13
925	Novak Djokovic	Grass	2021–2024	4	net_point_frequency_pct	412.000	2,996.000	13.752	13	13
926	Novak Djokovic	Grass	2021–2024	4	serve_volley_frequency_pct	93.000	1,444.000	6.440	13	13
927	Novak Djokovic	Grass	2021–2024	4	drop_shot_share_pct	106.000	5,356.000	1.979	13	13
928	Novak Djokovic	Grass	2021–2024	4	slice_share_pct	815.000	5,356.000	15.217	13	13
929	Novak Djokovic	Grass	2021–2024	4	short_rally_share_pct	1,926.000	2,892.000	66.598	13	13
930	Novak Djokovic	Grass	2021–2024	4	medium_rally_share_pct	620.000	2,892.000	21.438	13	13
931	Novak Djokovic	Grass	2021–2024	4	long_rally_share_pct	346.000	2,892.000	11.964	13	13
932	Novak Djokovic	Grass	2021–2024	4	deuce_wide_serve_share_pct	360.000	755.000	47.682	13	13
933	Novak Djokovic	Grass	2021–2024	4	deuce_body_serve_share_pct	107.000	755.000	14.172	13	13
934	Novak Djokovic	Grass	2021–2024	4	deuce_t_serve_share_pct	288.000	755.000	38.146	13	13

	player	surface	period	period_index	metric	numerator	denominator	estimate_pct	contributing_matches	cohort_matches
935	Novak Djokovic	Grass	2021–2024	4	ad_wide_serve_share_pct	256.000	686.000	37.318	13	13
936	Novak Djokovic	Grass	2021–2024	4	ad_body_serve_share_pct	118.000	686.000	17.201	13	13
937	Novak Djokovic	Grass	2021–2024	4	ad_t_serve_share_pct	312.000	686.000	45.481	13	13
938	Novak Djokovic	Hard	2021–2024	4	approach_frequency_pct	1,319.000	14,153.000	9.320	88	89
939	Novak Djokovic	Hard	2021–2024	4	net_point_frequency_pct	1,746.000	14,153.000	12.337	88	89
940	Novak Djokovic	Hard	2021–2024	4	serve_volley_frequency_pct	385.000	6,776.000	5.682	88	89
941	Novak Djokovic	Hard	2021–2024	4	drop_shot_share_pct	363.000	29,121.000	1.247	88	89
942	Novak Djokovic	Hard	2021–2024	4	slice_share_pct	3,232.000	29,121.000	11.099	88	89
943	Novak Djokovic	Hard	2021–2024	4	short_rally_share_pct	8,643.000	13,679.000	63.184	88	89
944	Novak Djokovic	Hard	2021–2024	4	medium_rally_share_pct	2,915.000	13,679.000	21.310	88	89
945	Novak Djokovic	Hard	2021–2024	4	long_rally_share_pct	2,121.000	13,679.000	15.506	88	89
946	Novak Djokovic	Hard	2021–2024	4	deuce_wide_serve_share_pct	1,755.000	3,561.000	49.284	88	89
947	Novak Djokovic	Hard	2021–2024	4	deuce_body_serve_share_pct	600.000	3,561.000	16.849	88	89
948	Novak Djokovic	Hard	2021–2024	4	deuce_t_serve_share_pct	1,206.000	3,561.000	33.867	88	89
949	Novak Djokovic	Hard	2021–2024	4	ad_wide_serve_share_pct	1,283.000	3,212.000	39.944	88	89
950	Novak Djokovic	Hard	2021–2024	4	ad_body_serve_share_pct	499.000	3,212.000	15.535	88	89
951	Novak Djokovic	Hard	2021–2024	4	ad_t_serve_share_pct	1,430.000	3,212.000	44.521	88	89
952	Novak Djokovic	Overall	2025–2026	5	approach_frequency_pct	694.000	7,262.000	9.557	42	43
953	Novak Djokovic	Overall	2025–2026	5	net_point_frequency_pct	997.000	7,414.000	13.448	43	43
954	Novak Djokovic	Overall	2025–2026	5	serve_volley_frequency_pct	170.000	3,672.000	4.630	43	43
955	Novak Djokovic	Overall	2025–2026	5	drop_shot_share_pct	314.000	14,987.000	2.095	43	43
956	Novak Djokovic	Overall	2025–2026	5	slice_share_pct	1,455.000	14,987.000	9.708	43	43
957	Novak Djokovic	Overall	2025–2026	5	short_rally_share_pct	4,480.000	7,196.000	62.257	43	43
958	Novak Djokovic	Overall	2025–2026	5	medium_rally_share_pct	1,650.000	7,196.000	22.929	43	43

	player	surface	period	period_index	metric	numerator	denominator	estimate_pct	contributing_matches	cohort_matches
959	Novak Djokovic	Overall	2025–2026	5	long_rally_share_pct	1,066.000	7,196.000	14.814	43	43
960	Novak Djokovic	Overall	2025–2026	5	deuce_wide_serve_share_pct	954.000	1,932.000	49.379	43	43
961	Novak Djokovic	Overall	2025–2026	5	deuce_body_serve_share_pct	270.000	1,932.000	13.975	43	43
962	Novak Djokovic	Overall	2025–2026	5	deuce_t_serve_share_pct	708.000	1,932.000	36.646	43	43
963	Novak Djokovic	Overall	2025–2026	5	ad_wide_serve_share_pct	706.000	1,740.000	40.575	43	43
964	Novak Djokovic	Overall	2025–2026	5	ad_body_serve_share_pct	238.000	1,740.000	13.678	43	43
965	Novak Djokovic	Overall	2025–2026	5	ad_t_serve_share_pct	796.000	1,740.000	45.747	43	43
966	Novak Djokovic	Clay	2025–2026	5	approach_frequency_pct	138.000	1,396.000	9.885	8	8
967	Novak Djokovic	Clay	2025–2026	5	net_point_frequency_pct	216.000	1,396.000	15.473	8	8
968	Novak Djokovic	Clay	2025–2026	5	serve_volley_frequency_pct	27.000	688.000	3.924	8	8
969	Novak Djokovic	Clay	2025–2026	5	drop_shot_share_pct	111.000	3,119.000	3.559	8	8
970	Novak Djokovic	Clay	2025–2026	5	slice_share_pct	233.000	3,119.000	7.470	8	8
971	Novak Djokovic	Clay	2025–2026	5	short_rally_share_pct	796.000	1,369.000	58.145	8	8
972	Novak Djokovic	Clay	2025–2026	5	medium_rally_share_pct	340.000	1,369.000	24.836	8	8
973	Novak Djokovic	Clay	2025–2026	5	long_rally_share_pct	233.000	1,369.000	17.020	8	8
974	Novak Djokovic	Clay	2025–2026	5	deuce_wide_serve_share_pct	178.000	363.000	49.036	8	8
975	Novak Djokovic	Clay	2025–2026	5	deuce_body_serve_share_pct	51.000	363.000	14.050	8	8
976	Novak Djokovic	Clay	2025–2026	5	deuce_t_serve_share_pct	134.000	363.000	36.915	8	8
977	Novak Djokovic	Clay	2025–2026	5	ad_wide_serve_share_pct	139.000	325.000	42.769	8	8
978	Novak Djokovic	Clay	2025–2026	5	ad_body_serve_share_pct	37.000	325.000	11.385	8	8
979	Novak Djokovic	Clay	2025–2026	5	ad_t_serve_share_pct	149.000	325.000	45.846	8	8
980	Novak Djokovic	Grass	2025–2026	5	approach_frequency_pct	141.000	1,098.000	12.842	5	5
981	Novak Djokovic	Grass	2025–2026	5	net_point_frequency_pct	174.000	1,098.000	15.847	5	5
982	Novak Djokovic	Grass	2025–2026	5	serve_volley_frequency_pct	59.000	509.000	11.591	5	5

	player	surface	period	period_index	metric	numerator	denominator	estimate_pct	contributing_matches	cohort_matches
983	Novak Djokovic	Grass	2025–2026	5	drop_shot_share_pct	42.000	2,119.000	1.982	5	5
984	Novak Djokovic	Grass	2025–2026	5	slice_share_pct	290.000	2,119.000	13.686	5	5
985	Novak Djokovic	Grass	2025–2026	5	short_rally_share_pct	711.000	1,069.000	66.511	5	5
986	Novak Djokovic	Grass	2025–2026	5	medium_rally_share_pct	233.000	1,069.000	21.796	5	5
987	Novak Djokovic	Grass	2025–2026	5	long_rally_share_pct	125.000	1,069.000	11.693	5	5
988	Novak Djokovic	Grass	2025–2026	5	deuce_wide_serve_share_pct	133.000	267.000	49.813	5	5
989	Novak Djokovic	Grass	2025–2026	5	deuce_body_serve_share_pct	28.000	267.000	10.487	5	5
990	Novak Djokovic	Grass	2025–2026	5	deuce_t_serve_share_pct	106.000	267.000	39.700	5	5
991	Novak Djokovic	Grass	2025–2026	5	ad_wide_serve_share_pct	88.000	242.000	36.364	5	5
992	Novak Djokovic	Grass	2025–2026	5	ad_body_serve_share_pct	34.000	242.000	14.050	5	5
993	Novak Djokovic	Grass	2025–2026	5	ad_t_serve_share_pct	120.000	242.000	49.587	5	5
994	Novak Djokovic	Hard	2025–2026	5	approach_frequency_pct	415.000	4,768.000	8.704	29	30
995	Novak Djokovic	Hard	2025–2026	5	net_point_frequency_pct	607.000	4,920.000	12.337	30	30
996	Novak Djokovic	Hard	2025–2026	5	serve_volley_frequency_pct	84.000	2,475.000	3.394	30	30
997	Novak Djokovic	Hard	2025–2026	5	drop_shot_share_pct	161.000	9,749.000	1.651	30	30
998	Novak Djokovic	Hard	2025–2026	5	slice_share_pct	932.000	9,749.000	9.560	30	30
999	Novak Djokovic	Hard	2025–2026	5	short_rally_share_pct	2,973.000	4,758.000	62.484	30	30
1000	Novak Djokovic	Hard	2025–2026	5	medium_rally_share_pct	1,077.000	4,758.000	22.636	30	30
1001	Novak Djokovic	Hard	2025–2026	5	long_rally_share_pct	708.000	4,758.000	14.880	30	30
1002	Novak Djokovic	Hard	2025–2026	5	deuce_wide_serve_share_pct	643.000	1,302.000	49.386	30	30
1003	Novak Djokovic	Hard	2025–2026	5	deuce_body_serve_share_pct	191.000	1,302.000	14.670	30	30
1004	Novak Djokovic	Hard	2025–2026	5	deuce_t_serve_share_pct	468.000	1,302.000	35.945	30	30
1005	Novak Djokovic	Hard	2025–2026	5	ad_wide_serve_share_pct	479.000	1,173.000	40.835	30	30
1006	Novak Djokovic	Hard	2025–2026	5	ad_body_serve_share_pct	167.000	1,173.000	14.237	30	30

	player	surface	period	period_index	metric	numerator	denominator	estimate_pct	contributing_matches	cohort_matches
1007	Novak Djokovic	Hard	2025–2026	5	ad_t_serve_share_pct	527.000	1,173.000	44.928	30	30

	player	surface	period	coverage_pct	small_sample	coverage_edge_period	valid_bootstraps	ci95_lower	ci95_upper
672	Novak Djokovic	Overall	2005–2008	94.000	False	True	2000	10.703	13.873
673	Novak Djokovic	Overall	2005–2008	94.000	False	True	2000	12.659	15.997
674	Novak Djokovic	Overall	2005–2008	98.000	False	True	2000	1.432	2.942
675	Novak Djokovic	Overall	2005–2008	100.000	False	True	2000	1.582	2.199
676	Novak Djokovic	Overall	2005–2008	100.000	False	True	2000	5.191	6.580
677	Novak Djokovic	Overall	2005–2008	100.000	False	True	2000	58.533	63.521
678	Novak Djokovic	Overall	2005–2008	100.000	False	True	2000	22.848	25.687
679	Novak Djokovic	Overall	2005–2008	100.000	False	True	2000	12.998	16.479
680	Novak Djokovic	Overall	2005–2008	100.000	False	True	2000	44.416	51.690
681	Novak Djokovic	Overall	2005–2008	100.000	False	True	2000	17.900	25.599
682	Novak Djokovic	Overall	2005–2008	100.000	False	True	2000	26.852	33.986
683	Novak Djokovic	Overall	2005–2008	100.000	False	True	2000	42.124	51.410
684	Novak Djokovic	Overall	2005–2008	100.000	False	True	2000	14.629	20.713
685	Novak Djokovic	Overall	2005–2008	100.000	False	True	2000	31.287	39.989
686	Novak Djokovic	Clay	2005–2008	91.667	True	True	2000	10.398	19.368
687	Novak Djokovic	Clay	2005–2008	91.667	True	True	2000	13.467	21.411
688	Novak Djokovic	Clay	2005–2008	91.667	True	True	2000	0.207	2.567
689	Novak Djokovic	Clay	2005–2008	100.000	True	True	2000	2.139	3.333
690	Novak Djokovic	Clay	2005–2008	100.000	True	True	2000	3.369	5.423
691	Novak Djokovic	Clay	2005–2008	100.000	True	True	2000	49.170	57.502
692	Novak Djokovic	Clay	2005–2008	100.000	True	True	2000	24.159	31.212
693	Novak Djokovic	Clay	2005–2008	100.000	True	True	2000	15.203	22.359
694	Novak Djokovic	Clay	2005–2008	100.000	True	True	2000	42.081	62.564
695	Novak Djokovic	Clay	2005–2008	100.000	True	True	2000	10.248	28.924
696	Novak Djokovic	Clay	2005–2008	100.000	True	True	2000	20.119	37.907
697	Novak Djokovic	Clay	2005–2008	100.000	True	True	2000	34.139	59.310
698	Novak Djokovic	Clay	2005–2008	100.000	True	True	2000	8.621	14.655
699	Novak Djokovic	Clay	2005–2008	100.000	True	True	2000	29.523	53.675
700	Novak Djokovic	Grass	2005–2008	100.000	True	True	1971	6.771	15.116
701	Novak Djokovic	Grass	2005–2008	100.000	True	True	1971	6.771	17.442
702	Novak Djokovic	Grass	2005–2008	100.000	True	True	1971	0.606	4.124
703	Novak Djokovic	Grass	2005–2008	100.000	True	True	1971	0.000	2.000
704	Novak Djokovic	Grass	2005–2008	100.000	True	True	1971	4.000	12.456
705	Novak Djokovic	Grass	2005–2008	100.000	True	True	1971	61.967	75.419

	player	surface	period	coverage_pct	small_sample	coverage_edge_period	valid_bootstraps	ci95_lower	ci95_upper
706	Novak Djokovic	Grass	2005–2008	100.000	True	True	1971	14.545	23.837
707	Novak Djokovic	Grass	2005–2008	100.000	True	True	1971	7.821	16.721
708	Novak Djokovic	Grass	2005–2008	100.000	True	True	1971	39.130	64.706
709	Novak Djokovic	Grass	2005–2008	100.000	True	True	1971	16.471	34.783
710	Novak Djokovic	Grass	2005–2008	100.000	True	True	1971	11.765	40.000
711	Novak Djokovic	Grass	2005–2008	100.000	True	True	1971	11.765	65.000
712	Novak Djokovic	Grass	2005–2008	100.000	True	True	1971	7.500	58.824
713	Novak Djokovic	Grass	2005–2008	100.000	True	True	1971	27.500	47.727
714	Novak Djokovic	Hard	2005–2008	94.118	False	True	2000	10.035	13.268
715	Novak Djokovic	Hard	2005–2008	94.118	False	True	2000	11.718	15.388
716	Novak Djokovic	Hard	2005–2008	100.000	False	True	2000	1.570	3.597
717	Novak Djokovic	Hard	2005–2008	100.000	False	True	2000	1.341	2.020
718	Novak Djokovic	Hard	2005–2008	100.000	False	True	2000	5.557	7.201
719	Novak Djokovic	Hard	2005–2008	100.000	False	True	2000	60.267	65.692
720	Novak Djokovic	Hard	2005–2008	100.000	False	True	2000	22.152	25.024
721	Novak Djokovic	Hard	2005–2008	100.000	False	True	2000	11.530	15.489
722	Novak Djokovic	Hard	2005–2008	100.000	False	True	2000	43.176	50.629
723	Novak Djokovic	Hard	2005–2008	100.000	False	True	2000	17.511	26.966
724	Novak Djokovic	Hard	2005–2008	100.000	False	True	2000	26.389	35.431
725	Novak Djokovic	Hard	2005–2008	100.000	False	True	2000	41.977	51.873
726	Novak Djokovic	Hard	2005–2008	100.000	False	True	2000	15.759	23.611
727	Novak Djokovic	Hard	2005–2008	100.000	False	True	2000	28.695	37.628
728	Novak Djokovic	Overall	2009–2012	93.396	False	False	2000	7.370	9.296
729	Novak Djokovic	Overall	2009–2012	97.170	False	False	2000	9.905	11.877
730	Novak Djokovic	Overall	2009–2012	98.113	False	False	2000	0.599	1.129
731	Novak Djokovic	Overall	2009–2012	100.000	False	False	2000	1.027	1.301
732	Novak Djokovic	Overall	2009–2012	100.000	False	False	2000	6.571	8.006
733	Novak Djokovic	Overall	2009–2012	100.000	False	False	2000	56.215	59.634
734	Novak Djokovic	Overall	2009–2012	100.000	False	False	2000	22.868	24.451
735	Novak Djokovic	Overall	2009–2012	100.000	False	False	2000	17.074	19.892
736	Novak Djokovic	Overall	2009–2012	100.000	False	False	2000	40.947	45.752
737	Novak Djokovic	Overall	2009–2012	100.000	False	False	2000	20.795	25.647
738	Novak Djokovic	Overall	2009–2012	100.000	False	False	2000	30.866	35.996
739	Novak Djokovic	Overall	2009–2012	100.000	False	False	2000	37.746	42.843
740	Novak Djokovic	Overall	2009–2012	100.000	False	False	2000	22.962	27.747
741	Novak Djokovic	Overall	2009–2012	100.000	False	False	2000	32.054	37.092
742	Novak Djokovic	Clay	2009–2012	92.308	True	False	2000	5.402	8.516
743	Novak Djokovic	Clay	2009–2012	96.154	True	False	2000	7.933	11.357
744	Novak Djokovic	Clay	2009–2012	96.154	True	False	2000	0.345	1.238

	player	surface	period	coverage_pct	small_sample	coverage_edge_period	valid_bootstraps	ci95_lower	ci95_upper
745	Novak Djokovic	Clay	2009–2012	100.000	True	False	2000	0.939	1.508
746	Novak Djokovic	Clay	2009–2012	100.000	True	False	2000	4.241	5.655
747	Novak Djokovic	Clay	2009–2012	100.000	True	False	2000	50.656	57.592
748	Novak Djokovic	Clay	2009–2012	100.000	True	False	2000	22.653	25.738
749	Novak Djokovic	Clay	2009–2012	100.000	True	False	2000	18.855	24.621
750	Novak Djokovic	Clay	2009–2012	100.000	True	False	2000	42.399	53.639
751	Novak Djokovic	Clay	2009–2012	100.000	True	False	2000	14.176	23.096
752	Novak Djokovic	Clay	2009–2012	100.000	True	False	2000	28.162	40.032
753	Novak Djokovic	Clay	2009–2012	100.000	True	False	2000	37.418	48.699
754	Novak Djokovic	Clay	2009–2012	100.000	True	False	2000	14.141	23.903
755	Novak Djokovic	Clay	2009–2012	100.000	True	False	2000	33.808	43.182
756	Novak Djokovic	Grass	2009–2012	81.818	True	False	1999	7.326	11.224
757	Novak Djokovic	Grass	2009–2012	100.000	True	False	2000	7.740	12.680
758	Novak Djokovic	Grass	2009–2012	100.000	True	False	2000	0.214	2.058
759	Novak Djokovic	Grass	2009–2012	100.000	True	False	2000	1.185	2.040
760	Novak Djokovic	Grass	2009–2012	100.000	True	False	2000	7.982	11.870
761	Novak Djokovic	Grass	2009–2012	100.000	True	False	2000	64.878	72.187
762	Novak Djokovic	Grass	2009–2012	100.000	True	False	2000	18.111	21.697
763	Novak Djokovic	Grass	2009–2012	100.000	True	False	2000	8.104	14.852
764	Novak Djokovic	Grass	2009–2012	100.000	True	False	2000	36.740	44.876
765	Novak Djokovic	Grass	2009–2012	100.000	True	False	2000	12.603	26.207
766	Novak Djokovic	Grass	2009–2012	100.000	True	False	2000	34.040	45.939
767	Novak Djokovic	Grass	2009–2012	100.000	True	False	2000	35.534	44.351
768	Novak Djokovic	Grass	2009–2012	100.000	True	False	2000	18.403	26.645
769	Novak Djokovic	Grass	2009–2012	100.000	True	False	2000	30.979	43.455
770	Novak Djokovic	Hard	2009–2012	95.652	False	False	2000	7.475	9.956
771	Novak Djokovic	Hard	2009–2012	97.101	False	False	2000	10.110	12.710
772	Novak Djokovic	Hard	2009–2012	98.551	False	False	2000	0.533	1.212
773	Novak Djokovic	Hard	2009–2012	100.000	False	False	2000	0.926	1.255
774	Novak Djokovic	Hard	2009–2012	100.000	False	False	2000	6.918	8.820
775	Novak Djokovic	Hard	2009–2012	100.000	False	False	2000	55.674	59.389
776	Novak Djokovic	Hard	2009–2012	100.000	False	False	2000	23.109	25.019
777	Novak Djokovic	Hard	2009–2012	100.000	False	False	2000	16.821	20.028
778	Novak Djokovic	Hard	2009–2012	100.000	False	False	2000	39.177	45.179
779	Novak Djokovic	Hard	2009–2012	100.000	False	False	2000	22.838	28.525
780	Novak Djokovic	Hard	2009–2012	100.000	False	False	2000	29.141	35.124
781	Novak Djokovic	Hard	2009–2012	100.000	False	False	2000	36.129	42.154
782	Novak Djokovic	Hard	2009–2012	100.000	False	False	2000	25.459	30.962
783	Novak Djokovic	Hard	2009–2012	100.000	False	False	2000	29.694	36.103

	player	surface	period	coverage_pct	small_sample	coverage_edge_period	valid_bootstraps	ci95_lower	ci95_upper
784	Novak Djokovic	Overall	2013–2016	95.455	False	False	2000	7.164	8.466
785	Novak Djokovic	Overall	2013–2016	99.242	False	False	2000	9.924	11.385
786	Novak Djokovic	Overall	2013–2016	100.000	False	False	2000	0.800	1.361
787	Novak Djokovic	Overall	2013–2016	100.000	False	False	2000	1.073	1.400
788	Novak Djokovic	Overall	2013–2016	100.000	False	False	2000	7.496	8.882
789	Novak Djokovic	Overall	2013–2016	100.000	False	False	2000	59.225	61.887
790	Novak Djokovic	Overall	2013–2016	100.000	False	False	2000	23.133	24.489
791	Novak Djokovic	Overall	2013–2016	100.000	False	False	2000	14.589	16.761
792	Novak Djokovic	Overall	2013–2016	100.000	False	False	2000	42.228	45.459
793	Novak Djokovic	Overall	2013–2016	100.000	False	False	2000	17.838	21.693
794	Novak Djokovic	Overall	2013–2016	100.000	False	False	2000	34.268	38.563
795	Novak Djokovic	Overall	2013–2016	100.000	False	False	2000	42.042	45.339
796	Novak Djokovic	Overall	2013–2016	100.000	False	False	2000	14.723	17.813
797	Novak Djokovic	Overall	2013–2016	100.000	False	False	2000	38.467	41.488
798	Novak Djokovic	Clay	2013–2016	80.952	True	False	2000	5.016	8.263
799	Novak Djokovic	Clay	2013–2016	95.238	True	False	2000	7.257	10.947
800	Novak Djokovic	Clay	2013–2016	100.000	True	False	2000	0.432	1.419
801	Novak Djokovic	Clay	2013–2016	100.000	True	False	2000	1.725	2.516
802	Novak Djokovic	Clay	2013–2016	100.000	True	False	2000	4.289	7.318
803	Novak Djokovic	Clay	2013–2016	100.000	True	False	2000	53.947	59.391
804	Novak Djokovic	Clay	2013–2016	100.000	True	False	2000	24.458	27.662
805	Novak Djokovic	Clay	2013–2016	100.000	True	False	2000	15.032	19.586
806	Novak Djokovic	Clay	2013–2016	100.000	True	False	2000	42.446	50.245
807	Novak Djokovic	Clay	2013–2016	100.000	True	False	2000	15.012	22.075
808	Novak Djokovic	Clay	2013–2016	100.000	True	False	2000	30.577	40.179
809	Novak Djokovic	Clay	2013–2016	100.000	True	False	2000	37.670	46.057
810	Novak Djokovic	Clay	2013–2016	100.000	True	False	2000	15.644	23.760
811	Novak Djokovic	Clay	2013–2016	100.000	True	False	2000	34.053	43.693
812	Novak Djokovic	Grass	2013–2016	91.667	True	False	2000	7.577	11.409
813	Novak Djokovic	Grass	2013–2016	100.000	True	False	2000	9.914	13.561
814	Novak Djokovic	Grass	2013–2016	100.000	True	False	2000	0.574	2.423
815	Novak Djokovic	Grass	2013–2016	100.000	True	False	2000	0.568	1.567
816	Novak Djokovic	Grass	2013–2016	100.000	True	False	2000	8.158	14.032
817	Novak Djokovic	Grass	2013–2016	100.000	True	False	2000	61.789	68.180
818	Novak Djokovic	Grass	2013–2016	100.000	True	False	2000	21.059	24.434
819	Novak Djokovic	Grass	2013–2016	100.000	True	False	2000	10.259	14.412
820	Novak Djokovic	Grass	2013–2016	100.000	True	False	2000	34.557	46.316
821	Novak Djokovic	Grass	2013–2016	100.000	True	False	2000	16.728	24.622
822	Novak Djokovic	Grass	2013–2016	100.000	True	False	2000	30.969	47.762

	player	surface	period	coverage_pct	small_sample	coverage_edge_period	valid_bootstraps	ci95_lower	ci95_upper
823	Novak Djokovic	Grass	2013–2016	100.000	True	False	2000	38.400	49.326
824	Novak Djokovic	Grass	2013–2016	100.000	True	False	2000	11.817	19.099
825	Novak Djokovic	Grass	2013–2016	100.000	True	False	2000	37.693	43.623
826	Novak Djokovic	Hard	2013–2016	98.990	False	False	2000	7.003	8.536
827	Novak Djokovic	Hard	2013–2016	100.000	False	False	2000	9.957	11.738
828	Novak Djokovic	Hard	2013–2016	100.000	False	False	2000	0.717	1.391
829	Novak Djokovic	Hard	2013–2016	100.000	False	False	2000	0.889	1.197
830	Novak Djokovic	Hard	2013–2016	100.000	False	False	2000	7.667	9.096
831	Novak Djokovic	Hard	2013–2016	100.000	False	False	2000	58.970	62.123
832	Novak Djokovic	Hard	2013–2016	100.000	False	False	2000	22.679	24.228
833	Novak Djokovic	Hard	2013–2016	100.000	False	False	2000	14.642	17.335
834	Novak Djokovic	Hard	2013–2016	100.000	False	False	2000	42.250	45.876
835	Novak Djokovic	Hard	2013–2016	100.000	False	False	2000	17.522	22.475
836	Novak Djokovic	Hard	2013–2016	100.000	False	False	2000	33.814	38.412
837	Novak Djokovic	Hard	2013–2016	100.000	False	False	2000	42.288	46.022
838	Novak Djokovic	Hard	2013–2016	100.000	False	False	2000	13.990	17.537
839	Novak Djokovic	Hard	2013–2016	100.000	False	False	2000	38.203	41.992
840	Novak Djokovic	Overall	2017–2020	97.468	False	False	2000	7.661	9.928
841	Novak Djokovic	Overall	2017–2020	100.000	False	False	2000	10.109	12.598
842	Novak Djokovic	Overall	2017–2020	100.000	False	False	2000	0.916	1.880
843	Novak Djokovic	Overall	2017–2020	100.000	False	False	2000	1.250	1.822
844	Novak Djokovic	Overall	2017–2020	100.000	False	False	2000	8.301	10.435
845	Novak Djokovic	Overall	2017–2020	100.000	False	False	2000	57.774	60.807
846	Novak Djokovic	Overall	2017–2020	100.000	False	False	2000	22.806	24.357
847	Novak Djokovic	Overall	2017–2020	100.000	False	False	2000	15.921	18.357
848	Novak Djokovic	Overall	2017–2020	100.000	False	False	2000	45.020	49.009
849	Novak Djokovic	Overall	2017–2020	100.000	False	False	2000	14.204	18.211
850	Novak Djokovic	Overall	2017–2020	100.000	False	False	2000	34.428	39.138
851	Novak Djokovic	Overall	2017–2020	100.000	False	False	2000	40.617	45.999
852	Novak Djokovic	Overall	2017–2020	100.000	False	False	2000	13.688	18.210
853	Novak Djokovic	Overall	2017–2020	100.000	False	False	2000	38.808	42.903
854	Novak Djokovic	Clay	2017–2020	100.000	True	False	2000	5.955	10.370
855	Novak Djokovic	Clay	2017–2020	100.000	True	False	2000	8.748	12.968
856	Novak Djokovic	Clay	2017–2020	100.000	True	False	2000	0.528	2.575
857	Novak Djokovic	Clay	2017–2020	100.000	True	False	2000	1.611	2.954
858	Novak Djokovic	Clay	2017–2020	100.000	True	False	2000	4.594	6.442
859	Novak Djokovic	Clay	2017–2020	100.000	True	False	2000	53.556	58.867
860	Novak Djokovic	Clay	2017–2020	100.000	True	False	2000	23.302	26.295
861	Novak Djokovic	Clay	2017–2020	100.000	True	False	2000	16.548	21.645

	player	surface	period	coverage_pct	small_sample	coverage_edge_period	valid_bootstraps	ci95_lower	ci95_upper
862	Novak Djokovic	Clay	2017–2020	100.000	True	False	2000	43.093	51.558
863	Novak Djokovic	Clay	2017–2020	100.000	True	False	2000	11.418	18.560
864	Novak Djokovic	Clay	2017–2020	100.000	True	False	2000	33.951	42.113
865	Novak Djokovic	Clay	2017–2020	100.000	True	False	2000	48.512	58.638
866	Novak Djokovic	Clay	2017–2020	100.000	True	False	2000	7.634	15.187
867	Novak Djokovic	Clay	2017–2020	100.000	True	False	2000	31.038	38.619
868	Novak Djokovic	Grass	2017–2020	100.000	True	False	2000	7.168	13.534
869	Novak Djokovic	Grass	2017–2020	100.000	True	False	2000	9.338	16.285
870	Novak Djokovic	Grass	2017–2020	100.000	True	False	2000	0.414	3.384
871	Novak Djokovic	Grass	2017–2020	100.000	True	False	2000	0.779	1.591
872	Novak Djokovic	Grass	2017–2020	100.000	True	False	2000	10.667	17.256
873	Novak Djokovic	Grass	2017–2020	100.000	True	False	2000	59.413	66.073
874	Novak Djokovic	Grass	2017–2020	100.000	True	False	2000	19.920	24.017
875	Novak Djokovic	Grass	2017–2020	100.000	True	False	2000	13.213	17.252
876	Novak Djokovic	Grass	2017–2020	100.000	True	False	2000	42.716	50.116
877	Novak Djokovic	Grass	2017–2020	100.000	True	False	2000	13.153	20.078
878	Novak Djokovic	Grass	2017–2020	100.000	True	False	2000	33.480	41.336
879	Novak Djokovic	Grass	2017–2020	100.000	True	False	2000	34.462	40.649
880	Novak Djokovic	Grass	2017–2020	100.000	True	False	2000	10.893	21.858
881	Novak Djokovic	Grass	2017–2020	100.000	True	False	2000	41.256	51.031
882	Novak Djokovic	Hard	2017–2020	95.556	False	False	2000	7.268	10.195
883	Novak Djokovic	Hard	2017–2020	100.000	False	False	2000	9.449	12.791
884	Novak Djokovic	Hard	2017–2020	100.000	False	False	2000	0.819	1.821
885	Novak Djokovic	Hard	2017–2020	100.000	False	False	2000	0.970	1.601
886	Novak Djokovic	Hard	2017–2020	100.000	False	False	2000	8.884	11.035
887	Novak Djokovic	Hard	2017–2020	100.000	False	False	2000	57.630	61.771
888	Novak Djokovic	Hard	2017–2020	100.000	False	False	2000	22.512	24.415
889	Novak Djokovic	Hard	2017–2020	100.000	False	False	2000	15.087	18.616
890	Novak Djokovic	Hard	2017–2020	100.000	False	False	2000	44.544	50.056
891	Novak Djokovic	Hard	2017–2020	100.000	False	False	2000	13.772	19.805
892	Novak Djokovic	Hard	2017–2020	100.000	False	False	2000	32.385	39.556
893	Novak Djokovic	Hard	2017–2020	100.000	False	False	2000	36.818	42.131
894	Novak Djokovic	Hard	2017–2020	100.000	False	False	2000	15.410	21.169
895	Novak Djokovic	Hard	2017–2020	100.000	False	False	2000	40.159	44.408
896	Novak Djokovic	Overall	2021–2024	99.301	False	False	2000	8.857	10.464
897	Novak Djokovic	Overall	2021–2024	99.301	False	False	2000	12.257	14.095
898	Novak Djokovic	Overall	2021–2024	99.301	False	False	2000	4.169	5.977
899	Novak Djokovic	Overall	2021–2024	99.301	False	False	2000	1.678	2.131
900	Novak Djokovic	Overall	2021–2024	99.301	False	False	2000	8.785	10.635

	player	surface	period	coverage_pct	small_sample	coverage_edge_period	valid_bootstraps	ci95_lower	ci95_upper
901	Novak Djokovic	Overall	2021–2024	99.301	False	False	2000	60.265	62.783
902	Novak Djokovic	Overall	2021–2024	99.301	False	False	2000	22.003	23.514
903	Novak Djokovic	Overall	2021–2024	99.301	False	False	2000	14.786	16.601
904	Novak Djokovic	Overall	2021–2024	99.301	False	False	2000	46.339	49.360
905	Novak Djokovic	Overall	2021–2024	99.301	False	False	2000	14.879	18.344
906	Novak Djokovic	Overall	2021–2024	99.301	False	False	2000	33.691	37.507
907	Novak Djokovic	Overall	2021–2024	99.301	False	False	2000	39.622	43.238
908	Novak Djokovic	Overall	2021–2024	99.301	False	False	2000	13.176	15.674
909	Novak Djokovic	Overall	2021–2024	99.301	False	False	2000	42.653	45.690
910	Novak Djokovic	Clay	2021–2024	100.000	False	False	2000	7.860	11.381
911	Novak Djokovic	Clay	2021–2024	100.000	False	False	2000	12.496	16.764
912	Novak Djokovic	Clay	2021–2024	100.000	False	False	2000	2.018	4.452
913	Novak Djokovic	Clay	2021–2024	100.000	False	False	2000	2.742	3.431
914	Novak Djokovic	Clay	2021–2024	100.000	False	False	2000	4.741	6.036
915	Novak Djokovic	Clay	2021–2024	100.000	False	False	2000	53.685	58.680
916	Novak Djokovic	Clay	2021–2024	100.000	False	False	2000	24.946	27.568
917	Novak Djokovic	Clay	2021–2024	100.000	False	False	2000	15.884	19.457
918	Novak Djokovic	Clay	2021–2024	100.000	False	False	2000	41.543	48.154
919	Novak Djokovic	Clay	2021–2024	100.000	False	False	2000	13.668	21.053
920	Novak Djokovic	Clay	2021–2024	100.000	False	False	2000	33.332	41.978
921	Novak Djokovic	Clay	2021–2024	100.000	False	False	2000	41.899	49.739
922	Novak Djokovic	Clay	2021–2024	100.000	False	False	2000	9.035	13.255
923	Novak Djokovic	Clay	2021–2024	100.000	False	False	2000	39.390	46.564
924	Novak Djokovic	Grass	2021–2024	100.000	True	False	2000	9.256	13.346
925	Novak Djokovic	Grass	2021–2024	100.000	True	False	2000	11.828	15.857
926	Novak Djokovic	Grass	2021–2024	100.000	True	False	2000	3.825	9.196
927	Novak Djokovic	Grass	2021–2024	100.000	True	False	2000	1.621	2.362
928	Novak Djokovic	Grass	2021–2024	100.000	True	False	2000	12.019	18.488
929	Novak Djokovic	Grass	2021–2024	100.000	True	False	2000	63.423	70.071
930	Novak Djokovic	Grass	2021–2024	100.000	True	False	2000	19.345	23.713
931	Novak Djokovic	Grass	2021–2024	100.000	True	False	2000	9.610	14.221
932	Novak Djokovic	Grass	2021–2024	100.000	True	False	2000	44.240	52.223
933	Novak Djokovic	Grass	2021–2024	100.000	True	False	2000	10.262	18.286
934	Novak Djokovic	Grass	2021–2024	100.000	True	False	2000	33.264	42.754
935	Novak Djokovic	Grass	2021–2024	100.000	True	False	2000	33.862	41.297
936	Novak Djokovic	Grass	2021–2024	100.000	True	False	2000	12.966	21.228
937	Novak Djokovic	Grass	2021–2024	100.000	True	False	2000	41.629	49.682
938	Novak Djokovic	Hard	2021–2024	98.876	False	False	2000	8.450	10.234
939	Novak Djokovic	Hard	2021–2024	98.876	False	False	2000	11.355	13.357

	player	surface	period	coverage_pct	small_sample	coverage_edge_period	valid_bootstraps	ci95_lower	ci95_upper
940	Novak Djokovic	Hard	2021–2024	98.876	False	False	2000	4.464	6.984
941	Novak Djokovic	Hard	2021–2024	98.876	False	False	2000	1.039	1.444
942	Novak Djokovic	Hard	2021–2024	98.876	False	False	2000	10.111	12.098
943	Novak Djokovic	Hard	2021–2024	98.876	False	False	2000	61.920	64.404
944	Novak Djokovic	Hard	2021–2024	98.876	False	False	2000	20.590	22.038
945	Novak Djokovic	Hard	2021–2024	98.876	False	False	2000	14.398	16.644
946	Novak Djokovic	Hard	2021–2024	98.876	False	False	2000	47.409	51.057
947	Novak Djokovic	Hard	2021–2024	98.876	False	False	2000	14.749	18.889
948	Novak Djokovic	Hard	2021–2024	98.876	False	False	2000	31.799	36.185
949	Novak Djokovic	Hard	2021–2024	98.876	False	False	2000	38.016	41.967
950	Novak Djokovic	Hard	2021–2024	98.876	False	False	2000	14.162	16.988
951	Novak Djokovic	Hard	2021–2024	98.876	False	False	2000	42.902	46.122
952	Novak Djokovic	Overall	2025–2026	97.674	False	True	2000	8.260	11.017
953	Novak Djokovic	Overall	2025–2026	100.000	False	True	2000	12.199	14.915
954	Novak Djokovic	Overall	2025–2026	100.000	False	True	2000	3.187	6.205
955	Novak Djokovic	Overall	2025–2026	100.000	False	True	2000	1.724	2.540
956	Novak Djokovic	Overall	2025–2026	100.000	False	True	2000	8.461	11.113
957	Novak Djokovic	Overall	2025–2026	100.000	False	True	2000	60.112	64.161
958	Novak Djokovic	Overall	2025–2026	100.000	False	True	2000	21.830	24.094
959	Novak Djokovic	Overall	2025–2026	100.000	False	True	2000	13.517	16.192
960	Novak Djokovic	Overall	2025–2026	100.000	False	True	2000	47.307	51.485
961	Novak Djokovic	Overall	2025–2026	100.000	False	True	2000	11.483	16.345
962	Novak Djokovic	Overall	2025–2026	100.000	False	True	2000	33.714	39.854
963	Novak Djokovic	Overall	2025–2026	100.000	False	True	2000	38.142	43.191
964	Novak Djokovic	Overall	2025–2026	100.000	False	True	2000	11.262	16.107
965	Novak Djokovic	Overall	2025–2026	100.000	False	True	2000	43.799	47.727
966	Novak Djokovic	Clay	2025–2026	100.000	True	True	2000	6.745	12.769
967	Novak Djokovic	Clay	2025–2026	100.000	True	True	2000	13.345	17.691
968	Novak Djokovic	Clay	2025–2026	100.000	True	True	2000	1.149	6.685
969	Novak Djokovic	Clay	2025–2026	100.000	True	True	2000	2.684	4.548
970	Novak Djokovic	Clay	2025–2026	100.000	True	True	2000	5.738	9.341
971	Novak Djokovic	Clay	2025–2026	100.000	True	True	2000	54.751	61.680
972	Novak Djokovic	Clay	2025–2026	100.000	True	True	2000	21.955	27.511
973	Novak Djokovic	Clay	2025–2026	100.000	True	True	2000	14.276	20.743
974	Novak Djokovic	Clay	2025–2026	100.000	True	True	2000	44.027	54.630
975	Novak Djokovic	Clay	2025–2026	100.000	True	True	2000	7.518	20.110
976	Novak Djokovic	Clay	2025–2026	100.000	True	True	2000	27.974	47.318
977	Novak Djokovic	Clay	2025–2026	100.000	True	True	2000	35.693	48.611
978	Novak Djokovic	Clay	2025–2026	100.000	True	True	2000	5.223	19.876

	player	surface	period	coverage_pct	small_sample	coverage_edge_period	valid_bootstraps	ci95_lower	ci95_upper
979	Novak Djokovic	Clay	2025–2026	100.000	True	True	2000	41.066	50.778
980	Novak Djokovic	Grass	2025–2026	100.000	True	True	1989	9.856	17.650
981	Novak Djokovic	Grass	2025–2026	100.000	True	True	1989	11.632	20.031
982	Novak Djokovic	Grass	2025–2026	100.000	True	True	1989	6.645	20.408
983	Novak Djokovic	Grass	2025–2026	100.000	True	True	1989	1.412	2.907
984	Novak Djokovic	Grass	2025–2026	100.000	True	True	1989	11.460	17.391
985	Novak Djokovic	Grass	2025–2026	100.000	True	True	1989	54.348	74.651
986	Novak Djokovic	Grass	2025–2026	100.000	True	True	1989	17.442	27.390
987	Novak Djokovic	Grass	2025–2026	100.000	True	True	1989	7.470	18.634
988	Novak Djokovic	Grass	2025–2026	100.000	True	True	1989	45.856	54.369
989	Novak Djokovic	Grass	2025–2026	100.000	True	True	1989	4.959	13.386
990	Novak Djokovic	Grass	2025–2026	100.000	True	True	1989	34.951	47.489
991	Novak Djokovic	Grass	2025–2026	100.000	True	True	1989	33.333	39.768
992	Novak Djokovic	Grass	2025–2026	100.000	True	True	1989	9.877	17.808
993	Novak Djokovic	Grass	2025–2026	100.000	True	True	1989	45.763	53.846
994	Novak Djokovic	Hard	2025–2026	96.667	True	True	2000	7.180	10.464
995	Novak Djokovic	Hard	2025–2026	100.000	False	True	2000	10.726	14.134
996	Novak Djokovic	Hard	2025–2026	100.000	False	True	2000	2.224	4.603
997	Novak Djokovic	Hard	2025–2026	100.000	False	True	2000	1.325	1.973
998	Novak Djokovic	Hard	2025–2026	100.000	False	True	2000	8.045	11.300
999	Novak Djokovic	Hard	2025–2026	100.000	False	True	2000	60.815	64.150
1000	Novak Djokovic	Hard	2025–2026	100.000	False	True	2000	21.460	23.745
1001	Novak Djokovic	Hard	2025–2026	100.000	False	True	2000	13.665	16.096
1002	Novak Djokovic	Hard	2025–2026	100.000	False	True	2000	46.753	52.057
1003	Novak Djokovic	Hard	2025–2026	100.000	False	True	2000	11.579	17.524
1004	Novak Djokovic	Hard	2025–2026	100.000	False	True	2000	32.543	39.623
1005	Novak Djokovic	Hard	2025–2026	100.000	False	True	2000	37.788	44.497
1006	Novak Djokovic	Hard	2025–2026	100.000	False	True	2000	11.026	17.046
1007	Novak Djokovic	Hard	2025–2026	100.000	False	True	2000	42.421	47.338

Equal-surface standardized trends; no estimate without all three surfaces:

▼ Complete evidence table: 252 rows × 10 columns

player	period_index	period	metric	standardized_pct	surfaces_available	small_sample	ci95_lower	ci95_upper	valid_bootstraps
Roger Federer	0	1998–2001	approach_frequency_pct	21.832	3	True	18.141	25.861	1975
Roger Federer	0	1998–2001	net_point_frequency_pct	22.389	3	True	17.967	26.483	1983
Roger Federer	0	1998–2001	serve_volley_frequency_pct	25.974	3	True	17.768	33.559	1983
Roger Federer	0	1998–2001	drop_shot_share_pct	0.358	3	True	0.212	0.533	1983
Roger Federer	0	1998–2001	slice_share_pct	16.325	3	True	14.145	18.768	1983
Roger Federer	0	1998–2001	short_rally_share_pct	72.616	3	True	69.032	77.364	1983
Roger Federer	0	1998–2001	medium_rally_share_pct	19.644	3	True	16.677	22.348	1983
Roger Federer	0	1998–2001	long_rally_share_pct	7.740	3	True	5.603	9.143	1983
Roger Federer	0	1998–2001	deuce_wide_serve_share_pct	35.782	3	True	31.117	41.510	1983
Roger Federer	0	1998–2001	deuce_body_serve_share_pct	19.118	3	True	14.600	22.813	1983
Roger Federer	0	1998–2001	deuce_t_serve_share_pct	45.099	3	True	41.128	48.755	1983
Roger Federer	0	1998–2001	ad_wide_serve_share_pct	58.656	3	True	53.459	62.366	1983
Roger Federer	0	1998–2001	ad_body_serve_share_pct	12.972	3	True	10.340	16.838	1983
Roger Federer	0	1998–2001	ad_t_serve_share_pct	28.372	3	True	25.631	31.259	1983
Roger Federer	1	2002–2005	approach_frequency_pct	18.193	3	True	16.669	19.787	2000
Roger Federer	1	2002–2005	net_point_frequency_pct	19.940	3	True	18.428	21.427	2000
Roger Federer	1	2002–2005	serve_volley_frequency_pct	18.810	3	True	15.562	22.428	2000

player	period_index	period	metric	standardized_pct	surfaces_available	small_sample	ci95_lower	ci95_upper	valid_bootstraps
Roger Federer	1	2002–2005	drop_shot_share_pct	0.698	3	True	0.577	0.834	2000
Roger Federer	1	2002–2005	slice_share_pct	19.015	3	True	17.936	20.185	2000
Roger Federer	1	2002–2005	short_rally_share_pct	69.408	3	True	67.788	71.009	2000
Roger Federer	1	2002–2005	medium_rally_share_pct	21.390	3	True	20.419	22.321	2000
Roger Federer	1	2002–2005	long_rally_share_pct	9.202	3	True	8.363	10.090	2000
Roger Federer	1	2002–2005	deuce_wide_serve_share_pct	38.813	3	True	37.261	40.442	2000
Roger Federer	1	2002–2005	deuce_body_serve_share_pct	18.326	3	True	16.009	20.635	2000
Roger Federer	1	2002–2005	deuce_t_serve_share_pct	42.862	3	True	40.466	45.310	2000
Roger Federer	1	2002–2005	ad_wide_serve_share_pct	56.475	3	True	54.358	58.772	2000
Roger Federer	1	2002–2005	ad_body_serve_share_pct	13.149	3	True	11.455	14.726	2000
Roger Federer	1	2002–2005	ad_t_serve_share_pct	30.376	3	True	28.639	32.157	2000
Roger Federer	2	2006–2009	approach_frequency_pct	13.801	3	True	12.479	14.950	2000
Roger Federer	2	2006–2009	net_point_frequency_pct	15.058	3	True	13.700	16.301	2000
Roger Federer	2	2006–2009	serve_volley_frequency_pct	5.040	3	True	4.273	5.814	2000
Roger Federer	2	2006–2009	drop_shot_share_pct	0.880	3	True	0.762	0.987	2000
Roger Federer	2	2006–2009	slice_share_pct	19.243	3	True	17.976	20.690	2000
Roger Federer	2	2006–2009	short_rally_share_pct	66.186	3	True	65.030	67.468	2000
Roger Federer	2	2006–2009	medium_rally_share_pct	22.645	3	True	21.786	23.402	2000

player	period_index	period	metric	standardized_pct	surfaces_available	small_sample	ci95_lower	ci95_upper	valid_bootstraps
Roger Federer	2	2006–2009	long_rally_share_pct	11.169	3	True	10.431	11.818	2000
Roger Federer	2	2006–2009	deuce_wide_serve_share_pct	39.444	3	True	36.990	41.685	2000
Roger Federer	2	2006–2009	deuce_body_serve_share_pct	16.739	3	True	14.939	18.659	2000
Roger Federer	2	2006–2009	deuce_t_serve_share_pct	43.817	3	True	41.409	46.261	2000
Roger Federer	2	2006–2009	ad_wide_serve_share_pct	51.250	3	True	49.485	53.024	2000
Roger Federer	2	2006–2009	ad_body_serve_share_pct	13.637	3	True	11.802	15.591	2000
Roger Federer	2	2006–2009	ad_t_serve_share_pct	35.113	3	True	33.402	36.852	2000
Roger Federer	3	2010–2013	approach_frequency_pct	12.547	3	True	11.549	13.571	2000
Roger Federer	3	2010–2013	net_point_frequency_pct	14.185	3	True	13.148	15.301	2000
Roger Federer	3	2010–2013	serve_volley_frequency_pct	4.391	3	True	3.637	5.153	2000
Roger Federer	3	2010–2013	drop_shot_share_pct	1.402	3	True	1.231	1.566	2000
Roger Federer	3	2010–2013	slice_share_pct	18.331	3	True	17.305	19.404	2000
Roger Federer	3	2010–2013	short_rally_share_pct	66.385	3	True	65.007	67.784	2000
Roger Federer	3	2010–2013	medium_rally_share_pct	22.990	3	True	22.156	23.908	2000
Roger Federer	3	2010–2013	long_rally_share_pct	10.625	3	True	9.750	11.460	2000
Roger Federer	3	2010–2013	deuce_wide_serve_share_pct	43.630	3	True	41.347	45.905	2000
Roger Federer	3	2010–2013	deuce_body_serve_share_pct	17.844	3	True	16.153	19.624	2000
Roger Federer	3	2010–2013	deuce_t_serve_share_pct	38.526	3	True	36.540	40.501	2000

player	period_index	period	metric	standardized_pct	surfaces_available	small_sample	ci95_lower	ci95_upper	valid_bootstraps
Roger Federer	3	2010–2013	ad_wide_serve_share_pct	50.056	3	True	47.738	52.335	2000
Roger Federer	3	2010–2013	ad_body_serve_share_pct	15.562	3	True	13.734	17.589	2000
Roger Federer	3	2010–2013	ad_t_serve_share_pct	34.382	3	True	32.104	36.575	2000
Roger Federer	4	2014–2017	approach_frequency_pct	14.017	3	True	12.868	15.095	2000
Roger Federer	4	2014–2017	net_point_frequency_pct	16.113	3	True	15.019	17.174	2000
Roger Federer	4	2014–2017	serve_volley_frequency_pct	10.599	3	True	9.356	11.896	2000
Roger Federer	4	2014–2017	drop_shot_share_pct	1.131	3	True	0.941	1.354	2000
Roger Federer	4	2014–2017	slice_share_pct	19.568	3	True	18.608	20.489	2000
Roger Federer	4	2014–2017	short_rally_share_pct	70.884	3	True	69.604	72.308	2000
Roger Federer	4	2014–2017	medium_rally_share_pct	21.252	3	True	20.356	22.154	2000
Roger Federer	4	2014–2017	long_rally_share_pct	7.864	3	True	7.067	8.666	2000
Roger Federer	4	2014–2017	deuce_wide_serve_share_pct	42.147	3	True	40.400	43.835	2000
Roger Federer	4	2014–2017	deuce_body_serve_share_pct	17.563	3	True	15.507	19.562	2000
Roger Federer	4	2014–2017	deuce_t_serve_share_pct	40.290	3	True	38.137	42.572	2000
Roger Federer	4	2014–2017	ad_wide_serve_share_pct	53.326	3	True	51.377	55.067	2000
Roger Federer	4	2014–2017	ad_body_serve_share_pct	11.976	3	True	10.711	13.336	2000
Roger Federer	4	2014–2017	ad_t_serve_share_pct	34.699	3	True	33.191	36.287	2000
Roger Federer	5	2018–2021	approach_frequency_pct	16.086	3	True	14.439	17.414	2000

player	period_index	period	metric	standardized_pct	surfaces_available	small_sample	ci95_lower	ci95_upper	valid_bootstraps
Roger Federer	5	2018–2021	net_point_frequency_pct	18.137	3	True	16.222	19.629	2000
Roger Federer	5	2018–2021	serve_volley_frequency_pct	12.507	3	True	10.441	14.422	2000
Roger Federer	5	2018–2021	drop_shot_share_pct	1.733	3	True	1.440	1.981	2000
Roger Federer	5	2018–2021	slice_share_pct	19.797	3	True	18.534	21.191	2000
Roger Federer	5	2018–2021	short_rally_share_pct	71.188	3	True	69.060	73.251	2000
Roger Federer	5	2018–2021	medium_rally_share_pct	20.489	3	True	19.231	21.833	2000
Roger Federer	5	2018–2021	long_rally_share_pct	8.324	3	True	7.339	9.496	2000
Roger Federer	5	2018–2021	deuce_wide_serve_share_pct	43.861	3	True	41.481	46.376	2000
Roger Federer	5	2018–2021	deuce_body_serve_share_pct	13.173	3	True	11.049	15.312	2000
Roger Federer	5	2018–2021	deuce_t_serve_share_pct	42.966	3	True	40.279	45.672	2000
Roger Federer	5	2018–2021	ad_wide_serve_share_pct	50.209	3	True	46.724	53.604	2000
Roger Federer	5	2018–2021	ad_body_serve_share_pct	11.775	3	True	9.760	14.596	2000
Roger Federer	5	2018–2021	ad_t_serve_share_pct	38.016	3	True	34.828	41.031	2000
Rafael Nadal	0	2003–2006	approach_frequency_pct	8.578	3	True	7.125	10.175	1999
Rafael Nadal	0	2003–2006	net_point_frequency_pct	10.911	3	True	9.246	12.778	1999
Rafael Nadal	0	2003–2006	serve_volley_frequency_pct	1.488	3	True	1.001	1.972	1999
Rafael Nadal	0	2003–2006	drop_shot_share_pct	1.168	3	True	0.909	1.466	1999
Rafael Nadal	0	2003–2006	slice_share_pct	8.941	3	True	7.453	10.440	1999

player	period_index	period	metric	standardized_pct	surfaces_available	small_sample	ci95_lower	ci95_upper	valid_bootstraps
Rafael Nadal	0	2003–2006	short_rally_share_pct	57.118	3	True	54.399	59.635	1999
Rafael Nadal	0	2003–2006	medium_rally_share_pct	25.985	3	True	24.365	27.779	1999
Rafael Nadal	0	2003–2006	long_rally_share_pct	16.897	3	True	15.047	18.999	1999
Rafael Nadal	0	2003–2006	deuce_wide_serve_share_pct	20.100	3	True	14.997	25.605	1999
Rafael Nadal	0	2003–2006	deuce_body_serve_share_pct	21.662	3	True	17.140	26.858	1999
Rafael Nadal	0	2003–2006	deuce_t_serve_share_pct	58.238	3	True	49.679	66.165	1999
Rafael Nadal	0	2003–2006	ad_wide_serve_share_pct	62.852	3	True	56.694	68.701	1999
Rafael Nadal	0	2003–2006	ad_body_serve_share_pct	20.161	3	True	16.101	24.364	1999
Rafael Nadal	0	2003–2006	ad_t_serve_share_pct	16.987	3	True	12.789	21.334	1999
Rafael Nadal	1	2007–2010	approach_frequency_pct	6.581	3	True	5.881	7.382	2000
Rafael Nadal	1	2007–2010	net_point_frequency_pct	8.779	3	True	8.041	9.694	2000
Rafael Nadal	1	2007–2010	serve_volley_frequency_pct	1.002	3	True	0.708	1.390	2000
Rafael Nadal	1	2007–2010	drop_shot_share_pct	0.801	3	True	0.649	0.962	2000
Rafael Nadal	1	2007–2010	slice_share_pct	11.532	3	True	10.348	13.007	2000
Rafael Nadal	1	2007–2010	short_rally_share_pct	59.770	3	True	58.323	61.291	2000
Rafael Nadal	1	2007–2010	medium_rally_share_pct	24.564	3	True	23.701	25.375	2000
Rafael Nadal	1	2007–2010	long_rally_share_pct	15.666	3	True	14.557	16.812	2000
Rafael Nadal	1	2007–2010	deuce_wide_serve_share_pct	18.629	3	True	16.189	21.042	2000

player	period_index	period	metric	standardized_pct	surfaces_available	small_sample	ci95_lower	ci95_upper	valid_bootstraps
Rafael Nadal	1	2007–2010	deuce_body_serve_share_pct	28.174	3	True	24.780	32.547	2000
Rafael Nadal	1	2007–2010	deuce_t_serve_share_pct	53.197	3	True	48.247	57.448	2000
Rafael Nadal	1	2007–2010	ad_wide_serve_share_pct	54.307	3	True	48.079	59.597	2000
Rafael Nadal	1	2007–2010	ad_body_serve_share_pct	23.987	3	True	20.142	28.349	2000
Rafael Nadal	1	2007–2010	ad_t_serve_share_pct	21.706	3	True	18.756	25.361	2000
Rafael Nadal	2	2011–2014	approach_frequency_pct	5.285	3	True	4.408	6.222	2000
Rafael Nadal	2	2011–2014	net_point_frequency_pct	7.802	3	True	6.849	8.807	2000
Rafael Nadal	2	2011–2014	serve_volley_frequency_pct	0.769	3	True	0.429	1.100	2000
Rafael Nadal	2	2011–2014	drop_shot_share_pct	0.611	3	True	0.450	0.800	2000
Rafael Nadal	2	2011–2014	slice_share_pct	12.565	3	True	11.174	13.836	2000
Rafael Nadal	2	2011–2014	short_rally_share_pct	61.575	3	True	59.364	63.712	2000
Rafael Nadal	2	2011–2014	medium_rally_share_pct	23.774	3	True	22.577	24.927	2000
Rafael Nadal	2	2011–2014	long_rally_share_pct	14.650	3	True	13.346	15.952	2000
Rafael Nadal	2	2011–2014	deuce_wide_serve_share_pct	19.171	3	True	16.812	21.569	2000
Rafael Nadal	2	2011–2014	deuce_body_serve_share_pct	36.084	3	True	33.374	38.816	2000
Rafael Nadal	2	2011–2014	deuce_t_serve_share_pct	44.745	3	True	41.306	48.395	2000
Rafael Nadal	2	2011–2014	ad_wide_serve_share_pct	50.558	3	True	46.658	54.149	2000
Rafael Nadal	2	2011–2014	ad_body_serve_share_pct	28.851	3	True	25.943	32.054	2000

player	period_index	period	metric	standardized_pct	surfaces_available	small_sample	ci95_lower	ci95_upper	valid_bootstraps
Rafael Nadal	2	2011–2014	ad_t_serve_share_pct	20.591	3	True	17.484	23.725	2000
Rafael Nadal	3	2015–2018	approach_frequency_pct	8.579	3	True	6.356	9.769	1969
Rafael Nadal	3	2015–2018	net_point_frequency_pct	10.791	3	True	9.834	11.466	1969
Rafael Nadal	3	2015–2018	serve_volley_frequency_pct	1.124	3	True	0.518	2.160	1969
Rafael Nadal	3	2015–2018	drop_shot_share_pct	1.208	3	True	1.073	1.366	1969
Rafael Nadal	3	2015–2018	slice_share_pct	9.679	3	True	8.630	10.417	1969
Rafael Nadal	3	2015–2018	short_rally_share_pct	62.774	3	True	59.258	66.809	1969
Rafael Nadal	3	2015–2018	medium_rally_share_pct	23.649	3	True	22.259	24.819	1969
Rafael Nadal	3	2015–2018	long_rally_share_pct	13.577	3	True	10.908	16.037	1969
Rafael Nadal	3	2015–2018	deuce_wide_serve_share_pct	30.576	3	True	26.381	33.731	1969
Rafael Nadal	3	2015–2018	deuce_body_serve_share_pct	23.193	3	True	21.180	25.763	1969
Rafael Nadal	3	2015–2018	deuce_t_serve_share_pct	46.232	3	True	43.789	48.637	1969
Rafael Nadal	3	2015–2018	ad_wide_serve_share_pct	47.998	3	True	43.908	52.072	1969
Rafael Nadal	3	2015–2018	ad_body_serve_share_pct	29.145	3	True	27.043	31.321	1969
Rafael Nadal	3	2015–2018	ad_t_serve_share_pct	22.857	3	True	18.711	25.925	1969
Rafael Nadal	4	2019–2022	approach_frequency_pct	6.630	3	True	5.663	7.818	1994
Rafael Nadal	4	2019–2022	net_point_frequency_pct	10.052	3	True	8.776	11.378	1994
Rafael Nadal	4	2019–2022	serve_volley_frequency_pct	2.474	3	True	1.728	3.614	1994

player	period_index	period	metric	standardized_pct	surfaces_available	small_sample	ci95_lower	ci95_upper	valid_bootstraps
Rafael Nadal	4	2019–2022	drop_shot_share_pct	1.103	3	True	0.860	1.313	1994
Rafael Nadal	4	2019–2022	slice_share_pct	10.244	3	True	8.981	11.496	1994
Rafael Nadal	4	2019–2022	short_rally_share_pct	62.947	3	True	61.244	64.591	1994
Rafael Nadal	4	2019–2022	medium_rally_share_pct	23.683	3	True	22.496	24.842	1994
Rafael Nadal	4	2019–2022	long_rally_share_pct	13.370	3	True	12.327	14.400	1994
Rafael Nadal	4	2019–2022	deuce_wide_serve_share_pct	30.089	3	True	25.938	34.293	1994
Rafael Nadal	4	2019–2022	deuce_body_serve_share_pct	20.293	3	True	18.118	22.583	1994
Rafael Nadal	4	2019–2022	deuce_t_serve_share_pct	49.618	3	True	43.901	54.844	1994
Rafael Nadal	4	2019–2022	ad_wide_serve_share_pct	46.796	3	True	42.817	50.651	1994
Rafael Nadal	4	2019–2022	ad_body_serve_share_pct	25.628	3	True	23.142	28.158	1994
Rafael Nadal	4	2019–2022	ad_t_serve_share_pct	27.576	3	True	24.886	30.600	1994
Rafael Nadal	5	2023–2024	approach_frequency_pct	Unavailable	2	True	Unavailable	Unavailable	0
Rafael Nadal	5	2023–2024	net_point_frequency_pct	Unavailable	2	True	Unavailable	Unavailable	0
Rafael Nadal	5	2023–2024	serve_volley_frequency_pct	Unavailable	2	True	Unavailable	Unavailable	0
Rafael Nadal	5	2023–2024	drop_shot_share_pct	Unavailable	2	True	Unavailable	Unavailable	0
Rafael Nadal	5	2023–2024	slice_share_pct	Unavailable	2	True	Unavailable	Unavailable	0
Rafael Nadal	5	2023–2024	short_rally_share_pct	Unavailable	2	True	Unavailable	Unavailable	0
Rafael Nadal	5	2023–2024	medium_rally_share_pct	Unavailable	2	True	Unavailable	Unavailable	0

player	period_index	period	metric	standardized_pct	surfaces_available	small_sample	ci95_lower	ci95_upper	valid_bootstraps
Rafael Nadal	5	2023–2024	long_rally_share_pct	Unavailable	2	True	Unavailable	Unavailable	0
Rafael Nadal	5	2023–2024	deuce_wide_serve_share_pct	Unavailable	2	True	Unavailable	Unavailable	0
Rafael Nadal	5	2023–2024	deuce_body_serve_share_pct	Unavailable	2	True	Unavailable	Unavailable	0
Rafael Nadal	5	2023–2024	deuce_t_serve_share_pct	Unavailable	2	True	Unavailable	Unavailable	0
Rafael Nadal	5	2023–2024	ad_wide_serve_share_pct	Unavailable	2	True	Unavailable	Unavailable	0
Rafael Nadal	5	2023–2024	ad_body_serve_share_pct	Unavailable	2	True	Unavailable	Unavailable	0
Rafael Nadal	5	2023–2024	ad_t_serve_share_pct	Unavailable	2	True	Unavailable	Unavailable	0
Novak Djokovic	0	2005–2008	approach_frequency_pct	12.244	3	True	10.390	14.612	1971
Novak Djokovic	0	2005–2008	net_point_frequency_pct	14.257	3	True	12.238	16.651	1971
Novak Djokovic	0	2005–2008	serve_volley_frequency_pct	1.910	3	True	1.237	2.734	1971
Novak Djokovic	0	2005–2008	drop_shot_share_pct	1.781	3	True	1.422	2.102	1971
Novak Djokovic	0	2005–2008	slice_share_pct	5.924	3	True	4.958	7.635	1971
Novak Djokovic	0	2005–2008	short_rally_share_pct	60.966	3	True	58.466	64.456	1971
Novak Djokovic	0	2005–2008	medium_rally_share_pct	23.865	3	True	21.807	25.556	1971
Novak Djokovic	0	2005–2008	long_rally_share_pct	15.169	3	True	13.001	16.925	1971
Novak Djokovic	0	2005–2008	deuce_wide_serve_share_pct	47.517	3	True	43.428	53.319	1971
Novak Djokovic	0	2005–2008	deuce_body_serve_share_pct	22.192	3	True	18.064	27.443	1971
Novak Djokovic	0	2005–2008	deuce_t_serve_share_pct	30.291	3	True	24.143	34.447	1971

player	period_index	period	metric	standardized_pct	surfaces_available	small_sample	ci95_lower	ci95_upper	valid_bootstraps
Novak Djokovic	0	2005–2008	ad_wide_serve_share_pct	46.621	3	True	35.989	53.839	1971
Novak Djokovic	0	2005–2008	ad_body_serve_share_pct	16.285	3	True	12.891	29.354	1971
Novak Djokovic	0	2005–2008	ad_t_serve_share_pct	37.094	3	True	31.656	42.676	1971
Novak Djokovic	1	2009–2012	approach_frequency_pct	8.269	3	True	7.366	9.190	1999
Novak Djokovic	1	2009–2012	net_point_frequency_pct	10.490	3	True	9.369	11.566	2000
Novak Djokovic	1	2009–2012	serve_volley_frequency_pct	0.895	3	True	0.555	1.278	2000
Novak Djokovic	1	2009–2012	drop_shot_share_pct	1.296	3	True	1.121	1.481	2000
Novak Djokovic	1	2009–2012	slice_share_pct	7.568	3	True	6.837	8.314	2000
Novak Djokovic	1	2009–2012	short_rally_share_pct	59.992	3	True	58.272	61.766	2000
Novak Djokovic	1	2009–2012	medium_rally_share_pct	22.779	3	True	21.900	23.606	2000
Novak Djokovic	1	2009–2012	long_rally_share_pct	17.230	3	True	15.654	18.769	2000
Novak Djokovic	1	2009–2012	deuce_wide_serve_share_pct	43.414	3	True	41.059	46.065	2000
Novak Djokovic	1	2009–2012	deuce_body_serve_share_pct	21.263	3	True	18.324	24.026	2000
Novak Djokovic	1	2009–2012	deuce_t_serve_share_pct	35.322	3	True	32.385	38.282	2000
Novak Djokovic	1	2009–2012	ad_wide_serve_share_pct	40.760	3	True	38.054	43.383	2000
Novak Djokovic	1	2009–2012	ad_body_serve_share_pct	23.349	3	True	20.981	25.708	2000
Novak Djokovic	1	2009–2012	ad_t_serve_share_pct	35.891	3	True	33.253	39.008	2000
Novak Djokovic	2	2013–2016	approach_frequency_pct	7.917	3	True	7.055	8.798	2000

player	period_index	period	metric	standardized_pct	surfaces_available	small_sample	ci95_lower	ci95_upper	valid_bootstraps
Novak Djokovic	2	2013–2016	net_point_frequency_pct	10.526	3	True	9.666	11.415	2000
Novak Djokovic	2	2013–2016	serve_volley_frequency_pct	1.139	3	True	0.769	1.513	2000
Novak Djokovic	2	2013–2016	drop_shot_share_pct	1.394	3	True	1.185	1.623	2000
Novak Djokovic	2	2013–2016	slice_share_pct	8.274	3	True	7.274	9.466	2000
Novak Djokovic	2	2013–2016	short_rally_share_pct	60.749	3	True	59.286	62.219	2000
Novak Djokovic	2	2013–2016	medium_rally_share_pct	24.099	3	True	23.294	24.950	2000
Novak Djokovic	2	2013–2016	long_rally_share_pct	15.152	3	True	14.058	16.276	2000
Novak Djokovic	2	2013–2016	deuce_wide_serve_share_pct	43.651	3	True	41.062	45.940	2000
Novak Djokovic	2	2013–2016	deuce_body_serve_share_pct	19.464	3	True	17.673	21.592	2000
Novak Djokovic	2	2013–2016	deuce_t_serve_share_pct	36.885	3	True	33.658	40.213	2000
Novak Djokovic	2	2013–2016	ad_wide_serve_share_pct	43.407	3	True	40.992	45.656	2000
Novak Djokovic	2	2013–2016	ad_body_serve_share_pct	16.765	3	True	14.962	18.833	2000
Novak Djokovic	2	2013–2016	ad_t_serve_share_pct	39.828	3	True	37.865	41.890	2000
Novak Djokovic	3	2017–2020	approach_frequency_pct	8.907	3	True	7.605	10.290	2000
Novak Djokovic	3	2017–2020	net_point_frequency_pct	11.491	3	True	10.056	12.989	2000
Novak Djokovic	3	2017–2020	serve_volley_frequency_pct	1.383	3	True	0.843	2.129	2000
Novak Djokovic	3	2017–2020	drop_shot_share_pct	1.542	3	True	1.281	1.846	2000
Novak Djokovic	3	2017–2020	slice_share_pct	9.699	3	True	8.603	11.033	2000

player	period_index	period	metric	standardized_pct	surfaces_available	small_sample	ci95_lower	ci95_upper	valid_bootstraps
Novak Djokovic	3	2017–2020	short_rally_share_pct	59.503	3	True	58.016	61.146	2000
Novak Djokovic	3	2017–2020	medium_rally_share_pct	23.478	3	True	22.567	24.329	2000
Novak Djokovic	3	2017–2020	long_rally_share_pct	17.019	3	True	15.773	18.248	2000
Novak Djokovic	3	2017–2020	deuce_wide_serve_share_pct	46.887	3	True	44.860	48.947	2000
Novak Djokovic	3	2017–2020	deuce_body_serve_share_pct	15.976	3	True	14.043	17.995	2000
Novak Djokovic	3	2017–2020	deuce_t_serve_share_pct	37.137	3	True	34.858	39.324	2000
Novak Djokovic	3	2017–2020	ad_wide_serve_share_pct	43.621	3	True	41.350	45.824	2000
Novak Djokovic	3	2017–2020	ad_body_serve_share_pct	15.317	3	True	13.026	17.781	2000
Novak Djokovic	3	2017–2020	ad_t_serve_share_pct	41.062	3	True	38.920	43.241	2000
Novak Djokovic	4	2021–2024	approach_frequency_pct	10.017	3	True	9.111	10.977	2000
Novak Djokovic	4	2021–2024	net_point_frequency_pct	13.541	3	True	12.531	14.623	2000
Novak Djokovic	4	2021–2024	serve_volley_frequency_pct	5.090	3	True	4.040	6.174	2000
Novak Djokovic	4	2021–2024	drop_shot_share_pct	2.099	3	True	1.916	2.278	2000
Novak Djokovic	4	2021–2024	slice_share_pct	10.551	3	True	9.414	11.680	2000
Novak Djokovic	4	2021–2024	short_rally_share_pct	61.978	3	True	60.578	63.500	2000
Novak Djokovic	4	2021–2024	medium_rally_share_pct	22.993	3	True	22.126	23.914	2000
Novak Djokovic	4	2021–2024	long_rally_share_pct	15.028	3	True	13.988	16.026	2000
Novak Djokovic	4	2021–2024	deuce_wide_serve_share_pct	47.308	3	True	45.452	49.264	2000

player	period_index	period	metric	standardized_pct	surfaces_available	small_sample	ci95_lower	ci95_upper	valid_bootstraps
Novak Djokovic	4	2021–2024	deuce_body_serve_share_pct	16.068	3	True	14.182	18.189	2000
Novak Djokovic	4	2021–2024	deuce_t_serve_share_pct	36.624	3	True	34.363	38.827	2000
Novak Djokovic	4	2021–2024	ad_wide_serve_share_pct	41.096	3	True	39.116	43.099	2000
Novak Djokovic	4	2021–2024	ad_body_serve_share_pct	14.609	3	True	12.972	16.190	2000
Novak Djokovic	4	2021–2024	ad_t_serve_share_pct	44.295	3	True	42.458	46.370	2000
Novak Djokovic	5	2025–2026	approach_frequency_pct	10.477	3	True	8.920	12.373	1989
Novak Djokovic	5	2025–2026	net_point_frequency_pct	14.552	3	True	13.006	16.334	1989
Novak Djokovic	5	2025–2026	serve_volley_frequency_pct	6.303	3	True	4.243	9.242	1989
Novak Djokovic	5	2025–2026	drop_shot_share_pct	2.397	3	True	2.019	2.871	1989
Novak Djokovic	5	2025–2026	slice_share_pct	10.239	3	True	9.163	11.637	1989
Novak Djokovic	5	2025–2026	short_rally_share_pct	62.380	3	True	58.026	65.657	1989
Novak Djokovic	5	2025–2026	medium_rally_share_pct	23.089	3	True	21.257	25.269	1989
Novak Djokovic	5	2025–2026	long_rally_share_pct	14.531	3	True	12.616	17.312	1989
Novak Djokovic	5	2025–2026	deuce_wide_serve_share_pct	49.411	3	True	47.111	51.935	1989
Novak Djokovic	5	2025–2026	deuce_body_serve_share_pct	13.069	3	True	10.134	15.608	1989
Novak Djokovic	5	2025–2026	deuce_t_serve_share_pct	37.520	3	True	33.802	41.813	1989
Novak Djokovic	5	2025–2026	ad_wide_serve_share_pct	39.989	3	True	37.275	42.480	1989
Novak Djokovic	5	2025–2026	ad_body_serve_share_pct	13.224	3	True	10.530	16.313	1989

player	period_index	period	metric	standardized_pct	surfaces_available	small_sample	ci95_lower	ci95_upper	valid_bootstraps
Novak Djokovic	5	2025–2026	ad_t_serve_share_pct	46.787	3	True	44.689	49.080	1989

First/last supported standardized-period contrasts, in percentage points (not performance gains):

▼ Complete evidence table: 42 rows × 9 columns

player	metric	first_supported_period	last_supported_period	change_percentage_points	ci95_lower	ci95_upper	valid_bootstraps	status
Roger Federer	approach_frequency_pct	1998–2001	2018–2021	-5.746	-10.071	-1.972	1975	small endpoint sample
Roger Federer	net_point_frequency_pct	1998–2001	2018–2021	-4.251	-8.687	0.380	1983	small endpoint sample
Roger Federer	serve_volley_frequency_pct	1998–2001	2018–2021	-13.468	-21.308	-5.072	1983	small endpoint sample
Roger Federer	drop_shot_share_pct	1998–2001	2018–2021	1.374	1.045	1.663	1983	small endpoint sample
Roger Federer	slice_share_pct	1998–2001	2018–2021	3.471	0.659	6.027	1983	small endpoint sample
Roger Federer	short_rally_share_pct	1998–2001	2018–2021	-1.428	-6.566	2.823	1983	small endpoint sample
Roger Federer	medium_rally_share_pct	1998–2001	2018–2021	0.845	-2.117	4.021	1983	small endpoint sample
Roger Federer	long_rally_share_pct	1998–2001	2018–2021	0.583	-1.241	2.912	1983	small endpoint sample
Roger Federer	deuce_wide_serve_share_pct	1998–2001	2018–2021	8.079	2.013	13.257	1983	small endpoint sample
Roger Federer	deuce_body_serve_share_pct	1998–2001	2018–2021	-5.945	-10.308	-0.943	1983	small endpoint sample
Roger Federer	deuce_t_serve_share_pct	1998–2001	2018–2021	-2.133	-6.782	2.589	1983	small endpoint sample
Roger Federer	ad_wide_serve_share_pct	1998–2001	2018–2021	-8.447	-13.710	-2.393	1983	small endpoint sample
Roger Federer	ad_body_serve_share_pct	1998–2001	2018–2021	-1.197	-5.530	2.511	1983	small endpoint sample
Roger Federer	ad_t_serve_share_pct	1998–2001	2018–2021	9.644	5.287	13.639	1983	small endpoint sample
Rafael Nadal	approach_frequency_pct	2003–2006	2019–2022	-1.948	-3.717	-0.116	1993	small endpoint sample
Rafael Nadal	net_point_frequency_pct	2003–2006	2019–2022	-0.859	-3.013	1.302	1993	small endpoint sample
Rafael Nadal	serve_volley_frequency_pct	2003–2006	2019–2022	0.986	0.107	2.221	1993	small endpoint sample

player	metric	first_supported_period	last_supported_period	change_percentage_points	ci95_lower	ci95_upper	valid_bootstraps	status
Rafael Nadal	drop_shot_share_pct	2003–2006	2019–2022	-0.065	-0.443	0.282	1993	small endpoint sample
Rafael Nadal	slice_share_pct	2003–2006	2019–2022	1.303	-0.608	3.317	1993	small endpoint sample
Rafael Nadal	short_rally_share_pct	2003–2006	2019–2022	5.828	2.895	8.987	1993	small endpoint sample
Rafael Nadal	medium_rally_share_pct	2003–2006	2019–2022	-2.302	-4.534	-0.272	1993	small endpoint sample
Rafael Nadal	long_rally_share_pct	2003–2006	2019–2022	-3.526	-5.871	-1.426	1993	small endpoint sample
Rafael Nadal	deuce_wide_serve_share_pct	2003–2006	2019–2022	9.989	3.236	16.639	1993	small endpoint sample
Rafael Nadal	deuce_body_serve_share_pct	2003–2006	2019–2022	-1.369	-7.070	3.566	1993	small endpoint sample
Rafael Nadal	deuce_t_serve_share_pct	2003–2006	2019–2022	-8.620	-18.281	1.372	1993	small endpoint sample
Rafael Nadal	ad_wide_serve_share_pct	2003–2006	2019–2022	-16.056	-23.016	-8.631	1993	small endpoint sample
Rafael Nadal	ad_body_serve_share_pct	2003–2006	2019–2022	5.467	0.596	10.382	1993	small endpoint sample
Rafael Nadal	ad_t_serve_share_pct	2003–2006	2019–2022	10.588	5.444	15.750	1993	small endpoint sample
Novak Djokovic	approach_frequency_pct	2005–2008	2025–2026	-1.767	-4.576	0.852	1961	small endpoint sample
Novak Djokovic	net_point_frequency_pct	2005–2008	2025–2026	0.295	-2.701	2.947	1961	small endpoint sample
Novak Djokovic	serve_volley_frequency_pct	2005–2008	2025–2026	4.393	2.094	7.408	1961	small endpoint sample
Novak Djokovic	drop_shot_share_pct	2005–2008	2025–2026	0.617	0.121	1.189	1961	small endpoint sample
Novak Djokovic	slice_share_pct	2005–2008	2025–2026	4.315	2.453	6.121	1961	small endpoint sample
Novak Djokovic	short_rally_share_pct	2005–2008	2025–2026	1.414	-4.370	5.824	1961	small endpoint sample
Novak Djokovic	medium_rally_share_pct	2005–2008	2025–2026	-0.776	-3.326	2.105	1961	small endpoint sample

player	metric	first_supported_period	last_supported_period	change_percentage_points	ci95_lower	ci95_upper	valid_bootstraps	status
Novak Djokovic	long_rally_share_pct	2005–2008	2025–2026	-0.638	-3.413	2.845	1961	small endpoint sample
Novak Djokovic	deuce_wide_serve_share_pct	2005–2008	2025–2026	1.894	-4.600	6.772	1961	small endpoint sample
Novak Djokovic	deuce_body_serve_share_pct	2005–2008	2025–2026	-9.123	-14.925	-4.133	1961	small endpoint sample
Novak Djokovic	deuce_t_serve_share_pct	2005–2008	2025–2026	7.229	1.548	14.675	1961	small endpoint sample
Novak Djokovic	ad_wide_serve_share_pct	2005–2008	2025–2026	-6.631	-14.319	4.353	1961	small endpoint sample
Novak Djokovic	ad_body_serve_share_pct	2005–2008	2025–2026	-3.062	-15.462	1.613	1961	small endpoint sample
Novak Djokovic	ad_t_serve_share_pct	2005–2008	2025–2026	9.693	3.858	15.788	1961	small endpoint sample

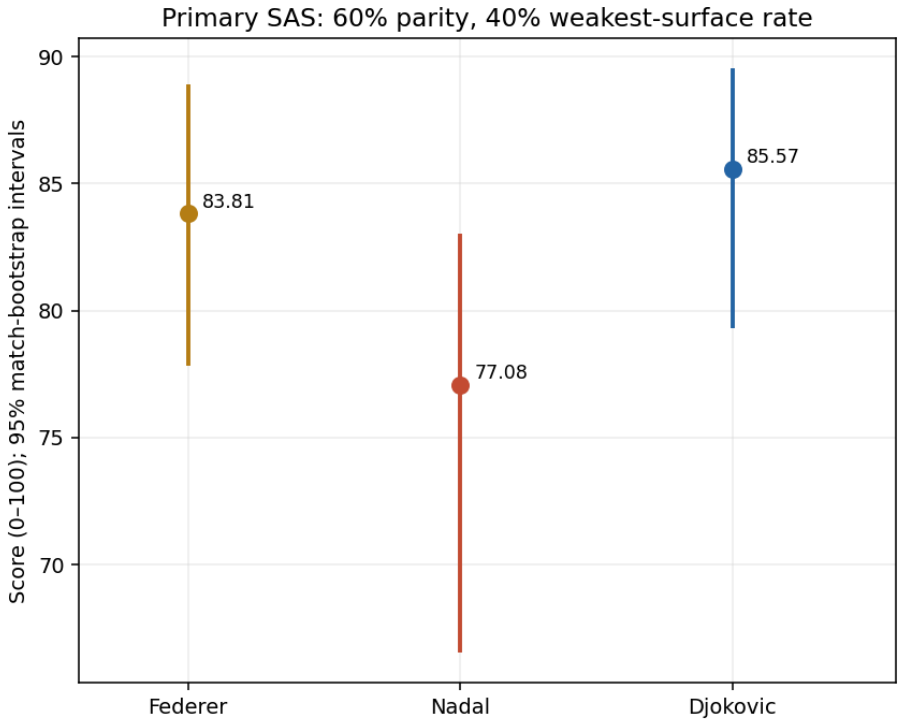
8.3 Adaptability and evolution charts

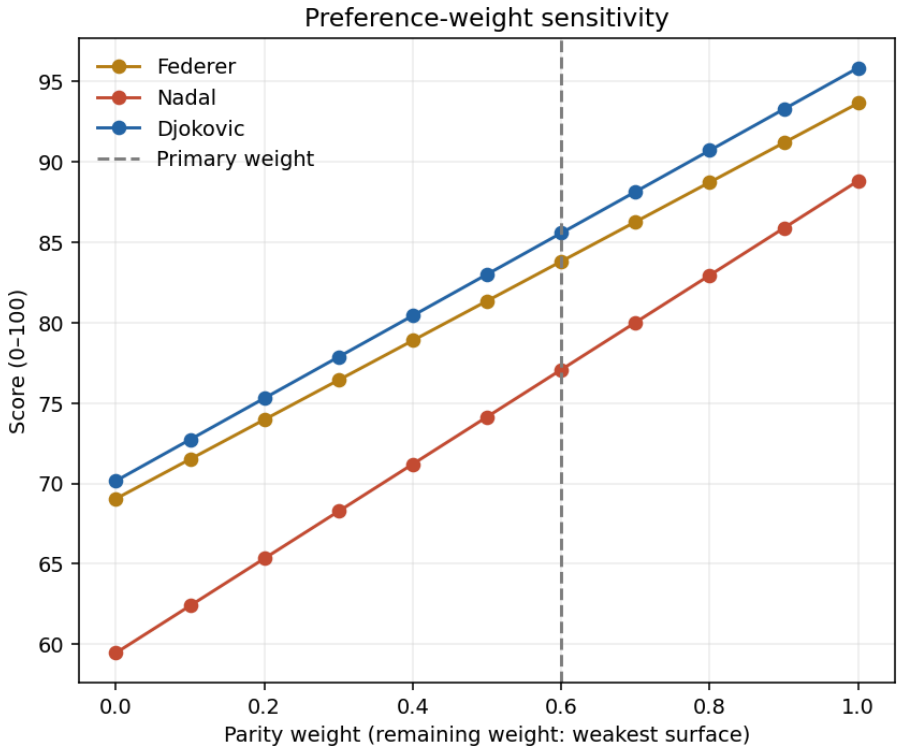
SAS intervals reflect uncertainty in the recorded outcomes; weight curves show definition sensitivity rather than probabilities of being greatest. Tactical solid lines are equal-surface standardized estimates and dashed lines are raw pooled estimates. Both use their own eligible samples; separation can reflect surface composition and source coverage. Within-surface panels and the evidence tables remain essential. Small-period samples are retained and flagged; no line is interpolated over a missing standardized estimate.

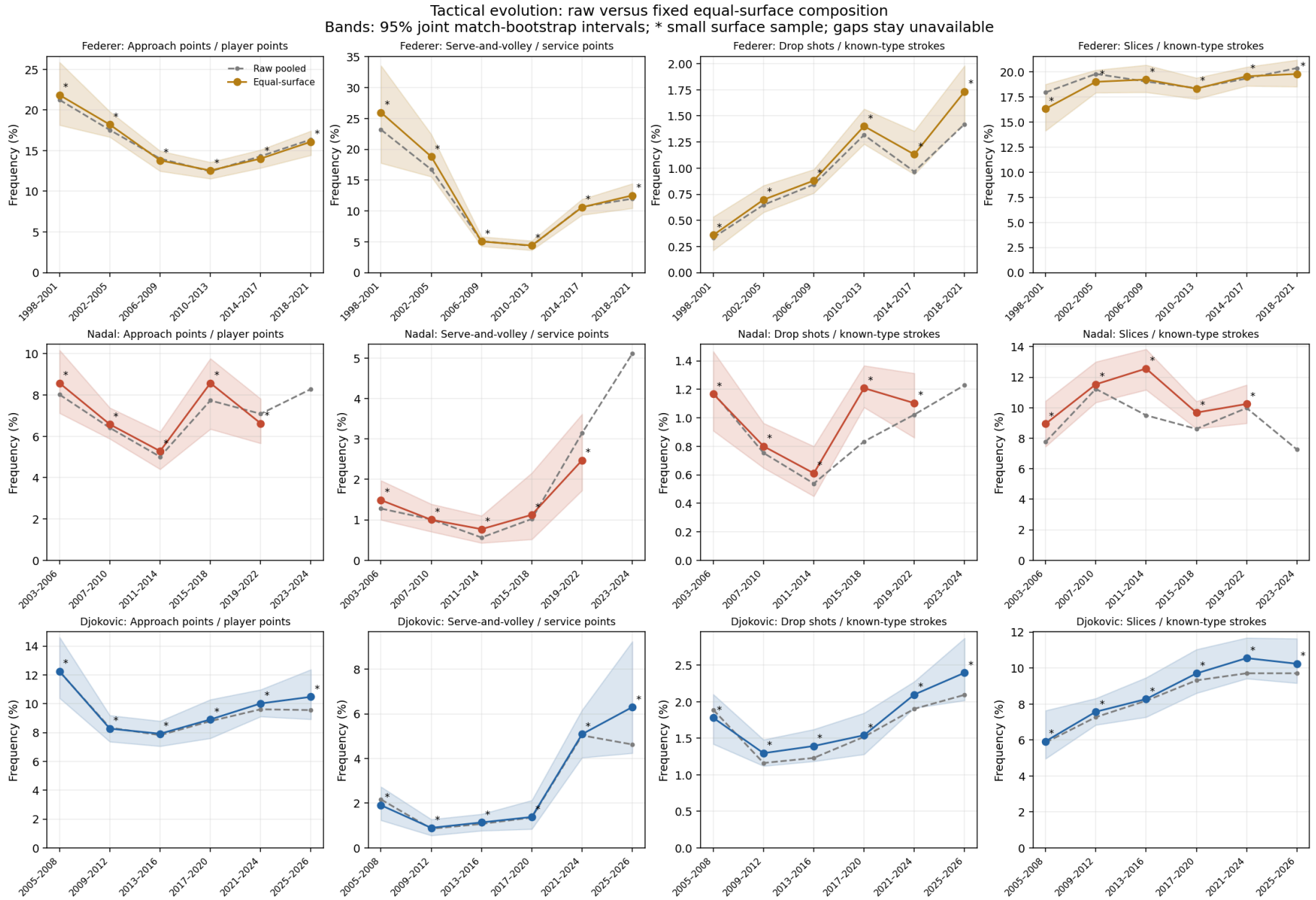
In [25]:
```
with plt.style.context("default"):
    fig, axes = plt.subplots(1, 2, figsize=(12, 5), constrained_layout=True)
    for index, player in enumerate(BIG_THREE):
        row = surface_adaptability_summary_v2.loc[surface_adaptability_summary_v2["player"].eq(player)].iloc[0]
        axes[0].scatter(index, row["surface_adaptability_score"], color=KPI_PLAYER_COLORS[player], s=60)
        axes[0].vlines(index, row["surface_adaptability_score_ci95_lower"], row["surface_adaptability_score_ci95_upper"], color=KPI_PLAYER_COLORS[player],
linewidth=2)
        axes[0].annotate(f"{row['surface_adaptability_score']:.2f}", (index, row["surface_adaptability_score"]), xytext=(7, 3), textcoords="offset points",
fontsize=9)
        curve = surface_adaptability_weight_sensitivity.loc[surface_adaptability_weight_sensitivity["player"].eq(player)]
        axes[1].plot(curve["parity_weight"], curve["score"], marker="o", color=KPI_PLAYER_COLORS[player], label=KPI_SHORT_NAMES[player])
    axes[0].set_xticks(range(3), [KPI_SHORT_NAMES[p] for p in BIG_THREE])
    axes[0].set_title("Primary SAS: 60% parity, 40% weakest-surface rate")
    axes[0].set_ylabel("Score (0–100); 95% match-bootstrap intervals")
    axes[0].set_xlim(-.4, 2.6)
    axes[1].axvline(.6, color="gray", linestyle="--", label="Primary weight")
    axes[1].set_xlabel("Parity weight (remaining weight: weakest surface)")
    axes[1].set_ylabel("Score (0–100)")
```

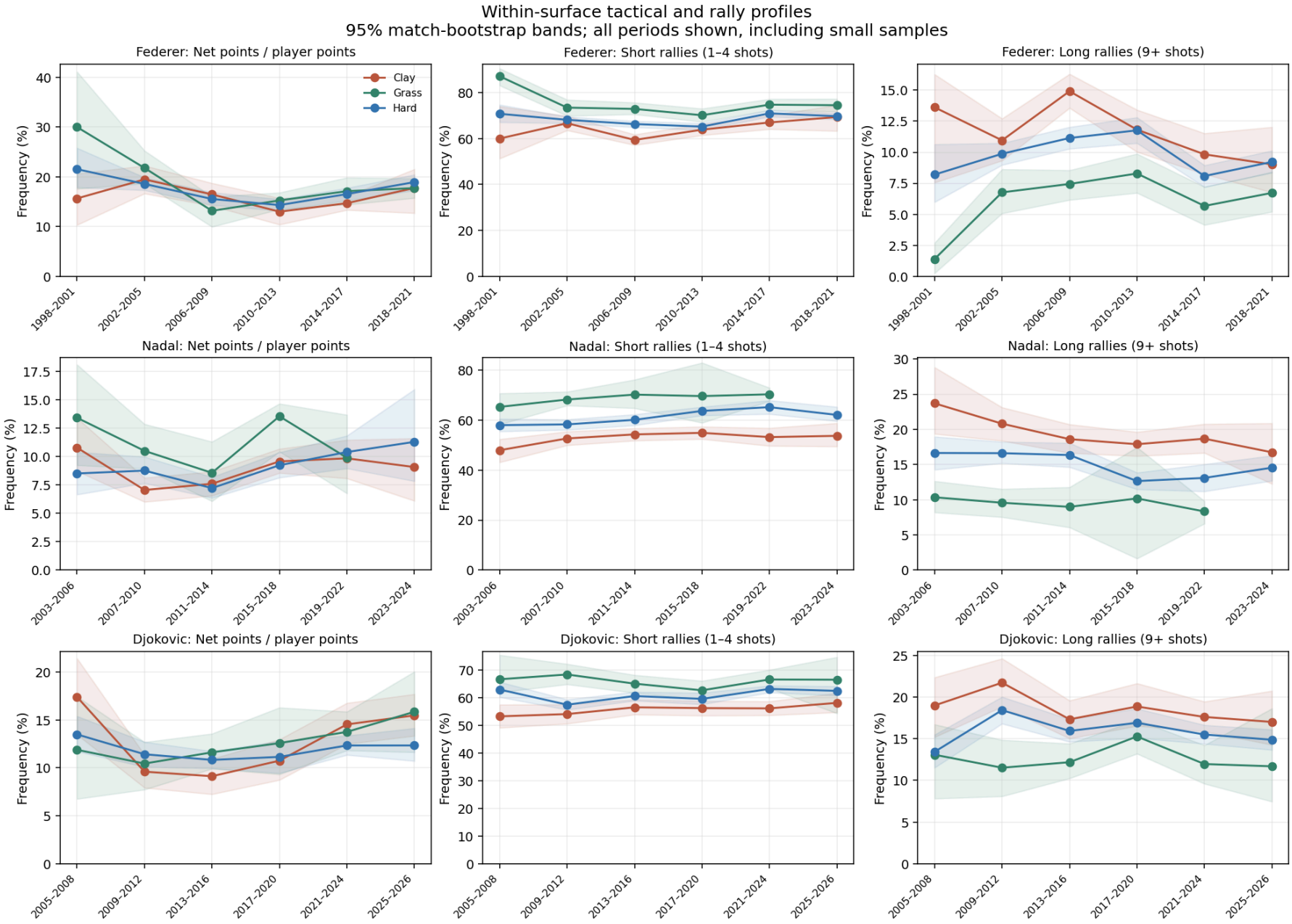
```
    axes[1].set_title("Preference-weight sensitivity")
    axes[1].legend(frameon=False)
    for ax in axes:
        ax.grid(alpha=.2)
    display_kpi_figure(fig)

    metrics = ["approach_frequency_pct", "serve_volley_frequency_pct", "drop_shot_share_pct", "slice_share_pct"]
    titles = ["Approach points / player points", "Serve-and-volley / service points", "Drop shots / known-type strokes", "Slices / known-type strokes"]
    fig, axes = plt.subplots(3, 4, figsize=(16, 11), constrained_layout=True)
    for row_index, player in enumerate(BIG_THREE):
        plan = tactical_period_plan.loc[tactical_period_plan["player"].eq(player)]
        x = plan["period_index"].to_numpy()
        for ax, metric, title in zip(axes[row_index], metrics, titles):
            raw = tactical_evolution_long.loc[tactical_evolution_long["player"].eq(player) & tactical_evolution_long["surface"].eq("Overall") &
tactical_evolution_long["metric"].eq(metric)].set_index("period_index").reindex(x)
            adjusted = tactical_standardized_trends.loc[tactical_standardized_trends["player"].eq(player) &
tactical_standardized_trends["metric"].eq(metric)].set_index("period_index").reindex(x)
            ax.plot(x, raw["estimate_pct"], linestyle="--", marker=".", color="gray", label="Raw pooled")
            ax.plot(x, adjusted["standardized_pct"], marker="o", color=KPI_PLAYER_COLORS[player], label="Equal-surface")
            ax.fill_between(x, adjusted["ci95_lower"].to_numpy(dtype=float), adjusted["ci95_upper"].to_numpy(dtype=float), color=KPI_PLAYER_COLORS[player],
alpha=.16)
            for xpos, (_, point) in zip(x, adjusted.iterrows()):
                if point["small_sample"] and pd.notna(point["standardized_pct"]):
                    ax.annotate("*", (xpos, point["standardized_pct"]), xytext=(3, 4), textcoords="offset points", fontsize=9)
            ax.set_xticks(x, plan["period"], rotation=45, ha="right", fontsize=8)
            ax.set_title(KPI_SHORT_NAMES[player]+": "+title, fontsize=9)
            ax.set_ylabel("Frequency (%)")
            ax.set_ylim(bottom=0)
            ax.grid(alpha=.2)
    axes[0, 0].legend(frameon=False, fontsize=8)
    fig.suptitle("Tactical evolution: raw versus fixed equal-surface composition\nBands: 95% joint match-bootstrap intervals; * small surface sample; gaps stay
unavailable", fontsize=13)
    display_kpi_figure(fig)

    fig, axes = plt.subplots(3, 3, figsize=(14, 10), constrained_layout=True)
    for row_index, player in enumerate(BIG_THREE):
        plan = tactical_period_plan.loc[tactical_period_plan["player"].eq(player)]
        x = plan["period_index"].to_numpy()
        for ax, metric, title in zip(axes[row_index], ["net_point_frequency_pct", "short_rally_share_pct", "long_rally_share_pct"],
            ["Net points / player points", "Short rallies (1–4 shots)", "Long rallies (9+ shots)"]):
            for surface, color in [("Clay", "#BA573C"), ("Grass", "#32836B"), ("Hard", "#3575B0")]:
                data = tactical_evolution_long.loc[tactical_evolution_long["player"].eq(player) & tactical_evolution_long["surface"].eq(surface) &
tactical_evolution_long["metric"].eq(metric)].set_index("period_index").reindex(x)
                ax.plot(x, data["estimate_pct"], marker="o", color=color, label=surface)
                ax.fill_between(x, data["ci95_lower"].to_numpy(dtype=float), data["ci95_upper"].to_numpy(dtype=float), color=color, alpha=.10)
            ax.set_xticks(x, plan["period"], rotation=45, ha="right", fontsize=8)
            ax.set_title(KPI_SHORT_NAMES[player]+": "+title, fontsize=10)
            ax.set_ylabel("Frequency (%)")
            ax.set_ylim(bottom=0)
            ax.grid(alpha=.2)
    axes[0, 0].legend(frameon=False, fontsize=8)
    fig.suptitle("Within-surface tactical and rally profiles\n95% match-bootstrap bands; all periods shown, including small samples", fontsize=13)
    display_kpi_figure(fig)

    fig, axes = plt.subplots(1, 3, figsize=(14, 5), constrained_layout=True)
    for ax, player in zip(axes, BIG_THREE):
        plan = tactical_period_plan.loc[tactical_period_plan["player"].eq(player)]
```

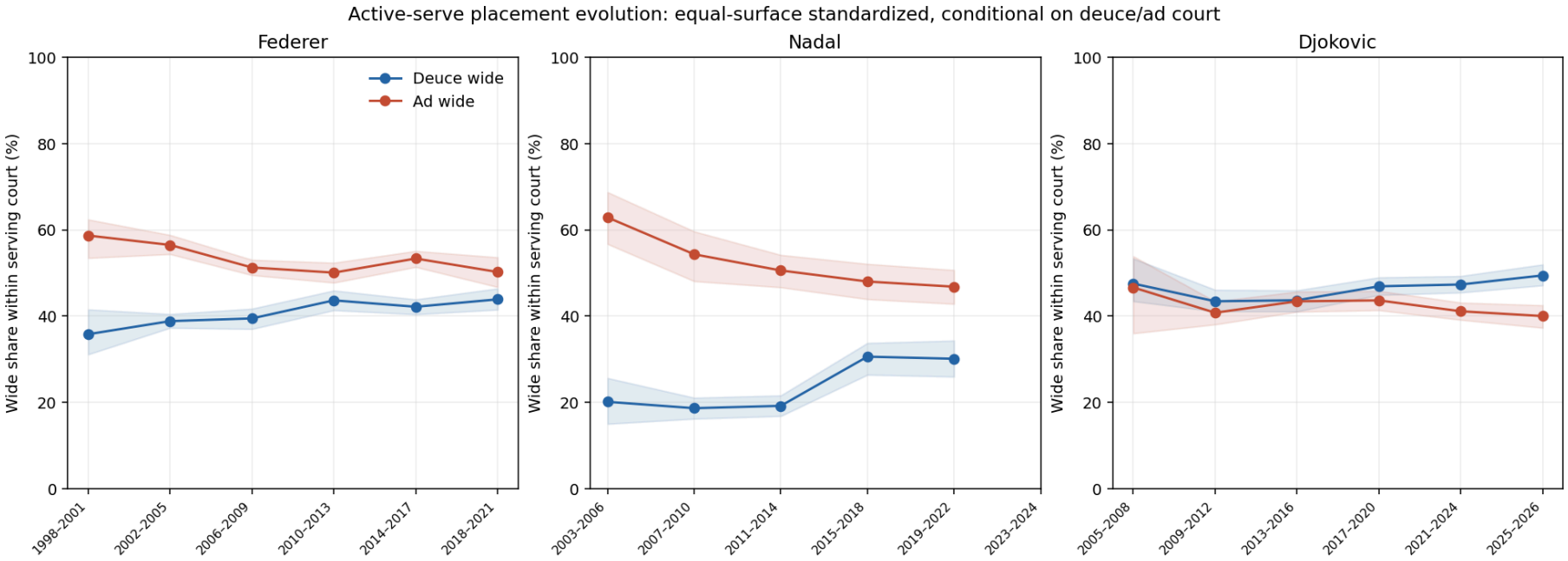
```
        for metric, label, color in [("deuce_wide_serve_share_pct", "Deuce wide", "#2766A5"), ("ad_wide_serve_share_pct", "Ad wide", "#C44E36")]:
            data = tactical_standardized_trends.loc[tactical_standardized_trends["player"].eq(player) &
tactical_standardized_trends["metric"].eq(metric)].set_index("period_index").reindex(plan["period_index"])
            ax.plot(plan["period_index"], data["standardized_pct"], marker="o", color=color, label=label)
            ax.fill_between(plan["period_index"].to_numpy(), data["ci95_lower"].to_numpy(dtype=float), data["ci95_upper"].to_numpy(dtype=float), color=color,
alpha=.12)
        ax.set_xticks(plan["period_index"], plan["period"], rotation=45, ha="right", fontsize=8)
        ax.set_title(KPI_SHORT_NAMES[player])
        ax.set_ylabel("Wide share within serving court (%)")
        ax.set_ylim(0, 100)
        ax.grid(alpha=.2)
    axes[0].legend(frameon=False)
    fig.suptitle("Active-serve placement evolution: equal-surface standardized, conditional on deuce/ad court", fontsize=12)
    display_kpi_figure(fig)
```






Tactical evolution: raw versus fixed equal-surface composition
Bands: 95% joint match-bootstrap intervals; * small surface sample; gaps stay unavailable
Federer: Approach points / player points
Federer: Serve-and-volley / service points
Federer: Drop shots / known-type strokes
Federer: Slices / known-type strokes
Raw pooled
Equal-surface
Nadal: Approach points / player points
Nadal: Serve-and-volley / service points
Nadal: Drop shots / known-type strokes
Nadal: Slices / known-type strokes
Djokovic: Approach points / player points
Djokovic: Serve-and-volley / service points
Djokovic: Drop shots / known-type strokes
Djokovic: Slices / known-type strokes
Frequency (%)
1998–2001
2002–2005
2006–2009
2010–2013
2014–2017
2018–2021
2003–2006
2007–2010
2011–2014
2015–2018
2019–2022
2023–2024
2005–2008
2009–2012
2013–2016
2017–2020
2021–2024
2025–2026


Within-surface tactical and rally profiles
95% match-bootstrap bands; all periods shown, including small samples
Federer: Net points / player points
Federer: Short rallies (1–4 shots)
Federer: Long rallies (9+ shots)
Nadal: Net points / player points
Nadal: Short rallies (1–4 shots)
Nadal: Long rallies (9+ shots)
Djokovic: Net points / player points
Djokovic: Short rallies (1–4 shots)
Djokovic: Long rallies (9+ shots)
Clay
Grass
Hard
Frequency (%)
1998–2001
2002–2005
2006–2009
2010–2013
2014–2017
2018–2021
2003–2006
2007–2010
2011–2014
2015–2018
2019–2022
2023–2024
2005–2008
2009–2012
2013–2016
2017–2020
2021–2024
2025–2026



8.4 Interpretation and limitations

SAS describes the declared blend of three-surface consistency and weakest-surface success in verified charted outcomes; it cannot establish causal adjustment skill when a player changes court surface. Tactical evolution reports component changes with coverage and surface context, not an innovation ranking. Source completeness, opponent mix, ageing, match selection, and sparse grass observations remain alternative explanations. The standardized series controls surface weights only, not all confounding.

The endpoint contrasts identify the actual periods being compared, including when sparse surface coverage prevents using a career-edge period. The raw, within-surface, and standardized series let readers see which changes survive a fixed surface composition. We can now carry these measured tendencies into the rivalries without turning change into an innovation ranking.

8.5 Evidence: cross-surface balance and supported tactical changes

In [26]:
```
sas_headline=surface_adaptability_summary_v2[["player","parity_score","weakest_surface","weakest_surface_win_pct",
    "surface_adaptability_score","surface_adaptability_score_ci95_lower","surface_adaptability_score_ci95_upper","known_outcome_matches"]]
report_table(sas_headline,caption="SAS: the declared balance/weakest-surface trade-off",expanded=True)
selected_changes=tactical_change_summary.loc[tactical_change_summary["metric"].isin(
    ["approach_frequency_pct","serve_volley_frequency_pct","drop_shot_share_pct","slice_share_pct","long_rally_share_pct"])]
```

```
report_table(selected_changes,caption="Selected supported endpoint contrasts: percentage-point changes, exploratory intervals",expanded=True)
lines=["**Observed result.**"]
for player in BIG_THREE:
    row=surface_adaptability_summary_v2.loc[surface_adaptability_summary_v2["player"].eq(player)].iloc[0]
    lines.append(f"- {player}: SAS {row.surface_adaptability_score:.2f}
({interval_text(row.surface_adaptability_score_ci95_lower,row.surface_adaptability_score_ci95_upper)}); weakest observed surface {row.weakest_surface},
{row.weakest_surface_win_pct:.2f}% known-outcome wins.")
for player,metric in [(BIG_THREE[0],"serve_volley_frequency_pct"),(BIG_THREE[1],"long_rally_share_pct"),(BIG_THREE[2],"slice_share_pct")]:
    row=tactical_change_summary.loc[tactical_change_summary["player"].eq(player)&tactical_change_summary["metric"].eq(metric)].iloc[0]
    if pd.notna(row.get("change_percentage_points")):
        lines.append(f"- {player}, {metric.replace('_',' ')}: {row.change_percentage_points:+.2f} percentage points from {row.first_supported_period} to
{row.last_supported_period} (95% interval {interval_text(row.ci95_lower,row.ci95_upper)}; {row.status}).")
lines.append("\n**Tennis interpretation.** The SAS weakest-surface term makes specialization costly under this definition, even when specialization brings
exceptional results on one surface. A change in net use, slice share, or rally length may reflect point-construction choices and changing opponents. It does not
demonstrate invention, improved tactics, or athletic decline.")
lines.append("\n**Alternative explanations.** Equal surface weights remove only one source of composition change. Opponents, career stage, tournament context,
and charting practices still change over time. The first/last supported periods can differ from career endpoints, and every selected endpoint contrast has its
own interval and sample warning. These exploratory intervals are not multiplicity-adjusted. The next section combines measured profiles with direct rivalry
evidence before asking how subjective weighting changes the ranking.")
display(Markdown("\n\n".join(lines)))
```

SAS: the declared balance/weakest-surface trade-off

player	parity_score	weakest_surface	weakest_surface_win_pct	surface_adaptability_score	surface_adaptability_score_ci95_lower	surface_adaptability_score_ci95_upper	known_outcome_matches
Roger Federer	93.652	Clay	69.048	83.810	77.811	88.900	709
Rafael Nadal	88.822	Grass	59.459	77.077	66.502	83.046	412
Novak Djokovic	95.850	Clay	70.161	85.575	79.299	89.564	526

Selected supported endpoint contrasts: percentage-point changes, exploratory intervals

	player	metric	first_supported_period	last_supported_period	change_percentage_points	ci95_lower	ci95_upper	valid_bootstraps	status
0	Roger Federer	approach_frequency_pct	1998–2001	2018–2021	-5.746	-10.071	-1.972	1975	small endpoint sample
2	Roger Federer	serve_volley_frequency_pct	1998–2001	2018–2021	-13.468	-21.308	-5.072	1983	small endpoint sample
3	Roger Federer	drop_shot_share_pct	1998–2001	2018–2021	1.374	1.045	1.663	1983	small endpoint sample
4	Roger Federer	slice_share_pct	1998–2001	2018–2021	3.471	0.659	6.027	1983	small endpoint sample
7	Roger Federer	long_rally_share_pct	1998–2001	2018–2021	0.583	-1.241	2.912	1983	small endpoint sample
14	Rafael Nadal	approach_frequency_pct	2003–2006	2019–2022	-1.948	-3.717	-0.116	1993	small endpoint sample
16	Rafael Nadal	serve_volley_frequency_pct	2003–2006	2019–2022	0.986	0.107	2.221	1993	small endpoint sample
17	Rafael Nadal	drop_shot_share_pct	2003–2006	2019–2022	-0.065	-0.443	0.282	1993	small endpoint sample
18	Rafael Nadal	slice_share_pct	2003–2006	2019–2022	1.303	-0.608	3.317	1993	small endpoint sample
21	Rafael Nadal	long_rally_share_pct	2003–2006	2019–2022	-3.526	-5.871	-1.426	1993	small endpoint sample
28	Novak Djokovic	approach_frequency_pct	2005–2008	2025–2026	-1.767	-4.576	0.852	1961	small endpoint sample
30	Novak Djokovic	serve_volley_frequency_pct	2005–2008	2025–2026	4.393	2.094	7.408	1961	small endpoint sample
31	Novak Djokovic	drop_shot_share_pct	2005–2008	2025–2026	0.617	0.121	1.189	1961	small endpoint sample
32	Novak Djokovic	slice_share_pct	2005–2008	2025–2026	4.315	2.453	6.121	1961	small endpoint sample
35	Novak Djokovic	long_rally_share_pct	2005–2008	2025–2026	-0.638	-3.413	2.845	1961	small endpoint sample

Observed result.

- Roger Federer: SAS 83.81 (77.81–88.90); weakest observed surface Clay, 69.05% known-outcome wins.
- Rafael Nadal: SAS 77.08 (66.50–83.05); weakest observed surface Grass, 59.46% known-outcome wins.
- Novak Djokovic: SAS 85.57 (79.30–89.56); weakest observed surface Clay, 70.16% known-outcome wins.
- Roger Federer, serve volley frequency pct: -13.47 percentage points from 1998–2001 to 2018–2021 (95% interval -21.31–-5.07; small endpoint sample).
- Rafael Nadal, long rally share pct: -3.53 percentage points from 2003–2006 to 2019–2022 (95% interval -5.87–-1.43; small endpoint sample).
- Novak Djokovic, slice share pct: +4.31 percentage points from 2005–2008 to 2025–2026 (95% interval 2.45–6.12; small endpoint sample).

Tennis interpretation. The SAS weakest-surface term makes specialization costly under this definition, even when specialization brings exceptional results on one surface. A change in net use, slice share, or rally length may reflect point-construction choices and changing opponents. It does not demonstrate invention, improved tactics, or athletic decline.

Alternative explanations. Equal surface weights remove only one source of composition change. Opponents, career stage, tournament context, and charting practices still change over time. The first/last supported periods can differ from career endpoints, and every selected endpoint contrast has its own interval and sample warning. These exploratory intervals are not multiplicity-adjusted. The next section combines measured profiles with direct rivalry evidence before asking how subjective weighting changes the ranking.

IX. Playing styles and preference-based Big Three comparison

Synthesis question: How do measured styles and direct matchups explain the comparison, and which conclusions depend on the reader's preferences? Direct rivalries can expose matchup differences that an overall sample obscures.

9.1 Evidence-based style profiles

Describe style with observed serving, returning, net activity, rally patterns, shot mix, and source-recorded winner/error rates. All overall profiles pool counts across each player's own charted window; surface profiles retain their own samples. Rates are not averaged across periods or surfaces. Source winner/error fields retain the source labels; their overlap with serve outcomes is not independently resolved, so they are not interpreted as independent aggression dimensions. Net-point success uses the NetPoints slice, not Approach. Missing values stay unavailable.

Profiles use raw rates and intervals rather than synthetic athleticism, acrobatics, or strength scores. Diversity is descriptive, not a reward for winning tennis. No fixed archetype is learned from three player summaries. Three pooled player summaries do not establish athleticism or validate intrinsic playing-style archetypes. The profiles below state measurable tendencies without claiming intrinsic player types or causal explanations.

In [27]:
```
profile_master = tactical_match_table.merge(
    analysis_dataframes["charting_m_stats_NetPoints"].loc[lambda x: x["row"].eq("NetPoints"), ["match_id", "player", "pts_won"]].rename(columns={"pts_won":
"net_points_won"}),
    on=["match_id", "player"], how="left", validate="one_to_one")
profile_rows = []
profile_tactical_metrics = ["approach_frequency_pct", "net_point_frequency_pct", "serve_volley_frequency_pct", "drop_shot_share_pct",
    "slice_share_pct", "short_rally_share_pct", "long_rally_share_pct"]
for player, surface in kpi_scope_groups:
    group_mask = profile_master["player"].eq(player)
    if surface != "Overall":
        group_mask &= profile_master["surface"].eq(surface)
    for metric in ["service_point_win_pct", "return_point_win_pct", "ace_pct"]:
        row = core_kpi_long.loc[core_kpi_long["player"].eq(player) & core_kpi_long["surface"].eq(surface) & core_kpi_long["metric"].eq(metric)].iloc[0]
        profile_rows.append({"player": player, "surface": surface, "metric": metric, "estimate": row["estimate"], "unit": "%",
            "numerator": row["numerator"], "denominator": row["denominator"], "contributing_matches": row["eligible_matches"],
            "ci95_lower": row["ci95_lower"], "ci95_upper": row["ci95_upper"], "small_sample": row["small_sample"]})
    profile_specs = {metric: tactical_metric_specs[metric] for metric in profile_tactical_metrics}
    profile_specs.update({"source_winner_rate_pct": ("winners", "source_player_points", "points"),
        "source_unforced_error_rate_pct": ("unforced", "source_player_points", "points"),
        "net_point_success_pct": ("net_points_won", "net_points_all", "events")})
    for metric, (num, den, basis) in profile_specs.items():
        a, b = profile_master[num], profile_master[den]
        mask = group_mask & a.notna() & b.notna() & np.isfinite(a) & np.isfinite(b) & a.ge(0) & b.gt(0) & a.le(b)
        if metric in tactical_masks:
            mask &= tactical_masks[metric]
        else:
            mask &= reconciled_mask(["service_points_parsed", "return_points_parsed"])
        sample = profile_master.loc[mask]
        numerator, denominator = sample[num].sum(min_count=1), sample[den].sum(min_count=1)
        matrix = np.zeros((len(bootstrap_match_ids), 2))
        for mid, x, y in sample[["match_id", num, den]].itertuples(index=False, name=None):
            matrix[bootstrap_match_index[mid]] = [x, y]
        totals = bootstrap_weights @ matrix
        with np.errstate(divide="ignore", invalid="ignore"):
            draws = np.where(totals[:, 1] > 0, 100*totals[:, 0]/totals[:, 1], np.nan)
```

```
        valid = draws[np.isfinite(draws)]
        profile_rows.append({"player": player, "surface": surface, "metric": metric,
            "estimate": 100*numerator/denominator if denominator > 0 else np.nan, "unit": "%", "numerator": numerator, "denominator": denominator,
            "contributing_matches": len(sample), "small_sample": pd.isna(denominator) or denominator < SMALL_SAMPLE_THRESHOLDS[basis],
            "ci95_lower": np.percentile(valid, 2.5) if len(valid) >= .95*BOOTSTRAP_REPLICATES else np.nan,
            "ci95_upper": np.percentile(valid, 97.5) if len(valid) >= .95*BOOTSTRAP_REPLICATES else np.nan})
    ground = get_shot_distribution(player, surface, "shot_type", SHOT_ALL_FAMILIES).loc["groundstroke"]
    profile_rows.append({"player": player, "surface": surface, "metric": "groundstroke_share_pct", "estimate": ground["share_pct"], "unit": "%",
        "numerator": ground["count"], "denominator": ground["classified_denominator"], "contributing_matches": ground["contributing_matches"],
        "ci95_lower": ground["ci95_lower"], "ci95_upper": ground["ci95_upper"], "small_sample": ground["classified_denominator"] <
SMALL_SAMPLE_THRESHOLDS["shots"]})
    for distribution, metric in [("shot_type", "shot_family_diversity"), ("source_direction_Total", "directional_diversity")]:
        row = shot_diversity_summary.loc[shot_diversity_summary["player"].eq(player) & shot_diversity_summary["surface"].eq(surface)
            & shot_diversity_summary["distribution"].eq(distribution)].iloc[0]
        profile_rows.append({"player": player, "surface": surface, "metric": metric, "estimate": row["normalized_entropy"], "unit": "index (0–1)",
            "numerator": np.nan, "denominator": row["classified_strokes"], "contributing_matches": row["contributing_matches"],
            "ci95_lower": row["ci95_lower"], "ci95_upper": row["ci95_upper"], "small_sample": row["small_sample"]})
style_profile_long = pd.DataFrame(profile_rows)
style_profile_wide = style_profile_long.pivot(index=["player", "surface"], columns="metric", values="estimate").reset_index()
style_cards = []
for row in style_profile_wide.itertuples(index=False):
    style_cards.append({"player": row.player, "surface": row.surface,
        "evidence_profile": f"Service points won {row.service_point_win_pct:.1f}%; return points won {row.return_point_win_pct:.1f}%. "
        f"Approach frequency {row.approach_frequency_pct:.1f}%; serve-and-volley {row.serve_volley_frequency_pct:.1f}%. "
        f"Short rallies {row.short_rally_share_pct:.1f}%; long rallies {row.long_rally_share_pct:.1f}%. "
        f"Groundstrokes {row.groundstroke_share_pct:.1f}% and slices {row.slice_share_pct:.1f}% of their stated classified stroke samples. "
        "See the evidence table for denominators and intervals; these are observed tendencies, not athleticism labels."})
playing_style_profiles = pd.DataFrame(style_cards)
assert not style_profile_long.duplicated(["player", "surface", "metric"]).any()
assert style_profile_long.loc[style_profile_long["unit"].eq("%"), "estimate"].dropna().between(0, 100).all()
print("Style profiles overall and by surface (raw units; no composite style score):")
report_table(style_profile_wide.round(3))
with pd.option_context("display.max_colwidth", None, "display.max_rows", None):
    report_table(playing_style_profiles)
    report_table(style_profile_long.round(4))
```

```
Style profiles overall and by surface (raw units; no composite style score):
```

metric	player	surface	ace_pct	approach_frequency_pct	directional_diversity	drop_shot_share_pct	groundstroke_share_pct	long_rally_share_pct	net_point_frequency_pct
	Novak Djokovic	Clay	4.891	8.810	0.839	2.375	87.652	18.656	12.350
	Novak Djokovic	Grass	9.272	10.336	0.823	1.436	80.801	12.700	12.547
	Novak Djokovic	Hard	7.612	8.866	0.827	1.226	85.160	16.143	11.708
	Novak Djokovic	Overall	7.195	9.047	0.830	1.547	85.295	16.275	11.971
	Rafael Nadal	Clay	2.609	5.869	0.861	1.052	89.413	19.402	8.611
	Rafael Nadal	Grass	7.008	8.390	0.859	1.124	79.027	9.484	10.901
	Rafael Nadal	Hard	4.786	6.920	0.856	0.553	84.399	14.778	8.955
	Rafael Nadal	Overall	4.277	6.728	0.859	0.813	85.980	15.831	9.074
	Roger Federer	Clay	7.295	14.597	0.868	1.520	76.505	12.243	16.270
	Roger Federer	Grass	11.886	16.196	0.866	0.880	67.682	6.632	17.680
	Roger Federer	Hard	10.523	15.157	0.856	0.811	71.360	10.063	16.908
	Roger Federer	Overall	10.178	15.248	0.860	0.969	71.843	9.823	16.938

metric	player	surface	net_point_success_pct	return_point_win_pct	serve_volley_frequency_pct	service_point_win_pct	short_rally_share_pct	shot_family_diversity	slice_share_pct
	Novak Djokovic	Clay	69.388	41.074	1.896	64.076	55.723	0.257	5.378
	Novak Djokovic	Grass	72.005	37.680	3.481	68.834	65.670	0.328	12.343
	Novak Djokovic	Hard	70.182	40.812	2.473	67.540	60.845	0.277	9.019
	Novak Djokovic	Overall	70.249	40.448	2.472	66.900	60.282	0.280	8.468
	Rafael Nadal	Clay	75.145	44.823	0.667	65.032	53.289	0.217	6.291
	Rafael Nadal	Grass	75.437	35.279	1.674	68.671	68.809	0.335	15.007
	Rafael Nadal	Hard	74.450	38.151	1.795	65.961	61.507	0.270	11.197
	Rafael Nadal	Overall	74.833	40.174	1.375	65.958	59.383	0.257	9.516
	Roger Federer	Clay	69.446	39.906	7.621	65.752	63.657	0.373	15.325
	Roger Federer	Grass	71.814	36.998	15.022	72.791	74.000	0.451	21.690
	Roger Federer	Hard	70.528	39.713	9.553	69.280	67.954	0.411	19.928
	Roger Federer	Overall	70.593	39.228	10.211	69.283	68.292	0.410	19.252

metric	player	surface	source_unforced_error_rate_pct	source_winner_rate_pct
	Novak Djokovic	Clay	17.266	16.267
	Novak Djokovic	Grass	13.882	16.333
	Novak Djokovic	Hard	15.414	15.638
	Novak Djokovic	Overall	15.636	15.878
	Rafael Nadal	Clay	14.480	15.074
	Rafael Nadal	Grass	11.817	17.101
	Rafael Nadal	Hard	14.042	14.713
	Rafael Nadal	Overall	13.921	15.139

metric	player	surface	source_unforced_error_rate_pct	source_winner_rate_pct
	Roger Federer	Clay	18.737	18.670
	Roger Federer	Grass	13.109	20.427
	Roger Federer	Hard	16.971	19.866
	Roger Federer	Overall	16.555	19.753

player	surface	evidence_profile
Novak Djokovic	Clay	Service points won 64.1%; return points won 41.1%. Approach frequency 8.8%; serve-and-volley 1.9%. Short rallies 55.7%; long rallies 18.7%. Groundstrokes 87.7% and slices 5.4% of their stated classified stroke samples. See the evidence table for denominators and intervals; these are observed tendencies, not athleticism labels.
Novak Djokovic	Grass	Service points won 68.8%; return points won 37.7%. Approach frequency 10.3%; serve-and-volley 3.5%. Short rallies 65.7%; long rallies 12.7%. Groundstrokes 80.8% and slices 12.3% of their stated classified stroke samples. See the evidence table for denominators and intervals; these are observed tendencies, not athleticism labels.
Novak Djokovic	Hard	Service points won 67.5%; return points won 40.8%. Approach frequency 8.9%; serve-and-volley 2.5%. Short rallies 60.8%; long rallies 16.1%. Groundstrokes 85.2% and slices 9.0% of their stated classified stroke samples. See the evidence table for denominators and intervals; these are observed tendencies, not athleticism labels.
Novak Djokovic	Overall	Service points won 66.9%; return points won 40.4%. Approach frequency 9.0%; serve-and-volley 2.5%. Short rallies 60.3%; long rallies 16.3%. Groundstrokes 85.3% and slices 8.5% of their stated classified stroke samples. See the evidence table for denominators and intervals; these are observed tendencies, not athleticism labels.
Rafael Nadal	Clay	Service points won 65.0%; return points won 44.8%. Approach frequency 5.9%; serve-and-volley 0.7%. Short rallies 53.3%; long rallies 19.4%. Groundstrokes 89.4% and slices 6.3% of their stated classified stroke samples. See the evidence table for denominators and intervals; these are observed tendencies, not athleticism labels.
Rafael Nadal	Grass	Service points won 68.7%; return points won 35.3%. Approach frequency 8.4%; serve-and-volley 1.7%. Short rallies 68.8%; long rallies 9.5%. Groundstrokes 79.0% and slices 15.0% of their stated classified stroke samples. See the evidence table for denominators and intervals; these are observed tendencies, not athleticism labels.
Rafael Nadal	Hard	Service points won 66.0%; return points won 38.2%. Approach frequency 6.9%; serve-and-volley 1.8%. Short rallies 61.5%; long rallies 14.8%. Groundstrokes 84.4% and slices 11.2% of their stated classified stroke samples. See the evidence table for denominators and intervals; these are observed tendencies, not athleticism labels.
Rafael Nadal	Overall	Service points won 66.0%; return points won 40.2%. Approach frequency 6.7%; serve-and-volley 1.4%. Short rallies 59.4%; long rallies 15.8%. Groundstrokes 86.0% and slices 9.5% of their stated classified stroke samples. See the evidence table for denominators and intervals; these are observed tendencies, not athleticism labels.
Roger Federer	Clay	Service points won 65.8%; return points won 39.9%. Approach frequency 14.6%; serve-and-volley 7.6%. Short rallies 63.7%; long rallies 12.2%. Groundstrokes 76.5% and slices 15.3% of their stated classified stroke samples. See the evidence table for denominators and intervals; these are observed tendencies, not athleticism labels.
Roger Federer	Grass	Service points won 72.8%; return points won 37.0%. Approach frequency 16.2%; serve-and-volley 15.0%. Short rallies 74.0%; long rallies 6.6%. Groundstrokes 67.7% and slices 21.7% of their stated classified stroke samples. See the evidence table for denominators and intervals; these are observed tendencies, not athleticism labels.
Roger Federer	Hard	Service points won 69.3%; return points won 39.7%. Approach frequency 15.2%; serve-and-volley 9.6%. Short rallies 68.0%; long rallies 10.1%. Groundstrokes 71.4% and slices 19.9% of their stated classified stroke samples. See the evidence table for denominators and intervals; these are observed tendencies, not athleticism labels.
Roger Federer	Overall	Service points won 69.3%; return points won 39.2%. Approach frequency 15.2%; serve-and-volley 10.2%. Short rallies 68.3%; long rallies 9.8%. Groundstrokes 71.8% and slices 19.3% of their stated classified stroke samples. See the evidence table for denominators and intervals; these are observed tendencies, not athleticism labels.

▼ Complete evidence table: 192 rows × 11 columns

player	surface	metric	estimate	unit	numerator	denominator	contributing_matches	ci95_lower	ci95_upper
Roger Federer	Overall	service_point_win_pct	69.283	%	40,906.000	59,042.000	712	68.743	69.800
Roger Federer	Overall	return_point_win_pct	39.228	%	24,853.000	63,355.000	711	38.679	39.763
Roger Federer	Overall	ace_pct	10.178	%	6,009.000	59,042.000	712	9.763	10.573
Roger Federer	Overall	approach_frequency_pct	15.248	%	18,029.000	118,238.000	687	14.724	15.765
Roger Federer	Overall	net_point_frequency_pct	16.938	%	20,492.000	120,981.000	703	16.416	17.472
Roger Federer	Overall	serve_volley_frequency_pct	10.211	%	5,931.000	58,086.000	702	9.329	11.106
Roger Federer	Overall	drop_shot_share_pct	0.969	%	2,135.000	220,248.000	722	0.906	1.039
Roger Federer	Overall	slice_share_pct	19.252	%	42,402.000	220,248.000	722	18.807	19.698
Roger Federer	Overall	short_rally_share_pct	68.292	%	82,408.000	120,670.000	722	67.708	68.890
Roger Federer	Overall	long_rally_share_pct	9.823	%	11,854.000	120,670.000	722	9.463	10.186
Roger Federer	Overall	source_winner_rate_pct	19.753	%	23,897.000	120,981.000	703	19.383	20.095
Roger Federer	Overall	source_unforced_error_rate_pct	16.555	%	20,029.000	120,981.000	703	16.180	16.926
Roger Federer	Overall	net_point_success_pct	70.593	%	14,466.000	20,492.000	703	69.922	71.256
Roger Federer	Overall	groundstroke_share_pct	71.843	%	158,256.000	220,280.000	722	71.354	72.332
Roger Federer	Overall	shot_family_diversity	0.410	index (0–1)	Unavailable	220,248.000	722	0.406	0.415
Roger Federer	Overall	directional_diversity	0.860	index (0–1)	Unavailable	132,560.000	719	0.857	0.863
Roger Federer	Clay	service_point_win_pct	65.752	%	7,238.000	11,008.000	127	64.415	67.065
Roger Federer	Clay	return_point_win_pct	39.906	%	4,679.000	11,725.000	129	38.759	41.054
Roger Federer	Clay	ace_pct	7.295	%	803.000	11,008.000	127	6.511	8.130
Roger Federer	Clay	approach_frequency_pct	14.597	%	3,178.000	21,772.000	122	13.394	15.762
Roger Federer	Clay	net_point_frequency_pct	16.270	%	3,646.000	22,410.000	126	15.052	17.470
Roger Federer	Clay	serve_volley_frequency_pct	7.621	%	825.000	10,825.000	125	6.140	9.437
Roger Federer	Clay	drop_shot_share_pct	1.520	%	696.000	45,781.000	130	1.342	1.704
Roger Federer	Clay	slice_share_pct	15.325	%	7,016.000	45,781.000	130	14.376	16.340

player	surface	metric	estimate	unit	numerator	denominator	contributing_matches	ci95_lower	ci95_upper
Roger Federer	Clay	short_rally_share_pct	63.657	%	14,298.000	22,461.000	130	62.253	65.159
Roger Federer	Clay	long_rally_share_pct	12.243	%	2,750.000	22,461.000	130	11.460	13.071
Roger Federer	Clay	source_winner_rate_pct	18.670	%	4,184.000	22,410.000	126	17.865	19.504
Roger Federer	Clay	source_unforced_error_rate_pct	18.737	%	4,199.000	22,410.000	126	17.870	19.659
Roger Federer	Clay	net_point_success_pct	69.446	%	2,532.000	3,646.000	126	67.782	71.163
Roger Federer	Clay	groundstroke_share_pct	76.505	%	35,034.000	45,793.000	130	75.409	77.592
Roger Federer	Clay	shot_family_diversity	0.373	index (0–1)	Unavailable	45,781.000	130	0.362	0.384
Roger Federer	Clay	directional_diversity	0.868	index (0–1)	Unavailable	29,093.000	130	0.861	0.875
Roger Federer	Grass	service_point_win_pct	72.791	%	8,090.000	11,114.000	119	71.591	73.921
Roger Federer	Grass	return_point_win_pct	36.998	%	4,493.000	12,144.000	119	35.838	38.219
Roger Federer	Grass	ace_pct	11.886	%	1,321.000	11,114.000	119	10.965	12.847
Roger Federer	Grass	approach_frequency_pct	16.196	%	3,581.000	22,110.000	114	14.811	17.620
Roger Federer	Grass	net_point_frequency_pct	17.680	%	4,112.000	23,258.000	119	16.284	19.115
Roger Federer	Grass	serve_volley_frequency_pct	15.022	%	1,624.000	10,811.000	117	11.874	18.535
Roger Federer	Grass	drop_shot_share_pct	0.880	%	309.000	35,122.000	119	0.751	1.012
Roger Federer	Grass	slice_share_pct	21.690	%	7,618.000	35,122.000	119	20.599	22.864
Roger Federer	Grass	short_rally_share_pct	74.000	%	16,804.000	22,708.000	119	72.661	75.488
Roger Federer	Grass	long_rally_share_pct	6.632	%	1,506.000	22,708.000	119	5.866	7.332
Roger Federer	Grass	source_winner_rate_pct	20.427	%	4,751.000	23,258.000	119	19.728	21.088
Roger Federer	Grass	source_unforced_error_rate_pct	13.110	%	3,049.000	23,258.000	119	12.307	13.936
Roger Federer	Grass	net_point_success_pct	71.814	%	2,953.000	4,112.000	119	70.089	73.508
Roger Federer	Grass	groundstroke_share_pct	67.682	%	23,774.000	35,126.000	119	66.490	68.866
Roger Federer	Grass	shot_family_diversity	0.451	index (0–1)	Unavailable	35,122.000	119	0.439	0.464
Roger Federer	Grass	directional_diversity	0.866	index (0–1)	Unavailable	18,665.000	117	0.858	0.873

player	surface	metric	estimate	unit	numerator	denominator	contributing_matches	ci95_lower	ci95_upper
Roger Federer	Hard	service_point_win_pct	69.279	%	25,578.000	36,920.000	466	68.656	69.928
Roger Federer	Hard	return_point_win_pct	39.713	%	15,681.000	39,486.000	463	39.032	40.400
Roger Federer	Hard	ace_pct	10.523	%	3,885.000	36,920.000	466	10.044	11.000
Roger Federer	Hard	approach_frequency_pct	15.157	%	11,270.000	74,356.000	451	14.529	15.799
Roger Federer	Hard	net_point_frequency_pct	16.908	%	12,734.000	75,313.000	458	16.298	17.541
Roger Federer	Hard	serve_volley_frequency_pct	9.553	%	3,482.000	36,450.000	460	8.659	10.451
Roger Federer	Hard	drop_shot_share_pct	0.811	%	1,130.000	139,345.000	473	0.740	0.883
Roger Federer	Hard	slice_share_pct	19.927	%	27,768.000	139,345.000	473	19.447	20.384
Roger Federer	Hard	short_rally_share_pct	67.954	%	51,306.000	75,501.000	473	67.280	68.553
Roger Federer	Hard	long_rally_share_pct	10.063	%	7,598.000	75,501.000	473	9.656	10.507
Roger Federer	Hard	source_winner_rate_pct	19.866	%	14,962.000	75,313.000	458	19.410	20.333
Roger Federer	Hard	source_unforced_error_rate_pct	16.971	%	12,781.000	75,313.000	458	16.508	17.403
Roger Federer	Hard	net_point_success_pct	70.528	%	8,981.000	12,734.000	458	69.708	71.366
Roger Federer	Hard	groundstroke_share_pct	71.360	%	99,448.000	139,361.000	473	70.816	71.922
Roger Federer	Hard	shot_family_diversity	0.410	index (0–1)	Unavailable	139,345.000	473	0.405	0.416
Roger Federer	Hard	directional_diversity	0.856	index (0–1)	Unavailable	84,802.000	472	0.852	0.859
Rafael Nadal	Overall	service_point_win_pct	65.958	%	22,394.000	33,952.000	416	65.216	66.661
Rafael Nadal	Overall	return_point_win_pct	40.174	%	14,059.000	34,995.000	411	39.417	40.958
Rafael Nadal	Overall	ace_pct	4.277	%	1,452.000	33,952.000	416	3.927	4.639
Rafael Nadal	Overall	approach_frequency_pct	6.728	%	4,453.000	66,183.000	398	6.344	7.141
Rafael Nadal	Overall	net_point_frequency_pct	9.074	%	6,124.000	67,487.000	406	8.671	9.506
Rafael Nadal	Overall	serve_volley_frequency_pct	1.375	%	464.000	33,754.000	414	1.153	1.627
Rafael Nadal	Overall	drop_shot_share_pct	0.813	%	1,234.000	151,781.000	423	0.744	0.886
Rafael Nadal	Overall	slice_share_pct	9.516	%	14,443.000	151,781.000	423	8.998	10.071
Rafael Nadal	Overall	short_rally_share_pct	59.383	%	40,800.000	68,706.000	423	58.443	60.336

player	surface	metric	estimate	unit	numerator	denominator	contributing_matches	ci95_lower	ci95_upper
Rafael Nadal	Overall	long_rally_share_pct	15.831	%	10,877.000	68,706.000	423	15.174	16.544
Rafael Nadal	Overall	source_winner_rate_pct	15.139	%	10,217.000	67,487.000	406	14.707	15.583
Rafael Nadal	Overall	source_unforced_error_rate_pct	13.921	%	9,395.000	67,487.000	406	13.495	14.345
Rafael Nadal	Overall	net_point_success_pct	74.833	%	4,585.000	6,127.000	407	73.595	76.054
Rafael Nadal	Overall	groundstroke_share_pct	85.980	%	130,511.000	151,792.000	423	85.401	86.547
Rafael Nadal	Overall	shot_family_diversity	0.257	index (0–1)	Unavailable	151,781.000	423	0.250	0.264
Rafael Nadal	Overall	directional_diversity	0.859	index (0–1)	Unavailable	105,412.000	422	0.853	0.866
Rafael Nadal	Clay	service_point_win_pct	65.032	%	7,902.000	12,151.000	161	63.865	66.242
Rafael Nadal	Clay	return_point_win_pct	44.823	%	5,606.000	12,507.000	157	43.628	46.027
Rafael Nadal	Clay	ace_pct	2.609	%	317.000	12,151.000	161	2.283	2.958
Rafael Nadal	Clay	approach_frequency_pct	5.869	%	1,398.000	23,820.000	153	5.343	6.436
Rafael Nadal	Clay	net_point_frequency_pct	8.611	%	2,073.000	24,075.000	155	8.000	9.259
Rafael Nadal	Clay	serve_volley_frequency_pct	0.667	%	81.000	12,151.000	161	0.476	0.879
Rafael Nadal	Clay	drop_shot_share_pct	1.052	%	662.000	62,929.000	166	0.938	1.183
Rafael Nadal	Clay	slice_share_pct	6.291	%	3,959.000	62,929.000	166	5.888	6.731
Rafael Nadal	Clay	short_rally_share_pct	53.289	%	13,351.000	25,054.000	166	52.043	54.624
Rafael Nadal	Clay	long_rally_share_pct	19.402	%	4,861.000	25,054.000	166	18.319	20.485
Rafael Nadal	Clay	source_winner_rate_pct	15.074	%	3,629.000	24,075.000	155	14.383	15.773
Rafael Nadal	Clay	source_unforced_error_rate_pct	14.480	%	3,486.000	24,075.000	155	13.809	15.150
Rafael Nadal	Clay	net_point_success_pct	75.144	%	1,560.000	2,076.000	156	72.832	77.339
Rafael Nadal	Clay	groundstroke_share_pct	89.413	%	56,272.000	62,935.000	166	88.889	89.929
Rafael Nadal	Clay	shot_family_diversity	0.217	index (0–1)	Unavailable	62,929.000	166	0.209	0.226
Rafael Nadal	Clay	directional_diversity	0.861	index (0–1)	Unavailable	44,589.000	165	0.849	0.874
Rafael Nadal	Grass	service_point_win_pct	68.671	%	2,832.000	4,124.000	39	66.614	70.880

player	surface	metric	estimate	unit	numerator	denominator	contributing_matches	ci95_lower	ci95_upper
Rafael Nadal	Grass	return_point_win_pct	35.279	%	1,553.000	4,402.000	40	33.326	37.481
Rafael Nadal	Grass	ace_pct	7.008	%	289.000	4,124.000	39	5.727	8.522
Rafael Nadal	Grass	approach_frequency_pct	8.390	%	705.000	8,403.000	39	7.005	9.634
Rafael Nadal	Grass	net_point_frequency_pct	10.901	%	916.000	8,403.000	39	9.561	12.227
Rafael Nadal	Grass	serve_volley_frequency_pct	1.674	%	67.000	4,003.000	38	1.127	2.311
Rafael Nadal	Grass	drop_shot_share_pct	1.124	%	158.000	14,053.000	39	0.855	1.402
Rafael Nadal	Grass	slice_share_pct	15.008	%	2,109.000	14,053.000	39	13.201	16.971
Rafael Nadal	Grass	short_rally_share_pct	68.809	%	5,652.000	8,214.000	39	66.464	71.284
Rafael Nadal	Grass	long_rally_share_pct	9.484	%	779.000	8,214.000	39	8.032	10.948
Rafael Nadal	Grass	source_winner_rate_pct	17.101	%	1,437.000	8,403.000	39	16.033	18.263
Rafael Nadal	Grass	source_unforced_error_rate_pct	11.817	%	993.000	8,403.000	39	10.665	13.003
Rafael Nadal	Grass	net_point_success_pct	75.437	%	691.000	916.000	39	72.688	78.251
Rafael Nadal	Grass	groundstroke_share_pct	79.027	%	11,108.000	14,056.000	39	76.971	80.934
Rafael Nadal	Grass	shot_family_diversity	0.335	index (0–1)	Unavailable	14,053.000	39	0.314	0.356
Rafael Nadal	Grass	directional_diversity	0.859	index (0–1)	Unavailable	9,114.000	40	0.845	0.872
Rafael Nadal	Hard	service_point_win_pct	65.961	%	11,660.000	17,677.000	216	64.905	66.999
Rafael Nadal	Hard	return_point_win_pct	38.151	%	6,900.000	18,086.000	214	37.286	39.033
Rafael Nadal	Hard	ace_pct	4.786	%	846.000	17,677.000	216	4.312	5.287
Rafael Nadal	Hard	approach_frequency_pct	6.920	%	2,350.000	33,960.000	206	6.389	7.499
Rafael Nadal	Hard	net_point_frequency_pct	8.955	%	3,135.000	35,009.000	212	8.396	9.551
Rafael Nadal	Hard	serve_volley_frequency_pct	1.796	%	316.000	17,600.000	215	1.416	2.213
Rafael Nadal	Hard	drop_shot_share_pct	0.553	%	414.000	74,799.000	218	0.484	0.628
Rafael Nadal	Hard	slice_share_pct	11.197	%	8,375.000	74,799.000	218	10.434	11.945
Rafael Nadal	Hard	short_rally_share_pct	61.507	%	21,797.000	35,438.000	218	60.422	62.500
Rafael Nadal	Hard	long_rally_share_pct	14.778	%	5,237.000	35,438.000	218	14.066	15.585

player	surface	metric	estimate	unit	numerator	denominator	contributing_matches	ci95_lower	ci95_upper
Rafael Nadal	Hard	source_winner_rate_pct	14.713	%	5,151.000	35,009.000	212	14.108	15.302
Rafael Nadal	Hard	source_unforced_error_rate_pct	14.042	%	4,916.000	35,009.000	212	13.419	14.661
Rafael Nadal	Hard	net_point_success_pct	74.450	%	2,334.000	3,135.000	212	72.802	76.143
Rafael Nadal	Hard	groundstroke_share_pct	84.399	%	63,131.000	74,801.000	218	83.649	85.162
Rafael Nadal	Hard	shot_family_diversity	0.270	index (0–1)	Unavailable	74,799.000	218	0.262	0.278
Rafael Nadal	Hard	directional_diversity	0.856	index (0–1)	Unavailable	51,709.000	217	0.849	0.864
Novak Djokovic	Overall	service_point_win_pct	66.900	%	30,359.000	45,380.000	549	66.295	67.517
Novak Djokovic	Overall	return_point_win_pct	40.448	%	19,038.000	47,068.000	547	39.861	41.041
Novak Djokovic	Overall	ace_pct	7.195	%	3,265.000	45,380.000	549	6.813	7.590
Novak Djokovic	Overall	approach_frequency_pct	9.047	%	8,135.000	89,923.000	533	8.656	9.452
Novak Djokovic	Overall	net_point_frequency_pct	11.970	%	10,996.000	91,859.000	545	11.537	12.403
Novak Djokovic	Overall	serve_volley_frequency_pct	2.473	%	1,122.000	45,380.000	549	2.146	2.819
Novak Djokovic	Overall	drop_shot_share_pct	1.547	%	3,127.000	202,184.000	552	1.454	1.645
Novak Djokovic	Overall	slice_share_pct	8.468	%	17,121.000	202,184.000	552	8.072	8.857
Novak Djokovic	Overall	short_rally_share_pct	60.282	%	54,432.000	90,296.000	552	59.606	60.984
Novak Djokovic	Overall	long_rally_share_pct	16.275	%	14,696.000	90,296.000	552	15.778	16.808
Novak Djokovic	Overall	source_winner_rate_pct	15.878	%	14,621.000	92,085.000	546	15.497	16.264
Novak Djokovic	Overall	source_unforced_error_rate_pct	15.636	%	14,398.000	92,085.000	546	15.219	16.071
Novak Djokovic	Overall	net_point_success_pct	70.249	%	7,752.000	11,035.000	547	69.294	71.239
Novak Djokovic	Overall	groundstroke_share_pct	85.295	%	172,474.000	202,209.000	552	84.851	85.733
Novak Djokovic	Overall	shot_family_diversity	0.280	index (0–1)	Unavailable	202,184.000	552	0.274	0.286
Novak Djokovic	Overall	directional_diversity	0.830	index (0–1)	Unavailable	138,901.000	549	0.827	0.834
Novak Djokovic	Clay	service_point_win_pct	64.076	%	6,826.000	10,653.000	127	62.781	65.415
Novak Djokovic	Clay	return_point_win_pct	41.074	%	4,468.000	10,878.000	127	39.924	42.227
Novak Djokovic	Clay	ace_pct	4.891	%	521.000	10,653.000	127	4.405	5.408

player	surface	metric	estimate	unit	numerator	denominator	contributing_matches	ci95_lower	ci95_upper
Novak Djokovic	Clay	approach_frequency_pct	8.809	%	1,813.000	20,580.000	122	7.875	9.803
Novak Djokovic	Clay	net_point_frequency_pct	12.351	%	2,633.000	21,319.000	126	11.296	13.499
Novak Djokovic	Clay	serve_volley_frequency_pct	1.896	%	202.000	10,653.000	127	1.414	2.495
Novak Djokovic	Clay	drop_shot_share_pct	2.375	%	1,237.000	52,084.000	129	2.145	2.621
Novak Djokovic	Clay	slice_share_pct	5.378	%	2,801.000	52,084.000	129	4.992	5.821
Novak Djokovic	Clay	short_rally_share_pct	55.724	%	11,873.000	21,307.000	129	54.468	57.144
Novak Djokovic	Clay	long_rally_share_pct	18.656	%	3,975.000	21,307.000	129	17.544	19.694
Novak Djokovic	Clay	source_winner_rate_pct	16.267	%	3,468.000	21,319.000	126	15.543	17.008
Novak Djokovic	Clay	source_unforced_error_rate_pct	17.266	%	3,681.000	21,319.000	126	16.359	18.144
Novak Djokovic	Clay	net_point_success_pct	69.388	%	1,836.000	2,646.000	127	67.490	71.347
Novak Djokovic	Clay	groundstroke_share_pct	87.652	%	45,665.000	52,098.000	129	86.947	88.285
Novak Djokovic	Clay	shot_family_diversity	0.258	index (0–1)	Unavailable	52,084.000	129	0.246	0.269
Novak Djokovic	Clay	directional_diversity	0.839	index (0–1)	Unavailable	36,480.000	129	0.831	0.849
Novak Djokovic	Grass	service_point_win_pct	68.834	%	4,172.000	6,061.000	58	67.294	70.305
Novak Djokovic	Grass	return_point_win_pct	37.680	%	2,404.000	6,380.000	58	35.970	39.470
Novak Djokovic	Grass	ace_pct	9.272	%	562.000	6,061.000	58	8.015	10.579
Novak Djokovic	Grass	approach_frequency_pct	10.336	%	1,225.000	11,852.000	55	9.265	11.496
Novak Djokovic	Grass	net_point_frequency_pct	12.547	%	1,561.000	12,441.000	58	11.349	13.825
Novak Djokovic	Grass	serve_volley_frequency_pct	3.481	%	211.000	6,061.000	58	2.336	4.793
Novak Djokovic	Grass	drop_shot_share_pct	1.436	%	338.000	23,536.000	58	1.210	1.655
Novak Djokovic	Grass	slice_share_pct	12.343	%	2,905.000	23,536.000	58	11.002	13.796
Novak Djokovic	Grass	short_rally_share_pct	65.670	%	7,927.000	12,071.000	58	63.881	67.466
Novak Djokovic	Grass	long_rally_share_pct	12.700	%	1,533.000	12,071.000	58	11.466	13.887
Novak Djokovic	Grass	source_winner_rate_pct	16.333	%	2,032.000	12,441.000	58	15.107	17.785
Novak Djokovic	Grass	source_unforced_error_rate_pct	13.882	%	1,727.000	12,441.000	58	12.816	14.998

player	surface	metric	estimate	unit	numerator	denominator	contributing_matches	ci95_lower	ci95_upper
Novak Djokovic	Grass	net_point_success_pct	72.005	%	1,124.000	1,561.000	58	69.644	74.253
Novak Djokovic	Grass	groundstroke_share_pct	80.801	%	19,019.000	23,538.000	58	79.116	82.384
Novak Djokovic	Grass	shot_family_diversity	0.328	index (0–1)	Unavailable	23,536.000	58	0.309	0.345
Novak Djokovic	Grass	directional_diversity	0.823	index (0–1)	Unavailable	15,240.000	58	0.813	0.831
Novak Djokovic	Hard	service_point_win_pct	67.540	%	19,361.000	28,666.000	364	66.769	68.291
Novak Djokovic	Hard	return_point_win_pct	40.812	%	12,166.000	29,810.000	362	40.034	41.575
Novak Djokovic	Hard	ace_pct	7.612	%	2,182.000	28,666.000	364	7.172	8.103
Novak Djokovic	Hard	approach_frequency_pct	8.866	%	5,097.000	57,491.000	356	8.409	9.350
Novak Djokovic	Hard	net_point_frequency_pct	11.708	%	6,802.000	58,099.000	361	11.196	12.236
Novak Djokovic	Hard	serve_volley_frequency_pct	2.473	%	709.000	28,666.000	364	2.062	2.888
Novak Djokovic	Hard	drop_shot_share_pct	1.226	%	1,552.000	126,564.000	365	1.134	1.316
Novak Djokovic	Hard	slice_share_pct	9.019	%	11,415.000	126,564.000	365	8.585	9.456
Novak Djokovic	Hard	short_rally_share_pct	60.845	%	34,632.000	56,918.000	365	60.048	61.618
Novak Djokovic	Hard	long_rally_share_pct	16.142	%	9,188.000	56,918.000	365	15.534	16.803
Novak Djokovic	Hard	source_winner_rate_pct	15.638	%	9,121.000	58,325.000	362	15.148	16.107
Novak Djokovic	Hard	source_unforced_error_rate_pct	15.414	%	8,990.000	58,325.000	362	14.884	15.949
Novak Djokovic	Hard	net_point_success_pct	70.182	%	4,792.000	6,828.000	362	68.940	71.444
Novak Djokovic	Hard	groundstroke_share_pct	85.160	%	107,790.000	126,573.000	365	84.660	85.686
Novak Djokovic	Hard	shot_family_diversity	0.277	index (0–1)	Unavailable	126,564.000	365	0.270	0.284
Novak Djokovic	Hard	directional_diversity	0.827	index (0–1)	Unavailable	87,181.000	362	0.823	0.831

	player	surface	metric	small_sample
	Roger Federer	Overall	service_point_win_pct	False
	Roger Federer	Overall	return_point_win_pct	False
	Roger Federer	Overall	ace_pct	False
	Roger Federer	Overall	approach_frequency_pct	False

player	surface	metric	small_sample
Roger Federer	Overall	net_point_frequency_pct	False
Roger Federer	Overall	serve_volley_frequency_pct	False
Roger Federer	Overall	drop_shot_share_pct	False
Roger Federer	Overall	slice_share_pct	False
Roger Federer	Overall	short_rally_share_pct	False
Roger Federer	Overall	long_rally_share_pct	False
Roger Federer	Overall	source_winner_rate_pct	False
Roger Federer	Overall	source_unforced_error_rate_pct	False
Roger Federer	Overall	net_point_success_pct	False
Roger Federer	Overall	groundstroke_share_pct	False
Roger Federer	Overall	shot_family_diversity	False
Roger Federer	Overall	directional_diversity	False
Roger Federer	Clay	service_point_win_pct	False
Roger Federer	Clay	return_point_win_pct	False
Roger Federer	Clay	ace_pct	False
Roger Federer	Clay	approach_frequency_pct	False
Roger Federer	Clay	net_point_frequency_pct	False
Roger Federer	Clay	serve_volley_frequency_pct	False
Roger Federer	Clay	drop_shot_share_pct	False
Roger Federer	Clay	slice_share_pct	False
Roger Federer	Clay	short_rally_share_pct	False
Roger Federer	Clay	long_rally_share_pct	False
Roger Federer	Clay	source_winner_rate_pct	False
Roger Federer	Clay	source_unforced_error_rate_pct	False
Roger Federer	Clay	net_point_success_pct	False
Roger Federer	Clay	groundstroke_share_pct	False
Roger Federer	Clay	shot_family_diversity	False
Roger Federer	Clay	directional_diversity	False
Roger Federer	Grass	service_point_win_pct	False
Roger Federer	Grass	return_point_win_pct	False
Roger Federer	Grass	ace_pct	False
Roger Federer	Grass	approach_frequency_pct	False
Roger Federer	Grass	net_point_frequency_pct	False
Roger Federer	Grass	serve_volley_frequency_pct	False
Roger Federer	Grass	drop_shot_share_pct	False
Roger Federer	Grass	slice_share_pct	False
Roger Federer	Grass	short_rally_share_pct	False
Roger Federer	Grass	long_rally_share_pct	False
Roger Federer	Grass	source_winner_rate_pct	False
Roger Federer	Grass	source_unforced_error_rate_pct	False

player	surface	metric	small_sample
Roger Federer	Grass	net_point_success_pct	False
Roger Federer	Grass	groundstroke_share_pct	False
Roger Federer	Grass	shot_family_diversity	False
Roger Federer	Grass	directional_diversity	False
Roger Federer	Hard	service_point_win_pct	False
Roger Federer	Hard	return_point_win_pct	False
Roger Federer	Hard	ace_pct	False
Roger Federer	Hard	approach_frequency_pct	False
Roger Federer	Hard	net_point_frequency_pct	False
Roger Federer	Hard	serve_volley_frequency_pct	False
Roger Federer	Hard	drop_shot_share_pct	False
Roger Federer	Hard	slice_share_pct	False
Roger Federer	Hard	short_rally_share_pct	False
Roger Federer	Hard	long_rally_share_pct	False
Roger Federer	Hard	source_winner_rate_pct	False
Roger Federer	Hard	source_unforced_error_rate_pct	False
Roger Federer	Hard	net_point_success_pct	False
Roger Federer	Hard	groundstroke_share_pct	False
Roger Federer	Hard	shot_family_diversity	False
Roger Federer	Hard	directional_diversity	False
Rafael Nadal	Overall	service_point_win_pct	False
Rafael Nadal	Overall	return_point_win_pct	False
Rafael Nadal	Overall	ace_pct	False
Rafael Nadal	Overall	approach_frequency_pct	False
Rafael Nadal	Overall	net_point_frequency_pct	False
Rafael Nadal	Overall	serve_volley_frequency_pct	False
Rafael Nadal	Overall	drop_shot_share_pct	False
Rafael Nadal	Overall	slice_share_pct	False
Rafael Nadal	Overall	short_rally_share_pct	False
Rafael Nadal	Overall	long_rally_share_pct	False
Rafael Nadal	Overall	source_winner_rate_pct	False
Rafael Nadal	Overall	source_unforced_error_rate_pct	False
Rafael Nadal	Overall	net_point_success_pct	False
Rafael Nadal	Overall	groundstroke_share_pct	False
Rafael Nadal	Overall	shot_family_diversity	False
Rafael Nadal	Overall	directional_diversity	False
Rafael Nadal	Clay	service_point_win_pct	False
Rafael Nadal	Clay	return_point_win_pct	False
Rafael Nadal	Clay	ace_pct	False
Rafael Nadal	Clay	approach_frequency_pct	False

player	surface	metric	small_sample
Rafael Nadal	Clay	net_point_frequency_pct	False
Rafael Nadal	Clay	serve_volley_frequency_pct	False
Rafael Nadal	Clay	drop_shot_share_pct	False
Rafael Nadal	Clay	slice_share_pct	False
Rafael Nadal	Clay	short_rally_share_pct	False
Rafael Nadal	Clay	long_rally_share_pct	False
Rafael Nadal	Clay	source_winner_rate_pct	False
Rafael Nadal	Clay	source_unforced_error_rate_pct	False
Rafael Nadal	Clay	net_point_success_pct	False
Rafael Nadal	Clay	groundstroke_share_pct	False
Rafael Nadal	Clay	shot_family_diversity	False
Rafael Nadal	Clay	directional_diversity	False
Rafael Nadal	Grass	service_point_win_pct	False
Rafael Nadal	Grass	return_point_win_pct	False
Rafael Nadal	Grass	ace_pct	False
Rafael Nadal	Grass	approach_frequency_pct	False
Rafael Nadal	Grass	net_point_frequency_pct	False
Rafael Nadal	Grass	serve_volley_frequency_pct	False
Rafael Nadal	Grass	drop_shot_share_pct	False
Rafael Nadal	Grass	slice_share_pct	False
Rafael Nadal	Grass	short_rally_share_pct	False
Rafael Nadal	Grass	long_rally_share_pct	False
Rafael Nadal	Grass	source_winner_rate_pct	False
Rafael Nadal	Grass	source_unforced_error_rate_pct	False
Rafael Nadal	Grass	net_point_success_pct	False
Rafael Nadal	Grass	groundstroke_share_pct	False
Rafael Nadal	Grass	shot_family_diversity	False
Rafael Nadal	Grass	directional_diversity	False
Rafael Nadal	Hard	service_point_win_pct	False
Rafael Nadal	Hard	return_point_win_pct	False
Rafael Nadal	Hard	ace_pct	False
Rafael Nadal	Hard	approach_frequency_pct	False
Rafael Nadal	Hard	net_point_frequency_pct	False
Rafael Nadal	Hard	serve_volley_frequency_pct	False
Rafael Nadal	Hard	drop_shot_share_pct	False
Rafael Nadal	Hard	slice_share_pct	False
Rafael Nadal	Hard	short_rally_share_pct	False
Rafael Nadal	Hard	long_rally_share_pct	False
Rafael Nadal	Hard	source_winner_rate_pct	False
Rafael Nadal	Hard	source_unforced_error_rate_pct	False

player	surface	metric	small_sample
Rafael Nadal	Hard	net_point_success_pct	False
Rafael Nadal	Hard	groundstroke_share_pct	False
Rafael Nadal	Hard	shot_family_diversity	False
Rafael Nadal	Hard	directional_diversity	False
Novak Djokovic	Overall	service_point_win_pct	False
Novak Djokovic	Overall	return_point_win_pct	False
Novak Djokovic	Overall	ace_pct	False
Novak Djokovic	Overall	approach_frequency_pct	False
Novak Djokovic	Overall	net_point_frequency_pct	False
Novak Djokovic	Overall	serve_volley_frequency_pct	False
Novak Djokovic	Overall	drop_shot_share_pct	False
Novak Djokovic	Overall	slice_share_pct	False
Novak Djokovic	Overall	short_rally_share_pct	False
Novak Djokovic	Overall	long_rally_share_pct	False
Novak Djokovic	Overall	source_winner_rate_pct	False
Novak Djokovic	Overall	source_unforced_error_rate_pct	False
Novak Djokovic	Overall	net_point_success_pct	False
Novak Djokovic	Overall	groundstroke_share_pct	False
Novak Djokovic	Overall	shot_family_diversity	False
Novak Djokovic	Overall	directional_diversity	False
Novak Djokovic	Clay	service_point_win_pct	False
Novak Djokovic	Clay	return_point_win_pct	False
Novak Djokovic	Clay	ace_pct	False
Novak Djokovic	Clay	approach_frequency_pct	False
Novak Djokovic	Clay	net_point_frequency_pct	False
Novak Djokovic	Clay	serve_volley_frequency_pct	False
Novak Djokovic	Clay	drop_shot_share_pct	False
Novak Djokovic	Clay	slice_share_pct	False
Novak Djokovic	Clay	short_rally_share_pct	False
Novak Djokovic	Clay	long_rally_share_pct	False
Novak Djokovic	Clay	source_winner_rate_pct	False
Novak Djokovic	Clay	source_unforced_error_rate_pct	False
Novak Djokovic	Clay	net_point_success_pct	False
Novak Djokovic	Clay	groundstroke_share_pct	False
Novak Djokovic	Clay	shot_family_diversity	False
Novak Djokovic	Clay	directional_diversity	False
Novak Djokovic	Grass	service_point_win_pct	False
Novak Djokovic	Grass	return_point_win_pct	False
Novak Djokovic	Grass	ace_pct	False
Novak Djokovic	Grass	approach_frequency_pct	False

player	surface	metric	small_sample
Novak Djokovic	Grass	net_point_frequency_pct	False
Novak Djokovic	Grass	serve_volley_frequency_pct	False
Novak Djokovic	Grass	drop_shot_share_pct	False
Novak Djokovic	Grass	slice_share_pct	False
Novak Djokovic	Grass	short_rally_share_pct	False
Novak Djokovic	Grass	long_rally_share_pct	False
Novak Djokovic	Grass	source_winner_rate_pct	False
Novak Djokovic	Grass	source_unforced_error_rate_pct	False
Novak Djokovic	Grass	net_point_success_pct	False
Novak Djokovic	Grass	groundstroke_share_pct	False
Novak Djokovic	Grass	shot_family_diversity	False
Novak Djokovic	Grass	directional_diversity	False
Novak Djokovic	Hard	service_point_win_pct	False
Novak Djokovic	Hard	return_point_win_pct	False
Novak Djokovic	Hard	ace_pct	False
Novak Djokovic	Hard	approach_frequency_pct	False
Novak Djokovic	Hard	net_point_frequency_pct	False
Novak Djokovic	Hard	serve_volley_frequency_pct	False
Novak Djokovic	Hard	drop_shot_share_pct	False
Novak Djokovic	Hard	slice_share_pct	False
Novak Djokovic	Hard	short_rally_share_pct	False
Novak Djokovic	Hard	long_rally_share_pct	False
Novak Djokovic	Hard	source_winner_rate_pct	False
Novak Djokovic	Hard	source_unforced_error_rate_pct	False
Novak Djokovic	Hard	net_point_success_pct	False
Novak Djokovic	Hard	groundstroke_share_pct	False
Novak Djokovic	Hard	shot_family_diversity	False
Novak Djokovic	Hard	directional_diversity	False

9.2 Rivalry context and consolidated scorecards

Head-to-head summaries use distinct eligible matches and verified outcomes; unknown endings remain counted separately. They cover charted meetings, not complete historical rivalry records, and are not additional inputs to the overall score. Surface results give context to specialization that an aggregate adaptability-focused model can under-reward.

The consolidated scorecards place performance, pressure, style, adaptability, and coverage together. SAS is overall only; its surface cells are intentionally unavailable. Tactical evolution points to the measured component changes in Section VIII, not an unsupported innovation score. Preference scores are displayed separately because they depend on chosen definitions, scales, and weights.

In [28]:
```
h2h_rows = []
for a,b in [(BIG_THREE[0], BIG_THREE[1]), (BIG_THREE[0], BIG_THREE[2]), (BIG_THREE[1], BIG_THREE[2])]:
    pair = canonical_match_results.loc[((canonical_match_results["player1"].eq(a) & canonical_match_results["player2"].eq(b))
        | (canonical_match_results["player1"].eq(b) & canonical_match_results["player2"].eq(a)))]
    for surface in ["Overall"]+PROJECT_SURFACES:
        sample = pair if surface == "Overall" else pair.loc[pair["surface"].eq(surface)]
        known = sample.loc[sample["winner"].notna()]
        matrix = np.zeros((len(bootstrap_match_ids),2))
        for mid, winner in known[["match_id", "winner"]].itertuples(index=False,name=None):
            matrix[bootstrap_match_index[mid]] = [int(winner==a),1]
        totals = bootstrap_weights @ matrix
        with np.errstate(divide="ignore", invalid="ignore"):
            draws = np.where(totals[:,1]>0,100*totals[:,0]/totals[:,1],np.nan)
        valid = draws[np.isfinite(draws)]
        h2h_rows.append({"player_a":a,"player_b":b,"surface":surface,"charted_meetings":len(sample),"known_outcomes":len(known),
            "unknown_outcomes":int(sample["winner"].isna().sum()),"a_verified_wins":int(known["winner"].eq(a).sum()),
            "b_verified_wins":int(known["winner"].eq(b).sum()),"a_win_pct":100*known["winner"].eq(a).mean() if len(known) else np.nan,
            "a_win_pct_ci95_lower":np.percentile(valid,2.5) if len(valid)>=.95*BOOTSTRAP_REPLICATES else np.nan,
            "a_win_pct_ci95_upper":np.percentile(valid,97.5) if len(valid)>=.95*BOOTSTRAP_REPLICATES else np.nan,
            "small_sample":len(known)<SMALL_SAMPLE_THRESHOLDS["matches"]})
validated_h2h_summary = pd.DataFrame(h2h_rows)
assert validated_h2h_summary["a_verified_wins"].add(validated_h2h_summary["b_verified_wins"]).eq(validated_h2h_summary["known_outcomes"]).all()
final_player_scorecard =
core_kpi_wide.merge(clutch_summary[["player","surface","clutch_index_full","clutch_index_full_ci95_lower","clutch_index_full_ci95_upper","common_ci_matches","st
atus"]].rename(columns={"status":"clutch_status"}),
    on=["player","surface"],how="left",validate="one_to_one")
style_columns = ["player","surface","approach_frequency_pct","net_point_frequency_pct","net_point_success_pct","serve_volley_frequency_pct",
    "short_rally_share_pct","long_rally_share_pct","groundstroke_share_pct","slice_share_pct","drop_shot_share_pct","source_winner_rate_pct",
    "source_unforced_error_rate_pct","shot_family_diversity","directional_diversity"]
final_player_scorecard = final_player_scorecard.merge(style_profile_wide[style_columns],on=["player","surface"],how="left",validate="one_to_one")
sas_scorecard =
surface_adaptability_summary_v2[["player","surface_adaptability_score","surface_adaptability_score_ci95_lower","surface_adaptability_score_ci95_upper"]].assign(
surface="Overall")
final_player_scorecard = final_player_scorecard.merge(sas_scorecard,on=["player","surface"],how="left",validate="one_to_one")
final_player_scorecard["tactical_evolution"] = "Supported component trends: Section VIII"
final_player_scorecard["tactical_innovation_composite"] = "Unavailable: invention, radar variation, and equipment change not established"
final_summary_fields = ["charted_matches","verified_wins","known_outcome_matches","unknown_outcome_matches","earliest_charted_date","latest_charted_date",

"handedness","weeks_at_no_1","match_win_pct","dominance_ratio","clutch_index_full","surface_adaptability_score","first_serve_win_pct","second_serve_win_pct",
    "ace_pct","double_fault_pct","return_point_win_pct","break_point_save_pct","break_point_conversion_pct","service_hold_pct","return_break_pct",

"approach_frequency_pct","serve_volley_frequency_pct","short_rally_share_pct","long_rally_share_pct","shot_family_diversity","directional_diversity","tactical_e
volution"]
final_overall_player_summary = final_player_scorecard.loc[final_player_scorecard["surface"].eq("Overall")].set_index("player").reindex(BIG_THREE)
[final_summary_fields].T
assert len(final_player_scorecard)==12
assert not final_player_scorecard.duplicated(["player","surface"]).any()
assert final_player_scorecard.loc[final_player_scorecard["surface"].ne("Overall"),"surface_adaptability_score"].isna().all()
print("Validated charted rivalry context:")
with pd.option_context("display.max_columns",None):
```

```
    report_table(validated_h2h_summary.round(3))
print("Overall consolidated player summary (percentages 0–100; DR and CI retain their own units):")
with pd.option_context("display.max_rows",None,"display.max_columns",None):
    report_table(final_overall_player_summary)
print("Full overall/surface scorecard:")
with pd.option_context("display.max_columns",None):
    report_table(final_player_scorecard.round(4))
```

```
Validated charted rivalry context:
```

surface	player_a	player_b	charted_meetings	known_outcomes	unknown_outcomes	a_verified_wins	b_verified_wins
Overall	Roger Federer	Rafael Nadal	35	35	0	13	22
Clay	Roger Federer	Rafael Nadal	16	16	0	2	14
Grass	Roger Federer	Rafael Nadal	4	4	0	3	1
Hard	Roger Federer	Rafael Nadal	15	15	0	8	7
Overall	Roger Federer	Novak Djokovic	47	45	2	19	26
Clay	Roger Federer	Novak Djokovic	8	7	1	3	4
Grass	Roger Federer	Novak Djokovic	4	4	0	1	3
Hard	Roger Federer	Novak Djokovic	35	34	1	15	19
Overall	Rafael Nadal	Novak Djokovic	52	49	3	21	28
Clay	Rafael Nadal	Novak Djokovic	24	23	1	14	9
Grass	Rafael Nadal	Novak Djokovic	4	3	1	1	2
Hard	Rafael Nadal	Novak Djokovic	24	23	1	6	17

surface	a_win_pct	a_win_pct_ci95_lower	a_win_pct_ci95_upper	small_sample
Overall	37.143	21.875	52.946	False
Clay	12.500	0.000	30.769	True
Grass	75.000	0.000	100.000	True
Hard	53.333	26.625	80.000	True
Overall	42.222	27.901	57.143	False
Clay	42.857	0.000	83.393	True
Grass	25.000	0.000	100.000	True
Hard	44.118	27.500	61.368	False
Overall	42.857	28.845	56.864	False
Clay	60.870	39.282	80.000	True
Grass	33.333	Unavailable	Unavailable	True
Hard	26.087	9.091	45.455	True

```
Overall consolidated player summary (percentages 0–100; DR and CI retain their own units):
```

player	Roger Federer	Rafael Nadal	Novak Djokovic
charted_matches	723	425	553
verified_wins	559	298	404
known_outcome_matches	709	412	526
unknown_outcome_matches	14	13	27
earliest_charted_date	1998-07-04 00:00:00	2003-05-15 00:00:00	2005-01-17 00:00:00
latest_charted_date	2021-07-05 00:00:00	2024-11-19 00:00:00	2026-03-11 00:00:00
handedness	Right-handed	Left-handed	Right-handed
weeks_at_no_1	Unavailable in supplied files	Unavailable in supplied files	Unavailable in supplied files
match_win_pct	78.843	72.330	76.806
dominance_ratio	1.277	1.183	1.221
clutch_index_full	0.639	0.608	0.643
surface_adaptability_score	83.810	77.077	85.575
first_serve_win_pct	76.994	70.554	73.271
second_serve_win_pct	56.457	55.787	54.934
ace_pct	10.178	4.277	7.195
double_fault_pct	2.198	2.317	2.817
return_point_win_pct	39.228	40.174	40.448
break_point_save_pct	65.794	63.858	63.921
break_point_conversion_pct	40.639	42.853	42.726
service_hold_pct	88.207	83.232	84.716
return_break_pct	25.683	28.314	28.138
approach_frequency_pct	15.248	6.728	9.047
serve_volley_frequency_pct	10.211	1.375	2.472
short_rally_share_pct	68.292	59.383	60.282
long_rally_share_pct	9.823	15.831	16.275
shot_family_diversity	0.410	0.257	0.280
directional_diversity	0.860	0.859	0.830
tactical_evolution	Supported component trends: Section VIII	Supported component trends: Section VIII	Supported component trends: Section VIII

Full overall/surface scorecard:

▼ Complete evidence table: 12 rows × 51 columns

player	surface	charted_matches	verified_wins	known_outcome_matches	unknown_outcome_matches	matches_with_overview	matches_with_points	earliest_charted_date
Roger Federer	Overall	723	559	709	14	722	722	1998-07-04
Roger Federer	Clay	130	87	126	4	130	130	1999-07-16
Roger Federer	Grass	119	97	116	3	119	119	1998-07-04
Roger Federer	Hard	474	375	467	7	473	473	1998-10-05
Rafael Nadal	Overall	425	298	412	13	424	423	2003-05-15
Rafael Nadal	Clay	166	139	164	2	165	166	2003-05-15
Rafael Nadal	Grass	40	22	37	3	40	39	2003-06-27
Rafael Nadal	Hard	219	137	211	8	219	218	2004-03-28
Novak Djokovic	Overall	553	404	526	27	550	552	2005-01-17
Novak Djokovic	Clay	129	87	124	5	128	129	2006-04-19
Novak Djokovic	Grass	58	44	56	2	58	58	2007-07-05
Novak Djokovic	Hard	366	273	346	20	364	365	2005-01-17

player	surface	latest_charted_date	handedness	handedness_resolution	handedness_conflicting_records	weeks_at_no_1	ace_pct	break_point_conversion_pct
Roger Federer	Overall	2021-07-05	Right-handed	consistent recognized source records	0	Unavailable in supplied files	10.178	40.639
Roger Federer	Clay	2021-06-05	Right-handed	consistent recognized source records	0	Unavailable in supplied files	7.295	40.557
Roger Federer	Grass	2021-07-05	Right-handed	consistent recognized source records	0	Unavailable in supplied files	11.886	39.719
Roger Federer	Hard	2021-03-10	Right-handed	consistent recognized source records	0	Unavailable in supplied files	10.523	40.902
Rafael Nadal	Overall	2024-11-19	Left-handed	documented source consensus; contradictions retained	3	Unavailable in supplied files	4.277	42.853
Rafael Nadal	Clay	2024-07-29	Left-handed	documented source consensus; contradictions retained	3	Unavailable in supplied files	2.609	45.841
Rafael Nadal	Grass	2022-07-04	Left-handed	documented source consensus; contradictions retained	3	Unavailable in supplied files	7.008	38.226
Rafael Nadal	Hard	2024-11-19	Left-handed	documented source consensus; contradictions retained	3	Unavailable in supplied files	4.786	40.926

player	surface	latest_charted_date	handedness	handedness_resolution	handedness_conflicting_records	weeks_at_no_1	ace_pct	break_point_conversion_pct
Novak Djokovic	Overall	2026-03-11	Right-handed	consistent recognized source records	0	Unavailable in supplied files	7.195	42.726
Novak Djokovic	Clay	2025-06-06	Right-handed	consistent recognized source records	0	Unavailable in supplied files	4.891	42.416
Novak Djokovic	Grass	2025-07-11	Right-handed	consistent recognized source records	0	Unavailable in supplied files	9.272	37.633
Novak Djokovic	Hard	2026-03-11	Right-handed	consistent recognized source records	0	Unavailable in supplied files	7.612	43.813

player	surface	break_point_save_pct	dominance_ratio	double_fault_pct	first_serve_in_pct	first_serve_win_pct	match_win_pct	point_win_pct
Roger Federer	Overall	65.794	1.277	2.198	62.452	76.994	78.843	53.755
Roger Federer	Clay	62.500	1.162	1.926	61.092	72.981	69.048	52.577
Roger Federer	Grass	71.816	1.360	1.584	65.404	79.736	83.621	54.096
Roger Federer	Hard	65.710	1.294	2.466	61.969	77.302	80.300	54.003
Rafael Nadal	Overall	63.858	1.183	2.317	68.877	70.554	72.330	52.751
Rafael Nadal	Clay	65.306	1.283	2.139	70.496	68.970	84.756	54.650
Rafael Nadal	Grass	67.755	1.129	1.940	69.108	73.649	59.459	51.647
Rafael Nadal	Hard	61.932	1.123	2.530	67.710	70.950	64.929	51.669
Novak Djokovic	Overall	63.921	1.221	2.817	65.253	73.271	76.806	53.422
Novak Djokovic	Clay	60.959	1.142	2.504	65.634	69.637	70.161	52.543
Novak Djokovic	Grass	68.449	1.209	2.821	66.986	76.478	78.571	52.845
Novak Djokovic	Hard	64.409	1.257	2.930	64.746	73.939	78.902	53.873

player	surface	return_break_pct	return_point_win_pct	second_serve_win_pct	service_hold_pct	service_point_win_pct	clutch_index_full	clutch_index_full_ci95_lower
Roger Federer	Overall	25.683	39.228	56.457	88.207	69.283	0.639	0.614
Roger Federer	Clay	27.009	39.906	54.401	82.450	65.752	0.600	0.541
Roger Federer	Grass	21.322	36.998	59.662	92.850	72.791	0.677	0.614
Roger Federer	Hard	26.630	39.713	56.207	88.462	69.279	0.641	0.609
Rafael Nadal	Overall	28.314	40.174	55.787	83.232	65.958	0.608	0.572
Rafael Nadal	Clay	37.954	44.823	55.621	81.859	65.032	0.671	0.614
Rafael Nadal	Grass	18.868	35.279	57.535	88.047	68.671	0.568	0.477
Rafael Nadal	Hard	23.921	38.151	55.501	83.038	65.961	0.577	0.529
Novak Djokovic	Overall	28.138	40.448	54.934	84.716	66.900	0.643	0.614
Novak Djokovic	Clay	29.588	41.074	53.455	79.629	64.076	0.616	0.564

player	surface	return_break_pct	return_point_win_pct	second_serve_win_pct	service_hold_pct	service_point_win_pct	clutch_index_full	clutch_index_full_ci95_lower
Novak Djokovic	Grass	21.996	37.680	53.323	87.964	68.834	0.675	0.596
Novak Djokovic	Hard	28.929	40.812	55.789	85.894	67.540	0.651	0.614

player	surface	clutch_index_full_ci95_upper	common_ci_matches	clutch_status	approach_frequency_pct	net_point_frequency_pct	net_point_success_pct	serve_volley_frequency_pct
Roger Federer	Overall	0.664	687	available	15.248	16.938	70.593	10.211
Roger Federer	Clay	0.659	122	available	14.597	16.270	69.446	7.621
Roger Federer	Grass	0.736	116	small sample	16.196	17.680	71.814	15.022
Roger Federer	Hard	0.674	449	available	15.157	16.908	70.528	9.553
Rafael Nadal	Overall	0.640	392	available	6.728	9.074	74.833	1.375
Rafael Nadal	Clay	0.721	153	available	5.869	8.611	75.144	0.667
Rafael Nadal	Grass	0.661	36	small sample	8.390	10.901	75.437	1.674
Rafael Nadal	Hard	0.625	203	available	6.920	8.955	74.450	1.796
Novak Djokovic	Overall	0.671	519	available	9.047	11.970	70.249	2.473
Novak Djokovic	Clay	0.677	122	available	8.809	12.351	69.388	1.896
Novak Djokovic	Grass	0.745	56	small sample	10.336	12.547	72.005	3.481
Novak Djokovic	Hard	0.688	341	available	8.866	11.708	70.182	2.473

player	surface	short_rally_share_pct	long_rally_share_pct	groundstroke_share_pct	slice_share_pct	drop_shot_share_pct	source_winner_rate_pct	source_unforced_error_rate_pct
Roger Federer	Overall	68.292	9.823	71.843	19.252	0.969	19.753	16.555
Roger Federer	Clay	63.657	12.243	76.505	15.325	1.520	18.670	18.737
Roger Federer	Grass	74.000	6.632	67.682	21.690	0.880	20.427	13.110
Roger Federer	Hard	67.954	10.063	71.360	19.927	0.811	19.866	16.971
Rafael Nadal	Overall	59.383	15.831	85.980	9.516	0.813	15.139	13.921
Rafael Nadal	Clay	53.289	19.402	89.413	6.291	1.052	15.074	14.480
Rafael Nadal	Grass	68.809	9.484	79.027	15.008	1.124	17.101	11.817
Rafael Nadal	Hard	61.507	14.778	84.399	11.197	0.553	14.713	14.042
Novak Djokovic	Overall	60.282	16.275	85.295	8.468	1.547	15.878	15.636
Novak Djokovic	Clay	55.724	18.656	87.652	5.378	2.375	16.267	17.266
Novak Djokovic	Grass	65.670	12.700	80.801	12.343	1.436	16.333	13.882
Novak Djokovic	Hard	60.845	16.142	85.160	9.019	1.226	15.638	15.414

player	surface	shot_family_diversity	directional_diversity	surface_adaptability_score	surface_adaptability_score_ci95_lower	surface_adaptability_score_ci95_upper	tactical_evolution	tactical_innovation_composite
Roger Federer	Overall	0.410	0.860	83.810	77.811	88.900	Supported component trends: Section VIII	Unavailable: invention, radar variation, and equipment change not established
Roger Federer	Clay	0.373	0.868	Unavailable	Unavailable	Unavailable	Supported component trends: Section VIII	Unavailable: invention, radar variation, and equipment change not established
Roger Federer	Grass	0.451	0.866	Unavailable	Unavailable	Unavailable	Supported component trends: Section VIII	Unavailable: invention, radar variation, and equipment change not established
Roger Federer	Hard	0.410	0.856	Unavailable	Unavailable	Unavailable	Supported component trends: Section VIII	Unavailable: invention, radar variation, and equipment change not established
Rafael Nadal	Overall	0.257	0.859	77.077	66.502	83.046	Supported component trends: Section VIII	Unavailable: invention, radar variation, and equipment change not established
Rafael Nadal	Clay	0.217	0.861	Unavailable	Unavailable	Unavailable	Supported component trends: Section VIII	Unavailable: invention, radar variation, and equipment change not established
Rafael Nadal	Grass	0.335	0.859	Unavailable	Unavailable	Unavailable	Supported component trends: Section VIII	Unavailable: invention, radar variation, and equipment change not established
Rafael Nadal	Hard	0.270	0.856	Unavailable	Unavailable	Unavailable	Supported component trends: Section VIII	Unavailable: invention, radar variation, and equipment change not established
Novak Djokovic	Overall	0.280	0.830	85.575	79.299	89.564	Supported component trends: Section VIII	Unavailable: invention, radar variation, and equipment change not established
Novak Djokovic	Clay	0.258	0.839	Unavailable	Unavailable	Unavailable	Supported component trends: Section VIII	Unavailable: invention, radar variation, and equipment change not established

player	surface	shot_family_diversity	directional_diversity	surface_adaptability_score	surface_adaptability_score_ci95_lower	surface_adaptability_score_ci95_upper	tactical_evolution	tactical_innovation_composite
Novak Djokovic	Grass	0.328	0.823	Unavailable	Unavailable	Unavailable	Supported component trends: Section VIII	Unavailable: invention, radar variation, and equipment change not established
Novak Djokovic	Hard	0.277	0.827	Unavailable	Unavailable	Unavailable	Supported component trends: Section VIII	Unavailable: invention, radar variation, and equipment change not established

9.2.1 Evidence: direct matchups and source-issue disposition

In [29]:
```
h2h_overall=validated_h2h_summary.loc[validated_h2h_summary["surface"].eq("Overall")]
report_table(h2h_overall,caption="Direct rivalries in charted meetings; a/b order is explicit",expanded=True)
lines=["**Observed rivalry result.**"]
for row in h2h_overall.itertuples(index=False):
    lines.append(f"- {row.player_a} vs {row.player_b}: {row.a_verified_wins}–{row.b_verified_wins} in {row.known_outcomes} verified outcomes;
{row.unknown_outcomes} unknown of {row.charted_meetings} charted meetings. The first player's win rate is {row.a_win_pct:.1f}%
({interval_text(row.a_win_pct_ci95_lower,row.a_win_pct_ci95_upper,1)}).")
lines.append("\n**Tennis interpretation.** Overall performance against an uneven field and direct success against a particular rival answer different questions.
Serve/return matchups and surface composition are plausible explanations for different ordering; the surface rivalry table provides the measured context. These
are not complete historical rivalry records, and tiny grass samples make some intervals extremely wide or unavailable.")
lines.append("\n**Alternative explanations.** Rivalry meetings are selected by tournament progression and career overlap. The same match can inform two player
profiles, so the paired bootstrap retains both observations. H2H is contextual evidence and receives no additional composite weight, avoiding another
overlapping input.")
display(Markdown("\n\n".join(lines)))
# Disposition distinguishes genuine limitations from failed software invariants.
source_issue_disposition=pd.DataFrame([
    {"issue":"Equivalent hand labels/whitespace","status":"Resolved","affected_count":int(handedness_source_records["raw_representation"].str.contains("
").sum()),"handling":"Unambiguous profile normalization; raw values retained"},
    {"issue":"Contradictory handedness codes","status":"Known limitation","affected_count":len(handedness_conflicts),"handling":"Documented >=99% player-profile
consensus; full exceptions retained; source directions not recomputed"},
    {"issue":"Conflicting match identifiers","status":"Known limitation","affected_count":len(conflicted_match_ids),"handling":"Entire affected match
quarantined across sources"},
    {"issue":"Exact duplicate point rows","status":"Resolved","affected_count":int(sum(len(audit_exact_duplicate_rows[k]) for k in
point_source_keys)),"handling":"Exact repeats removed; retained source provenance"},
    {"issue":"Conflicting point keys","status":"Known limitation","affected_count":len(point_conflict_keys),"handling":"All affected matches quarantined; no
arbitrary row selection"},
    {"issue":"Conflicting statistical player-match keys","status":"Known limitation","affected_count":len(all_player_match_conflicts),"handling":"Shared
statistical exclusion; none of the affected keys belongs to the Big Three"},
    {"issue":"Unknown Big Three outcomes","status":"Known limitation","affected_count":int(canonical_master["winner"].isna().sum()),"handling":"Missing, never
losses; valid unrelated statistics retained"},
    {"issue":"Source/point field discrepancies","status":"Known limitation","affected_count":len(canonical_reconciliation_flags),"handling":"Metric-specific
eligibility; source values retained without forced reconciliation"},
    {"issue":"Contradictory best-of metadata","status":"Known limitation","affected_count":len(canonical_format_review),"handling":"No historical override;
contradicted outcome/deciding contexts excluded"},
    {"issue":"Invalid regular-game groups","status":"Known
limitation","affected_count":int((~kpi_games["is_tiebreak_point"]&~kpi_games["eligible_regular_game"]).sum()),"handling":"Hold/break requires valid starts,
```

```
transitions, single server and terminal game"},
    {"issue":"Unvalidated tiebreak groups","status":"Known
limitation","affected_count":int((~validated_tiebreaks["tb_complete_validated"]).sum()),"handling":"Excluded from completed-event cohorts; missing not
fabricated"},
    {"issue":"Unknown/other Big Three surfaces","status":"Known
limitation","affected_count":int((~canonical_master["surface"].isin(PROJECT_SURFACES)).sum()),"handling":"Retained in Overall; SAS restricted to named three-
surface universe"},
    {"issue":"Unavailable ServeInfluence values and other numeric flags","status":"Known
limitation","affected_count":len(numeric_quality_flags),"handling":"Explicit flags; unavailable entries do not become zero; not headline model inputs"},
    {"issue":"SnV source/code discrepancies","status":"Known limitation","affected_count":len(snv_code_source_review),"handling":"Excluded from affected SnV
rates; explicit code/complement recovery is separate"},
    {"issue":"Missing ranking history and complete career results","status":"Known limitation","affected_count":3,"handling":"No extrapolation; ranking weeks
remain nullable/unavailable"},
])
report_table(source_issue_disposition,caption="Issue disposition: retained source limitations are distinct from software failures",expanded=True)
```

Direct rivalries in charted meetings; a/b order is explicit

	surface	player_a	player_b	charted_meetings	known_outcomes	unknown_outcomes	a_verified_wins	b_verified_wins
0	Overall	Roger Federer	Rafael Nadal	35	35	0	13	22
4	Overall	Roger Federer	Novak Djokovic	47	45	2	19	26
8	Overall	Rafael Nadal	Novak Djokovic	52	49	3	21	28

	surface	a_win_pct	a_win_pct_ci95_lower	a_win_pct_ci95_upper	small_sample
0	Overall	37.143	21.875	52.946	False
4	Overall	42.222	27.901	57.143	False
8	Overall	42.857	28.845	56.864	False

Observed rivalry result.

- Roger Federer vs Rafael Nadal: 13–22 in 35 verified outcomes; 0 unknown of 35 charted meetings. The first player's win rate is 37.1% (21.9–52.9).
- Roger Federer vs Novak Djokovic: 19–26 in 45 verified outcomes; 2 unknown of 47 charted meetings. The first player's win rate is 42.2% (27.9–57.1).
- Rafael Nadal vs Novak Djokovic: 21–28 in 49 verified outcomes; 3 unknown of 52 charted meetings. The first player's win rate is 42.9% (28.8–56.9).

Tennis interpretation. Overall performance against an uneven field and direct success against a particular rival answer different questions. Serve/return matchups and surface composition are plausible explanations for different ordering; the surface rivalry table provides the measured context. These are not complete historical rivalry records, and tiny grass samples make some intervals extremely wide or unavailable.

Alternative explanations. Rivalry meetings are selected by tournament progression and career overlap. The same match can inform two player profiles, so the paired bootstrap retains both observations. H2H is contextual evidence and receives no additional composite weight, avoiding another overlapping input.

Issue disposition: retained source limitations are distinct from software failures

issue	status	affected_count	handling
Equivalent hand labels/whitespace	Resolved	3	Unambiguous profile normalization; raw values retained
Contradictory handedness codes	Known limitation	3	Documented >=99% player-profile consensus; full exceptions retained; source directions not recomputed
Conflicting match identifiers	Known limitation	1	Entire affected match quarantined across sources
Exact duplicate point rows	Resolved	1625	Exact repeats removed; retained source provenance
Conflicting point keys	Known limitation	486	All affected matches quarantined; no arbitrary row selection
Conflicting statistical player-match keys	Known limitation	16	Shared statistical exclusion; none of the affected keys belongs to the Big Three
Unknown Big Three outcomes	Known limitation	54	Missing, never losses; valid unrelated statistics retained
Source/point field discrepancies	Known limitation	91	Metric-specific eligibility; source values retained without forced reconciliation
Contradictory best-of metadata	Known limitation	13	No historical override; contradicted outcome/deciding contexts excluded
Invalid regular-game groups	Known limitation	24	Hold/break requires valid starts, transitions, single server and terminal game
Unvalidated tiebreak groups	Known limitation	7	Excluded from completed-event cohorts; missing not fabricated
Unknown/other Big Three surfaces	Known limitation	0	Retained in Overall; SAS restricted to named three-surface universe
Unavailable ServeInfluence values and other numeric flags	Known limitation	12138	Explicit flags; unavailable entries do not become zero; not headline model inputs
SnV source/code discrepancies	Known limitation	10	Excluded from affected SnV rates; explicit code/complement recovery is separate
Missing ranking history and complete career results	Known limitation	3	No extrapolation; ranking weeks remain nullable/unavailable

9.3 Declared comparison model, scaling, and dependence

The overall comparison uses four validated measures: known-outcome match win rate, DR, full common-cohort CI, and primary SAS. Totals are excluded because charting exposure differs; athleticism, style diversity, tactical change, unsupported innovation inputs, and unavailable ranking weeks are not scored. Metrics retain their own disclosed eligible samples; only the CI's components share its internal common cohort. The comparison retains each player's individual observed window.

Primary fixed scaling: win rate and CI already lie in [0,1]; SAS is divided by 100; DR is mapped monotonically to **DR / (1 + DR)**. This transformation is a disclosed scoring scale, not a new measured tennis KPI. No fitted reference population or sample extrema are needed. A weighted arithmetic mean is displayed ×100. Equal weights mean equal influence for equal changes on these fixed scales, not equal empirical variance or independent information. DR overlaps serving/returning performance; CI includes serving outcomes; SAS includes win rates. These are overlapping perspectives, not four independent votes.

Five declared scenarios emphasize balance, winning, dominance, clutch, or adaptability. Sensitivity includes a 0.1-step four-weight simplex grid, leave-one-metric-out scores, and an alternative within-Big-Three min–max point-estimate scale. Min–max sensitivity uses only these three players and can amplify small differences; it is not the primary score and has no separate uncertainty claim. Grid-leading share describes the chosen preference grid, **not a probability of greatness**.

All primary score draws reuse the same joint match bootstrap, with eligibility and scales fixed and derived metrics recomputed from their component draws. First-place frequencies are conditional bootstrap rank frequencies, not posterior probabilities that a player is the GOAT. Ties split rank credit. Score-difference intervals and conditional pairwise frequencies are shown directly; none repairs missing outcomes or chart-selection bias. Separate illustrative surface comparisons omit SAS and equally weight surface win rate, transformed DR, and full surface CI; sparse CI components remain flagged.

In [30]:
```
COMPARISON_METRICS = ["winning", "dominance", "clutch", "adaptability"]
COMPARISON_SCENARIOS = {
    "Balanced": {"winning": .25, "dominance": .25, "clutch": .25, "adaptability": .25},
    "Winning priority": {"winning": .55, "dominance": .15, "clutch": .15, "adaptability": .15},
    "Dominance priority": {"winning": .20, "dominance": .50, "clutch": .15, "adaptability": .15},
    "Clutch priority": {"winning": .20, "dominance": .15, "clutch": .50, "adaptability": .15},
    "Adaptability priority": {"winning": .20, "dominance": .15, "clutch": .15, "adaptability": .50},
}
comparison_weights = pd.DataFrame(COMPARISON_SCENARIOS).T.rename_axis("scenario").reset_index()
assert comparison_weights[COMPARISON_METRICS].ge(0).all().all()
assert np.allclose(comparison_weights[COMPARISON_METRICS].sum(axis=1), 1)

# Recover already-computed core bootstrap metrics; no second resampling or clipping.
core_metric_draws = {}
for index, metric, columns in bootstrap_descriptors:
```

```
    if metric not in ["match_win_pct", "dominance_ratio"]:
        continue
    info = core_kpi_long.loc[index]
    with np.errstate(divide="ignore", invalid="ignore"):
        if metric == "dominance_ratio":
            rw, rp, sw, sp = [bootstrap_totals[:, column] for column in columns]
            loss = 1-sw/sp
            values = np.where((rp > 0) & (sp > 0) & (loss > 0), (rw/rp)/loss, np.nan)
        else:
            wins, matches = [bootstrap_totals[:, column] for column in columns]
            values = np.where(matches > 0, wins/matches, np.nan)
    core_metric_draws[(info["player"], info["surface"], metric)] = values
comparison_input_rows, comparison_draw_arrays = [], []
for player in BIG_THREE:
    core = core_kpi_overall.loc[core_kpi_overall["player"].eq(player)].iloc[0]
    ci = clutch_overall.loc[clutch_overall["player"].eq(player)].iloc[0]
    sas = surface_adaptability_summary_v2.loc[surface_adaptability_summary_v2["player"].eq(player)].iloc[0]
    raw = [core["match_win_pct"]/100, core["dominance_ratio"], ci["clutch_index_full"], sas["surface_adaptability_score"]]
    scaled = [raw[0], raw[1]/(1+raw[1]), raw[2], raw[3]/100]
    dr_draws = core_metric_draws[(player, "Overall", "dominance_ratio")]
    draws = np.column_stack([core_metric_draws[(player, "Overall", "match_win_pct")], dr_draws/(1+dr_draws),
        clutch_bootstrap_distributions[(player, "Overall")], adaptability_bootstraps[player]["sas"]/100])
    comparison_draw_arrays.append(draws)
    counts = [int(core["known_outcome_matches"]), int(core_kpi_long.loc[core_kpi_long["player"].eq(player)
        & core_kpi_long["surface"].eq("Overall") & core_kpi_long["metric"].eq("dominance_ratio"), "eligible_matches"].iloc[0]),
        int(ci["common_ci_matches"]), int(sas["known_outcome_matches"])]
    for metric, value, score_value, count in zip(COMPARISON_METRICS, raw, scaled, counts):
        comparison_input_rows.append({"player": player, "metric": metric, "raw_value": value, "fixed_scale_value": score_value,
            "contributing_matches": count, "earliest_charted_date": core["earliest_charted_date"], "latest_charted_date": core["latest_charted_date"]})
comparison_inputs = pd.DataFrame(comparison_input_rows)
comparison_points = comparison_inputs.pivot(index="player", columns="metric", values="fixed_scale_value").reindex(index=BIG_THREE,
columns=COMPARISON_METRICS).to_numpy()
comparison_raw_points = comparison_inputs.pivot(index="player", columns="metric", values="raw_value").reindex(index=BIG_THREE,
columns=COMPARISON_METRICS).to_numpy()
comparison_draws = np.stack(comparison_draw_arrays, axis=1)  # replicate × player × metric
comparison_valid_draw = np.isfinite(comparison_draws).all(axis=(1, 2))
comparison_draws_valid = comparison_draws[comparison_valid_draw]
if comparison_valid_draw.sum() < .95*BOOTSTRAP_REPLICATES:
    raise ValueError("Too few fully paired score draws; investigate metric coverage before ranking")
assert np.isfinite(comparison_points).all()
assert ((comparison_points >= 0) & (comparison_points <= 1)).all()

def fractional_rank_frequencies(scores):
    # Each player occupies a span of ranks when tied; split credit evenly over that span.
    greater = (scores[:, None, :] > scores[:, :, None]+1e-12).sum(axis=2)
    ties = np.isclose(scores[:, None, :], scores[:, :, None], rtol=0, atol=1e-12).sum(axis=2)
    return np.column_stack([np.mean(np.where((greater < rank) & (rank <= greater+ties), 1/ties, 0), axis=0)
        for rank in range(1, scores.shape[1]+1)])

scenario_rows, contribution_rows, score_difference_rows = [], [], []
comparison_score_bootstraps = {}
for scenario, weights in COMPARISON_SCENARIOS.items():
    w = np.array([weights[metric] for metric in COMPARISON_METRICS])
    points = 100*(comparison_points @ w)
    draws = 100*(comparison_draws_valid @ w)
    comparison_score_bootstraps[scenario] = draws
    rank_frequencies = fractional_rank_frequencies(draws)
```

```
    for index, player in enumerate(BIG_THREE):
        scenario_rows.append({"scenario": scenario, "player": player, "score": points[index],
            "score_boot_median": np.median(draws[:, index]), "ci95_lower": np.percentile(draws[:, index], 2.5),
            "ci95_upper": np.percentile(draws[:, index], 97.5), "point_estimate_rank": int(1+(points > points[index]+1e-12).sum()),
            "bootstrap_first_place_pct": 100*rank_frequencies[index, 0], "bootstrap_second_place_pct": 100*rank_frequencies[index, 1],
            "bootstrap_third_place_pct": 100*rank_frequencies[index, 2], "paired_valid_bootstraps": len(draws)})
        for metric_index, metric in enumerate(COMPARISON_METRICS):
            contribution_rows.append({"scenario": scenario, "player": player, "metric": metric, "weight": w[metric_index],
                "fixed_scale_value": comparison_points[index, metric_index], "score_contribution": 100*w[metric_index]*comparison_points[index, metric_index]})
    for a, b in [(0, 1), (0, 2), (1, 2)]:
        difference = draws[:, a]-draws[:, b]
        score_difference_rows.append({"scenario": scenario, "comparison": BIG_THREE[a]+" minus "+BIG_THREE[b],
            "point_score_difference": points[a]-points[b], "ci95_lower": np.percentile(difference, 2.5), "ci95_upper": np.percentile(difference, 97.5),
            "bootstrap_a_ahead_pct": 100*np.mean((difference > 1e-12)+.5*np.isclose(difference, 0, rtol=0, atol=1e-12))})
comparison_scores = pd.DataFrame(scenario_rows)
comparison_contributions = pd.DataFrame(contribution_rows)
comparison_pairwise_differences = pd.DataFrame(score_difference_rows)

# Full nonnegative 0.1-step simplex, including single-metric corners; not a probability distribution over true preferences.
grid_rows = []
for a in range(11):
    for b in range(11-a):
        for c in range(11-a-b):
            w = np.array([a, b, c, 10-a-b-c])/10
            scores = 100*(comparison_points @ w)
            leaders = np.isclose(scores, scores.max(), rtol=0, atol=1e-12)
            row = {metric+"_weight": w[i] for i, metric in enumerate(COMPARISON_METRICS)}
            row["leaders"] = "; ".join(p for p, lead in zip(BIG_THREE, leaders) if lead)
            for i, player in enumerate(BIG_THREE):
                row[player+"_score"] = scores[i]
                row[player+"_lead_credit"] = int(leaders[i])/leaders.sum()
            grid_rows.append(row)
comparison_weight_grid = pd.DataFrame(grid_rows)
comparison_grid_summary = pd.DataFrame([{"player": p, "grid_settings": len(comparison_weight_grid),
    "fractional_grid_lead_pct": 100*comparison_weight_grid[p+"_lead_credit"].mean()} for p in BIG_THREE])

raw_min, raw_max = comparison_raw_points.min(axis=0), comparison_raw_points.max(axis=0)
span = raw_max-raw_min
alternative_minmax_points = np.divide(comparison_raw_points-raw_min, span, out=np.full_like(comparison_raw_points, .5), where=span > 0)
normalization_audit = comparison_inputs.copy()
for j, metric in enumerate(COMPARISON_METRICS):
    normalization_audit.loc[normalization_audit["metric"].eq(metric), "three_player_min"] = raw_min[j]
    normalization_audit.loc[normalization_audit["metric"].eq(metric), "three_player_max"] = raw_max[j]
    for i, player in enumerate(BIG_THREE):
        normalization_audit.loc[normalization_audit["metric"].eq(metric) & normalization_audit["player"].eq(player), "alternative_minmax_value"] =
alternative_minmax_points[i, j]
scaling_sensitivity_rows = []
for scenario, weights in COMPARISON_SCENARIOS.items():
    w = np.array([weights[m] for m in COMPARISON_METRICS])
    values = 100*(alternative_minmax_points @ w)
    for i, player in enumerate(BIG_THREE):
        scaling_sensitivity_rows.append({"scenario": scenario, "player": player, "alternative_minmax_score": values[i],
            "alternative_point_rank": int(1+(values > values[i]+1e-12).sum())})
comparison_scaling_sensitivity = pd.DataFrame(scaling_sensitivity_rows)
leave_one_out_rows = []
for omitted in COMPARISON_METRICS:
    w = np.array([0 if m == omitted else 1/3 for m in COMPARISON_METRICS])
```

```
    values = 100*(comparison_points @ w)
    for i, player in enumerate(BIG_THREE):
        leave_one_out_rows.append({"omitted_metric": omitted, "player": player, "score": values[i],
            "point_rank": int(1+(values > values[i]+1e-12).sum())})
comparison_leave_one_out = pd.DataFrame(leave_one_out_rows)

# Illustrative surface scores use three metrics only; adaptability has no single-surface interpretation.
surface_comparison_rows = []
for surface in PROJECT_SURFACES:
    point_rows, draw_rows = [], []
    for player in BIG_THREE:
        core = core_kpi_by_surface.loc[core_kpi_by_surface["player"].eq(player) & core_kpi_by_surface["surface"].eq(surface)].iloc[0]
        ci = clutch_by_surface.loc[clutch_by_surface["player"].eq(player) & clutch_by_surface["surface"].eq(surface)].iloc[0]
        dr = core_metric_draws[(player, surface, "dominance_ratio")]
        point_rows.append([core["match_win_pct"]/100, core["dominance_ratio"]/(1+core["dominance_ratio"]), ci["clutch_index_full"]])
        draw_rows.append(np.column_stack([core_metric_draws[(player, surface, "match_win_pct")], dr/(1+dr), clutch_bootstrap_distributions[(player, surface)]]))
    points = np.array(point_rows).mean(axis=1)*100
    draws = np.stack(draw_rows, axis=1).mean(axis=2)*100
    valid = np.isfinite(draws).all(axis=1)
    frequencies = fractional_rank_frequencies(draws[valid]) if valid.any() else np.full((3,3), np.nan)
    for i, player in enumerate(BIG_THREE):
        ci = clutch_by_surface.loc[clutch_by_surface["player"].eq(player) & clutch_by_surface["surface"].eq(surface)].iloc[0]
        surface_comparison_rows.append({"player": player, "surface": surface, "illustrative_score": points[i], "small_ci_sample": ci["small_sample"],
            "ci95_lower": np.percentile(draws[valid, i], 2.5) if valid.sum() >= .95*BOOTSTRAP_REPLICATES else np.nan,
            "ci95_upper": np.percentile(draws[valid, i], 97.5) if valid.sum() >= .95*BOOTSTRAP_REPLICATES else np.nan,
            "bootstrap_first_place_pct": 100*frequencies[i, 0], "paired_valid_bootstraps": int(valid.sum())})
surface_preference_comparison = pd.DataFrame(surface_comparison_rows)

# Assertions cover ties, decomposition, probability mass, score bounds, and paired resampling.
assert np.allclose(fractional_rank_frequencies(np.ones((5, 3))), np.full((3,3), 1/3))
assert np.allclose(fractional_rank_frequencies(np.tile([3,2,1], (5,1))), np.eye(3))
assert len(comparison_weight_grid) == 286
assert comparison_scores["score"].between(0, 100).all()
assert np.allclose(comparison_scores.groupby("scenario")["bootstrap_first_place_pct"].sum(), 100)
assert np.allclose(comparison_scores[["bootstrap_first_place_pct", "bootstrap_second_place_pct", "bootstrap_third_place_pct"]].sum(axis=1), 100)
assert np.allclose(comparison_grid_summary["fractional_grid_lead_pct"].sum(), 100)
recomputed = comparison_contributions.groupby(["scenario", "player"])["score_contribution"].sum()
assert np.allclose(recomputed.sort_index(), comparison_scores.set_index(["scenario", "player"])["score"].sort_index())
print("Primary fixed-scale inputs, eligible samples, and alternative three-player min-max scale:")
report_table(normalization_audit.round(5))
print("Declared scenario weights:")
report_table(comparison_weights)
print("Primary scores and conditional bootstrap rank frequencies:")
with pd.option_context("display.max_columns", None):
    report_table(comparison_scores.round(4))
print("Score contributions (sum to each primary score):")
with pd.option_context("display.max_rows", None):
    report_table(comparison_contributions.round(5))
print("Direct score differences; bootstrap-ahead frequency is not a hypothesis-test p-value:")
with pd.option_context("display.max_colwidth", None):
    report_table(comparison_pairwise_differences.round(4))
print("Preference-grid lead shares: a summary of this chosen grid, not a probability of greatest ability:")
report_table(comparison_grid_summary.round(3))
with pd.option_context("display.max_rows", None):
    report_table(comparison_weight_grid.round(4))
print("Alternative scaling sensitivity (point estimates only):")
report_table(comparison_scaling_sensitivity.round(4))
```

```
print("Leave-one-metric-out sensitivity (equal remaining weights):")
report_table(comparison_leave_one_out.round(4))
print("Illustrative within-surface comparison; equal winning/DR/CI weights, SAS omitted:")
report_table(surface_preference_comparison.round(4))

scenario_leader_audit=pd.DataFrame([{"scenario":scenario,
    "primary_fixed_scale_leader":"; ".join(comparison_scores.loc[comparison_scores.scenario.eq(scenario)&comparison_scores.point_estimate_rank.eq(1),"player"]),
    "alternative_minmax_leader":"; 
".join(comparison_scaling_sensitivity.loc[comparison_scaling_sensitivity.scenario.eq(scenario)&comparison_scaling_sensitivity.alternative_point_rank.eq(1),"play
er"]),
    "interpretation":"Different normalization is sensitivity, not an inconsistent primary ranking"}
    for scenario in COMPARISON_SCENARIOS])
report_table(scenario_leader_audit,caption="Which ranking is being reported? Primary versus alternative scaling",expanded=True)
```

Primary fixed-scale inputs, eligible samples, and alternative three-player min-max scale:

player	metric	raw_value	fixed_scale_value	contributing_matches	earliest_charted_date	latest_charted_date	three_player_min	three_player_max	alternative_minmax_value
Roger Federer	winning	0.788	0.788	709	1998-07-04	2021-07-05	0.723	0.788	1.000
Roger Federer	dominance	1.277	0.561	703	1998-07-04	2021-07-05	1.183	1.277	1.000
Roger Federer	clutch	0.639	0.639	687	1998-07-04	2021-07-05	0.608	0.643	0.889
Roger Federer	adaptability	83.810	0.838	709	1998-07-04	2021-07-05	77.077	85.575	0.792
Rafael Nadal	winning	0.723	0.723	412	2003-05-15	2024-11-19	0.723	0.788	0.000
Rafael Nadal	dominance	1.183	0.542	406	2003-05-15	2024-11-19	1.183	1.277	0.000
Rafael Nadal	clutch	0.608	0.608	392	2003-05-15	2024-11-19	0.608	0.643	0.000
Rafael Nadal	adaptability	77.077	0.771	412	2003-05-15	2024-11-19	77.077	85.575	0.000
Novak Djokovic	winning	0.768	0.768	526	2005-01-17	2026-03-11	0.723	0.788	0.687
Novak Djokovic	dominance	1.221	0.550	546	2005-01-17	2026-03-11	1.183	1.277	0.406
Novak Djokovic	clutch	0.643	0.643	519	2005-01-17	2026-03-11	0.608	0.643	1.000
Novak Djokovic	adaptability	85.575	0.856	526	2005-01-17	2026-03-11	77.077	85.575	1.000

Declared scenario weights:

scenario	winning	dominance	clutch	adaptability
Balanced	0.250	0.250	0.250	0.250
Winning priority	0.550	0.150	0.150	0.150
Dominance priority	0.200	0.500	0.150	0.150
Clutch priority	0.200	0.150	0.500	0.150
Adaptability priority	0.200	0.150	0.150	0.500

Primary scores and conditional bootstrap rank frequencies:

player	scenario	score	score_boot_median	ci95_lower	ci95_upper	point_estimate_rank	bootstrap_first_place_pct
Roger Federer	Balanced	70.653	70.626	68.415	72.804	1	58.600
Rafael Nadal	Balanced	66.093	65.930	62.488	68.917	3	0.200
Novak Djokovic	Balanced	70.406	70.231	67.909	72.458	2	41.200
Roger Federer	Winning priority	73.930	73.892	71.591	76.240	1	71.650
Rafael Nadal	Winning priority	68.588	68.496	64.955	71.733	3	0.200
Novak Djokovic	Winning priority	72.966	72.866	70.213	75.491	2	28.150
Roger Federer	Dominance priority	65.964	65.937	64.408	67.501	1	73.300
Rafael Nadal	Dominance priority	62.241	62.140	59.759	64.273	3	0.000
Novak Djokovic	Dominance priority	65.328	65.234	63.589	66.834	2	26.700
Roger Federer	Clutch priority	68.691	68.689	66.561	70.855	1	54.500
Rafael Nadal	Clutch priority	64.541	64.420	61.320	67.331	3	0.500
Novak Djokovic	Clutch priority	68.576	68.468	66.180	70.808	2	45.000
Roger Federer	Adaptability priority	75.668	75.641	72.315	78.683	2	48.450
Rafael Nadal	Adaptability priority	70.249	70.020	64.629	74.049	3	0.300
Novak Djokovic	Adaptability priority	76.035	75.720	72.382	78.631	1	51.250

player	bootstrap_second_place_pct	bootstrap_third_place_pct	paired_valid_bootstraps
Roger Federer	40.650	0.750	2000
Rafael Nadal	2.000	97.800	2000
Novak Djokovic	57.350	1.450	2000
Roger Federer	28.050	0.300	2000
Rafael Nadal	2.550	97.250	2000
Novak Djokovic	69.400	2.450	2000
Roger Federer	26.400	0.300	2000
Rafael Nadal	1.950	98.050	2000
Novak Djokovic	71.650	1.650	2000
Roger Federer	44.350	1.150	2000
Rafael Nadal	2.300	97.200	2000
Novak Djokovic	53.350	1.650	2000
Roger Federer	50.000	1.550	2000
Rafael Nadal	2.850	96.850	2000
Novak Djokovic	47.150	1.600	2000

Score contributions (sum to each primary score):

▼ Complete evidence table: 60 rows × 6 columns

scenario	player	metric	weight	fixed_scale_value	score_contribution
Balanced	Roger Federer	winning	0.250	0.788	19.711
Balanced	Roger Federer	dominance	0.250	0.561	14.021
Balanced	Roger Federer	clutch	0.250	0.639	15.969
Balanced	Roger Federer	adaptability	0.250	0.838	20.953
Balanced	Rafael Nadal	winning	0.250	0.723	18.083
Balanced	Rafael Nadal	dominance	0.250	0.542	13.549
Balanced	Rafael Nadal	clutch	0.250	0.608	15.192
Balanced	Rafael Nadal	adaptability	0.250	0.771	19.269
Balanced	Novak Djokovic	winning	0.250	0.768	19.202
Balanced	Novak Djokovic	dominance	0.250	0.550	13.746
Balanced	Novak Djokovic	clutch	0.250	0.643	16.065
Balanced	Novak Djokovic	adaptability	0.250	0.856	21.394
Winning priority	Roger Federer	winning	0.550	0.788	43.364
Winning priority	Roger Federer	dominance	0.150	0.561	8.413
Winning priority	Roger Federer	clutch	0.150	0.639	9.581
Winning priority	Roger Federer	adaptability	0.150	0.838	12.572
Winning priority	Rafael Nadal	winning	0.550	0.723	39.782
Winning priority	Rafael Nadal	dominance	0.150	0.542	8.130
Winning priority	Rafael Nadal	clutch	0.150	0.608	9.115
Winning priority	Rafael Nadal	adaptability	0.150	0.771	11.562
Winning priority	Novak Djokovic	winning	0.550	0.768	42.243
Winning priority	Novak Djokovic	dominance	0.150	0.550	8.247
Winning priority	Novak Djokovic	clutch	0.150	0.643	9.639
Winning priority	Novak Djokovic	adaptability	0.150	0.856	12.836
Dominance priority	Roger Federer	winning	0.200	0.788	15.769
Dominance priority	Roger Federer	dominance	0.500	0.561	28.043
Dominance priority	Roger Federer	clutch	0.150	0.639	9.581
Dominance priority	Roger Federer	adaptability	0.150	0.838	12.572
Dominance priority	Rafael Nadal	winning	0.200	0.723	14.466
Dominance priority	Rafael Nadal	dominance	0.500	0.542	27.099

scenario	player	metric	weight	fixed_scale_value	score_contribution
Dominance priority	Rafael Nadal	clutch	0.150	0.608	9.115
Dominance priority	Rafael Nadal	adaptability	0.150	0.771	11.562
Dominance priority	Novak Djokovic	winning	0.200	0.768	15.361
Dominance priority	Novak Djokovic	dominance	0.500	0.550	27.491
Dominance priority	Novak Djokovic	clutch	0.150	0.643	9.639
Dominance priority	Novak Djokovic	adaptability	0.150	0.856	12.836
Clutch priority	Roger Federer	winning	0.200	0.788	15.769
Clutch priority	Roger Federer	dominance	0.150	0.561	8.413
Clutch priority	Roger Federer	clutch	0.500	0.639	31.937
Clutch priority	Roger Federer	adaptability	0.150	0.838	12.572
Clutch priority	Rafael Nadal	winning	0.200	0.723	14.466
Clutch priority	Rafael Nadal	dominance	0.150	0.542	8.130
Clutch priority	Rafael Nadal	clutch	0.500	0.608	30.384
Clutch priority	Rafael Nadal	adaptability	0.150	0.771	11.562
Clutch priority	Novak Djokovic	winning	0.200	0.768	15.361
Clutch priority	Novak Djokovic	dominance	0.150	0.550	8.247
Clutch priority	Novak Djokovic	clutch	0.500	0.643	32.131
Clutch priority	Novak Djokovic	adaptability	0.150	0.856	12.836
Adaptability priority	Roger Federer	winning	0.200	0.788	15.769
Adaptability priority	Roger Federer	dominance	0.150	0.561	8.413
Adaptability priority	Roger Federer	clutch	0.150	0.639	9.581
Adaptability priority	Roger Federer	adaptability	0.500	0.838	41.905
Adaptability priority	Rafael Nadal	winning	0.200	0.723	14.466
Adaptability priority	Rafael Nadal	dominance	0.150	0.542	8.130
Adaptability priority	Rafael Nadal	clutch	0.150	0.608	9.115
Adaptability priority	Rafael Nadal	adaptability	0.500	0.771	38.539
Adaptability priority	Novak Djokovic	winning	0.200	0.768	15.361
Adaptability priority	Novak Djokovic	dominance	0.150	0.550	8.247
Adaptability priority	Novak Djokovic	clutch	0.150	0.643	9.639
Adaptability priority	Novak Djokovic	adaptability	0.500	0.856	42.787

Direct score differences; bootstrap-ahead frequency is not a hypothesis-test p-value:

scenario	comparison	point_score_difference	ci95_lower	ci95_upper	bootstrap_a_ahead_pct
Balanced	Roger Federer minus Rafael Nadal	4.560	0.877	8.762	99.050
Balanced	Roger Federer minus Novak Djokovic	0.247	-2.972	3.698	58.800
Balanced	Rafael Nadal minus Novak Djokovic	-4.313	-8.649	-0.396	1.450
Winning priority	Roger Federer minus Rafael Nadal	5.341	1.255	9.765	99.550
Winning priority	Roger Federer minus Novak Djokovic	0.963	-2.638	4.717	71.800
Winning priority	Rafael Nadal minus Novak Djokovic	-4.378	-9.147	-0.029	2.500
Dominance priority	Roger Federer minus Rafael Nadal	3.723	1.066	6.651	99.700
Dominance priority	Roger Federer minus Novak Djokovic	0.636	-1.680	3.080	73.300
Dominance priority	Rafael Nadal minus Novak Djokovic	-3.086	-6.142	-0.280	1.650
Clutch priority	Roger Federer minus Rafael Nadal	4.149	0.416	8.167	98.650
Clutch priority	Roger Federer minus Novak Djokovic	0.115	-3.130	3.479	54.700
Clutch priority	Rafael Nadal minus Novak Djokovic	-4.034	-8.168	-0.105	1.950
Adaptability priority	Roger Federer minus Rafael Nadal	5.418	0.306	11.959	98.200
Adaptability priority	Roger Federer minus Novak Djokovic	-0.367	-4.699	4.562	48.700
Adaptability priority	Rafael Nadal minus Novak Djokovic	-5.786	-12.155	-0.448	1.650

`Preference-grid lead shares: a summary of this chosen grid, not a probability of greatest ability:`

player	grid_settings	fractional_grid_lead_pct
Roger Federer	286	62.587
Rafael Nadal	286	0.000
Novak Djokovic	286	37.413

▼ Complete evidence table: 286 rows × 11 columns

winning_weight	dominance_weight	clutch_weight	adaptability_weight	leaders	Roger Federer_score	Roger Federer_lead_credit
0.000	0.000	0.000	1.000	Novak Djokovic	83.810	0.000
0.000	0.000	0.100	0.900	Novak Djokovic	81.817	0.000
0.000	0.000	0.200	0.800	Novak Djokovic	79.823	0.000
0.000	0.000	0.300	0.700	Novak Djokovic	77.830	0.000
0.000	0.000	0.400	0.600	Novak Djokovic	75.836	0.000
0.000	0.000	0.500	0.500	Novak Djokovic	73.843	0.000
0.000	0.000	0.600	0.400	Novak Djokovic	71.849	0.000
0.000	0.000	0.700	0.300	Novak Djokovic	69.856	0.000
0.000	0.000	0.800	0.200	Novak Djokovic	67.862	0.000
0.000	0.000	0.900	0.100	Novak Djokovic	65.868	0.000
0.000	0.000	1.000	0.000	Novak Djokovic	63.875	0.000
0.000	0.100	0.000	0.900	Novak Djokovic	81.038	0.000
0.000	0.100	0.100	0.800	Novak Djokovic	79.044	0.000
0.000	0.100	0.200	0.700	Novak Djokovic	77.051	0.000
0.000	0.100	0.300	0.600	Novak Djokovic	75.057	0.000
0.000	0.100	0.400	0.500	Novak Djokovic	73.064	0.000
0.000	0.100	0.500	0.400	Novak Djokovic	71.070	0.000
0.000	0.100	0.600	0.300	Novak Djokovic	69.076	0.000
0.000	0.100	0.700	0.200	Novak Djokovic	67.083	0.000
0.000	0.100	0.800	0.100	Novak Djokovic	65.090	0.000
0.000	0.100	0.900	0.000	Novak Djokovic	63.096	0.000
0.000	0.200	0.000	0.800	Novak Djokovic	78.265	0.000
0.000	0.200	0.100	0.700	Novak Djokovic	76.272	0.000
0.000	0.200	0.200	0.600	Novak Djokovic	74.278	0.000
0.000	0.200	0.300	0.500	Novak Djokovic	72.285	0.000
0.000	0.200	0.400	0.400	Novak Djokovic	70.291	0.000
0.000	0.200	0.500	0.300	Novak Djokovic	68.298	0.000
0.000	0.200	0.600	0.200	Novak Djokovic	66.304	0.000
0.000	0.200	0.700	0.100	Novak Djokovic	64.311	0.000
0.000	0.200	0.800	0.000	Novak Djokovic	62.317	0.000
0.000	0.300	0.000	0.700	Novak Djokovic	75.493	0.000
0.000	0.300	0.100	0.600	Novak Djokovic	73.499	0.000
0.000	0.300	0.200	0.500	Novak Djokovic	71.506	0.000
0.000	0.300	0.300	0.400	Novak Djokovic	69.512	0.000
0.000	0.300	0.400	0.300	Novak Djokovic	67.519	0.000
0.000	0.300	0.500	0.200	Novak Djokovic	65.525	0.000
0.000	0.300	0.600	0.100	Novak Djokovic	63.532	0.000

winning_weight	dominance_weight	clutch_weight	adaptability_weight	leaders	Roger Federer_score	Roger Federer_lead_credit
0.000	0.300	0.700	0.000	Roger Federer	61.538	1.000
0.000	0.400	0.000	0.600	Novak Djokovic	72.720	0.000
0.000	0.400	0.100	0.500	Novak Djokovic	70.727	0.000
0.000	0.400	0.200	0.400	Novak Djokovic	68.733	0.000
0.000	0.400	0.300	0.300	Novak Djokovic	66.740	0.000
0.000	0.400	0.400	0.200	Novak Djokovic	64.746	0.000
0.000	0.400	0.500	0.100	Roger Federer	62.753	1.000
0.000	0.400	0.600	0.000	Roger Federer	60.759	1.000
0.000	0.500	0.000	0.500	Novak Djokovic	69.948	0.000
0.000	0.500	0.100	0.400	Novak Djokovic	67.954	0.000
0.000	0.500	0.200	0.300	Novak Djokovic	65.961	0.000
0.000	0.500	0.300	0.200	Roger Federer	63.967	1.000
0.000	0.500	0.400	0.100	Roger Federer	61.974	1.000
0.000	0.500	0.500	0.000	Roger Federer	59.980	1.000
0.000	0.600	0.000	0.400	Novak Djokovic	67.175	0.000
0.000	0.600	0.100	0.300	Roger Federer	65.182	1.000
0.000	0.600	0.200	0.200	Roger Federer	63.188	1.000
0.000	0.600	0.300	0.100	Roger Federer	61.195	1.000
0.000	0.600	0.400	0.000	Roger Federer	59.201	1.000
0.000	0.700	0.000	0.300	Roger Federer	64.403	1.000
0.000	0.700	0.100	0.200	Roger Federer	62.409	1.000
0.000	0.700	0.200	0.100	Roger Federer	60.416	1.000
0.000	0.700	0.300	0.000	Roger Federer	58.422	1.000
0.000	0.800	0.000	0.200	Roger Federer	61.630	1.000
0.000	0.800	0.100	0.100	Roger Federer	59.637	1.000
0.000	0.800	0.200	0.000	Roger Federer	57.643	1.000
0.000	0.900	0.000	0.100	Roger Federer	58.858	1.000
0.000	0.900	0.100	0.000	Roger Federer	56.864	1.000
0.000	1.000	0.000	0.000	Roger Federer	56.085	1.000
0.100	0.000	0.000	0.900	Novak Djokovic	83.313	0.000
0.100	0.000	0.100	0.800	Novak Djokovic	81.320	0.000
0.100	0.000	0.200	0.700	Novak Djokovic	79.326	0.000
0.100	0.000	0.300	0.600	Novak Djokovic	77.333	0.000
0.100	0.000	0.400	0.500	Novak Djokovic	75.339	0.000
0.100	0.000	0.500	0.400	Novak Djokovic	73.346	0.000
0.100	0.000	0.600	0.300	Novak Djokovic	71.352	0.000
0.100	0.000	0.700	0.200	Novak Djokovic	69.359	0.000
0.100	0.000	0.800	0.100	Novak Djokovic	67.365	0.000
0.100	0.000	0.900	0.000	Novak Djokovic	65.372	0.000

winning_weight	dominance_weight	clutch_weight	adaptability_weight	leaders	Roger Federer_score	Roger Federer_lead_credit
0.100	0.100	0.000	0.800	Novak Djokovic	80.541	0.000
0.100	0.100	0.100	0.700	Novak Djokovic	78.547	0.000
0.100	0.100	0.200	0.600	Novak Djokovic	76.554	0.000
0.100	0.100	0.300	0.500	Novak Djokovic	74.560	0.000
0.100	0.100	0.400	0.400	Novak Djokovic	72.567	0.000
0.100	0.100	0.500	0.300	Novak Djokovic	70.573	0.000
0.100	0.100	0.600	0.200	Novak Djokovic	68.580	0.000
0.100	0.100	0.700	0.100	Novak Djokovic	66.586	0.000
0.100	0.100	0.800	0.000	Roger Federer	64.593	1.000
0.100	0.200	0.000	0.700	Novak Djokovic	77.769	0.000
0.100	0.200	0.100	0.600	Novak Djokovic	75.775	0.000
0.100	0.200	0.200	0.500	Novak Djokovic	73.781	0.000
0.100	0.200	0.300	0.400	Novak Djokovic	71.788	0.000
0.100	0.200	0.400	0.300	Novak Djokovic	69.794	0.000
0.100	0.200	0.500	0.200	Novak Djokovic	67.801	0.000
0.100	0.200	0.600	0.100	Roger Federer	65.807	1.000
0.100	0.200	0.700	0.000	Roger Federer	63.814	1.000
0.100	0.300	0.000	0.600	Novak Djokovic	74.996	0.000
0.100	0.300	0.100	0.500	Novak Djokovic	73.002	0.000
0.100	0.300	0.200	0.400	Novak Djokovic	71.009	0.000
0.100	0.300	0.300	0.300	Novak Djokovic	69.016	0.000
0.100	0.300	0.400	0.200	Roger Federer	67.022	1.000
0.100	0.300	0.500	0.100	Roger Federer	65.028	1.000
0.100	0.300	0.600	0.000	Roger Federer	63.035	1.000
0.100	0.400	0.000	0.500	Novak Djokovic	72.224	0.000
0.100	0.400	0.100	0.400	Novak Djokovic	70.230	0.000
0.100	0.400	0.200	0.300	Roger Federer	68.237	1.000
0.100	0.400	0.300	0.200	Roger Federer	66.243	1.000
0.100	0.400	0.400	0.100	Roger Federer	64.249	1.000
0.100	0.400	0.500	0.000	Roger Federer	62.256	1.000
0.100	0.500	0.000	0.400	Roger Federer	69.451	1.000
0.100	0.500	0.100	0.300	Roger Federer	67.457	1.000
0.100	0.500	0.200	0.200	Roger Federer	65.464	1.000
0.100	0.500	0.300	0.100	Roger Federer	63.471	1.000
0.100	0.500	0.400	0.000	Roger Federer	61.477	1.000
0.100	0.600	0.000	0.300	Roger Federer	66.679	1.000
0.100	0.600	0.100	0.200	Roger Federer	64.685	1.000
0.100	0.600	0.200	0.100	Roger Federer	62.691	1.000
0.100	0.600	0.300	0.000	Roger Federer	60.698	1.000

winning_weight	dominance_weight	clutch_weight	adaptability_weight	leaders	Roger Federer_score	Roger Federer_lead_credit
0.100	0.700	0.000	0.200	Roger Federer	63.906	1.000
0.100	0.700	0.100	0.100	Roger Federer	61.913	1.000
0.100	0.700	0.200	0.000	Roger Federer	59.919	1.000
0.100	0.800	0.000	0.100	Roger Federer	61.134	1.000
0.100	0.800	0.100	0.000	Roger Federer	59.140	1.000
0.100	0.900	0.000	0.000	Roger Federer	58.361	1.000
0.200	0.000	0.000	0.800	Novak Djokovic	82.817	0.000
0.200	0.000	0.100	0.700	Novak Djokovic	80.823	0.000
0.200	0.000	0.200	0.600	Novak Djokovic	78.830	0.000
0.200	0.000	0.300	0.500	Novak Djokovic	76.836	0.000
0.200	0.000	0.400	0.400	Novak Djokovic	74.843	0.000
0.200	0.000	0.500	0.300	Novak Djokovic	72.849	0.000
0.200	0.000	0.600	0.200	Novak Djokovic	70.856	0.000
0.200	0.000	0.700	0.100	Novak Djokovic	68.862	0.000
0.200	0.000	0.800	0.000	Roger Federer	66.869	1.000
0.200	0.100	0.000	0.700	Novak Djokovic	80.044	0.000
0.200	0.100	0.100	0.600	Novak Djokovic	78.051	0.000
0.200	0.100	0.200	0.500	Novak Djokovic	76.057	0.000
0.200	0.100	0.300	0.400	Novak Djokovic	74.064	0.000
0.200	0.100	0.400	0.300	Novak Djokovic	72.070	0.000
0.200	0.100	0.500	0.200	Novak Djokovic	70.077	0.000
0.200	0.100	0.600	0.100	Roger Federer	68.083	1.000
0.200	0.100	0.700	0.000	Roger Federer	66.090	1.000
0.200	0.200	0.000	0.600	Novak Djokovic	77.272	0.000
0.200	0.200	0.100	0.500	Novak Djokovic	75.278	0.000
0.200	0.200	0.200	0.400	Novak Djokovic	73.285	0.000
0.200	0.200	0.300	0.300	Novak Djokovic	71.291	0.000
0.200	0.200	0.400	0.200	Roger Federer	69.298	1.000
0.200	0.200	0.500	0.100	Roger Federer	67.304	1.000
0.200	0.200	0.600	0.000	Roger Federer	65.311	1.000
0.200	0.300	0.000	0.500	Novak Djokovic	74.499	0.000
0.200	0.300	0.100	0.400	Novak Djokovic	72.506	0.000
0.200	0.300	0.200	0.300	Roger Federer	70.512	1.000
0.200	0.300	0.300	0.200	Roger Federer	68.519	1.000
0.200	0.300	0.400	0.100	Roger Federer	66.525	1.000
0.200	0.300	0.500	0.000	Roger Federer	64.532	1.000
0.200	0.400	0.000	0.400	Roger Federer	71.727	1.000
0.200	0.400	0.100	0.300	Roger Federer	69.733	1.000
0.200	0.400	0.200	0.200	Roger Federer	67.740	1.000

winning_weight	dominance_weight	clutch_weight	adaptability_weight	leaders	Roger Federer_score	Roger Federer_lead_credit
0.200	0.400	0.300	0.100	Roger Federer	65.746	1.000
0.200	0.400	0.400	0.000	Roger Federer	63.753	1.000
0.200	0.500	0.000	0.300	Roger Federer	68.954	1.000
0.200	0.500	0.100	0.200	Roger Federer	66.961	1.000
0.200	0.500	0.200	0.100	Roger Federer	64.967	1.000
0.200	0.500	0.300	0.000	Roger Federer	62.974	1.000
0.200	0.600	0.000	0.200	Roger Federer	66.182	1.000
0.200	0.600	0.100	0.100	Roger Federer	64.188	1.000
0.200	0.600	0.200	0.000	Roger Federer	62.195	1.000
0.200	0.700	0.000	0.100	Roger Federer	63.409	1.000
0.200	0.700	0.100	0.000	Roger Federer	61.416	1.000
0.200	0.800	0.000	0.000	Roger Federer	60.637	1.000
0.300	0.000	0.000	0.700	Novak Djokovic	82.320	0.000
0.300	0.000	0.100	0.600	Novak Djokovic	80.327	0.000
0.300	0.000	0.200	0.500	Novak Djokovic	78.333	0.000
0.300	0.000	0.300	0.400	Novak Djokovic	76.340	0.000
0.300	0.000	0.400	0.300	Novak Djokovic	74.346	0.000
0.300	0.000	0.500	0.200	Roger Federer	72.353	1.000
0.300	0.000	0.600	0.100	Roger Federer	70.359	1.000
0.300	0.000	0.700	0.000	Roger Federer	68.365	1.000
0.300	0.100	0.000	0.600	Novak Djokovic	79.548	0.000
0.300	0.100	0.100	0.500	Novak Djokovic	77.554	0.000
0.300	0.100	0.200	0.400	Novak Djokovic	75.561	0.000
0.300	0.100	0.300	0.300	Roger Federer	73.567	1.000
0.300	0.100	0.400	0.200	Roger Federer	71.574	1.000
0.300	0.100	0.500	0.100	Roger Federer	69.580	1.000
0.300	0.100	0.600	0.000	Roger Federer	67.587	1.000
0.300	0.200	0.000	0.500	Novak Djokovic	76.775	0.000
0.300	0.200	0.100	0.400	Roger Federer	74.782	1.000
0.300	0.200	0.200	0.300	Roger Federer	72.788	1.000
0.300	0.200	0.300	0.200	Roger Federer	70.795	1.000
0.300	0.200	0.400	0.100	Roger Federer	68.801	1.000
0.300	0.200	0.500	0.000	Roger Federer	66.808	1.000
0.300	0.300	0.000	0.400	Roger Federer	74.003	1.000
0.300	0.300	0.100	0.300	Roger Federer	72.009	1.000
0.300	0.300	0.200	0.200	Roger Federer	70.016	1.000
0.300	0.300	0.300	0.100	Roger Federer	68.022	1.000
0.300	0.300	0.400	0.000	Roger Federer	66.029	1.000
0.300	0.400	0.000	0.300	Roger Federer	71.230	1.000

winning_weight	dominance_weight	clutch_weight	adaptability_weight	leaders	Roger Federer_score	Roger Federer_lead_credit
0.300	0.400	0.100	0.200	Roger Federer	69.237	1.000
0.300	0.400	0.200	0.100	Roger Federer	67.243	1.000
0.300	0.400	0.300	0.000	Roger Federer	65.250	1.000
0.300	0.500	0.000	0.200	Roger Federer	68.458	1.000
0.300	0.500	0.100	0.100	Roger Federer	66.464	1.000
0.300	0.500	0.200	0.000	Roger Federer	64.471	1.000
0.300	0.600	0.000	0.100	Roger Federer	65.685	1.000
0.300	0.600	0.100	0.000	Roger Federer	63.692	1.000
0.300	0.700	0.000	0.000	Roger Federer	62.913	1.000
0.400	0.000	0.000	0.600	Novak Djokovic	81.823	0.000
0.400	0.000	0.100	0.500	Novak Djokovic	79.830	0.000
0.400	0.000	0.200	0.400	Roger Federer	77.836	1.000
0.400	0.000	0.300	0.300	Roger Federer	75.843	1.000
0.400	0.000	0.400	0.200	Roger Federer	73.849	1.000
0.400	0.000	0.500	0.100	Roger Federer	71.856	1.000
0.400	0.000	0.600	0.000	Roger Federer	69.862	1.000
0.400	0.100	0.000	0.500	Roger Federer	79.051	1.000
0.400	0.100	0.100	0.400	Roger Federer	77.057	1.000
0.400	0.100	0.200	0.300	Roger Federer	75.064	1.000
0.400	0.100	0.300	0.200	Roger Federer	73.070	1.000
0.400	0.100	0.400	0.100	Roger Federer	71.077	1.000
0.400	0.100	0.500	0.000	Roger Federer	69.083	1.000
0.400	0.200	0.000	0.400	Roger Federer	76.278	1.000
0.400	0.200	0.100	0.300	Roger Federer	74.285	1.000
0.400	0.200	0.200	0.200	Roger Federer	72.291	1.000
0.400	0.200	0.300	0.100	Roger Federer	70.298	1.000
0.400	0.200	0.400	0.000	Roger Federer	68.304	1.000
0.400	0.300	0.000	0.300	Roger Federer	73.506	1.000
0.400	0.300	0.100	0.200	Roger Federer	71.513	1.000
0.400	0.300	0.200	0.100	Roger Federer	69.519	1.000
0.400	0.300	0.300	0.000	Roger Federer	67.525	1.000
0.400	0.400	0.000	0.200	Roger Federer	70.734	1.000
0.400	0.400	0.100	0.100	Roger Federer	68.740	1.000
0.400	0.400	0.200	0.000	Roger Federer	66.746	1.000
0.400	0.500	0.000	0.100	Roger Federer	67.961	1.000
0.400	0.500	0.100	0.000	Roger Federer	65.968	1.000
0.400	0.600	0.000	0.000	Roger Federer	65.189	1.000
0.500	0.000	0.000	0.500	Roger Federer	81.327	1.000
0.500	0.000	0.100	0.400	Roger Federer	79.333	1.000

winning_weight	dominance_weight	clutch_weight	adaptability_weight	leaders	Roger Federer_score	Roger Federer_lead_credit
0.500	0.000	0.200	0.300	Roger Federer	77.340	1.000
0.500	0.000	0.300	0.200	Roger Federer	75.346	1.000
0.500	0.000	0.400	0.100	Roger Federer	73.353	1.000
0.500	0.000	0.500	0.000	Roger Federer	71.359	1.000
0.500	0.100	0.000	0.400	Roger Federer	78.554	1.000
0.500	0.100	0.100	0.300	Roger Federer	76.561	1.000
0.500	0.100	0.200	0.200	Roger Federer	74.567	1.000
0.500	0.100	0.300	0.100	Roger Federer	72.574	1.000
0.500	0.100	0.400	0.000	Roger Federer	70.580	1.000
0.500	0.200	0.000	0.300	Roger Federer	75.782	1.000
0.500	0.200	0.100	0.200	Roger Federer	73.788	1.000
0.500	0.200	0.200	0.100	Roger Federer	71.795	1.000
0.500	0.200	0.300	0.000	Roger Federer	69.801	1.000
0.500	0.300	0.000	0.200	Roger Federer	73.009	1.000
0.500	0.300	0.100	0.100	Roger Federer	71.016	1.000
0.500	0.300	0.200	0.000	Roger Federer	69.022	1.000
0.500	0.400	0.000	0.100	Roger Federer	70.237	1.000
0.500	0.400	0.100	0.000	Roger Federer	68.243	1.000
0.500	0.500	0.000	0.000	Roger Federer	67.464	1.000
0.600	0.000	0.000	0.400	Roger Federer	80.830	1.000
0.600	0.000	0.100	0.300	Roger Federer	78.837	1.000
0.600	0.000	0.200	0.200	Roger Federer	76.843	1.000
0.600	0.000	0.300	0.100	Roger Federer	74.850	1.000
0.600	0.000	0.400	0.000	Roger Federer	72.856	1.000
0.600	0.100	0.000	0.300	Roger Federer	78.058	1.000
0.600	0.100	0.100	0.200	Roger Federer	76.064	1.000
0.600	0.100	0.200	0.100	Roger Federer	74.071	1.000
0.600	0.100	0.300	0.000	Roger Federer	72.077	1.000
0.600	0.200	0.000	0.200	Roger Federer	75.285	1.000
0.600	0.200	0.100	0.100	Roger Federer	73.292	1.000
0.600	0.200	0.200	0.000	Roger Federer	71.298	1.000
0.600	0.300	0.000	0.100	Roger Federer	72.513	1.000
0.600	0.300	0.100	0.000	Roger Federer	70.519	1.000
0.600	0.400	0.000	0.000	Roger Federer	69.740	1.000
0.700	0.000	0.000	0.300	Roger Federer	80.333	1.000
0.700	0.000	0.100	0.200	Roger Federer	78.340	1.000
0.700	0.000	0.200	0.100	Roger Federer	76.346	1.000
0.700	0.000	0.300	0.000	Roger Federer	74.353	1.000
0.700	0.100	0.000	0.200	Roger Federer	77.561	1.000

winning_weight	dominance_weight	clutch_weight	adaptability_weight	leaders	Roger Federer_score	Roger Federer_lead_credit
0.700	0.100	0.100	0.100	Roger Federer	75.567	1.000
0.700	0.100	0.200	0.000	Roger Federer	73.574	1.000
0.700	0.200	0.000	0.100	Roger Federer	74.788	1.000
0.700	0.200	0.100	0.000	Roger Federer	72.795	1.000
0.700	0.300	0.000	0.000	Roger Federer	72.016	1.000
0.800	0.000	0.000	0.200	Roger Federer	79.837	1.000
0.800	0.000	0.100	0.100	Roger Federer	77.843	1.000
0.800	0.000	0.200	0.000	Roger Federer	75.850	1.000
0.800	0.100	0.000	0.100	Roger Federer	77.064	1.000
0.800	0.100	0.100	0.000	Roger Federer	75.071	1.000
0.800	0.200	0.000	0.000	Roger Federer	74.292	1.000
0.900	0.000	0.000	0.100	Roger Federer	79.340	1.000
0.900	0.000	0.100	0.000	Roger Federer	77.347	1.000
0.900	0.100	0.000	0.000	Roger Federer	76.568	1.000
1.000	0.000	0.000	0.000	Roger Federer	78.843	1.000

Rafael Nadal_score	Rafael Nadal_lead_credit	Novak Djokovic_score	Novak Djokovic_lead_credit
77.077	0.000	85.575	1.000
75.446	0.000	83.443	1.000
73.815	0.000	81.312	1.000
72.185	0.000	79.181	1.000
70.554	0.000	77.049	1.000
68.923	0.000	74.918	1.000
67.292	0.000	72.787	1.000
65.661	0.000	70.655	1.000
64.030	0.000	68.524	1.000
62.400	0.000	66.393	1.000
60.769	0.000	64.261	1.000
74.789	0.000	82.515	1.000
73.158	0.000	80.384	1.000
71.527	0.000	78.253	1.000
69.897	0.000	76.121	1.000
68.266	0.000	73.990	1.000
66.635	0.000	71.859	1.000
65.004	0.000	69.728	1.000
63.373	0.000	67.596	1.000
61.742	0.000	65.465	1.000
60.112	0.000	63.334	1.000

Rafael Nadal_score	Rafael Nadal_lead_credit	Novak Djokovic_score	Novak Djokovic_lead_credit
72.501	0.000	79.456	1.000
70.870	0.000	77.325	1.000
69.239	0.000	75.194	1.000
67.609	0.000	73.062	1.000
65.978	0.000	70.931	1.000
64.347	0.000	68.800	1.000
62.716	0.000	66.668	1.000
61.085	0.000	64.537	1.000
59.454	0.000	62.406	1.000
70.213	0.000	76.397	1.000
68.582	0.000	74.266	1.000
66.951	0.000	72.134	1.000
65.321	0.000	70.003	1.000
63.690	0.000	67.872	1.000
62.059	0.000	65.740	1.000
60.428	0.000	63.609	1.000
58.797	0.000	61.478	0.000
67.925	0.000	73.338	1.000
66.294	0.000	71.206	1.000
64.663	0.000	69.075	1.000
63.033	0.000	66.944	1.000
61.402	0.000	64.812	1.000
59.771	0.000	62.681	0.000
58.140	0.000	60.550	0.000
65.637	0.000	70.278	1.000
64.006	0.000	68.147	1.000
62.375	0.000	66.016	1.000
60.745	0.000	63.885	0.000
59.114	0.000	61.753	0.000
57.483	0.000	59.622	0.000
63.349	0.000	67.219	1.000
61.718	0.000	65.088	0.000
60.087	0.000	62.957	0.000
58.457	0.000	60.825	0.000
56.826	0.000	58.694	0.000
61.061	0.000	64.160	0.000
59.430	0.000	62.029	0.000
57.799	0.000	59.897	0.000
56.169	0.000	57.766	0.000
58.773	0.000	61.101	0.000

Rafael Nadal_score	Rafael Nadal_lead_credit	Novak Djokovic_score	Novak Djokovic_lead_credit
57.142	0.000	58.969	0.000
55.511	0.000	56.838	0.000
56.485	0.000	58.041	0.000
54.854	0.000	55.910	0.000
54.197	0.000	54.982	0.000
76.602	0.000	84.698	1.000
74.972	0.000	82.567	1.000
73.341	0.000	80.435	1.000
71.710	0.000	78.304	1.000
70.079	0.000	76.172	1.000
68.448	0.000	74.041	1.000
66.817	0.000	71.910	1.000
65.187	0.000	69.778	1.000
63.556	0.000	67.647	1.000
61.925	0.000	65.516	1.000
74.314	0.000	81.639	1.000
72.683	0.000	79.507	1.000
71.053	0.000	77.376	1.000
69.422	0.000	75.245	1.000
67.791	0.000	73.113	1.000
66.160	0.000	70.982	1.000
64.529	0.000	68.851	1.000
62.899	0.000	66.719	1.000
61.268	0.000	64.588	0.000
72.026	0.000	78.579	1.000
70.395	0.000	76.448	1.000
68.765	0.000	74.317	1.000
67.134	0.000	72.185	1.000
65.503	0.000	70.054	1.000
63.872	0.000	67.923	1.000
62.241	0.000	65.791	0.000
60.611	0.000	63.660	0.000
69.738	0.000	75.520	1.000
68.108	0.000	73.389	1.000
66.477	0.000	71.257	1.000
64.846	0.000	69.126	1.000
63.215	0.000	66.995	0.000
61.584	0.000	64.864	0.000
59.953	0.000	62.732	0.000
67.450	0.000	72.461	1.000

Rafael Nadal_score	Rafael Nadal_lead_credit	Novak Djokovic_score	Novak Djokovic_lead_credit
65.820	0.000	70.329	1.000
64.189	0.000	68.198	0.000
62.558	0.000	66.067	0.000
60.927	0.000	63.935	0.000
59.296	0.000	61.804	0.000
65.162	0.000	69.402	0.000
63.532	0.000	67.270	0.000
61.901	0.000	65.139	0.000
60.270	0.000	63.008	0.000
58.639	0.000	60.876	0.000
62.874	0.000	66.342	0.000
61.244	0.000	64.211	0.000
59.613	0.000	62.080	0.000
57.982	0.000	59.948	0.000
60.586	0.000	63.283	0.000
58.956	0.000	61.152	0.000
57.325	0.000	59.020	0.000
58.298	0.000	60.224	0.000
56.668	0.000	58.093	0.000
56.010	0.000	57.165	0.000
76.128	0.000	83.821	1.000
74.497	0.000	81.690	1.000
72.866	0.000	79.558	1.000
71.235	0.000	77.427	1.000
69.604	0.000	75.296	1.000
67.974	0.000	73.164	1.000
66.343	0.000	71.033	1.000
64.712	0.000	68.902	1.000
63.081	0.000	66.770	0.000
73.840	0.000	80.762	1.000
72.209	0.000	78.630	1.000
70.578	0.000	76.499	1.000
68.947	0.000	74.368	1.000
67.316	0.000	72.236	1.000
65.686	0.000	70.105	1.000
64.055	0.000	67.974	0.000
62.424	0.000	65.842	0.000
71.552	0.000	77.703	1.000
69.921	0.000	75.571	1.000
68.290	0.000	73.440	1.000

Rafael Nadal_score	Rafael Nadal_lead_credit	Novak Djokovic_score	Novak Djokovic_lead_credit
66.659	0.000	71.308	1.000
65.028	0.000	69.177	0.000
63.398	0.000	67.046	0.000
61.767	0.000	64.915	0.000
69.264	0.000	74.643	1.000
67.633	0.000	72.512	1.000
66.002	0.000	70.381	0.000
64.371	0.000	68.249	0.000
62.740	0.000	66.118	0.000
61.109	0.000	63.987	0.000
66.976	0.000	71.584	0.000
65.345	0.000	69.453	0.000
63.714	0.000	67.321	0.000
62.083	0.000	65.190	0.000
60.452	0.000	63.059	0.000
64.688	0.000	68.525	0.000
63.057	0.000	66.393	0.000
61.426	0.000	64.262	0.000
59.795	0.000	62.131	0.000
62.400	0.000	65.466	0.000
60.769	0.000	63.334	0.000
59.138	0.000	61.203	0.000
60.112	0.000	62.406	0.000
58.481	0.000	60.275	0.000
57.824	0.000	59.347	0.000
75.653	0.000	82.944	1.000
74.022	0.000	80.813	1.000
72.391	0.000	78.681	1.000
70.760	0.000	76.550	1.000
69.130	0.000	74.419	1.000
67.499	0.000	72.287	0.000
65.868	0.000	70.156	0.000
64.237	0.000	68.025	0.000
73.365	0.000	79.885	1.000
71.734	0.000	77.754	1.000
70.103	0.000	75.622	1.000
68.472	0.000	73.491	0.000
66.842	0.000	71.360	0.000
65.211	0.000	69.228	0.000
63.580	0.000	67.097	0.000

Rafael Nadal_score	Rafael Nadal_lead_credit	Novak Djokovic_score	Novak Djokovic_lead_credit
71.077	0.000	76.826	1.000
69.446	0.000	74.694	0.000
67.815	0.000	72.563	0.000
66.184	0.000	70.432	0.000
64.554	0.000	68.300	0.000
62.923	0.000	66.169	0.000
68.789	0.000	73.766	0.000
67.158	0.000	71.635	0.000
65.527	0.000	69.504	0.000
63.897	0.000	67.372	0.000
62.266	0.000	65.241	0.000
66.501	0.000	70.707	0.000
64.870	0.000	68.576	0.000
63.239	0.000	66.445	0.000
61.608	0.000	64.313	0.000
64.213	0.000	67.648	0.000
62.582	0.000	65.517	0.000
60.951	0.000	63.385	0.000
61.925	0.000	64.589	0.000
60.294	0.000	62.457	0.000
59.637	0.000	61.529	0.000
75.178	0.000	82.067	1.000
73.547	0.000	79.936	1.000
71.917	0.000	77.805	0.000
70.286	0.000	75.673	0.000
68.655	0.000	73.542	0.000
67.024	0.000	71.411	0.000
65.393	0.000	69.279	0.000
72.890	0.000	79.008	0.000
71.259	0.000	76.877	0.000
69.629	0.000	74.745	0.000
67.998	0.000	72.614	0.000
66.367	0.000	70.483	0.000
64.736	0.000	68.351	0.000
70.602	0.000	75.949	0.000
68.971	0.000	73.817	0.000
67.341	0.000	71.686	0.000
65.710	0.000	69.555	0.000
64.079	0.000	67.424	0.000
68.314	0.000	72.889	0.000

Rafael Nadal_score	Rafael Nadal_lead_credit	Novak Djokovic_score	Novak Djokovic_lead_credit
66.683	0.000	70.758	0.000
65.053	0.000	68.627	0.000
63.422	0.000	66.496	0.000
66.026	0.000	69.830	0.000
64.395	0.000	67.699	0.000
62.765	0.000	65.568	0.000
63.738	0.000	66.771	0.000
62.108	0.000	64.640	0.000
61.450	0.000	63.712	0.000
74.704	0.000	81.190	0.000
73.073	0.000	79.059	0.000
71.442	0.000	76.928	0.000
69.811	0.000	74.796	0.000
68.180	0.000	72.665	0.000
66.549	0.000	70.534	0.000
72.416	0.000	78.131	0.000
70.785	0.000	76.000	0.000
69.154	0.000	73.868	0.000
67.523	0.000	71.737	0.000
65.892	0.000	69.606	0.000
70.128	0.000	75.072	0.000
68.497	0.000	72.941	0.000
66.866	0.000	70.809	0.000
65.235	0.000	68.678	0.000
67.840	0.000	72.013	0.000
66.209	0.000	69.881	0.000
64.578	0.000	67.750	0.000
65.552	0.000	68.953	0.000
63.921	0.000	66.822	0.000
63.264	0.000	65.894	0.000
74.229	0.000	80.314	0.000
72.598	0.000	78.182	0.000
70.967	0.000	76.051	0.000
69.336	0.000	73.919	0.000
67.706	0.000	71.788	0.000
71.941	0.000	77.254	0.000
70.310	0.000	75.123	0.000
68.679	0.000	72.992	0.000
67.048	0.000	70.860	0.000
69.653	0.000	74.195	0.000

Rafael Nadal_score	Rafael Nadal_lead_credit	Novak Djokovic_score	Novak Djokovic_lead_credit
68.022	0.000	72.064	0.000
66.391	0.000	69.932	0.000
67.365	0.000	71.136	0.000
65.734	0.000	69.004	0.000
65.077	0.000	68.076	0.000
73.754	0.000	79.437	0.000
72.123	0.000	77.305	0.000
70.493	0.000	75.174	0.000
68.862	0.000	73.043	0.000
71.466	0.000	76.377	0.000
69.835	0.000	74.246	0.000
68.204	0.000	72.115	0.000
69.178	0.000	73.318	0.000
67.547	0.000	71.187	0.000
66.890	0.000	70.259	0.000
73.279	0.000	78.560	0.000
71.649	0.000	76.428	0.000
70.018	0.000	74.297	0.000
70.992	0.000	75.501	0.000
69.361	0.000	73.369	0.000
68.704	0.000	72.441	0.000
72.805	0.000	77.683	0.000
71.174	0.000	75.552	0.000
70.517	0.000	74.624	0.000
72.330	0.000	76.806	0.000

Alternative scaling sensitivity (point estimates only):

scenario	player	alternative_minmax_score	alternative_point_rank
Balanced	Roger Federer	92.042	1
Balanced	Rafael Nadal	0.000	3
Balanced	Novak Djokovic	77.321	2
Winning priority	Roger Federer	95.225	1
Winning priority	Rafael Nadal	0.000	3
Winning priority	Novak Djokovic	73.880	2
Dominance priority	Roger Federer	95.225	1
Dominance priority	Rafael Nadal	0.000	3
Dominance priority	Novak Djokovic	64.025	2
Clutch priority	Roger Federer	91.353	1
Clutch priority	Rafael Nadal	0.000	3
Clutch priority	Novak Djokovic	84.828	2
Adaptability priority	Roger Federer	87.957	1
Adaptability priority	Rafael Nadal	0.000	3
Adaptability priority	Novak Djokovic	84.828	2

Leave-one-metric-out sensitivity (equal remaining weights):

omitted_metric	player	score	point_rank
winning	Roger Federer	67.924	2
winning	Rafael Nadal	64.014	3
winning	Novak Djokovic	68.273	1
dominance	Roger Federer	75.510	2
dominance	Rafael Nadal	70.059	3
dominance	Novak Djokovic	75.547	1
clutch	Roger Federer	72.913	1
clutch	Rafael Nadal	67.868	3
clutch	Novak Djokovic	72.454	2
adaptability	Roger Federer	66.268	1
adaptability	Rafael Nadal	62.432	3
adaptability	Novak Djokovic	65.350	2

Illustrative within-surface comparison; equal winning/DR/CI weights, SAS omitted:

player	surface	illustrative_score	small_ci_sample	ci95_lower	ci95_upper	bootstrap_first_place_pct	paired_valid_bootstraps
Roger Federer	Clay	60.921	False	56.421	65.411	0.100	2000
Rafael Nadal	Clay	69.355	False	66.038	72.525	99.350	2000
Novak Djokovic	Clay	61.701	False	57.463	66.269	0.550	2000
Roger Federer	Grass	69.644	True	65.058	73.888	75.200	2000
Rafael Nadal	Grass	56.419	True	48.617	64.551	0.250	2000
Novak Djokovic	Grass	66.939	True	60.714	72.438	24.550	2000
Roger Federer	Hard	66.949	False	64.669	69.257	58.400	2000
Rafael Nadal	Hard	58.514	False	54.882	62.208	0.000	2000
Novak Djokovic	Hard	66.576	False	63.979	69.065	41.600	2000

Which ranking is being reported? Primary versus alternative scaling

scenario	primary_fixed_scale_leader	alternative_minmax_leader	interpretation
Balanced	Roger Federer	Roger Federer	Different normalization is sensitivity, not an inconsistent primary ranking
Winning priority	Roger Federer	Roger Federer	Different normalization is sensitivity, not an inconsistent primary ranking
Dominance priority	Roger Federer	Roger Federer	Different normalization is sensitivity, not an inconsistent primary ranking
Clutch priority	Roger Federer	Roger Federer	Different normalization is sensitivity, not an inconsistent primary ranking
Adaptability priority	Novak Djokovic	Roger Federer	Different normalization is sensitivity, not an inconsistent primary ranking

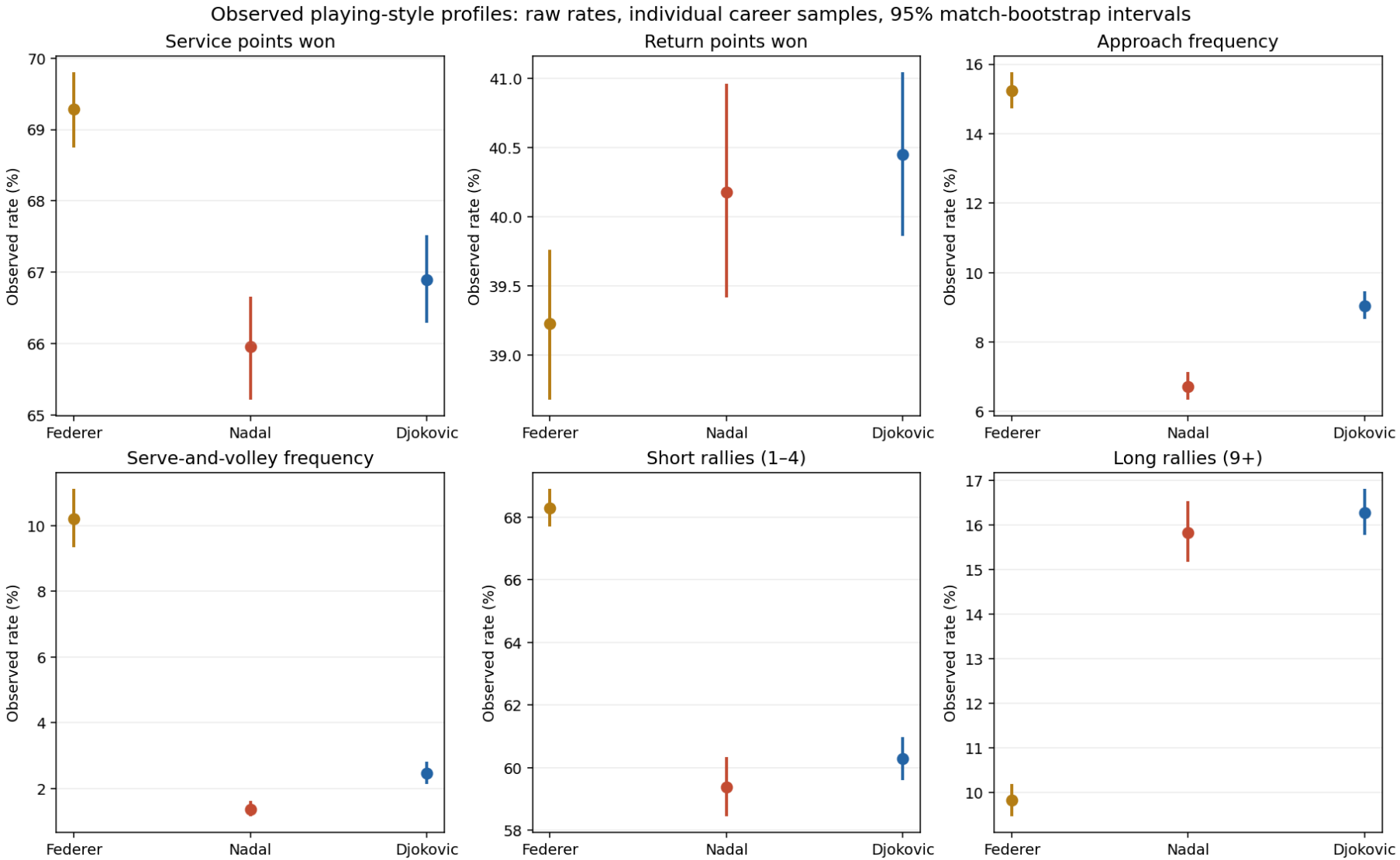
9.4 Visual comparison and conclusions

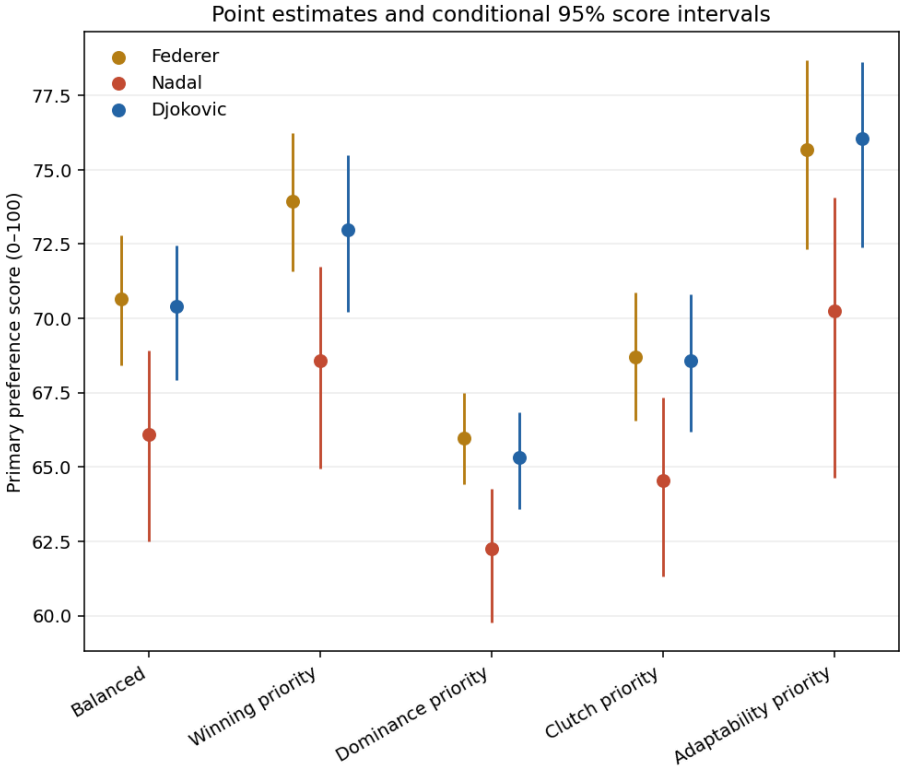
Style panels display raw measurements and match-bootstrap intervals. Scenario scores combine the declared four fixed scales, while contribution bars expose each metric's role. The weight sweep varies one metric's weight and divides the remainder equally among the other three. These are examples of preferences, not an exhaustive value system for greatness.

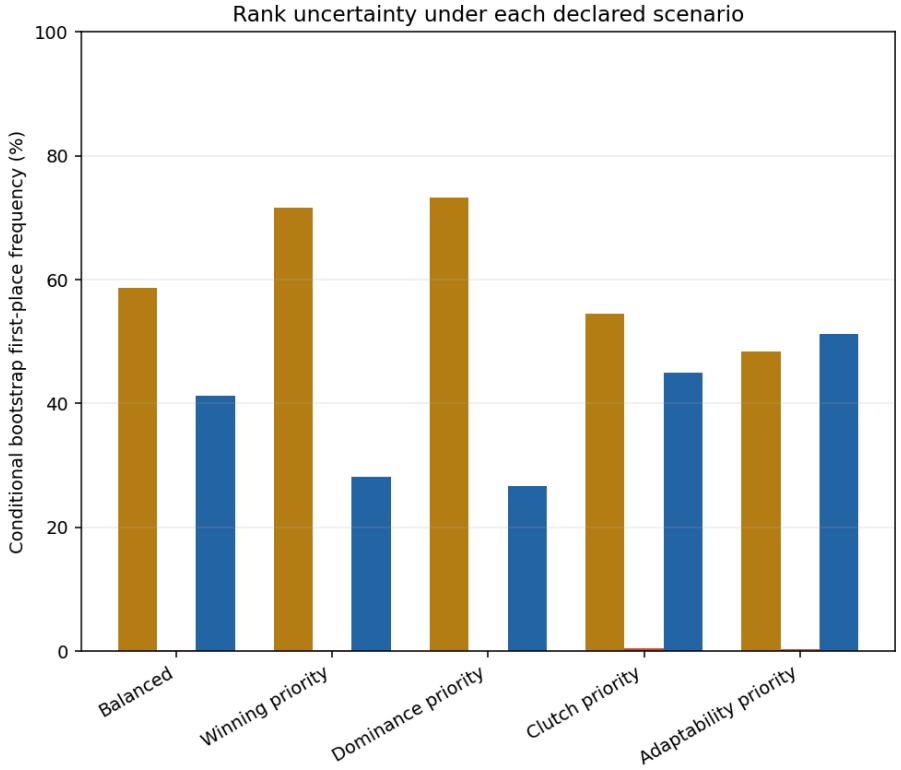
In [31]:
```
with plt.style.context("default"):
    fig,axes=plt.subplots(2,3,figsize=(13,8),constrained_layout=True)
    for ax,metric,title in zip(axes.flat,
["service_point_win_pct","return_point_win_pct","approach_frequency_pct","serve_volley_frequency_pct","short_rally_share_pct","long_rally_share_pct"],
        ["Service points won","Return points won","Approach frequency","Serve-and-volley frequency","Short rallies (1–4)","Long rallies (9+)"]):
        for i,player in enumerate(BIG_THREE):

row=style_profile_long.loc[style_profile_long["player"].eq(player)&style_profile_long["surface"].eq("Overall")&style_profile_long["metric"].eq(metric)].iloc[0]
            ax.scatter(i,row["estimate"],color=KPI_PLAYER_COLORS[player],s=50)
            ax.vlines(i,row["ci95_lower"],row["ci95_upper"],color=KPI_PLAYER_COLORS[player],linewidth=2)
        ax.set_xticks(range(3),[KPI_SHORT_NAMES[p] for p in BIG_THREE]);ax.set_ylabel("Observed rate (%)");ax.set_title(title);ax.grid(axis="y",alpha=.2)
```

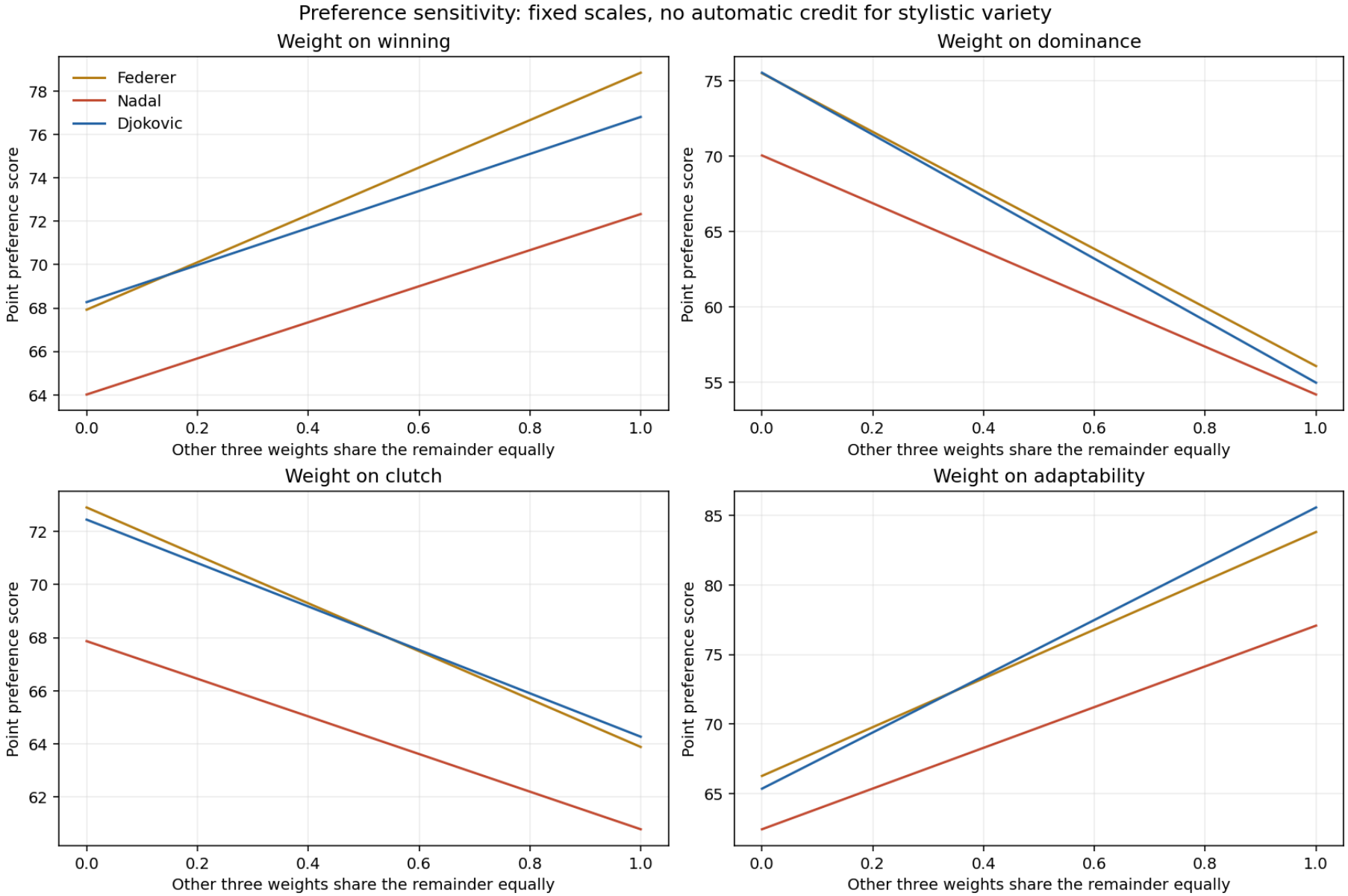
```
    fig.suptitle("Observed playing-style profiles: raw rates, individual career samples, 95% match-bootstrap intervals",fontsize=13)
    display_kpi_figure(fig)

    fig,axes=plt.subplots(1,2,figsize=(14,6),constrained_layout=True)
    scenarios=list(COMPARISON_SCENARIOS)
    for i,player in enumerate(BIG_THREE):
        rows=comparison_scores.loc[comparison_scores["player"].eq(player)].set_index("scenario").reindex(scenarios)
        x=np.arange(len(scenarios))+(i-1)*.16
        axes[0].scatter(x,rows["score"],color=KPI_PLAYER_COLORS[player],label=KPI_SHORT_NAMES[player],s=45)
        for xpos,(_,row) in zip(x,rows.iterrows()):
            axes[0].vlines(xpos,row["ci95_lower"],row["ci95_upper"],color=KPI_PLAYER_COLORS[player])
        axes[1].bar(np.arange(len(scenarios))+(i-
1)*.25,rows["bootstrap_first_place_pct"],width=.25,color=KPI_PLAYER_COLORS[player],label=KPI_SHORT_NAMES[player])
    for ax in axes:
        ax.set_xticks(range(len(scenarios)),scenarios,rotation=30,ha="right");ax.grid(axis="y",alpha=.2)
    axes[0].set_ylabel("Primary preference score (0–100)");axes[0].set_title("Point estimates and conditional 95% score
intervals");axes[0].legend(frameon=False, loc="upper left")
    axes[1].set_ylabel("Conditional bootstrap first-place frequency (%)");axes[1].set_ylim(0,100);axes[1].set_title("Rank uncertainty under each declared
scenario")
    display_kpi_figure(fig)

    fig,axes=plt.subplots(2,2,figsize=(12,8),constrained_layout=True)
    weight_sweep_rows=[]
    for ax,emphasis in zip(axes.flat,COMPARISON_METRICS):
        for i,player in enumerate(BIG_THREE):
            values=[]
            for weight in np.linspace(0,1,21):
                w=np.array([weight if m==emphasis else (1-weight)/3 for m in COMPARISON_METRICS])
                value=100*comparison_points[i]@w
                values.append(value)
                weight_sweep_rows.append({"emphasized_metric":emphasis,"emphasis_weight":weight,"player":player,"score":value})
            ax.plot(np.linspace(0,1,21),values,color=KPI_PLAYER_COLORS[player],label=KPI_SHORT_NAMES[player])
        ax.set_title("Weight on "+emphasis);ax.set_xlabel("Other three weights share the remainder equally");ax.set_ylabel("Point preference
score");ax.grid(alpha=.2)
    axes.flat[0].legend(frameon=False)
    fig.suptitle("Preference sensitivity: fixed scales, no automatic credit for stylistic variety",fontsize=13)
    comparison_weight_sweeps=pd.DataFrame(weight_sweep_rows)
    display_kpi_figure(fig)

    fig,axes=plt.subplots(1,2,figsize=(13,5),constrained_layout=True)
    colors=["#4C78A8","#F58518","#54A24B","#B279A2"]
    bottom=np.zeros(3)
    for metric,color in zip(COMPARISON_METRICS,colors):

values=comparison_contributions.loc[comparison_contributions["scenario"].eq("Balanced")&comparison_contributions["metric"].eq(metric)].set_index("player").reind
ex(BIG_THREE)["score_contribution"].to_numpy()
        axes[0].bar(range(3),values,bottom=bottom,color=color,label=metric);bottom+=values
    axes[0].set_xticks(range(3),[KPI_SHORT_NAMES[p] for p in BIG_THREE]);axes[0].set_ylabel("Balanced score
contribution");axes[0].set_ylim(0,100);axes[0].legend(frameon=False, loc="upper left");axes[0].set_title("What builds the balanced score?")
    for i,player in enumerate(BIG_THREE):
        rows=surface_preference_comparison.loc[surface_preference_comparison["player"].eq(player)].set_index("surface").reindex(PROJECT_SURFACES)
        x=np.arange(3)+(i-1)*.16
        axes[1].scatter(x,rows["illustrative_score"],color=KPI_PLAYER_COLORS[player],label=KPI_SHORT_NAMES[player])
        for xpos,(_,row) in zip(x,rows.iterrows()):
            axes[1].vlines(xpos,row["ci95_lower"],row["ci95_upper"],color=KPI_PLAYER_COLORS[player])
            if row["small_ci_sample"]:
                axes[1].annotate("*",(xpos,row["illustrative_score"]),xytext=(3,4),textcoords="offset points")
```

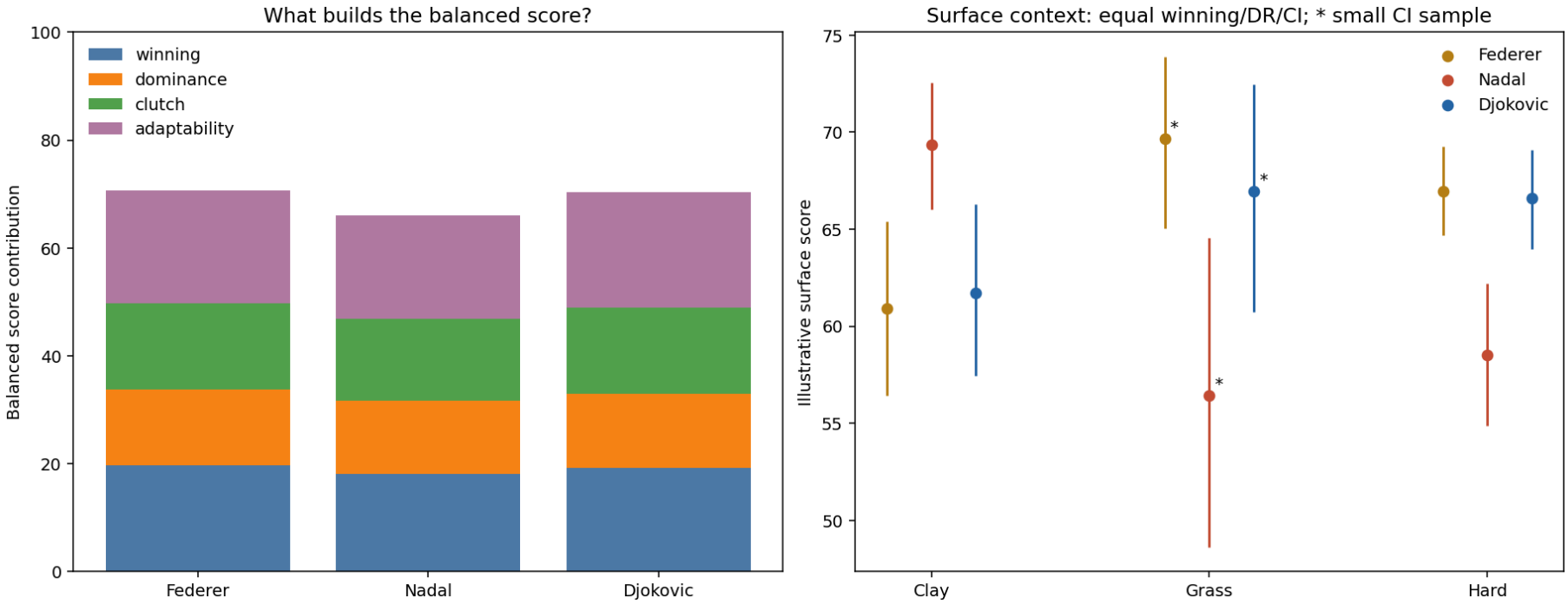
```
    axes[1].set_xticks(range(3),PROJECT_SURFACES);axes[1].set_ylabel("Illustrative surface score");axes[1].set_title("Surface context: equal winning/DR/CI; *
small CI sample");axes[1].legend(frameon=False)
    display_kpi_figure(fig)
```




Point estimates and conditional 95% score intervals
Federer
Nadal
Djokovic
Primary preference score (0–100)
77.5
75.0
72.5
70.0
67.5
65.0
62.5
60.0
Balanced
Winning priority
Dominance priority
Clutch priority
Adaptability priority


Rank uncertainty under each declared scenario
Conditional bootstrap first-place frequency (%)
100
80
60
40
20
0
Balanced
Winning priority
Dominance priority
Clutch priority
Adaptability priority


Preference sensitivity: fixed scales, no automatic credit for stylistic variety
Weight on winning
Weight on dominance
Weight on clutch
Weight on adaptability
Federer
Nadal
Djokovic
Point preference score
Other three weights share the remainder equally



Executive conclusion

The evidence now asks us to make a choice about values. Protecting serve, winning return points, succeeding in pressure events, and maintaining results across surfaces are related achievements, but they are not interchangeable. A declared weighting scheme can organize that choice; it cannot settle it for every reader. The synthesis below is generated from this run, including its direct score-difference interval and scenario leaders.

In [32]:
```
summary_columns=
["player","handedness","verified_wins","unknown_outcome_matches","match_win_pct","dominance_ratio","clutch_index_full","surface_adaptability_score"]
report_table(final_player_scorecard.loc[final_player_scorecard["surface"].eq("Overall"),summary_columns],caption="Executive scorecard: charted evidence,
individual coverage windows",expanded=True)
balanced=comparison_scores.loc[comparison_scores["scenario"].eq("Balanced")].sort_values("score",ascending=False)
lines=["**Answer to the research question.** The available evidence supports different strengths, rather than a single value-free definition of greatness."]
for player in BIG_THREE:
    core=core_kpi_overall.loc[core_kpi_overall["player"].eq(player)].iloc[0]
    style=style_profile_wide.loc[style_profile_wide["player"].eq(player)&style_profile_wide["surface"].eq("Overall")].iloc[0]
    lines.append(f"- **{player}:** observed match wins {core.match_win_pct:.2f}%, DR {core.dominance_ratio:.3f}; approach frequency
{style.approach_frequency_pct:.2f}% and long-rally share {style.long_rally_share_pct:.2f}%. These profile rates use their separately disclosed denominators;
they describe tennis played, not intrinsic athleticism.")
```

```
serve_leader=style_profile_wide.loc[style_profile_wide.surface.eq("Overall")].sort_values("service_point_win_pct",ascending=False).iloc[0]
return_leader=style_profile_wide.loc[style_profile_wide.surface.eq("Overall")].sort_values("return_point_win_pct",ascending=False).iloc[0]
clay_leader=core_kpi_by_surface.loc[core_kpi_by_surface.surface.eq("Clay")].sort_values("match_win_pct",ascending=False).iloc[0]
lines.append(f"\n**Three different routes through the evidence.** {serve_leader.player} has the highest observed service-point success
({serve_leader.service_point_win_pct:.2f}%), alongside approach frequency {serve_leader.approach_frequency_pct:.2f}% and short-rally share
{serve_leader.short_rally_share_pct:.2f}%. That combination is consistent with a serving-and-attacking profile, without proving that attacking caused the wins.
{return_leader.player} leads observed return-point success ({return_leader.return_point_win_pct:.2f}%); the baseline/rally evidence supplies context, not an
inferred movement or athleticism score. {clay_leader.player} leads the clay sample ({int(clay_leader.verified_wins)}/{int(clay_leader.known_outcome_matches)}
verified wins, {clay_leader.match_win_pct:.2f}%). A balance-focused score can understate the value a reader places on that specialization.")
lines.append("\n**Conditional synthesis.** Serving/returning strength, pressure-event outcomes, surface balance, and style must be kept distinct. The surface
table shows specialization that an overall balance metric can under-reward. The direct rivalries can order players differently from their broader charted
schedules. Tactical change documents a changing game, without awarding innovation points.")
for row in balanced.itertuples(index=False):
    lines.append(f"- Balanced preference: {row.player} {row.score:.2f} (95% interval {interval_text(row.ci95_lower,row.ci95_upper)}), first in
{row.bootstrap_first_place_pct:.1f}% of paired bootstrap draws.")
paired=comparison_pairwise_differences.loc[comparison_pairwise_differences["scenario"].eq("Balanced")&comparison_pairwise_differences["comparison"].eq("Roger
Federer minus Novak Djokovic")].iloc[0]
lines.append(f"\nFederer minus Djokovic is **{paired.point_score_difference:+.2f} score points**, with a paired 95% difference interval **
{interval_text(paired.ci95_lower,paired.ci95_upper)}**. This directly quantifies their separation; the narrow point lead is not a decisive statistical victory.
The conditional rank frequency is not a probability that either player is the greatest in history.")
leaders=comparison_scores.loc[comparison_scores["point_estimate_rank"].eq(1),["scenario","player"]]
lines.append("\n**What we value changes the answer:** "+"; ".join(f"{r.scenario}: {r.player}" for r in leaders.itertuples(index=False))+". Primary metrics use
fixed scales and declared weights. Alternative normalization and leave-one-metric-out results are sensitivity analyses, not competing discovered truths.")
lines.append(f"\nThe pressure models yield {int(pressure_pairwise.p_value_holm_six_tests.lt(.05).sum())} Holm-significant player contrasts out of six. That does
not prove equal pressure ability. CI remains a subjective blend of observed outcomes, and shot variety remains descriptive.")
lines.append("\n**What remains unresolved:** chart selection, uncertain endings, different opponents and career stages, unmeasured injuries/era effects, and
missing complete-career achievements. Resampling quantifies conditional uncertainty within this evidence; it cannot make the sample representative. A reader can
reasonably value a different balance of achievements, specialization, and consistency. This report makes that disagreement inspectable rather than declaring an
objective all-history winner.")
executive_findings=pd.DataFrame([
    {"finding":"Overall match-win leader","player":core_kpi_overall.sort_values("match_win_pct",ascending=False).iloc[0].player},
    {"finding":"Overall DR leader","player":core_kpi_overall.sort_values("dominance_ratio",ascending=False).iloc[0].player},
    {"finding":"Primary SAS leader","player":surface_adaptability_summary_v2.sort_values("surface_adaptability_score",ascending=False).iloc[0].player},
    {"finding":"Clay match-win
leader","player":core_kpi_by_surface.loc[core_kpi_by_surface.surface.eq("Clay")].sort_values("match_win_pct",ascending=False).iloc[0].player}])
display(Markdown("\n\n".join(lines)))
```

Executive scorecard: charted evidence, individual coverage windows

	player	handedness	verified_wins	unknown_outcome_matches	match_win_pct	dominance_ratio	clutch_index_full	surface_adaptability_score
0	Roger Federer	Right-handed	559	14	78.843	1.277	0.639	83.810
4	Rafael Nadal	Left-handed	298	13	72.330	1.183	0.608	77.077
8	Novak Djokovic	Right-handed	404	27	76.806	1.221	0.643	85.575

Answer to the research question. The available evidence supports different strengths, rather than a single value-free definition of greatness.

- **Roger Federer:** observed match wins 78.84%, DR 1.277; approach frequency 15.25% and long-rally share 9.82%. These profile rates use their separately disclosed denominators; they describe tennis played, not intrinsic athleticism.
- **Rafael Nadal:** observed match wins 72.33%, DR 1.183; approach frequency 6.73% and long-rally share 15.83%. These profile rates use their separately disclosed denominators; they describe tennis played, not intrinsic athleticism.
- **Novak Djokovic:** observed match wins 76.81%, DR 1.221; approach frequency 9.05% and long-rally share 16.28%. These profile rates use their separately disclosed denominators; they describe tennis played, not intrinsic athleticism.

Three different routes through the evidence. Roger Federer has the highest observed service-point success (69.28%), alongside approach frequency 15.25% and short-rally share 68.29%. That combination is consistent with a serving-and-attacking profile, without proving that attacking caused the wins. Novak Djokovic leads observed return-point success (40.45%); the baseline/rally evidence supplies context, not an inferred movement or athleticism score. Rafael Nadal leads the clay sample (139/164 verified wins, 84.76%). A balance-focused score can understate the value a reader places on that specialization.

Conditional synthesis. Serving/returning strength, pressure-event outcomes, surface balance, and style must be kept distinct. The surface table shows specialization that an overall balance metric can under-reward. The direct rivalries can order players differently from their broader charted schedules. Tactical change documents a changing game, without awarding innovation points.

- Balanced preference: Roger Federer 70.65 (95% interval 68.42–72.80), first in 58.6% of paired bootstrap draws.
- Balanced preference: Novak Djokovic 70.41 (95% interval 67.91–72.46), first in 41.2% of paired bootstrap draws.
- Balanced preference: Rafael Nadal 66.09 (95% interval 62.49–68.92), first in 0.2% of paired bootstrap draws.

Federer minus Djokovic is **+0.25 score points**, with a paired 95% difference interval **-2.97–3.70**. This directly quantifies their separation; the narrow point lead is not a decisive statistical victory. The conditional rank frequency is not a probability that either player is the greatest in history.

What we value changes the answer: Balanced: Roger Federer; Winning priority: Roger Federer; Dominance priority: Roger Federer; Clutch priority: Roger Federer; Adaptability priority: Novak Djokovic. Primary metrics use fixed scales and declared weights. Alternative normalization and leave-one-metric-out results are sensitivity analyses, not competing discovered truths.

The pressure models yield 0 Holm-significant player contrasts out of six. That does not prove equal pressure ability. CI remains a subjective blend of observed outcomes, and shot variety remains descriptive.

What remains unresolved: chart selection, uncertain endings, different opponents and career stages, unmeasured injuries/era effects, and missing complete-career achievements. Resampling quantifies conditional uncertainty within this evidence; it cannot make the sample representative. A reader can reasonably value a different balance of achievements, specialization, and consistency. This report makes that disagreement inspectable rather than declaring an objective all-history winner.

X. Reproduction, Tableau handoff, and release checks

10.1 Reproduce the saved report

Open `goat.ipynb` from the project root so `Raw Data` resolves beside it, select a Python environment with the packages listed in Section II, then **Restart Kernel and Run All**. All transformations, validation, visualizations, and export code are in this notebook. Source CSVs are never overwritten. The local dictionary and workbook explain chart semantics; the analysis uses no external datasets or network calls.

The version table and installation command below record the executed environment, not a universal compatibility guarantee. Each source is fingerprinted with SHA-256; the aggregate fingerprint identifies this input snapshot. Re-execution with different files can change results, denominators, and confidence intervals. The generated executive conclusion is authoritative for that run. The report uses seed 42 and 2,000 shared match-bootstrap draws.

Interpretation limits remain: chart selection, different opponents and career stages, uncertain endings, source disagreements, missing ranking history, descriptive opponent proxies, and conditional bootstrap inference. Court illustrations show categories, not tracking. More tactical change or variety is not automatically better performance. Unknown and not-applicable values remain missing, never zero-filled for presentation.

In [33]:
```
import hashlib
import io
import json

REPORT_VERSION = "1.2"
REPORT_SNAPSHOT_DATE = "2026-10-17"  # Report preparation date supplied by the project session.
WRITE_TABLEAU_EXPORTS = True  # Final release: write the validated Tableau CSV bundle.
```

```
TABLEAU_EXPORT_DIR = Path("Tableau Exports")
source_manifest_rows = []
source_paths = [raw_data_dir / name for name in csv_filenames] + [raw_data_dir/"data_dictionary.txt", raw_data_dir/"MatchChart 0.3.2.xlsm"]
for path in sorted(source_paths, key=lambda item: item.as_posix()):
    digest = hashlib.sha256()
    with path.open("rb") as stream:
        for block in iter(lambda: stream.read(1024*1024), b""):
            digest.update(block)
    key = path.stem.replace("-", "_")
    source_manifest_rows.append({"relative_path": path.as_posix(), "bytes": path.stat().st_size,
        "sha256": digest.hexdigest(), "raw_rows": len(raw_dataframes_unfiltered[key]) if key in raw_dataframes_unfiltered else pd.NA,
        "role": "analysis CSV" if path.suffix == ".csv" else "local interpretation reference"})
source_manifest = pd.DataFrame(source_manifest_rows)
source_snapshot_sha256 = hashlib.sha256("\n".join(
    source_manifest["relative_path"]+" "+source_manifest["sha256"]).encode("utf-8")).hexdigest()
reproduction_requirements = " ".join(package+"=="+version(package) for package in ["pandas","numpy","matplotlib","seaborn","scipy","scikit-
learn","ipython","ipykernel","nbformat","nbclient"])
print("Executed package installation command (use in your chosen environment):")
print("python -m pip install "+reproduction_requirements)
print("Input snapshot SHA-256:", source_snapshot_sha256)
report_table(source_manifest)
```

```
Executed package installation command (use in your chosen environment):
python -m pip install pandas==2.2.2 numpy==1.26.4 matplotlib==3.8.4 seaborn==0.13.2 scipy==1.13.0 scikit-learn==1.4.2 ipython==8.5.0 ipykernel==6.15.3 nbformat==
5.5.0 nbclient==0.6.8
Input snapshot SHA-256: 1b253ea57274935af4e7102b6e83d97d534b8fc74b6cc1ed48effcc8c258eab7
```

relative_path	bytes	sha256	raw_rows	role
Raw Data/MatchChart 0.3.2.xlsm	482316	46e2349eee512296a86170449f6e463a6be91be9261a0c7b6b5d5a25c006729f	<NA>	local interpretation reference
Raw Data/charting-m-matches.csv	1141817	d8dce1ad8d12c04efa01da587441b54ece8e810053066052831224f632efe80	7566	analysis CSV
Raw Data/charting-m-points-2010s.csv	36475702	2719cdc136b64698f24b360cf4f60302e11add24971c2fdd585e1b729178db8c	357229	analysis CSV
Raw Data/charting-m-points-2020s.csv	56590483	2cd43f73e0530a47ea4c02b99dae40177ca6d58a8ccf9189358eb05dffb4be9a	547478	analysis CSV
Raw Data/charting-m-points-to-2009.csv	38025307	12613c7fc4d7acc6d4577ec33554a4308497a5e18669560688f1a2e00704fc72	379569	analysis CSV
Raw Data/charting-m-stats-KeyPointsReturn.csv	5351478	817cebdd70f81a57b699cbd15a89313f1e76c7bfccae73e639a8dce101a5e697	60464	analysis CSV
Raw Data/charting-m-stats-KeyPointsServe.csv	5832013	a2cec9bfe6cb98545895ea100c4fd77900c8febfb9d34c6d63326999f5dcd8f0	60464	analysis CSV
Raw Data/charting-m-stats-NetPoints.csv	6184262	239540c77767512921153a05d8e320450524b6a06976c5bf0d496268a2dee0e8	59396	analysis CSV
Raw Data/charting-m-stats-Overview.csv	6651805	9b08d763e1209dec276206998b96823731f0351ab9b86d25415a7e3dd313290c	56850	analysis CSV
Raw Data/charting-m-stats-Rally.csv	10974130	39fca5a5f930a7a3444d00533016baa764ec37ce43d0a06aa07b713f38d6b498	96706	analysis CSV
Raw Data/charting-m-stats-ReturnDepth.csv	25626585	2a014083db4ba0daaa56f8cc95c6c473058464f5396a8c4dc0c7a4070581132e	270002	analysis CSV
Raw Data/charting-m-stats-ReturnOutcomes.csv	2989479 1	6573360d5629d4ccbe4fa35269f8090aca2797fdb895ed95695d9e9514d0f1ea	310496	analysis CSV
Raw Data/charting-m-stats-ServeBasics.csv	448266 4	5336ad5a021b482c50529ccc78043518b3ce75ca011ef1562a27683c45fffb0d	45341	analysis CSV
Raw Data/charting-m-stats-ServeDirection.csv	4663213	991a437b8af0d69176671e91f9c6c0429fc6060bc6fb0cb741d88ab071bc14fe	45341	analysis CSV
Raw Data/charting-m-stats-ServeInfluence.csv	4123579	abeabf1751d6fb89656d68603c936515d15095929bdcca2adfc0b89788f09b57	30235	analysis CSV
Raw Data/charting-m-stats-ShotDirOutcomes.csv	16221892	46f1862bc3f77af62a58e4250f05bb86469e9c59ecfd0f0174fe669a4e84b4f8	174687	analysis CSV
Raw Data/charting-m-stats-ShotDirection.csv	5263874	9934dffead433d3853b65ff65eda41017bb80623f2a890238a93209e43397043	59997	analysis CSV
Raw Data/charting-m-stats-SnV.csv	5323961	015b46349823b1af14b75c54c5001b940bd1bb5db7e02031ede106a781d89fe3	51300	analysis CSV

relative_path	bytes	sha256	raw_rows	role
Raw Data/charting-m-stats-SvBreakSplit.csv	16575752	293d8495765885892e05a1aebc6ce25273b53bfe709f014fc09c32babf65646a	164871	analysis CSV
Raw Data/charting-m-stats-SvBreakTotal.csv	15346433	298ff58e2259194ba19f0f8ebb75c6295a2887f6c6c1f7989f720a2ccb04516b	164871	analysis CSV
Raw Data/data_dictionary.txt	2649	e60154eb0289e4e0e11221e62c60dcfc94e9d6e4d4a889aa065a7c17bf42b238	<NA>	local interpretation reference

10.2 Tableau data model and export contract

The registry below defines each CSV's grain, unique key, and use. All registered tables are also retained as notebook DataFrames, and their evidence is displayed in the report. **Exports are enabled for this release.** Set `WRITE_TABLEAU_EXPORTS = False` for a notebook-only rerun; this switch controls file writing and does not change analytical results. Rerun the registry/export cell after running the analysis to refresh the bundle. The export directory is `Tableau Exports`; it contains generated tables, a field catalog, and a manifest, not independent analysis code.

Use `players` as a dimension and **relationships rather than physical joins between multiple fact tables**. Scope values are `Overall`, `Clay`, `Grass`, or `Hard`; never add Overall to its component surfaces. Period facts relate through `player_id` plus `period_index`, and scope-specific facts through `player_id` plus `scope`. Categories, metrics, scenarios, and replicate IDs complete the grains listed below. H2H has two role-playing player keys. Standardized tactical trends relate to players/periods and must remain distinct from observed surface facts.

Percentages are 0–100 unless the unit explicitly says fraction/index; CI and entropy are 0–1, SAS and preference scores 0–100, and DR is a ratio. Format using the supplied units. Do not SUM percentages, scores, CI, entropy, or ranks across rows. At a complete grain use ATTR/MIN for precomputed rates; where pooling is meaningful, use summed numerators divided by summed denominators from the same metric/eligibility sample. Unknown values export as blank cells; ranking weeks are a nullable numeric field with a separate availability label.

For a first dashboard, use: (1) the consolidated overall/surface scorecard; (2) shot type and conditional direction shares with coverage; (3) raw versus standardized tactical trends with period counts; (4) primary scenario scores with uncertainty. Keep coverage and small-sample flags visible in tooltips. Existing intervals and rank frequencies belong to their saved scenarios and cannot be reused for arbitrary dashboard weights.

For custom weight parameters, `comparison_inputs_wide` contains the fixed-scale player inputs. With nonnegative weights and a positive total, calculate `100 × (w_win × winning + w_dr × dominance + w_ci × clutch + w_sas × adaptability) / (w_win + w_dr + w_ci + w_sas)`. The paired `comparison_bootstrap_inputs` uses exactly the same four columns per replicate/player if custom uncertainty is implemented: retain all three players within each replicate and split tie credit. Treat bootstrap first-place frequency as conditional uncertainty, not a probability of being the greatest in history.

10.2.1 Machine-readable metric contracts

The implementation below records the definitions already used in Sections V–IX. It adds traceability without changing formulas, eligibility, weights, or resampling.

In [34]:
```
contract_rows=[]
def register_metric(analysis,metric,formula,unit,inputs,aggregation,uncertainty,interpretation):
    contract_rows.append(dict(analysis=analysis,metric=metric,formula=formula,unit=unit,eligible_inputs=inputs,
        aggregation=aggregation,uncertainty=uncertainty,interpretation=interpretation))
for metric,(num,den,basis,fields) in rate_specs.items():
    eligible="kpi_master filtered by kpi_masks[metric]; see core_kpi_long for counts and coverage"
    register_metric("performance",metric,f"100 × sum({num}) / sum({den})","percent (0–100)",eligible,
        "pooled counts; not mean match percentages","2,000 joint match-bootstrap percentile intervals; >=95% finite draws",
        "sample match outcomes" if metric=="match_win_pct" else "descriptive eligible point/event rate; source reconciliation is KPI-specific")
register_metric("performance","dominance_ratio","(sum(return_pts_won)/sum(return_pts)) / (1-sum(total_service_wins)/sum(serve_pts))","ratio",
    "kpi_masks['dominance_ratio']: same reconciled serve/return matches","pooled counts on common DR sample","shared match bootstrap; zero loss denominator
unavailable","return strength relative to service-point losses")
for component,(num,den,weight) in CI_COMPONENTS.items():
    register_metric("pressure",component+"_component",f"sum({num})/sum({den})","fraction (0–1)",
        "clutch_match_events common_ci_eligible; component-only evidence is separately named","common cohort, pooled component counts","shared match
bootstrap","different events; not a mental-toughness measure")
register_metric("pressure","clutch_index_full","0.50*bp_component + 0.30*tb_component + 0.20*deciding_component","index (0–1)",
    "common CI cohort; all three positive component denominators required","weighted component rates","recomputed in paired match draws","declared project-
specific pressure preference")
for metric,formula in [("parity_score","100*(1-sqrt(mean_s((p_s-p)^2)))"),("weakest_surface_win_pct","100*min_s(p_s)"),
    ("surface_adaptability_score","0.60*parity_score + 0.40*weakest_surface_win_pct")]:
    register_metric("adaptability",metric,formula,"score/percent (0–100)","verified outcomes on Clay, Grass, Hard; all surfaces required",
        "p pooled across same three-surface universe; surface rates separately pooled","all quantities recomputed in shared match draws","project-specific
balance/weakest-surface measure; not causal adaptation")
for distribution,(_,categories,entropy_categories) in distribution_inputs.items():
    register_metric("shot_distribution",distribution,"100*category_count/sum(category_counts)","percent (0–100)",
        "distribution_inputs[distribution]; fixed categories; separate classified denominator and coverage","pooled category counts","shared match-bootstrap
intervals","conditional category mix; no measured tracking")
    if entropy_categories is not None:
        register_metric("shot_diversity",distribution,"-sum(p_k*log(p_k))/log(K)","normalized entropy (0–1)",
            f"same distribution; fixed K={len(entropy_categories)}; unknown types excluded from entropy","pooled known-category counts; zero categories
retained",
            "shared match-bootstrap intervals","descriptive variety; no effectiveness credit")
for metric,(num,den,basis) in tactical_metric_specs.items():
    register_metric("tactical_evolution",metric,f"100*sum({num})/sum({den})","percent (0–100)",
        "tactical_match_table filtered by tactical_masks[metric], period and scope","pooled within period/surface; standardized mean of all 3 surface rates",
        "shared match bootstrap; first/last supported-period difference intervals are exploratory","change in observed frequency, not innovation or
```

```
improvement")
for metric,num,den in [("source_winner_rate_pct","winners","source_player_points"),("source_unforced_error_rate_pct","unforced","source_player_points"),
    ("net_point_success_pct","net_points_won","net_points_all")]:
    register_metric("style",metric,f"100*sum({num})/sum({den})","percent (0–100)",
        "profile_master; valid finite counts; source/point reconciliation; NetPoints slice for success","pooled eligible counts",
        "shared match bootstrap","source winner/error labels retained as supplied, not a complete aggression or athleticism measure")
register_metric("rivalry","a_win_pct","100*a_verified_wins/known_outcomes","percent (0–100)",
    "distinct canonical_match_results for each pair/scope; unknown outcomes counted separately","pooled verified meetings",
    "shared match bootstrap; interval unavailable if <95% finite draws","charted rivalry evidence, not complete historical H2H")
register_metric("preference","scenario_score","100*sum(w_m*z_m), z=[win fraction, DR/(1+DR), full CI, SAS/100]","score (0–100)",
    "validated per-metric inputs; no requirement for common eligibility across all four","declared scenario weights; fixed scales",
    "same fully paired match draws across players, metrics and scenarios; ties split rank credit","subjective comparison; overlapping metrics; no probability of
historical greatness")
metric_contract=pd.DataFrame(contract_rows)
assert not metric_contract.duplicated(["analysis","metric"]).any()
report_table(metric_contract,caption="Complete metric contract: formula, unit, eligible inputs, aggregation, uncertainty and meaning")
```

▼ Complete metric contract: formula, unit, eligible inputs, aggregation, uncertainty and meaning

analysis	metric	formula	unit	eligible_inputs	aggregation	uncertainty	interpretation
performance	match_win_pct	100 × sum(verified_win_count) / sum(known_outcome)	percent (0–100)	kpi_master filtered by kpi_masks[metric]; see core_kpi_long for counts and coverage	pooled counts; not mean match percentages	2,000 joint match-bootstrap percentile intervals; >=95% finite draws	sample match outcomes
performance	point_win_pct	100 × sum(points_won) / sum(points_played)	percent (0–100)	kpi_master filtered by kpi_masks[metric]; see core_kpi_long for counts and coverage	pooled counts; not mean match percentages	2,000 joint match-bootstrap percentile intervals; >=95% finite draws	descriptive eligible point/event rate; source reconciliation is KPI-specific
performance	first_serve_in_pct	100 × sum(first_in) / sum(serve_pts)	percent (0–100)	kpi_master filtered by kpi_masks[metric]; see core_kpi_long for counts and coverage	pooled counts; not mean match percentages	2,000 joint match-bootstrap percentile intervals; >=95% finite draws	descriptive eligible point/event rate; source reconciliation is KPI-specific
performance	first_serve_win_pct	100 × sum(first_service_wins) / sum(first_in)	percent (0–100)	kpi_master filtered by kpi_masks[metric]; see core_kpi_long for counts and coverage	pooled counts; not mean match percentages	2,000 joint match-bootstrap percentile intervals; >=95% finite draws	descriptive eligible point/event rate; source reconciliation is KPI-specific
performance	second_serve_win_pct	100 × sum(second_service_wins) / sum(second_in)	percent (0–100)	kpi_master filtered by kpi_masks[metric]; see core_kpi_long for counts and coverage	pooled counts; not mean match percentages	2,000 joint match-bootstrap percentile intervals; >=95% finite draws	descriptive eligible point/event rate; source reconciliation is KPI-specific
performance	service_point_win_pct	100 × sum(total_service_wins) / sum(serve_pts)	percent (0–100)	kpi_master filtered by kpi_masks[metric]; see core_kpi_long for counts and coverage	pooled counts; not mean match percentages	2,000 joint match-bootstrap percentile intervals; >=95% finite draws	descriptive eligible point/event rate; source reconciliation is KPI-specific
performance	return_point_win_pct	100 × sum(return_pts_won) / sum(return_pts)	percent (0–100)	kpi_master filtered by kpi_masks[metric]; see core_kpi_long for counts and coverage	pooled counts; not mean match percentages	2,000 joint match-bootstrap percentile intervals; >=95% finite draws	descriptive eligible point/event rate; source reconciliation is KPI-specific
performance	ace_pct	100 × sum(aces) / sum(serve_pts)	percent (0–100)	kpi_master filtered by kpi_masks[metric]; see core_kpi_long for counts and coverage	pooled counts; not mean match percentages	2,000 joint match-bootstrap percentile intervals; >=95% finite draws	descriptive eligible point/event rate; source reconciliation is KPI-specific
performance	double_fault_pct	100 × sum(dfs) / sum(serve_pts)	percent (0–100)	kpi_master filtered by kpi_masks[metric]; see core_kpi_long for counts and coverage	pooled counts; not mean match percentages	2,000 joint match-bootstrap percentile intervals; >=95% finite draws	descriptive eligible point/event rate; source reconciliation is KPI-specific

analysis	metric	formula	unit	eligible_inputs	aggregation	uncertainty	interpretation
performance	break_point_save_pct	100 × sum(bp_saved_parsed) / sum(bp_faced_parsed)	percent (0–100)	kpi_master filtered by kpi_masks[metric]; see core_kpi_long for counts and coverage	pooled counts; not mean match percentages	2,000 joint match-bootstrap percentile intervals; >=95% finite draws	descriptive eligible point/event rate; source reconciliation is KPI-specific
performance	break_point_conversion_pct	100 × sum(bp_converted_parsed) / sum(bp_opportunities_parsed)	percent (0–100)	kpi_master filtered by kpi_masks[metric]; see core_kpi_long for counts and coverage	pooled counts; not mean match percentages	2,000 joint match-bootstrap percentile intervals; >=95% finite draws	descriptive eligible point/event rate; source reconciliation is KPI-specific
performance	service_hold_pct	100 × sum(service_holds) / sum(service_games)	percent (0–100)	kpi_master filtered by kpi_masks[metric]; see core_kpi_long for counts and coverage	pooled counts; not mean match percentages	2,000 joint match-bootstrap percentile intervals; >=95% finite draws	descriptive eligible point/event rate; source reconciliation is KPI-specific
performance	return_break_pct	100 × sum(return_breaks) / sum(return_games)	percent (0–100)	kpi_master filtered by kpi_masks[metric]; see core_kpi_long for counts and coverage	pooled counts; not mean match percentages	2,000 joint match-bootstrap percentile intervals; >=95% finite draws	descriptive eligible point/event rate; source reconciliation is KPI-specific
performance	dominance_ratio	(sum(return_pts_won)/sum(return_pts)) / (1-sum(total_service_wins)/sum(serve_pts))	ratio	kpi_masks['dominance_ratio']: same reconciled serve/return matches	pooled counts on common DR sample	shared match bootstrap; zero loss denominator unavailable	return strength relative to service-point losses
pressure	bp_component	sum(bp_saved)/sum(bp_faced)	fraction (0–1)	clutch_match_events common_ci_eligible; component-only evidence is separately named	common cohort, pooled component counts	shared match bootstrap	different events; not a mental-toughness measure
pressure	tb_component	sum(tiebreaks_won)/sum(tiebreaks_played)	fraction (0–1)	clutch_match_events common_ci_eligible; component-only evidence is separately named	common cohort, pooled component counts	shared match bootstrap	different events; not a mental-toughness measure
pressure	deciding_component	sum(deciding_sets_won)/sum(deciding_sets_played)	fraction (0–1)	clutch_match_events common_ci_eligible; component-only evidence is separately named	common cohort, pooled component counts	shared match bootstrap	different events; not a mental-toughness measure
pressure	clutch_index_full	0.50*bp_component + 0.30*tb_component + 0.20*deciding_component	index (0–1)	common CI cohort; all three positive component denominators required	weighted component rates	recomputed in paired match draws	declared project-specific pressure preference

analysis	metric	formula	unit	eligible_inputs	aggregation	uncertainty	interpretation
adaptability	parity_score	100*(1-sqrt(mean_s((p_s-p)^2)))	score/percent (0–100)	verified outcomes on Clay, Grass, Hard; all surfaces required	p pooled across same three-surface universe; surface rates separately pooled	all quantities recomputed in shared match draws	project-specific balance/weakest-surface measure; not causal adaptation
adaptability	weakest_surface_win_pct	100*min_s(p_s)	score/percent (0–100)	verified outcomes on Clay, Grass, Hard; all surfaces required	p pooled across same three-surface universe; surface rates separately pooled	all quantities recomputed in shared match draws	project-specific balance/weakest-surface measure; not causal adaptation
adaptability	surface_adaptability_score	0.60*parity_score + 0.40*weakest_surface_win_pct	score/percent (0–100)	verified outcomes on Clay, Grass, Hard; all surfaces required	p pooled across same three-surface universe; surface rates separately pooled	all quantities recomputed in shared match draws	project-specific balance/weakest-surface measure; not causal adaptation
shot_distribution	shot_type	100*category_count/sum(category_counts)	percent (0–100)	distribution_inputs[distribution]; fixed categories; separate classified denominator and coverage	pooled category counts	shared match-bootstrap intervals	conditional category mix; no measured tracking
shot_diversity	shot_type	-sum(p_k*log(p_k))/log(K)	normalized entropy (0–1)	same distribution; fixed K=9; unknown types excluded from entropy	pooled known-category counts; zero categories retained	shared match-bootstrap intervals	descriptive variety; no effectiveness credit
shot_distribution	target_all_attempts	100*category_count/sum(category_counts)	percent (0–100)	distribution_inputs[distribution]; fixed categories; separate classified denominator and coverage	pooled category counts	shared match-bootstrap intervals	conditional category mix; no measured tracking
shot_diversity	target_all_attempts	-sum(p_k*log(p_k))/log(K)	normalized entropy (0–1)	same distribution; fixed K=3; unknown types excluded from entropy	pooled known-category counts; zero categories retained	shared match-bootstrap intervals	descriptive variety; no effectiveness credit
shot_distribution	target_non_error_strokes	100*category_count/sum(category_counts)	percent (0–100)	distribution_inputs[distribution]; fixed categories; separate classified denominator and coverage	pooled category counts	shared match-bootstrap intervals	conditional category mix; no measured tracking

analysis	metric	formula	unit	eligible_inputs	aggregation	uncertainty	interpretation
shot_diversity	target_non_error_strokes	-sum(p_k*log(p_k))/log(K)	normalized entropy (0–1)	same distribution; fixed K=3; unknown types excluded from entropy	pooled known-category counts; zero categories retained	shared match-bootstrap intervals	descriptive variety; no effectiveness credit
shot_distribution	target_returns	100*category_count/sum(category_counts)	percent (0–100)	distribution_inputs[distribution]; fixed categories; separate classified denominator and coverage	pooled category counts	shared match-bootstrap intervals	conditional category mix; no measured tracking
shot_diversity	target_returns	-sum(p_k*log(p_k))/log(K)	normalized entropy (0–1)	same distribution; fixed K=3; unknown types excluded from entropy	pooled known-category counts; zero categories retained	shared match-bootstrap intervals	descriptive variety; no effectiveness credit
shot_distribution	active_serve_target	100*category_count/sum(category_counts)	percent (0–100)	distribution_inputs[distribution]; fixed categories; separate classified denominator and coverage	pooled category counts	shared match-bootstrap intervals	conditional category mix; no measured tracking
shot_distribution	return_depth_target	100*category_count/sum(category_counts)	percent (0–100)	distribution_inputs[distribution]; fixed categories; separate classified denominator and coverage	pooled category counts	shared match-bootstrap intervals	conditional category mix; no measured tracking
shot_distribution	source_direction_Total	100*category_count/sum(category_counts)	percent (0–100)	distribution_inputs[distribution]; fixed categories; separate classified denominator and coverage	pooled category counts	shared match-bootstrap intervals	conditional category mix; no measured tracking
shot_diversity	source_direction_Total	-sum(p_k*log(p_k))/log(K)	normalized entropy (0–1)	same distribution; fixed K=5; unknown types excluded from entropy	pooled known-category counts; zero categories retained	shared match-bootstrap intervals	descriptive variety; no effectiveness credit
shot_distribution	source_direction_F	100*category_count/sum(category_counts)	percent (0–100)	distribution_inputs[distribution]; fixed categories; separate classified denominator and coverage	pooled category counts	shared match-bootstrap intervals	conditional category mix; no measured tracking

analysis	metric	formula	unit	eligible_inputs	aggregation	uncertainty	interpretation
shot_diversity	source_direction_F	-sum(p_k*log(p_k))/log(K)	normalized entropy (0–1)	same distribution; fixed K=5; unknown types excluded from entropy	pooled known-category counts; zero categories retained	shared match-bootstrap intervals	descriptive variety; no effectiveness credit
shot_distribution	source_direction_B	100*category_count/sum(category_counts)	percent (0–100)	distribution_inputs[distribution]; fixed categories; separate classified denominator and coverage	pooled category counts	shared match-bootstrap intervals	conditional category mix; no measured tracking
shot_diversity	source_direction_B	-sum(p_k*log(p_k))/log(K)	normalized entropy (0–1)	same distribution; fixed K=5; unknown types excluded from entropy	pooled known-category counts; zero categories retained	shared match-bootstrap intervals	descriptive variety; no effectiveness credit
shot_distribution	source_direction_S	100*category_count/sum(category_counts)	percent (0–100)	distribution_inputs[distribution]; fixed categories; separate classified denominator and coverage	pooled category counts	shared match-bootstrap intervals	conditional category mix; no measured tracking
shot_diversity	source_direction_S	-sum(p_k*log(p_k))/log(K)	normalized entropy (0–1)	same distribution; fixed K=5; unknown types excluded from entropy	pooled known-category counts; zero categories retained	shared match-bootstrap intervals	descriptive variety; no effectiveness credit
tactical_evolution	approach_frequency_pct	100*sum(approach_points)/sum(source_player_points)	percent (0–100)	tactical_match_table filtered by tactical_masks[metric], period and scope	pooled within period/surface; standardized mean of all 3 surface rates	shared match bootstrap; first/last supported-period difference intervals are exploratory	change in observed frequency, not innovation or improvement
tactical_evolution	net_point_frequency_pct	100*sum(net_points_all)/sum(source_player_points)	percent (0–100)	tactical_match_table filtered by tactical_masks[metric], period and scope	pooled within period/surface; standardized mean of all 3 surface rates	shared match bootstrap; first/last supported-period difference intervals are exploratory	change in observed frequency, not innovation or improvement
tactical_evolution	serve_volley_frequency_pct	100*sum(snv_points)/sum(serve_pts)	percent (0–100)	tactical_match_table filtered by tactical_masks[metric], period and scope	pooled within period/surface; standardized mean of all 3 surface rates	shared match bootstrap; first/last supported-period difference intervals are exploratory	change in observed frequency, not innovation or improvement
tactical_evolution	drop_shot_share_pct	100*sum(drop_count)/sum(known_family)	percent (0–100)	tactical_match_table filtered by tactical_masks[metric], period and scope	pooled within period/surface; standardized mean of all 3 surface rates	shared match bootstrap; first/last supported-period difference intervals are exploratory	change in observed frequency, not innovation or improvement

analysis	metric	formula	unit	eligible_inputs	aggregation	uncertainty	interpretation
tactical_evolution	slice_share_pct	100*sum(slice_count)/sum(known_family)	percent (0–100)	tactical_match_table filtered by tactical_masks[metric], period and scope	pooled within period/surface; standardized mean of all 3 surface rates	shared match bootstrap; first/last supported-period difference intervals are exploratory	change in observed frequency, not innovation or improvement
tactical_evolution	short_rally_share_pct	100*sum(short_rallies)/sum(known_rallies)	percent (0–100)	tactical_match_table filtered by tactical_masks[metric], period and scope	pooled within period/surface; standardized mean of all 3 surface rates	shared match bootstrap; first/last supported-period difference intervals are exploratory	change in observed frequency, not innovation or improvement
tactical_evolution	medium_rally_share_pct	100*sum(medium_rallies)/sum(known_rallies)	percent (0–100)	tactical_match_table filtered by tactical_masks[metric], period and scope	pooled within period/surface; standardized mean of all 3 surface rates	shared match bootstrap; first/last supported-period difference intervals are exploratory	change in observed frequency, not innovation or improvement
tactical_evolution	long_rally_share_pct	100*sum(long_rallies)/sum(known_rallies)	percent (0–100)	tactical_match_table filtered by tactical_masks[metric], period and scope	pooled within period/surface; standardized mean of all 3 surface rates	shared match bootstrap; first/last supported-period difference intervals are exploratory	change in observed frequency, not innovation or improvement
tactical_evolution	deuce_wide_serve_share_pct	100*sum(deuce_wide_serves)/sum(deuce_classified_serves)	percent (0–100)	tactical_match_table filtered by tactical_masks[metric], period and scope	pooled within period/surface; standardized mean of all 3 surface rates	shared match bootstrap; first/last supported-period difference intervals are exploratory	change in observed frequency, not innovation or improvement
tactical_evolution	deuce_body_serve_share_pct	100*sum(deuce_body_serves)/sum(deuce_classified_serves)	percent (0–100)	tactical_match_table filtered by tactical_masks[metric], period and scope	pooled within period/surface; standardized mean of all 3 surface rates	shared match bootstrap; first/last supported-period difference intervals are exploratory	change in observed frequency, not innovation or improvement
tactical_evolution	deuce_t_serve_share_pct	100*sum(deuce_t_serves)/sum(deuce_classified_serves)	percent (0–100)	tactical_match_table filtered by tactical_masks[metric], period and scope	pooled within period/surface; standardized mean of all 3 surface rates	shared match bootstrap; first/last supported-period difference intervals are exploratory	change in observed frequency, not innovation or improvement
tactical_evolution	ad_wide_serve_share_pct	100*sum(ad_wide_serves)/sum(ad_classified_serves)	percent (0–100)	tactical_match_table filtered by tactical_masks[metric], period and scope	pooled within period/surface; standardized mean of all 3 surface rates	shared match bootstrap; first/last supported-period difference intervals are exploratory	change in observed frequency, not innovation or improvement
tactical_evolution	ad_body_serve_share_pct	100*sum(ad_body_serves)/sum(ad_classified_serves)	percent (0–100)	tactical_match_table filtered by tactical_masks[metric], period and scope	pooled within period/surface; standardized mean of all 3 surface rates	shared match bootstrap; first/last supported-period difference intervals are exploratory	change in observed frequency, not innovation or improvement
tactical_evolution	ad_t_serve_share_pct	100*sum(ad_t_serves)/sum(ad_classified_serves)	percent (0–100)	tactical_match_table filtered by tactical_masks[metric], period and scope	pooled within period/surface; standardized mean of all 3 surface rates	shared match bootstrap; first/last supported-period difference intervals are exploratory	change in observed frequency, not innovation or improvement

analysis	metric	formula	unit	eligible_inputs	aggregation	uncertainty	interpretation
style	source_winner_rate_pct	100*sum(winners)/sum(source_player_points)	percent (0–100)	profile_master; valid finite counts; source/point reconciliation; NetPoints slice for success	pooled eligible counts	shared match bootstrap	source winner/error labels retained as supplied, not a complete aggression or athleticism measure
style	source_unforced_error_rate_pct	100*sum(unforced)/sum(source_player_points)	percent (0–100)	profile_master; valid finite counts; source/point reconciliation; NetPoints slice for success	pooled eligible counts	shared match bootstrap	source winner/error labels retained as supplied, not a complete aggression or athleticism measure
style	net_point_success_pct	100*sum(net_points_won)/sum(net_points_all)	percent (0–100)	profile_master; valid finite counts; source/point reconciliation; NetPoints slice for success	pooled eligible counts	shared match bootstrap	source winner/error labels retained as supplied, not a complete aggression or athleticism measure
rivalry	a_win_pct	100*a_verified_wins/known_outcomes	percent (0–100)	distinct canonical_match_results for each pair/scope; unknown outcomes counted separately	pooled verified meetings	shared match bootstrap; interval unavailable if <95% finite draws	charted rivalry evidence, not complete historical H2H
preference	scenario_score	100*sum(w_m*z_m), z=[win fraction, DR/(1+DR), full CI, SAS/100]	score (0–100)	validated per-metric inputs; no requirement for common eligibility across all four	declared scenario weights; fixed scales	same fully paired match draws across players, metrics and scenarios; ties split rank credit	subjective comparison; overlapping metrics; no probability of historical greatness

In [35]:

```
def export_field_unit(table,column,frame):
    if column=="bytes":return "bytes"
    if column=="raw_rows":return "source rows"
    if pd.api.types.is_bool_dtype(frame[column]):return "boolean flag"
    if not pd.api.types.is_numeric_dtype(frame[column]):
        return "ISO date (YYYY-MM-DD)" if "date" in column else "categorical/text attribute"
    if column=="weeks_at_no_1":return "weeks; unavailable in supplied sources"
    if column in ["estimate","ci95_lower","ci95_upper"]:
        if table in ["performance_metrics","style_profiles"]:return "row-specific: see unit column"
        if table=="clutch_metrics":return "fraction (0–1)"
        if table=="shot_diversity":return "normalized entropy (0–1)"
        if table in ["pressure_model_contrasts","pressure_context_effects"]:return "odds ratio"
        if table=="tactical_changes":return "percentage-point change"
        if table=="score_differences":return "preference-score-point difference"
        return "percentage or score (0–100), as table contract states"
    if "p_value" in column:return "p-value (0–1)"
    if "odds_ratio" in column:return "odds ratio"
    if column in ["raw_value","three_player_min","three_player_max"]:return "row-specific: see raw_unit column"
    if column in
["winning","dominance","clutch","adaptability","fixed_scale_value","alternative_minmax_value","weight","parity_weight","weakest_surface_weight"]:return
"fraction / fixed scale (0–1)"
    if "rank" in column:return "ordinal rank"
    if column in ["period_index","replicate_id"]:return "integer identifier; not additive"
```

```
    if column.endswith("year") or column=="nominal_years":return "calendar year" if column!="nominal_years" else "years"
    if column in
["bp_component","tb_component","deciding_component","clutch_index_full","clutch_index_full_ci95_lower","clutch_index_full_ci95_upper","component_only_rate","com
mon_ci_rate"]:return "fraction (0–1)"
    if "diversity" in column or column=="normalized_entropy":return "normalized entropy (0–1)"
    if column=="dominance_ratio":return "ratio"
    if column=="change_percentage_points":return "percentage-point change"
    if column=="point_score_difference":return "preference-score-point difference"
    if "pct" in column or column in ["standardized_pct","estimate_pct"]:return "percent (0–100)"
    if "score" in column or "parity" in column or column=="equal_surface_anchor_sas_sensitivity":
        if column=="max_absolute_score_per_observation":return "absolute likelihood score / observation"
        return "score (0–100)"
    if column in ["minimum_fitted_probability","maximum_fitted_probability"]:return "probability (0–1)"
    if column=="minimum_covariance_eigenvalue":return "coefficient covariance eigenvalue (log-odds parameterization)"
    if column in ["numerator","denominator"]:return "pooled eligible count; blank for derived composites/ratios without a single count"
    return "count (see field name and table grain)"

# Every table has an explicit grain, no inf values, deterministic player IDs, and validated CSV round-trips.
PLAYER_IDS = {"Roger Federer":"federer", "Rafael Nadal":"nadal", "Novak Djokovic":"djokovic"}

def tableau_frame(frame):
    result = frame.copy()
    for column in ["player", "player_a", "player_b"]:
        if column in result:
            id_column = "player_id" if column == "player" else column+"_id"
            result[id_column] = result[column].map(PLAYER_IDS)
            if result[id_column].isna().any():
                raise ValueError(f"Unrecognized player in {column}")
    if "surface" in result:
        result = result.rename(columns={"surface":"scope"})
    for column in result:
        if pd.api.types.is_datetime64_any_dtype(result[column]):
            result[column] = result[column].dt.strftime("%Y-%m-%d")
    if "weeks_at_no_1" in result:
        result["weeks_at_no_1_availability"] = result["weeks_at_no_1"]
        result["weeks_at_no_1"] = pd.Series(pd.NA, index=result.index, dtype="Float64")
    return result.reset_index(drop=True)

players_export = core_kpi_coverage.loc[core_kpi_coverage["surface"].eq("Overall"),
    ["player","handedness","handedness_resolution","handedness_conflicting_records","earliest_charted_date","latest_charted_date","weeks_at_no_1"]]
clutch_export_rows=[]
for row in clutch_summary.itertuples(index=False):
    for name,num,den in [("bp_component","bp_saved","bp_faced"),("tb_component","tiebreaks_won","tiebreaks_played"),
                         ("deciding_component","deciding_sets_won","deciding_sets_played"),("clutch_index_full",None,None)]:
        clutch_export_rows.append({"player":row.player,"surface":row.surface,"metric":name,"estimate":getattr(row,name),
            "unit":"fraction (0–1)","numerator":getattr(row,num) if num else np.nan,"denominator":getattr(row,den) if den else np.nan,
            "common_ci_matches":row.common_ci_matches,"cohort_matches":row.cohort_matches,"small_sample":row.small_sample,
            "ci95_lower":getattr(row,name+"_ci95_lower"),"ci95_upper":getattr(row,name+"_ci95_upper")})
bootstrap_export = pd.DataFrame(comparison_draws_valid.reshape(-1,4),columns=COMPARISON_METRICS)
bootstrap_export["replicate_id"] = np.repeat(np.arange(1,len(comparison_draws_valid)+1),len(BIG_THREE))
bootstrap_export["player"] = np.tile(BIG_THREE,len(comparison_draws_valid))
inputs_wide_export = pd.DataFrame(comparison_points,columns=COMPARISON_METRICS).assign(player=BIG_THREE)
weights_long_export = comparison_weights.melt(id_vars="scenario",value_vars=COMPARISON_METRICS,var_name="metric",value_name="weight")

# table name: (dataframe, full unique key, dashboard use / unit notes)
export_specs = {
 "players":(players_export,["player_id"],"Player dimension; charted coverage dates; ranking weeks blank with availability label"),
```

```
 "coverage":(core_kpi_coverage,["player_id","scope"],"Observed player/scope coverage and known/unknown outcomes"),
 "performance_metrics":(core_kpi_long,["player_id","scope","metric"],"Percentages 0–100 or DR ratio as unit states; matched counts and bootstrap intervals"),
 "clutch_metrics":(pd.DataFrame(clutch_export_rows),["player_id","scope","metric"],"Common-cohort components and full CI, fractions 0–1"),
 "clutch_component_evidence":(clutch_component_evidence,["player_id","scope","component"],"Component-specific versus common-cohort rates, fractions 0–1"),
 "pressure_event_coverage":(pressure_event_coverage,["model","player_id","scope"],"Observed eligible pressure points and match clusters; percentages 0–100"),
 "pressure_model_contrasts":(pressure_pairwise,["model","comparison"],"Direct conditional odds ratios, Wald intervals, Holm-adjusted p-values"),
 "shot_distribution":(shot_distribution_long,["player_id","scope","distribution","category"],"Categorical counts and shares 0–100; no tracking coordinates"),
 "shot_diversity":(shot_diversity_summary,["player_id","scope","distribution"],"Normalized entropy 0–1; fixed K and classified denominator"),
 "shot_coverage":(shot_coverage_summary,["player_id","scope"],"Sequence/type/target/depth coverage, not inferred missing strokes"),
 "surface_adaptability":(surface_adaptability_summary_v2.assign(surface="Overall"),["player_id","scope"],"Primary SAS/parity 0–100; weakest surface evidence and
intervals"),
 "adaptability_weights":(surface_adaptability_weight_sensitivity,["player_id","parity_weight"],"Point definition-sensitivity scores; weights are fractions"),
 "periods":(tactical_period_plan,["player_id","period_index"],"Individual four-year bin definitions and observed dates; sort by period_index"),
 "tactical_trends":(tactical_evolution_long,["player_id","scope","period_index","metric"],"Raw pooled and within-surface rates 0–100, counts, eligible
samples"),
 "tactical_standardized":(tactical_standardized_trends,["player_id","period_index","metric"],"Equal-surface estimates 0–100; not an observed Overall scope"),
 "tactical_changes":(tactical_change_summary,["player_id","metric"],"Exploratory first/last supported-period changes in percentage points"),
 "style_profiles":(style_profile_long,["player_id","scope","metric"],"Raw percentages 0–100 or diversity indices 0–1; unit column applies"),
 "comparison_inputs":(normalization_audit.assign(raw_unit=normalization_audit["metric"].map({"winning":"fraction (0–1)","dominance":"ratio","clutch":"fraction
(0–1)","adaptability":"score (0–100)"})),["player_id","metric"],"Raw values, primary fixed scales 0–1, alternative three-player min–max sensitivity"),
 "comparison_inputs_wide":(inputs_wide_export,["player_id"],"Four primary fixed scales 0–1 for custom point weights"),
 "comparison_bootstrap_inputs":(bootstrap_export,["replicate_id","player_id"],"Joint paired scaled draws; do not break replicate/player dependence"),
 "preference_scores":(comparison_scores,["scenario","player_id"],"Scenario scores 0–100, rank frequencies 0–100 and conditional intervals"),
 "preference_weights":(weights_long_export,["scenario","metric"],"Scenario weights are nonnegative fractions summing to one"),
 "preference_contributions":(comparison_contributions,["scenario","player_id","metric"],"Contributions sum to scenario/player score, not independent
achievements"),
 "score_differences":(comparison_pairwise_differences,["scenario","comparison"],"Paired score differences and bootstrap-ahead frequency, not p-values"),
 "surface_preference_scores":(surface_preference_comparison,["player_id","scope"],"Illustrative equal winning/DR/CI score 0–100, SAS omitted"),
 "head_to_head":(validated_h2h_summary,["player_a_id","player_b_id","scope"],"Charted rivalry outcomes; unknown matches retained, rates 0–100"),
 "scorecard":(final_player_scorecard,["player_id","scope"],"Consolidated wide KPIs; units follow named metrics; SAS only Overall"),
 "source_manifest":(source_manifest,["relative_path"],"Local input filenames, byte counts and SHA-256 fingerprints"),
 "audit_summary":(audit_summary,["dataset"],"Raw = duplicates removed + quarantined + eligible; counts are not rates"),
 "player_metadata_evidence":(player_metadata,["player_id"],"Source-consensus player profile; supporting/conflicting/missing record counts"),
 "handedness_raw_values":(handedness_raw_values,["player_id","source_column","raw_representation"],"Every original hand label, preserved via repr; record
counts"),
 "handedness_exceptions":(handedness_exceptions,["match_id","player_id","source_column"],"Contradictory or missing hand records; no silent source overrides"),
 "issue_disposition":(source_issue_disposition,["issue"],"Resolved versus known source limitations; scope-specific counts are not additive"),
 "metric_contract":(metric_contract,["analysis","metric"],"Formula, unit, eligible analytical inputs, pooling and uncertainty contract"),
 "pressure_diagnostics":(pressure_numerical_diagnostics,["model"],"Numerical fit diagnostics; probabilities 0–1, score residual per observation"),
 "pressure_context_effects":(pressure_context_effects,["model"],"Conditional odds ratio for +10 percentage points of opponent proxy; not causal"),
 "audit_review_register":(audit_review_register,["issue"],"Review scopes and metric-specific handling; flags are not all corrupt matches"),
}
tableau_exports, registry_rows, field_rows, csv_payloads = {}, [], [], {}
for name,(frame,key,notes) in export_specs.items():
    exported=tableau_frame(frame)
    if exported[key].isna().any().any() or exported.duplicated(key).any():
        raise AssertionError(f"Invalid export key: {name}")
    for column in exported.select_dtypes(include="number"):
        if np.isinf(exported[column].to_numpy(dtype=float,na_value=np.nan)).any():
            raise AssertionError(f"Infinite numeric export value: {name}.{column}")
    with np.errstate(invalid="ignore"):
        csv_text=exported.to_csv(index=False,na_rep="",lineterminator="\n",float_format="%.12g")
    parsed=pd.read_csv(io.StringIO(csv_text),dtype=str,keep_default_na=False)
    if len(parsed)!=len(exported) or list(parsed.columns)!=list(exported.columns):
        raise AssertionError(f"CSV schema/row round-trip failed: {name}")
    if parsed.duplicated(key).any():
        raise AssertionError(f"CSV key round-trip failed: {name}")
```

```
    for column in exported.select_dtypes(include="number"):
        if pd.api.types.is_bool_dtype(exported[column]):
            continue
        expected=exported[column].to_numpy(dtype=float,na_value=np.nan)
        actual=pd.to_numeric(parsed[column].where(parsed[column].ne(""),np.nan),errors="coerce").to_numpy(dtype=float)
        if not np.allclose(expected,actual,rtol=1e-10,atol=1e-10,equal_nan=True):
            raise AssertionError(f"CSV numeric round-trip failed: {name}.{column}")
    for column in exported:
        missing=exported[column].isna().to_numpy()
        if not np.array_equal(missing,parsed[column].eq("").to_numpy()):
            raise AssertionError(f"CSV missingness round-trip failed: {name}.{column}")
        if not pd.api.types.is_numeric_dtype(exported[column]) or pd.api.types.is_bool_dtype(exported[column]):
            expected_text=exported[column].astype("string").fillna("").astype(str).to_numpy()
            if not np.array_equal(expected_text,parsed[column].to_numpy()):
                raise AssertionError(f"CSV text/boolean round-trip failed: {name}.{column}")
    tableau_exports[name]=exported
    csv_payloads[name]=csv_text
    registry_rows.append({"table":name,"rows":len(exported),"columns":len(exported.columns),"unique_key":", ".join(key),
        "notes":notes,"csv_sha256":hashlib.sha256(csv_text.encode("utf-8")).hexdigest()})
    for column in exported:
        role="key" if column in key else "measure" if pd.api.types.is_numeric_dtype(exported[column]) else "attribute"
        field_rows.append({"table":name,"field":column,"pandas_dtype":str(exported[column].dtype),"role":role,
            "missing_cells":int(exported[column].isna().sum()),"unit":export_field_unit(name,column,exported),"table_units_and_use":notes})
tableau_export_registry=pd.DataFrame(registry_rows)
tableau_field_catalog=pd.DataFrame(field_rows)
export_manifest={"report_version":REPORT_VERSION,"report_snapshot_date":REPORT_SNAPSHOT_DATE,"input_snapshot_sha256":source_snapshot_sha256,
    "random_seed":RANDOM_SEED,"bootstrap_replicates":BOOTSTRAP_REPLICATES,"player_ids":PLAYER_IDS,"percentage_scale":"0-100 unless unit states fraction/index",
    "missing_values":"empty CSV cells; unknown and not-applicable are not zeros","tables":tableau_export_registry.to_dict(orient="records")}
if WRITE_TABLEAU_EXPORTS:
    TABLEAU_EXPORT_DIR.mkdir(parents=True,exist_ok=True)
    for name,csv_text in csv_payloads.items():
        (TABLEAU_EXPORT_DIR/(name+".csv")).write_text(csv_text,encoding="utf-8")
    tableau_field_catalog.to_csv(TABLEAU_EXPORT_DIR/"field_catalog.csv",index=False)
    tableau_export_registry.to_csv(TABLEAU_EXPORT_DIR/"table_registry.csv",index=False)
    (TABLEAU_EXPORT_DIR/"manifest.json").write_text(json.dumps(export_manifest,ensure_ascii=False,indent=2)+"\n",encoding="utf-8")
    # Verify the actual on-disk files, including the manifest, not just memory payloads.
    written_manifest=json.loads((TABLEAU_EXPORT_DIR/"manifest.json").read_text(encoding="utf-8"))
    assert written_manifest==export_manifest
    for record in written_manifest["tables"]:
        path=TABLEAU_EXPORT_DIR/(record["table"]+".csv")
        assert hashlib.sha256(path.read_bytes()).hexdigest()==record["csv_sha256"]
        actual=pd.read_csv(path,dtype=str,keep_default_na=False)
        expected=pd.read_csv(io.StringIO(csv_payloads[record["table"]]),dtype=str,keep_default_na=False)
        pd.testing.assert_frame_equal(actual,expected)
    for name,expected in [("field_catalog",tableau_field_catalog),("table_registry",tableau_export_registry)]:
        actual=pd.read_csv(TABLEAU_EXPORT_DIR/(name+".csv"),dtype=str,keep_default_na=False)
        target=pd.read_csv(io.StringIO(expected.to_csv(index=False)),dtype=str,keep_default_na=False)
        pd.testing.assert_frame_equal(actual,target)
    print(f"Wrote {len(csv_payloads)} data CSVs, field_catalog.csv, table_registry.csv and manifest.json to {TABLEAU_EXPORT_DIR}.")
else:
    print(f"Validated {len(csv_payloads)} optional data CSVs in memory; no export files written. Set WRITE_TABLEAU_EXPORTS=True to materialize them.")
report_table(tableau_export_registry,caption="Tableau table registry: complete grain and unit contracts")
report_table(tableau_field_catalog,caption="Tableau field catalog: all columns, roles, dtypes and missingness")
```

```
Wrote 37 data CSVs, field_catalog.csv, table_registry.csv and manifest.json to Tableau Exports.
```

Tableau table registry: complete grain and unit contracts

table	rows	columns	unique_key	notes	csv_sha256
players	3	9	player_id	Player dimension; charted coverage dates; ranking weeks blank with availability label	4bb928957496499e96834df470d9f560d971c9f37272381d96d936e48b5cb41d
coverage	12	16	player_id, scope	Observed player/scope coverage and known/unknown outcomes	251e0afdf2567892750e036dd2d1c61545046ab51d1b64edba74a419b43fe439
performance_metrics	168	22	player_id, scope, metric	Percentages 0–100 or DR ratio as unit states; matched counts and bootstrap intervals	b4b56b02aef74a3a230eb7acdb70a8591574fe4b8a3ac9cfcba06fe1366e9050
clutch_metrics	48	13	player_id, scope, metric	Common-cohort components and full CI, fractions 0–1	9eae8bfaf994e8874c4dca2b5051e3dff33a715c69127bd8ec75a747bd9b3638
clutch_component_evidence	36	11	player_id, scope, component	Component-specific versus common-cohort rates, fractions 0–1	33af6acf8482f944c77b5992314fd3102f85a18b253e8ced2db0db10f1d068e7
pressure_event_coverage	18	8	model, player_id, scope	Observed eligible pressure points and match clusters; percentages 0–100	73272bf57d2f70b877fa4c7afaebe40927f36e7b33bba723ed74005e461a5317
pressure_model_contrasts	6	7	model, comparison	Direct conditional odds ratios, Wald intervals, Holm-adjusted p-values	7694a553ba40cf56a42a202676403f8a00e08ad5dcb92c5916ca8e202a51e0a6
shot_distribution	648	12	player_id, scope, distribution, category	Categorical counts and shares 0–100; no tracking coordinates	d687601c1e5907102221086ef4c68680ff99c5d2f1da8de40eeee887534fd90a
shot_diversity	96	12	player_id, scope, distribution	Normalized entropy 0–1; fixed K and classified denominator	87bfc7159115d6569ea7ebb7450da44e2e7df273cd867234d6a045018c011d9e
shot_coverage	12	24	player_id, scope	Sequence/type/target/depth coverage, not inferred missing strokes	ff71f735ac0ed2fc20b3fd5d57652a9e453799966349db1ad8c066ad74695333
surface_adaptability	3	30	player_id, scope	Primary SAS/parity 0–100; weakest surface evidence and intervals	43bb2b22166a00c3cd53d47ec759858b0c607eae2c641c6c4320dd1c049f3649
adaptability_weights	33	6	player_id, parity_weight	Point definition-sensitivity scores; weights are fractions	92bb2872c26fcdc02f85c712027b861fdfb0238912bc1b05ce13b5f168edc33c
periods	18	11	player_id, period_index	Individual four-year bin definitions and observed dates; sort by period_index	4c9f5e2f84564607d937a7587940a7edf112b41773f04087b3c2b77135d7d0c8
tactical_trends	1008	17	player_id, scope, period_index, metric	Raw pooled and within-surface rates 0–100, counts, eligible samples	be631aeedd04de2886add878d8a59c06536d2fa774c036b1979780a6c73d2b9e
tactical_standardized	252	11	player_id, period_index, metric	Equal-surface estimates 0–100; not an observed Overall scope	f850f786b896cb500de7040a043a891a5ad2c673bed471244af4a9b00176f674
tactical_changes	42	10	player_id, metric	Exploratory first/last supported-period changes in percentage points	ef96a44b1e487e786059ec7c9909970f33dc1d45b53ef40d58822f11360206fb
style_profiles	192	12	player_id, scope, metric	Raw percentages 0–100 or diversity indices 0–1; unit column applies	a151864c7f1c2d528e56ec773e276d6540d264e444f1fa825a30cd0ec150d3ca

table	rows	columns	unique_key	notes	csv_sha256
comparison_inputs	12	12	player_id, metric	Raw values, primary fixed scales 0–1, alternative three-player min–max sensitivity	7bba8b513ba97702a2c89832ebaef9623b7df836e2d4fb3a810b17ffa588372c
comparison_inputs_wide	3	6	player_id	Four primary fixed scales 0–1 for custom point weights	69f908b4babd2a9b57469571e368ac9f821c90ab146f57b0ed7516e5af5ef519
comparison_bootstrap_inputs	6000	7	replicate_id, player_id	Joint paired scaled draws; do not break replicate/player dependence	4bd338d74b0d2b42f9d4cea0728fc7a0cf573b02a43c693a69247736381c9e4d
preference_scores	15	12	scenario, player_id	Scenario scores 0–100, rank frequencies 0–100 and conditional intervals	e388cfbb2af6724e434f4cbd79b007c43769effe6182ca3a42d196db8a9cf12a
preference_weights	20	3	scenario, metric	Scenario weights are nonnegative fractions summing to one	13ec6ac55060e77e96c7257541ae1bbbd8e432c737b713516779988c2ffc7ee7
preference_contributions	60	7	scenario, player_id, metric	Contributions sum to scenario/player score, not independent achievements	6b9b0af5cec1c571f567ed68642a270570bea628feda39b05653a17da87d8f1b
score_differences	15	6	scenario, comparison	Paired score differences and bootstrap-ahead frequency, not p-values	9e46c8e6ef220a3118b2854a949e8e6d1adb25baed09cf8594354a923101156a
surface_preference_scores	9	9	player_id, scope	Illustrative equal winning/DR/CI score 0–100, SAS omitted	9cf18ace2a997b3e0bfa930ce9c3d74049d7e53fb924504cceb97ad7852b4f59
head_to_head	12	14	player_a_id, player_b_id, scope	Charted rivalry outcomes; unknown matches retained, rates 0–100	c088a61745348770ccfc492a94386fc87150ff63d0e04b912e690b15904b3199
scorecard	12	53	player_id, scope	Consolidated wide KPIs; units follow named metrics; SAS only Overall	dd95be437ea3e3599686863d26c2b38d822e7cead1ad4a19c768e45b82111197
source_manifest	21	5	relative_path	Local input filenames, byte counts and SHA-256 fingerprints	58e6cbbc011c0fc4eb4ff6dcbe08c64a2155c97867691b2a32d64e7fdcad4b86
audit_summary	19	10	dataset	Raw = duplicates removed + quarantined + eligible; counts are not rates	3e9a3bfc3352204bd7635cbaaa9b6377e0dcf815ed14947acc59112a5a1020d1
player_metadata_evidence	3	11	player_id	Source-consensus player profile; supporting/conflicting/missing record counts	63145e8d48442a81722ddb25970585ca1ddf53896f2884e11ecfbae86f05284d
handedness_raw_values	14	5	player_id, source_column, raw_representation	Every original hand label, preserved via repr; record counts	94369fe2f0676080cd874b997d56cd1b4396a7247cc2ff704e200f67c53457fc
handedness_exceptions	7	8	match_id, player_id, source_column	Contradictory or missing hand records; no silent source overrides	74c24cb6947f0e1f429ba19f3b235fb4b171ecdc887658e93d64bc254797987c
issue_disposition	15	4	issue	Resolved versus known source limitations; scope-specific counts are not additive	fda4376c784274bf9d5de8d962d7bc12e016846cfe2e6748b63e79c256592094
metric_contract	58	8	analysis, metric	Formula, unit, eligible analytical inputs, pooling and uncertainty contract	511bd77ae66f581997a352769abcc46b44a459d547a388c2ed72b8e8de68ec54
pressure_diagnostics	2	7	model	Numerical fit diagnostics; probabilities 0–1, score residual per observation	3352b076272b732d140dc7da6050af3c045b8a28e194c1a88ac84d360fba6de2

table	rows	columns	unique_key	notes	csv_sha256
pressure_context_effects	2	6	model	Conditional odds ratio for +10 percentage points of opponent proxy; not causal	bef20e2721fc59cfb24a02ac1383aec9833aaedc4d39c43a8180db49ff1f0699
audit_review_register	4	3	issue	Review scopes and metric-specific handling; flags are not all corrupt matches	54a3eb35830a713e75a9f0fd5687a7e76c38483fa1babf9e4703a246e4c9b4e0

▼ Tableau field catalog: all columns, roles, dtypes and missingness

table	field	pandas_dtype	role	missing_cells	unit	table_units_and_use
players	player	object	attribute	0	categorical/text attribute	Player dimension; charted coverage dates; ranking weeks blank with availability label
players	handedness	object	attribute	0	categorical/text attribute	Player dimension; charted coverage dates; ranking weeks blank with availability label
players	handedness_resolution	object	attribute	0	categorical/text attribute	Player dimension; charted coverage dates; ranking weeks blank with availability label
players	handedness_conflicting_records	int64	measure	0	count (see field name and table grain)	Player dimension; charted coverage dates; ranking weeks blank with availability label
players	earliest_charted_date	object	attribute	0	ISO date (YYYY-MM-DD)	Player dimension; charted coverage dates; ranking weeks blank with availability label
players	latest_charted_date	object	attribute	0	ISO date (YYYY-MM-DD)	Player dimension; charted coverage dates; ranking weeks blank with availability label
players	weeks_at_no_1	Float64	measure	3	weeks; unavailable in supplied sources	Player dimension; charted coverage dates; ranking weeks blank with availability label
players	player_id	object	key	0	categorical/text attribute	Player dimension; charted coverage dates; ranking weeks blank with availability label
players	weeks_at_no_1_availability	object	attribute	0	categorical/text attribute	Player dimension; charted coverage dates; ranking weeks blank with availability label
coverage	player	object	attribute	0	categorical/text attribute	Observed player/scope coverage and known/unknown outcomes
coverage	scope	object	key	0	categorical/text attribute	Observed player/scope coverage and known/unknown outcomes
coverage	charted_matches	int64	measure	0	count (see field name and table grain)	Observed player/scope coverage and known/unknown outcomes
coverage	verified_wins	int64	measure	0	count (see field name and table grain)	Observed player/scope coverage and known/unknown outcomes
coverage	known_outcome_matches	int64	measure	0	count (see field name and table grain)	Observed player/scope coverage and known/unknown outcomes
coverage	unknown_outcome_matches	int64	measure	0	count (see field name and table grain)	Observed player/scope coverage and known/unknown outcomes
coverage	matches_with_overview	int64	measure	0	count (see field name and table grain)	Observed player/scope coverage and known/unknown outcomes
coverage	matches_with_points	int64	measure	0	count (see field name and table grain)	Observed player/scope coverage and known/unknown outcomes

table	field	pandas_dtype	role	missing_cells	unit	table_units_and_use
coverage	earliest_charted_date	object	attribute	0	ISO date (YYYY-MM-DD)	Observed player/scope coverage and known/unknown outcomes
coverage	latest_charted_date	object	attribute	0	ISO date (YYYY-MM-DD)	Observed player/scope coverage and known/unknown outcomes
coverage	handedness	object	attribute	0	categorical/text attribute	Observed player/scope coverage and known/unknown outcomes
coverage	handedness_resolution	object	attribute	0	categorical/text attribute	Observed player/scope coverage and known/unknown outcomes
coverage	handedness_conflicting_records	int64	measure	0	count (see field name and table grain)	Observed player/scope coverage and known/unknown outcomes
coverage	weeks_at_no_1	Float64	measure	12	weeks; unavailable in supplied sources	Observed player/scope coverage and known/unknown outcomes
coverage	player_id	object	key	0	categorical/text attribute	Observed player/scope coverage and known/unknown outcomes
coverage	weeks_at_no_1_availability	object	attribute	0	categorical/text attribute	Observed player/scope coverage and known/unknown outcomes
performance_metrics	player	object	attribute	0	categorical/text attribute	Percentages 0–100 or DR ratio as unit states; matched counts and bootstrap intervals
performance_metrics	scope	object	key	0	categorical/text attribute	Percentages 0–100 or DR ratio as unit states; matched counts and bootstrap intervals
performance_metrics	metric	object	key	0	categorical/text attribute	Percentages 0–100 or DR ratio as unit states; matched counts and bootstrap intervals
performance_metrics	estimate	float64	measure	0	row-specific: see unit column	Percentages 0–100 or DR ratio as unit states; matched counts and bootstrap intervals
performance_metrics	unit	object	attribute	0	categorical/text attribute	Percentages 0–100 or DR ratio as unit states; matched counts and bootstrap intervals
performance_metrics	numerator	float64	measure	12	pooled eligible count; blank for derived composites/ratios without a single count	Percentages 0–100 or DR ratio as unit states; matched counts and bootstrap intervals
performance_metrics	denominator	float64	measure	12	pooled eligible count; blank for derived composites/ratios without a single count	Percentages 0–100 or DR ratio as unit states; matched counts and bootstrap intervals

table	field	pandas_dtype	role	missing_cells	unit	table_units_and_use
performance_metrics	eligible_matches	int64	measure	0	count (see field name and table grain)	Percentages 0–100 or DR ratio as unit states; matched counts and bootstrap intervals
performance_metrics	cohort_matches	int64	measure	0	count (see field name and table grain)	Percentages 0–100 or DR ratio as unit states; matched counts and bootstrap intervals
performance_metrics	coverage_pct	float64	measure	0	percent (0–100)	Percentages 0–100 or DR ratio as unit states; matched counts and bootstrap intervals
performance_metrics	matches_with_point_reconciliation	object	attribute	72	categorical/text attribute	Percentages 0–100 or DR ratio as unit states; matched counts and bootstrap intervals
performance_metrics	small_sample	bool	measure	0	boolean flag	Percentages 0–100 or DR ratio as unit states; matched counts and bootstrap intervals
performance_metrics	near_zero_dr_denominator	bool	measure	0	boolean flag	Percentages 0–100 or DR ratio as unit states; matched counts and bootstrap intervals
performance_metrics	return_points_won_dr	float64	measure	156	count (see field name and table grain)	Percentages 0–100 or DR ratio as unit states; matched counts and bootstrap intervals
performance_metrics	return_points_dr	float64	measure	156	count (see field name and table grain)	Percentages 0–100 or DR ratio as unit states; matched counts and bootstrap intervals
performance_metrics	service_points_won_dr	float64	measure	156	count (see field name and table grain)	Percentages 0–100 or DR ratio as unit states; matched counts and bootstrap intervals
performance_metrics	service_points_dr	float64	measure	156	count (see field name and table grain)	Percentages 0–100 or DR ratio as unit states; matched counts and bootstrap intervals
performance_metrics	valid_bootstrap_replicates	int64	measure	0	count (see field name and table grain)	Percentages 0–100 or DR ratio as unit states; matched counts and bootstrap intervals
performance_metrics	ci95_lower	float64	measure	0	row-specific: see unit column	Percentages 0–100 or DR ratio as unit states; matched counts and bootstrap intervals
performance_metrics	ci95_upper	float64	measure	0	row-specific: see unit column	Percentages 0–100 or DR ratio as unit states; matched counts and bootstrap intervals

table	field	pandas_dtype	role	missing_cells	unit	table_units_and_use
performance_metrics	status	object	attribute	0	categorical/text attribute	Percentages 0–100 or DR ratio as unit states; matched counts and bootstrap intervals
performance_metrics	player_id	object	key	0	categorical/text attribute	Percentages 0–100 or DR ratio as unit states; matched counts and bootstrap intervals
clutch_metrics	player	object	attribute	0	categorical/text attribute	Common-cohort components and full CI, fractions 0–1
clutch_metrics	scope	object	key	0	categorical/text attribute	Common-cohort components and full CI, fractions 0–1
clutch_metrics	metric	object	key	0	categorical/text attribute	Common-cohort components and full CI, fractions 0–1
clutch_metrics	estimate	float64	measure	0	fraction (0–1)	Common-cohort components and full CI, fractions 0–1
clutch_metrics	unit	object	attribute	0	categorical/text attribute	Common-cohort components and full CI, fractions 0–1
clutch_metrics	numerator	float64	measure	12	pooled eligible count; blank for derived composites/ratios without a single count	Common-cohort components and full CI, fractions 0–1
clutch_metrics	denominator	float64	measure	12	pooled eligible count; blank for derived composites/ratios without a single count	Common-cohort components and full CI, fractions 0–1
clutch_metrics	common_ci_matches	int64	measure	0	count (see field name and table grain)	Common-cohort components and full CI, fractions 0–1
clutch_metrics	cohort_matches	int64	measure	0	count (see field name and table grain)	Common-cohort components and full CI, fractions 0–1
clutch_metrics	small_sample	bool	measure	0	boolean flag	Common-cohort components and full CI, fractions 0–1
clutch_metrics	ci95_lower	float64	measure	0	fraction (0–1)	Common-cohort components and full CI, fractions 0–1
clutch_metrics	ci95_upper	float64	measure	0	fraction (0–1)	Common-cohort components and full CI, fractions 0–1
clutch_metrics	player_id	object	key	0	categorical/text attribute	Common-cohort components and full CI, fractions 0–1
clutch_component_evidence	player	object	attribute	0	categorical/text attribute	Component-specific versus common-cohort rates, fractions 0–1
clutch_component_evidence	scope	object	key	0	categorical/text attribute	Component-specific versus common-cohort rates, fractions 0–1

table	field	pandas_dtype	role	missing_cells	unit	table_units_and_use
clutch_component_evidence	component	object	key	0	categorical/text attribute	Component-specific versus common-cohort rates, fractions 0–1
clutch_component_evidence	component_eligible_matches	int64	measure	0	count (see field name and table grain)	Component-specific versus common-cohort rates, fractions 0–1
clutch_component_evidence	component_wins	float64	measure	0	count (see field name and table grain)	Component-specific versus common-cohort rates, fractions 0–1
clutch_component_evidence	component_events	float64	measure	0	count (see field name and table grain)	Component-specific versus common-cohort rates, fractions 0–1
clutch_component_evidence	component_only_rate	float64	measure	0	fraction (0–1)	Component-specific versus common-cohort rates, fractions 0–1
clutch_component_evidence	common_ci_wins	float64	measure	0	count (see field name and table grain)	Component-specific versus common-cohort rates, fractions 0–1
clutch_component_evidence	common_ci_events	float64	measure	0	count (see field name and table grain)	Component-specific versus common-cohort rates, fractions 0–1
clutch_component_evidence	common_ci_rate	float64	measure	0	fraction (0–1)	Component-specific versus common-cohort rates, fractions 0–1
clutch_component_evidence	player_id	object	key	0	categorical/text attribute	Component-specific versus common-cohort rates, fractions 0–1
pressure_event_coverage	model	object	key	0	categorical/text attribute	Observed eligible pressure points and match clusters; percentages 0–100
pressure_event_coverage	player	object	attribute	0	categorical/text attribute	Observed eligible pressure points and match clusters; percentages 0–100
pressure_event_coverage	scope	string	key	0	categorical/text attribute	Observed eligible pressure points and match clusters; percentages 0–100
pressure_event_coverage	observations	int64	measure	0	count (see field name and table grain)	Observed eligible pressure points and match clusters; percentages 0–100
pressure_event_coverage	successes	int64	measure	0	count (see field name and table grain)	Observed eligible pressure points and match clusters; percentages 0–100
pressure_event_coverage	match_clusters	int64	measure	0	count (see field name and table grain)	Observed eligible pressure points and match clusters; percentages 0–100
pressure_event_coverage	observed_success_pct	float64	measure	0	percent (0–100)	Observed eligible pressure points and match clusters; percentages 0–100
pressure_event_coverage	player_id	object	key	0	categorical/text attribute	Observed eligible pressure points and match clusters; percentages 0–100
pressure_model_contrasts	model	object	key	0	categorical/text attribute	Direct conditional odds ratios, Wald intervals, Holm-adjusted p-values

table	field	pandas_dtype	role	missing_cells	unit	table_units_and_use
pressure_model_contrasts	comparison	object	key	0	categorical/text attribute	Direct conditional odds ratios, Wald intervals, Holm-adjusted p-values
pressure_model_contrasts	odds_ratio	float64	measure	0	odds ratio	Direct conditional odds ratios, Wald intervals, Holm-adjusted p-values
pressure_model_contrasts	ci95_lower	float64	measure	0	odds ratio	Direct conditional odds ratios, Wald intervals, Holm-adjusted p-values
pressure_model_contrasts	ci95_upper	float64	measure	0	odds ratio	Direct conditional odds ratios, Wald intervals, Holm-adjusted p-values
pressure_model_contrasts	p_value_unadjusted	float64	measure	0	p-value (0–1)	Direct conditional odds ratios, Wald intervals, Holm-adjusted p-values
pressure_model_contrasts	p_value_holm_six_tests	float64	measure	0	p-value (0–1)	Direct conditional odds ratios, Wald intervals, Holm-adjusted p-values
shot_distribution	player	object	attribute	0	categorical/text attribute	Categorical counts and shares 0–100; no tracking coordinates
shot_distribution	scope	object	key	0	categorical/text attribute	Categorical counts and shares 0–100; no tracking coordinates
shot_distribution	distribution	object	key	0	categorical/text attribute	Categorical counts and shares 0–100; no tracking coordinates
shot_distribution	category	object	key	0	categorical/text attribute	Categorical counts and shares 0–100; no tracking coordinates
shot_distribution	count	int64	measure	0	count (see field name and table grain)	Categorical counts and shares 0–100; no tracking coordinates
shot_distribution	classified_denominator	int64	measure	0	count (see field name and table grain)	Categorical counts and shares 0–100; no tracking coordinates
shot_distribution	share_pct	float64	measure	0	percent (0–100)	Categorical counts and shares 0–100; no tracking coordinates
shot_distribution	contributing_matches	int64	measure	0	count (see field name and table grain)	Categorical counts and shares 0–100; no tracking coordinates
shot_distribution	valid_bootstraps	int64	measure	0	count (see field name and table grain)	Categorical counts and shares 0–100; no tracking coordinates
shot_distribution	ci95_lower	float64	measure	0	percentage or score (0–100), as table contract states	Categorical counts and shares 0–100; no tracking coordinates
shot_distribution	ci95_upper	float64	measure	0	percentage or score (0–100), as table contract states	Categorical counts and shares 0–100; no tracking coordinates
shot_distribution	player_id	object	key	0	categorical/text attribute	Categorical counts and shares 0–100; no tracking coordinates

table	field	pandas_dtype	role	missing_cells	unit	table_units_and_use
shot_diversity	player	object	attribute	0	categorical/text attribute	Normalized entropy 0–1; fixed K and classified denominator
shot_diversity	scope	object	key	0	categorical/text attribute	Normalized entropy 0–1; fixed K and classified denominator
shot_diversity	distribution	object	key	0	categorical/text attribute	Normalized entropy 0–1; fixed K and classified denominator
shot_diversity	normalized_entropy	float64	measure	0	normalized entropy (0–1)	Normalized entropy 0–1; fixed K and classified denominator
shot_diversity	fixed_categories_K	int64	measure	0	count (see field name and table grain)	Normalized entropy 0–1; fixed K and classified denominator
shot_diversity	classified_strokes	int64	measure	0	count (see field name and table grain)	Normalized entropy 0–1; fixed K and classified denominator
shot_diversity	contributing_matches	int64	measure	0	count (see field name and table grain)	Normalized entropy 0–1; fixed K and classified denominator
shot_diversity	small_sample	bool	measure	0	boolean flag	Normalized entropy 0–1; fixed K and classified denominator
shot_diversity	valid_bootstraps	int64	measure	0	count (see field name and table grain)	Normalized entropy 0–1; fixed K and classified denominator
shot_diversity	ci95_lower	float64	measure	0	normalized entropy (0–1)	Normalized entropy 0–1; fixed K and classified denominator
shot_diversity	ci95_upper	float64	measure	0	normalized entropy (0–1)	Normalized entropy 0–1; fixed K and classified denominator
shot_diversity	player_id	object	key	0	categorical/text attribute	Normalized entropy 0–1; fixed K and classified denominator
shot_coverage	player	object	attribute	0	categorical/text attribute	Sequence/type/target/depth coverage, not inferred missing strokes
shot_coverage	scope	object	key	0	categorical/text attribute	Sequence/type/target/depth coverage, not inferred missing strokes
shot_coverage	cohort_matches	int64	measure	0	count (see field name and table grain)	Sequence/type/target/depth coverage, not inferred missing strokes
shot_coverage	matches_with_parsed_strokes	int64	measure	0	count (see field name and table grain)	Sequence/type/target/depth coverage, not inferred missing strokes
shot_coverage	recorded_player_points	int64	measure	0	count (see field name and table grain)	Sequence/type/target/depth coverage, not inferred missing strokes
shot_coverage	points_with_sequence_codes	int64	measure	0	count (see field name and table grain)	Sequence/type/target/depth coverage, not inferred missing strokes

table	field	pandas_dtype	role	missing_cells	unit	table_units_and_use
shot_coverage	sequence_code_coverage_pct	float64	measure	0	percent (0–100)	Sequence/type/target/depth coverage, not inferred missing strokes
shot_coverage	attempted_rally_strokes	int64	measure	0	count (see field name and table grain)	Sequence/type/target/depth coverage, not inferred missing strokes
shot_coverage	error_attempts	int64	measure	0	count (see field name and table grain)	Sequence/type/target/depth coverage, not inferred missing strokes
shot_coverage	unknown_type_strokes	int64	measure	0	count (see field name and table grain)	Sequence/type/target/depth coverage, not inferred missing strokes
shot_coverage	unknown_type_pct	float64	measure	0	percent (0–100)	Sequence/type/target/depth coverage, not inferred missing strokes
shot_coverage	known_wing_strokes	int64	measure	0	count (see field name and table grain)	Sequence/type/target/depth coverage, not inferred missing strokes
shot_coverage	forehand_share_known_wing_pct	float64	measure	0	percent (0–100)	Sequence/type/target/depth coverage, not inferred missing strokes
shot_coverage	classified_targets	int64	measure	0	count (see field name and table grain)	Sequence/type/target/depth coverage, not inferred missing strokes
shot_coverage	target_direction_coverage_pct	float64	measure	0	percent (0–100)	Sequence/type/target/depth coverage, not inferred missing strokes
shot_coverage	parsed_return_attempts	int64	measure	0	count (see field name and table grain)	Sequence/type/target/depth coverage, not inferred missing strokes
shot_coverage	returns_with_known_depth	int64	measure	0	count (see field name and table grain)	Sequence/type/target/depth coverage, not inferred missing strokes
shot_coverage	return_depth_coverage_pct	float64	measure	0	percent (0–100)	Sequence/type/target/depth coverage, not inferred missing strokes
shot_coverage	returns_with_depth_and_direction	int64	measure	0	count (see field name and table grain)	Sequence/type/target/depth coverage, not inferred missing strokes
shot_coverage	recorded_service_points	int64	measure	0	count (see field name and table grain)	Sequence/type/target/depth coverage, not inferred missing strokes
shot_coverage	classified_active_serve_targets	int64	measure	0	count (see field name and table grain)	Sequence/type/target/depth coverage, not inferred missing strokes
shot_coverage	active_serve_target_coverage_pct	float64	measure	0	percent (0–100)	Sequence/type/target/depth coverage, not inferred missing strokes
shot_coverage	source_Total_direction_matches	int64	measure	0	count (see field name and table grain)	Sequence/type/target/depth coverage, not inferred missing strokes
shot_coverage	player_id	object	key	0	categorical/text attribute	Sequence/type/target/depth coverage, not inferred missing strokes

table	field	pandas_dtype	role	missing_cells	unit	table_units_and_use
surface_adaptability	player	object	attribute	0	categorical/text attribute	Primary SAS/parity 0–100; weakest surface evidence and intervals
surface_adaptability	known_outcome_matches	int64	measure	0	count (see field name and table grain)	Primary SAS/parity 0–100; weakest surface evidence and intervals
surface_adaptability	unknown_outcome_matches	int64	measure	0	count (see field name and table grain)	Primary SAS/parity 0–100; weakest surface evidence and intervals
surface_adaptability	win_rate_overall_pct	float64	measure	0	percent (0–100)	Primary SAS/parity 0–100; weakest surface evidence and intervals
surface_adaptability	parity_score	float64	measure	0	score (0–100)	Primary SAS/parity 0–100; weakest surface evidence and intervals
surface_adaptability	weakest_surface_win_pct	float64	measure	0	percent (0–100)	Primary SAS/parity 0–100; weakest surface evidence and intervals
surface_adaptability	weakest_surface	object	attribute	0	categorical/text attribute	Primary SAS/parity 0–100; weakest surface evidence and intervals
surface_adaptability	surface_adaptability_score	float64	measure	0	score (0–100)	Primary SAS/parity 0–100; weakest surface evidence and intervals
surface_adaptability	equal_surface_anchor_sas_sensitivity	float64	measure	0	score (0–100)	Primary SAS/parity 0–100; weakest surface evidence and intervals
surface_adaptability	small_surface_sample	bool	measure	0	boolean flag	Primary SAS/parity 0–100; weakest surface evidence and intervals
surface_adaptability	Clay_known_matches	int64	measure	0	count (see field name and table grain)	Primary SAS/parity 0–100; weakest surface evidence and intervals
surface_adaptability	Clay_wins	int64	measure	0	count (see field name and table grain)	Primary SAS/parity 0–100; weakest surface evidence and intervals
surface_adaptability	Clay_win_pct	float64	measure	0	percent (0–100)	Primary SAS/parity 0–100; weakest surface evidence and intervals
surface_adaptability	Grass_known_matches	int64	measure	0	count (see field name and table grain)	Primary SAS/parity 0–100; weakest surface evidence and intervals
surface_adaptability	Grass_wins	int64	measure	0	count (see field name and table grain)	Primary SAS/parity 0–100; weakest surface evidence and intervals
surface_adaptability	Grass_win_pct	float64	measure	0	percent (0–100)	Primary SAS/parity 0–100; weakest surface evidence and intervals
surface_adaptability	Hard_known_matches	int64	measure	0	count (see field name and table grain)	Primary SAS/parity 0–100; weakest surface evidence and intervals
surface_adaptability	Hard_wins	int64	measure	0	count (see field name and table grain)	Primary SAS/parity 0–100; weakest surface evidence and intervals

table	field	pandas_dtype	role	missing_cells	unit	table_units_and_use
surface_adaptability	Hard_win_pct	float64	measure	0	percent (0–100)	Primary SAS/parity 0–100; weakest surface evidence and intervals
surface_adaptability	parity_score_valid_bootstraps	int64	measure	0	score (0–100)	Primary SAS/parity 0–100; weakest surface evidence and intervals
surface_adaptability	parity_score_ci95_lower	float64	measure	0	score (0–100)	Primary SAS/parity 0–100; weakest surface evidence and intervals
surface_adaptability	parity_score_ci95_upper	float64	measure	0	score (0–100)	Primary SAS/parity 0–100; weakest surface evidence and intervals
surface_adaptability	weakest_surface_win_pct_valid_bootstraps	int64	measure	0	percent (0–100)	Primary SAS/parity 0–100; weakest surface evidence and intervals
surface_adaptability	weakest_surface_win_pct_ci95_lower	float64	measure	0	percent (0–100)	Primary SAS/parity 0–100; weakest surface evidence and intervals
surface_adaptability	weakest_surface_win_pct_ci95_upper	float64	measure	0	percent (0–100)	Primary SAS/parity 0–100; weakest surface evidence and intervals
surface_adaptability	surface_adaptability_score_valid_bootstraps	int64	measure	0	score (0–100)	Primary SAS/parity 0–100; weakest surface evidence and intervals
surface_adaptability	surface_adaptability_score_ci95_lower	float64	measure	0	score (0–100)	Primary SAS/parity 0–100; weakest surface evidence and intervals
surface_adaptability	surface_adaptability_score_ci95_upper	float64	measure	0	score (0–100)	Primary SAS/parity 0–100; weakest surface evidence and intervals
surface_adaptability	scope	object	key	0	categorical/text attribute	Primary SAS/parity 0–100; weakest surface evidence and intervals
surface_adaptability	player_id	object	key	0	categorical/text attribute	Primary SAS/parity 0–100; weakest surface evidence and intervals
adaptability_weights	player	object	attribute	0	categorical/text attribute	Point definition-sensitivity scores; weights are fractions
adaptability_weights	parity_weight	float64	key	0	fraction / fixed scale (0–1)	Point definition-sensitivity scores; weights are fractions
adaptability_weights	weakest_surface_weight	float64	measure	0	fraction / fixed scale (0–1)	Point definition-sensitivity scores; weights are fractions
adaptability_weights	score	float64	measure	0	score (0–100)	Point definition-sensitivity scores; weights are fractions
adaptability_weights	rank_within_weight	float64	measure	0	ordinal rank	Point definition-sensitivity scores; weights are fractions
adaptability_weights	player_id	object	key	0	categorical/text attribute	Point definition-sensitivity scores; weights are fractions

table	field	pandas_dtype	role	missing_cells	unit	table_units_and_use
periods	player	object	attribute	0	categorical/text attribute	Individual four-year bin definitions and observed dates; sort by period_index
periods	period_index	int64	key	0	integer identifier; not additive	Individual four-year bin definitions and observed dates; sort by period_index
periods	period	object	attribute	0	categorical/text attribute	Individual four-year bin definitions and observed dates; sort by period_index
periods	start_year	int64	measure	0	calendar year	Individual four-year bin definitions and observed dates; sort by period_index
periods	end_year	int64	measure	0	calendar year	Individual four-year bin definitions and observed dates; sort by period_index
periods	nominal_years	int64	measure	0	years	Individual four-year bin definitions and observed dates; sort by period_index
periods	coverage_edge	bool	measure	0	boolean flag	Individual four-year bin definitions and observed dates; sort by period_index
periods	first_charted_date	object	attribute	0	ISO date (YYYY-MM-DD)	Individual four-year bin definitions and observed dates; sort by period_index
periods	last_charted_date	object	attribute	0	ISO date (YYYY-MM-DD)	Individual four-year bin definitions and observed dates; sort by period_index
periods	charted_matches	int64	measure	0	count (see field name and table grain)	Individual four-year bin definitions and observed dates; sort by period_index
periods	player_id	object	key	0	categorical/text attribute	Individual four-year bin definitions and observed dates; sort by period_index
tactical_trends	player	object	attribute	0	categorical/text attribute	Raw pooled and within-surface rates 0–100, counts, eligible samples
tactical_trends	period_index	int64	key	0	integer identifier; not additive	Raw pooled and within-surface rates 0–100, counts, eligible samples
tactical_trends	period	object	attribute	0	categorical/text attribute	Raw pooled and within-surface rates 0–100, counts, eligible samples
tactical_trends	scope	object	key	0	categorical/text attribute	Raw pooled and within-surface rates 0–100, counts, eligible samples
tactical_trends	metric	object	key	0	categorical/text attribute	Raw pooled and within-surface rates 0–100, counts, eligible samples
tactical_trends	numerator	float64	measure	14	pooled eligible count; blank for derived composites/ratios without a single count	Raw pooled and within-surface rates 0–100, counts, eligible samples
tactical_trends	denominator	float64	measure	14	pooled eligible count; blank for derived composites/ratios without a single count	Raw pooled and within-surface rates 0–100, counts, eligible samples

table	field	pandas_dtype	role	missing_cells	unit	table_units_and_use
tactical_trends	estimate_pct	float64	measure	14	percent (0–100)	Raw pooled and within-surface rates 0–100, counts, eligible samples
tactical_trends	contributing_matches	int64	measure	0	count (see field name and table grain)	Raw pooled and within-surface rates 0–100, counts, eligible samples
tactical_trends	cohort_matches	int64	measure	0	count (see field name and table grain)	Raw pooled and within-surface rates 0–100, counts, eligible samples
tactical_trends	coverage_pct	float64	measure	14	percent (0–100)	Raw pooled and within-surface rates 0–100, counts, eligible samples
tactical_trends	small_sample	bool	measure	0	boolean flag	Raw pooled and within-surface rates 0–100, counts, eligible samples
tactical_trends	coverage_edge_period	bool	measure	0	boolean flag	Raw pooled and within-surface rates 0–100, counts, eligible samples
tactical_trends	valid_bootstraps	int64	measure	0	count (see field name and table grain)	Raw pooled and within-surface rates 0–100, counts, eligible samples
tactical_trends	ci95_lower	float64	measure	14	percentage or score (0–100), as table contract states	Raw pooled and within-surface rates 0–100, counts, eligible samples
tactical_trends	ci95_upper	float64	measure	14	percentage or score (0–100), as table contract states	Raw pooled and within-surface rates 0–100, counts, eligible samples
tactical_trends	player_id	object	key	0	categorical/text attribute	Raw pooled and within-surface rates 0–100, counts, eligible samples
tactical_standardized	player	object	attribute	0	categorical/text attribute	Equal-surface estimates 0–100; not an observed Overall scope
tactical_standardized	period_index	int64	key	0	integer identifier; not additive	Equal-surface estimates 0–100; not an observed Overall scope
tactical_standardized	period	object	attribute	0	categorical/text attribute	Equal-surface estimates 0–100; not an observed Overall scope
tactical_standardized	metric	object	key	0	categorical/text attribute	Equal-surface estimates 0–100; not an observed Overall scope
tactical_standardized	standardized_pct	float64	measure	14	percent (0–100)	Equal-surface estimates 0–100; not an observed Overall scope
tactical_standardized	surfaces_available	int64	measure	0	count (see field name and table grain)	Equal-surface estimates 0–100; not an observed Overall scope
tactical_standardized	small_sample	bool	measure	0	boolean flag	Equal-surface estimates 0–100; not an observed Overall scope
tactical_standardized	ci95_lower	float64	measure	14	percentage or score (0–100), as table contract states	Equal-surface estimates 0–100; not an observed Overall scope

table	field	pandas_dtype	role	missing_cells	unit	table_units_and_use
tactical_standardized	ci95_upper	float64	measure	14	percentage or score (0–100), as table contract states	Equal-surface estimates 0–100; not an observed Overall scope
tactical_standardized	valid_bootstraps	int64	measure	0	count (see field name and table grain)	Equal-surface estimates 0–100; not an observed Overall scope
tactical_standardized	player_id	object	key	0	categorical/text attribute	Equal-surface estimates 0–100; not an observed Overall scope
tactical_changes	player	object	attribute	0	categorical/text attribute	Exploratory first/last supported-period changes in percentage points
tactical_changes	metric	object	key	0	categorical/text attribute	Exploratory first/last supported-period changes in percentage points
tactical_changes	first_supported_period	object	attribute	0	categorical/text attribute	Exploratory first/last supported-period changes in percentage points
tactical_changes	last_supported_period	object	attribute	0	categorical/text attribute	Exploratory first/last supported-period changes in percentage points
tactical_changes	change_percentage_points	float64	measure	0	percentage-point change	Exploratory first/last supported-period changes in percentage points
tactical_changes	ci95_lower	float64	measure	0	percentage-point change	Exploratory first/last supported-period changes in percentage points
tactical_changes	ci95_upper	float64	measure	0	percentage-point change	Exploratory first/last supported-period changes in percentage points
tactical_changes	valid_bootstraps	int64	measure	0	count (see field name and table grain)	Exploratory first/last supported-period changes in percentage points
tactical_changes	status	object	attribute	0	categorical/text attribute	Exploratory first/last supported-period changes in percentage points
tactical_changes	player_id	object	key	0	categorical/text attribute	Exploratory first/last supported-period changes in percentage points
style_profiles	player	object	attribute	0	categorical/text attribute	Raw percentages 0–100 or diversity indices 0–1; unit column applies
style_profiles	scope	object	key	0	categorical/text attribute	Raw percentages 0–100 or diversity indices 0–1; unit column applies
style_profiles	metric	object	key	0	categorical/text attribute	Raw percentages 0–100 or diversity indices 0–1; unit column applies
style_profiles	estimate	float64	measure	0	row-specific: see unit column	Raw percentages 0–100 or diversity indices 0–1; unit column applies
style_profiles	unit	object	attribute	0	categorical/text attribute	Raw percentages 0–100 or diversity indices 0–1; unit column applies

table	field	pandas_dtype	role	missing_cells	unit	table_units_and_use
style_profiles	numerator	float64	measure	24	pooled eligible count; blank for derived composites/ratios without a single count	Raw percentages 0–100 or diversity indices 0–1; unit column applies
style_profiles	denominator	float64	measure	0	pooled eligible count; blank for derived composites/ratios without a single count	Raw percentages 0–100 or diversity indices 0–1; unit column applies
style_profiles	contributing_matches	int64	measure	0	count (see field name and table grain)	Raw percentages 0–100 or diversity indices 0–1; unit column applies
style_profiles	ci95_lower	float64	measure	0	row-specific: see unit column	Raw percentages 0–100 or diversity indices 0–1; unit column applies
style_profiles	ci95_upper	float64	measure	0	row-specific: see unit column	Raw percentages 0–100 or diversity indices 0–1; unit column applies
style_profiles	small_sample	bool	measure	0	boolean flag	Raw percentages 0–100 or diversity indices 0–1; unit column applies
style_profiles	player_id	object	key	0	categorical/text attribute	Raw percentages 0–100 or diversity indices 0–1; unit column applies
comparison_inputs	player	object	attribute	0	categorical/text attribute	Raw values, primary fixed scales 0–1, alternative three-player min–max sensitivity
comparison_inputs	metric	object	key	0	categorical/text attribute	Raw values, primary fixed scales 0–1, alternative three-player min–max sensitivity
comparison_inputs	raw_value	float64	measure	0	row-specific: see raw_unit column	Raw values, primary fixed scales 0–1, alternative three-player min–max sensitivity
comparison_inputs	fixed_scale_value	float64	measure	0	fraction / fixed scale (0–1)	Raw values, primary fixed scales 0–1, alternative three-player min–max sensitivity
comparison_inputs	contributing_matches	int64	measure	0	count (see field name and table grain)	Raw values, primary fixed scales 0–1, alternative three-player min–max sensitivity
comparison_inputs	earliest_charted_date	object	attribute	0	ISO date (YYYY-MM-DD)	Raw values, primary fixed scales 0–1, alternative three-player min–max sensitivity
comparison_inputs	latest_charted_date	object	attribute	0	ISO date (YYYY-MM-DD)	Raw values, primary fixed scales 0–1, alternative three-player min–max sensitivity
comparison_inputs	three_player_min	float64	measure	0	row-specific: see raw_unit column	Raw values, primary fixed scales 0–1, alternative three-player min–max sensitivity
comparison_inputs	three_player_max	float64	measure	0	row-specific: see raw_unit column	Raw values, primary fixed scales 0–1, alternative three-player min–max sensitivity
comparison_inputs	alternative_minmax_value	float64	measure	0	fraction / fixed scale (0–1)	Raw values, primary fixed scales 0–1, alternative three-player min–max sensitivity
comparison_inputs	raw_unit	object	attribute	0	categorical/text attribute	Raw values, primary fixed scales 0–1, alternative three-player min–max sensitivity

table	field	pandas_dtype	role	missing_cells	unit	table_units_and_use
comparison_inputs	player_id	object	key	0	categorical/text attribute	Raw values, primary fixed scales 0–1, alternative three-player min–max sensitivity
comparison_inputs_wide	winning	float64	measure	0	fraction / fixed scale (0–1)	Four primary fixed scales 0–1 for custom point weights
comparison_inputs_wide	dominance	float64	measure	0	fraction / fixed scale (0–1)	Four primary fixed scales 0–1 for custom point weights
comparison_inputs_wide	clutch	float64	measure	0	fraction / fixed scale (0–1)	Four primary fixed scales 0–1 for custom point weights
comparison_inputs_wide	adaptability	float64	measure	0	fraction / fixed scale (0–1)	Four primary fixed scales 0–1 for custom point weights
comparison_inputs_wide	player	object	attribute	0	categorical/text attribute	Four primary fixed scales 0–1 for custom point weights
comparison_inputs_wide	player_id	object	key	0	categorical/text attribute	Four primary fixed scales 0–1 for custom point weights
comparison_bootstrap_inputs	winning	float64	measure	0	fraction / fixed scale (0–1)	Joint paired scaled draws; do not break replicate/player dependence
comparison_bootstrap_inputs	dominance	float64	measure	0	fraction / fixed scale (0–1)	Joint paired scaled draws; do not break replicate/player dependence
comparison_bootstrap_inputs	clutch	float64	measure	0	fraction / fixed scale (0–1)	Joint paired scaled draws; do not break replicate/player dependence
comparison_bootstrap_inputs	adaptability	float64	measure	0	fraction / fixed scale (0–1)	Joint paired scaled draws; do not break replicate/player dependence
comparison_bootstrap_inputs	replicate_id	int64	key	0	integer identifier; not additive	Joint paired scaled draws; do not break replicate/player dependence
comparison_bootstrap_inputs	player	object	attribute	0	categorical/text attribute	Joint paired scaled draws; do not break replicate/player dependence
comparison_bootstrap_inputs	player_id	object	key	0	categorical/text attribute	Joint paired scaled draws; do not break replicate/player dependence
preference_scores	scenario	object	key	0	categorical/text attribute	Scenario scores 0–100, rank frequencies 0–100 and conditional intervals
preference_scores	player	object	attribute	0	categorical/text attribute	Scenario scores 0–100, rank frequencies 0–100 and conditional intervals
preference_scores	score	float64	measure	0	score (0–100)	Scenario scores 0–100, rank frequencies 0–100 and conditional intervals
preference_scores	score_boot_median	float64	measure	0	score (0–100)	Scenario scores 0–100, rank frequencies 0–100 and conditional intervals

table	field	pandas_dtype	role	missing_cells	unit	table_units_and_use
preference_scores	ci95_lower	float64	measure	0	percentage or score (0–100), as table contract states	Scenario scores 0–100, rank frequencies 0–100 and conditional intervals
preference_scores	ci95_upper	float64	measure	0	percentage or score (0–100), as table contract states	Scenario scores 0–100, rank frequencies 0–100 and conditional intervals
preference_scores	point_estimate_rank	int64	measure	0	ordinal rank	Scenario scores 0–100, rank frequencies 0–100 and conditional intervals
preference_scores	bootstrap_first_place_pct	float64	measure	0	percent (0–100)	Scenario scores 0–100, rank frequencies 0–100 and conditional intervals
preference_scores	bootstrap_second_place_pct	float64	measure	0	percent (0–100)	Scenario scores 0–100, rank frequencies 0–100 and conditional intervals
preference_scores	bootstrap_third_place_pct	float64	measure	0	percent (0–100)	Scenario scores 0–100, rank frequencies 0–100 and conditional intervals
preference_scores	paired_valid_bootstraps	int64	measure	0	count (see field name and table grain)	Scenario scores 0–100, rank frequencies 0–100 and conditional intervals
preference_scores	player_id	object	key	0	categorical/text attribute	Scenario scores 0–100, rank frequencies 0–100 and conditional intervals
preference_weights	scenario	object	key	0	categorical/text attribute	Scenario weights are nonnegative fractions summing to one
preference_weights	metric	object	key	0	categorical/text attribute	Scenario weights are nonnegative fractions summing to one
preference_weights	weight	float64	measure	0	fraction / fixed scale (0–1)	Scenario weights are nonnegative fractions summing to one
preference_contributions	scenario	object	key	0	categorical/text attribute	Contributions sum to scenario/player score, not independent achievements
preference_contributions	player	object	attribute	0	categorical/text attribute	Contributions sum to scenario/player score, not independent achievements
preference_contributions	metric	object	key	0	categorical/text attribute	Contributions sum to scenario/player score, not independent achievements
preference_contributions	weight	float64	measure	0	fraction / fixed scale (0–1)	Contributions sum to scenario/player score, not independent achievements
preference_contributions	fixed_scale_value	float64	measure	0	fraction / fixed scale (0–1)	Contributions sum to scenario/player score, not independent achievements
preference_contributions	score_contribution	float64	measure	0	score (0–100)	Contributions sum to scenario/player score, not independent achievements
preference_contributions	player_id	object	key	0	categorical/text attribute	Contributions sum to scenario/player score, not independent achievements

table	field	pandas_dtype	role	missing_cells	unit	table_units_and_use
score_differences	scenario	object	key	0	categorical/text attribute	Paired score differences and bootstrap-ahead frequency, not p-values
score_differences	comparison	object	key	0	categorical/text attribute	Paired score differences and bootstrap-ahead frequency, not p-values
score_differences	point_score_difference	float64	measure	0	preference-score-point difference	Paired score differences and bootstrap-ahead frequency, not p-values
score_differences	ci95_lower	float64	measure	0	preference-score-point difference	Paired score differences and bootstrap-ahead frequency, not p-values
score_differences	ci95_upper	float64	measure	0	preference-score-point difference	Paired score differences and bootstrap-ahead frequency, not p-values
score_differences	bootstrap_a_ahead_pct	float64	measure	0	percent (0–100)	Paired score differences and bootstrap-ahead frequency, not p-values
surface_preference_scores	player	object	attribute	0	categorical/text attribute	Illustrative equal winning/DR/CI score 0–100, SAS omitted
surface_preference_scores	scope	object	key	0	categorical/text attribute	Illustrative equal winning/DR/CI score 0–100, SAS omitted
surface_preference_scores	illustrative_score	float64	measure	0	score (0–100)	Illustrative equal winning/DR/CI score 0–100, SAS omitted
surface_preference_scores	small_ci_sample	bool	measure	0	boolean flag	Illustrative equal winning/DR/CI score 0–100, SAS omitted
surface_preference_scores	ci95_lower	float64	measure	0	percentage or score (0–100), as table contract states	Illustrative equal winning/DR/CI score 0–100, SAS omitted
surface_preference_scores	ci95_upper	float64	measure	0	percentage or score (0–100), as table contract states	Illustrative equal winning/DR/CI score 0–100, SAS omitted
surface_preference_scores	bootstrap_first_place_pct	float64	measure	0	percent (0–100)	Illustrative equal winning/DR/CI score 0–100, SAS omitted
surface_preference_scores	paired_valid_bootstraps	int64	measure	0	count (see field name and table grain)	Illustrative equal winning/DR/CI score 0–100, SAS omitted
surface_preference_scores	player_id	object	key	0	categorical/text attribute	Illustrative equal winning/DR/CI score 0–100, SAS omitted
head_to_head	player_a	object	attribute	0	categorical/text attribute	Charted rivalry outcomes; unknown matches retained, rates 0–100
head_to_head	player_b	object	attribute	0	categorical/text attribute	Charted rivalry outcomes; unknown matches retained, rates 0–100
head_to_head	scope	object	key	0	categorical/text attribute	Charted rivalry outcomes; unknown matches retained, rates 0–100

table	field	pandas_dtype	role	missing_cells	unit	table_units_and_use
head_to_head	charted_meetings	int64	measure	0	count (see field name and table grain)	Charted rivalry outcomes; unknown matches retained, rates 0–100
head_to_head	known_outcomes	int64	measure	0	count (see field name and table grain)	Charted rivalry outcomes; unknown matches retained, rates 0–100
head_to_head	unknown_outcomes	int64	measure	0	count (see field name and table grain)	Charted rivalry outcomes; unknown matches retained, rates 0–100
head_to_head	a_verified_wins	int64	measure	0	count (see field name and table grain)	Charted rivalry outcomes; unknown matches retained, rates 0–100
head_to_head	b_verified_wins	int64	measure	0	count (see field name and table grain)	Charted rivalry outcomes; unknown matches retained, rates 0–100
head_to_head	a_win_pct	float64	measure	0	percent (0–100)	Charted rivalry outcomes; unknown matches retained, rates 0–100
head_to_head	a_win_pct_ci95_lower	float64	measure	1	percent (0–100)	Charted rivalry outcomes; unknown matches retained, rates 0–100
head_to_head	a_win_pct_ci95_upper	float64	measure	1	percent (0–100)	Charted rivalry outcomes; unknown matches retained, rates 0–100
head_to_head	small_sample	bool	measure	0	boolean flag	Charted rivalry outcomes; unknown matches retained, rates 0–100
head_to_head	player_a_id	object	key	0	categorical/text attribute	Charted rivalry outcomes; unknown matches retained, rates 0–100
head_to_head	player_b_id	object	key	0	categorical/text attribute	Charted rivalry outcomes; unknown matches retained, rates 0–100
scorecard	player	object	attribute	0	categorical/text attribute	Consolidated wide KPIs; units follow named metrics; SAS only Overall
scorecard	scope	object	key	0	categorical/text attribute	Consolidated wide KPIs; units follow named metrics; SAS only Overall
scorecard	charted_matches	int64	measure	0	count (see field name and table grain)	Consolidated wide KPIs; units follow named metrics; SAS only Overall
scorecard	verified_wins	int64	measure	0	count (see field name and table grain)	Consolidated wide KPIs; units follow named metrics; SAS only Overall
scorecard	known_outcome_matches	int64	measure	0	count (see field name and table grain)	Consolidated wide KPIs; units follow named metrics; SAS only Overall
scorecard	unknown_outcome_matches	int64	measure	0	count (see field name and table grain)	Consolidated wide KPIs; units follow named metrics; SAS only Overall
scorecard	matches_with_overview	int64	measure	0	count (see field name and table grain)	Consolidated wide KPIs; units follow named metrics; SAS only Overall

table	field	pandas_dtype	role	missing_cells	unit	table_units_and_use
scorecard	matches_with_points	int64	measure	0	count (see field name and table grain)	Consolidated wide KPIs; units follow named metrics; SAS only Overall
scorecard	earliest_charted_date	object	attribute	0	ISO date (YYYY-MM-DD)	Consolidated wide KPIs; units follow named metrics; SAS only Overall
scorecard	latest_charted_date	object	attribute	0	ISO date (YYYY-MM-DD)	Consolidated wide KPIs; units follow named metrics; SAS only Overall
scorecard	handedness	object	attribute	0	categorical/text attribute	Consolidated wide KPIs; units follow named metrics; SAS only Overall
scorecard	handedness_resolution	object	attribute	0	categorical/text attribute	Consolidated wide KPIs; units follow named metrics; SAS only Overall
scorecard	handedness_conflicting_records	int64	measure	0	count (see field name and table grain)	Consolidated wide KPIs; units follow named metrics; SAS only Overall
scorecard	weeks_at_no_1	Float64	measure	12	weeks; unavailable in supplied sources	Consolidated wide KPIs; units follow named metrics; SAS only Overall
scorecard	ace_pct	float64	measure	0	percent (0–100)	Consolidated wide KPIs; units follow named metrics; SAS only Overall
scorecard	break_point_conversion_pct	float64	measure	0	percent (0–100)	Consolidated wide KPIs; units follow named metrics; SAS only Overall
scorecard	break_point_save_pct	float64	measure	0	percent (0–100)	Consolidated wide KPIs; units follow named metrics; SAS only Overall
scorecard	dominance_ratio	float64	measure	0	ratio	Consolidated wide KPIs; units follow named metrics; SAS only Overall
scorecard	double_fault_pct	float64	measure	0	percent (0–100)	Consolidated wide KPIs; units follow named metrics; SAS only Overall
scorecard	first_serve_in_pct	float64	measure	0	percent (0–100)	Consolidated wide KPIs; units follow named metrics; SAS only Overall
scorecard	first_serve_win_pct	float64	measure	0	percent (0–100)	Consolidated wide KPIs; units follow named metrics; SAS only Overall
scorecard	match_win_pct	float64	measure	0	percent (0–100)	Consolidated wide KPIs; units follow named metrics; SAS only Overall
scorecard	point_win_pct	float64	measure	0	percent (0–100)	Consolidated wide KPIs; units follow named metrics; SAS only Overall
scorecard	return_break_pct	float64	measure	0	percent (0–100)	Consolidated wide KPIs; units follow named metrics; SAS only Overall
scorecard	return_point_win_pct	float64	measure	0	percent (0–100)	Consolidated wide KPIs; units follow named metrics; SAS only Overall

table	field	pandas_dtype	role	missing_cells	unit	table_units_and_use
scorecard	second_serve_win_pct	float64	measure	0	percent (0–100)	Consolidated wide KPIs; units follow named metrics; SAS only Overall
scorecard	service_hold_pct	float64	measure	0	percent (0–100)	Consolidated wide KPIs; units follow named metrics; SAS only Overall
scorecard	service_point_win_pct	float64	measure	0	percent (0–100)	Consolidated wide KPIs; units follow named metrics; SAS only Overall
scorecard	clutch_index_full	float64	measure	0	fraction (0–1)	Consolidated wide KPIs; units follow named metrics; SAS only Overall
scorecard	clutch_index_full_ci95_lower	float64	measure	0	fraction (0–1)	Consolidated wide KPIs; units follow named metrics; SAS only Overall
scorecard	clutch_index_full_ci95_upper	float64	measure	0	fraction (0–1)	Consolidated wide KPIs; units follow named metrics; SAS only Overall
scorecard	common_ci_matches	int64	measure	0	count (see field name and table grain)	Consolidated wide KPIs; units follow named metrics; SAS only Overall
scorecard	clutch_status	object	attribute	0	categorical/text attribute	Consolidated wide KPIs; units follow named metrics; SAS only Overall
scorecard	approach_frequency_pct	float64	measure	0	percent (0–100)	Consolidated wide KPIs; units follow named metrics; SAS only Overall
scorecard	net_point_frequency_pct	float64	measure	0	percent (0–100)	Consolidated wide KPIs; units follow named metrics; SAS only Overall
scorecard	net_point_success_pct	float64	measure	0	percent (0–100)	Consolidated wide KPIs; units follow named metrics; SAS only Overall
scorecard	serve_volley_frequency_pct	float64	measure	0	percent (0–100)	Consolidated wide KPIs; units follow named metrics; SAS only Overall
scorecard	short_rally_share_pct	float64	measure	0	percent (0–100)	Consolidated wide KPIs; units follow named metrics; SAS only Overall
scorecard	long_rally_share_pct	float64	measure	0	percent (0–100)	Consolidated wide KPIs; units follow named metrics; SAS only Overall
scorecard	groundstroke_share_pct	float64	measure	0	percent (0–100)	Consolidated wide KPIs; units follow named metrics; SAS only Overall
scorecard	slice_share_pct	float64	measure	0	percent (0–100)	Consolidated wide KPIs; units follow named metrics; SAS only Overall
scorecard	drop_shot_share_pct	float64	measure	0	percent (0–100)	Consolidated wide KPIs; units follow named metrics; SAS only Overall
scorecard	source_winner_rate_pct	float64	measure	0	percent (0–100)	Consolidated wide KPIs; units follow named metrics; SAS only Overall

table	field	pandas_dtype	role	missing_cells	unit	table_units_and_use
scorecard	source_unforced_error_rate_pct	float64	measure	0	percent (0–100)	Consolidated wide KPIs; units follow named metrics; SAS only Overall
scorecard	shot_family_diversity	float64	measure	0	normalized entropy (0–1)	Consolidated wide KPIs; units follow named metrics; SAS only Overall
scorecard	directional_diversity	float64	measure	0	normalized entropy (0–1)	Consolidated wide KPIs; units follow named metrics; SAS only Overall
scorecard	surface_adaptability_score	float64	measure	9	score (0–100)	Consolidated wide KPIs; units follow named metrics; SAS only Overall
scorecard	surface_adaptability_score_ci95_lower	float64	measure	9	score (0–100)	Consolidated wide KPIs; units follow named metrics; SAS only Overall
scorecard	surface_adaptability_score_ci95_upper	float64	measure	9	score (0–100)	Consolidated wide KPIs; units follow named metrics; SAS only Overall
scorecard	tactical_evolution	object	attribute	0	categorical/text attribute	Consolidated wide KPIs; units follow named metrics; SAS only Overall
scorecard	tactical_innovation_composite	object	attribute	0	categorical/text attribute	Consolidated wide KPIs; units follow named metrics; SAS only Overall
scorecard	player_id	object	key	0	categorical/text attribute	Consolidated wide KPIs; units follow named metrics; SAS only Overall
scorecard	weeks_at_no_1_availability	object	attribute	0	categorical/text attribute	Consolidated wide KPIs; units follow named metrics; SAS only Overall
source_manifest	relative_path	object	key	0	categorical/text attribute	Local input filenames, byte counts and SHA-256 fingerprints
source_manifest	bytes	int64	measure	0	bytes	Local input filenames, byte counts and SHA-256 fingerprints
source_manifest	sha256	object	attribute	0	categorical/text attribute	Local input filenames, byte counts and SHA-256 fingerprints
source_manifest	raw_rows	object	attribute	2	source rows	Local input filenames, byte counts and SHA-256 fingerprints
source_manifest	role	object	attribute	0	categorical/text attribute	Local input filenames, byte counts and SHA-256 fingerprints
audit_summary	dataset	object	key	0	categorical/text attribute	Raw = duplicates removed + quarantined + eligible; counts are not rates
audit_summary	key	object	attribute	0	categorical/text attribute	Raw = duplicates removed + quarantined + eligible; counts are not rates
audit_summary	raw_rows	int64	measure	0	source rows	Raw = duplicates removed + quarantined + eligible; counts are not rates

table	field	pandas_dtype	role	missing_cells	unit	table_units_and_use
audit_summary	exact_duplicates_removed	int64	measure	0	count (see field name and table grain)	Raw = duplicates removed + quarantined + eligible; counts are not rates
audit_summary	missing_key_rows	int64	measure	0	count (see field name and table grain)	Raw = duplicates removed + quarantined + eligible; counts are not rates
audit_summary	conflicting_key_rows	int64	measure	0	count (see field name and table grain)	Raw = duplicates removed + quarantined + eligible; counts are not rates
audit_summary	conflicting_keys	int64	measure	0	count (see field name and table grain)	Raw = duplicates removed + quarantined + eligible; counts are not rates
audit_summary	quarantined_rows	int64	measure	0	count (see field name and table grain)	Raw = duplicates removed + quarantined + eligible; counts are not rates
audit_summary	eligible_rows	int64	measure	0	count (see field name and table grain)	Raw = duplicates removed + quarantined + eligible; counts are not rates
audit_summary	rows_after_shared_quarantine	int64	measure	0	count (see field name and table grain)	Raw = duplicates removed + quarantined + eligible; counts are not rates
player_metadata_evidence	player	object	attribute	0	categorical/text attribute	Source-consensus player profile; supporting/conflicting/missing record counts
player_metadata_evidence	hand_code	object	attribute	0	categorical/text attribute	Source-consensus player profile; supporting/conflicting/missing record counts
player_metadata_evidence	handedness	object	attribute	0	categorical/text attribute	Source-consensus player profile; supporting/conflicting/missing record counts
player_metadata_evidence	supporting_records	int64	measure	0	count (see field name and table grain)	Source-consensus player profile; supporting/conflicting/missing record counts
player_metadata_evidence	recognized_records	int64	measure	0	count (see field name and table grain)	Source-consensus player profile; supporting/conflicting/missing record counts
player_metadata_evidence	conflicting_records	int64	measure	0	count (see field name and table grain)	Source-consensus player profile; supporting/conflicting/missing record counts
player_metadata_evidence	missing_or_unrecognized_records	int64	measure	0	count (see field name and table grain)	Source-consensus player profile; supporting/conflicting/missing record counts
player_metadata_evidence	consensus_pct	float64	measure	0	percent (0–100)	Source-consensus player profile; supporting/conflicting/missing record counts
player_metadata_evidence	resolution_status	object	attribute	0	categorical/text attribute	Source-consensus player profile; supporting/conflicting/missing record counts
player_metadata_evidence	evidence_source	object	attribute	0	categorical/text attribute	Source-consensus player profile; supporting/conflicting/missing record counts
player_metadata_evidence	player_id	object	key	0	categorical/text attribute	Source-consensus player profile; supporting/conflicting/missing record counts

table	field	pandas_dtype	role	missing_cells	unit	table_units_and_use
handedness_raw_values	player	object	attribute	0	categorical/text attribute	Every original hand label, preserved via repr; record counts
handedness_raw_values	source_column	object	key	0	categorical/text attribute	Every original hand label, preserved via repr; record counts
handedness_raw_values	raw_representation	object	key	0	categorical/text attribute	Every original hand label, preserved via repr; record counts
handedness_raw_values	records	int64	measure	0	count (see field name and table grain)	Every original hand label, preserved via repr; record counts
handedness_raw_values	player_id	object	key	0	categorical/text attribute	Every original hand label, preserved via repr; record counts
handedness_exceptions	match_id	object	key	0	categorical/text attribute	Contradictory or missing hand records; no silent source overrides
handedness_exceptions	player	object	attribute	0	categorical/text attribute	Contradictory or missing hand records; no silent source overrides
handedness_exceptions	source_column	object	key	0	categorical/text attribute	Contradictory or missing hand records; no silent source overrides
handedness_exceptions	raw_representation	object	attribute	0	categorical/text attribute	Contradictory or missing hand records; no silent source overrides
handedness_exceptions	normalized_hand	object	attribute	4	categorical/text attribute	Contradictory or missing hand records; no silent source overrides
handedness_exceptions	hand_code	object	attribute	0	categorical/text attribute	Contradictory or missing hand records; no silent source overrides
handedness_exceptions	exception_type	object	attribute	0	categorical/text attribute	Contradictory or missing hand records; no silent source overrides
handedness_exceptions	player_id	object	key	0	categorical/text attribute	Contradictory or missing hand records; no silent source overrides
issue_disposition	issue	object	key	0	categorical/text attribute	Resolved versus known source limitations; scope-specific counts are not additive
issue_disposition	status	object	attribute	0	categorical/text attribute	Resolved versus known source limitations; scope-specific counts are not additive
issue_disposition	affected_count	int64	measure	0	count (see field name and table grain)	Resolved versus known source limitations; scope-specific counts are not additive
issue_disposition	handling	object	attribute	0	categorical/text attribute	Resolved versus known source limitations; scope-specific counts are not additive
metric_contract	analysis	object	key	0	categorical/text attribute	Formula, unit, eligible analytical inputs, pooling and uncertainty contract

table	field	pandas_dtype	role	missing_cells	unit	table_units_and_use
metric_contract	metric	object	key	0	categorical/text attribute	Formula, unit, eligible analytical inputs, pooling and uncertainty contract
metric_contract	formula	object	attribute	0	categorical/text attribute	Formula, unit, eligible analytical inputs, pooling and uncertainty contract
metric_contract	unit	object	attribute	0	categorical/text attribute	Formula, unit, eligible analytical inputs, pooling and uncertainty contract
metric_contract	eligible_inputs	object	attribute	0	categorical/text attribute	Formula, unit, eligible analytical inputs, pooling and uncertainty contract
metric_contract	aggregation	object	attribute	0	categorical/text attribute	Formula, unit, eligible analytical inputs, pooling and uncertainty contract
metric_contract	uncertainty	object	attribute	0	categorical/text attribute	Formula, unit, eligible analytical inputs, pooling and uncertainty contract
metric_contract	interpretation	object	attribute	0	categorical/text attribute	Formula, unit, eligible analytical inputs, pooling and uncertainty contract
pressure_diagnostics	model	object	key	0	categorical/text attribute	Numerical fit diagnostics; probabilities 0–1, score residual per observation
pressure_diagnostics	observations	int64	measure	0	count (see field name and table grain)	Numerical fit diagnostics; probabilities 0–1, score residual per observation
pressure_diagnostics	match_clusters	int64	measure	0	count (see field name and table grain)	Numerical fit diagnostics; probabilities 0–1, score residual per observation
pressure_diagnostics	max_absolute_score_per_observation	float64	measure	0	absolute likelihood score / observation	Numerical fit diagnostics; probabilities 0–1, score residual per observation
pressure_diagnostics	minimum_fitted_probability	float64	measure	0	probability (0–1)	Numerical fit diagnostics; probabilities 0–1, score residual per observation
pressure_diagnostics	maximum_fitted_probability	float64	measure	0	probability (0–1)	Numerical fit diagnostics; probabilities 0–1, score residual per observation
pressure_diagnostics	minimum_covariance_eigenvalue	float64	measure	0	coefficient covariance eigenvalue (log-odds parameterization)	Numerical fit diagnostics; probabilities 0–1, score residual per observation
pressure_context_effects	model	object	key	0	categorical/text attribute	Conditional odds ratio for +10 percentage points of opponent proxy; not causal
pressure_context_effects	opponent_proxy_change	object	attribute	0	categorical/text attribute	Conditional odds ratio for +10 percentage points of opponent proxy; not causal
pressure_context_effects	conditional_odds_ratio	float64	measure	0	odds ratio	Conditional odds ratio for +10 percentage points of opponent proxy; not causal
pressure_context_effects	ci95_lower	float64	measure	0	odds ratio	Conditional odds ratio for +10 percentage points of opponent proxy; not causal

table	field	pandas_dtype	role	missing_cells	unit	table_units_and_use
pressure_context_effects	ci95_upper	float64	measure	0	odds ratio	Conditional odds ratio for +10 percentage points of opponent proxy; not causal
pressure_context_effects	interpretation	object	attribute	0	categorical/text attribute	Conditional odds ratio for +10 percentage points of opponent proxy; not causal
audit_review_register	issue	object	key	0	categorical/text attribute	Review scopes and metric-specific handling; flags are not all corrupt matches
audit_review_register	count	int64	measure	0	count (see field name and table grain)	Review scopes and metric-specific handling; flags are not all corrupt matches
audit_review_register	next_action	object	attribute	0	categorical/text attribute	Review scopes and metric-specific handling; flags are not all corrupt matches

10.3 Release validation and source traceability

The checks below verify input preservation, output grains and bounds, paired rank-frequency mass, export round-trips, and required unavailable fields. Metric-specific review flags are retained and handled by eligibility; a passed check does not erase source uncertainty or selection bias. A fresh-kernel execution must finish every cell without an error before saving the report. Detailed evidence remains embedded even when its presentation panel is collapsed.

To publish, share this notebook with its saved outputs. Reproducing it requires the unchanged `Raw Data` inputs identified by the manifest; viewing its tables and figures does not require execution. To build Tableau, use the validated release bundle or regenerate it using the documented switch and use the registry's relationships and units. All analysis and export logic remains here.

10.3.1 Independent numerical verification

Execution checks establish that code runs. The following audit separately reaggregates original Overview counts, enumerates score-defined pressure events, checks tiebreak score accumulation, recomputes clustered covariance through linear solves, and verifies bootstrap/scenario arithmetic. It conditions on the published eligibility rules rather than silently creating a different cohort. Full check-level evidence remains embedded below.

In [36]:
```
# Independent release audit: reaggregate source counts and reconstruct checks by another route.
independent_checks=[]
def independent_check(name,condition,detail):
    if not bool(condition):raise AssertionError("Independent audit: "+name)
    independent_checks.append({"check":name,"status":"PASS","evidence":detail})
```

```
raw_overview=raw_dataframes_unfiltered["charting_m_stats_Overview"]
raw_overview=raw_overview.loc[raw_overview["set"].astype(str).str.casefold().eq("total")].drop_duplicates().set_index(["match_id","player"])
independent_check("Source Overview Total is unique for analytical participants",not
raw_overview.loc[raw_overview.index.get_level_values("player").isin(BIG_THREE)].index.duplicated().any(),"Original Total rows, not the enriched KPI table")
raw_source_formulas={
    "first_serve_in_pct":("first_in","serve_pts"),"first_serve_win_pct":("first_won","first_in"),
    "second_serve_win_pct":("second_won","second_in"),"service_point_win_pct":("__service_wins","serve_pts"),
    "return_point_win_pct":("return_pts_won","return_pts"),"ace_pct":("aces","serve_pts"),"double_fault_pct":("dfs","serve_pts")}
raw_overview["__service_wins"]=raw_overview["first_won"]+raw_overview["second_won"]
# These explicit score strings independently identify ordinary break points in server-relative notation.
raw_bp_mask=canonical_points["Pts"].astype(str).str.strip().str.upper().isin(["0-40","15-40","30-40","40-AD","40-A"])
raw_bp_mask|=canonical_points["Final TB?"].eq("N")&canonical_points["Pts"].eq("40-40")
raw_bp_mask&=~canonical_points["is_tiebreak_point"]
independent_check("Break-point direction is receiver-leading",np.array_equal(raw_bp_mask,canonical_points["is_break_point"]),"Explicit server-relative score
strings; no player-1 score assumption")
verification_rows=[]
for player,surface in kpi_scope_groups:
    cohort=canonical_master.loc[canonical_master.player.eq(player)]
    if surface!="Overall":cohort=cohort.loc[cohort.surface.eq(surface)]
    known=cohort.loc[cohort.winner.notna()]
    reported=core_kpi_coverage.loc[core_kpi_coverage.player.eq(player)&core_kpi_coverage.surface.eq(surface)].iloc[0]
    wins=int(known.winner.eq(player).sum());losses=int(known.winner.ne(player).sum())
    independent_check("Outcome accounting: "+player+" / "+surface,
        len(cohort)==reported.charted_matches and len(known)==reported.known_outcome_matches and wins==reported.verified_wins and
wins+losses+int(reported.unknown_outcome_matches)==len(cohort),
        f"{wins} wins + {losses} losses + {int(reported.unknown_outcome_matches)} unknown = {len(cohort)} charted matches")
    for metric,(num,den,_,_) in rate_specs.items():
        sample=kpi_master.loc[kpi_master.player.eq(player)&kpi_masks[metric]]
        if surface!="Overall":sample=sample.loc[sample.surface.eq(surface)]
        keys=pd.MultiIndex.from_frame(sample[["match_id","player"]])
        if metric in raw_source_formulas:
            a,b=raw_source_formulas[metric];original=raw_overview.loc[keys]
            numerator=original[a].sum(min_count=1);denominator=original[b].sum(min_count=1)
        elif metric=="match_win_pct":numerator=wins;denominator=len(known)
        elif metric in ["point_win_pct","break_point_save_pct","break_point_conversion_pct"]:
            points=canonical_points.loc[canonical_points.match_id.isin(sample.match_id)&
(canonical_points.player1.eq(player)|canonical_points.player2.eq(player))]
            own_number=np.where(points.player1.eq(player),1,2);won=points.PtWinner.to_numpy()==own_number
            serving=points.Svr.to_numpy()==own_number
            if metric=="point_win_pct":eligible=np.ones(len(points),dtype=bool)
            else:eligible=raw_bp_mask.loc[points.index].to_numpy()&points.sequence_known.to_numpy()&(serving if metric=="break_point_save_pct" else ~serving)
            numerator=int((won&eligible).sum());denominator=int(eligible.sum())
        else:
            games=kpi_games.loc[kpi_games.eligible_regular_game&kpi_games.match_id.isin(sample.match_id)]
            games=games.loc[games.server_name.eq(player) if metric=="service_hold_pct" else games.returner_name.eq(player)]
            numerator=int(games.point_winner.eq(player).sum());denominator=len(games)
        estimate=100*numerator/denominator if denominator>0 else np.nan
        result=evidence_metric(player,metric,surface)
        independent_check("Source ratio: "+player+" / "+surface+" / "+metric,
            np.isclose(estimate,result.estimate,equal_nan=True) and np.isclose(numerator,result.numerator,equal_nan=True) and
np.isclose(denominator,result.denominator,equal_nan=True),"Independent counts on documented KPI-specific eligible match keys")
        verification_rows.append({"player":player,"surface":surface,"metric":metric,"numerator":numerator,"denominator":denominator,"estimate":estimate})
    dr_sample=kpi_master.loc[kpi_master.player.eq(player)&kpi_masks["dominance_ratio"]]
    if surface!="Overall":dr_sample=dr_sample.loc[dr_sample.surface.eq(surface)]
    original=raw_overview.loc[pd.MultiIndex.from_frame(dr_sample[["match_id","player"]])]
    rw,rp,sw,sp=[original[field].sum() for field in ["return_pts_won","return_pts","__service_wins","serve_pts"]]
    dr=rw/rp/((sp-sw)/sp)
```

```
    independent_check("DR from original common-sample counts: "+player+" /
"+surface,np.isclose(dr,evidence_metric(player,"dominance_ratio",surface).estimate),"Return win fraction divided by service loss fraction; no averaging of match
ratios")
independent_metric_verification=pd.DataFrame(verification_rows)
# Each game must keep set/game state fixed until its terminal point.
regular_ids=pd.MultiIndex.from_frame(kpi_games.loc[kpi_games.eligible_regular_game,game_keys])
regular_points=canonical_points.loc[pd.MultiIndex.from_frame(canonical_points[game_keys]).isin(regular_ids)]
state_nunique=regular_points.groupby(game_keys)[["Set1","Set2","Gm1","Gm2","Svr"]].nunique()
independent_check("Eligible service games have stable state and server",state_nunique.eq(1).all().all(),"All points in eligible games, not terminal rows alone")
# In a valid TB, player-relative prepoint scores equal cumulative points already won.
tb_verify=tb_points.loc[pd.MultiIndex.from_frame(tb_points[tb_keys]).isin(valid_tb_keys)].copy()
for number in [1,2]:
    pre=np.where(tb_verify.Svr.eq(number),tb_verify.score_server_numeric,tb_verify.score_receiver_numeric)
    won=tb_verify.PtWinner.eq(number).astype(int)
    cumulative=won.groupby([tb_verify[k] for k in tb_keys]).cumsum()-won
    independent_check("Completed TB score accumulation: player "+str(number),np.array_equal(pre,cumulative.to_numpy()),"Independent cumulative point wins
through server changes")
for row in clutch_summary.itertuples(index=False):
    sample=clutch_match_events.loc[clutch_match_events.player.eq(row.player)&clutch_match_events.common_ci_eligible]
    if row.surface!="Overall":sample=sample.loc[sample.surface.eq(row.surface)]
    components=[]
    for _,(num,den,_) in CI_COMPONENTS.items():
        a,b=sample[num].sum(min_count=1),sample[den].sum(min_count=1)
        components.append(a/b if b>0 else np.nan)
    independent_check("Full CI cohort/scale: "+row.player+" / "+row.surface,
        len(sample)==row.common_ci_matches and np.isclose(np.dot([.5,.3,.2],components),row.clutch_index_full,equal_nan=True),"Pooled components on one common
cohort; proportions, not percentages")
valid_deciders=canonical_sets.loc[canonical_sets.deciding_set_validated&~canonical_sets["Final
TB?"].eq("S")].merge(canonical_match_results[["match_id","winner"]],on="match_id",validate="many_to_one")
valid_deciders=valid_deciders.loc[valid_deciders.winner.notna()]
independent_check("Known deciding-set winner agrees with match winner",valid_deciders.set_winner.eq(valid_deciders.winner).all(),"Completed supported deciding
sets only; match tiebreaks excluded")
expected_weights=np.random.default_rng(RANDOM_SEED).multinomial(len(bootstrap_match_ids),np.full(len(bootstrap_match_ids),1/len(bootstrap_match_ids)),size=BOOTS
TRAP_REPLICATES)
independent_check("Bootstrap seed reproduces every draw",np.array_equal(bootstrap_weights,expected_weights) and
np.all(bootstrap_weights.sum(axis=1)==len(bootstrap_match_ids)),"One shared weight per match ID retains both rivalry observations")
del expected_weights
for result in core_kpi_long.itertuples():
    descriptor=bootstrap_descriptors[result.Index];columns=descriptor[2]
    values=bootstrap_totals[:,columns]
    with np.errstate(divide="ignore",invalid="ignore"):
        draws=100*values[:,0]/values[:,1] if result.metric!="dominance_ratio" else (values[:,0]/values[:,1])/((values[:,3]-values[:,2])/values[:,3])
    valid=draws[np.isfinite(draws)]
    q=np.percentile(valid,[2.5,97.5]) if len(valid)>=.95*BOOTSTRAP_REPLICATES else [np.nan,np.nan]
    independent_check("Bootstrap interval: "+result.player+" / "+result.surface+" / "+result.metric,
        len(valid)==result.valid_bootstrap_replicates and np.allclose(q,[result.ci95_lower,result.ci95_upper],equal_nan=True),"Recomputed percentile endpoints,
including unavailable sparse intervals")
for kind,result in pressure_models.items():
    X=result["design"].to_numpy();y=result["events"].success.to_numpy();p=1/(1+np.exp(-X@result["beta"]))
    information=X.T@(X*(p*(1-p))[:,None])
    clusters=pd.factorize(result["events"].match_id,sort=True)[0]
    sums=np.zeros((clusters.max()+1,X.shape[1]));np.add.at(sums,clusters,X*(y-p)[:,None])
    influence=np.linalg.solve(information,sums.T);G,N,K=len(sums),len(y),X.shape[1]
    covariance=(influence@influence.T)*(G/(G-1))*((N-1)/(N-K))
    independent_check("Clustered covariance independently recomputed: "+kind,np.allclose(covariance,result["covariance"],rtol=1e-8,atol=1e-10),"Solve-based
match-score sandwich; rivalry TB rows share one cluster")
    independent_check("Logistic score equation and design: "+kind,np.max(np.abs(X.T@(y-p)))/N<1e-5 and np.linalg.matrix_rank(X)==K,"Unpenalized explanatory
```

```
logit; no global intercept")
    for row in pressure_pairwise.loc[pressure_pairwise.model.eq(kind)].itertuples():
        a,b=row.comparison.split(" / ");columns=list(result["design"])
        v=np.zeros(K);v[columns.index(PLAYER_PRESSURE_TERMS[a])]=1;v[columns.index(PLAYER_PRESSURE_TERMS[b])]=-1
        delta=v@result["beta"];se=np.sqrt(v@covariance@v)
        independent_check("Pressure contrast: "+kind+" / "+row.comparison,
            np.allclose(np.exp([delta,delta-1.96*se,delta+1.96*se]),[row.odds_ratio,row.ci95_lower,row.ci95_upper]),"Coefficient difference and full covariance,
not separate marginal intervals")
    # One record per opponent/current match is sufficient to test leave-current-match-out context.
    for row in result["events"].drop_duplicates(["match_id","opponent"]).itertuples():
        other=opponent_overview.loc[opponent_overview.player.eq(row.opponent)&opponent_overview.match_id.ne(row.match_id)]
        a,b=("return_pts_won","return_pts") if kind=="BP save" else ("context_point_wins","context_points")
        expected=other[a].sum()/other[b].sum() if other.match_id.nunique()>=10 and other[b].sum()>0 else np.nan
        if not np.isclose(expected,row.opponent_strength,equal_nan=True):raise AssertionError("Opponent proxy includes wrong match/context")
    independent_check("Opponent context excludes current match: "+kind,True,"Direct reaggregation of all other eligible local Overview matches; retrospective,
not forecasting")
for row in shot_diversity_summary.itertuples():
    counts,categories,known_categories=distribution_inputs[row.distribution]
    selected=counts.loc[counts.player.eq(row.player)]
    if row.surface!="Overall":selected=selected.loc[selected.surface.eq(row.surface)]
    vector=selected.groupby("category")["count"].sum().reindex(known_categories,fill_value=0).to_numpy(dtype=float)
    probs=vector[vector>0]/vector.sum() if vector.sum()>0 else np.array([])
    entropy=-sum(float(p)*__import__('math').log(float(p)) for p in probs)/__import__('math').log(len(known_categories)) if len(probs) else np.nan
    independent_check("Entropy from category counts: "+row.player+" / "+row.surface+" /
"+row.distribution,np.isclose(entropy,row.normalized_entropy,equal_nan=True),"Fixed K; unknown types excluded; no performance interpretation")
for row in surface_adaptability_summary_v2.itertuples():
    sample=canonical_master.loc[canonical_master.player.eq(row.player)&canonical_master.surface.isin(PROJECT_SURFACES)&canonical_master.winner.notna()]
    rates=sample.assign(won=sample.winner.eq(row.player)).groupby("surface").won.mean().reindex(PROJECT_SURFACES).to_numpy()
    anchor=sample.winner.eq(row.player).mean();sas=60*(1-np.sqrt(np.mean((rates-anchor)**2)))+40*rates.min()
    independent_check("SAS from verified surface outcomes: "+row.player,np.isclose(sas,row.surface_adaptability_score),"Three named surfaces; pooled anchor,
60/40 declared weights")
for row in tactical_evolution_long.itertuples():
    eligible=tactical_match_table.player.eq(row.player)&tactical_match_table.period_index.eq(row.period_index)&tactical_masks[row.metric]
    if row.surface!="Overall":eligible&=tactical_match_table.surface.eq(row.surface)
    a,b,_=tactical_metric_specs[row.metric];sample=tactical_match_table.loc[eligible]
    expected=100*sample[a].sum()/sample[b].sum() if len(sample) and sample[b].sum()>0 else np.nan
    if not np.isclose(expected,row.estimate_pct,equal_nan=True) or len(sample)!=row.contributing_matches:raise AssertionError("Tactical period
pooling/eligibility mismatch")
independent_check("All tactical period rates reaggregate",True,"Individual four-year bins; explicit masks and numerator/denominator pooling")
for row in tactical_standardized_trends.itertuples():

group=tactical_evolution_long.loc[tactical_evolution_long.player.eq(row.player)&tactical_evolution_long.period_index.eq(row.period_index)&tactical_evolution_lon
g.metric.eq(row.metric)&tactical_evolution_long.surface.isin(PROJECT_SURFACES)]
    expected=group.estimate_pct.mean() if group.estimate_pct.notna().sum()==3 else np.nan
    if not np.isclose(expected,row.standardized_pct,equal_nan=True):raise AssertionError("Surface standardization mismatch")
independent_check("Standardized trends require all surfaces",True,"Fixed 1/3 surface weights; absent grass not interpolated")
for row in validated_h2h_summary.itertuples():
    pair=canonical_match_results.loc[((canonical_match_results.player1.eq(row.player_a)&canonical_match_results.player2.eq(row.player_b))|
(canonical_match_results.player1.eq(row.player_b)&canonical_match_results.player2.eq(row.player_a)))]
    if row.surface!="Overall":pair=pair.loc[pair.surface.eq(row.surface)]
    known=pair.loc[pair.winner.notna()]
    if len(pair)!=row.charted_meetings or len(known)!=row.known_outcomes or known.winner.eq(row.player_a).sum()!=row.a_verified_wins:raise AssertionError("H2H
participant/surface mismatch")
independent_check("All charted H2H pairs and surface splits verified",True,"Distinct matches; unknown outcomes retained separately; not official career
records")
for scenario,weights in COMPARISON_SCENARIOS.items():
    w=np.array([weights[m] for m in COMPARISON_METRICS]);point=100*np.sum(comparison_points*w,axis=1)
```

```
    draw=100*np.sum(comparison_draws_valid*w,axis=2)
    saved=comparison_scores.loc[comparison_scores.scenario.eq(scenario)].set_index("player").reindex(BIG_THREE)
    independent_check("Preference arithmetic: "+scenario,np.allclose(point,saved.score) and np.allclose(np.percentile(draw,
[2.5,97.5],axis=0).T,saved[["ci95_lower","ci95_upper"]]),"Unchanged fixed scales, declared weights, and fully paired draws")
    maximum=draw.max(axis=1,keepdims=True);ties=np.isclose(draw,maximum,rtol=0,atol=1e-12);freq=100*(ties/ties.sum(axis=1,keepdims=True)).mean(axis=0)
    independent_check("First-place frequencies: "+scenario,np.allclose(freq,saved.bootstrap_first_place_pct),"Independently counted maxima with fractional tie
credit")
    for a,b in [(0,1),(0,2),(1,2)]:

row=comparison_pairwise_differences.loc[comparison_pairwise_differences.scenario.eq(scenario)&comparison_pairwise_differences.comparison.eq(BIG_THREE[a]+" minus
"+BIG_THREE[b])].iloc[0]
        if not np.isclose(row.point_score_difference,point[a]-point[b]) or not np.allclose([row.ci95_lower,row.ci95_upper],np.percentile(draw[:,a]-draw[:,b],
[2.5,97.5])):raise AssertionError("Paired score difference mismatch")
independent_check("All paired scenario differences verified",True,"Differences within the same draws; no subtraction of marginal interval endpoints")
independent_audit_results=pd.DataFrame(independent_checks)
report_table(independent_audit_results,caption="Independent source/count/model verification: every check and its evidence")
print(f"Independent verification passed: {len(independent_audit_results)} checks; source counts, units, event cohorts, model covariance, resampling and
scoring.")
```

▼ **Independent source/count/model verification: every check and its evidence**

check	status	evidence
Source Overview Total is unique for analytical participants	PASS	Original Total rows, not the enriched KPI table
Break-point direction is receiver-leading	PASS	Explicit server-relative score strings; no player-1 score assumption
Outcome accounting: Roger Federer / Overall	PASS	559 wins + 150 losses + 14 unknown = 723 charted matches
Source ratio: Roger Federer / Overall / match_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Roger Federer / Overall / point_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Roger Federer / Overall / first_serve_in_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Roger Federer / Overall / first_serve_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Roger Federer / Overall / second_serve_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Roger Federer / Overall / service_point_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Roger Federer / Overall / return_point_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Roger Federer / Overall / ace_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Roger Federer / Overall / double_fault_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Roger Federer / Overall / break_point_save_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Roger Federer / Overall / break_point_conversion_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Roger Federer / Overall / service_hold_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Roger Federer / Overall / return_break_pct	PASS	Independent counts on documented KPI-specific eligible match keys
DR from original common-sample counts: Roger Federer / Overall	PASS	Return win fraction divided by service loss fraction; no averaging of match ratios
Outcome accounting: Roger Federer / Clay	PASS	87 wins + 39 losses + 4 unknown = 130 charted matches
Source ratio: Roger Federer / Clay / match_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Roger Federer / Clay / point_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Roger Federer / Clay / first_serve_in_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Roger Federer / Clay / first_serve_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Roger Federer / Clay / second_serve_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Roger Federer / Clay / service_point_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Roger Federer / Clay / return_point_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Roger Federer / Clay / ace_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Roger Federer / Clay / double_fault_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Roger Federer / Clay / break_point_save_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Roger Federer / Clay / break_point_conversion_pct	PASS	Independent counts on documented KPI-specific eligible match keys

check	status	evidence
Source ratio: Roger Federer / Clay / service_hold_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Roger Federer / Clay / return_break_pct	PASS	Independent counts on documented KPI-specific eligible match keys
DR from original common-sample counts: Roger Federer / Clay	PASS	Return win fraction divided by service loss fraction; no averaging of match ratios
Outcome accounting: Roger Federer / Grass	PASS	97 wins + 19 losses + 3 unknown = 119 charted matches
Source ratio: Roger Federer / Grass / match_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Roger Federer / Grass / point_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Roger Federer / Grass / first_serve_in_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Roger Federer / Grass / first_serve_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Roger Federer / Grass / second_serve_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Roger Federer / Grass / service_point_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Roger Federer / Grass / return_point_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Roger Federer / Grass / ace_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Roger Federer / Grass / double_fault_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Roger Federer / Grass / break_point_save_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Roger Federer / Grass / break_point_conversion_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Roger Federer / Grass / service_hold_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Roger Federer / Grass / return_break_pct	PASS	Independent counts on documented KPI-specific eligible match keys
DR from original common-sample counts: Roger Federer / Grass	PASS	Return win fraction divided by service loss fraction; no averaging of match ratios
Outcome accounting: Roger Federer / Hard	PASS	375 wins + 92 losses + 7 unknown = 474 charted matches
Source ratio: Roger Federer / Hard / match_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Roger Federer / Hard / point_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Roger Federer / Hard / first_serve_in_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Roger Federer / Hard / first_serve_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Roger Federer / Hard / second_serve_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Roger Federer / Hard / service_point_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Roger Federer / Hard / return_point_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Roger Federer / Hard / ace_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Roger Federer / Hard / double_fault_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Roger Federer / Hard / break_point_save_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Roger Federer / Hard / break_point_conversion_pct	PASS	Independent counts on documented KPI-specific eligible match keys

check	status	evidence
Source ratio: Roger Federer / Hard / service_hold_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Roger Federer / Hard / return_break_pct	PASS	Independent counts on documented KPI-specific eligible match keys
DR from original common-sample counts: Roger Federer / Hard	PASS	Return win fraction divided by service loss fraction; no averaging of match ratios
Outcome accounting: Rafael Nadal / Overall	PASS	298 wins + 114 losses + 13 unknown = 425 charted matches
Source ratio: Rafael Nadal / Overall / match_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Rafael Nadal / Overall / point_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Rafael Nadal / Overall / first_serve_in_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Rafael Nadal / Overall / first_serve_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Rafael Nadal / Overall / second_serve_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Rafael Nadal / Overall / service_point_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Rafael Nadal / Overall / return_point_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Rafael Nadal / Overall / ace_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Rafael Nadal / Overall / double_fault_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Rafael Nadal / Overall / break_point_save_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Rafael Nadal / Overall / break_point_conversion_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Rafael Nadal / Overall / service_hold_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Rafael Nadal / Overall / return_break_pct	PASS	Independent counts on documented KPI-specific eligible match keys
DR from original common-sample counts: Rafael Nadal / Overall	PASS	Return win fraction divided by service loss fraction; no averaging of match ratios
Outcome accounting: Rafael Nadal / Clay	PASS	139 wins + 25 losses + 2 unknown = 166 charted matches
Source ratio: Rafael Nadal / Clay / match_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Rafael Nadal / Clay / point_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Rafael Nadal / Clay / first_serve_in_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Rafael Nadal / Clay / first_serve_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Rafael Nadal / Clay / second_serve_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Rafael Nadal / Clay / service_point_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Rafael Nadal / Clay / return_point_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Rafael Nadal / Clay / ace_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Rafael Nadal / Clay / double_fault_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Rafael Nadal / Clay / break_point_save_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Rafael Nadal / Clay / break_point_conversion_pct	PASS	Independent counts on documented KPI-specific eligible match keys

check	status	evidence
Source ratio: Rafael Nadal / Clay / service_hold_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Rafael Nadal / Clay / return_break_pct	PASS	Independent counts on documented KPI-specific eligible match keys
DR from original common-sample counts: Rafael Nadal / Clay	PASS	Return win fraction divided by service loss fraction; no averaging of match ratios
Outcome accounting: Rafael Nadal / Grass	PASS	22 wins + 15 losses + 3 unknown = 40 charted matches
Source ratio: Rafael Nadal / Grass / match_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Rafael Nadal / Grass / point_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Rafael Nadal / Grass / first_serve_in_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Rafael Nadal / Grass / first_serve_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Rafael Nadal / Grass / second_serve_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Rafael Nadal / Grass / service_point_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Rafael Nadal / Grass / return_point_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Rafael Nadal / Grass / ace_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Rafael Nadal / Grass / double_fault_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Rafael Nadal / Grass / break_point_save_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Rafael Nadal / Grass / break_point_conversion_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Rafael Nadal / Grass / service_hold_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Rafael Nadal / Grass / return_break_pct	PASS	Independent counts on documented KPI-specific eligible match keys
DR from original common-sample counts: Rafael Nadal / Grass	PASS	Return win fraction divided by service loss fraction; no averaging of match ratios
Outcome accounting: Rafael Nadal / Hard	PASS	137 wins + 74 losses + 8 unknown = 219 charted matches
Source ratio: Rafael Nadal / Hard / match_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Rafael Nadal / Hard / point_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Rafael Nadal / Hard / first_serve_in_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Rafael Nadal / Hard / first_serve_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Rafael Nadal / Hard / second_serve_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Rafael Nadal / Hard / service_point_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Rafael Nadal / Hard / return_point_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Rafael Nadal / Hard / ace_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Rafael Nadal / Hard / double_fault_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Rafael Nadal / Hard / break_point_save_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Rafael Nadal / Hard / break_point_conversion_pct	PASS	Independent counts on documented KPI-specific eligible match keys

check	status	evidence
Source ratio: Rafael Nadal / Hard / service_hold_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Rafael Nadal / Hard / return_break_pct	PASS	Independent counts on documented KPI-specific eligible match keys
DR from original common-sample counts: Rafael Nadal / Hard	PASS	Return win fraction divided by service loss fraction; no averaging of match ratios
Outcome accounting: Novak Djokovic / Overall	PASS	404 wins + 122 losses + 27 unknown = 553 charted matches
Source ratio: Novak Djokovic / Overall / match_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Novak Djokovic / Overall / point_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Novak Djokovic / Overall / first_serve_in_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Novak Djokovic / Overall / first_serve_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Novak Djokovic / Overall / second_serve_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Novak Djokovic / Overall / service_point_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Novak Djokovic / Overall / return_point_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Novak Djokovic / Overall / ace_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Novak Djokovic / Overall / double_fault_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Novak Djokovic / Overall / break_point_save_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Novak Djokovic / Overall / break_point_conversion_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Novak Djokovic / Overall / service_hold_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Novak Djokovic / Overall / return_break_pct	PASS	Independent counts on documented KPI-specific eligible match keys
DR from original common-sample counts: Novak Djokovic / Overall	PASS	Return win fraction divided by service loss fraction; no averaging of match ratios
Outcome accounting: Novak Djokovic / Clay	PASS	87 wins + 37 losses + 5 unknown = 129 charted matches
Source ratio: Novak Djokovic / Clay / match_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Novak Djokovic / Clay / point_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Novak Djokovic / Clay / first_serve_in_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Novak Djokovic / Clay / first_serve_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Novak Djokovic / Clay / second_serve_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Novak Djokovic / Clay / service_point_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Novak Djokovic / Clay / return_point_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Novak Djokovic / Clay / ace_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Novak Djokovic / Clay / double_fault_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Novak Djokovic / Clay / break_point_save_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Novak Djokovic / Clay / break_point_conversion_pct	PASS	Independent counts on documented KPI-specific eligible match keys

check	status	evidence
Source ratio: Novak Djokovic / Clay / service_hold_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Novak Djokovic / Clay / return_break_pct	PASS	Independent counts on documented KPI-specific eligible match keys
DR from original common-sample counts: Novak Djokovic / Clay	PASS	Return win fraction divided by service loss fraction; no averaging of match ratios
Outcome accounting: Novak Djokovic / Grass	PASS	44 wins + 12 losses + 2 unknown = 58 charted matches
Source ratio: Novak Djokovic / Grass / match_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Novak Djokovic / Grass / point_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Novak Djokovic / Grass / first_serve_in_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Novak Djokovic / Grass / first_serve_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Novak Djokovic / Grass / second_serve_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Novak Djokovic / Grass / service_point_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Novak Djokovic / Grass / return_point_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Novak Djokovic / Grass / ace_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Novak Djokovic / Grass / double_fault_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Novak Djokovic / Grass / break_point_save_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Novak Djokovic / Grass / break_point_conversion_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Novak Djokovic / Grass / service_hold_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Novak Djokovic / Grass / return_break_pct	PASS	Independent counts on documented KPI-specific eligible match keys
DR from original common-sample counts: Novak Djokovic / Grass	PASS	Return win fraction divided by service loss fraction; no averaging of match ratios
Outcome accounting: Novak Djokovic / Hard	PASS	273 wins + 73 losses + 20 unknown = 366 charted matches
Source ratio: Novak Djokovic / Hard / match_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Novak Djokovic / Hard / point_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Novak Djokovic / Hard / first_serve_in_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Novak Djokovic / Hard / first_serve_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Novak Djokovic / Hard / second_serve_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Novak Djokovic / Hard / service_point_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Novak Djokovic / Hard / return_point_win_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Novak Djokovic / Hard / ace_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Novak Djokovic / Hard / double_fault_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Novak Djokovic / Hard / break_point_save_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Novak Djokovic / Hard / break_point_conversion_pct	PASS	Independent counts on documented KPI-specific eligible match keys

check	status	evidence
Source ratio: Novak Djokovic / Hard / service_hold_pct	PASS	Independent counts on documented KPI-specific eligible match keys
Source ratio: Novak Djokovic / Hard / return_break_pct	PASS	Independent counts on documented KPI-specific eligible match keys
DR from original common-sample counts: Novak Djokovic / Hard	PASS	Return win fraction divided by service loss fraction; no averaging of match ratios
Eligible service games have stable state and server	PASS	All points in eligible games, not terminal rows alone
Completed TB score accumulation: player 1	PASS	Independent cumulative point wins through server changes
Completed TB score accumulation: player 2	PASS	Independent cumulative point wins through server changes
Full CI cohort/scale: Roger Federer / Overall	PASS	Pooled components on one common cohort; proportions, not percentages
Full CI cohort/scale: Roger Federer / Clay	PASS	Pooled components on one common cohort; proportions, not percentages
Full CI cohort/scale: Roger Federer / Grass	PASS	Pooled components on one common cohort; proportions, not percentages
Full CI cohort/scale: Roger Federer / Hard	PASS	Pooled components on one common cohort; proportions, not percentages
Full CI cohort/scale: Rafael Nadal / Overall	PASS	Pooled components on one common cohort; proportions, not percentages
Full CI cohort/scale: Rafael Nadal / Clay	PASS	Pooled components on one common cohort; proportions, not percentages
Full CI cohort/scale: Rafael Nadal / Grass	PASS	Pooled components on one common cohort; proportions, not percentages
Full CI cohort/scale: Rafael Nadal / Hard	PASS	Pooled components on one common cohort; proportions, not percentages
Full CI cohort/scale: Novak Djokovic / Overall	PASS	Pooled components on one common cohort; proportions, not percentages
Full CI cohort/scale: Novak Djokovic / Clay	PASS	Pooled components on one common cohort; proportions, not percentages
Full CI cohort/scale: Novak Djokovic / Grass	PASS	Pooled components on one common cohort; proportions, not percentages
Full CI cohort/scale: Novak Djokovic / Hard	PASS	Pooled components on one common cohort; proportions, not percentages
Known deciding-set winner agrees with match winner	PASS	Completed supported deciding sets only; match tiebreaks excluded
Bootstrap seed reproduces every draw	PASS	One shared weight per match ID retains both rivalry observations
Bootstrap interval: Roger Federer / Overall / match_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Roger Federer / Overall / point_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Roger Federer / Overall / first_serve_in_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Roger Federer / Overall / first_serve_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Roger Federer / Overall / second_serve_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Roger Federer / Overall / service_point_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Roger Federer / Overall / return_point_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Roger Federer / Overall / ace_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Roger Federer / Overall / double_fault_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Roger Federer / Overall / break_point_save_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals

check	status	evidence
Bootstrap interval: Roger Federer / Overall / break_point_conversion_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Roger Federer / Overall / service_hold_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Roger Federer / Overall / return_break_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Roger Federer / Overall / dominance_ratio	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Roger Federer / Clay / match_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Roger Federer / Clay / point_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Roger Federer / Clay / first_serve_in_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Roger Federer / Clay / first_serve_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Roger Federer / Clay / second_serve_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Roger Federer / Clay / service_point_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Roger Federer / Clay / return_point_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Roger Federer / Clay / ace_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Roger Federer / Clay / double_fault_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Roger Federer / Clay / break_point_save_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Roger Federer / Clay / break_point_conversion_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Roger Federer / Clay / service_hold_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Roger Federer / Clay / return_break_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Roger Federer / Clay / dominance_ratio	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Roger Federer / Grass / match_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Roger Federer / Grass / point_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Roger Federer / Grass / first_serve_in_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Roger Federer / Grass / first_serve_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Roger Federer / Grass / second_serve_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Roger Federer / Grass / service_point_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Roger Federer / Grass / return_point_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Roger Federer / Grass / ace_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Roger Federer / Grass / double_fault_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Roger Federer / Grass / break_point_save_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Roger Federer / Grass / break_point_conversion_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals

check	status	evidence
Bootstrap interval: Roger Federer / Grass / service_hold_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Roger Federer / Grass / return_break_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Roger Federer / Grass / dominance_ratio	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Roger Federer / Hard / match_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Roger Federer / Hard / point_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Roger Federer / Hard / first_serve_in_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Roger Federer / Hard / first_serve_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Roger Federer / Hard / second_serve_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Roger Federer / Hard / service_point_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Roger Federer / Hard / return_point_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Roger Federer / Hard / ace_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Roger Federer / Hard / double_fault_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Roger Federer / Hard / break_point_save_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Roger Federer / Hard / break_point_conversion_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Roger Federer / Hard / service_hold_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Roger Federer / Hard / return_break_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Roger Federer / Hard / dominance_ratio	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Rafael Nadal / Overall / match_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Rafael Nadal / Overall / point_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Rafael Nadal / Overall / first_serve_in_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Rafael Nadal / Overall / first_serve_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Rafael Nadal / Overall / second_serve_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Rafael Nadal / Overall / service_point_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Rafael Nadal / Overall / return_point_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Rafael Nadal / Overall / ace_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Rafael Nadal / Overall / double_fault_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Rafael Nadal / Overall / break_point_save_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Rafael Nadal / Overall / break_point_conversion_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Rafael Nadal / Overall / service_hold_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Rafael Nadal / Overall / return_break_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals

check	status	evidence
Bootstrap interval: Rafael Nadal / Overall / dominance_ratio	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Rafael Nadal / Clay / match_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Rafael Nadal / Clay / point_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Rafael Nadal / Clay / first_serve_in_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Rafael Nadal / Clay / first_serve_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Rafael Nadal / Clay / second_serve_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Rafael Nadal / Clay / service_point_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Rafael Nadal / Clay / return_point_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Rafael Nadal / Clay / ace_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Rafael Nadal / Clay / double_fault_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Rafael Nadal / Clay / break_point_save_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Rafael Nadal / Clay / break_point_conversion_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Rafael Nadal / Clay / service_hold_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Rafael Nadal / Clay / return_break_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Rafael Nadal / Clay / dominance_ratio	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Rafael Nadal / Grass / match_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Rafael Nadal / Grass / point_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Rafael Nadal / Grass / first_serve_in_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Rafael Nadal / Grass / first_serve_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Rafael Nadal / Grass / second_serve_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Rafael Nadal / Grass / service_point_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Rafael Nadal / Grass / return_point_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Rafael Nadal / Grass / ace_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Rafael Nadal / Grass / double_fault_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Rafael Nadal / Grass / break_point_save_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Rafael Nadal / Grass / break_point_conversion_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Rafael Nadal / Grass / service_hold_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Rafael Nadal / Grass / return_break_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Rafael Nadal / Grass / dominance_ratio	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Rafael Nadal / Hard / match_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Rafael Nadal / Hard / point_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals

check	status	evidence
Bootstrap interval: Rafael Nadal / Hard / first_serve_in_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Rafael Nadal / Hard / first_serve_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Rafael Nadal / Hard / second_serve_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Rafael Nadal / Hard / service_point_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Rafael Nadal / Hard / return_point_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Rafael Nadal / Hard / ace_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Rafael Nadal / Hard / double_fault_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Rafael Nadal / Hard / break_point_save_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Rafael Nadal / Hard / break_point_conversion_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Rafael Nadal / Hard / service_hold_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Rafael Nadal / Hard / return_break_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Rafael Nadal / Hard / dominance_ratio	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Novak Djokovic / Overall / match_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Novak Djokovic / Overall / point_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Novak Djokovic / Overall / first_serve_in_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Novak Djokovic / Overall / first_serve_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Novak Djokovic / Overall / second_serve_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Novak Djokovic / Overall / service_point_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Novak Djokovic / Overall / return_point_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Novak Djokovic / Overall / ace_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Novak Djokovic / Overall / double_fault_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Novak Djokovic / Overall / break_point_save_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Novak Djokovic / Overall / break_point_conversion_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Novak Djokovic / Overall / service_hold_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Novak Djokovic / Overall / return_break_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Novak Djokovic / Overall / dominance_ratio	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Novak Djokovic / Clay / match_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Novak Djokovic / Clay / point_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Novak Djokovic / Clay / first_serve_in_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Novak Djokovic / Clay / first_serve_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Novak Djokovic / Clay / second_serve_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals

check	status	evidence
Bootstrap interval: Novak Djokovic / Clay / service_point_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Novak Djokovic / Clay / return_point_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Novak Djokovic / Clay / ace_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Novak Djokovic / Clay / double_fault_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Novak Djokovic / Clay / break_point_save_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Novak Djokovic / Clay / break_point_conversion_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Novak Djokovic / Clay / service_hold_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Novak Djokovic / Clay / return_break_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Novak Djokovic / Clay / dominance_ratio	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Novak Djokovic / Grass / match_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Novak Djokovic / Grass / point_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Novak Djokovic / Grass / first_serve_in_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Novak Djokovic / Grass / first_serve_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Novak Djokovic / Grass / second_serve_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Novak Djokovic / Grass / service_point_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Novak Djokovic / Grass / return_point_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Novak Djokovic / Grass / ace_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Novak Djokovic / Grass / double_fault_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Novak Djokovic / Grass / break_point_save_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Novak Djokovic / Grass / break_point_conversion_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Novak Djokovic / Grass / service_hold_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Novak Djokovic / Grass / return_break_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Novak Djokovic / Grass / dominance_ratio	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Novak Djokovic / Hard / match_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Novak Djokovic / Hard / point_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Novak Djokovic / Hard / first_serve_in_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Novak Djokovic / Hard / first_serve_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Novak Djokovic / Hard / second_serve_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Novak Djokovic / Hard / service_point_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Novak Djokovic / Hard / return_point_win_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals

check	status	evidence
Bootstrap interval: Novak Djokovic / Hard / ace_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Novak Djokovic / Hard / double_fault_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Novak Djokovic / Hard / break_point_save_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Novak Djokovic / Hard / break_point_conversion_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Novak Djokovic / Hard / service_hold_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Novak Djokovic / Hard / return_break_pct	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Bootstrap interval: Novak Djokovic / Hard / dominance_ratio	PASS	Recomputed percentile endpoints, including unavailable sparse intervals
Clustered covariance independently recomputed: BP save	PASS	Solve-based match-score sandwich; rivalry TB rows share one cluster
Logistic score equation and design: BP save	PASS	Unpenalized explanatory logit; no global intercept
Pressure contrast: BP save / Roger Federer / Rafael Nadal	PASS	Coefficient difference and full covariance, not separate marginal intervals
Pressure contrast: BP save / Roger Federer / Novak Djokovic	PASS	Coefficient difference and full covariance, not separate marginal intervals
Pressure contrast: BP save / Rafael Nadal / Novak Djokovic	PASS	Coefficient difference and full covariance, not separate marginal intervals
Opponent context excludes current match: BP save	PASS	Direct reaggregation of all other eligible local Overview matches; retrospective, not forecasting
Clustered covariance independently recomputed: TB point	PASS	Solve-based match-score sandwich; rivalry TB rows share one cluster
Logistic score equation and design: TB point	PASS	Unpenalized explanatory logit; no global intercept
Pressure contrast: TB point / Roger Federer / Rafael Nadal	PASS	Coefficient difference and full covariance, not separate marginal intervals
Pressure contrast: TB point / Roger Federer / Novak Djokovic	PASS	Coefficient difference and full covariance, not separate marginal intervals
Pressure contrast: TB point / Rafael Nadal / Novak Djokovic	PASS	Coefficient difference and full covariance, not separate marginal intervals
Opponent context excludes current match: TB point	PASS	Direct reaggregation of all other eligible local Overview matches; retrospective, not forecasting
Entropy from category counts: Roger Federer / Overall / shot_type	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Roger Federer / Clay / shot_type	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Roger Federer / Grass / shot_type	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Roger Federer / Hard / shot_type	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Rafael Nadal / Overall / shot_type	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Rafael Nadal / Clay / shot_type	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Rafael Nadal / Grass / shot_type	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Rafael Nadal / Hard / shot_type	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Novak Djokovic / Overall / shot_type	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Novak Djokovic / Clay / shot_type	PASS	Fixed K; unknown types excluded; no performance interpretation

check	status	evidence
Entropy from category counts: Novak Djokovic / Grass / shot_type	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Novak Djokovic / Hard / shot_type	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Roger Federer / Overall / target_all_attempts	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Roger Federer / Clay / target_all_attempts	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Roger Federer / Grass / target_all_attempts	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Roger Federer / Hard / target_all_attempts	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Rafael Nadal / Overall / target_all_attempts	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Rafael Nadal / Clay / target_all_attempts	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Rafael Nadal / Grass / target_all_attempts	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Rafael Nadal / Hard / target_all_attempts	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Novak Djokovic / Overall / target_all_attempts	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Novak Djokovic / Clay / target_all_attempts	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Novak Djokovic / Grass / target_all_attempts	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Novak Djokovic / Hard / target_all_attempts	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Roger Federer / Overall / target_non_error_strokes	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Roger Federer / Clay / target_non_error_strokes	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Roger Federer / Grass / target_non_error_strokes	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Roger Federer / Hard / target_non_error_strokes	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Rafael Nadal / Overall / target_non_error_strokes	PASS	Fixed K; unknown types excluded; no performance interpretation

check	status	evidence
Entropy from category counts: Rafael Nadal / Clay / target_non_error_strokes	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Rafael Nadal / Grass / target_non_error_strokes	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Rafael Nadal / Hard / target_non_error_strokes	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Novak Djokovic / Overall / target_non_error_strokes	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Novak Djokovic / Clay / target_non_error_strokes	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Novak Djokovic / Grass / target_non_error_strokes	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Novak Djokovic / Hard / target_non_error_strokes	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Roger Federer / Overall / target_returns	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Roger Federer / Clay / target_returns	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Roger Federer / Grass / target_returns	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Roger Federer / Hard / target_returns	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Rafael Nadal / Overall / target_returns	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Rafael Nadal / Clay / target_returns	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Rafael Nadal / Grass / target_returns	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Rafael Nadal / Hard / target_returns	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Novak Djokovic / Overall / target_returns	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Novak Djokovic / Clay / target_returns	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Novak Djokovic / Grass / target_returns	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Novak Djokovic / Hard / target_returns	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Roger Federer / Overall / source_direction_Total	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Roger Federer / Clay / source_direction_Total	PASS	Fixed K; unknown types excluded; no performance interpretation

check	status	evidence
Entropy from category counts: Roger Federer / Grass / source_direction_Total	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Roger Federer / Hard / source_direction_Total	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Rafael Nadal / Overall / source_direction_Total	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Rafael Nadal / Clay / source_direction_Total	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Rafael Nadal / Grass / source_direction_Total	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Rafael Nadal / Hard / source_direction_Total	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Novak Djokovic / Overall / source_direction_Total	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Novak Djokovic / Clay / source_direction_Total	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Novak Djokovic / Grass / source_direction_Total	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Novak Djokovic / Hard / source_direction_Total	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Roger Federer / Overall / source_direction_F	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Roger Federer / Clay / source_direction_F	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Roger Federer / Grass / source_direction_F	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Roger Federer / Hard / source_direction_F	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Rafael Nadal / Overall / source_direction_F	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Rafael Nadal / Clay / source_direction_F	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Rafael Nadal / Grass / source_direction_F	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Rafael Nadal / Hard / source_direction_F	PASS	Fixed K; unknown types excluded; no performance interpretation

check	status	evidence
Entropy from category counts: Novak Djokovic / Overall / source_direction_F	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Novak Djokovic / Clay / source_direction_F	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Novak Djokovic / Grass / source_direction_F	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Novak Djokovic / Hard / source_direction_F	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Roger Federer / Overall / source_direction_B	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Roger Federer / Clay / source_direction_B	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Roger Federer / Grass / source_direction_B	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Roger Federer / Hard / source_direction_B	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Rafael Nadal / Overall / source_direction_B	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Rafael Nadal / Clay / source_direction_B	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Rafael Nadal / Grass / source_direction_B	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Rafael Nadal / Hard / source_direction_B	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Novak Djokovic / Overall / source_direction_B	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Novak Djokovic / Clay / source_direction_B	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Novak Djokovic / Grass / source_direction_B	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Novak Djokovic / Hard / source_direction_B	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Roger Federer / Overall / source_direction_S	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Roger Federer / Clay / source_direction_S	PASS	Fixed K; unknown types excluded; no performance interpretation

check	status	evidence
Entropy from category counts: Roger Federer / Grass / source_direction_S	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Roger Federer / Hard / source_direction_S	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Rafael Nadal / Overall / source_direction_S	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Rafael Nadal / Clay / source_direction_S	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Rafael Nadal / Grass / source_direction_S	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Rafael Nadal / Hard / source_direction_S	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Novak Djokovic / Overall / source_direction_S	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Novak Djokovic / Clay / source_direction_S	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Novak Djokovic / Grass / source_direction_S	PASS	Fixed K; unknown types excluded; no performance interpretation
Entropy from category counts: Novak Djokovic / Hard / source_direction_S	PASS	Fixed K; unknown types excluded; no performance interpretation
SAS from verified surface outcomes: Roger Federer	PASS	Three named surfaces; pooled anchor, 60/40 declared weights
SAS from verified surface outcomes: Rafael Nadal	PASS	Three named surfaces; pooled anchor, 60/40 declared weights
SAS from verified surface outcomes: Novak Djokovic	PASS	Three named surfaces; pooled anchor, 60/40 declared weights
All tactical period rates reaggregate	PASS	Individual four-year bins; explicit masks and numerator/denominator pooling
Standardized trends require all surfaces	PASS	Fixed 1/3 surface weights; absent grass not interpolated
All charted H2H pairs and surface splits verified	PASS	Distinct matches; unknown outcomes retained separately; not official career records
Preference arithmetic: Balanced	PASS	Unchanged fixed scales, declared weights, and fully paired draws
First-place frequencies: Balanced	PASS	Independently counted maxima with fractional tie credit
Preference arithmetic: Winning priority	PASS	Unchanged fixed scales, declared weights, and fully paired draws
First-place frequencies: Winning priority	PASS	Independently counted maxima with fractional tie credit
Preference arithmetic: Dominance priority	PASS	Unchanged fixed scales, declared weights, and fully paired draws
First-place frequencies: Dominance priority	PASS	Independently counted maxima with fractional tie credit
Preference arithmetic: Clutch priority	PASS	Unchanged fixed scales, declared weights, and fully paired draws

check	status	evidence
First-place frequencies: Clutch priority	PASS	Independently counted maxima with fractional tie credit
Preference arithmetic: Adaptability priority	PASS	Unchanged fixed scales, declared weights, and fully paired draws
First-place frequencies: Adaptability priority	PASS	Independently counted maxima with fractional tie credit
All paired scenario differences verified	PASS	Differences within the same draws; no subtraction of marginal interval endpoints

```
Independent verification passed: 492 checks; source counts, units, event cohorts, model covariance, resampling and scoring.
```

In [37]:

```
release_checks=[]
def release_check(name,condition):
    release_checks.append({"check":name,"status":"PASS" if bool(condition) else "FAIL"})
    if not condition:
        raise AssertionError(name)

# Re-read each CSV to verify that the preserved original frames were not changed in memory.
for filename in csv_filenames:
    key=Path(filename).stem.replace("-","_")
    original_again=pd.read_csv(raw_data_dir/filename,low_memory=False)
    pd.testing.assert_frame_equal(raw_dataframes_unfiltered[key],original_again)
release_check("All 19 original input frames equal freshly read CSVs",len(raw_dataframes_unfiltered)==19)
release_check("Shared audit accounting and hard key checks pass",integrity_check_results["failures"].sum()==0)
release_check("Canonical player-match spine is unique",not canonical_master.duplicated(["match_id","player"]).any())
release_check("Unknown outcomes stay missing rather than losses",canonical_master.loc[canonical_master["winner"].isna(),"is_win"].isna().all())
release_check("Core performance keys are unique",not core_kpi_long.duplicated(["player","surface","metric"]).any())
release_check("Full CI remains within 0–1",clutch_summary["clutch_index_full"].dropna().between(0,1).all())
release_check("SAS remains within 0–100",surface_adaptability_summary_v2["surface_adaptability_score"].dropna().between(0,100).all())
release_check("Standardized trends require every
surface",tactical_standardized_trends.loc[tactical_standardized_trends["surfaces_available"].lt(3),"standardized_pct"].isna().all())
release_check("Primary scenario rank frequencies sum to 100",np.allclose(comparison_scores.groupby("scenario")["bootstrap_first_place_pct"].sum(),100))
release_check("Consolidated scorecard has 12 unique player/scope rows",len(final_player_scorecard)==12 and not
final_player_scorecard.duplicated(["player","surface"]).any())
release_check("Single-surface SAS is not
fabricated",final_player_scorecard.loc[final_player_scorecard["surface"].ne("Overall"),"surface_adaptability_score"].isna().all())
release_check("Ranking weeks remain nullable numeric with unavailable status",tableau_exports["players"]["weeks_at_no_1"].isna().all())
release_check("All registered export tables passed unique-key and CSV round-trip checks",len(tableau_exports)==len(export_specs)==len(csv_payloads))
release_check("All 21 local sources have recorded fingerprints",len(source_manifest)==21 and source_manifest["sha256"].str.len().eq(64).all())

expected_hands={"Roger Federer":"Right-handed","Rafael Nadal":"Left-handed","Novak Djokovic":"Right-handed"}
release_check("Player profile handedness regression",player_metadata.set_index("player")["handedness"].to_dict()==expected_hands)
release_check("All raw handedness conflicts retained",len(handedness_conflicts)==int(player_metadata["conflicting_records"].sum()))
for table_name,table in [("coverage",core_kpi_coverage),("scorecard",final_player_scorecard),("players export",tableau_exports["players"]),("scorecard
export",tableau_exports["scorecard"])]:
    release_check("Handedness consistent: "+table_name,table["handedness"].eq(table["player"].map(expected_hands)).all())
ratio_rows=core_kpi_long.loc[core_kpi_long["unit"].eq("%")&core_kpi_long["estimate"].notna()]
release_check("Headline rate numerators/denominators reproduce estimates",ratio_rows["denominator"].gt(0).all() and
ratio_rows["numerator"].between(0,ratio_rows["denominator"]).all() and
np.allclose(ratio_rows["estimate"],100*ratio_rows["numerator"]/ratio_rows["denominator"]))
for player in BIG_THREE:
    total=core_kpi_coverage.loc[core_kpi_coverage.player.eq(player)&core_kpi_coverage.surface.eq("Overall")].iloc[0]
    pieces=core_kpi_coverage.loc[core_kpi_coverage.player.eq(player)&core_kpi_coverage.surface.isin(PROJECT_SURFACES)]
```

```
    other=canonical_master.loc[canonical_master.player.eq(player)&~canonical_master.surface.isin(PROJECT_SURFACES)]
    release_check("Surface + other scope accounting: "+player,int(pieces.charted_matches.sum())+len(other)==total.charted_matches)
release_check("CI component counts bounded and common cohort respected",all(clutch_summary[num].dropna().le(clutch_summary[den].dropna()).all() for num,den,_ in
CI_COMPONENTS.values()) and clutch_summary.common_ci_matches.le(clutch_summary.cohort_matches).all())
release_check("Pressure observations have unique player-point keys",not bp_model_events.duplicated(["match_id","Pt","player"]).any() and not
tb_model_events.duplicated(["match_id","Pt","player"]).any())
release_check("Completed tiebreaks have valid score progression",validated_tiebreaks.loc[validated_tiebreaks.tb_complete_validated,"all_tb_rows_valid"].all())
release_check("Hold/break games satisfy completion and transition rules",kpi_games.loc[kpi_games.eligible_regular_game,"all_transitions_valid"].all() and not
kpi_games.loc[kpi_games.eligible_regular_game,"is_tiebreak_point"].any())
release_check("Paired bootstrap coverage and dimensions",comparison_draws_valid.shape==(int(comparison_valid_draw.sum()),len(BIG_THREE),len(COMPARISON_METRICS))
and comparison_valid_draw.sum()>=.95*BOOTSTRAP_REPLICATES)
release_check("Composite contributions reproduce primary
scores",np.allclose(comparison_contributions.groupby(["scenario","player"]).score_contribution.sum().sort_index(),comparison_scores.set_index(["scenario","playe
r"]).score.sort_index()))
release_check("Saved opening claims match current computed leaders",executive_findings.player.tolist()==[BIG_THREE[0],BIG_THREE[0],BIG_THREE[2],BIG_THREE[1]]
and comparison_scores.loc[comparison_scores.scenario.eq("Balanced")].sort_values("score",ascending=False).iloc[0].player==BIG_THREE[0] and
comparison_scores.loc[comparison_scores.scenario.eq("Adaptability priority")].sort_values("score",ascending=False).iloc[0].player==BIG_THREE[2] and not
pressure_pairwise.p_value_holm_six_tests.lt(.05).any())
release_check("Narrow balanced lead remains uncertain in paired difference",paired.ci95_lower<0<paired.ci95_upper and abs(paired.point_score_difference)<1)
release_check("All export player IDs reference the player dimension",all(set(frame[col].dropna()).issubset(set(PLAYER_IDS.values())) for frame in
tableau_exports.values() for col in ["player_id","player_a_id","player_b_id"] if col in frame))
release_check("All export scopes are valid",all(frame["scope"].isin(["Overall"]+PROJECT_SURFACES).all() for frame in tableau_exports.values() if "scope" in
frame))
release_check("Period facts reference registered individual
periods",all(set(map(tuple,frame[["player_id","period_index"]].itertuples(index=False,name=None))).issubset(set(tableau_exports["periods"]
[["player_id","period_index"]].itertuples(index=False,name=None))) for frame in tableau_exports.values() if "period_index" in frame))
release_check("No infinite numeric output in registered tables",all(not
np.isinf(frame.select_dtypes(include="number").to_numpy(dtype=float,na_value=np.nan)).any() for frame in tableau_exports.values()))
release_check("No unhandled blocking source defect",not source_issue_disposition.status.eq("Blocking defect").any())
release_check("Export bundle verified on disk when enabled",not WRITE_TABLEAU_EXPORTS or
all(hashlib.sha256((TABLEAU_EXPORT_DIR/(name+".csv")).read_bytes()).hexdigest()==tableau_export_registry.set_index("table").loc[name,"csv_sha256"] for name in
tableau_exports))


release_check("Independent source/count/model verification passes",independent_audit_results.status.eq("PASS").all() and len(independent_audit_results)>0)
release_check("Export directory agrees with current registry",not WRITE_TABLEAU_EXPORTS or {p.name for p in TABLEAU_EXPORT_DIR.glob("*.csv")}=={name+".csv" for
name in tableau_exports}|{"field_catalog.csv","table_registry.csv"})
release_check("Pressure score residuals and covariance are well behaved",pressure_numerical_diagnostics.max_absolute_score_per_observation.lt(1e-5).all() and
pressure_numerical_diagnostics.minimum_covariance_eigenvalue.gt(0).all())
release_check("Shared statistical conflicts exclude no Big Three player",len(conflicted_big_three_player_matches)==0)
release_validation=pd.DataFrame(release_checks)

report_table(release_validation,expanded=True)
report_run_manifest=pd.DataFrame([{"report_version":REPORT_VERSION,"snapshot_preparation_date":REPORT_SNAPSHOT_DATE,
    "input_snapshot_sha256":source_snapshot_sha256,"seed":RANDOM_SEED,"bootstrap_replicates":BOOTSTRAP_REPLICATES,
    "original_source_tables":len(raw_dataframes_unfiltered),"canonical_player_matches":len(canonical_master),
    "source_review_flags_retained":len(numeric_quality_flags),"tableau_data_tables":len(tableau_exports),
    "export_files_written":WRITE_TABLEAU_EXPORTS,"release_assertions_passed":len(release_validation)}])
report_table(report_run_manifest,expanded=True)
print("All release assertions passed. Analysis tables and static figures are saved in the notebook; input review flags and inferential limits remain
disclosed.")

ANALYTICAL_RESULT_NAMES = {
    "performance":["performance_focus","surface_evidence","core_kpi_long","core_kpi_coverage","core_kpi_wide","core_kpi_overall","core_kpi_by_surface"],
    "pressure":
["pressure_headline","clutch_summary","clutch_component_evidence","clutch_overall","clutch_by_surface","pressure_model_diagnostics","pressure_model_coefficients
","pressure_pairwise","pressure_reference_profiles","pressure_event_coverage","pressure_numerical_diagnostics","pressure_context_effects"],
```

```
    "shots":["shot_headline","shot_distribution_long","shot_diversity_summary","shot_coverage_summary"],
    "adaptability/evolution":
["sas_headline","selected_changes","surface_adaptability_summary_v2","surface_adaptability_weight_sensitivity","tactical_period_plan","tactical_evolution_long",
"tactical_standardized_trends","tactical_change_summary"],
    "style":["style_profile_long","style_profile_wide","playing_style_profiles"],
    "comparison":
["executive_findings","scenario_leader_audit","comparison_inputs","comparison_weights","comparison_scores","comparison_contributions","comparison_pairwise_diffe
rences","comparison_weight_grid","comparison_grid_summary","comparison_scaling_sensitivity","comparison_leave_one_out","surface_preference_comparison","comparis
on_weight_sweeps","normalization_audit"],
    "rivalry/scorecard":["h2h_overall","validated_h2h_summary","final_player_scorecard","final_overall_player_summary"]}
analytical_table_inventory=pd.DataFrame([{"section":section,"table_or_view":name,"rows":len(globals()[name]),"columns":len(globals()[name].columns)}
    for section,names in ANALYTICAL_RESULT_NAMES.items() for name in names])
report_table(analytical_table_inventory,caption="Named analytical result tables/views; long/wide views are not independent analyses")
print(f"Analytical result tables/views: {len(analytical_table_inventory)}. Export datasets: {len(tableau_exports)}; CSV files written when enabled:
{len(tableau_exports)+2} (data plus field catalog/registry). Figures and rendered table displays are counted separately from the saved notebook during release
review.")
```

check	status
All 19 original input frames equal freshly read CSVs	PASS
Shared audit accounting and hard key checks pass	PASS
Canonical player-match spine is unique	PASS
Unknown outcomes stay missing rather than losses	PASS
Core performance keys are unique	PASS
Full CI remains within 0–1	PASS
SAS remains within 0–100	PASS
Standardized trends require every surface	PASS
Primary scenario rank frequencies sum to 100	PASS
Consolidated scorecard has 12 unique player/scope rows	PASS
Single-surface SAS is not fabricated	PASS
Ranking weeks remain nullable numeric with unavailable status	PASS
All registered export tables passed unique-key and CSV round-trip checks	PASS
All 21 local sources have recorded fingerprints	PASS
Player profile handedness regression	PASS
All raw handedness conflicts retained	PASS
Handedness consistent: coverage	PASS
Handedness consistent: scorecard	PASS
Handedness consistent: players export	PASS
Handedness consistent: scorecard export	PASS
Headline rate numerators/denominators reproduce estimates	PASS
Surface + other scope accounting: Roger Federer	PASS
Surface + other scope accounting: Rafael Nadal	PASS
Surface + other scope accounting: Novak Djokovic	PASS
CI component counts bounded and common cohort respected	PASS
Pressure observations have unique player-point keys	PASS
Completed tiebreaks have valid score progression	PASS
Hold/break games satisfy completion and transition rules	PASS
Paired bootstrap coverage and dimensions	PASS
Composite contributions reproduce primary scores	PASS
Saved opening claims match current computed leaders	PASS
Narrow balanced lead remains uncertain in paired difference	PASS

check	status
All export player IDs reference the player dimension	PASS
All export scopes are valid	PASS
Period facts reference registered individual periods	PASS
No infinite numeric output in registered tables	PASS
No unhandled blocking source defect	PASS
Export bundle verified on disk when enabled	PASS
Independent source/count/model verification passes	PASS
Export directory agrees with current registry	PASS
Pressure score residuals and covariance are well behaved	PASS
Shared statistical conflicts exclude no Big Three player	PASS

report_version	snapshot_preparation_date	input_snapshot_sha256	seed	bootstrap_replicates	original_source_tables	canonical_player_matches
1.2	2026-10-17	1b253ea57274935af4e7102b6e83d97d534b8fc74b6cc1ed48effcc8c258eab7	42	2000	19	1701

source_review_flags_retained	tableau_data_tables	export_files_written	release_assertions_passed
12138	37	True	42

```
All release assertions passed. Analysis tables and static figures are saved in the notebook; input review flags and inferential limits remain disclosed.
```

▼ Named analytical result tables/views; long/wide views are not independent analyses

section	table_or_view	rows	columns
performance	performance_focus	18	9
performance	surface_evidence	9	8
performance	core_kpi_long	168	21
performance	core_kpi_coverage	12	14
performance	core_kpi_wide	12	28
performance	core_kpi_overall	3	28
performance	core_kpi_by_surface	9	28
pressure	pressure_headline	3	11
pressure	clutch_summary	12	31
pressure	clutch_component_evidence	36	10
pressure	clutch_overall	3	31
pressure	clutch_by_surface	9	31
pressure	pressure_model_diagnostics	2	12
pressure	pressure_model_coefficients	23	8
pressure	pressure_pairwise	6	7
pressure	pressure_reference_profiles	9	10
pressure	pressure_event_coverage	18	7
pressure	pressure_numerical_diagnostics	2	7
pressure	pressure_context_effects	2	6
shots	shot_headline	3	10
shots	shot_distribution_long	648	11
shots	shot_diversity_summary	96	11
shots	shot_coverage_summary	12	23
adaptability/evolution	sas_headline	3	8
adaptability/evolution	selected_changes	15	9
adaptability/evolution	surface_adaptability_summary_v2	3	28
adaptability/evolution	surface_adaptability_weight_sensitivity	33	5
adaptability/evolution	tactical_period_plan	18	10
adaptability/evolution	tactical_evolution_long	1008	16
adaptability/evolution	tactical_standardized_trends	252	10

section	table_or_view	rows	columns
adaptability/evolution	tactical_change_summary	42	9
style	style_profile_long	192	11
style	style_profile_wide	12	18
style	playing_style_profiles	12	3
comparison	executive_findings	4	2
comparison	scenario_leader_audit	5	4
comparison	comparison_inputs	12	7
comparison	comparison_weights	5	5
comparison	comparison_scores	15	11
comparison	comparison_contributions	60	6
comparison	comparison_pairwise_differences	15	6
comparison	comparison_weight_grid	286	11
comparison	comparison_grid_summary	3	3
comparison	comparison_scaling_sensitivity	15	4
comparison	comparison_leave_one_out	12	4
comparison	surface_preference_comparison	9	8
comparison	comparison_weight_sweeps	252	4
comparison	normalization_audit	12	10
rivalry/scorecard	h2h_overall	3	12
rivalry/scorecard	validated_h2h_summary	12	12
rivalry/scorecard	final_player_scorecard	12	51
rivalry/scorecard	final_overall_player_summary	28	3

Analytical result tables/views: 52. Export datasets: 37; CSV files written when enabled: 39 (data plus field catalog/registry). Figures and rendered table displays are counted separately from the saved notebook during release review.